YMNOTES · DIGESTIVE · MID-MODULE

Questions — Answer Key & Study Guide

Version 2 · for the teacher · every answer links to its line in the book

مفتاح الإجابات ودليل المذاكرة — النسخة الثانية للمدرّس

How to use · طريقة الاستخدام

- **Green row = correct answer.** Tap it: the book opens at the exact line, highlighted in green.
- At the bottom of that book page tap the green **< A·1** chip to return to the question.
- Each card: why it is right, why each other option is wrong, and a one-line **Tell the students** pearl.
- Every card shows its subject tag and the original chapter title.

- **الصف الأخضر** = **الإجابة الصحيحة**. اضغط عليه فيفتح الكتاب عند السطر نفسه مظللاً بالأخضر.
- أسفل صفحة الكتاب اضغط الشريحة الخضراء **A·1 <** للرجوع إلى السؤال.
- في كل سؤال: لماذا صحيحة، ولماذا باقي الاختيارات خطأ، ونقطة **قل للطلاب**.
- كل سؤال عليه التخصص وعنوان الفصل الأصلي.

SURGERY جراحة · INTERNAL MEDICINE باطنة · MICROBIOLOGY ميكروبيولوجي · PARASITOLOGY باراسيتولوجي · PHARMACOLOGY فارماكولوجي

Group index · فهرس المجموعات

Part 1 — Digestive System — Internal Medicine & Surgery · الجهاز الهضمي — باطنة وجراحة

A B C D E F G H I J K L M N O P Q R
S T U V W X Y Z AA AB AC AD AE AF AG AH AI AJ
AK AL AM AN AO AP AQ AR AS AT AU AV AW AX AY AZ

Part 2 — Microbiology · ميكروبيولوجي

BA BB BC

Part 3 — Parasitology · باراسيتولوجي

BD BE BF

Part 4 — Pharmacology · فارماكولوجي

BG BH BI BJ BK BL BM

Open the book ↗ Teacher notes ↗

PART 1

Digestive System — Internal Medicine & Surgery — الجهاز الهضمي — باطنة وجراحة

Group A

A·1 INTERNAL MEDICINE باطنة — Digestive System — Chapter 03: The Stomach and Duodenum

What is the most common cause of serious upper GI bleeding?

A. Esophageal varices
- × Second to PUD overall; it dominates only in cirrhosis.

أقل شيوعاً عموماً، تسود فقط في تليف الكبد.

B. Erosive gastritis
- × Causes bleeding but less often than ulcers.

تسبب نزيفاً لكن أقل من القرحة.

C. Peptic ulcer disease ✓ Book p. 50 ↗

D. Mallory-Weiss tear
- × A minority of cases, after forceful retching.

نسبة قليلة، بعد قيء شديد.

Why: Peptic ulcer disease (NSAID/aspirin and H. pylori related) is the leading cause of serious upper GI bleeding.

لماذا: القرحة الهضمية (بسبب NSAIDs أو H. pylori) هي أشهر سبب لنزيف الجهاز الهضمي العلوي الخطير.

★ **Tell the students:** In cirrhotics (and in Egypt) varices are the commonest cause of MASSIVE hematemesis.

قل للطلاب: في مرضى التليف (ومصر) الدوالي هي أشهر سبب للقيء الدموي الغزير.

A·2 SURGERY جراحة — Digestive System — Chapter 03: The Stomach and Duodenum

What is the mechanism of death in postpartum acute gastric dilatation?

A. Pulmonary embolism
- × Not the mechanism of AGD death.

ليست آلية الوفاة في AGD.

B. Aspiration of gastric contents
- × Aspiration can occur but is not the described mechanism.

قد تحدث لكنها ليست الآلية الأساسية.

C. Acute renal failure
- × Renal failure is a late result of shock.

الفشل الكلوي نتيجة متأخرة للصدمة.

D. Ischemia and necrosis of the stomach ✓ Book p. 68 ↗

Why: The stomach distends, wall perfusion falls, then ischemic necrosis and perforation cause collapse and death.

لماذا: المعدة تتمدد فيقل تروية الجدار ثم يحدث نخر إقفاري وانثقاب وانهيار دوراني فالوفاة.

★ **Tell the students:** Mortality is 80-100% once ischemia or perforation develops, so decompress early.

قل للطلاب: الوفيات 80-100% بعد الإقفار أو الانثقاب، لذا يجب تفريغ المعدة مبكراً.

A·3 INTERNAL MEDICINE باطنة

Digestive System — Chapter 03: The Stomach and Duodenum · Digestive System — Chapter 04: Pathology of GIT Tumours

Which gastric ulcer is most liable to be associated with stomach cancer?

A. Gastric ulcer associated with atrophic gastritis ✓ Book p. 62 ↗

B. Duodenal ulcer without gastric inflammation
× Duodenal ulcers are almost never malignant.
قرحة الاثني عشر نادراً ما تتسرطن.

C. Gastric ulcer caused by acute stress
× Acute stress ulcers are not premalignant.
قرح الإجهاد الحادة ليست سابقة للسرطان.

D. Gastric ulcer with normal surrounding mucosa
× Without a mucosal field change the risk is lower.
بدون تغيّر مخاطي حولها يقل الخطر.

Why: Atrophic (chronic) gastritis with intestinal metaplasia is the soil for gastric adenocarcinoma.
لماذا: التهاب المعدة الضموري مع الحؤول المعوي هو الأرضية المؤدية لسرطان المعدة.

★ **Tell the students:** Every gastric ulcer must be biopsied and followed until healed to exclude cancer.
قل للطلاب: كل قرحة معدة تؤخذ منها عينة وتتابع حتى الشفاء لاستبعاد السرطان.

Group B

B·1 SURGERY جراحة Hernia book — Chapter 09: Hernia Complications

What is a Richter's hernia?

A. Herniation of the entire circumference of the bowel
× Whole circumference is a complete (ordinary) hernia.
كامل محيط الأمعاء = فتق عادي.

B. Herniation of anti-mesenteric border of the bowel ✓ Hernia book — not in this file

C. Herniation of the mesenteric border only
× Not the mesenteric side.
ليس جهة المساريق.

D. Herniation of the stomach through the diaphragm
× That is a hiatus hernia.
هذا فتق حجابي.

Why: Richter's hernia traps only the ANTI-mesenteric border of the bowel wall, so the lumen stays open.
لماذا: فتق ريختر يحبس فقط الحافة المقابلة للمساريق من جدار الأمعاء فتبقى التجويف مفتوحاً.

★ **Tell the students:** It can strangulate and perforate WITHOUT obstruction, so a normal bowel pattern does not exclude it. Also: Littre = Meckel's, Amyand = appendix, Maydl = W-loop.
قل للطلاب: قد يختنق وينثقب بدون انسداد، فلا تطمئن لانتظام الإخراج. أيضاً: Littre ميكل، Amyand الزائدة، Maydl حلقتان في الكيس.

B·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

A patient with suspected celiac disease has normal total IgA and is still eating gluten. Which initial antibody test is generally preferred?

A. tTG-IgA ✓ Book p. 91 ↗

B. Anti-centromere antibody
× Scleroderma (CREST). مرتبط بالسكليرودرما.

C. IgG antimitochondrial antibody
× AMA = primary biliary cholangitis. AMA = التهاب الصفراوية الأولي.

D. IgM rheumatoid factor
× RF = rheumatoid arthritis. RF = الروماتويد.

Why: tTG-IgA, with total IgA, is the single preferred first test in adults who are still eating gluten.
لماذا: tTG-IgA مع قياس IgA الكلي هو الاختبار الأول المفضل لدى البالغين الذين ما زالوا يأكلون الجلوتين.

★ **Tell the students:** If IgA-deficient use IgG-based tests (tTG-IgG or DGP-IgG). The test is valid only on a gluten diet.
قل للطلاب: لو نقص IgA نستخدم اختبارات IgG. والاختبار صالح فقط أثناء تناول الجلوتين.

B·3 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

What is the most common indication for surgery in peptic ulcer disease?

A. Uncomplicated heartburn
× Heartburn without complications is treated medically. الحرقان غير المصحوب بمضاعفات علاجه دوائي.

B. Perforated ulcer ✓ Nearest: p. 53 ↗

C. Asymptomatic H. pylori infection
× Asymptomatic H. pylori needs eradication, not surgery. H. pylori بدون أعراض يُعالج بالاستئصال لا بالجراحة.

D. Recurrent bleeding ulcer
× A relative indication, only after failed endoscopic control. استطباب نسبي بعد فشل التحكم المنظاري.

Why: Perforation is the commonest indication for emergency ulcer surgery.
لماذا: الانثقاب هو أشهر سبب للجراحة الطارئة في القرحة.

★ **Tell the students:** Anterior duodenal ulcers perforate; posterior ones bleed (gastroduodenal artery).
قل للطلاب: قرحة الجدار الأمامي تنثقب، والخلفية تنزف (الشريان المعدي الاثني عشري).

B·4 INTERNAL MEDICINE باطنة

Digestive System — Chapter 02: The Esophagus · Digestive System — Chapter 04: Pathology of GIT Tumours

Barrett esophagus predisposes to which type of cancer?

A. Gastrointestinal stromal tumor
× GIST arises from cells of Cajal.
GIST ينشأ من خلايا كاجال.

B. Adenocarcinoma ✓ Book p. 71 ↗

C. Small-cell carcinoma
× Small-cell carcinoma is not linked to Barrett's.
الخلايا الصغيرة غير مرتبط بباريت.

D. Squamous cell carcinoma
× SCC links to smoking, alcohol and achalasia.
الحرشفي يرتبط بالتدخين والكحول والأكالازيا.

Why: Barrett's (columnar metaplasia) predisposes to adenocarcinoma, especially in the lower third.
لماذا: مريء باريت (تحوّل عمودي) يهيّئ للسرطان الغدي خاصة في الثلث السفلي.

★ **Tell the students:** Adenocarcinoma = lower third/GEJ (West); SCC = middle third (worldwide).
قل للطلاب: الغدي = الثلث السفلي/GEJ، والحرشفي = الثلث الأوسط.

Group C

C·1 SURGERY جراحة Digestive System — Chapter 11: Diverticulosis of the Colon

Which diet is associated with diverticulosis?

A. Low residue diet ✓ Book p. 146 ↗

B. High-protein diet
× Protein is not the factor; fibre is.
البروتين ليس العامل، بل قلة الألياف.

C. High-fat diet
× Fat is more linked to colon cancer.
الدهون ترتبط أكثر بسرطان القولون.

D. High-residue diet
× High residue protects.
الغذاء كثير الألياف وقائي.

Why: A low-residue (low-fibre) diet raises colonic intraluminal pressure and favors diverticula.
لماذا: الغذاء قليل الألياف يرفع ضغط القولون الداخلي ويساعد على تكوّن الرتوج.

★ **Tell the students:** Diverticulosis is usually asymptomatic, mostly sigmoid, and the rectum never has it.
قل للطلاب: الرتوج غالباً بلا أعراض وأغلبها في السيني، والمستقيم لا يصاب أبداً.

C·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 03: The Stomach and Duodenum

Which finding indicates shock in a patient with GI bleeding?

A. BP 110/70 mmHg with pulse 80 bpm
× Normal vitals.
علامات حيوية طبيعية.

B. BP 140/90 mmHg with pulse 60 bpm
× Hypertension with bradycardia is not shock.
ارتفاع ضغط مع بطء نبض ليس صدمة.

C. BP 90/60 mmHg with pulse 110 bpm ✓ Book p. 50 ↗

D. BP 130/80 mmHg with pulse 75 bpm
× Normal vitals.
علامات حيوية طبيعية.

Why: Hypotension (90/60) with tachycardia (110) means hypovolemic shock.

لماذا: هبوط الضغط (90/60) مع سرعة النبض (110) يدل على صدمة نقص حجم.

★ **Tell the students:** Shock index (pulse/systolic BP) above 1 = significant bleed; beta-blockers can mask tachycardia.

قل للطلاب: مؤشر الصدمة (النبض ÷ الانقباضي) فوق 1 = نزف كبير؛ حاصرات بيتا تخفي تسرع القلب.

C·3 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Which statement about stomach cancer is correct?

A. Barium meal is the definitive staging test
× Barium shows a defect; biopsy by endoscopy is definitive.
الباريوم يظهر عيباً فقط، والتشخيص بالمنظار والخزعة.

B. Serum CEA alone confirms the diagnosis and stage
× CEA/CA 72-4 are for follow-up, not diagnosis.
الواسمات للمتابعة لا للتشخيص.

C. Ultrasound is a basic tool important in staging ✓ Book p. 65 ↗

D. CT cannot detect distant metastases
× CT does detect liver, lung and nodal metastases.
الأشعة المقطعية تكشف نقائل الكبد والرئة والعقد.

Why: Abdominal ultrasound is a basic staging tool (liver metastases, ascites); endoscopy gives the diagnosis.

لماذا: الموجات فوق الصوتية أداة أساسية في التقسيم (نقائل الكبد والاستسقاء)؛ والمنظار يعطي التشخيص.

★ **Tell the students:** Staging = CT chest/abdomen +/- EUS +/- staging laparoscopy. CA 72-4 is the key marker for recurrence.

قل للطلاب: التقسيم = CT صدر وبطن ± EUS ± منظار بطن. و CA 72-4 أهم واسم للنكس.

C·4 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

What is the best management for esophageal cancer with liver metastasis?

A. Curative esophagectomy
× Curative resection is not for M1 disease.
الاستئصال الشافي لا يناسب النقائل.

B. Definitive chemoradiotherapy with curative intent
× Curative-intent chemoradiation is for locoregional disease.
العلاج الشافي للمرض الموضعي فقط.

C. Palliative stenting and supportive care ✓ Book p. 29 ↗

D. Endoscopic mucosal resection
× EMR is only for early mucosal (T1) cancer.
EMR فقط للسرطان المبكر المخاطي.

Why: Liver metastasis means incurable disease, so palliation (stent, supportive care, RT/chemo) is the plan.
لماذا: نقائل الكبد تعني مرضاً غير قابل للشفاء، فالعلاج تلطيفي (دعامة ودعم).

★ **Tell the students:** Stents for dysphagia: Celestin and Souttar tubes, or self-expanding metal stents.
قل للطلاب: دعامات البلع: أنابيب Celestin وSouttar أو الدعامات المعدنية المتمددة.

C·5 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

What can cause exudative diarrhea?

A. Rapid intestinal transit
× Rapid transit = motility diarrhea.
السرعة = إسهال حركي.

B. Unabsorbed lactose
× Unabsorbed lactose = osmotic.
اللاكتوز غير الممتص = أسموزي.

C. Excessive water intake
× Not a pathological mechanism.
ليس آلية مرضية.

D. Ulceration of the mucosa ✓ Book p. 80 ↗

Why: Exudative diarrhea comes from an inflamed, ulcerated mucosa leaking mucus, blood and pus.
لماذا: الإسهال الإفرازي الالتهابي ينتج عن مخاطية ملتهبة متقرحة تسرّب مخاطاً ودماً وصديداً.

★ **Tell the students:** Examples: UC, Crohn's, Salmonella, Shigella, C. difficile. Stool has WBCs/blood.
قل للطلاب: أمثلة: UC وكرون والسالمونيلا والشيجيلا. البراز فيه كرات بيضاء ودم.

Group D

D·1 INTERNAL MEDICINE باطنة — Digestive System — Chapter 03: The Stomach and Duodenum

> Which noninvasive test relies on bacterial conversion of labeled urea to labeled carbon dioxide?

A. Fecal elastase test
× Fecal elastase tests the pancreas. — فحص البنكرياس.

B. Urea breath test ✓ Book p. 40 ↗

C. Hydrogen breath test for lactose
× Lactase deficiency test. — اختبار نقص اللاكتيز.

D. Serum calcium test
× Irrelevant. — غير مرتبط.

Why: The 13C-urea breath test detects bacterial urease: labeled urea becomes labeled CO2 in the breath.
لماذا: اختبار التنفس باليوريا: بكتيريا H. pylori تحوّل اليوريا الموسومة إلى CO2 موسوم.

★ **Tell the students:** Sensitivity 90%, specificity 96%. Stop PPIs 2 weeks and antibiotics 4 weeks before.
قل للطلاب: الحساسية 90% والنوعية 96%. أوقف الـPPI أسبوعين والمضادات 4 أسابيع قبله.

D·2 INTERNAL MEDICINE باطنة

Digestive System — Chapter 02: The Esophagus · Digestive System — Chapter 04: Pathology of GIT Tumours

> Esophageal squamous cell carcinoma is associated with which factors?

A. H. pylori and celiac disease
× Not linked to the esophagus. — لا علاقة بالمريء.

B. Obesity and Barrett's esophagus
× Obesity and Barrett's lead to adenocarcinoma. — السمنة وباريت للغدي.

C. Lactose intolerance and smoking
× Lactose intolerance is unrelated. — عدم تحمل اللاكتوز غير مرتبط.

D. Alcohol and smoking ✓ Book p. 71 ↗

Why: Alcohol and smoking are the main risk factors for esophageal squamous cell carcinoma.
لماذا: الكحول والتدخين أهم عوامل خطر السرطان الحرشفي للمريء.

★ **Tell the students:** Other SCC risks: achalasia, caustic stricture, hot drinks, low vitamins/zinc.
قل للطلاب: عوامل أخرى: الأكالازيا والتضيق الكاوي والمشروبات الساخنة ونقص الفيتامينات.

D·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 06: Constipation (including IBS)

What is the best definition of constipation?

A. Urgency with fecal incontinence
× Fecal incontinence.
سلس برازي.

B. Infrequent bowel movements with difficulty passing stool ✓ Book p. 96 ↗

C. Daily soft stools without straining
× Normal bowel habit.
إخراج طبيعي.

D. Frequent loose stools with urgency
× Diarrhea.
إسهال.

Why: Constipation = infrequent stools with difficult passage, straining or hard stools.

لماذا: الإمساك = إخراج متباعد مع صعوبة أو إجهاد أو براز صلب.

★ **Tell the students:** Rome: at least 2 of: fewer than 3 stools/week, straining, hard stool, incomplete evacuation.

قل للطلاب: معايير روما: اثنان على الأقل: أقل من 3 مرات/أسبوع، إجهاد، براز صلب، شعور بعدم الإفراغ.

Group E

E·1 SURGERY جراحة Hernia book — Chapter 06: Incisional and Recurrent Hernias

What is the most important factor for incisional hernia recurrence?

A. Large defect size ✓ Hernia book — not in this file

B. Duration of preoperative fasting
× Fasting time is irrelevant.
مدة الصيام غير مؤثرة.

C. Patient's blood group
× Blood group is irrelevant.
فصيلة الدم غير مؤثرة.

D. Time of day of surgery
× Time of surgery is irrelevant.
وقت العملية غير مؤثر.

Why: Defect size is the strongest anatomical determinant of recurrence among the options.

لماذا: حجم الفتحة هو أقوى عامل تشريحي للنكس بين الاختيارات.

★ **Tell the students:** Other risks: wound infection, obesity, smoking, tension, no mesh. Sublay mesh lowers recurrence.

قل للطلاب: عوامل أخرى: عدوى الجرح والسمنة والتدخين والشد وعدم استخدام الشبكة.

E·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 02: The Esophagus

Untreated chronic reflux can lead to which structural complication?

A. Appendiceal abscess
× Unrelated to reflux. غير مرتبط.

B. Esophageal stricture ✓ Book p. 15 ↗

C. Ileal atresia
× Congenital. خلقي.

D. Femoral hernia
× A different hernia type. نوع آخر من الفتق.

Why: Chronic acid reflux causes fibrosis and an esophageal (peptic) stricture.

لماذا: الارتجاع المزمن يسبب تليفاً وتضيقاً بالمريء.

★ **Tell the students:** GERD complications: stricture, Barrett's, bleeding/ulcer, adenocarcinoma (not SCC).

قل للطلاب: مضاعفات GERD: تضيق، باريت، نزف/قرحة، سرطان غدي (وليس الحرشفي).

E·3 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

What is the most common type of hiatus hernia?

A. Mixed type hiatus hernia
× Mixed type is rarer. المختلط أقل شيوعاً.

B. Sliding hiatus hernia ✓ Book p. 17 ↗

C. Type IV hiatus hernia
× Type IV has other organs too, and is rare. النوع IV نادر ويحوي أعضاء أخرى.

D. Paraesophageal hiatus hernia
× Rolling type is about 10%. نحو 10% فقط.

Why: Sliding hiatus hernia (type I) accounts for about 85% of hiatus hernias.

لماذا: الفتق الحجابي الانزلاقي (النوع I) يمثل نحو 85% من الفتوق الحجابية.

★ **Tell the students:** Sliding = reflux. Rolling (paraesophageal) = obstruction/strangulation, so repair even if silent.

قل للطلاب: الانزلاقي = ارتجاع. الدحرجي = انسداد/اختناق لذا يُصلح حتى لو صامت.

E·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 03: The Stomach and Duodenum

What is melena?

A. Passage of pale stools due to bile duct obstruction
× Pale stools = obstructive jaundice.
براز شاحب = انسداد صفراوي.

B. Passage of black tarry stools due to altered blood in the gut ✓ Book p. 4 ↗

C. Passage of fresh red blood from the rectum
× Fresh rectal blood = hematochezia.
دم أحمر طازج = hematochezia.

D. Passage of fatty stools due to malabsorption
× Fatty stools = steatorrhea.
براز دهني = steatorrhea.

Why: Melena is black tarry stool from altered blood, usually bleeding proximal to the right colon.
لماذا: Melena براز أسود قطراني بسبب دم متحول، عادة من نزف أعلى القولون الأيمن.

★ **Tell the students:** It needs more than 50 mL of blood. Massive upper bleeding can even give fresh blood per rectum with shock.
قل للطلاب: يحتاج أكثر من 50 مل دم. النزف العلوي الغزير قد يعطي دماً أحمر مع صدمة.

Group F

F·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Chronic diarrhea is clinically defined as symptoms lasting more than how many days?

A. 7 days
× 7 days is acute.
7 أيام = حاد.

B. 30 days ✓ Book p. 77 ↗

C. 14 days
× 14 days is acute/persistent.
14 يوماً = حاد/مستمر.

D. 90 days
× Far longer than required.
أطول بكثير من المطلوب.

Why: Diarrhea lasting more than 4 weeks (about 30 days) is chronic.
لماذا: الإسهال لأكثر من 4 أسابيع (نحو 30 يوماً) يعتبر مزمناً.

★ **Tell the students:** Most chronic diarrhea is NON-infectious; about 80% of acute diarrhea is infectious.
قل للطلاب: معظم الإسهال المزمن غير معدٍ؛ ونحو 80% من الحاد معدٍ.

F·2 INTERNAL MEDICINE باطنة

Digestive System — Chapter 02: The Esophagus · Digestive System — Chapter 04: Pathology of GIT Tumours

The cancer associated with a small proportion of Barrett's esophagus cases is most often which histologic type?

A. Cutaneous melanoma
× Not related. غير مرتبط.

B. Esophageal adenocarcinoma ✓ Book p. 71 ↗

C. Renal cell carcinoma
× Not related. غير مرتبط.

D. Hepatocellular carcinoma
× Not related. غير مرتبط.

Why: Esophageal adenocarcinoma is the cancer that arises from Barrett's metaplasia.

لماذا: السرطان الغدي للمريء هو الذي ينشأ من تحوّل باريت.

★ **Tell the students:** Only a small proportion of Barrett's progresses, hence surveillance endoscopy.

قل للطلاب: نسبة قليلة فقط تتحول، لذلك يلزم منظار متابعة.

F·3 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

Which finding makes esophageal carcinoma unresectable?

A. Mild mucosal erythema
× Minor finding. علامة بسيطة.

B. Tracheobronchial invasion ✓ Book p. 29 ↗

C. A tumor confined to the mucosa
× Early mucosal disease is resectable/endoscopic. المرض المخاطي المبكر قابل للاستئصال.

D. A small superficial lesion
× Small superficial lesions are resectable. الآفة الصغيرة السطحية قابلة للاستئصال.

Why: Invasion of the trachea/bronchus (T4) makes the tumor unresectable.

لماذا: غزو القصبة الهوائية/الشعب (T4) يجعل الورم غير قابل للاستئصال.

★ **Tell the students:** Unresectable also: aorta invasion, distant metastasis, celiac or distant nodes.

قل للطلاب: غير قابل أيضاً: غزو الأبهر ونقائل بعيدة وعقد بعيدة.

F·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

Which pattern of Crohn's disease is most common?

A. Disease in the ileum and cecum ✓ Book p. 116 ↗

B. Disease limited to the rectum
× Rectal-only is typical of UC proctitis.
المستقيم فقط يميز UC.

C. Isolated disease of the esophagus
× Rare.
نادر.

D. Disease limited to the stomach
× Rare.
نادر.

Why: The commonest Crohn's pattern is ileocecal disease (about 40%).
لماذا: أشهر نمط لكرون هو اللفائفي-الأعوري (نحو 40%).

★ **Tell the students:** Three patterns: ileocecal (40%), small bowel only, colon only (25%).
قل للطلاب: ثلاثة أنماط: لفائفي-أعوري (40%)، أمعاء دقيقة فقط، قولون فقط (25%).

F·5 SURGERY جراحة Digestive System — Chapter 10: Small Intestine and Colon Surgery

Which statement about bleeding from Meckel's diverticulum is true?

A. It is caused by variceal rupture
× Variceal bleeding is portal hypertension.
نزف الدوالي = ارتفاع الضغط البابي.

B. It indicates ulceration of the adjacent segment of ileum ✓ Book p. 143 ↗

C. It indicates a bleeding gastric polyp
× Not a gastric polyp.
ليس من سليلة معدية.

D. It is due to appendiceal inflammation
× Appendicitis does not bleed.
التهاب الزائدة لا ينزف.

Why: Meckel's bleeding comes from ulceration of the adjacent ileum, due to acid from ectopic gastric mucosa.
لماذا: نزف ميكل ينتج عن تقرح اللفائفي المجاور بسبب حمض الغشاء المعدي الهاجر.

★ **Tell the students:** Rule of 2s; painless bleeding in a child under 5; Tc-99m pertechnetate scan is most accurate.
قل للطلاب: قاعدة الـ2؛ نزف بلا ألم عند طفل أقل من 5؛ أدق فحص Tc-99m.

Group G

G·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

> Which laboratory finding distinguishes bacterial overgrowth from tropical sprue?

A. Low serum folate with high B12
× Reversed pattern.
عكس الصحيح.

B. Normal or high serum folate with low B12 ✓ Book p. 88 ↗

C. High serum folate and high B12
× Not seen.
لا يحدث.

D. Low serum folate and low B12
× Low both = tropical sprue.
كلاهما منخفض = tropical sprue.

Why: In bacterial overgrowth B12 is LOW but folate is normal/high; in tropical sprue both are low.

لماذا: في فرط النمو البكتيري B12 منخفض والفولات طبيعي/مرتفع؛ وفي Tropical sprue كلاهما منخفض.

★ **Tell the students:** Bacteria consume B12 and make folate.

قل للطلاب: البكتيريا تستهلك B12 وتصنع الفولات.

G·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

> An inflammatory bowel condition is limited to the colon and usually affects its inner lining. Which condition fits this pattern?

A. Achalasia
× Esophageal motility disorder.
اضطراب حركة المريء.

B. Lactase deficiency
× An enzyme deficiency.
نقص إنزيم.

C. Crohn's disease of the mouth
× Crohn's is transmural and can hit any site.
كرون يصيب كل الطبقات وأي موضع.

D. Ulcerative colitis ✓ Book p. 116 ↗

Why: UC is limited to the colon and is mucosal (continuous from the rectum).

لماذا: UC يقتصر على القولون ويصيب الغشاء المخاطي (متصل من المستقيم).

★ **Tell the students:** UC = mucosal + continuous + rectum first; Crohn's = transmural + skip lesions.

قل للطلاب: UC مخاطي ومتصل ويبدأ بالمستقيم؛ كرون كل الطبقات وبقع متفرقة.

G·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 03: The Stomach and Duodenum

A patient has epigastric pain, antral erythema, and chronic inflammation. What is the most likely cause?

A. Cytomegalovirus infection
× CMV is seen in immunocompromised patients. CMV لدى المناعة المنخفضة.

B. Helicobacter pylori infection ✓ Book p. 38 ↗

C. Autoimmune gastritis
× Autoimmune gastritis affects body/fundus. المناعي يصيب الجسم/القاع.

D. Acute pancreatitis
× Pancreatitis does not cause antral erythema. التهاب البنكرياس لا يسبب احمرار الغار.

Why: Antral-predominant chronic gastritis with epigastric pain is the classic picture of H. pylori.
لماذا: التهاب معدة مزمن يغلب على الغار مع ألم شرسوفي هو الصورة الكلاسيكية لـ H. pylori.

★ **Tell the students:** H. pylori = antrum (duodenal ulcer); autoimmune = body/fundus (pernicious anemia).
قل للطلاب: H. pylori = الغار؛ المناعي = الجسم والقاع (أنيميا خبيثة).

Group H

H·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

What causes diarrhea in lactase deficiency?

A. Osmotic effect of unabsorbed lactose ✓ Book p. 87 ↗

B. Increased intestinal secretion of chloride
× Secretory mechanism. آلية إفرازية.

C. Reduced bile acid production
× Steatorrhea mechanism. آلية الإسهال الدهني.

D. Ulceration of the intestinal mucosa
× Exudative mechanism. آلية التهابية.

Why: Unabsorbed lactose is osmotically active and pulls water into the lumen (osmotic diarrhea).
لماذا: اللاكتوز غير الممتص يسحب الماء إلى التجويف (إسهال أسموزي).

★ **Tell the students:** Osmotic diarrhea stops on fasting and responds to a lactose-free diet.
قل للطلاب: الإسهال الأسموزي يتوقف بالصيام ويتحسن بنظام خالٍ من اللاكتوز.

H·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 06: Constipation (including IBS)

Which is an alarm feature in constipation?

A. Iron-deficiency anemia ✓ Book p. 97 ↗

B. Occasional bloating
× Common, benign. شائع وحميد.

C. Hard stools
× Part of constipation itself. جزء من الإمساك نفسه.

D. Straining during defecation
× Part of constipation itself. جزء من الإمساك نفسه.

Why: Anemia (iron deficiency) is a red flag that mandates colorectal cancer work-up.

لماذا: الأنيميا (نقص حديد) علامة إنذار تستوجب البحث عن سرطان القولون.

★ **Tell the students:** Red flags: bleeding, weight loss, anemia, change in stool caliber, rectal mass, age over 50.

قل للطلاب: علامات الإنذار: نزف، فقدان وزن، أنيميا، تغير قطر البراز، كتلة مستقيم، عمر فوق 50.

H·3 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

What is the surgical approach for duodenal ulcer bleeding that is not controlled endoscopically?

A. Nissen fundoplication
× Nissen is for reflux. Nissen لعلاج الارتجاع.

B. Wedge resection of the gastric fundus
× Fundal wedge is for other lesions. لآفات أخرى.

C. Exploration and over-sewing of the bleeding ulcer base ✓ Book p. 57 ↗

D. Vagotomy alone without controlling the bleeding vessel
× Vagotomy alone does not stop a bleeding vessel. قطع المبهم وحده لا يوقف النزف.

Why: Open the duodenum and under-run (over-sew) the ulcer base to control the vessel.

لماذا: فتح الاثني عشر وخياطة قاع القرحة للسيطرة على الوعاء النازف.

★ **Tell the students:** Bleeding suspicious gastric ulcer: partial gastrectomy.

قل للطلاب: قرحة معدة مشتبه بها: استئصال جزئي للمعدة.

H·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

> Colonoscopy reveals separated areas of inflammation with normal-appearing intervening mucosa. Which disease pattern is most consistent?

A. Continuous ulcerative colitis only
× UC is continuous. — UC متصل.

B. Simple reflux
× Reflux is esophageal. — الارتجاع مريئي.

C. Lactose malabsorption
× Lactose malabsorption has a normal mucosa. — سوء امتصاص اللاكتوز مخاطية طبيعية.

D. Patchy Crohn's disease ✓ Book p. 116 ↗

Why: Skip areas of inflammation with normal mucosa in between are typical of Crohn's.

لماذا: مناطق التهاب متفرقة بينها مخاطية طبيعية (skip lesions) مميزة لكرون.

★ **Tell the students:** Crohn's: skip lesions, cobblestone, transmural, granulomas, perianal disease.

قل للطلاب: كرون: skip lesions، cobblestone، كل الطبقات، حبيبومات، آفات حول الشرج.

Group I

I·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

> A 28-year-old woman presents with bloody diarrhea and continuous rectal inflammation. What is the most likely diagnosis?

A. Crohn's disease
× Crohn's is patchy with perianal disease. — كرون متقطع مع آفات شرجية.

B. Ischemic colitis
× Ischemic colitis affects older patients. — الإقفاري لكبار السن.

C. Ulcerative colitis ✓ Book p. 116 ↗

D. Infectious colitis
× Infectious colitis is short and acute. — المعدي حاد وقصير.

Why: Bloody diarrhea with continuous rectal inflammation in a young adult is ulcerative colitis.

لماذا: إسهال دموي مع التهاب مستقيمي متصل لدى شاب = التهاب القولون التقرحي.

★ **Tell the students:** UC peaks at 15-25 and again at 55-65 years.

قل للطلاب: UC له ذروتان: 15-25 سنة و55-65 سنة.

I·2 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

A patient's pyloric obstruction has not responded to dilation. Which statement about management is correct?

A. Perform colectomy
× Colectomy is unrelated. غير مرتبط.

B. Continue dilation indefinitely without reassessment
× Repeating dilation without reassessment is wrong. تكرار التوسيع بلا إعادة تقييم خطأ.

C. Faulty decision → patient needs vagotomy and gastrojejunostomy ✓ Book p. 59 ↗

D. Perform fundoplication alone
× Fundoplication treats reflux. لعلاج الارتجاع.

Why: A fibrotic pyloric obstruction that fails dilation needs surgical drainage (vagotomy + gastrojejunostomy).
لماذا: انسداد البواب التليفي الذي فشل معه التوسيع يحتاج تصريفاً جراحياً (قطع مبهم + مفاغرة معدية صائمية).

★ **Tell the students:** Add post-operative PPI; partial gastrectomy if the stomach is hugely dilated.
قل للطلاب: يضاف PPI بعد العملية؛ واستئصال جزئي لو المعدة متمددة جداً.

I·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 01: GI Symptoms and Investigations

Which investigation is the choice for diagnosing and treating upper GI disorders?

A. Abdominal CT scan
× CT cannot treat. CT لا يعالج.

B. Barium swallow
× Barium only images. الباريوم تصوير فقط.

C. Esophagogastroduodenoscopy (EGD) ✓ Book p. 5 ↗

D. Esophageal manometry
× Manometry is diagnostic only. للتشخيص فقط.

Why: Upper endoscopy (EGD) both diagnoses (biopsy) and treats (hemostasis, dilation) upper GI disorders.
لماذا: المنظار العلوي يشخص (خزعة) ويعالج (إيقاف نزف وتوسيع) أمراض الجهاز الهضمي العلوي.

★ **Tell the students:** Gastroscopy is more sensitive than barium meal and allows biopsy.
قل للطلاب: المنظار أدق من الباريوم ويسمح بالخزعة.

I·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 06: Constipation (including IBS)

> Recurrent abdominal pain associated with altered bowel habits occurs without visible structural damage. What broad disorder category fits?

A. A mandatory malignancy diagnosis
× It is benign. حميد.

B. Mechanical obstruction in every case
× No mechanical obstruction. لا انسداد ميكانيكي.

C. Disorder of gut–brain interaction ✓ Book p. 101 ↗

D. A primary kidney disorder
× Not renal. ليس كلوياً.

Why: IBS is a disorder of gut-brain interaction: symptoms without structural damage.

لماذا: IBS اضطراب في التفاعل الدماغي المعوي: أعراض بدون تلف بنيوي.

★ **Tell the students:** Diagnosis is clinical (Rome criteria); routine tests are normal.

قل للطلاب: التشخيص إكلينيكي (معايير روما) والتحاليل الروتينية طبيعية.

I·5 INTERNAL MEDICINE باطنة

Digestive System — Chapter 04: Pathology of GIT Tumours · Digestive System — Chapter 14: Polyposis of Colon

> Which type of colonic polyp may become malignant?

A. Hyperplastic polyp
× Hyperplastic: little or no risk. فرط التنسج: خطر ضئيل.

B. Adenomatous polyp ✓ Book p. 190 ↗

C. Juvenile hamartomatous polyp
× Hamartoma: low risk. هامارتوم: خطر منخفض.

D. Inflammatory polyp
× Inflammatory polyps are not neoplastic. ليست نيوبلازمية.

Why: Adenomatous polyps (tubular, villous) are neoplastic and can become malignant.

لماذا: الأورام الغدية (أنبوبية، زغابية) نيوبلازمية وقد تتسرطن.

★ **Tell the students:** Villous > tubular in malignant potential; polyp size above 2 cm is high risk.

قل للطلاب: الزغابي أخطر من الأنبوبي؛ الحجم فوق 2 سم عالي الخطورة.

Group J

J·1 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

A male patient presents with severe epigastric distension and vomiting and has a history of bulimia nervosa. What is the expected diagnosis?

A. Superior mesenteric artery syndrome ✓ Book p. 68 ↗

B. Distal pyloric obstruction on top of ulcer
× Needs an ulcer history.
تحتاج تاريخ قرحة.

C. Gastric volvulus
× Volvulus is acute, with pain and failed NG passage.
الانفتال حاد وألم وصعوبة تمرير الأنبوب.

D. Acute pancreatitis due to recurrent vomiting
× Pancreatitis presents with back pain and high amylase.
التهاب البنكرياس بألم ظهر وأميليز مرتفع.

Why: Among the options, SMA syndrome is the one linked to eating disorders (weight loss) and gives epigastric distension and vomiting.
لماذا: بين الاختيارات، متلازمة SMA هي المرتبطة باضطرابات الأكل (فقدان وزن) وتعطي انتفاخاً وقيئاً.

★ **Tell the students:** Wording issue: the book lists bulimia as a functional cause of ACUTE GASTRIC DILATATION, which is not offered here.
قل للطلاب: ملاحظة: الكتاب يذكر الـbulimia كسبب وظيفي لـAcute Gastric Dilatation وهو غير موجود بالاختيارات.

Question note: Question flaw: expected answer is acute gastric dilatation (not listed).
ملاحظة على السؤال: السؤال به عيب: الإجابة المتوقعة acute gastric dilatation وهي غير مذكورة.

J·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

What is the characteristic histology of celiac disease?

A. Villous hypertrophy with crypt atrophy
× Reversed.
عكسه.

B. Villous atrophy with crypt hyperplasia ✓ Nearest: p. 89 ↗

C. Normal villi with increased gastric pits
× Not celiac.
غير صحيح.

D. Granulomas without villous changes
× Granulomas suggest Crohn's.
الحبيبومات تدل على كرون.

Why: Celiac biopsy shows villous atrophy with crypt hyperplasia and intraepithelial lymphocytes.
لماذا: خزعة السيلياك: ضمور الزغابات مع فرط تنسج الخبايا وزيادة اللمفاويات.

★ **Tell the students:** Biopsy is done on a gluten diet; the target is the duodenum (Marsh classification).
قل للطلاب: الخزعة تؤخذ والمريض على الجلوتين من الاثني عشر (Marsh).

J·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 01: GI Symptoms and Investigations

For bleeding thought to arise in the small intestine, which swallowed device records images as it passes through the gut?

A. Nasogastric suction tube
× NG tube suctions only. للشفط فقط.

B. Electrocardiograph lead
× ECG leads record the heart. تسجل القلب.

C. Rectal thermometer
× Not an imaging device. ليس جهاز تصوير.

D. Capsule endoscope ✓ Book p. 7 ↗

Why: A swallowed capsule endoscope images the whole small bowel as it passes.

لماذا: كبسولة المنظار تُبتلع وتصور الأمعاء الدقيقة بالكامل أثناء مرورها.

★ **Tell the students:** Used for obscure GI bleeding after negative gastroscopy and colonoscopy.

قل للطلاب: يستخدم للنزف الغامض بعد منظار علوي وسفلي سلبيين.

Group K

K·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which is the most specific serology for celiac disease?

A. IgG anti-mitochondrial antibody
× AMA = primary biliary cholangitis. AMA = التهاب الصفراوية الأولي.

B. IgA anti-gliadin antibody
× Anti-gliadin is less specific. Anti-gliadin أقل نوعية.

C. IgM anti-endomysial antibody
× Endomysial antibody is IgA, not IgM. EMA من نوع IgA وليس IgM.

D. IgA anti-transglutaminase antibody ✓ Book p. 89 ↗

Why: IgA anti-tTG is the most specific/practical serology; EMA-IgA is also highly specific, but the option says IgM.

لماذا: IgA anti-tTG هو الأدق والأعملي؛ و EMA من نوع IgA أيضاً نوعيته عالية لكن الاختيار كتب IgM.

★ **Tell the students:** EMA-IgA ~100% specific but operator-dependent; tTG-IgA is the first-line test.

قل للطلاب: EMA-IgA نوعيته ~100% لكنه يعتمد على الفاحص؛ وtTG-IgA هو الاختبار الأول.

K·2 INTERNAL MEDICINE باطنة

Digestive System — Chapter 02: The Esophagus · Digestive System — Chapter 04: Pathology of GIT Tumours

Achalasia predisposes to which type of cancer?

A. Adenocarcinoma
× Adenocarcinoma follows Barrett's. الغدي يتبع باريت.

B. Gastrointestinal stromal tumor
× GIST is mesenchymal. GIST ورم لحمي.

C. Small-cell carcinoma
× Not related. غير مرتبط.

D. Squamous cell carcinoma ✓ Book p. 71 ↗

Why: Achalasia causes food stasis and chronic irritation, leading to squamous cell carcinoma.

لماذا: الأكالازيا تسبب ركود الطعام وتهيجاً مزمناً يؤدي للسرطان الحرشفي.

★ **Tell the students:** The cancer risk persists even after successful myotomy, so keep watching.

قل للطلاب: الخطر يستمر حتى بعد شق العضلة، فتستمر المتابعة.

K·3 SURGERY جراحة Digestive System — Chapter 12: Appendix

An untreated inflamed appendix can result in which complication?

A. Primary achalasia
× An esophageal condition. حالة مريئية.

B. Peritonitis or an abscess ✓ Book p. 156 ↗

C. Lactase persistence
× Not a complication. ليست مضاعفة.

D. Barrett's esophagus
× Esophageal metaplasia. تحول مريئي.

Why: Untreated appendicitis progresses to appendicular mass/abscess, perforation and peritonitis.

لماذا: التهاب الزائدة غير المعالج يتطور لكتلة/خراج ثم انثقاب والتهاب بريتوني.

★ **Tell the students:** Late: portal pyemia (pylephlebitis) and liver abscess.

قل للطلاب: مضاعفات متأخرة: تعفن وريد بابي وخراج كبدي.

K·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

What is the dominant symptom in ulcerative colitis?

A. Constipation
× Constipation is not typical.
الإمساك ليس نمطياً.

B. Painless jaundice
× Painless jaundice suggests pancreatic head cancer.
اليرقان غير المؤلم يوحي بسرطان رأس البنكرياس.

C. Steatorrhea
× Steatorrhea = malabsorption.
الدهنية = سوء امتصاص.

D. Bloody diarrhea ✓ Book p. 116 ↗

Why: Diarrhea with blood is the dominant symptom of ulcerative colitis.

لماذا: الإسهال الدموي هو العرض الأساسي في التهاب القولون التقرحي.

★ **Tell the students:** UC: bloody diarrhea + tenesmus; Crohn's: non-bloody diarrhea + pain + weight loss.

قل للطلاب: UC: دموي + زحير؛ كرون: غير دموي + ألم + فقدان وزن.

Group L

L·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

What percentage of fluid is absorbed in the small intestine?

A. 99%
× 99% is the whole gut (small + colon).
99% هو إجمالي الجهاز كله.

B. 50%
× Too low.
قليل جداً.

C. 10%
× 10% is what reaches the colon.
10% هو ما يصل للقولون.

D. 90% ✓ Book p. 77 ↗

Why: About 9 L enters the small bowel daily and 90% is absorbed there.

لماذا: يدخل نحو 9 لتر يومياً للأمعاء الدقيقة ويُمتص منها 90%.

★ **Tell the students:** Colon receives 1-2 L, absorbs about 90%, leaving 100-200 mL in stool.

قل للطلاب: القولون يستقبل 1-2 لتر ويمتص 90% ويبقى 100-200 مل في البراز.

L·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 02: The Esophagus

60-yo, 4-year intermittent dysphagia, CXR: absent fundic air bubble. TRUE:

A. Esophageal motility disorder highly suspected ✓ Book p. 22 ↗

B. pH study highly indicated
× pH study is for reflux.
فحص الـpH للارتجاع.

C. Surgery is first line
× Confirm diagnosis first; exclude cancer before treatment.
نؤكد التشخيص أولاً ونستبعد السرطان قبل العلاج.

D. Endoscopy commonly reveals tight stricture
× Endoscopy shows a dilated esophagus; a tight stricture suggests another cause.
المنظار يظهر مريئاً متسعاً، والتضيق يوحي بسبب آخر.

Why: Intermittent dysphagia with an absent gastric air bubble strongly suggests an esophageal motility disorder (achalasia).

لماذا: عسر بلع متقطع مع غياب فقاعة الغاز المعدية يوحي بقوة باضطراب حركة المريء (achalasia).

★ **Tell the students:** Barium: bird's beak; manometry = gold standard; endoscopy to exclude pseudoachalasia (cancer).

قل للطلاب: الباريوم: منقار الطائر؛ المانومتري = المعيار الذهبي؛ المنظار لاستبعاد السرطان.

L·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

The D-xylose test is used for:

A. Diagnosing C. difficile
× C. difficile is diagnosed by toxin assay.
يشخص بفحص السم.

B. Diagnosing acute infectious diarrhea
× Acute infection needs stool culture.
العدوى الحادة بمزرعة براز.

C. Differentiating small-bowel from pancreatic malabsorption ✓ Nearest: p. 93 ↗

D. Testing gastric acid secretion
× Gastric acid is tested differently.
حموضة المعدة تفحص بطريقة أخرى.

Why: D-xylose is absorbed passively without pancreatic enzymes: low in mucosal disease, normal in pancreatic insufficiency.

لماذا: D-xylose يُمتص دون إنزيمات البنكرياس: منخفض في أمراض المخاطية وطبيعي في قصور البنكرياس.

★ **Tell the students:** Not printed in the book; the nearest page is the malabsorption tests page.

قل للطلاب: غير مذكور نصاً في الكتاب؛ أقرب صفحة هي صفحة اختبارات سوء الامتصاص.

L·4 SURGERY جراحة Digestive System — Chapter 12: Appendix

The most common position of the appendix is:

A. Pelvic
× Pelvic is second (20-30%). الحوضي ثاني الأشهر.

B. Pre-ileal
× Rare. نادر.

C. Sub-hepatic
× Rare. نادر.

D. Retrocecal (~65%) ✓ Book p. 153 ↗

Why: The retrocecal position (60-70%) is the commonest.

لماذا: الوضع خلف الأعور (60-70%) هو الأشهر.

★ **Tell the students:** Retrocecal appendicitis gives a positive psoas sign; pelvic gives rectal signs and diarrhea.

قل للطلاب: خلف الأعور: psoas sign إيجابية؛ الحوضي: أعراض مستقيمية وإسهال.

L·5 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which autoantibody is found in celiac disease?

A. Anti-endomysial / anti-tTG ✓ Book p. 89 ↗

B. ANCA
× ANCA: UC / PSC. ANCA: UC / PSC.

C. ASCA
× ASCA: Crohn's. ASCA: كرون.

D. ANA
× ANA: SLE / autoimmune hepatitis. ANA: الذئبة / التهاب كبد مناعي.

Why: Anti-endomysial and anti-tTG antibodies are the celiac autoantibodies.

لماذا: الأجسام المضادة لـ endomysium وtTG هي الأجسام الذاتية للسيلياك.

★ **Tell the students:** Remember: ASCA for Crohn's, pANCA for UC.

قل للطلاب: تذكّر: ASCA لكرون، وpANCA للـUC.

Group M

M·1 INTERNAL MEDICINE باطنة — Digestive System — Chapter 05: Diarrhea and Malabsorption

> In malabsorption, which test evaluates ileal function (bile acid absorption)?

A. Hydrogen breath test
× Hydrogen breath: lactase deficiency / SIBO. — اختبار الهيدروجين: اللاكتيز / SIBO.

B. SeHCAT (selenium-homotaurocholic acid) test ✓ Book p. 93 ↗

C. Fecal elastase
× Fecal elastase: pancreas. — البنكرياس.

D. Secretin stimulation test
× Secretin test: pancreas. — البنكرياس.

Why: SeHCAT measures bile-acid retention, a test of ileal function.

لماذا: SeHCAT يقيس احتباس الأحماض الصفراوية وهو اختبار لوظيفة اللفائفي.

★ **Tell the students:** Low retention at 7 days = bile acid malabsorption (ileal Crohn's or resection).

قل للطلاب: احتباس منخفض بعد 7 أيام = سوء امتصاص أحماض صفراوية.

M·2 INTERNAL MEDICINE باطنة

Digestive System — Chapter 02: The Esophagus · Digestive System — Chapter 04: Pathology of GIT Tumours

> Barrett oesophagus is predisposed to:

A. Leiomyosarcoma
× Not linked. — غير مرتبط.

B. Squamous cell carcinoma
× SCC = smoking/alcohol/achalasia. — الحرشفي = تدخين/كحول/أكالازيا.

C. Lymphoma
× Not linked. — غير مرتبط.

D. Adenocarcinoma ✓ Book p. 71 ↗

Why: Barrett's esophagus predisposes to adenocarcinoma.

لماذا: مريء باريت يهيّئ للسرطان الغدي.

★ **Tell the students:** Metaplasia, dysplasia, then adenocarcinoma.

قل للطلاب: تحول، ثم خلل تنسج، ثم سرطان غدي.

M·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

47-y-old woman with chronic diarrhea + bloating (worse with wheat) + lethargy + weight loss + anemia. Most likely diagnosis?

A. Lactose intolerance
× Lactose intolerance: milk-related, no weight loss or anemia.
مرتبط باللبن، بلا نقص وزن أو أنيميا.

B. Ulcerative colitis
× UC is bloody.
UC دموي.

C. Celiac disease ✓ Book p. 89 ↗

D. Crohn's disease
× Crohn's has RLQ pain and no wheat link.
كرون ألم يمين سفلي بلا علاقة بالقمح.

Why: Chronic diarrhea, bloating after wheat, anemia and weight loss in an adult = celiac disease.

لماذا: إسهال مزمن وانتفاخ مع القمح وأنيميا ونقص وزن لدى بالغ = السيلياك.

★ **Tell the students:** Iron-deficiency anemia unresponsive to iron should trigger celiac testing.

قل للطلاب: أنيميا نقص حديد لا تستجيب للحديد تستدعي فحص السيلياك.

Group N

N·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which autoantibody is found in celiac disease?

A. ANA
× ANA: SLE.
ANA: الذئبة.

B. ANCA
× ANCA: UC/PSC.
ANCA: UC/PSC.

C. EMA (anti-endomysial Ab) ✓ Book p. 89 ↗

D. ASCA
× ASCA: Crohn's.
ASCA: كرون.

Why: EMA (anti-endomysial) is a celiac autoantibody.

لماذا: EMA من الأجسام الذاتية الخاصة بالسيلياك.

★ **Tell the students:** Same question repeated with another order; learn: tTG/EMA = celiac.

قل للطلاب: نفس السؤال بترتيب مختلف؛ احفظ: tTG/EMA = سيلياك.

N·2 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

Obese 44-yo severe heartburn — best tool for hiatus hernia:

A. CT upper abdomen
× CT is for complicated/para-esophageal cases. للحالات المعقدة.

B. Esophageal motility
× Manometry assesses motility. للحركة.

C. Abdominal US
× US cannot show it. لا تظهره.

D. Barium meal in Anti-Trendelenburg (head-down) ✓ Book p. 19 ↗

Why: A barium meal is the investigation that demonstrates a hiatus hernia (sliding type).

لماذا: وجبة الباريوم هي الفحص الذي يظهر الفتق الحجابي (الانزلاقي).

★ **Tell the students:** For a hiatus hernia the patient swallows barium head-down (Trendelenburg); a sliding hernia reduces when upright.

قل للطلاب: في الفتق الحجابي يبلع المريض الباريوم والرأس لأسفل (Trendelenburg)؛ والانزلاقي يرجع عند الوقوف.

Question note: Option D says 'Anti-Trendelenburg (head-down)': contradictory, since head-down is Trendelenburg. D is still the only barium option.

ملاحظة على السؤال: الاختيار D مكتوب 'Anti-Trendelenburg (رأس لأسفل)' وهذا متناقض (رأس لأسفل = Trendelenburg). ومع ذلك D هو الباريوم الوحيد.

N·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

Which diagnostic test shows the cobblestone appearance and aphthous ulcers in Crohn's disease?

A. Abdominal ultrasound
× US lacks mucosal detail. بلا تفاصيل مخاطية.

B. MRI head
× Irrelevant. غير مرتبط.

C. Barium follow-through / small-bowel enema ✓ Book p. 118 ↗

D. Abdominal CT
× CT shows wall thickening, not aphthae. يظهر سماكة الجدار لا القرح الصغيرة.

Why: Barium follow-through / small-bowel enema shows cobblestoning and aphthous ulcers of Crohn's.

لماذا: الباريوم المتتابع/حقنة الأمعاء الدقيقة تظهر cobblestone والقرح القلاعية في كرون.

★ **Tell the students:** Crohn's signs: string sign, skip lesions, cobblestone, fistulas.

قل للطلاب: علامات كرون: string sign وskip lesions وcobblestone وناسور.

N·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which is CORRECT regarding tropical sprue?

A. Involves only proximal small bowel
× It affects the whole small bowel, proximal and distal. يصيب الأمعاء كلها القريبة والبعيدة.

B. No resemblance to celiac disease
× It resembles celiac clinically. يشبه السيلياك إكلينيكياً.

C. Immune-mediated disease
× Celiac is the immune-mediated one. السيلياك هو المناعي.

D. Caused by an infectious (bacterial) etiology ✓ Book p. 92 ↗

Why: Tropical sprue is infectious: bacterial overgrowth (coliforms) in people living in or visiting the tropics.
لماذا: Tropical sprue معدٍ: فرط نمو بكتيري (coliforms) لدى سكان أو زوار المناطق الاستوائية.

★ **Tell the students:** Treat with tetracycline for a prolonged course + folic acid ± B12.
قل للطلاب: العلاج: tetracycline لفترة طويلة + حمض الفوليك ± B12.

Group O

O·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

In malabsorption, which test assesses ileal function?

A. Hydrogen breath test
× Hydrogen breath test: lactase/SIBO. اللاكتيز/SIBO.

B. SeHCAT (bile acid absorption) ✓ Book p. 93 ↗

C. Secretin stimulation
× Secretin: pancreas. البنكرياس.

D. Fecal elastase
× Fecal elastase: pancreas. البنكرياس.

Why: SeHCAT (bile-acid absorption) assesses ileal function.
لماذا: SeHCAT (امتصاص الأحماض الصفراوية) يقيّم وظيفة اللفائفي.

★ **Tell the students:** Mnemonic: Ileum = SeHCAT, Pancreas = secretin/elastase.
قل للطلاب: تذكّر: اللفائفي = SeHCAT، البنكرياس = secretin/elastase.

O·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

A 26-year-old has had eight months of crampy abdominal pain, diarrhea, weight loss, aphthous ulcers, conjunctivitis, and right lower quadrant tenderness. What is the most likely diagnosis?

A. Celiac disease
× Celiac: wheat-related, no RLQ mass/tenderness.
مرتبط بالقمح ولا ألم يمين سفلي.

B. Lactose intolerance
× Lactose intolerance: no weight loss.
لا فقدان وزن.

C. Crohn disease ✓ Book p. 116 ↗

D. Ulcerative colitis
× UC: bloody, rectal, rarely aphthae in the mouth.
دموي ومستقيمي.

Why: Chronic crampy pain, diarrhea, weight loss, aphthous ulcers, conjunctivitis and RLQ tenderness = Crohn's.

لماذا: ألم مزمن وإسهال ونقص وزن وقرح قلاعية والتهاب ملتحمة وألم يمين سفلي = كرون.

★ **Tell the students:** Extra-intestinal: aphthous ulcers, uveitis, arthritis, erythema nodosum, PSC.

قل للطلاب: خارج الأمعاء: قرح قلاعية، التهاب العنبية، مفاصل، erythema nodosum.

O·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

In Crohn's, B12 deficiency occurs from involvement of:

A. Proximal ileum
× The proximal ileum does not absorb B12.
اللفائفي القريب لا يمتص B12.

B. Terminal ileum ✓ Book p. 117 ↗

C. Jejunum
× Jejunum absorbs iron, folate.
الصائم يمتص الحديد والفولات.

D. Colon
× The colon does not absorb B12.
القولون لا يمتص B12.

Why: Vitamin B12 is absorbed in the terminal ileum, so ileal Crohn's causes B12 deficiency.

لماذا: فيتامين B12 يُمتص في اللفائفي الانتهائي، لذا كرون اللفائفي يسبب نقصه.

★ **Tell the students:** Terminal ileal disease: B12 deficiency, bile-acid diarrhea (cholestyramine), gallstones, oxalate stones.

قل للطلاب: اللفائفي الانتهائي: نقص B12، إسهال أحماض صفراوية، حصوات مرارة وأوكسالات.

O·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

A 47-year-old has chronic diarrhea and steatorrhea after eating wheat, fatigue, and iron-deficiency anemia. What is the most likely diagnosis?

A. Crohn disease
× Crohn's: RLQ pain, ileal disease.
كرون: ألم يمين سفلي.

B. Ulcerative colitis
× UC: bloody diarrhea.
UC: إسهال دموي.

C. Celiac disease ✓ Book p. 89 ↗

D. Lactose intolerance
× Lactose intolerance does not cause steatorrhea or anemia.
لا يسبب دهنياً أو أنيميا.

Why: Steatorrhea and iron-deficiency anemia that follow wheat intake = celiac disease.

لماذا: إسهال دهني وأنيميا نقص حديد مع القمح = السيلياك.

★ **Tell the students:** Celiac malabsorption: iron, folate, calcium, fat-soluble vitamins.

قل للطلاب: سوء امتصاص: حديد، فولات، كالسيوم، فيتامينات ذائبة في الدهون.

O·5 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which test diagnoses bacterial overgrowth as a cause of malabsorption?

A. Hydrogen breath test (glucose/lactulose) ✓ Book p. 93 ↗

B. Fecal elastase
× Fecal elastase tests the pancreas.
للبنكرياس.

C. Schilling test
× Schilling is obsolete and not diagnostic.
قديم وغير تشخيصي.

D. D-xylose test
× D-xylose: mucosal disease.
أمراض المخاطية.

Why: A glucose or lactulose hydrogen breath test diagnoses small-bowel bacterial overgrowth.

لماذا: اختبار الهيدروجين بالجلوكوز أو اللاكتولوز يشخص فرط النمو البكتيري.

★ **Tell the students:** The gold standard is culture of small-bowel aspirate (≥10^5 organisms).

قل للطلاب: المعيار الذهبي مزرعة شفط الأمعاء الدقيقة.

Group P

P·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

> Which part of the small bowel is responsible for vitamin B12 absorption?

A. Jejunum
× Jejunum: iron, folate and nutrients.
الصائم: حديد وفولات.

B. Cecum
× The cecum absorbs water.
الأعور يمتص الماء.

C. Terminal ileum ✓ Nearest: p. 117 ↗

D. Duodenum
× Duodenum: iron, calcium.
الاثني عشر: حديد وكالسيوم.

Why: The terminal ileum has the receptors for the B12-intrinsic factor complex.
لماذا: اللفائفي الانتهائي فيه مستقبلات معقد B12 مع العامل الداخلي.

★ **Tell the students:** Resection or disease of >60 cm of terminal ileum causes B12 deficiency.
قل للطلاب: استئصال أو مرض فوق 60 سم من اللفائفي يسبب نقص B12.

P·2 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

> Infant with CHPS — expected US finding:

A. Small cystic mass at pyloric area
× The hypertrophied muscle is solid (an 'olive').
العضلة المتضخمة صلبة (زيتونة).

B. Absent pylorus
× The pylorus is present, only thickened.
البواب موجود لكنه سميك.

C. Pyloric channel length elongated (>16 mm) ✓ Book p. 61 ↗

D. Pyloric wall thickness < 3 mm
× Normal thickness; CHPS is >3 mm.
الطبيعي أقل؛ وCHPS أكبر من 3 مم.

Why: Ultrasound in CHPS shows muscle thickness >3 mm and channel length >15 mm (elongated).
لماذا: الموجات في CHPS: سمك العضلة أكثر من 3 مم وطول القناة أكثر من 15 مم.

★ **Tell the students:** Projectile non-bilious vomiting at 2-8 weeks, firstborn male, hypochloremic alkalosis.
قل للطلاب: قيء قذفي غير صفراوي عمر 2-8 أسابيع، ذكر بكر، قلاء نقص كلور.

P·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Bloody diarrhea can occur in all of the following EXCEPT:

A. Ischemic colitis
× Ischemic colitis causes bloody stools.
الإقفاري يسبب دماً.

B. Diverticulitis
× Diverticular bleeding is typical.
نزف الرتوج شائع.

C. Celiac disease ✓ Book p. 80 ↗

D. C. difficile infection
× Severe C. difficile colitis can be bloody.
التهاب C. difficile الشديد قد يكون دموياً.

Why: Celiac disease causes non-bloody malabsorptive diarrhea.

لماذا: السيلياك يسبب إسهال سوء امتصاص بدون دم.

★ **Tell the students:** Bloody diarrhea = inflammatory/invasive: UC, Crohn's, Shigella, amoeba, EHEC.

قل للطلاب: الدموي = التهابي/غزوي: UC، كرون، شيجيلا، أميبا، EHEC.

Group Q

Q·1 SURGERY جراحة

Digestive System — Chapter 03: The Stomach and Duodenum · Digestive System — Chapter 04: Pathology of GIT Tumours

The following is a characteristic finding in malignant pyloric channel carcinoma:

A. CA 72-4 has higher specificity in this type of cancer
× CA 72-4 is for follow-up, not a feature of the tumor.
واسم متابعة وليس صفة للورم.

B. Epigastric pain radiates to the back
× Back-radiating pain suggests posterior invasion.
ألم الظهر يوحي بغزو خلفي.

C. GIST is the most common type
× Adenocarcinoma, not GIST, is commonest.
الأشهر السرطان الغدي لا GIST.

D. The stomach shadow in a barium meal usually reaches the pelvis ✓ Book p. 59 ↗

Why: A pyloric-channel cancer causes gastric outlet obstruction: the stomach on barium is hugely dilated, may reach the pelvis.

لماذا: سرطان القناة البوابية يسبب انسداد مخرج المعدة: تتمدد بشدة وقد يصل ظلها للحوض.

★ **Tell the students:** Gastric outlet obstruction: succussion splash, projectile vomiting of old food, hypochloremic alkalosis.

قل للطلاب: انسداد المخرج: صوت ارتجاج، قيء طعام قديم، قلاء نقص كلور.

Q·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which statement about tropical sprue is correct?

A. Immune-mediated disease
× Celiac is the immune-mediated disease. السيلياك هو المناعي.

B. No resemblance to celiac disease
× It resembles celiac in symptoms. يشبه السيلياك في الأعراض.

C. Responds to prolonged antibiotics + folate/B12 ✓ Book p. 92 ↗

D. Involves only proximal small bowel
× It injures all levels of the small intestine. يصيب كل مستويات الأمعاء الدقيقة.

Why: Tropical sprue responds to prolonged tetracycline plus folic acid (and B12 if needed).

لماذا: Tropical sprue يستجيب لـ tetracycline لفترة طويلة مع حمض الفوليك (وB12 عند اللزوم).

★ **Tell the students:** Biopsy: subtotal patchy villous atrophy, proximal and distal.

قل للطلاب: الخزعة: ضمور زغابي جزئي متقطع قريب وبعيد.

Q·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

D-xylose absorption test is used for:

A. Diagnosing acute infectious diarrhea
× Infectious diarrhea is diagnosed by stool tests. بفحوص البراز.

B. Differentiating small bowel vs hepatic causes of malabsorption
× It does not test hepatic causes. لا يفحص الكبد.

C. Diagnosing C. difficile
× C. difficile needs a toxin assay. يحتاج فحص السم.

D. Differentiating small bowel (mucosal) vs pancreatic causes of malabsorption ✓ Nearest: p. 93 ↗

Why: D-xylose separates mucosal (abnormal) from pancreatic (normal) malabsorption.

لماذا: D-xylose يفرق بين المخاطي (غير طبيعي) والبنكرياسي (طبيعي).

★ **Tell the students:** Not printed in the book; nearest page is the malabsorption tests page.

قل للطلاب: غير مذكور في الكتاب نصاً؛ أقرب صفحة هي اختبارات سوء الامتصاص.

Q·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

A 19-year-old has chronic diarrhea, bloating, and weight loss that worsen with wheat, along with iron-deficiency anemia and flattened villi on duodenal biopsy. Which specific diagnostic test is indicated?

A. Anti-tissue transglutaminase (anti-tTG) IgA ✓ Book p. 89 ↗

B. Schilling test
× Schilling tests B12 absorption. يفحص امتصاص B12.

C. Secretin test
× Secretin tests the pancreas. للبنكرياس.

D. D-xylose absorption test
× D-xylose is a general absorption test. اختبار امتصاص عام.

Why: Anti-tTG IgA is the specific serologic test for celiac disease.
لماذا: Anti-tTG IgA هو الاختبار المصلي النوعي للسيلياك.

★ **Tell the students:** Flat villi + positive tTG on gluten = diagnosis; start a gluten-free diet.
قل للطلاب: ضمور الزغابات + tTG إيجابي = تشخيص؛ ثم نظام خالٍ من الجلوتين.

Group R

R·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which test is used to diagnose bacterial overgrowth as a cause of malabsorption?

A. D-xylose test
× D-xylose: mucosal disease. أمراض المخاطية.

B. Hydrogen breath test ✓ Book p. 93 ↗

C. Schilling test
× Schilling: B12 absorption. امتصاص B12.

D. Fecal elastase
× Fecal elastase: pancreas. البنكرياس.

Why: The hydrogen breath test (glucose/lactulose) diagnoses bacterial overgrowth.
لماذا: اختبار الهيدروجين (جلوكوز/لاكتولوز) يشخص فرط النمو البكتيري.

★ **Tell the students:** Sensitivity of the 50 g glucose breath test is about 90%.
قل للطلاب: حساسية اختبار الجلوكوز 50 جم نحو 90%.

R·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 06: Constipation (including IBS)

Which food group is specifically listed among high-FODMAP foods to limit in the GI book?

A. Plain water
× Water is FODMAP-free. خالٍ من FODMAP.

B. Onions and garlic ✓ Book p. 104 ↗

C. Egg whites
× Egg whites are low FODMAP. منخفض FODMAP.

D. Lean poultry
× Lean poultry is low FODMAP. منخفض FODMAP.

Why: Onions and garlic are high-FODMAP foods to limit.

لماذا: البصل والثوم من الأطعمة العالية FODMAP ويجب تقليلها.

★ **Tell the students:** FODMAPs = fermentable oligo-, di-, mono-saccharides and polyols (wheat, beans, lactose, apples).

قل للطلاب: FODMAPs = سكريات قابلة للتخمر (قمح، فول، لاكتوز، تفاح).

R·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 06: Constipation (including IBS)

Which statement best reflects how IBS is diagnosed as a disorder of gut–brain interaction?

A. By endoscopy in every patient
× Endoscopy is not needed in every patient. ليس لكل المرضى.

B. By elevated liver enzymes
× Liver enzymes are normal. إنزيمات الكبد طبيعية.

C. Primarily by clinical criteria such as Rome guidelines when routine tests are usually normal ✓ Book p. 101 ↗

D. By a specific structural lesion on imaging
× There is no structural lesion. لا آفة بنيوية.

Why: IBS is diagnosed clinically (Rome criteria); routine tests are normal.

لماذا: IBS يُشخص إكلينيكياً (معايير روما) والفحوص الروتينية طبيعية.

★ **Tell the students:** A positive diagnosis, not a diagnosis of exclusion; investigate only if red flags.

قل للطلاب: تشخيص إيجابي وليس بالاستبعاد؛ نفحص فقط عند وجود علامات إنذار.

R·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Mucus and blood leaking from an actively inflamed intestinal mucosa are characteristic of which diarrhea mechanism?

A. Secretory
× Secretory: watery, no pus or blood.
مائي بلا دم أو صديد.

B. Exudative ✓ Book p. 80 ↗

C. Osmotic
× Osmotic stops with fasting.
يتوقف بالصيام.

D. Altered motility
× Motility: rapid or slow transit.
تغير زمن العبور.

Why: Blood, mucus and pus from inflamed mucosa define exudative diarrhea.
لماذا: الدم والمخاط والصديد من مخاطية ملتهبة يعرّف الإسهال الإفرازي الالتهابي (exudative).

★ **Tell the students:** Secretory diarrhea continues during fasting, osmotic stops.
قل للطلاب: الإفرازي يستمر أثناء الصيام والأسموزي يتوقف.

R·5 INTERNAL MEDICINE باطنة Digestive System — Chapter 06: Constipation (including IBS)

Which factor can alter gut motility and processing through the gut–brain axis in IBS?

A. Portal hypertension only
× Portal hypertension is organic.
عضوي.

B. A fixed intestinal stricture
× A fixed stricture is structural.
بنيوي.

C. Gallstones only
× Gallstones are structural.
بنيوي.

D. Stress and anxiety ✓ Book p. 101 ↗

Why: Stress and anxiety alter gut motility and processing through the gut-brain axis.
لماذا: التوتر والقلق يغيران حركة الأمعاء ومعالجة الإشارات عبر محور الأمعاء-الدماغ.

★ **Tell the students:** IBS: visceral hypersensitivity, altered motility, dysbiosis and the gut-brain axis.
قل للطلاب: IBS: فرط حساسية، تغير حركة، خلل بكتيري ومحور الأمعاء-الدماغ.

Group S

S·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which is the normal range of stool frequency stated in the GI book?

A. Four to six times daily
× 4-6/day is frequent.
4-6 يومياً كثير.

B. Three times a week to three times a day ✓ Book p. 77 ↗

C. Once every two weeks to twice a week
× Once every 2 weeks is constipation.
كل أسبوعين إمساك.

D. Once daily only
× Normal varies widely.
الطبيعي واسع.

Why: Normal stool frequency is 3 times a week to 3 times a day.

لماذا: التكرار الطبيعي من 3 مرات أسبوعياً إلى 3 مرات يومياً.

★ **Tell the students:** Daily bowel opening is not required for a normal habit.

قل للطلاب: الإخراج اليومي ليس شرطاً للطبيعي.

S·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Diarrhea lasting more than four weeks should be classified and evaluated as:

A. Acute diarrhea
× Acute is under 2-3 weeks.
الحاد أقل من 2-3 أسابيع.

B. Chronic diarrhea ✓ Book p. 77 ↗

C. Food poisoning only
× Food poisoning is short and acute.
حاد وقصير.

D. A normal variation
× Not a normal variant.
ليس طبيعياً.

Why: Diarrhea lasting more than 4 weeks is chronic and needs evaluation.

لماذا: الإسهال لأكثر من 4 أسابيع مزمن ويحتاج تقييماً.

★ **Tell the students:** Most chronic diarrhea is non-infectious.

قل للطلاب: معظم الإسهال المزمن غير معدٍ.

S·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which ion is the major actively absorbed ion in the intestine according to the GI book?

A. Bicarbonate
× Secreted/absorbed passively. لا يُمتص بنشاط رئيسي.

B. Potassium
× Mostly secreted into stool. يُفقد بالبراز.

C. Sodium ✓ Book p. 78 ↗

D. Chloride
× Cl- is the major ion secreted. الكلوريد هو الذي يُفرز.

Why: Na+ is the major ion actively absorbed; Cl- is the major ion secreted.

لماذا: الصوديوم هو الأيون الرئيسي الذي يُمتص بنشاط، والكلوريد هو الذي يُفرز.

★ **Tell the students:** Na-glucose co-transport is the basis of oral rehydration solution.

قل للطلاب: نقل الصوديوم مع الجلوكوز هو أساس محلول الإماهة الفموي.

Group T

T·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 01: GI Symptoms and Investigations

After negative gastroscopy and colonoscopy, which test may be used to investigate obscure GI bleeding from the small bowel?

A. ERCP
× ERCP is for bile/pancreatic ducts. للقنوات الصفراوية.

B. Liver biopsy
× Liver biopsy is irrelevant. غير مرتبط.

C. Wireless capsule endoscopy ✓ Book p. 7 ↗

D. Plain abdominal X-ray
× Plain X-ray lacks mucosal detail. لا تفاصيل مخاطية.

Why: Capsule endoscopy is used for obscure GI bleeding after negative gastroscopy and colonoscopy.

لماذا: المنظار الكبسولي يستخدم للنزف الغامض بعد منظار علوي وسفلي سلبيين.

★ **Tell the students:** Positive capsule findings are treated with balloon-assisted enteroscopy.

قل للطلاب: نتائج الكبسولة الإيجابية تعالج بمنظار البالون.

T·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 06: Constipation (including IBS)

Which statement about probiotics for IBS matches the GI book?

A. They are diagnostic rather than therapeutic
× Not diagnostic. ليست تشخيصية.

B. There is no routine recommendation for probiotics ✓ Book p. 105 ↗

C. They replace all dietary measures
× They do not replace diet. لا تغني عن النظام الغذائي.

D. They are routinely recommended for every patient
× Contradicts the book. يخالف الكتاب.

Why: The book states there is no routine recommendation for probiotics in IBS.

لماذا: الكتاب يذكر أنه لا توصية روتينية بالـprobiotics في IBS.

★ **Tell the students:** Evidence is inconsistent; focus on diet, fibre, antispasmodics and gut-directed therapy.

قل للطلاب: الأدلة غير ثابتة؛ التركيز على الغذاء والألياف ومضادات التشنج.

T·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Approximately what proportion of acute diarrhea is attributed to infection in the GI book?

A. 30%
× Too low. قليل.

B. 10%
× Too low. قليل.

C. 80% ✓ Book p. 77 ↗

D. Nearly 100%
× Not all are infectious. ليس كله معدياً.

Why: Approximately 80% of acute diarrhea is infectious (viruses, bacteria, parasites).

لماذا: نحو 80% من الإسهال الحاد معدٍ (فيروسات وبكتيريا وطفيليات).

★ **Tell the students:** Chronic diarrhea is mostly non-infectious.

قل للطلاب: المزمن غالباً غير معدٍ.

T·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 06: Constipation (including IBS)

Which IBS subtype is characterized by predominantly loose stools?

A. IBS-D ✓ Book p. 103 ↗

B. IBS-C
× IBS-C: hard stools. IBS-C: براز صلب.

C. IBS-U
× IBS-U: unclassified. غير مصنف.

D. IBS-M
× IBS-M: mixed. مختلط.

Why: IBS-D is the subtype with predominantly loose stools.

لماذا: IBS-D هو النوع الذي يغلب عليه البراز السائب.

★ **Tell the students:** Bristol stool scale decides the subtype.

قل للطلاب: مقياس بريستول يحدد النوع.

Group U

U·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

With low dietary fibre intake, which stool output is listed as meeting the GI book's diarrhea definition?

A. More than 1 kg per day
× Far above the cut-off. أعلى بكثير من الحد.

B. More than 50 g per day
× Within normal. ضمن الطبيعي.

C. More than 100 g per day
× Still within normal limits. ما زال طبيعياً.

D. More than 200 g per day ✓ Book p. 3 ↗

Why: On a low-fibre diet, stool above 200 g/day meets the definition of diarrhea.

لماذا: مع نظام قليل الألياف، براز أكثر من 200 جم يومياً يعد إسهالاً.

★ **Tell the students:** Diarrhea = increased frequency, volume and fluidity.

قل للطلاب: الإسهال = زيادة التكرار والكمية والسيولة.

U·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 06: Constipation (including IBS)

IBS with both constipation and diarrhea is classified as:

A. IBS-U
× IBS-U: unclassifiable. غير مصنف.

B. IBS-C
× IBS-C: constipation only. إمساك فقط.

C. IBS-M ✓ Book p. 103 ↗

D. IBS-D
× IBS-D: diarrhea only. إسهال فقط.

Why: IBS with both constipation and diarrhea is IBS-M (mixed).

لماذا: IBS مع إمساك وإسهال معاً هو IBS-M (مختلط).

★ **Tell the students:** Subtype counts by days with abnormal stool form.

قل للطلاب: التصنيف حسب شكل البراز في الأيام المصابة.

U·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 03: The Stomach and Duodenum

Which statement about anticoagulants and acute GI bleeding is most accurate?

A. They do not usually cause bleeding by themselves, but can worsen bleeding from another lesion ✓ Book p. 50 ↗

B. They commonly cause ulcers directly
× They do not commonly cause ulcers. لا تسبب القرح عادة.

C. They only increase lower GI bleeding
× They affect upper and lower bleeding. تؤثر على العلوي والسفلي.

D. They prevent bleeding from peptic ulcers
× They never protect against bleeding. لا تحمي من النزف.

Why: Anticoagulants usually do not cause a bleeding lesion themselves, but they worsen bleeding from another lesion.

لماذا: مضادات التجلط عادة لا تسبب آفة نازفة بنفسها لكنها تزيد شدة النزف من آفة أخرى.

★ **Tell the students:** In life-threatening bleeding, reverse anticoagulation (vitamin K / PCC).

قل للطلاب: عند النزف المهدد للحياة يُعكس تأثير المضاد (فيتامين K / PCC).

U·4 SURGERY جراحة Digestive System — Chapter 04: Pathology of GIT Tumours

Early gastric carcinoma is defined by invasion limited to which layers?

A. Mucosa and submucosa ✓ Book p. 72 ↗

B. Serosa only
× Serosal invasion is advanced.
الغزو المصلي متقدم.

C. Any depth if there are no symptoms
× Depth defines it, not symptoms.
العمق هو المعيار لا الأعراض.

D. Muscularis propria only
× Muscle wall invasion = late gastric cancer.
غزو العضلة = متأخر.

Why: Early gastric cancer is confined to the mucosa and submucosa (regardless of nodal status).
لماذا: سرطان المعدة المبكر محصور في المخاطية وتحت المخاطية (بغض النظر عن العقد).

★ **Tell the students:** Early disease: excellent prognosis after resection; EMR/ESD only if small, mucosal, no ulcer.
قل للطلاب: المبكر: تشخيص ممتاز بعد الاستئصال؛ EMR/ESD فقط للصغير المخاطي.

U·5 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Which electrolyte disturbance is listed as a risk factor for functional acute gastric dilatation?

A. Hypercalcemia only
× Not listed.
غير مذكور.

B. Severe hypokalemia ✓ Book p. 68 ↗

C. Hypophosphatemia only
× Not listed.
غير مذكور.

D. Isolated hypernatremia
× Hypernatremia is not listed.
غير مذكور.

Why: Severe hypokalemia is a listed metabolic risk factor for functional acute gastric dilatation.
لماذا: نقص البوتاسيوم الشديد عامل خطر مذكور للتمدد الوظيفي الحاد للمعدة.

★ **Tell the students:** Also: DKA, hypomagnesemia, eating disorders, CPR, opioids and anticholinergics.
قل للطلاب: أيضاً: DKA ونقص المغنيسيوم واضطرابات الأكل والأفيونات ومضادات الكولين.

Group V

V·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 06: Constipation (including IBS)

The core clinical presentation of irritable bowel syndrome is:

A. Persistent jaundice
× Jaundice is not IBS. — ليس IBS.

B. Recurrent abdominal pain related to defecation with altered bowel habits ✓ Book p. 101 ↗

C. Painless rectal bleeding
× Painless rectal bleeding suggests piles/cancer. — يوحي بالبواسير/سرطان.

D. Progressive dysphagia
× Progressive dysphagia is an alarm feature. — علامة إنذار.

Why: IBS = recurrent abdominal pain related to defecation with altered bowel habits.

لماذا: IBS = ألم بطني متكرر مرتبط بالإخراج مع تغير نمط الإخراج.

★ **Tell the students:** Rome: pain at least 1 day/week for 3 months, linked to defecation, with stool change.

قل للطلاب: روما: ألم يوم أسبوعياً لـ3 أشهر مرتبط بالإخراج مع تغير البراز.

V·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

Which stool test is especially useful as a marker of intestinal inflammation in suspected IBD?

A. Fecal occult fat
× Fecal fat: steatorrhea. — الدهنية.

B. Fecal elastase
× Fecal elastase: pancreas. — البنكرياس.

C. Fecal calprotectin ✓ Book p. 118 ↗

D. Stool pH
× Stool pH: carbohydrate malabsorption. — سوء امتصاص كربوهيدرات.

Why: Fecal calprotectin (neutrophil protein) marks intestinal inflammation in IBD.

لماذا: Calprotectin بروتين من العدلات يدل على التهاب الأمعاء في IBD.

★ **Tell the students:** Normal calprotectin in a patient with diarrhea points to IBS rather than IBD.

قل للطلاب: طبيعي مع إسهال يوحي بـ IBS لا IBD.

V·3 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

> Which finding may indicate that conservative management of acute gastric dilatation is succeeding?

A. Pneumoperitoneum on CT
× Pneumoperitoneum = perforation, surgery. انثقاب، جراحة.

B. Persistent severe pain after NG decompression
× Persistent pain = failure. فشل.

C. New peritonitis
× Peritonitis = surgery. جراحة.

D. Pain resolves after NG decompression and imaging shows no ischemia or perforation ✓ Book p. 68 ↗

Why: If NG decompression relieves pain and imaging shows no ischemia or perforation, conservative management is working.

لماذا: إذا خففت الشفط الأنفي المعدي الألم ولا يوجد إقفار أو انثقاب فالعلاج التحفظي ناجح.

★ **Tell the students:** Continue fasting, IV fluids, IV PPI and serial abdominal exams.

قل للطلاب: صيام وسوائل وريدية وPPI وفحوص بطن متكررة.

Group W

W·1 INTERNAL MEDICINE باطنة

Digestive System — Chapter 01: GI Symptoms and Investigations · Digestive System — Chapter 03: The Stomach and Duodenum

> Which symptom is an alarm feature that warrants investigation in a patient with dyspepsia?

A. Progressive dysphagia ✓ Book p. 2 ↗

B. Intermittent flatulence
× Flatulence is functional. وظيفي.

C. Belching
× Belching is functional. وظيفي.

D. Occasional bloating
× Occasional bloating is functional. وظيفي.

Why: Progressive dysphagia is an alarm feature in dyspepsia and mandates endoscopy.

لماذا: عسر البلع المتزايد علامة إنذار في عسر الهضم وتستلزم منظاراً.

★ **Tell the students:** Alarms: dysphagia, weight loss, anemia, vomiting, bleeding, age over 55 new-onset.

قل للطلاب: الإنذار: عسر بلع، نقص وزن، أنيميا، قيء، نزف، بدء الأعراض فوق 55.

W·2 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Which finding is an indication for emergency laparotomy in acute gastric dilatation?

A. Failure of NG decompression ✓ Book p. 68 ↗

B. Successful NG decompression with pain relief
× Success means conservative care. النجاح يعني علاجاً تحفظياً.

C. Mild isolated bloating
× Mild bloating does not need surgery. لا يحتاج جراحة.

D. Normal gastric wall enhancement
× Normal wall enhancement means a viable stomach. جدار حي.

Why: Failure of NG decompression is an indication for emergency laparotomy.

لماذا: فشل التفريغ بالأنبوب الأنفي المعدي استطباب لفتح البطن الطارئ.

★ **Tell the students:** Other indications: pneumoperitoneum, peritonitis, septic shock.

قل للطلاب: استطبابات أخرى: هواء حر، التهاب بريتوني، صدمة إنتانية.

W·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 06: Constipation (including IBS)

Which dietary approach is described in the GI book for managing IBS symptoms?

A. Exclusive parenteral nutrition
× Parenteral nutrition is not needed. غير مطلوب.

B. Low-FODMAP diet ✓ Book p. 104 ↗

C. Lifelong gluten elimination for every patient
× Gluten elimination is not for every patient. ليس لكل المرضى.

D. High-FODMAP diet
× High FODMAP worsens symptoms. يزيد الأعراض.

Why: A low-FODMAP diet is the dietary approach for IBS in the book.

لماذا: نظام منخفض FODMAP هو النهج الغذائي لـ IBS في الكتاب.

★ **Tell the students:** Soluble fibre helps; insoluble fibre may worsen bloating.

قل للطلاب: الألياف الذائبة تفيد والغير ذائبة قد تزيد الانتفاخ.

W·4 SURGERY جراحة Digestive System — Chapter 04: Pathology of GIT Tumours

The most common gross type of esophageal carcinoma described in the GI book is:

A. Exophytic ✓ Book p. 71 ↗

B. Scirrhous
× Scirrhous (annular) is the least frequent. الأقل شيوعاً.

C. Multicystic
× Not a gross type. ليس نوعاً.

D. Flat infiltrating
× Not the commonest. ليس الأشهر.

Why: The exophytic (polypoid/fungating) type is the commonest gross type.

لماذا: النوع النامي نحو التجويف (polypoid/fungating) هو الأشهر.

★ **Tell the students:** Gross types: exophytic > ulcerative > infiltrating (stricturing).

قل للطلاب: الأنواع: نامٍ > متقرح > متسلل.

Group X

X·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

How much fluid is delivered to the small intestine daily on a normal diet, according to the GI book?

A. About 1 L
× 1-2 L reaches the colon. 1-2 لتر يصل للقولون.

B. About 3 L
× Too low. قليل.

C. About 20 L
× Too high. كثير.

D. About 9 L ✓ Book p. 77 ↗

Why: About 9 L of fluid enters the small intestine daily.

لماذا: نحو 9 لتر من السوائل تدخل الأمعاء الدقيقة يومياً.

★ **Tell the students:** Sources: diet plus saliva, gastric, biliary, pancreatic and intestinal secretions.

قل للطلاب: المصادر: الغذاء + اللعاب والمعدة والصفراء والبنكرياس والأمعاء.

X·2 SURGERY جراحة

Digestive System — Chapter 04: Pathology of GIT Tumours · Digestive System — Chapter 14: Polyposis of Colon

A villous adenoma of the colon is typically:

A. A non-neoplastic inflammatory lesion
× It is a neoplasm. ورم حقيقي.

B. Multiple and pedunculated
× Multiple pedunculated = tubular adenoma. المتعدد المعنّق = أنبوبي.

C. Single, large, sessile, and broad-based ✓ Book p. 73 ↗

D. Always smaller than a tubular adenoma
× Usually larger. عادة أكبر.

Why: A villous adenoma is typically a single, large, sessile, broad-based lesion.

لماذا: الورم الغدي الزغابي عادة مفرد كبير جالس عريض القاعدة.

★ **Tell the students:** Villous has the highest malignant potential; may cause secretory diarrhea with hypokalemia.

قل للطلاب: الزغابي أعلى خطورة؛ وقد يسبب إسهالاً مع نقص بوتاسيوم.

X·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which statement about fluid absorption in the small intestine is correct?

A. No fluid is absorbed before the colon
× Most fluid is absorbed before the colon. معظمها يُمتص قبل القولون.

B. The small intestine normally absorbs only 10%
× 10% only reaches the colon. 10% فقط يصل للقولون.

C. All fluid is excreted in stool
× Only 100-200 mL is excreted. 100-200 مل فقط تخرج.

D. About 90% of the delivered fluid is absorbed there ✓ Book p. 77 ↗

Why: About 90% of the delivered fluid is absorbed in the small intestine.

لماذا: نحو 90% من السوائل الواصلة يُمتص في الأمعاء الدقيقة.

★ **Tell the students:** The colon can absorb all that reaches it, so diarrhea means colonic capacity is exceeded.

قل للطلاب: القولون يستوعب ما يصله؛ الإسهال يعني تجاوز قدرته.

X·4 SURGERY جراحة Digestive System — Chapter 04: Pathology of GIT Tumours

Which gastric polyp type has essentially no carcinoma risk in the GI book?

A. Fundic gland polyp ✓ Book p. 72 ↗

B. Hyperplastic polyp
× Hyperplastic: slight risk.
خطر ضئيل.

C. Villous polyp
× Villous types are neoplastic.
نيوبلازمي.

D. Adenomatous polyp
× Adenomatous: moderate risk.
خطر متوسط.

Why: The risk of carcinoma is nil in fundic gland polyps (slight in hyperplastic, moderate in adenomatous).

لماذا: خطر السرطان معدوم في fundic gland polyps (ضئيل في hyperplastic ومتوسط في adenomatous).

★ **Tell the students:** Gastric polyps: hyperplastic 70%, fundic gland 20%, adenomatous 10%.

قل للطلاب: التوزيع: hyperplastic 70%، fundic 20%، adenomatous 10%.

X·5 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

In acute gastric dilatation, contrast CT is especially useful to assess gastric dimensions and:

A. Portal vein pressure
× Portal pressure is not measured.
لا يقيس ضغط البابي.

B. Wall enhancement and the position of the NG tube ✓ Book p. 68 ↗

C. Appendiceal diameter
× Appendiceal diameter is irrelevant.
غير مرتبط.

D. Gallbladder ejection fraction
× Gallbladder EF is irrelevant.
غير مرتبط.

Why: CT with IV contrast defines gastric size, wall enhancement and NG-tube position.

لماذا: CT بالصبغة يحدد حجم المعدة وتروية الجدار ووضع الأنبوب.

★ **Tell the students:** CT with contrast is the gold standard for AGD.

قل للطلاب: CT بالصبغة هو المعيار الذهبي.

Group Y

Y·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 01: GI Symptoms and Investigations

Difficulty swallowing is termed:

A. Dyspepsia
× Dyspepsia = upper abdominal discomfort.
عسر هضم.

B. Odynophagia
× Odynophagia = painful swallowing.
بلع مؤلم.

C. Dysphagia ✓ Book p. 3 ↗

D. Regurgitation
× Regurgitation = effortless return of food.
ارتجاع طعام.

Why: Dysphagia = difficulty swallowing.

لماذا: Dysphagia = صعوبة البلع.

★ **Tell the students:** Progressive dysphagia + weight loss: cancer until proven otherwise.

قل للطلاب: عسر بلع متزايد + نقص وزن: سرطان حتى يثبت العكس.

Y·2 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

A serious complication of acute gastric dilatation is:

A. Gastric perforation with hemodynamic collapse ✓ Book p. 68 ↗

B. Esophageal varices
× Varices: portal hypertension.
ضغط بابي.

C. Painless jaundice
× Painless jaundice: pancreatic cancer.
سرطان البنكرياس.

D. Appendiceal rupture
× Not related.
غير مرتبط.

Why: Gastric ischemia, perforation and hemodynamic collapse are the serious complications.

لماذا: الإقفار والانثقاب والانهيار الدوراني هي المضاعفات الخطيرة.

★ **Tell the students:** Mortality 80-100% after ischemia or perforation.

قل للطلاب: الوفيات 80-100%.

Y·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which mechanism best describes secretory diarrhea?

A. Mechanical bowel obstruction
× Obstruction is not a diarrhea mechanism.
ليست آلية إسهال.

B. Loss of pancreatic lipase
× Lipase loss gives steatorrhea.
إسهال دهني.

C. Increased intestinal chloride and water secretion ✓ Book p. 79 ↗

D. Unabsorbed solutes remaining in the lumen
× Unabsorbed solutes = osmotic.
أسموزي.

Why: Secretory diarrhea results from increased chloride and water secretion.

لماذا: الإسهال الإفرازي (secretory) ينتج عن زيادة إفراز الكلوريد والماء.

★ **Tell the students:** Large volume (>1 L), watery, persists on fasting, no pus: cholera, VIPoma.

قل للطلاب: حجم كبير، مائي، يستمر بالصيام، بلا صديد: كوليرا، VIPoma.

Group Z

Z·1 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

The first supportive step for a patient with acute gastric dilatation is:

A. Immediate oral feeding
× Oral feeding worsens dilatation.
يزيد التمدد.

B. Elective colonoscopy
× Not indicated.
غير مناسب.

C. Long-term antacids alone
× Antacids alone are inadequate.
غير كافية.

D. Nasogastric decompression and IV fluid resuscitation ✓ Book p. 68 ↗

Why: First step: nasogastric decompression and IV fluid/electrolyte resuscitation.

لماذا: الخطوة الأولى: تفريغ المعدة بأنبوب أنفي معدي مع سوائل وريدية وأملاح.

★ **Tell the students:** Add IV PPI and keep the patient fasting.

قل للطلاب: أضف PPI وريدي مع الصيام.

Z·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Compared with acute diarrhea, most causes of chronic diarrhea are:

A. Always parasitic
× Not always parasitic. ليست دائماً طفيلية.

B. Due to viruses only
× Not viral only. ليست فيروسية فقط.

C. Noninfectious ✓ Book p. 77 ↗

D. Due to food allergy only
× Not only food allergy. ليست حساسية فقط.

Why: Most causes of chronic diarrhea are noninfectious.

لماذا: معظم أسباب الإسهال المزمن غير معدية.

★ **Tell the students:** Contrast: about 80% of acute diarrhea is infectious.

قل للطلاب: المقابل: 80% من الحاد معدٍ.

Z·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

What is the primary mechanism of osmotic diarrhea?

A. Reduced colonic motility
× That is altered motility. اضطراب حركة.

B. Active chloride secretion
× Active chloride secretion = secretory. إفرازي.

C. Mucosal bleeding
× Mucosal bleeding = exudative. التهابي.

D. Poorly absorbed osmotically active solutes in the bowel lumen ✓ Book p. 79 ↗

Why: Osmotic diarrhea results from poorly absorbed, osmotically active solutes in the lumen.

لماذا: الإسهال الأسموزي ينتج عن مواد نشطة أسموزياً غير ممتصة داخل التجويف.

★ **Tell the students:** Stops with fasting; e.g. lactase deficiency, lactulose, magnesium laxatives.

قل للطلاب: يتوقف بالصيام؛ مثل نقص اللاكتيز واللاكتولوز.

Z·4 SURGERY جراحة Digestive System — Chapter 13: Intestinal Obstruction

Treatment of early intussusception in children:

A. Antibiotics only
× Antibiotics alone do not reduce it. لا تردّه.

B. Immediate surgical resection always
× Surgery only if reduction fails or complicated. الجراحة عند الفشل أو التعقيد.

C. IV fluids only
× Fluids alone are inadequate. غير كافية.

D. Conservative pneumatic/hydrostatic enema reduction (success rate >80% when treated early) ✓ Book p. 171 ↗

Why: Early uncomplicated intussusception is reduced by hydrostatic/pneumatic enema (success above 80%).
لماذا: الانغلاف المبكر غير المعقد يُرد بالحقنة المائية/الهوائية (نجاح فوق 80%).

★ **Tell the students:** Contraindicated with peritonitis or perforation. Give fluids and antibiotics first.
قل للطلاب: ممنوعة عند التهاب بريتوني أو انثقاب. تُعطى سوائل ومضادات أولاً.

Group AA

AA·1 SURGERY جراحة Digestive System — Chapter 12: Appendix

Why is acute appendicitis in elderly and children usually complicated?

A. The appendix is always retrocecal in them
× Position is not the reason. ليس بسبب الوضع.

B. The omentum and peritoneal cavity are less efficient in localizing the disease ✓ Book p. 154 ↗

C. Antibiotic response is poor in these ages
× Response to antibiotics is not poor. الاستجابة ليست ضعيفة.

D. The appendix is less vascular in these age groups
× Not the explanation. ليس التفسير.

E. They tolerate pain better and present late only
× They present late but because of blunted signs. يتأخرون بسبب غياب العلامات.

Why: In children and the elderly the omentum and peritoneum localize infection poorly, so perforation is common.
لماذا: عند الأطفال وكبار السن الثرب والبريتون أقل كفاءة في تحديد الالتهاب فيحدث الانثقاب.

★ **Tell the students:** Short omentum in children; atherosclerotic thrombosis plus blunted signs in the elderly.
قل للطلاب: ثرب قصير عند الأطفال؛ وتجلط وعائي مع علامات خفيفة عند كبار السن.

AA·2 SURGERY جراحة

Digestive System — Chapter 01: GI Symptoms and Investigations · Digestive System — Chapter 38: Acute Abdomen

On an erect plain X-ray, air under the diaphragm indicates:

A. Normal
× Never normal.
ليس طبيعياً.

B. Ileus
× Ileus shows dilated loops.
ileus يظهر أمعاء متمددة.

C. Perforated viscus (pneumoperitoneum) → surgical exploration ✓ Book p. 432 ↗

D. Simple SBO
× Simple SBO shows air-fluid levels.
SBO يظهر مستويات سائل-هواء.

Why: Air under the diaphragm on an erect film means perforated viscus (pneumoperitoneum): explore surgically.

لماذا: هواء تحت الحجاب في الأشعة الواقفة = انثقاب حشوي (pneumoperitoneum): استكشاف جراحي.

★ **Tell the students:** Take the film erect; free air may be absent in about a third of perforated ulcers.

قل للطلاب: تؤخذ واقفاً؛ وقد لا يظهر الهواء في ثلث حالات القرحة المثقوبة.

AA·3 SURGERY جراحة Digestive System — Chapter 13: Intestinal Obstruction

When assessing intestinal obstruction, what is the correct sequence?

A. No need to distinguish between types
× Types must be distinguished.
يجب التفريق.

B. CT, followed by anything else
× Clinical assessment comes first.
الإكلينيكي أولاً.

C. Mechanical vs functional → cause → simple vs complicated → management ✓ Book p. 163 ↗

D. Immediate surgery
× Not all need immediate surgery.
ليس كلها تحتاج جراحة فورية.

Why: Sequence: mechanical vs functional, then cause, then simple vs strangulated, then management.

لماذا: التسلسل: ميكانيكي أم وظيفي، ثم السبب، ثم بسيط أم مختنق، ثم العلاج.

★ **Tell the students:** Also classify: acute vs chronic, partial vs complete.

قل للطلاب: ويصنف: حاد/مزمن وجزئي/كامل.

AA·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

Which is NOT a barium enema finding in Ulcerative Colitis?

A. Loss of haustrations (lead-pipe colon)
× Lead-pipe colon is UC. .UC

B. Shortened colon
× Shortened colon is UC. .UC

C. Granular mucosa
× Granular mucosa is UC. .UC

D. Skip lesions with cobblestoning ✓ Nearest: p. 118 ↗

E. Pseudopolyps
× Pseudopolyps are UC. .UC

Why: Skip lesions with cobblestoning are Crohn's; UC gives continuous granular mucosa.

لماذا: Skip lesions مع cobblestone من صفات كرون؛ أما UC فمتصل ومخاطيته حُبيبية.

★ **Tell the students:** UC barium: loss of haustra, granularity, pseudopolyps, short colon.

قل للطلاب: باريوم UC: اختفاء الثنيات، حبيبية، pseudopolyps، قولون قصير.

AA·5 SURGERY جراحة Digestive System — Chapter 13: Intestinal Obstruction

In acute intestinal obstruction, signs of peritonitis indicate:

A. Sigmoid volvulus resolved
× Not a sign of resolution. ليست علامة تحسن.

B. Strangulation ✓ Book p. 165 ↗

C. Pseudo-obstruction
× Pseudo-obstruction is non-mechanical. غير ميكانيكي.

D. Paralytic ileus
× Ileus is painless and quiet. ileus صامت.

E. Simple obstruction
× Simple obstruction has no peritonism. البسيط بلا التهاب بريتوني.

Why: Peritonitis in obstruction signals strangulation (ischemic bowel).

لماذا: علامات التهاب البريتون في الانسداد تعني اختناق (إقفار الأمعاء).

★ **Tell the students:** Constant pain, fever, tachycardia, leukocytosis = operate urgently.

قل للطلاب: ألم مستمر وحمى وتسرع قلب وارتفاع كرات = جراحة عاجلة.

Group AB

AB·1 SURGERY جراحة Digestive System — Chapter 13: Intestinal Obstruction

> Infantile intussusception usually occurs during:

A. 2–5 years
× Less common.
أقل شيوعاً.

B. First year of life (esp. 6–12 months) ✓ Nearest: p. 170 ↗

C. First month of life
× Rare in the first month.
نادر.

D. Adolescence
× Usually has a lead point.
غالباً له نقطة قيادة.

E. School age
× Less common.
أقل شيوعاً.

Why: Infantile intussusception peaks in the first year of life, especially 6-12 months.
لماذا: الانغلاف عند الرضع يحدث في السنة الأولى خاصة 6-12 شهراً.

★ **Tell the students:** Older children: look for a lead point (Meckel, polyp, lymphoma).
قل للطلاب: الأكبر سناً: ابحث عن نقطة قيادة (ميكل، سليلة، ورم).

AB·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

> A 25-year-old man with recurrent discharging peri-anal opening, weight loss, bloody diarrhea, tenesmus, and normal-looking rectum on proctoscopy. Most likely diagnosis:

A. Cryptoglandular anal fistula
× Cryptoglandular fistula has no systemic features.
بلا أعراض جهازية.

B. Crohn's disease ✓ Book p. 116 ↗

C. Ulcerative colitis
× UC always involves the rectum.
UC يصيب المستقيم دائماً.

D. TB proctitis
× TB proctitis is rare.
نادر.

E. Rectal cancer
× Rectal cancer would look abnormal.
سيظهر غير طبيعي.

Why: Recurrent perianal fistula, weight loss, bloody diarrhea with a normal-looking rectum = Crohn's (rectal sparing).
لماذا: ناسور حول الشرج متكرر ونقص وزن وإسهال دموي مع مستقيم طبيعي = كرون (المستقيم سليم).

★ **Tell the students:** Crohn's: perianal disease, skip lesions, rectal sparing.
قل للطلاب: كرون: آفات شرجية وskip lesions والمستقيم سليم.

AB·3 SURGERY جراحة Digestive System — Chapter 38: Acute Abdomen

Which is NOT a feature of generalized peritonitis?

A. Board-like abdomen
× A feature. من علاماته.

B. Silent abdomen on auscultation
× A feature. من علاماته.

C. Severe colicky abdominal pain ✓ Book p. 55 ↗

D. Diffuse tenderness and rigidity
× A feature. من علاماته.

E. Fever and tachycardia
× A feature. من علاماته.

Why: Generalized peritonitis gives constant pain with board-like rigidity, not colicky pain.
لماذا: التهاب البريتون العام يعطي ألماً مستمراً مع بطن خشبي وليس مغصاً.

★ **Tell the students:** Patient lies still in peritonitis; colic makes the patient roll (obstruction).
قل للطلاب: مريض الالتهاب البريتوني ساكن، والمغص يجعله يتقلب (انسداد).

Group AC

AC·1 SURGERY جراحة Digestive System — Chapter 13: Intestinal Obstruction

The nature of pain in small-intestinal obstruction:

A. Constant dull ache
× Constant ache suggests strangulation. مستمر يوحي بالاختناق.

B. Colicky, sharp, periumbilical pain that coincides with peristaltic waves ✓ Book p. 164 ↗

C. Burning epigastric
× Epigastric burning suggests ulcer. قرحة.

D. Radiating to back
× Back pain suggests pancreas/aorta. بنكرياس/أبهر.

Why: SBO pain is colicky, periumbilical and coincides with peristaltic waves.
لماذا: ألم انسداد الأمعاء الدقيقة مغصي حول السرة ويتزامن مع الحركات الدودية.

★ **Tell the students:** Colic that becomes constant = strangulation or closed loop.
قل للطلاب: المغص الذي يصبح مستمراً = اختناق أو closed loop.

AC·2 SURGERY جراحة Digestive System — Chapter 12: Appendix

A corroborative sign of acute appendicitis is:

A. Rovsing's sign — RIF pain on LIF pressure ✓ Book p. 155 ↗

B. Cullen's sign
× Cullen's = periumbilical bruising (pancreatitis). كدمة سرة (التهاب بنكرياس).

C. Relief of pain with thigh extension
× Psoas sign is pain on extension. psoas ألم عند البسط.

D. Relief of pain with internal rotation of the right thigh
× Obturator sign is pain on internal rotation. obturator ألم عند التدوير الداخلي.

E. Decrease of pain with testicular elevation
× Testicular elevation relief suggests epididymitis (Prehn). يوحي بالتهاب البربخ.

Why: Rovsing's sign: pressure in the LIF causes pain in the RIF.

لماذا: علامة Rovsing: الضغط في الحفرة اليسرى يسبب ألماً في اليمنى.

★ **Tell the students:** Also: McBurney point tenderness and Blumberg (rebound).

قل للطلاب: أيضاً: McBurney وBlumberg.

AC·3 SURGERY جراحة Digestive System — Chapter 08: Inflammatory Bowel Disease

Indication for total proctocolectomy in Ulcerative Colitis includes all EXCEPT:

A. Toxic megacolon not responding to medical therapy
× Indication. استطباب.

B. Occasional bouts of colic and diarrhea ✓ Book p. 121 ↗

C. Evidence of malignancy/dysplasia
× Indication. استطباب.

D. Massive uncontrolled hemorrhage
× Indication. استطباب.

E. Failure of medical treatment
× Indication. استطباب.

Why: Occasional colic and diarrhea are controlled medically and do not need colectomy.

لماذا: نوبات مغص وإسهال عرضية تُعالج دوائياً ولا تستدعي جراحة.

★ **Tell the students:** Total proctocolectomy + ileal pouch (IPAA) cures UC.

قل للطلاب: استئصال القولون والمستقيم مع جيب لفائفي يشفي UC.

AC·4 SURGERY جراحة Digestive System — Chapter 13: Intestinal Obstruction

A child presents with colicky pain, a right iliac fossa mass, and red currant jelly stools:

A. Intussusception (ileocolic — the most common form in children) ✓ Book p. 170 ↗

B. Volvulus
× Volvulus: bilious vomiting, no jelly stool. قيء صفراوي بلا براز جيلي.

C. Meckel's
× Meckel's bleeds painlessly. ينزف بلا ألم.

D. Appendicitis
× Appendicitis: no jelly stool. لا براز جيلي.

Why: Colicky pain, RIF mass and red currant jelly stool = ileocolic intussusception.

لماذا: مغص وكتلة يمين سفلي وبراز جيلي = انغلاف لفائفي-قولوني.

★ **Tell the students:** Triad: colicky pain, sausage-shaped mass, currant jelly stool; Dance sign.

قل للطلاب: الثلاثي: مغص، كتلة نقانقية، براز جيلي؛ Dance sign.

Group AD

AD·1 SURGERY جراحة Digestive System — Chapter 11: Diverticulosis of the Colon

The usual presentation of uncomplicated diverticulosis is:

A. Fever and throbbing pain
× Fever and pain = diverticulitis. التهاب رتجي.

B. Bleeding per rectum
× Bleeding is a complication. مضاعفة.

C. Absolute constipation
× Absolute constipation = obstruction. انسداد.

D. Asymptomatic ✓ Nearest: p. 146 ↗

E. Severe LIF pain with rigidity
× Rigidity = perforation/peritonitis. انثقاب/التهاب بريتون.

Why: Uncomplicated diverticulosis is usually asymptomatic.

لماذا: الرتوج غير المصحوبة بمضاعفات غالباً بلا أعراض.

★ **Tell the students:** About 80% never develop symptoms; 10-25% develop diverticulitis.

قل للطلاب: نحو 80% بلا أعراض؛ و10-25% يتطور لالتهاب.

AD·2 SURGERY جراحة Digestive System — Chapter 13: Intestinal Obstruction

The commonest cause of intestinal obstruction in children is:

A. Paralytic ileus
× Ileus is functional. وظيفي.

B. Neoplasm
× Neoplasm is an adult cause. للكبار.

C. Intussusception ✓ Nearest: p. 170 ↗

D. Strangulated hernia
× Hernia is the 2nd common cause. ثاني الأسباب.

E. Adhesions
× Adhesions are the adult cause. للكبار.

Why: Intussusception is the commonest cause of intestinal obstruction in infants and young children.
لماذا: الانغلاف أشهر سبب لانسداد الأمعاء عند الرضع والأطفال الصغار.

★ **Tell the students:** Adults: adhesions 60%, hernias 20%, tumors.
قل للطلاب: الكبار: التصاقات 60% وفتوق 20% وأورام.

AD·3 SURGERY جراحة Digestive System — Chapter 11: Diverticulosis of the Colon

The best investigation for uncomplicated diverticular disease is:

A. Angiography
× Angiography is for active bleeding. للنزف النشط.

B. Colonoscopy
× Colonoscopy is risky when inflamed. خطر عند الالتهاب.

C. Abdominal US
× US is for acute complications. للمضاعفات الحادة.

D. Double-contrast barium enema ✓ Book p. 9 ↗

E. Plain X-ray
× Plain X-ray is not diagnostic. غير تشخيصي.

Why: Exam key: double-contrast barium enema shows diverticula best in uncomplicated disease.
لماذا: حسب مفتاح الامتحان: الحقنة الباريوم المزدوجة تظهر الرتوج في الحالات غير المعقدة.

★ **Tell the students:** Modern practice: CT/colonoscopy; barium enema is avoided in acute diverticulitis.
قل للطلاب: عملياً: CT أو منظار؛ ولا تُجرى حقنة الباريوم في الالتهاب الحاد.

Question note: Answer follows the exam key. The book (p.9) says barium enema has largely been replaced by colonoscopy and CT colonography.
ملاحظة على السؤال: الإجابة حسب مفتاح الامتحان. والكتاب (ص 9) يذكر أن حقنة الباريوم حل محلها المنظار وCT colonography.

AD·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

A finding consistent with Ulcerative Colitis on colonoscopy is:

A. Cobblestoning and deep fissures
× Cobblestone/fissures = Crohn's. كرون.

B. Transmural (full-thickness) inflammation
× Transmural = Crohn's. كرون.

C. Terminal ileum involvement
× Terminal ileum = Crohn's. كرون.

D. Confluent disease with rectal involvement and no skip areas ✓ Nearest: p. 117 ↗

E. Rectal sparing with skip areas
× Rectal sparing = Crohn's. كرون.

Why: UC is continuous (confluent) from the rectum, without skip areas.

لماذا: UC متصل من المستقيم بدون skip areas.

★ **Tell the students:** UC: mucosal, continuous, rectum first; backwash ileitis is rare.

قل للطلاب: UC: مخاطي ومتصل ويبدأ من المستقيم.

AD·5 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Investigation of choice for CHPS (congenital hypertrophic pyloric stenosis):

A. Abdominal ultrasound (pyloric muscle >3 mm, length >15 mm) ✓ Book p. 61 ↗

B. Upper endoscopy
× Endoscopy is not first line. ليس خط أول.

C. Plain X-ray
× Plain X-ray only shows a dilated stomach. يظهر معدة متمددة فقط.

D. Barium meal
× Barium is a second-line test. ثانوي.

Why: Abdominal ultrasound is the investigation of choice in CHPS (muscle >3 mm, length >15 mm).

لماذا: الموجات فوق الصوتية هي الفحص المفضل لـ CHPS (سمك >3 مم وطول >15 مم).

★ **Tell the students:** Correct hypochloremic alkalosis before Ramstedt pyloromyotomy.

قل للطلاب: صحح القلاء قبل عملية Ramstedt.

Group AE

AE·1 INTERNAL MEDICINE باطنة — Digestive System — Chapter 03: The Stomach and Duodenum

A 32-y-old with large-volume hematemesis. Most important FIRST step?

A. Nasogastric tube insertion
× NG tube is not first. — ليس الأول.

B. Fluid resuscitation (2 large-bore IV lines) ✓ Book p. 50 ↗

C. Full history and examination
× History comes after stabilization. — بعد الاستقرار.

D. Upper GI endoscopy
× Endoscopy after resuscitation. — بعد الإنعاش.

Why: First step: ABC resuscitation with 2 large-bore IV lines and fluids.

لماذا: الخطوة الأولى: الإنعاش (ABC) بخطين وريديين واسعين وسوائل.

★ **Tell the students:** Then cross-match, stop NSAIDs, NPO, endoscopy within 24 h.

قل للطلاب: ثم تحليل الدم وإيقاف NSAIDs وصيام ومنظار خلال 24 ساعة.

AE·2 SURGERY جراحة — Hernia book — Chapter 02: Inguinal Hernias

Regarding inguinal hernia and inferior epigastric vessels:

A. Both lateral
× Cannot both be lateral. — لا يمكن أن يكونا جانبيين.

B. Both medial
× Cannot both be medial. — لا يمكن أن يكونا إنسيين.

C. Indirect is medial, Direct is lateral
× Reversed. — معكوس.

D. Indirect is lateral, Direct is medial ✓ Hernia book — not in this file

Why: Indirect hernia is lateral and direct is medial to the inferior epigastric vessels.

لماذا: الفتق غير المباشر جانبي والمباشر إنسي بالنسبة للأوعية الشرسوفية السفلية.

★ **Tell the students:** Mnemonic: 'MD's don't lie': Medial = Direct. Direct sits in Hesselbach's triangle.

قل للطلاب: تذكّر: Medial = Direct؛ والمباشر في مثلث Hesselbach.

AE·3 SURGERY جراحة Hernia book — Chapter 05: Umbilical Hernias

The coverings of a true umbilical hernia sac in adults include:

A. None
× It has coverings. — له أغطية.

B. Rectus abdominis alone
× Rectus lies lateral to the defect. — العضلة المستقيمة جانب الفتحة.

C. Skin + external oblique aponeurosis only
× Those are inguinal hernia layers. — هذه طبقات الفتق الأربي.

D. Skin, umbilical fascia, fascia transversalis, extraperitoneal fat, peritoneum ✓ Hernia book — not in this file

Why: A true umbilical hernia sac is covered by skin, umbilical fascia, fascia transversalis, extraperitoneal fat and peritoneum.

لماذا: كيس الفتق السري الحقيقي يغطيه جلد وfascia السرة وfascia transversalis وشحم خارج البريتون وبريتون.

★ **Tell the students:** Adult para-umbilical hernias pass through the linea alba above/below the umbilicus.

قل للطلاب: الفتق جانب السري عند البالغين عبر الخط الأبيض فوق أو تحت السرة.

Group AF

AF·1 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

An elderly woman with a 'benign' ulcer on greater curvature 5 cm proximal to antrum, still with anemia + guaiac-positive stool after 3 months of medical therapy. Next step?

A. Partial gastrectomy (suspect malignancy) ✓ Book p. 57 ↗

B. Total gastrectomy
× Total gastrectomy is more than needed for this site. — أكثر من المطلوب لهذا الموضع.

C. Continue medical therapy
× Failure to heal means more of the same is wrong. — الاستمرار في العلاج الدوائي خطأ.

D. H. pylori eradication only
× Eradication alone is inadequate. — الاستئصال الجرثومي وحده غير كاف.

Why: A 'benign' gastric ulcer that fails 3 months of therapy with anemia and occult blood is malignant until proven otherwise: gastrectomy.

لماذا: قرحة معدة 'حميدة' لم تلتئم بعد 3 أشهر مع أنيميا ودم خفي = تُعتبر خبيثة: استئصال معدة.

★ **Tell the students:** Every gastric ulcer: biopsy, and re-biopsy until healed.

قل للطلاب: كل قرحة معدة: خزعة وإعادة خزعة حتى الالتئام.

AF·2 SURGERY جراحة Hernia book — Chapter 05: Umbilical Hernias

The following is NOT a cause of para-umbilical hernia recurrence:

A. Wound infection
× Infection causes recurrence.
العدوى سبب.

B. Inlay (sublay) mesh repair ✓ Hernia book — not in this file

C. Uncontrolled ascites / ↑ IAP
× Ascites/raised pressure causes recurrence.
الاستسقاء وزيادة الضغط سبب.

D. Absorbable sutures
× Absorbable sutures fail early.
الخيوط الماصة تفشل مبكراً.

Why: Sublay (inlay) mesh repair lowers recurrence; it is not a cause.
لماذا: إصلاح الشبكة (sublay) يقلل النكس وليس سبباً له.

★ **Tell the students:** Retrorectus (sublay) mesh gives the lowest recurrence rates.
قل للطلاب: الشبكة خلف المستقيمة أقل نكس.

AF·3 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

An 'incisura ulcer' of the stomach is classified as:

A. Type 2 (gastric + duodenal)
× Type II = gastric + duodenal ulcer.
النوع II = معدة + اثني عشر.

B. Type 3 (prepyloric)
× Type III = prepyloric.
النوع III = قبل البواب.

C. Type 1 (lesser curve, normal acid) ✓ Book p. 43 ↗

D. Type 4 (near GE junction)
× Type IV = near GE junction.
النوع IV = قرب GEJ.

Why: Incisura ulcers lie on the lesser curve: Johnson type I (normal or low acid).
لماذا: قرحة incisura على الانحناء الصغير: Johnson النوع I (حمض طبيعي أو منخفض).

★ **Tell the students:** Modified Johnson: I lesser curve, II gastric + DU, III prepyloric, IV juxtacardiac.
قل للطلاب: I صغرى، II معدة + اثني عشر، III قبل البواب، IV بجوار الفؤاد.

AF·4 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

47-yo with bleeding DU, pulse 90, BP 100/70, failed medical + endoscopic control. Best surgical option:

A. Excision of ulcer alone
× Excision alone does not control the vessel. الاستئصال وحده لا يسيطر.

B. Total gastrectomy
× Total gastrectomy is excessive. مبالغ فيه.

C. Over-sewing of the bleeding vessel + truncal vagotomy + pyloroplasty ✓ Book p. 57 ↗

D. Continue transfusion only
× Transfusion alone does not stop it. لا يوقف النزف.

Why: After failed medical and endoscopic control: over-sew the bleeding vessel + truncal vagotomy + pyloroplasty.

لماذا: بعد فشل الدواء والمنظار: خياطة الوعاء النازف + قطع المبهم الجذعي + رأب البواب.

★ **Tell the students:** Indications: shock despite measures, recurrent bleed, elderly atherosclerotic, visible vessel.
قل للطلاب: استطبابات: صدمة، نزف متكرر، مسن مصاب بتصلب، وعاء ظاهر.

Group AG

AG·1 SURGERY جراحة Hernia book — Chapter 06: Incisional and Recurrent Hernias

Causes of hernia recurrence include all EXCEPT:

A. Repair under tension
× Tension causes recurrence. الشد سبب.

B. Tissue (herniorrhaphy) repair
× Tissue repair has high recurrence. نكس مرتفع.

C. Wound infection
× Infection causes recurrence. العدوى سبب.

D. Tension-free Lichtenstein mesh ✓ Hernia book — not in this file

Why: Tension-free Lichtenstein mesh repair reduces recurrence.

لماذا: إصلاح Lichtenstein بالشبكة بدون شد يقلل النكس.

★ **Tell the students:** Mesh recurrence about 1%; tissue repair up to 10%.
قل للطلاب: نكس الشبكة ~1% والإصلاح النسيجي حتى 10%.

AG·2 SURGERY جراحة Digestive System — Chapter 11: Diverticulosis of the Colon

One of the following sites NEVER develops diverticulosis:

A. Rectum ✓ Nearest: p. 146 ↗

B. Sigmoid colon
× The sigmoid is the commonest site. السيني الأشهر.

C. Cecum
× Right-sided disease occurs (especially in Asia). اليمنى تحدث خاصة في آسيا.

D. Transverse colon
× Can occur. ممكنة.

Why: Diverticula spare the rectum (continuous outer longitudinal muscle there).

لماذا: الرتوج لا تصيب المستقيم (الطبقة الطولية مكتملة هناك).

★ **Tell the students:** Diverticulosis: sigmoid in about 95%.

قل للطلاب: السيني في نحو 95%.

AG·3 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

Heller's cardiomyotomy is best performed by:

A. None
× It is performed. تُجرى.

B. Laparoscopy (with partial fundoplication) ✓ Book p. 24 ↗

C. Upper endoscopy
× Endoscopic myotomy is POEM, a different operation. المنظار POEM عملية مختلفة.

D. Open thoracotomy
× Open thoracotomy is no longer preferred. لم تعد مفضلة.

Why: Heller cardiomyotomy is done laparoscopically with a partial fundoplication.

لماذا: عملية Heller تُجرى بالمنظار مع fundoplication جزئي.

★ **Tell the students:** The partial wrap (Dor/Toupet) prevents post-operative reflux.

قل للطلاب: اللفة الجزئية تمنع الارتجاع.

AG·4 SURGERY جراحة Digestive System — Chapter 12: Appendix

Why is appendicitis more dangerous in old age?

A. Fever is always high
× Fever is often low-grade.
الحرارة غالباً خفيفة.

B. Omental ability to encapsulate is reduced + rapid vascular thrombosis ✓ Book p. 154 ↗

C. Pain is severe
× Pain is often blunted.
الألم غالباً خفيف.

D. Appendix is absent
× The appendix is present.
الزائدة موجودة.

Why: In the elderly the omentum fails to wall off the appendix and vascular thrombosis causes early gangrene.
لماذا: عند المسنين الثرب لا يحيط بالزائدة والتجلط الوعائي يسبب غرغرينا مبكرة.

★ **Tell the students:** Elderly: atypical, late, high perforation and mortality.
قل للطلاب: المسن: أعراض غير نمطية ومتأخر وانثقاب ووفيات أعلى.

AG·5 SURGERY جراحة Hernia book — Chapter 03: Femoral Hernia

The structure most commonly responsible for strangulation of a femoral hernia is:

A. Pectineus
× Pectineus forms the floor.
يكوّن القاع.

B. Inguinal ligament
× Inguinal ligament is anterior.
أمامي.

C. Lacunar (Gimbernat's) ligament ✓ Hernia book — not in this file

D. Cooper's ligament
× Cooper's ligament is posterior.
خلفي.

Why: The sharp medial lacunar (Gimbernat's) ligament most commonly strangulates a femoral hernia.
لماذا: الرباط الحاد الإنسي lacunar (Gimbernat) هو الأكثر اختناقاً للفتق الفخذي.

★ **Tell the students:** Femoral hernia: women, high risk of strangulation, Richter's type common.
قل للطلاب: نساء، خطر اختناق عالٍ، وغالباً Richter.

Group AH

AH·1 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

> Female on medical Tx for chronic DU develops GOO; endoscope can't pass duodenum. Best surgical option:

A. Vagotomy + pyloroplasty alone (usually inadequate for fibrotic stenosis)
- × Vagotomy + pyloroplasty is inadequate for fibrotic stenosis.

غير كاف للتضيق التليفي.

B. Barium meal follow-up
- × Follow-up barium does not treat it.

لا يعالجه.

C. Distal gastrectomy + gastro-jejunostomy (Billroth II) ± vagotomy ✓ Book p. 59 ↗

D. Endoscopic dilatation only
- × Dilatation has already failed or is not possible.

التوسيع فشل أو غير ممكن.

Why: Fibrotic stenosis that stops the scope needs resection (distal gastrectomy + gastrojejunostomy, Billroth II) ± vagotomy.

لماذا: التضيق التليفي الذي يمنع المنظار يحتاج استئصالاً (distal gastrectomy + gastrojejunostomy) ± قطع المبهم.

★ **Tell the students:** Choose GJ ± vagotomy or antrectomy; add PPI afterwards.

قل للطلاب: اختر GJ ± vagotomy أو antrectomy مع PPI.

AH·2 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

> 55-year-old on NSAIDs for arthritis develops severe epigastric pain with board-like rigidity; erect CXR shows air under the right dome of diaphragm. Diagnosis:

A. Ruptured hepatic abscess
- × No free air.

لا هواء حر.

B. Acute pancreatitis
- × Pancreatitis rarely causes free air.

نادراً ما يسبب هواءً حراً.

C. Perforated peptic ulcer ✓ Book p. 55 ↗

D. Acute cholecystitis
- × Cholecystitis: RUQ pain, no free air.

ألم يمين علوي بلا هواء.

Why: NSAID user + board-like rigidity + free air under the diaphragm = perforated peptic ulcer.

لماذا: مستخدم NSAIDs + بطن خشبي + هواء تحت الحجاب = قرحة هضمية مثقوبة.

★ **Tell the students:** Resuscitate, then Graham omental patch (laparoscopic or open).

قل للطلاب: إنعاش ثم Graham omental patch.

AH·3 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

Ivor Lewis operation refers to:

A. Laparotomy + right thoracotomy: gastric pull-up with intra-thoracic esophago-gastric anastomosis ✓ Book p. 31 ↗

B. POEM
× POEM is endoscopic myotomy. شق عضلي بالمنظار.

C. Total gastrectomy
× Total gastrectomy is another operation. عملية أخرى.

D. Transhiatal esophagectomy
× Transhiatal avoids thoracotomy. بدون فتح صدر.

Why: Ivor Lewis = laparotomy + right thoracotomy with intrathoracic esophagogastric anastomosis.
لماذا: Ivor Lewis = فتح بطن + فتح صدر أيمن مع مفاغرة داخل الصدر.

★ **Tell the students:** McKeown = three-field (neck, chest, abdomen).
قل للطلاب: McKeown = ثلاثة حقول.

Group AI

AI·1 SURGERY جراحة Hernia book — Chapter 02: Inguinal Hernias

'Bubonocele' refers to an inguinal hernia that:

A. Contains bladder
× Bladder in the sac = sliding/cystocele. مثانة = انزلاقي.

B. Is irreducible
× Irreducible = incarcerated. محبوس.

C. Is confined to the inguinal canal (does not pass external ring) ✓ Hernia book — not in this file

D. Extends into the scrotum
× Reaches the scrotum = complete/scrotal. كامل/صفني.

Why: A bubonocele is an inguinal hernia confined to the inguinal canal (does not pass the external ring).
لماذا: Bubonocele فتق أربي محصور داخل القناة ولا يتجاوز الحلقة الخارجية.

★ **Tell the students:** Incomplete hernia: bubonocele; complete: reaches the scrotum.
قل للطلاب: ناقص: bubonocele؛ كامل: يصل للصفن.

AI·2 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Billroth II operation means:

A. Highly selective vagotomy
× Highly selective vagotomy is acid surgery. جراحة حمض.

B. Lower radical gastrectomy + gastro-duodenal anastomosis
× Gastroduodenal anastomosis = Billroth I. .Billroth I =

C. Total gastrectomy + Roux-en-Y esophago-jejunostomy
× Total gastrectomy + Roux-en-Y esophagojejunostomy. استئصال كلي مع Roux-en-Y.

D. Lower radical gastrectomy + gastro-jejunostomy ✓ Book p. 66 ↗

Why: Billroth II = lower (distal) radical gastrectomy with gastrojejunostomy.

لماذا: Billroth II = استئصال المعدة السفلي مع مفاغرة معدية صائمية.

★ **Tell the students:** Billroth I = gastroduodenostomy; II = gastrojejunostomy.

قل للطلاب: I = معدي اثني عشري، II = معدي صائمي.

AI·3 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

The lower third of the esophagus is supplied by:

A. Short gastric vessels
× Short gastrics supply the fundus. القاع.

B. Right gastric vessels
× Right gastric supplies the lesser curve. الانحناء الصغير.

C. Left gastro-epiploic vessels
× Left gastroepiploic supplies the greater curve. الانحناء الكبير.

D. Left gastric vessels ✓ Nearest: p. 27 ↗

Why: The lower third of the esophagus is supplied by the left gastric vessels.

لماذا: الثلث السفلي للمريء يتغذى من الأوعية المعدية اليسرى.

★ **Tell the students:** Upper third: inferior thyroid; middle: aorta/bronchial branches; lower: left gastric + inferior phrenic.

قل للطلاب: العلوي: الدرقية السفلية؛ الأوسط: فروع الأبهر؛ السفلي: المعدية اليسرى.

AI·4 SURGERY جراحة Hernia book — Chapter 09: Hernia Complications

A hernia is called Littre's hernia when its contents include:

A. Meckel's diverticulum ✓ Hernia book — not in this file

B. Bladder wall
× Bladder wall = sliding/cystocele. مثانة.

C. Ovary
× Ovary: not named. ليس له اسم.

D. Appendix
× Appendix = Amyand's. الزائدة = Amyand.

Why: Littre's hernia contains a Meckel's diverticulum.

لماذا: فتق Littre يحتوي رتج ميكل.

★ **Tell the students:** Richter = partial wall, Maydl = W-loop, Amyand = appendix.

قل للطلاب: Richter جزئي، Maydl حلقة W، Amyand الزائدة.

Group AJ

AJ·1 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Patient with long ulcer dyspepsia vomits large amounts of old undigested food; epigastric distension. Diagnosis:

A. Cancer stomach fungating
× Cancer has a short history and weight loss. السرطان تاريخ قصير ونقص وزن.

B. Gastric outlet obstruction due to chronic DU (fibrotic pyloric stenosis) ✓ Book p. 57 ↗

C. Acute gastritis
× Acute gastritis does not obstruct. لا يسد.

D. Achalasia
× Achalasia affects the esophagus. يصيب المريء.

Why: Long ulcer history with vomiting of old food and distension = gastric outlet obstruction from fibrotic duodenal ulcer stenosis.

لماذا: تاريخ قرحة طويل مع قيء طعام قديم وانتفاخ = انسداد مخرج المعدة من تضيق ليفي.

★ **Tell the students:** Succussion splash, visible peristalsis; hypochloremic hypokalemic alkalosis.

قل للطلاب: صوت ارتجاج ودودية مرئية وقلاء نقص كلور وبوتاسيوم.

AJ·2 SURGERY جراحة Digestive System — Chapter 10: Small Intestine and Colon Surgery

The most accurate investigation to diagnose Meckel's diverticulum is:

A. CT abdomen
× CT is non-specific. غير نوعي.

B. Angiography
× Angiography only in brisk bleeding. عند نزف غزير فقط.

C. Tc-99m pertechnetate (Meckel's) scan ✓ Book p. 143 ↗

D. Abdominal US
× US is non-diagnostic. غير تشخيصي.

Why: Tc-99m pertechnetate scan (Meckel's scan) is the most accurate test.

لماذا: مسح Tc-99m (Meckel's scan) هو الأدق.

★ **Tell the students:** The tracer is taken up by ectopic gastric mucosa.

قل للطلاب: الصبغة تُلتقط بواسطة الغشاء المعدي الهاجر.

AJ·3 SURGERY جراحة Hernia book — Chapter 07: Sliding Hernia

Sliding hernia does NOT occur in:

A. Bubonocele (small hernia within canal) ✓ Hernia book — not in this file

B. Elderly patients
× Common in the elderly. شائع عند المسنين.

C. Large hernias
× Common in large hernias. شائع في الكبيرة.

D. Scrotal hernia
× Scrotal hernias are often sliding. الصفنية غالباً انزلاقية.

Why: Sliding hernias are large, long-standing or scrotal; a small bubonocele confined to the canal is not sliding.

لماذا: الفتق الانزلاقي يكون كبيراً أو مزمناً أو صفنياً؛ و bubonocele الصغير داخل القناة ليس انزلاقياً.

★ **Tell the students:** Sliding: cecum on the right, sigmoid on the left forms part of the sac wall.

قل للطلاب: الأعور يمين والسيني يسار جزء من الجدار.

AJ·4 SURGERY جراحة Digestive System — Chapter 01: GI Symptoms and Investigations

Barium swallow does NOT show:

A. Anatomical distortion
× Shows distortion. يظهر التشوه.

B. Mucosal irregularity
× Shows irregularity. يظهر عدم الانتظام.

C. Regional lymph nodes in cancer ✓ Book p. 9 ↗

D. Luminal narrowing
× Shows narrowing. يظهر التضيق.

Why: Barium swallow shows luminal contour (distortion, irregularity, narrowing), not lymph nodes.

لماذا: الباريوم يظهر الشكل الداخلي (تشوه وعدم انتظام وتضيق) ولا يظهر العقد.

★ **Tell the students:** Lymph nodes need CT/EUS.

قل للطلاب: العقد تحتاج CT أو EUS.

AJ·5 SURGERY جراحة

Digestive System — Chapter 02: The Esophagus · Digestive System — Chapter 04: Pathology of GIT Tumours

Squamous cell carcinoma of the esophagus most commonly occurs in:

A. Upper thoracic esophagus
× Less common. أقل.

B. Cervical esophagus
× Less common. أقل.

C. Middle thoracic esophagus ✓ Book p. 71 ↗

D. Lower thoracic esophagus / GEJ
× Lower third/GEJ = adenocarcinoma. السفلي/GEJ = غدي.

Why: About 50% of esophageal squamous carcinomas are in the middle third.

لماذا: نحو 50% من السرطان الحرشفي في الثلث الأوسط.

★ **Tell the students:** SCC: middle third; adenocarcinoma: lower third.

قل للطلاب: الحرشفي: الأوسط؛ الغدي: السفلي.

Group AK

AK·1 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

Which is NOT a predisposing factor for esophageal cancer?

A. Achalasia
× Achalasia is a risk factor. عامل خطر.

B. Columnar metaplasia (Barrett's)
× Barrett's is the most important risk factor. الأهم.

C. Chronic reflux esophagitis
× Long-standing reflux is a risk factor. عامل خطر.

D. Long-standing esophageal varices ✓ Book p. 26 ↗

Why: Esophageal varices are not a predisposing factor for cancer.

لماذا: دوالي المريء ليست عامل خطر للسرطان.

★ **Tell the students:** Risks: smoking, alcohol, achalasia, reflux/Barrett's, caustic stricture, hot drinks.

قل للطلاب: العوامل: تدخين، كحول، أكالازيا، ارتجاع/باريت، تضيق كاوي، مشروبات ساخنة.

AK·2 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

In esophageal cancer, investigation for mediastinal + LN invasion:

A. Bronchoscopy
× Bronchoscopy checks airway invasion only. تفحص غزو القصبة فقط.

B. Endoluminal ultrasonography (EUS) ✓ Book p. 28 ↗

C. Esophagoscopy
× Esophagoscopy: biopsy and diagnosis. للخزعة والتشخيص.

D. Barium swallow
× Barium shows the lumen only. يظهر التجويف فقط.

Why: Endoluminal ultrasound (EUS) shows tumor invasion of the wall, mediastinum and nodes.

لماذا: الموجات الداخلية (EUS) تظهر غزو الجدار والمنصف والعقد.

★ **Tell the students:** EUS = local T and N stage; CT/PET = distant metastasis.

قل للطلاب: EUS للمرحلة الموضعية؛ CT/PET للنقائل.

AK·3 SURGERY جراحة General oncology — not covered in the Digestive book

Radiotherapy as 1st-line gives results similar to surgery in:

A. Pancreatic exocrine carcinoma
× Surgery is the only potential cure. الجراحة هي العلاج الشافي.

B. Barrett's adenocarcinoma
× Surgery/EMR is the standard. الجراحة/EMR هي المعيار.

C. RCC
× RCC is radio-resistant. مقاوم للأشعة.

D. HCC
× HCC: resection, ablation or transplant. استئصال أو كي أو زراعة.

E. SCC of skin ✓ Not in the book

Why: Skin squamous cell carcinoma is treated equally well by radiotherapy as a first-line option.
لماذا: سرطان الجلد الحرشفي يعالج بالأشعة كخط أول بنتائج تماثل الجراحة.

★ **Tell the students:** Radiotherapy is equal to surgery also in laryngeal, anal canal and cervical cancers.
قل للطلاب: الأشعة تساوي الجراحة أيضاً في الحنجرة وقناة الشرج وعنق الرحم.

Question note: General oncology: no matching page in this book.
ملاحظة على السؤال: أورام عامة: لا توجد صفحة مطابقة في الكتاب.

Group AL

AL·1 SURGERY جراحة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which bariatric procedure induces MALABSORPTION?

A. Total gastrectomy
× Total gastrectomy is not a bariatric operation. ليست عملية سمنة.

B. Sleeve gastrectomy
× Sleeve is restrictive only. تقييدية فقط.

C. Roux-en-Y gastric bypass ✓ Book p. 86 ↗

D. Adjustable gastric band
× Band is restrictive only. تقييدي فقط.

Why: Roux-en-Y gastric bypass is restrictive and malabsorptive; sleeve and band are purely restrictive.
لماذا: Roux-en-Y gastric bypass تقييدي وماص للسوء امتصاص؛ أما sleeve وband فتقييديان فقط.

★ **Tell the students:** Mostly malabsorptive: biliopancreatic diversion / duodenal switch.
قل للطلاب: الأكثر امتصاصياً: BPD/duodenal switch.

AL·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 03: The Stomach and Duodenum

All the following predispose to stress ulceration of the stomach EXCEPT:

A. Acute pancreatitis
× Acute pancreatitis predisposes. تهيّئ.

B. Major burn (Curling's ulcer)
× Burns: Curling's ulcer. .Curling

C. Deep vein thrombosis ✓ Book p. 36 ↗

D. Head injury (Cushing's ulcer)
× Head injury: Cushing's ulcer. .Cushing

E. Septic shock
× Septic shock predisposes. تهيّئ.

Why: DVT does not predispose to stress ulcers.

لماذا: الجلطة الوريدية العميقة لا تسبب قرح الإجهاد.

★ **Tell the students:** Give PPI/H2 prophylaxis in ICU: ventilation over 48 h, coagulopathy.

قل للطلاب: وقاية بـ PPI/H2 في العناية: تنفس صناعي فوق 48 ساعة واضطراب تجلط.

AL·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

The INCORRECT statement about secretory diarrhea is:

A. Diarrhea continues while fasting for 24–48 h
× True: continues while fasting. صحيح: يستمر مع الصيام.

B. Lactose intolerance is a classic example ✓ Book p. 80 ↗

C. Stool volume is usually large (>1 L/day)
× True: large volume (>1 L). صحيح: حجم كبير.

D. Stools do not contain pus or blood
× True: no pus or blood. صحيح: لا دم أو صديد.

Why: Lactose intolerance is osmotic diarrhea, not secretory (that is why the statement is incorrect).

لماذا: عدم تحمل اللاكتوز إسهال أسموزي وليس إفرازياً (لذلك العبارة خاطئة).

★ **Tell the students:** Secretory: cholera, VIPoma, carcinoid, stimulant laxatives.

قل للطلاب: كوليرا وVIPoma وكارسينويد ومسهلات.

AL·4 SURGERY جراحة Digestive System — Chapter 04: Pathology of GIT Tumours

The most common benign tumor of the esophagus is:

A. Lipoma
× Rarer. أندر.

B. Hemangioma
× Rarer. أندر.

C. Fibroma
× Rarer. أندر.

D. Leiomyoma ✓ Book p. 71 ↗

Why: Leiomyoma is the commonest benign esophageal tumor.
لماذا: الورم العضلي الأملس أشهر ورم حميد بالمريء.

★ **Tell the students:** Submucosal and smooth on barium; do not biopsy (bleeding).
قل للطلاب: تحت المخاطية وملساء؛ لا خزعة (نزف).

Group AM

AM·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

Features of ulcerative colitis include:

A. Mouth-to-anus distribution
× Mouth-to-anus = Crohn's. كرون.

B. Transmural inflammation
× Transmural = Crohn's. كرون.

C. Marked pseudopolyps ✓ Nearest: p. 117 ↗

D. Skip lesions
× Skip lesions = Crohn's. كرون.

Why: Pseudopolyps are marked in ulcerative colitis.
لماذا: الـpseudopolyps واضحة في التهاب القولون التقرحي.

★ **Tell the students:** UC: continuous, mucosal, crypt abscesses, rectum always, cancer risk.
قل للطلاب: UC: متصل ومخاطي وخراجات الخبايا والمستقيم دائماً وخطر سرطان.

AM·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which is associated with exudative diarrhea?

A. Celiac disease
× Celiac: malabsorptive diarrhea.
سوء امتصاص.

B. Salmonella infection ✓ Book p. 80 ↗

C. Lactase deficiency
× Lactase deficiency: osmotic.
أسموزي.

D. Vibrio cholerae
× Cholera: secretory.
إفرازي.

Why: Salmonella is invasive: exudative (inflammatory) diarrhea with pus/blood.

لماذا: السالمونيلا غازية: إسهال التهابي مع صديد/دم.

★ **Tell the students:** Cholera = secretory (toxin raises cAMP); Salmonella/Shigella = exudative.

قل للطلاب: الكوليرا إفرازية (cAMP)؛ السالمونيلا/الشيجيلا التهابية.

AM·3 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Indications for surgery in peptic ulcer include all EXCEPT:

A. Pyloric stenosis
× Indication.
استطباب.

B. Uncomplicated acute ulcer responsive to PPI ✓ Book p. 53 ↗

C. Perforation
× Indication.
استطباب.

D. Bleeding with failed endoscopic control
× Indication.
استطباب.

Why: An uncomplicated acute ulcer that responds to PPI is treated medically, not surgically.

لماذا: قرحة حادة غير مصحوبة بمضاعفات وتستجيب للـPPI تعالج دوائياً.

★ **Tell the students:** Surgery for complications only: perforation, uncontrolled bleeding, GOO, suspected cancer.

قل للطلاب: الجراحة للمضاعفات فقط: انثقاب، نزف غير مسيطر، انسداد، اشتباه سرطان.

AM·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

Morphological features of Crohn's disease include:

A. Broad-based ulcers
× Broad shallow ulcers = UC. .UC

B. Skip lesions ✓ Nearest: p. 117 ↗

C. Marked pseudopolyps
× Marked pseudopolyps = UC. .UC

D. Left-sided disease
× Left-sided disease = UC. .UC

Why: Skip lesions are a morphological hallmark of Crohn's.

لماذا: Skip lesions ميزة شكلية لكرون.

★ **Tell the students:** Crohn's: skip lesions, cobblestone, fissures, granulomas, strictures.

قل للطلاب: كرون: skip وcobblestone وشقوق وحبيبومات وتضيقات.

AM·5 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Which is NOT a relative indication for surgery in bleeding peptic ulcer?

A. Elderly patient
× Relative indication: elderly rebleed poorly tolerated. استطباب نسبي: المسن لا يتحمل النزف.

B. Recurrence of bleeding after endoscopic control
× Indication: recurrent bleeding. استطباب: نزف متكرر.

C. Long history of peptic ulcer ✓ Book p. 57 ↗

D. Patient with associated serious disease
× Indication: serious associated disease. استطباب: مرض مصاحب شديد.

Why: A long ulcer history alone is not an indication for surgery in bleeding ulcer.

لماذا: تاريخ قرحة طويل وحده ليس استطباباً للجراحة في النزف.

★ **Tell the students:** Also: visible vessel on endoscopy, persistent shock.

قل للطلاب: أيضاً: وعاء ظاهر وصدمة مستمرة.

Group AN

AN·1 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Regarding congenital hypertrophic pyloric stenosis, TRUE statement:

A. Presents in first 2 days of life
× Presents at 2-8 weeks, not the first 2 days. يظهر عمر 2-8 أسابيع.

B. Bilious vomiting
× Vomiting is non-bilious. القيء غير صفراوي.

C. Diagnosed by ultrasound ✓ Book p. 61 ↗

D. Metabolic acidosis
× It causes hypochloremic metabolic ALKALOSIS. يسبب قلاء.

E. More common in females
× More common in males (4:1). أكثر في الذكور.

Why: CHPS is diagnosed by ultrasound (muscle >3 mm, channel >15 mm).
لماذا: CHPS يشخص بالموجات (سمك >3 مم وطول >15 مم).

★ **Tell the students:** First-born male, projectile vomiting, olive mass; Ramstedt pyloromyotomy.
قل للطلاب: ذكر بكر، قيء قذفي، كتلة زيتونة؛ Ramstedt.

AN·2 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Patients with chronic pyloric stenosis characteristically develop:

A. Respiratory alkalosis
× It is metabolic, not respiratory. أيضي وليس تنفسي.

B. Hyperkalemia and hypernatremia
× Hypo- (not hyper-) kalemia and natremia. نقص وليس زيادة.

C. Hypochloremic hypokalemic metabolic alkalosis ✓ Book p. 60 ↗

D. Metabolic acidosis with hyperkalemia
× Alkalosis, not acidosis. قلاء وليس حماض.

Why: Persistent vomiting loses HCl and K+: hypochloremic, hypokalemic metabolic alkalosis.
لماذا: القيء المستمر يفقد HCl وK+: قلاء أيضي مع نقص كلور وبوتاسيوم.

★ **Tell the students:** Paradoxical aciduria; correct with saline and KCl.
قل للطلاب: حموضة بول متناقضة؛ تصحح بالملح والبوتاسيوم.

AN·3 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Surgery for duodenal ulcer is indicated if:

A. Repeated ulcerative attacks more than twice per year
× Recurrences are managed medically.
تُعالج دوائياً.

B. Repeated vomiting of partially digested food ✓ Book p. 53 ↗

C. Single attack of melena in a young patient
× A single melena in a young patient is managed conservatively.
تحفظي.

D. Failure to respond to medical treatment for three weeks
× 3 weeks is too short to call it failure.
3 أسابيع قصيرة جداً.

Why: Repeated vomiting of partly digested food means gastric outlet obstruction, an indication for surgery.

لماذا: تكرار قيء طعام غير مهضوم = انسداد مخرج المعدة، استطباب جراحي.

★ **Tell the students:** Complicated PUD: perforation, bleeding, obstruction, intractability.

قل للطلاب: المضاعفات: انثقاب، نزف، انسداد، عدم استجابة.

Group AO

AO·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 02: The Esophagus

Regarding complications of gastroesophageal reflux, one of the following is FALSE:

A. Squamous cell carcinoma ✓ Book p. 15 ↗

B. Esophageal stricture
× True complication.
مضاعفة.

C. Shortening of esophagus
× True: reflux shortens the esophagus.
صحيح: يقصّر المريء.

D. Metaplasia & dysplasia of lower esophageal mucosa
× True: metaplasia and dysplasia.
صحيح.

Why: GERD leads to adenocarcinoma (via Barrett's), not squamous cell carcinoma.

لماذا: الـGERD يؤدي لسرطان غدي (عبر باريت) وليس حرشفي.

★ **Tell the students:** Complications: stricture, Barrett's, bleeding, short esophagus, adenocarcinoma.

قل للطلاب: تضيق، باريت، نزف، مريء قصير، سرطان غدي.

AO·2 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

Complications of anti-reflux surgery for GERD include all the following EXCEPT:

A. Gas bloat syndrome
× True: gas-bloat (tight 360 wrap). صحيح.

B. Pneumothorax
× True: pleural injury during hiatal dissection. صحيح: إصابة الجنب.

C. Dumping syndrome ✓ Book p. 21 ↗

D. Recurrent reflux
× True: recurrent reflux/wrap slippage. صحيح.

E. Dysphagia
× True: acute dysphagia (tight wrap). صحيح.

Why: Dumping syndrome follows gastric resection or bypass, not anti-reflux surgery.

لماذا: Dumping يتبع استئصال المعدة أو التحويلة وليس عمليات الارتجاع.

★ **Tell the students:** Gas-bloat: can't belch; managed by diet and simethicone.

قل للطلاب: لا يستطيع التجشؤ؛ نظام غذائي وsimethicone.

AO·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

64-year-old diabetic on omeprazole, chronic oily diarrhea, macrocytic anemia, mild hypoalbuminemia. Most likely diagnosis?

A. Hemochromatosis
× Hemochromatosis: diabetes + skin pigment, no steatorrhea. سكري وتصبغ بلا إسهال دهني.

B. Crohn's disease
× Crohn's is ileal with pain. ألم ولفائفي.

C. Bacterial overgrowth syndrome ✓ Book p. 88 ↗

D. Intestinal lymphoma
× Lymphoma: weight loss, nodes. نقص وزن وعقد.

E. Chronic pancreatitis
× Pancreatitis does not give B12 deficiency. لا يسبب نقص B12.

Why: Diabetes (dysmotility) + omeprazole (hypochlorhydria) lead to bacterial overgrowth: steatorrhea and B12-deficiency macrocytic anemia.

لماذا: السكري (قصور حركة) + omeprazole (قلة الحمض) تؤدي لفرط نمو بكتيري: إسهال دهني وأنيميا B12.

★ **Tell the students:** Stasis (DM, scleroderma) + hypochlorhydria (PPI, atrophic gastritis) = SIBO.

قل للطلاب: ركود + قلة حمض = SIBO.

AO·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 03: The Stomach and Duodenum

Gastrin-secreting G cells in the stomach are located in the:

A. Fundus
× Fundus: parietal and chief cells. خلايا جدارية ورئيسية.

B. Pyloric canal only
× Not limited to the canal. ليست في القناة فقط.

C. Body
× Body: parietal and chief cells. خلايا جدارية ورئيسية.

D. Pyloric antrum ✓ Book p. 35 ↗

Why: Gastrin-secreting G cells lie in the pyloric antrum.

لماذا: خلايا G المفرزة للغاسترين في الغار البوابي.

★ **Tell the students:** Antral gastritis (H. pylori) lowers somatostatin, raising gastrin and acid.

قل للطلاب: التهاب الغار يرفع الغاسترين والحمض.

Group AP

AP·1 SURGERY جراحة

Digestive System — Chapter 03: The Stomach and Duodenum · Digestive System — Chapter 04: Pathology of GIT Tumours

All the following about linitis plastica are true EXCEPT:

A. The stomach is stiff (leather-bottle appearance)
× True: stiff leather-bottle stomach. صحيح: معدة جلدية.

B. The stomach is narrowed
× True: narrowed stomach. صحيح: ضيقة.

C. The prognosis is poor
× True: poor prognosis. صحيح.

D. The tumor may not be visible on the mucosa
× True: mucosa may look normal. صحيح: قد تبدو المخاطية طبيعية.

E. It is a lymphoma of the stomach ✓ Book p. 72 ↗

Why: Linitis plastica is a diffuse infiltrating adenocarcinoma, not a lymphoma.

لماذا: Linitis plastica سرطان غدي منتشر وليس لمفوما.

★ **Tell the students:** Diffuse signet-ring cells; barium shows a rigid tube.

قل للطلاب: خلايا signet-ring؛ الباريوم يظهر أنبوباً جامداً.

AP·2 SURGERY جراحة

Digestive System — Chapter 02: The Esophagus · Digestive System — Chapter 04: Pathology of GIT Tumours

Which of the following is NOT a risk factor for the development of esophageal malignancy?

A. Excessive intake of alcoholic spirits
× Alcohol is a risk. عامل خطر.

B. Esophageal metaplasia
× Metaplasia is a risk. عامل خطر.

C. Smoking
× Smoking is a risk. عامل خطر.

D. Blood group O ✓ Book p. 71 ↗

Why: Blood group O is not a risk factor for esophageal cancer (blood group A links to gastric cancer).

لماذا: فصيلة O ليست عامل خطر (فصيلة A ترتبط بسرطان المعدة).

★ **Tell the students:** Gastric cancer: blood group A, H. pylori, pernicious anemia.

قل للطلاب: سرطان المعدة: فصيلة A وH. pylori وأنيميا خبيثة.

AP·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 03: The Stomach and Duodenum

All are useful to prevent PUD recurrence EXCEPT:

A. Avoid frequent small meals ✓ Nearest: p. 43 ↗

B. Stop smoking
× Stopping smoking helps. يفيد.

C. Avoid alcohol
× Avoiding alcohol helps. يفيد.

D. Avoid NSAIDs
× Avoiding NSAIDs helps. يفيد.

Why: Frequent small meals do not prevent recurrence; smoking, alcohol and NSAIDs do.

لماذا: الوجبات الصغيرة المتكررة لا تمنع النكس، بينما التدخين والكحول والـNSAIDs تسببه.

★ **Tell the students:** Test and treat H. pylori; add PPI if NSAIDs are unavoidable.

قل للطلاب: افحص وعالج H. pylori؛ وأضف PPI إذا لزم NSAIDs.

Question note: Lifestyle advice is not printed in the book; nearest page is the PUD page.

ملاحظة على السؤال: نصائح الحياة غير مكتوبة؛ أقرب صفحة هي القرحة.

AP·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 02: The Esophagus

Regarding Barrett's esophagus:

A. Occurs in 40% of cases of GERD
× Only about 10-15% of GERD patients. نحو 10-15% فقط.

B. Regular endoscopic surveillance is mandatory ✓ Nearest: p. 16 ↗

C. Best treated with proton pump inhibitors
× PPIs control reflux but do not remove Barrett's. تضبط الارتجاع لكن لا تزيله.

D. Predisposes to squamous cell carcinoma
× It predisposes to adenocarcinoma. يهيّئ للغدي.

E. It is a squamous metaplasia of the lower third
× It is columnar, not squamous, metaplasia. تحول عمودي وليس حرشفي.

Why: Barrett's needs regular endoscopic surveillance (dysplasia and cancer risk).
لماذا: مريء باريت يحتاج منظار متابعة دورياً (خطر الخلل والسرطان).

★ **Tell the students:** Dysplasia: EMR or radiofrequency ablation.
قل للطلاب: الخلل التنسجي: EMR أو RFA.

AP·5 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

In bleeding per rectum, which is TRUE?

A. Meckel's is the commonest cause in adults >60 y
× Meckel's causes bleeding in children. عند الأطفال.

B. Massive bleeding never occurs in diverticular disease
× Diverticular bleeding can be massive. قد يكون غزيراً.

C. Commonest cause is anal lesions e.g. hemorrhoids ✓ Book p. 52 ↗

D. Angiodysplasia usually affects the left side of the colon
× Angiodysplasia is usually right-sided. غالباً في القولون الأيمن.

Why: Anorectal lesions (piles) are the commonest cause of bleeding per rectum.
لماذا: الآفات الشرجية (البواسير) أشهر سبب للنزف الشرجي.

★ **Tell the students:** Massive lower GI bleeding: diverticula and angiodysplasia (elderly).
قل للطلاب: النزف السفلي الغزير: رتوج وتوسع أوعية.

Group AQ

AQ·1 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

In malignant tumors of the esophagus, all are true EXCEPT:

A. Are commonest at the upper third ✓ Book p. 71 ↗

B. Occur most commonly in males
× True: males more. صحيح.

C. Spread commonly by lymphatics
× True: early lymphatic spread. صحيح.

D. Are mostly squamous cell carcinoma
× True: mostly SCC. صحيح.

Why: Esophageal cancers are commonest in the middle third (not the upper third).

لماذا: سرطانات المريء أشهر في الثلث الأوسط وليس العلوي.

★ **Tell the students:** No serosa and rich lymphatics lead to early spread.

قل للطلاب: غياب المصلية وغنى اللمف يعطي انتشاراً مبكراً.

AQ·2 SURGERY جراحة

Digestive System — Chapter 02: The Esophagus · Digestive System — Chapter 04: Pathology of GIT Tumours

Which of the following conditions are precancerous?

A. Corrosive stricture
× Premalignant (but not alone). ما قبل سرطانية.

B. Barrett's esophagus
× Premalignant (but not alone). ما قبل سرطانية.

C. Achalasia
× Premalignant (but not alone). ما قبل سرطانية.

D. All of the above ✓ Book p. 26 ↗

Why: Corrosive stricture, Barrett's and achalasia are all premalignant.

لماذا: التضيق الكاوي وباريت والأكالازيا كلها ما قبل سرطانية.

★ **Tell the students:** Also reflux esophagitis and Plummer-Vinson syndrome.

قل للطلاب: أيضاً التهاب الارتجاع وPlummer-Vinson.

AQ·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

The following features characterize UC EXCEPT:

A. Crypt abscess and cryptitis
× UC feature: crypt abscess and cryptitis. من سمات UC.

B. Pseudopolyps
× UC feature: pseudopolyps. من سمات UC.

C. Superficial mucosal ulceration
× UC feature: superficial ulceration. من سمات UC.

D. Stricture formation in chronic cases ✓ Nearest: p. 117 ↗

Why: Strictures are typical of Crohn's; in UC they are rare and should raise suspicion of cancer.

لماذا: التضيقات مميزة لكرون؛ وفي UC نادرة وتثير الشك بالسرطان.

★ **Tell the students:** UC stricture = think cancer.

قل للطلاب: تضيق في UC = فكّر بالسرطان.

Group AR

AR·1 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

About complications of chronic peptic ulcers, all are true EXCEPT:

A. Posterior duodenal ulcers tend to cause substantial bleeding
× True: posterior DU bleeds. صحيح: الخلفية تنزف.

B. Benign gastric ulcers have a 50% tendency to malignant transformation ✓ Book p. 53 ↗

C. Anterior duodenal ulcers tend to perforate
× True: anterior DU perforates. صحيح: الأمامية تنثقب.

D. The primary treatment of bleeding ulcers is usually non-surgical
× True: bleeding is primarily managed non-surgically. صحيح: غير جراحي أساساً.

E. Fibrosis of a chronic duodenal ulcer can cause pyloric obstruction
× True: fibrosis causes pyloric obstruction. صحيح.

Why: Malignant transformation of benign gastric ulcers is uncommon; '50%' is false.

لماذا: التحول الخبيث للقرحة الحميدة غير شائع؛ و'50%' خاطئة.

★ **Tell the students:** Always biopsy gastric ulcers: ulcerated cancer can mimic a benign ulcer.

قل للطلاب: اخزع كل قرحة معدة فقد يكون سرطاناً متقرحاً.

AR·2 SURGERY جراحة

Digestive System — Chapter 03: The Stomach and Duodenum · Digestive System — Chapter 23: Portal Hypertension

Which is the definitive treatment for a patient with actively bleeding esophageal varices?

A. Blood transfusion
× Supportive only.
داعم فقط.

B. Endoscopic band ligation ✓ Book p. 315 ↗

C. Balloon tamponade
× Balloon tamponade is a temporary bridge.
مؤقت.

D. Vasopressin
× Vasoactive drugs are adjuncts.
مساعد.

Why: Endoscopic variceal band ligation is the preferred definitive endoscopic treatment of bleeding varices.

لماذا: ربط الدوالي بالمنظار هو العلاج المنظاري الشافي المفضل.

★ **Tell the students:** Also: terlipressin/octreotide, ceftriaxone, EVL within 12 h; TIPS if failure.

قل للطلاب: أيضاً: terlipressin وceftriaxone وEVL خلال 12 ساعة؛ وTIPS عند الفشل.

AR·3 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

In perforated duodenal ulcer, all are true EXCEPT:

A. Produces abdominal tenderness most marked in the epigastrium
× True: epigastric tenderness.
صحيح.

B. Usually accompanied by leukocytosis
× True: leukocytosis.
صحيح.

C. Occurs most frequently in women ✓ Book p. 54 ↗

D. Usually preceded by exacerbation of ulcer symptoms
× True: usually preceded by ulcer symptoms.
صحيح: تسبقه أعراض قرحة.

Why: Perforated duodenal ulcer is commoner in MEN, not women.

لماذا: انثقاب قرحة الاثني عشر أشهر في الرجال وليس النساء.

★ **Tell the students:** Sudden severe epigastric pain, board-like rigidity, loss of liver dullness, free air.

قل للطلاب: ألم شرسوفي مفاجئ، بطن خشبي، اختفاء بلادة الكبد، هواء حر.

AR·4 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

About acute gastric dilatation, TRUE statement:

A. Treated by gastrectomy
× Treated by decompression; gastrectomy only for necrosis.
العلاج بالتفريغ؛ والاستئصال عند النخر فقط.

B. Hypovolemia is prominent ✓ Book p. 68 ↗

C. Rare postoperative complication
× It is not rare after surgery or trauma.
ليس نادراً بعد الجراحة أو الإصابة.

D. Causes hyperkalemia
× Hypokalemia is the risk, not hyperkalemia.
نقص البوتاسيوم هو الخطر.

E. Stomach shows hyperperistalsis
× The stomach is atonic, not hyperperistaltic.
المعدة مشلولة وليست مفرطة الحركة.

Why: Fluid sequestration in the dilated stomach produces prominent hypovolemia.
لماذا: تجمع السوائل في المعدة المتمددة يسبب نقص حجم واضح.

★ **Tell the students:** NG decompression + IV fluids + potassium replacement.
قل للطلاب: تفريغ أنفي معدي + سوائل + تعويض بوتاسيوم.

Group AS

AS·1 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

During exploration for acute abdomen a hole in the anterior wall of the duodenum with turbid bile is found. Surgical steps include all EXCEPT:

A. Sub-hepatic, sub-diaphragmatic and pelvic drains
× Done: drains.
تُجرى.

B. Thorough peritoneal lavage
× Done: peritoneal lavage.
يُجرى.

C. Vagotomy and gastric drainage in the acute setting ✓ Book p. 56 ↗

D. Omental patch over the hole (Graham patch)
× Done: Graham omental patch.
يُجرى.

Why: Definitive acid surgery (vagotomy and drainage) is not done in the acute perforated setting.
لماذا: الجراحة الحمضية الشافية (قطع المبهم) لا تُجرى في الانثقاب الحاد.

★ **Tell the students:** Then PPI and H. pylori eradication.
قل للطلاب: ثم PPI واستئصال H. pylori.

AS·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 08: Inflammatory Bowel Disease

In Ulcerative colitis all are true EXCEPT:

A. Runs a steady progressive course with no remission ✓ Book p. 121 ↗

B. Presents with frequent watery blood-stained diarrhea
× True: bloody watery diarrhea. صحيح.

C. Incidence of malignancy is proportional to extent of colonic involvement
× True: cancer risk rises with extent and duration. صحيح.

D. Toxic megacolon should be treated with surgical resection
× True: toxic megacolon needs colectomy if medical therapy fails. صحيح: يحتاج استئصال عند فشل الدواء.

Why: UC is relapsing and remitting; a steady course without remission is false.

لماذا: UC ينتكس ويهدأ؛ ولا يسير بلا فترات هدوء.

★ **Tell the students:** Surveillance colonoscopy starts 8-10 years after onset.

قل للطلاب: منظار متابعة بعد 8-10 سنوات من البداية.

AS·3 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

As regards staging of cancer stomach, which is TRUE?

A. Invasion of up to 7 regional L.N. is considered N3
× N3 is more than 15 nodes (7-15 = N2). N3 أكثر من 15 عقدة.

B. Metastases to bone and liver are of equal incidence in advanced stage
× Liver metastases are much commoner than bone. الكبد أكثر بكثير.

C. Serosa is invaded only in T4
× Per the book, serosal invasion is T3; T4 = adjacent organs. حسب الكتاب: المصلية T3 وT4 الأعضاء المجاورة.

D. Enlarged supraclavicular L.N. is evident in advanced cases ✓ Book p. 64 ↗

Why: Enlarged left supraclavicular node (Troisier's sign) is seen in advanced gastric cancer.

لماذا: تضخم العقدة فوق الترقوة اليسرى (Troisier) في السرطان المتقدم.

★ **Tell the students:** Book TNM: N1 = 1-6, N2 = 7-15, N3 = >15 nodes.

قل للطلاب: TNM بالكتاب: N1 = 1-6، N2 = 7-15، N3 = >15.

AS·4 SURGERY جراحة

Digestive System — Chapter 03: The Stomach and Duodenum · Digestive System — Chapter 38: Acute Abdomen

> A 47-year-old man presents with several hours of severe abdominal pain, board-like rigidity, shifting dullness and loss of liver dullness, after NSAID intake for backache. Best management is:

A. IV ranitidine and observation
× Conservative care is only for selected sealed leaks. التحفظي لحالات مختارة.

B. Immediate surgery (after resuscitation) ✓ Book p. 55 ↗

C. Ryle tube insertion only
× NG tube alone is inadequate. غير كاف.

D. Immediate blood transfusion
× Transfusion alone does not treat perforation. لا يعالج الانثقاب.

Why: Board-like rigidity with loss of liver dullness = perforated ulcer: resuscitate then operate.

لماذا: بطن خشبي مع اختفاء بلادة الكبد = قرحة مثقوبة: إنعاش ثم جراحة.

★ **Tell the students:** Erect X-ray: air under the diaphragm.

قل للطلاب: أشعة واقف: هواء تحت الحجاب.

AS·5 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

> The least common complication of ACUTE peptic ulcer disease is:

A. Pyloric stenosis ✓ Book p. 53 ↗

B. Microcytic hypochromic anemia
× Chronic occult loss, common. فقد مزمن شائع.

C. Hemorrhage
× Common acute complication. مضاعفة حادة شائعة.

D. Perforation
× Common acute complication. مضاعفة حادة شائعة.

Why: Acute PUD complications are bleeding and perforation; pyloric stenosis is a chronic, late complication.

لماذا: مضاعفات القرحة الحادة النزف والانثقاب؛ والتضيق مضاعفة مزمنة ومتأخرة.

★ **Tell the students:** Chronic: stenosis, penetration, malignancy (rare).

قل للطلاب: المزمنة: تضيق، اختراق، تسرطن (نادر).

Group AT

AT·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 03: The Stomach and Duodenum

> A 50-year-old man with 2-year dyspepsia has a small pyloric gastric ulcer on endoscopy. Biopsy shows chronic inflammation, and the ulcer is not responding to PPI. Most likely diagnosis:

A. Intestinal hyperplasia of the ulcer mucosa
× Not a diagnosis. ليس تشخيصاً.

B. Inherited APC gene mutation
× APC mutation = FAP. .FAP

C. Helicobacter pylori infection ✓ Book p. 38 ↗

D. Vitamin B12 deficiency
× B12 deficiency = fundic autoimmune gastritis. التهاب مناعي بالقاع.

Why: H. pylori causes antral chronic inflammation and non-healing pyloric ulcers.

لماذا: H. pylori يسبب التهاباً مزمناً بالغار وقرحاً بوابية متأخرة الالتئام.

★ **Tell the students:** Non-healing ulcer: exclude NSAIDs, H. pylori, gastrinoma and cancer (biopsy).

قل للطلاب: قرحة لا تلتئم: استبعد NSAIDs وH. pylori وغاسترينوما وسرطان.

AT·2 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

> Regarding corrosive esophageal injury:

A. Early corticosteroids are recommended ✓ Nearest: p. 16 ↗

B. Acids cause more injuries than alkalies
× Alkalis injure the esophagus more; acids hit the stomach. القلويات أضر للمريء والأحماض للمعدة.

C. Perforation is not a complication
× Perforation is a feared complication. الانثقاب مضاعفة خطيرة.

D. Endoscopic dilatation should be started immediately
× Dilatation is delayed about 3 weeks. التوسيع يؤجل نحو 3 أسابيع.

Why: Per the curriculum key, early corticosteroids (with antibiotics) are recommended to reduce stricture formation.

لماذا: حسب مفتاح المنهج: الكورتيزون المبكر (مع المضادات) لتقليل التضيق.

★ **Tell the students:** Modern guidelines do not recommend routine steroids; no emesis, NPO, endoscopy within 24 h.

قل للطلاب: الإرشادات الحديثة لا توصي به؛ لا قيء، صيام، منظار خلال 24 ساعة.

Question note: Answer follows the exam key; practice has changed.

ملاحظة على السؤال: الإجابة حسب مفتاح الامتحان؛ والممارسة الحديثة اختلفت.

AT·3 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

For early carcinoma of the lower third of the esophagus, the best hope for cure is:

A. Gastrostomy
× Gastrostomy is for feeding only. للتغذية فقط.

B. Radiotherapy
× Radiotherapy is not first-line for cure here. ليس خط أول للشفاء.

C. Chemotherapy
× Chemotherapy is adjuvant/palliative. مساعد/تلطيفي.

D. Esophagectomy and gastric pull-up ✓ Book p. 31 ↗

E. Endoscopic stenting
× Stenting is palliative. تلطيفي.

Why: Esophagectomy with gastric pull-up gives the best chance of cure in early lower-third cancer.
لماذا: استئصال المريء مع رفع المعدة أفضل فرصة للشفاء في السرطان المبكر بالثلث السفلي.

★ **Tell the students:** T1 mucosal disease may be treated by EMR/ESD.
قل للطلاب: المخاطي T1 قد يعالج بـ EMR/ESD.

Group AU

AU·1 SURGERY جراحة

Digestive System — Chapter 04: Pathology of GIT Tumours · Digestive System — Chapter 15: Colorectal Cancer

Hematogenous spread of colonic adenocarcinoma is primarily to:

A. Lungs
× Lung is second (and in low rectal cancer). الثاني.

B. Brain
× Less common. أقل.

C. Bones
× Less common. أقل.

D. Liver ✓ Book p. 198 ↗

Why: Colonic cancer spreads hematogenously to the liver first (portal circulation).
لماذا: سرطان القولون ينتشر دموياً أولاً للكبد (الدورة البابية).

★ **Tell the students:** Low rectal cancer may reach the lungs first via the systemic rectal veins.
قل للطلاب: سرطان المستقيم السفلي قد يصل للرئة أولاً.

AU·2 SURGERY جراحة Digestive System — Chapter 04: Pathology of GIT Tumours

Cell of origin of GIST:

A. G cells
× G cells secrete gastrin. تفرز الغاسترين.

B. Peptic cells
× Chief (peptic) cells make pepsinogen. تفرز البيبسينوجين.

C. Cells of Cajal ✓ Book p. 75 ↗

D. Mucus cells
× Mucus cells: not GIST. ليست GIST.

E. Parietal cells
× Parietal cells make acid. تفرز الحمض.

Why: GIST arises from the interstitial cells of Cajal (pacemaker cells).

لماذا: GIST ينشأ من الخلايا البينية لكاجال (منظمة الحركة).

★ **Tell the students:** GIST: KIT/CD117 positive; imatinib.

قل للطلاب: GIST: KIT/CD117 ؛العلاج imatinib.

AU·3 SURGERY جراحة Diaphragm anatomy — not covered in the Digestive book

Which structure does NOT pass through one of the three main diaphragmatic openings?

A. Left gastric artery
× Esophageal branches of the left gastric pass through the esophageal opening. فروعها المريئية تمر من فتحة المريء.

B. Esophagus
× Esophageal opening (T10). فتحة المريء (T10).

C. Aorta
× Aortic opening (T12). فتحة الأبهر (T12).

D. Inferior vena cava
× Caval opening (T8). فتحة الأجوف (T8).

E. Left phrenic nerve ✓ Not in the book

Why: The left phrenic nerve pierces the left dome of the diaphragm; it does not use the caval, esophageal or aortic openings.

لماذا: العصب الحجابي الأيسر يخترق القبة اليسرى ولا يمر من الفتحات الثلاث.

★ **Tell the students:** T8: IVC + right phrenic; T10: esophagus + vagi; T12: aorta + thoracic duct + azygos.

قل للطلاب: T8: الأجوف؛ T10: المريء والمبهمان؛ T12: الأبهر والقناة الصدرية.

Question note: Anatomy question; not covered in this book.

ملاحظة على السؤال: سؤال تشريح؛ غير موجود في الكتاب.

AU·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 03: The Stomach and Duodenum

Regarding gastrointestinal bleeding, one of the following is FALSE:

A. Hematemesis arises distal to the ligament of Treitz ✓ Book p. 4 ↗

B. Melena usually signifies upper GI bleeding
× True: melena usually means upper GI bleeding. صحيح.

C. Endoscopy is the main investigation for upper GI bleeding
× True: endoscopy is the main test. صحيح.

D. Hematemesis is due to bleeding proximal to the ligament of Treitz
× True: hematemesis is proximal to Treitz. صحيح.

Why: Hematemesis arises PROXIMAL to the ligament of Treitz, so the statement is false.
لماذا: القيء الدموي يأتي من قبل (أعلى) رباط Treitz لذا الجملة خاطئة.

★ **Tell the students:** Treitz = duodenojejunal junction; upper GI bleed is above it.
قل للطلاب: Treitz = اتصال الاثني عشر بالصائم.

Group AV

AV·1 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

5-year survival after esophagectomy for EARLY esophageal cancer:

A. 80–90% ✓ Not in the book

B. 60–70%
× Too low for early disease. منخفض جداً للمبكر.

C. <40%
× That is advanced disease. للمتقدم.

D. 70–80%
× Slightly below the key. أقل قليلاً من المفتاح.

E. >90%
× Overstated. مبالغ فيه.

Why: Five-year survival after esophagectomy for early esophageal cancer is about 80-90% (exam key).
لماذا: البقاء 5 سنوات بعد الاستئصال للمرض المبكر نحو 80-90% (حسب مفتاح الامتحان).

★ **Tell the students:** The book gives only the overall figure: 5-year cure about 10%.
قل للطلاب: الكتاب يذكر الإجمالي فقط: نحو 10%.

Question note: Not stated in the book; lower confidence.
ملاحظة على السؤال: غير مذكور بالكتاب؛ ثقة أقل.

AV·2 SURGERY جراحة

Digestive System — Chapter 02: The Esophagus · Digestive System — Chapter 04: Pathology of GIT Tumours

The carcinoma commonly arising in Barrett's esophagus is:

A. Squamous cell carcinoma
× SCC: smoking/alcohol. تدخين/كحول.

B. Adenocarcinoma ✓ Book p. 71 ↗

C. Adenosquamous carcinoma
× Not typical. غير نمطي.

D. Transitional cell carcinoma
× Not typical. غير نمطي.

Why: Adenocarcinoma commonly arises in Barrett's esophagus.

لماذا: السرطان الغدي يظهر شائعاً في مريء باريت.

★ **Tell the students:** Barrett's: metaplasia, dysplasia, adenocarcinoma.

قل للطلاب: باريت: تحول ثم خلل ثم سرطان غدي.

AV·3 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

All the following about pyloric obstruction in adults are true EXCEPT:

A. Upper GI endoscopy is indicated
× True: endoscopy is indicated. صحيح.

B. Vomiting is characteristically bilious ✓ Book p. 58 ↗

C. May cause tetany
× True: alkalosis lowers ionized calcium (tetany). صحيح: يسبب تكزّز.

D. May cause metabolic alkalosis
× True: metabolic alkalosis. صحيح: قلاء.

Why: In adult pyloric obstruction the vomit is NON-bilious (obstruction is proximal to the ampulla).

لماذا: في انسداد البواب عند البالغين القيء غير صفراوي (الانسداد قبل الأمبولة).

★ **Tell the students:** Vomit of old, undigested food without bile.

قل للطلاب: قيء طعام قديم بلا صفراء.

AV·4 INTERNAL MEDICINE باطنة Digestive System — Chapter 02: The Esophagus

In achalasia of the cardia, all are true EXCEPT:

A. There is impaired peristalsis at the lower esophageal sphincter
× Book: failure of LES relaxation and loss of peristalsis.
الكتاب: فشل ارتخاء LES وفقد التمعج.

B. May be complicated by esophageal carcinoma
× True: esophageal cancer risk.
صحيح: خطر السرطان.

C. Characterized by 'bird's beak' appearance on contrast radiography
× True: bird's beak on barium.
صحيح: منقار الطائر.

D. Dysphagia is equal for fluids and solids ✓ Book p. 21 ↗

Why: Per the book, achalasia dysphagia is greater for liquids than solids, so 'equal' is the exception.

لماذا: حسب الكتاب: في achalasia البلع أصعب للسوائل أكثر من المواد الصلبة فعبارة 'متساوٍ' هي الاستثناء.

★ **Tell the students:** Manometry: high resting LES pressure and aperistalsis (gold standard).

قل للطلاب: المانومتري: ضغط LES مرتفع وغياب التمعج.

Question note: Option A is loosely worded; the book (p.24) also says 'equal dysphagia' for motility disorders in general.

ملاحظة على السؤال: الاختيار A مصاغ بشكل فضفاض؛ والكتاب (24) يذكر 'عسر بلع متساوٍ' في اضطرابات الحركة عامة.

AV·5 SURGERY جراحة

Digestive System — Chapter 03: The Stomach and Duodenum · Digestive System — Chapter 04: Pathology of GIT Tumours

All increase risk of gastric cancer EXCEPT:

A. Smoking
× Risk factor.
عامل خطر.

B. H. pylori infection
× Risk factor.
عامل خطر.

C. Pernicious anemia
× Risk factor.
عامل خطر.

D. Prolonged H2 blocker intake ✓ Book p. 62 ↗

E. Distal gastrectomy (stump)
× Stump cancer risk.
خطر سرطان الجذع.

Why: Prolonged H2-blocker use is not a risk factor for gastric cancer.

لماذا: استخدام حاصرات H2 لفترة طويلة ليس عامل خطر.

★ **Tell the students:** Risks: H. pylori, nitrites/smoked food, salt, pernicious anemia, blood group A, polyps.

قل للطلاب: العوامل: H. pylori، النيتريت والمدخن، الملح، أنيميا خبيثة، فصيلة A.

Group AW

AW·1 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Complications of duodenal ulcer include:

A. Hour-glass stomach
- × Hour-glass stomach is a gastric ulcer complication.

من مضاعفات قرحة المعدة.

B. Malignant transformation
- × Malignant change is for gastric ulcer.

لقرحة المعدة.

C. Gastric outlet obstruction ✓ Book p. 57 ↗

D. Obstructive jaundice
- × Not a typical DU complication.

ليست نمطية.

Why: Gastric outlet obstruction from fibrotic stenosis is a complication of duodenal ulcer.

لماذا: انسداد مخرج المعدة بسبب التليف مضاعفة لقرحة الاثني عشر.

★ **Tell the students:** DU: posterior bleeds, anterior perforates, scarring obstructs.

قل للطلاب: الخلفية تنزف والأمامية تنثقب والتليف يسد.

AW·2 SURGERY جراحة Digestive System — Chapter 04: Pathology of GIT Tumours

About gastric lymphoma, all TRUE EXCEPT:

A. Better prognosis than adenocarcinoma
- × True: better prognosis than adenocarcinoma.

صحيح.

B. Chemotherapy improves outcome
- × True: chemotherapy helps.

صحيح.

C. Surgery when possible gives best results
- × True (exam key): surgery when possible gives good results.

صحيح (حسب المفتاح).

D. Usually Hodgkin type ✓ Book p. 42 ↗

E. Less frequent than adenocarcinoma
- × True: less frequent than adenocarcinoma.

صحيح.

Why: Gastric lymphoma is usually B-cell non-Hodgkin (MALT or DLBCL), not Hodgkin.

لماذا: لمفوما المعدة غالباً non-Hodgkin من الخلايا B (MALT) وليست Hodgkin.

★ **Tell the students:** Early MALToma: H. pylori eradication can cure.

قل للطلاب: MALToma المبكر: استئصال H. pylori قد يشفي.

AW·3 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

Boerhaave's syndrome refers to:

A. Mucosal tear at GE junction after forceful vomiting
× Mucosal tear = Mallory-Weiss.
.Mallory-Weiss =

B. Iatrogenic perforation during endoscopy
× Iatrogenic perforation is a different entity.
كيان مختلف.

C. Corrosive-induced perforation
× Corrosive injury is different.
مختلف.

D. Full-thickness spontaneous esophageal perforation after forceful vomiting ✓ Not in the book

Why: Boerhaave's syndrome is spontaneous full-thickness esophageal rupture after forceful vomiting.
لماذا: متلازمة Boerhaave هي تمزق المريء الكامل الطبقات تلقائياً بعد قيء شديد.

★ **Tell the students:** Mackler triad: vomiting, chest pain, subcutaneous emphysema. Urgent surgery or stent.
قل للطلاب: ثلاثي Mackler: قيء وألم صدر وانتفاخ تحت الجلد.

Question note: Not in this book.
ملاحظة على السؤال: غير موجود في الكتاب.

Group AX

AX·1 SURGERY جراحة

Digestive System — Chapter 03: The Stomach and Duodenum · Digestive System — Chapter 04: Pathology of GIT Tumours

Transcoelomic spread of gastric carcinoma goes mainly to:

A. Liver
× Liver: hematogenous.
دموي.

B. Ovary (Krukenberg) ✓ Book p. 62 ↗

C. Intestine
× Not the main site.
ليس الأشهر.

D. Kidney
× Not the main site.
ليس الأشهر.

Why: Transcoelomic spread of gastric cancer goes to the ovary (Krukenberg tumor).
لماذا: الانتشار عبر التجويف البطني لسرطان المعدة يصل للمبيض (ورم Krukenberg).

★ **Tell the students:** Also Douglas pouch (Blumer shelf) and umbilicus (Sister Mary Joseph).
قل للطلاب: أيضاً كيس دوغلاس وسرة Sister Mary Joseph.

AX·2 SURGERY جراحة Digestive System — Chapter 02: The Esophagus

67-year-old with dysphagia + hoarseness. Endoscopy: middle 1/3 esophageal cancer + right vocal cord paralysis. Best management:

A. Tracheostomy
× Tracheostomy is not indicated. غير مطلوب.

B. Esophagectomy through right 5th ICS
× Unresectable. غير قابل للاستئصال.

C. Esophagectomy through left thoraco-abdominal incision
× Unresectable. غير قابل للاستئصال.

D. Esophageal stent ✓ Book p. 29 ↗

Why: Vocal-cord paralysis means recurrent laryngeal nerve invasion (T4b): unresectable, so palliate (stent).

لماذا: شلل الحبل الصوتي = غزو العصب الحنجري الراجع (T4b): غير قابل للاستئصال فالتلطيف (دعامة).

★ **Tell the students:** Add palliative chemo/RT as appropriate.

قل للطلاب: مع علاج كيماوي/إشعاعي تلطيفي.

AX·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 03: The Stomach and Duodenum

All the following weaken the gastric mucosal barrier EXCEPT:

A. Prostaglandin analogues ✓ Book p. 35 ↗

B. Duodeno-gastric reflux
× Bile reflux weakens it. ارتجاع الصفراء يضعفه.

C. H. pylori
× H. pylori weakens it. يضعفه.

D. Smoking
× Smoking weakens it. يضعفه.

E. Shock
× Shock (ischemia) weakens it. الصدمة تضعفه.

Why: Prostaglandin analogues strengthen the mucosal barrier; they do not weaken it.

لماذا: نظائر البروستاجلاندين تقوّي حاجز المخاطية ولا تضعفه.

★ **Tell the students:** Barrier = mucus-bicarbonate layer + PGE2 + blood flow.

قل للطلاب: الحاجز = المخاط والبيكربونات وPGE2 والتروية.

AX·4 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

Regarding gastric peptic ulcer classification (Modified Johnson), which is FALSE?

A. Type I ulcers occur in the lesser curve with normal or subnormal acid secretion
× True: type I, lesser curve, low/normal acid.
صحيح.

B. Type II ulcers are located in the antrum only ✓ Nearest: p. 43 ↗

C. Malignant transformation is rare
× Accepted statement: malignant change is uncommon.
عبارة مقبولة: التسرطن غير شائع.

D. Type II is usually associated with duodenal ulcer
× True: type II goes with DU.
صحيح.

Why: Johnson type II is a gastric ulcer WITH a duodenal ulcer, not an antral-only ulcer.

لماذا: Johnson النوع II قرحة معدة مع قرحة اثني عشر وليست بالغار فقط.

★ **Tell the students:** I lesser curve (low acid), II gastric + DU, III prepyloric, IV near GEJ.

قل للطلاب: I صغرى، II معدة + اثني عشر، III قبل البواب، IV قرب GEJ.

Group AY

AY·1 INTERNAL MEDICINE باطنة Digestive System — Chapter 02: The Esophagus

Complications of gastroesophageal reflux include:

A. Barrett's esophagus
× A complication.
مضاعفة.

B. Stricture
× A complication.
مضاعفة.

C. Reflux esophagitis
× A complication.
مضاعفة.

D. Hematemesis
× A complication.
مضاعفة.

E. All the above ✓ Book p. 15 ↗

Why: All listed are complications of GERD: Barrett's, stricture, esophagitis and hematemesis.

لماذا: كلها مضاعفات للـGERD: باريت وتضيق والتهاب وقيء دموي.

★ **Tell the students:** Also short esophagus and adenocarcinoma.

قل للطلاب: أيضاً مريء قصير وسرطان غدي.

AY·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Which test is usually NORMAL in bacterial overgrowth syndrome?

A. D-Xylose absorption ✓ Book p. 93 ↗

B. Stage II Schilling test
× Stage II stays abnormal; corrected only by antibiotics.
يبقى غير طبيعي ويصحح بالمضادات.

C. Fecal fat (24h)
× Fecal fat is increased.
دهون البراز مرتفعة.

D. Jejunal aspirate culture
× Culture is positive.
المزرعة إيجابية.

E. Lactulose breath test
× Breath test is positive.
اختبار التنفس إيجابي.

Why: The D-xylose test is the one that may stay normal (intact mucosa); culture, breath test, fecal fat and Schilling stage II are abnormal.

لماذا: اختبار D-xylose قد يبقى طبيعياً (المخاطية سليمة)؛ أما المزرعة واختبار التنفس ودهون البراز وSchilling II فغير طبيعية.

★ **Tell the students:** SIBO: low B12 with normal/high folate; gold standard = jejunal aspirate culture.

قل للطلاب: SIBO: B12 منخفض وفولات طبيعي/مرتفع؛ المعيار مزرعة الصائم.

Question note: D-xylose is not printed in the book; chosen by elimination.

ملاحظة على السؤال: D-xylose غير مذكور بالكتاب؛ اخترنا بالاستبعاد.

AY·3 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

Disaccharidase deficiency is an example of:

A. Exudative diarrhea
× Exudative: inflammation.
التهابي.

B. Osmotic diarrhea ✓ Book p. 79 ↗

C. Motility diarrhea
× Motility: transit change.
حركي.

D. Secretory diarrhea
× Secretory: toxins/hormones.
إفرازي.

Why: Disaccharidase (e.g. lactase) deficiency causes osmotic diarrhea.

لماذا: نقص إنزيمات السكريات الثنائية (كاللاكتيز) يسبب إسهالاً أسموزياً.

★ **Tell the students:** Osmotic gap is high and the diarrhea stops with fasting.

قل للطلاب: الفجوة الأسموزية عالية ويتوقف بالصيام.

AY·4 SURGERY جراحة

Digestive System — Chapter 03: The Stomach and Duodenum · Digestive System — Chapter 04: Pathology of GIT Tumours

Regarding gastric carcinoma, all are true EXCEPT:

A. Rarely affects the antrum of the stomach ✓ Book p. 62 ↗

B. Troisier's sign carries a bad prognosis
× True: Troisier's sign means a bad prognosis. صحيح.

C. May complicate pernicious anemia
× True: pernicious anemia predisposes. صحيح.

D. May be associated with dietary nitrosamines and smoked foods
× True: nitrosamines and smoked foods. صحيح.

Why: The antrum/pylorus is the commonest site of gastric carcinoma (about 60%).

لماذا: الغار/البواب هو أشهر مكان لسرطان المعدة (نحو 60%).

★ **Tell the students:** Sites: antrum 60%, fundus 25%, body 15%.

قل للطلاب: المواضع: الغار 60% والقاع 25% والجسم 15%.

AY·5 SURGERY جراحة

Digestive System — Chapter 03: The Stomach and Duodenum · Digestive System — Chapter 04: Pathology of GIT Tumours

A recent Japanese immigrant with 3-month indigestion + gastric mass. Most likely diagnosis:

A. Chronic peptic ulcer
× An ulcer is not a 'mass'. القرحة ليست كتلة.

B. Multiple acute stress ulcers
× Stress ulcers occur in critical illness. تحدث في المرض الحرج.

C. Peutz-Jeghers syndrome
× Peutz-Jeghers: hamartomatous polyps. سلائل هامارتومية.

D. Gastric adenocarcinoma ✓ Book p. 72 ↗

E. Chronic gastritis
× Gastritis does not produce a mass. لا يعطي كتلة.

Why: A gastric mass with dyspepsia in a recent Japanese immigrant points to gastric adenocarcinoma (high-incidence region).

لماذا: كتلة معدة مع عسر هضم عند مهاجر ياباني حديثاً = سرطان المعدة الغدي (منطقة عالية الحدوث).

★ **Tell the students:** High-incidence: Japan, Chile, Iceland.

قل للطلاب: المناطق العالية: اليابان وتشيلي وآيسلندا.

Group AZ

AZ·1 SURGERY جراحة

Digestive System — Chapter 03: The Stomach and Duodenum · Digestive System — Chapter 04: Pathology of GIT Tumours

> Which is NOT a predisposing factor for gastric cancer?

A. Foods with preservatives
× Risk: preservatives/nitrites. عامل خطر.

B. Gastric polyp
× Risk: adenomatous polyp. عامل خطر.

C. H. pylori
× Risk: H. pylori. عامل خطر.

D. Biliary gastritis
× Risk: bile reflux gastritis. عامل خطر.

E. Iron deficiency anemia ✓ Book p. 62 ↗

Why: Iron-deficiency anemia is a consequence, not a predisposing factor; pernicious anemia is the predisposing one.

لماذا: أنيميا نقص الحديد نتيجة وليست عامل خطر؛ والأنيميا الخبيثة هي المهيئة.

★ **Tell the students:** Also blood group A and partial gastrectomy.

قل للطلاب: أيضاً فصيلة A واستئصال المعدة الجزئي.

AZ·2 INTERNAL MEDICINE باطنة Digestive System — Chapter 05: Diarrhea and Malabsorption

> What differentiates malabsorption in tropical sprue vs bacterial overgrowth?

A. Serum calcium
× Both malabsorb calcium. الاثنان يسببان سوء امتصاص.

B. Serum folate ✓ Book p. 88 ↗

C. Serum B12
× B12 is low in both. B12 منخفض في الاثنين.

D. Serum iron
× Iron is low in both. الحديد منخفض في الاثنين.

Why: Serum folate: normal/high in bacterial overgrowth but low in tropical sprue.

لماذا: الفولات: طبيعي/مرتفع في فرط النمو البكتيري لكنه منخفض في Tropical sprue.

★ **Tell the students:** Bacteria synthesize folate and consume B12.

قل للطلاب: البكتيريا تصنع الفولات وتستهلك B12.

AZ·3 SURGERY جراحة Digestive System — Chapter 03: The Stomach and Duodenum

All are frequent complications of chronic PUD EXCEPT:

A. Carcinomatous change (only in gastric, very rare in duodenal) ✓ Book p. 53 ↗

B. Bleeding
× Frequent: bleeding. متكرر.

C. Scarring deformity (stenosis)
× Frequent: scarring stenosis. متكرر.

D. Perforation
× Frequent: perforation. متكرر.

Why: Carcinomatous change is rare (gastric ulcer only), so it is not a frequent complication.

لماذا: التسرطن نادر (لقرحة المعدة فقط) فليس مضاعفة متكررة.

★ **Tell the students:** Chronic PUD: bleeding, perforation, stenosis, penetration.

قل للطلاب: المزمنة: نزف وانثقاب وتضيق واختراق.

PART 2

Microbiology ميكروبيولوجي

Group BA

BA·1 MICROBIOLOGY ميكروبيولوجي Digestive System — Chapter 07: Foodborne Infection

Salmonella spp. have the following biochemical reactions EXCEPT:

A. Ferment lactose ✓ Book p. 107 ↗

B. Urease negative
× True: urease negative. صحيح: urease سلبي.

C. Indole negative
× True: indole negative. صحيح: indole سلبي.

D. Ferment glucose
× True: ferments glucose. صحيح: تخمّر الجلوكوز.

Why: Salmonella is a NON-lactose fermenter (pale colonies on MacConkey).

لماذا: السالمونيلا لا تخمّر اللاكتوز (مستعمرات شاحبة على MacConkey).

★ **Tell the students:** Salmonella: H2S positive, motile, oxidase negative; grow on SS agar, selenite broth.

قل للطلاب: H2S موجب ومتحركة؛ تنمو على SS وselenite.

BA·2 MICROBIOLOGY ميكروبيولوجي Digestive System — Chapter 07: Foodborne Infection

In a retirement-home outbreak, what is the primary reservoir of infection?

A. Birds
× Birds: Campylobacter, Salmonella. الطيور: Campylobacter وSalmonella.

B. Cattle
× Cattle: Brucella, E. coli O157. الأبقار: Brucella وE. coli.

C. Rodents
× Rodents: Leptospira, S. typhimurium. القوارض: Leptospira وS. typhimurium.

D. Human ✓ Nearest: p. 107 ↗

Why: Institutional (retirement home) outbreaks spread person to person, so the reservoir is human.

لماذا: الفاشيات في دور المسنين تنتقل من شخص لآخر فالمستودع هو الإنسان.

★ **Tell the students:** The stem names no organism. Think norovirus, Shigella, S. typhi carriers, C. difficile.

قل للطلاب: السؤال لا يذكر الميكروب. فكّر: Norovirus وShigella وحاملي S. typhi وC. difficile.

Question note: Question gives no organism; the best answer by setting is human.

ملاحظة على السؤال: السؤال لا يذكر الميكروب؛ الأنسب حسب السياق: الإنسان.

BA·3 MICROBIOLOGY ميكروبيولوجي Digestive System — Chapter 07: Foodborne Infection

Which culture medium is used for curved gram-negative bacilli growing at 42°C?

A. Thayer-Martin medium
× Thayer-Martin: Neisseria. لـ Neisseria.

B. MacConkey agar
× MacConkey: enterics. للمعويات.

C. Skirrow's media ✓ Book p. 110 ↗

D. Lowenstein-Jensen medium
× Lowenstein-Jensen: M. tuberculosis. للسل.

Why: Skirrow's medium (selective, with antibiotics, at 42 C) grows Campylobacter.

لماذا: وسط Skirrow الانتقائي بالمضادات عند 42 م ينمّي Campylobacter.

★ **Tell the students:** Campylobacter: curved 'gull-wing', oxidase +, microaerophilic, thermophilic.

قل للطلاب: منحنية، microaerophilic، oxidase +، تحب الحرارة.

Group BB

BB·1 MICROBIOLOGY ميكروبيولوجي Digestive System — Chapter 07: Foodborne Infection

Which organism would NOT benefit from prompt antibiotic treatment?

A. C. botulinum ✓ Book p. 113 ↗

B. C. perfringens
× The book gives no such statement for it; it is self-limited.
الكتاب لا يذكر ذلك؛ المرض محدود ذاتياً.

C. B. anthracis
× Anthrax needs prompt antibiotics.
الجمرة تحتاج مضاداً فورياً.

D. C. difficile
× C. difficile needs vancomycin/fidaxomicin.
يحتاج vancomycin/fidaxomicin.

Why: Botulism is toxin-mediated: treatment is antitoxin and respiratory support; antibiotics are not used.
لماذا: التسمم الوشيقي بسبب السم: العلاج مضاد السم ودعم التنفس ولا تُستخدم المضادات.

★ **Tell the students:** Antitoxin works only before toxin binds the neuromuscular junction (within ~12 h).
قل للطلاب: مضاد السم فعال قبل ارتباط السم (خلال ~12 ساعة).

Question note: Both botulism and perfringens food poisoning are toxin-mediated; the book states botulism needs no antibiotics.
ملاحظة على السؤال: كلاهما سمّي المنشأ؛ والكتاب ينص على أن الوشيقية لا تعالج بالمضادات.

BB·2 MICROBIOLOGY ميكروبيولوجي Digestive System — Chapter 07: Foodborne Infection

Bacillus cereus is the etiological agent of all of the following EXCEPT:

A. "Long-incubation" or diarrheal form of food poisoning
× True: diarrheal form, 8-16 h, heat-labile toxin.
صحيح: النوع الإسهالي 8-16 ساعة.

B. Non bloody diarrhea
× True: non-bloody diarrhea.
صحيح: بلا دم.

C. Cutaneous anthrax ✓ Book p. 112 ↗

D. "Short-incubation" or emetic form of food poisoning
× True: emetic form, 1-6 h, preformed toxin.
صحيح: النوع المقيّئ 1-6 ساعات.

Why: Cutaneous anthrax is caused by B. anthracis, not B. cereus.
لماذا: الجمرة الجلدية سببها B. anthracis وليس B. cereus.

★ **Tell the students:** Emetic form: fried/stored rice (preformed heat-stable toxin).
قل للطلاب: المقيّئ: الأرز المخزن.

BB·3 MICROBIOLOGY ميكروبيولوجي Digestive System — Chapter 05: Diarrhea and Malabsorption

> Which stool study is specifically recommended for hospital-acquired diarrhea or diarrhea after recent antibiotics?

A. Fecal occult blood only
× FOBT detects blood only. للدم فقط.

B. Ova and parasites only
× O&P only finds parasites. للطفيليات.

C. Clostridioides difficile toxin testing ✓ Book p. 83 ↗

D. Stool elastase
× Elastase tests the pancreas. للبنكرياس.

Why: Hospital-acquired or post-antibiotic diarrhea calls for C. difficile toxin testing.
لماذا: الإسهال داخل المستشفى أو بعد المضادات يتطلب فحص سم C. difficile.

★ **Tell the students:** Treat with oral vancomycin or fidaxomicin and stop the triggering antibiotic.
قل للطلاب: العلاج vancomycin فموي أو fidaxomicin مع إيقاف المضاد.

BB·4 MICROBIOLOGY ميكروبيولوجي Digestive System — Chapter 07: Foodborne Infection

> Rodents are reservoirs of Leptospira which colonize their:

A. Eye
× Not the site. ليس الموضع.

B. Skin
× Not the site. ليس الموضع.

C. Spleen
× Not the site. ليس الموضع.

D. Kidneys ✓ Book p. 113 ↗

Why: Rodents carry Leptospira in the renal tubules and shed it in urine.
لماذا: القوارض تحمل Leptospira في الكلى وتفرزه في البول.

★ **Tell the students:** Humans get infected via skin breaks in water contaminated by animal urine (Weil's disease).
قل للطلاب: ينتقل عبر جروح الجلد بماء ملوث ببول الحيوان (Weil).

Group BC

BC·1 MICROBIOLOGY ميكروبيولوجي Digestive System — Chapter 07: Foodborne Infection

Salmonella causing food poisoning include:

A. Salmonella typhi
× S. typhi: enteric (typhoid) fever.
حمى تيفية.

B. Salmonella typhimurium ✓ Book p. 110 ↗

C. Salmonella paratyphi A
× Paratyphi A: paratyphoid.
باراتيفويد.

D. Salmonella paratyphi B
× Paratyphi B: paratyphoid.
باراتيفويد.

Why: S. typhimurium (and S. enteritidis) cause food poisoning; S. typhi and paratyphi cause enteric fever.
لماذا: S. typhimurium (وenteritidis) تسبب تسمماً غذائياً؛ وtyphi وparatyphi تسبب الحمى المعوية.

★ **Tell the students:** Food poisoning: 12-72 h, self-limited, no antibiotics.
قل للطلاب: 12-72 ساعة، محدود ذاتياً بلا مضادات.

BC·2 MICROBIOLOGY ميكروبيولوجي Digestive System — Chapter 07: Foodborne Infection

The laboratory test of choice to diagnose brucellosis is:

A. Gram stain
× Gram stain is of no significance.
صبغة جرام بلا قيمة.

B. Direct culture on nutrient agar
× Needs enriched media, not plain agar.
تحتاج وسطاً مغذياً.

C. Blood film
× Not diagnostic.
غير تشخيصي.

D. Blood culture ✓ Book p. 111 ↗

Why: Blood culture is the method of choice early in brucellosis (kept up to 4-6 weeks).
لماذا: مزرعة الدم هي الاختيار المبكر في البروسيلا (تحضن حتى 4-6 أسابيع).

★ **Tell the students:** Serology (agglutination, two samples) is the main laboratory method.
قل للطلاب: السيرولوجي هو الطريقة الرئيسية.

BC·3 MICROBIOLOGY ميكروبيولوجي Digestive System — Chapter 05: Diarrhea and Malabsorption

A 32-year-old has had diarrhea for two weeks, sometimes with dysentery, after taking antibiotics for sinusitis before symptoms began. Which organism is most likely?

A. Clostridium difficile ✓ Book p. 83 ↗

B. Staphylococcus aureus
× S. aureus toxin acts within hours.
سم خلال ساعات.

C. Salmonella typhi
× S. typhi: fever, not antibiotic-linked.
حمى وليس مرتبطاً بالمضاد.

D. Campylobacter jejuni
× Campylobacter is from poultry.
من الدواجن.

Why: Diarrhea (with dysentery) after antibiotics for sinusitis = Clostridium difficile.

لماذا: إسهال (قد يكون دموياً) بعد مضاد حيوي = Clostridium difficile.

★ **Tell the students:** Triggers: clindamycin, cephalosporins, fluoroquinolones; pseudomembranous colitis.

قل للطلاب: المحفزات: clindamycin، سيفالوسبورين، فلوروكينولون.

BC·4 MICROBIOLOGY ميكروبيولوجي Digestive System — Chapter 07: Foodborne Infection

Culture of Campylobacter is done under the following condition:

A. Strict anaerobe
× Not strict anaerobe.
ليس لاهوائياً.

B. Micro-aerophilic and capnophilic ✓ Book p. 110 ↗

C. Strict aerobe
× Not strict aerobe.
ليس هوائياً.

D. Facultative anaerobe
× Not facultative.
ليس اختيارياً.

Why: Campylobacter is cultured under micro-aerophilic, capnophilic (10% CO2) conditions at 42 C.

لماذا: Campylobacter يُزرع في ظروف micro-aerophilic وcapnophilic (CO2 10%) عند 42 م.

★ **Tell the students:** Oxidase +, catalase +, does not use carbohydrates.

قل للطلاب: oxidase + وcatalase + ولا يستخدم الكربوهيدرات.

BC·5 MICROBIOLOGY ميكروبيولوجي Digestive System — Chapter 07: Foodborne Infection

An extracellular virulence factor of Campylobacter is:

A. Flagella
× Cellular component.
مكوّن خلوي.

B. Adhesins
× Cellular component.
مكوّن خلوي.

C. Enterotoxins (LT) ✓ Book p. 110 ↗

D. Endotoxin (LPS)
× Endotoxin (LPS) is cellular.
الـLPS مكوّن خلوي.

Why: Enterotoxins (LT) and cytotoxins are Campylobacter's extracellular virulence factors.

لماذا: السموم المعوية (LT) والسموم الخلوية هي عوامل الضراوة خارج الخلوية لـ Campylobacter.

★ **Tell the students:** Complications: Guillain-Barre, Reiter's, reactive arthritis.

قل للطلاب: مضاعفات: Guillain-Barre وReiter وتهاب مفاصل تفاعلي.

PART 3

Parasitology باراسيتولوجي

Group BD

BD·1 PARASITOLOGY باراسيتولوجي Digestive System — Chapter 05: Diarrhea and Malabsorption

What is the most common infectious cause of chronic diarrhea?

A. Shigella dysenteriae
× Shigella: acute dysentery. زحار حاد.

B. Salmonella enterica
× Salmonella: acute gastroenteritis. التهاب حاد.

C. Giardia lamblia ✓ Book p. 83 ↗

D. Norovirus
× Norovirus: acute, self-limited. حاد ومحدود.

Why: Giardia lamblia is the commonest infectious cause of chronic diarrhea.
لماذا: Giardia lamblia أشهر سبب معدٍ للإسهال المزمن.

★ **Tell the students:** Giardia: cysts in stool, foul fatty diarrhea; treat with metronidazole or tinidazole.
قل للطلاب: Giardia: أكياس في البراز وإسهال دهني كريه؛ metronidazole أو tinidazole.

BD·2 PARASITOLOGY باراسيتولوجي Digestive System — Chapter 09: Parasitic GI Diseases

Which snail is the intermediate host of Schistosoma japonicum in the GI book's table?

A. Lymnaea species
× Lymnaea: Fasciola. لـ Fasciola.

B. Biomphalaria alexandrina
× Biomphalaria: S. mansoni. لـ S. mansoni.

C. No snail host
× Every schistosome needs a snail. كل البلهارسيا تحتاج قوقعة.

D. Onchomelania species ✓ Book p. 123 ↗

Why: Oncomelania is the snail host of Schistosoma japonicum.
لماذا: Oncomelania هو العائل الوسيط لـ S. japonicum.

★ **Tell the students:** Egypt: Biomphalaria alexandrina = S. mansoni; Bulinus truncatus = S. haematobium.
قل للطلاب: في مصر: Biomphalaria = mansoni؛ Bulinus = haematobium.

BD·3 PARASITOLOGY باراسيتولوجي Digestive System — Chapter 09: Parasitic GI Diseases

Which Schistosoma life-cycle stage hatches from eggs in fresh water and penetrates the snail?

A. Cercaria
× Cercariae leave the snail to infect humans. تخرج من القوقعة لتصيب الإنسان.

B. Miracidium ✓ Book p. 124 ↗

C. Adult female
× Adults live in human veins. تعيش في أوردة الإنسان.

D. Schistosomulum
× Schistosomulum is the stage after skin penetration. المرحلة بعد اختراق الجلد.

Why: Miracidia hatch from eggs in fresh water and penetrate the snail.

لماذا: الميراسيديا تفقس من البيض في الماء العذب وتخترق القوقعة.

★ **Tell the students:** Egg, miracidium, snail sporocyst, cercaria, human skin, schistosomulum, adult.

قل للطلاب: بيضة، ميراسيديا، سبوروسيست، سركاريا، جلد، schistosomulum، ناضجة.

Group BE

BE·1 PARASITOLOGY باراسيتولوجي Digestive System — Chapter 09: Parasitic GI Diseases

What is the mode of infection of Taenia saginata?

A. Ingestion of undercooked meat ✓ Book p. 127 ↗

B. Ingestion of contaminated water
× Water does not transmit it. الماء لا ينقلها.

C. Ingestion of raw pork
× Raw pork = T. solium. لحم الخنزير = T. solium.

D. Penetration of larvae through the skin
× Skin penetration: hookworm, schistosomes. اختراق الجلد: الانكلستوما والبلهارسيا.

Why: T. saginata is acquired by eating undercooked beef containing cysticercus bovis.

لماذا: T. saginata تنتقل بأكل لحم بقري ناقص الطهو يحوي cysticercus bovis.

★ **Tell the students:** T. solium eggs can cause cysticercosis; T. saginata cannot.

قل للطلاب: بيض T. solium يسبب cysticercosis أما saginata فلا.

BE·2 PARASITOLOGY باراسيتولوجي Digestive System — Chapter 09: Parasitic GI Diseases

What is the diagnostic stage of Schistosoma mansoni detected in stool?

A. Eggs ✓ Book p. 124 ↗

B. Adult worms
× Adults live in the veins. تعيش في الأوردة.

C. Miracidia
× Miracidia hatch in water. تفقس في الماء.

D. Cercariae
× Cercariae are infective, not diagnostic. معدية وليست تشخيصية.

Why: The embryonated eggs in stool are the diagnostic stage.

لماذا: البيض الجنيني في البراز هو المرحلة التشخيصية.

★ **Tell the students:** S. mansoni egg: lateral spine; haematobium: terminal spine (urine).

قل للطلاب: mansoni شوكة جانبية؛ haematobium شوكة طرفية (بول).

BE·3 PARASITOLOGY باراسيتولوجي Digestive System — Chapter 09: Parasitic GI Diseases

Cysticercus bovis is the infective stage of which parasite?

A. Taenia saginata ✓ Book p. 128 ↗

B. Echinococcus granulosus
× Echinococcus: hydatid cyst. كيس hydatid.

C. Diphyllobothrium latum
× Diphyllobothrium: plerocercoid in fish. في السمك.

D. Taenia solium
× T. solium: cysticercus cellulosae. cellulosae.

Why: Cysticercus bovis is the larval stage of Taenia saginata in cattle.

لماذا: Cysticercus bovis هو يرقة T. saginata في الأبقار.

★ **Tell the students:** C. bovis has no hooks; C. cellulosae has hooks.

قل للطلاب: bovis بلا خطاطيف وcellulosae بخطاطيف.

BE·4 PARASITOLOGY باراسيتولوجي Digestive System — Chapter 09: Parasitic GI Diseases

Which stage of Schistosoma infects humans after exposure to contaminated fresh water?

A. Egg
× Eggs are diagnostic. تشخيصية.

B. Cercaria ✓ Book p. 124 ↗

C. Miracidium
× Miracidia infect snails. تصيب القواقع.

D. Adult worm
× Adults live in veins. تعيش في الأوردة.

Why: Cercariae penetrate the skin during exposure to contaminated fresh water.

لماذا: السركاريا تخترق الجلد عند التعرض للماء الملوث.

★ **Tell the students:** The tail is shed, forming a schistosomulum.

قل للطلاب: يُفقد الذيل فتتكون schistosomulum.

Group BF

BF·1 PARASITOLOGY باراسيتولوجي Digestive System — Chapter 09: Parasitic GI Diseases

A 12-year-old Egyptian boy with a history of swimming in irrigation canals presents with blood and mucus in the stool. Stool examination reveals parasitic eggs. What is the treatment?

A. Albendazole
× Albendazole: nematodes/hydatid. للديدان الأسطوانية.

B. Praziquantel ✓ Nearest: p. 126 ↗

C. Niclosamide
× Niclosamide: tapeworms. للشريطيات.

D. Mebendazole
× Mebendazole: nematodes. للديدان الأسطوانية.

Why: Blood and mucus in stool after canal exposure = Schistosoma mansoni; praziquantel is the drug of choice.

لماذا: دم ومخاط في البراز بعد التعرض لماء الترع = S. mansoni؛ praziquantel هو الدواء المختار.

★ **Tell the students:** The book names praziquantel on the Taenia page (p.128); it is also the schistosomiasis drug of choice.

قل للطلاب: الكتاب يذكر praziquantel في صفحة (128) Taenia؛ وهو أيضاً دواء البلهارسيا.

BF·2 PARASITOLOGY باراسيتولوجي Digestive System — Chapter 09: Parasitic GI Diseases

Where do adult Schistosoma haematobium worms reside?

A. Vesical venous plexus ✓ Book p. 125 ↗

B. Inferior mesenteric venous plexus
× Inferior mesenteric veins: S. mansoni. .mansoni

C. Pulmonary arteries
× Not a habitat. ليس موطناً.

D. Bile ducts
× Bile ducts: Fasciola/Clonorchis. .Fasciola/Clonorchis

Why: S. haematobium adults live in the vesical venous plexus.

لماذا: ديدان S. haematobium البالغة تعيش في الضفيرة الوريدية المثانية.

★ **Tell the students:** Hence hematuria; S. mansoni gives intestinal and hepatosplenic disease.

قل للطلاب: لذلك البول الدموي؛ و mansoni يسبب أعراضاً معوية وكبدية.

BF·3 PARASITOLOGY باراسيتولوجي Digestive System — Chapter 09: Parasitic GI Diseases

What is the intermediate host of Schistosoma mansoni in the GI book's life-cycle table?

A. Mosquitoes
× Mosquitoes: malaria, filaria. للملاريا والفيلاريا.

B. Humans
× Humans are the definitive host. العائل النهائي.

C. Biomphalaria alexandrina ✓ Book p. 123 ↗

D. Onchomelania species
× Oncomelania: S. japonicum. لـ japonicum.

Why: Biomphalaria alexandrina is the intermediate host of S. mansoni.

لماذا: Biomphalaria alexandrina العائل الوسيط لـ S. mansoni.

★ **Tell the students:** Reservoir for mansoni: monkeys and rodents.

قل للطلاب: المستودع: القرود والقوارض.

BF·4 PARASITOLOGY باراسيتولوجي Digestive System — Chapter 09: Parasitic GI Diseases

The diagnostic stage of Ascaris lumbricoides is:

A. Cercaria
× Cercaria: schistosomes. للبلهارسيا.

B. Encysted metacercaria
× Metacercaria: Fasciola. لـ Fasciola.

C. Egg (in stool) ✓ Nearest: p. 130 ↗

D. Egg and cercaria
× Cercaria is not an Ascaris stage. ليست مرحلة للأسكاريس.

Why: The diagnostic stage of Ascaris is the egg in stool.

لماذا: المرحلة التشخيصية للأسكاريس هي البيضة في البراز.

★ **Tell the students:** Infection by swallowing embryonated eggs; larval lung phase (Loeffler).

قل للطلاب: العدوى ببلع بيض جنيني؛ مرحلة رئوية (Loeffler).

PART 4
Pharmacology فارماكولوجي

Group BG

BG·1 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

What is the mechanism of action of vonoprazan?

A. Irreversibly inhibits the proton pump
× Irreversible = PPIs. غير عكوس = PPIs.

B. Inhibits gastric prostaglandin synthesis
× Prostaglandin inhibition = NSAIDs. تثبيط البروستاجلاندين = NSAIDs.

C. Reversibly blocks proton pump potassium-binding site ✓ Book p. 46 ↗

D. Competitively blocks H2 receptors
× H2 blockade = famotidine. حجب H2 = famotidine.

Why: Vonoprazan is a P-CAB: it reversibly blocks the K+ site of the proton pump (active and resting pumps).
لماذا: Vonoprazan من فئة P-CAB: يحجب بشكل عكوس موقع البوتاسيوم في مضخة البروتون (النشطة والخاملة).

★ **Tell the students:** P-CAB: not a prodrug, acid-stable, no acid activation needed, works before or after meals, T1/2 6-9 h, max effect in 1 day.
قل للطلاب: P-CAB ليس دواءً سالفاً، ثابت في الحمض، لا يحتاج تنشيطاً، يؤخذ قبل/بعد الأكل، نصف عمره 6-9 ساعات، أقصى تأثير في يوم.

BG·2 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Which combination targets the underlying cause of an H. pylori-associated ulcer?

A. A laxative with a diuretic
× Nonsense combination. تركيبة غير منطقية.

B. A proton pump inhibitor with two or more antibiotics ✓ Book p. 48 ↗

C. Acid suppression alone
× Heals the ulcer but it relapses. يشفي القرحة لكنها تعود.

D. Antacids alone
× Symptomatic relief only. تخفيف أعراض فقط.

Why: A PPI plus two or more antibiotics eradicates H. pylori, which removes the cause and cuts recurrence.
لماذا: PPI مع مضادين حيويين أو أكثر يقضي على H. pylori فيزيل السبب ويقلل النكس.

★ **Tell the students:** Test of cure: urea breath test after 4 weeks off antibiotics and 2 weeks off PPI.
قل للطلاب: اختبار الشفاء: Urea breath test بعد 4 أسابيع من المضادات و2 أسبوع من الـPPI.

BG·3 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Which drug most commonly causes upper GI bleeding?

A. Calcium carbonate
× Antacid. مضاد حموضة.

B. Paracetamol
× Paracetamol harms the liver, not the stomach. الباراسيتامول يضر الكبد لا المعدة.

C. Famotidine
× Famotidine treats ulcers. Famotidine يعالج القرح.

D. Aspirin ✓ Book p. 50 ↗

Why: Aspirin and NSAIDs are the commonest drugs causing ulcers and upper GI bleeding.
لماذا: الأسبرين والـNSAIDs أشهر أدوية تسبب القرح ونزف الجهاز الهضمي العلوي.

★ **Tell the students:** Anticoagulants worsen bleeding from another lesion; they do not usually cause it themselves.
قل للطلاب: مضادات التجلط تزيد النزف من آفة أخرى ولا تسببه عادة بنفسها.

Group BH

BH·1 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 02: The Esophagus

When is the best time to take a PPI for GERD?

A. Immediately after the first meal
× Food lowers bioavailability by about 50%. الأكل يقلل الامتصاص نحو 50%.

B. At bedtime after dinner
× Pumps are quiet while fasting overnight. المضخات خاملة أثناء الصيام الليلي.

C. Only when symptoms begin
× Needs repeated dosing; PRN is ineffective. يحتاج جرعات متكررة وليس عند اللزوم.

D. 30–60 minutes before first meal ✓ Book p. 45 ↗

Why: PPIs only block ACTIVE pumps, so give them 30-60 min before the first meal.
لماذا: الـPPI يثبط المضخات النشطة فقط، فيؤخذ قبل أول وجبة بـ30-60 دقيقة.

★ **Tell the students:** Maximal effect after 3-5 days; vonoprazan (P-CAB) does not need this timing.
قل للطلاب: أقصى تأثير بعد 3-5 أيام؛ وvonoprazan لا يحتاج هذا التوقيت.

BH·2 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

How do NSAIDs cause gastric ulcers?

A. By blocking histamine H2 receptors
× H2 blockers are treatment, not a cause. حاصرات H2 علاج لا سبب.

B. Inhibition of prostaglandin synthesis ✓ Book p. 36 ↗

C. By increasing gastric mucus production
× NSAIDs reduce mucus. ينقصون المخاط.

D. By increasing prostaglandin synthesis
× They decrease PG synthesis. ينقصون تصنيع PG.

Why: NSAIDs inhibit COX-1, so PGE2 falls, mucus/bicarbonate drop and acid output rises.

لماذا: الـNSAIDs تثبط COX-1 فينقص PGE2 والمخاط والبيكربونات ويزداد الحمض.

★ **Tell the students:** Misoprostol (PGE1 analogue) or a PPI protects high-risk NSAID users.

قل للطلاب: Misoprostol (نظير PGE1) أو PPI يحمي المرضى عاليي الخطورة.

BH·3 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

What is the mechanism of action of omeprazole?

A. Reversibly blocks the proton pump potassium-binding site
× That is vonoprazan (P-CAB). هذا vonoprazan.

B. Irreversibly inhibits H^+/K^+ ATPase in parietal cells ✓ Book p. 45 ↗

C. Blocks muscarinic receptors on parietal cells
× Muscarinic block is anticholinergic action. حجب المسكارينيك دواء آخر.

D. Competitively blocks H2 receptors
× H2 blockade is famotidine. حجب H2 = famotidine.

Why: Omeprazole is a prodrug converted in acid to an agent that irreversibly inhibits H+/K+ ATPase.

لماذا: Omeprazole دواء سالف يتحول بالحمض ويثبط H+/K+ ATPase بشكل غير عكوس.

★ **Tell the students:** Long action (24 h) because new pumps take 24 h to be made, though T1/2 is only 1-2 h.

قل للطلاب: تأثيره 24 ساعة لأن بناء مضخات جديدة يحتاج 24 ساعة رغم أن نصف عمره 1-2 ساعة.

BH·4 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 02: The Esophagus

What is the drug of choice for moderate to severe GERD?

A. Sucralfate alone
× Sucralfate is weak and not first line.
ضعيف وليس خط أول.

B. Proton pump inhibitors ✓ Book p. 47 ↗

C. Antacids alone
× Antacids give symptom relief only.
تسكين أعراض فقط.

D. Antispasmodics
× Antispasmodics do not treat reflux.
لا تعالج الارتجاع.

Why: PPIs are the drugs of choice for GERD (and moderate-severe disease).
لماذا: مثبطات مضخة البروتون هي الدواء المفضل للـGERD (خاصة المتوسط والشديد).

★ **Tell the students:** H2 blockers are added at night for nocturnal acid breakthrough.
قل للطلاب: تضاف حاصرات H2 ليلاً لتغطية الإفراز الليلي.

Group BI

BI·1 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

What is an eradication therapy for H. pylori?

A. Triple therapy (PPI + amoxicillin + clarithromycin) ✓ Book p. 48 ↗

B. Antacid and sucralfate
× No antibiotic, so no eradication.
بدون مضاد حيوي فلا استئصال.

C. PPI alone
× PPI alone does not eradicate.
PPI وحده لا يستأصل.

D. Amoxicillin and metronidazole without acid suppression
× Antibiotics need acid suppression to work well.
المضادات تحتاج تثبيط الحمض لتعمل جيداً.

Why: Triple therapy (PPI + amoxicillin + clarithromycin) for 10-14 days is an eradication regimen.
لماذا: العلاج الثلاثي (PPI + amoxicillin + clarithromycin) لمدة 10-14 يوماً هو برنامج استئصال.

★ **Tell the students:** With clarithromycin resistance, bismuth quadruple therapy is preferred empirically.
قل للطلاب: مع مقاومة clarithromycin يفضل العلاج الرباعي بالبزموث.

BI·2 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 06: Constipation (including IBS)

How can peppermint oil help some patients with IBS according to the GI book?

A. It relaxes GI smooth muscle ✓ Book p. 105 ↗

B. It acts as a bile acid
× Not a bile acid. ليس حمضاً صفراوياً.

C. It increases gastric acid secretion
× It does not increase acid. لا يزيد الحمض.

D. It eradicates H. pylori
× Not an antibacterial for H. pylori. لا يستأصل H. pylori.

Why: Peppermint oil relaxes GI smooth muscle and relieves spasm.

لماذا: زيت النعناع يرخي العضلات الملساء ويخفف التشنج.

★ **Tell the students:** Also helpful in IBS pain: tricyclics (amitriptyline, nortriptyline).

قل للطلاب: مفيد أيضاً لألم IBS: مضادات الاكتئاب ثلاثية الحلقات.

BI·3 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 06: Constipation (including IBS)

Which treatment listed in the GI book is directed at IBS-D?

A. Lubiprostone
× Lubiprostone is for IBS-C. لـ IBS-C.

B. Mesalazine
× Mesalazine is for UC. لـ UC.

C. Pyloroplasty
× Pyloroplasty is gastric surgery. جراحة معدة.

D. Rifaximin ✓ Book p. 105 ↗

Why: Rifaximin, a non-absorbed antibiotic, treats the diarrhea-predominant type (IBS-D).

لماذا: Rifaximin مضاد حيوي غير ممتص يعالج النوع الإسهالي (IBS-D).

★ **Tell the students:** IBS-C: lubiprostone (chloride channel activator), prucalopride (5-HT4 agonist).

قل للطلاب: IBS-C: lubiprostone وprucalopride.

BI·4 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 06: Constipation (including IBS)

Which medication group may help relieve IBS-associated abdominal pain?

A. Proton pump inhibitors for every patient
× PPIs do not treat IBS pain. لا تعالج ألم IBS.

B. Tricyclic antidepressants ✓ Book p. 105 ↗

C. Loop diuretics
× Diuretics are irrelevant. غير مرتبطة.

D. Oral anticoagulants
× Anticoagulants are irrelevant. غير مرتبطة.

Why: Low-dose tricyclics (amitriptyline, nortriptyline) relieve IBS abdominal pain.

لماذا: مضادات الاكتئاب ثلاثية الحلقات بجرعة صغيرة تخفف ألم IBS.

★ **Tell the students:** They also slow transit, so they suit IBS-D.

قل للطلاب: تبطئ الحركة فتناسب IBS-D.

BI·5 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 06: Constipation (including IBS)

Which drug class is represented by lubiprostone in IBS-C treatment?

A. Chloride channel activator ✓ Book p. 105 ↗

B. Proton pump inhibitor
× PPI is an acid blocker. حاصر حمض.

C. Bile acid sequestrant
× Bile acid sequestrants cause constipation. تسبب الإمساك.

D. H2 receptor blocker
× H2 blockers: famotidine, etc. famotidine وغيره.

Why: Lubiprostone is a chloride channel activator used in IBS-C.

لماذا: Lubiprostone منشط لقنوات الكلوريد يستخدم في IBS-C.

★ **Tell the students:** Other IBS-C agent: prucalopride (5-HT4 agonist, prokinetic).

قل للطلاب: دواء آخر: prucalopride (محفز 5-HT4).

Group BJ

BJ·1 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Which medication group can contribute to functional acute gastric dilatation by suppressing GI motility?

A. Bile acid binders
× Bile acid binders do not suppress motility. لا تثبط الحركة.

B. Oral iron only
× Oral iron: not listed. غير مذكور.

C. High-dose anticholinergics ✓ Book p. 68 ↗

D. Prokinetic agents
× Prokinetics increase motility. تزيد الحركة.

Why: High-dose anticholinergics (also opioids, anesthetics) suppress motility and can cause functional AGD.
لماذا: مضادات الكولين بجرعات عالية (وكذلك الأفيونات والتخدير) تثبط الحركة وتسبب AGD وظيفياً.

★ **Tell the students:** Drug-induced atony is reversible; stop the drug and decompress.
قل للطلاب: الشلل الدوائي عكوس؛ أوقف الدواء وفرّغ المعدة.

BJ·2 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Which of the following is NOT true concerning bismuth subsalicylate?

A. It should be given for more than 4 months ✓ Book p. 48 ↗

B. It is used in quadruple anti-H. pylori regimen
× True: part of quadruple therapy. صحيح: ضمن الرباعي.

C. It acts as mucosal protective agent
× True: mucosal protective. صحيح: حامٍ للمخاطية.

D. It may cause neurotoxicity & encephalopathy
× True: encephalopathy/neurotoxicity. صحيح: سمية عصبية.

Why: Bismuth should NOT be given for more than 2 months (neurotoxicity risk); 'more than 4 months' is the false statement.
لماذا: البزموث لا يعطى أكثر من شهرين (خطر سمية عصبية)؛ و'أكثر من 4 أشهر' هي الجملة الخاطئة.

★ **Tell the students:** Not restarted within 1 year; black stools and tongue are harmless.
قل للطلاب: لا يعاد خلال سنة؛ وسواد البراز واللسان غير ضار.

BJ·3 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Which of the following phases of acid secretion is most likely affected by H2 receptor antagonists?

A. Gastrin stimulated acid secretion
× Gastrin/ACh pathways are only partly inhibited.
مسار الغاسترين والأسيتيل كولين يتثبط جزئياً.

B. Both nocturnal and meal stimulated phases
× Not both; the meal phase is less affected.
ليس الاثنين؛ الوجبات أقل تأثراً.

C. Nocturnal acid secretion ✓ Book p. 46 ↗

D. Meal stimulated acid secretion
× Meal phase is the weaker effect.
الأضعف تأثراً.

Why: H2 blockers potently inhibit nocturnal/fasting acid secretion and much less the meal-stimulated phase.
لماذا: حاصرات H2 تثبط بقوة الإفراز الليلي/الصيامي وأقل الإفراز بعد الوجبات.

★ **Tell the students:** Complete block of histamine-mediated, partial block of ACh/gastrin-mediated secretion; tolerance develops.
قل للطلاب: حجب كامل لمسار الهستامين وجزئي للغاسترين والأسيتيل كولين؛ ويحدث تحمل.

Group BK

BK·1 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Which of the following is NOT correct concerning misoprostol?

A. Contraindicated during pregnancy as it may induce abortion
× True: contraindicated in pregnancy (abortion).
صحيح: ممنوع في الحمل.

B. A cytoprotective agent used in all types of peptic ulcer if H2 blockers fail ✓ Book p. 48 ↗

C. Diarrhea and abdominal cramps are its most common adverse effects
× True: diarrhea and colic are the commonest ADRs.
صحيح: الإسهال والمغص أشهر آثاره.

D. PGE analog that increases mucosal blood flow, mucus & bicarbonate
× True: PGE1 analog, raises mucus, bicarbonate and blood flow.
صحيح: نظير PGE1 يزيد المخاط والبيكربونات والتروية.

Why: Misoprostol is selective for NSAID-induced gastric ulcer when PPIs fail; it is not used for all peptic ulcers.
لماذا: Misoprostol مخصص لقرحة المعدة الناتجة عن NSAIDs عند فشل الـPPI وليس لكل القرح.

★ **Tell the students:** Used prophylactically with NSAIDs or corticosteroids.
قل للطلاب: يستخدم وقائياً مع NSAIDs أو الكورتيزون.

BK·2 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Which of the following medications is mostly included in anti-H. pylori regimens?

A. Doxycycline
× Tetracycline, not doxycycline, is used in quadruple therapy.
يستخدم tetracycline وليس doxycycline.

B. Misoprostol
× Misoprostol is a protectant.
حامٍ للمخاطية.

C. Clarithromycin ✓ Book p. 48 ↗

D. Bismuth subcitrate
× Bismuth is only in quadruple regimens.
فقط في الرباعي.

Why: Clarithromycin (with amoxicillin and a PPI) is the antibiotic most used in H. pylori regimens.
لماذا: Clarithromycin (مع amoxicillin وPPI) أكثر مضاد يدخل في برامج H. pylori.

★ **Tell the students:** Clarithromycin resistance: replace with bismuth quadruple therapy.
قل للطلاب: مع المقاومة: العلاج الرباعي بالبزموث.

BK·3 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Which of the following is a key advantage of P-CABs compared to proton pump inhibitors (PPIs)?

A. P-CABs require less frequent dosing than PPIs
× Both are dosed once daily.
الاثنان مرة يومياً.

B. P-CABs are more effective antisecretory drugs regardless of food intake ✓ Book p. 46 ↗

C. P-CABs are less effective in treating Helicobacter pylori infections
× They are MORE effective in eradication regimens.
أكثر فعالية في الاستئصال.

D. P-CABs have a longer duration of action
× PPIs act long (irreversible binding).
الـPPI تأثيره أطول (ارتباط غير عكوس).

Why: P-CABs are acid-stable and need no acid activation, so they work regardless of food intake.
لماذا: P-CABs ثابتة في الحمض ولا تحتاج تنشيطاً فتعمل بغض النظر عن الأكل.

★ **Tell the students:** Maximum suppression in 1 day (PPI: 3-5 days); T1/2 6-9 h.
قل للطلاب: أقصى تثبيط في يوم (PPI: 3-5 أيام).

BK·4 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Which of the following mucosal protective agents has anti-H. pylori effect?

A. Sucralfate
× Sucralfate has no anti-H. pylori effect. لا تأثير مضاد لـ H. pylori.

B. Misoprostol
× Misoprostol: PGE1 analog only. نظير PGE1 فقط.

C. Bismuth subcitrate ✓ Book p. 48 ↗

D. Bicarbonate
× Bicarbonate is an antacid. مضاد حموضة.

Why: Colloidal bismuth (subcitrate) is the protective agent with an anti-H. pylori effect.

لماذا: البزموث الغرواني (subcitrate) هو الحامي الذي له تأثير مضاد لـ H. pylori.

★ **Tell the students:** Bismuth: chelates proteins in ulcer base; black stools; not more than 2 months.

قل للطلاب: يرتبط ببروتينات القرحة؛ سواد البراز؛ لا يزيد عن شهرين.

Group BL

BL·1 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Systemic alkalosis could be an adverse effect of long term use of which of the following antacids?

A. Aluminum hydroxide
× Aluminum hydroxide: constipation. إمساك.

B. Magnesium oxide
× MgO is a non-absorbed local antacid. غير ممتص.

C. Magnesium hydroxide
× Mg hydroxide: diarrhea. إسهال.

D. Sodium bicarbonate ✓ Book p. 49 ↗

Why: Sodium bicarbonate is absorbed and causes systemic alkalosis with long-term use.

لماذا: بيكربونات الصوديوم تُمتص وتسبب قلاء جهازياً عند الاستخدام الطويل.

★ **Tell the students:** NaHCO3 also: rebound acidity, sodium load, alkaline urine/stones. CaCO3: milk-alkali syndrome.

قل للطلاب: أيضاً: ارتداد الحموضة وحمل صوديوم وحصوات. CaCO3: milk-alkali.

BL·2 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

A 45-year-old woman with peptic ulcer disease is treated with misoprostol. Which of the following is a common adverse effect?

A. Hypertension
× Not typical.
غير نمطي.

B. Diarrhea ✓ Book p. 48 ↗

C. Constipation
× Constipation is typical of aluminum antacids.
للألومنيوم.

D. Headache
× Not the commonest.
ليس الأشهر.

Why: Diarrhea and colic are the commonest adverse effects of misoprostol.
لماذا: الإسهال والمغص أشهر الآثار الجانبية لـ misoprostol.

★ **Tell the students:** Contraindicated in pregnancy: uterine contractions and vaginal bleeding.
قل للطلاب: ممنوع في الحمل: انقباضات ونزف مهبلي.

BL·3 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Which of the following is correct concerning pharmacokinetics of H2-receptor antagonists?

A. 50% of the dose of ranitidine is destroyed by gastric acidity
× The acid-decomposition figure applies to famotidine, not ranitidine.
نسبة التحلل بالحمض تخص famotidine.

B. They are contraindicated in mild renal dysfunction due to renal excretion
× Only dose reduction is needed if creatinine clearance falls.
يلزم فقط تقليل الجرعة.

C. 50% of the dose of famotidine is lost by hepatic first pass effect
× Famotidine is decomposed by acid; ranitidine has first-pass loss.
famotidine يتحلل بالحمض؛ وranitidine فقد كبدي.

D. Most of them have low to moderate bioavailability except nizatidine ✓ Book p. 46 ↗

Why: Most H2 blockers have only ~50% bioavailability; nizatidine is the exception (~100%, no first-pass).
لماذا: معظم حاصرات H2 توافرها الحيوي ~50% عدا nizatidine (~100% بلا first-pass).

★ **Tell the students:** Cimetidine: enzyme inhibitor, antiandrogenic; the others avoid these effects.
قل للطلاب: Cimetidine مثبط إنزيمي ومضاد أندروجين؛ والبقية لا.

BL·4 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

A 50-year-old woman with GERD is prescribed famotidine. Which is the primary mechanism of action of famotidine?

A. Blocking histamine H2 receptors in the stomach ✓ Book p. 46 ↗

B. Inhibition of the proton pump
× Proton pump inhibition = PPIs. .PPIs

C. Blocking muscarinic receptors on parietal cells
× Muscarinic block = anticholinergics. مضادات الكولين.

D. Neutralization of gastric acid
× Neutralization = antacids. مضادات الحموضة.

Why: Famotidine competitively and reversibly blocks histamine H2 receptors on parietal cells.

لماذا: Famotidine يحجب مستقبلات H2 للهستامين على الخلايا الجدارية بشكل تنافسي عكوس.

★ **Tell the students:** Famotidine is the most potent and has no enzyme-inhibiting effect.

قل للطلاب: الأقوى ولا يثبط الإنزيمات.

BL·5 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Which of the following antacids has the most rapid and potent effect in hyperacidity?

A. Sodium bicarbonate ✓ Book p. 49 ↗

B. Magnesium oxide
× MgO: slower. أبطأ.

C. Aluminum hydroxide
× Al hydroxide: slow, constipating. بطيء ويسبب إمساكاً.

D. Magnesium hydroxide
× Mg hydroxide: potent but not as rapid. قوي لكن أبطأ.

Why: Sodium bicarbonate has the most rapid onset and is potent (pH above 7).

لماذا: بيكربونات الصوديوم الأسرع بداية والأقوى (pH فوق 7).

★ **Tell the students:** Fast but not preferred: systemic alkalosis and sodium load.

قل للطلاب: سريع لكنه غير مفضل: قلاء وحمل صوديوم.

Group BM

BM·1 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

Which of the following antacids is mostly preferred for hyperacidity?

A. Both Mg and Al salts together ✓ Book p. 49 ↗

B. Sodium bicarbonate
× Systemic alkalosis and Na load.
قلاء جهازي وحمل صوديوم.

C. Aluminum hydroxide
× Alone it constipates.
وحده يسبب إمساكاً.

D. Magnesium hydroxide
× Alone it causes diarrhea.
وحده يسبب إسهالاً.

Why: Al and Mg salts (alone or combined, e.g. magaldrate) are preferred: no CO2 release or systemic alkalosis.

لماذا: أملاح Al وMg (منفردة أو معاً مثل magaldrate) هي المفضلة: بلا CO2 أو قلاء جهازي.

★ **Tell the students:** Combination cancels the bowel effects: Al constipates, Mg gives diarrhea.

قل للطلاب: الجمع يلغي التأثيرات: Al إمساك وMg إسهال.

BM·2 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 03: The Stomach and Duodenum

NSAIDs cause acute gastritis through:

A. Decreased PGE2 ✓ Book p. 35 ↗

B. Increased histamine
× Histamine is not the mechanism.
ليس الهستامين.

C. Decreased CCK
× CCK is not involved.
غير مشارك.

D. Increased PGE2
× Prostaglandins fall, not rise.
تنقص.

Why: NSAIDs inhibit COX, decreasing PGE2, so mucus and bicarbonate protection falls.

لماذا: الـNSAIDs تثبط COX فينقص PGE2 وتقل حماية المخاط والبيكربونات.

★ **Tell the students:** Acute gastritis causes: NSAIDs, alcohol, stress, early H. pylori.

قل للطلاب: أسباب: NSAIDs وكحول وإجهاد وH. pylori.

BM·3 PHARMACOLOGY فارماكولوجي Digestive System — Chapter 02: The Esophagus

Which drug increases the risk of GERD?

A. High-protein diet
× Fat, chocolate, coffee, alcohol worsen it; protein does not.
الدهون والكافيين تزيده؛ البروتين لا.

B. Baclofen
× Baclofen reduces transient LES relaxations.
يقلل ارتخاء LES العابر.

C. β-blockers
× Beta-blockers do not typically cause reflux.
لا تسبب الارتجاع عادة.

D. Calcium channel blockers ✓ Book p. 14 ↗

Why: Calcium-channel blockers lower LES pressure and promote reflux.

لماذا: حاصرات الكالسيوم تخفض ضغط LES وتزيد الارتجاع.

★ **Tell the students:** LES-relaxing drugs: nitrates, CCB, anticholinergics, theophylline, progesterone.

قل للطلاب: أدوية مرخية للـLES: نيترات وCCB ومضادات كولين وtheophylline.

Teacher notes

Questions to double-check · أسئلة تحتاج مراجعتك

J·1 A male patient presents with severe epigastric distension and vomiting and has a histor...
Question flaw: expected answer is acute gastric dilatation (not listed).
السؤال به عيب: الإجابة المتوقعة acute gastric dilatation وهي غير مذكورة.

N·2 Obese 44-yo severe heartburn — best tool for hiatus hernia:
Option D says 'Anti-Trendelenburg (head-down)': contradictory, since head-down is Trendelenburg. D is still the only barium option.
الاختيار D مكتوب 'Anti-Trendelenburg (رأس لأسفل)' وهذا متناقض (رأس لأسفل = Trendelenburg). ومع ذلك D هو الباريوم الوحيد.

AD·3 The best investigation for uncomplicated diverticular disease is:
Answer follows the exam key. The book (p.9) says barium enema has largely been replaced by colonoscopy and CT colonography.
الإجابة حسب مفتاح الامتحان. والكتاب (ص 9) يذكر أن حقنة الباريوم حل محلها المنظار وCT colonography.

AK·3 Radiotherapy as 1st-line gives results similar to surgery in:
General oncology: no matching page in this book.
أورام عامة: لا توجد صفحة مطابقة في الكتاب.

AP·3 All are useful to prevent PUD recurrence EXCEPT:
Lifestyle advice is not printed in the book; nearest page is the PUD page.
نصائح الحياة غير مكتوبة؛ أقرب صفحة هي القرحة.

AT·2 Regarding corrosive esophageal injury:
Answer follows the exam key; practice has changed.
الإجابة حسب مفتاح الامتحان؛ والممارسة الحديثة اختلفت.

AU·3 Which structure does NOT pass through one of the three main diaphragmatic openings?
Anatomy question; not covered in this book.
سؤال تشريح؛ غير موجود في الكتاب.

AV·1 5-year survival after esophagectomy for EARLY esophageal cancer:
Not stated in the book; lower confidence.
غير مذكور بالكتاب؛ ثقة أقل.

AV·4 In achalasia of the cardia, all are true EXCEPT:
Option A is loosely worded; the book (p.24) also says 'equal dysphagia' for motility disorders in general.
الاختيار A مصاغ بشكل فضفاض؛ والكتاب (24) يذكر 'عسر بلع متساوٍ' في اضطرابات الحركة عامة.

AW·3 Boerhaave's syndrome refers to:
Not in this book.
غير موجود في الكتاب.

AY·2 Which test is usually NORMAL in bacterial overgrowth syndrome?
D-xylose is not printed in the book; chosen by elimination.
D-xylose غير مذكور بالكتاب؛ اخترنا بالاستبعاد.

BA·2 In a retirement-home outbreak, what is the primary reservoir of infection?
Question gives no organism; the best answer by setting is human.
السؤال لا يذكر الميكروب؛ الأنسب حسب السياق: الإنسان.

BB·1 Which organism would NOT benefit from prompt antibiotic treatment?
Both botulism and perfringens food poisoning are toxin-mediated; the book states botulism needs no antibiotics.
كلاهما سمّي المنشأ؛ والكتاب ينص على أن الوشيقية لا تعالج بالمضادات.

Not in the supplied book · غير موجودة في الكتاب

Hernia questions come from the separate **Hernia book** (not supplied); a few others are general anatomy/oncology. They are answered from standard references. · أسئلة الفتق من كتاب الفتق المنفصل (غير مرفق) وبعضها تشريح/أورام عامة.

B·1 What is a Richter's hernia?

E·1 What is the most important factor for incisional hernia recurrence?

AE·2 Regarding inguinal hernia and inferior epigastric vessels:

AE·3 The coverings of a true umbilical hernia sac in adults include:

AF·2 The following is NOT a cause of para-umbilical hernia recurrence:

AG·1 Causes of hernia recurrence include all EXCEPT:

AG·5 The structure most commonly responsible for strangulation of a femoral hernia is:

AI·1 'Bubonocele' refers to an inguinal hernia that:

AI·4 A hernia is called Littre's hernia when its contents include:

AJ·3 Sliding hernia does NOT occur in:

Nearest page only · أقرب صفحة فقط

The exact sentence is not printed in the book (or sits inside a figure), so the link opens the closest topic page without a highlight. · الجملة نفسها غير مكتوبة بالكتاب فالرابط يفتح أقرب صفحة بدون تظليل.

B·3 What is the most common indication for surgery in peptic ulcer disease?

J·2 What is the characteristic histology of celiac disease?

L·3 The D-xylose test is used for:

P·1 Which part of the small bowel is responsible for vitamin B12 absorption?

Q·3 D-xylose absorption test is used for:

AA·4 Which is NOT a barium enema finding in Ulcerative Colitis?

AB·1 Infantile intussusception usually occurs during:

AD·1 The usual presentation of uncomplicated diverticulosis is:

AD·2 The commonest cause of intestinal obstruction in children is:

AD·4 A finding consistent with Ulcerative Colitis on colonoscopy is:

AG·2 One of the following sites NEVER develops diverticulosis:

AI·3 The lower third of the esophagus is supplied by:

AM·1 Features of ulcerative colitis include:

AM·4 Morphological features of Crohn's disease include:

AP·3 All are useful to prevent PUD recurrence EXCEPT:

AP·4 Regarding Barrett's esophagus:

AQ·3 The following features characterize UC EXCEPT:

AT·2 Regarding corrosive esophageal injury:

AX·4 Regarding gastric peptic ulcer classification (Modified Johnson), which is FALSE?

BA·2 In a retirement-home outbreak, what is the primary reservoir of infection?

BF·1 A 12-year-old Egyptian boy with a history of swimming in irrigation canals presents wit...

BF·4 The diagnostic stage of Ascaris lumbricoides is:

What was changed versus the original file · ما تغيّر عن الملف الأصلي

Questions are regrouped (3-4-5 cycle) into 4 parts: Internal Medicine & Surgery, Microbiology, Parasitology, Pharmacology. 27 drug-focused questions were collected into the Pharmacology part (the Giardia question moved to Parasitology). Book images were compressed to ~150 dpi to keep the file phone-friendly; text stays sharp.

أعدنا تجميع الأسئلة (دورة 3-4-5) في 4 أجزاء. نُقل 27 سؤالاً دوائياً إلى جزء الفارماكولوجي، ونُقل سؤال Giardia إلى الباراسيتولوجي. ضُغطت صور الكتاب إلى ~150 dpi لتخفيف الحجم والنص يبقى واضحاً.

YMnotes

YMNOTES MEDICAL

Digestive System

Hepatogastroenterology

Hepatogastrointestinal Surgery

Most efficient personal study steps
1) Hear body while following
2) Read once with AI
3) Highlight

Do video after feeling a boundary is there.

Version 3 STUDY EDITION

Clear structure. Connected understanding.

Hepato- = Liver → الكبد

Gastro = Stomach → المعدة

Entero = intestine → الأمعاء

-ology = study → دراسة

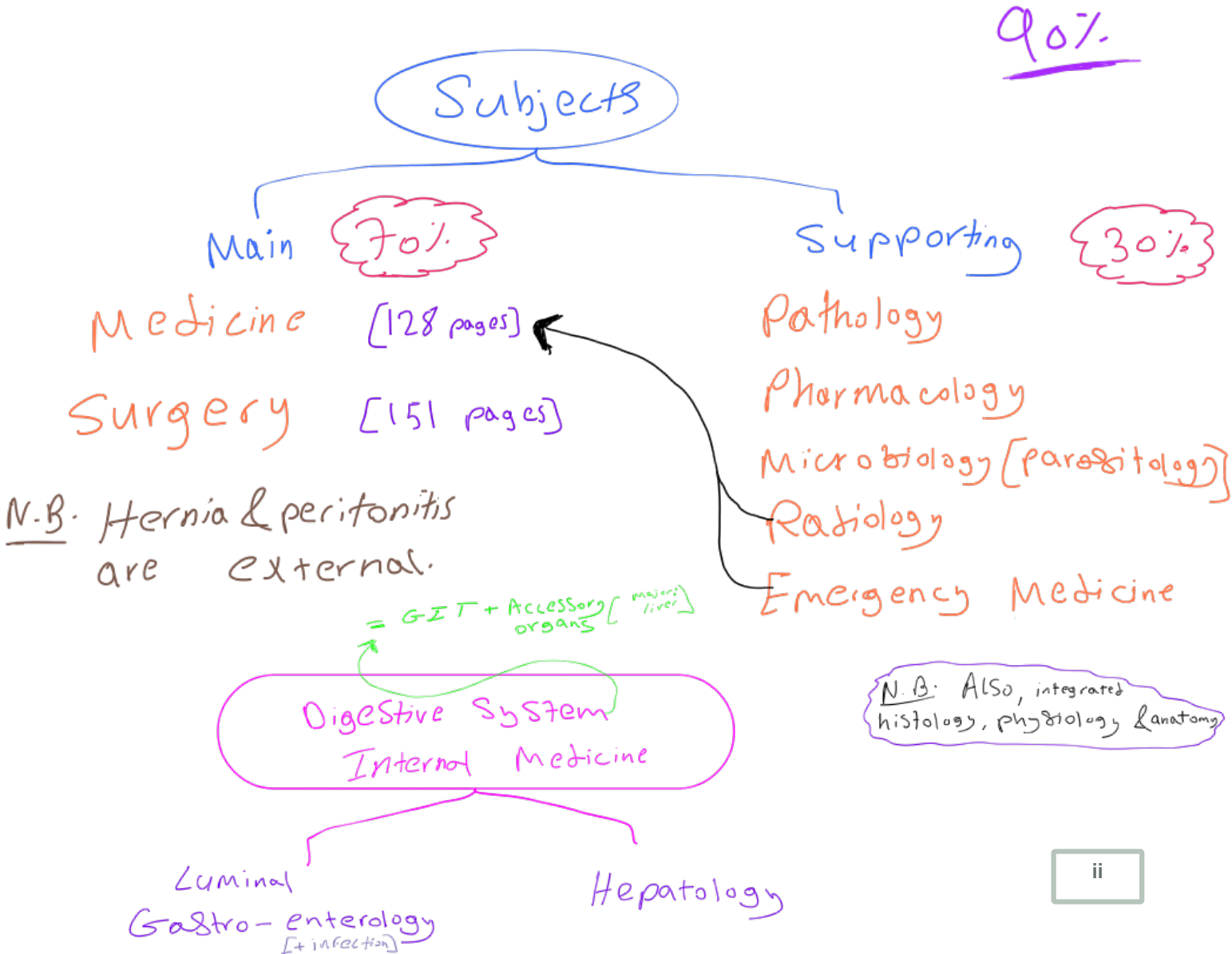
YMnotes

YMNOTES MEDICAL

باطنة Hepatogastroenterology and جراحة Hepatogastrointestinal Surgery

= (Green)

Pale-green highlights identify corrected wording. A fine green side rule identifies restored text grouping. The end-of-book corrections register records these changes and source restorations.



DIGESTIVE SYSTEM / VERSION 3

Interactive Contents

Tap a chapter to jump directly to it.

01	GI Symptoms and Investigations Internal Medicine · Radiology / Investigation	10	1
02	The Esophagus Internal Medicine · Surgery	19	10
03	The Stomach and Duodenum Internal Medicine · Pharmacology · Surgery	31	33
04	Pathology of GIT Tumours Pathology · Surgery	16	70
05	Diarrhea and Malabsorption Internal Medicine	16	76
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14	Polyposis of Colon Surgery · Pathology	6	189
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Chapters 01–20 · Page numbers refer to the book body.

Interactive Contents

Tap a chapter to jump directly to it.

21	Autoimmune Hepatitis Internal Medicine	274
22	Liver Cirrhosis Internal Medicine · Pathology	280
23	Portal Hypertension Internal Medicine · Surgery	308
24	Hydatid Cyst Parasitology · Surgery	318
25	Pyogenic Liver Abscess Surgery · Internal Medicine · Parasitology	324
26	Hepatocellular Carcinoma Surgery · Pathology	332
27	Ascites Internal Medicine	337
28	Renal Impairment in Cirrhosis Internal Medicine	345
29	Jaundice Internal Medicine	349
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Shortest chapter

405

Chapters 21–40 · Page numbers refer to the book body.

Average Pages Length 405 ÷ 40 = 10.125 pages

old version calculation

Gastrointestinal Clinical Features and Investigations

Notes at the end of this PDF (version 2)

CHAPTER 01

Introduction

01

Learning objectives

The student will

1) Know the various symptoms of gastrointestinal diseases

2) Understand the basic mechanism underlying each of the symptoms of gastrointestinal tract

3) Understand the basic diagnostic tools for gastrointestinal diseases

Symptoms and signs of gastrointestinal diseases

Alarm symptoms- Red flag symptoms:

- Dysphagia.
- Weight loss.
- Persistant vomiting.
- Anorexia.
- Haematemesis or melaena.

NB: Patients with these features have a higher possibility of significant gastrointestinal pathology and should be investigated properly.

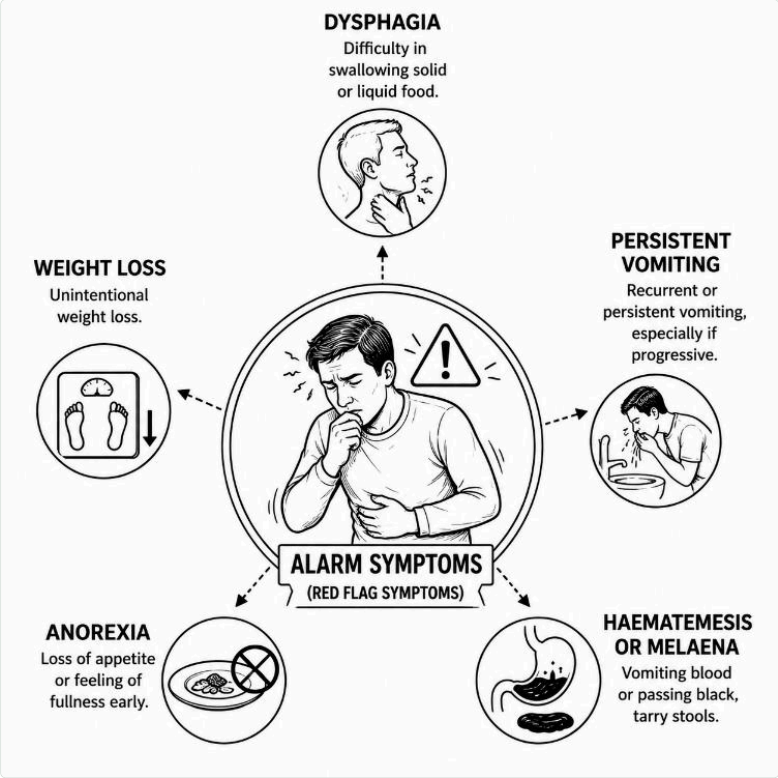


Figure 1.1 — Alarm symptoms (red flag symptoms)

- Stomatitis : It is an inflammation in the mouth from any cause.

- Angular stomatitis : is inflammation of the corners of the mouth .
- Dysphagia : Difficulty of swallowing.
- Odynophagia : Painful swallowing.
- Dyspepsia : Describe a number of upper abdominal symptoms such as heartburn, acidity, pain or discomfort, nausea, wind, fullness or belching.
- Vomiting : causes of vomiting
- Any Gastrointestinal disease.
- Infections: viral and bacterial.
- CNS diseases: Increase Intracranial pressure , migraine.
- Metabolic causes : Chronic renal failure , diabetic Ketoacidosis.
- Drugs : Digoxin.
- Reflex : Myocardial infarction.
- Psychogenic.
- Pregnancy.
- Alcohol.

Diarrhea & Constipation

Diarrhea

- Common complaint not always due to organic disease.
  - Defined as a change in bowel habit with:
  - Increase Frequency.
  - Increase Amount.
  - Decrease Consistency (loose/watery stool).
  - More than 200 g stool/day with low dietary fiber considered diarrhea.
  - Clinical:
  - Large-volume watery stool → usually organic cause
  - Bloody diarrhea → usually colonic pathology
  - Fever + acute diarrhea for several days → usually infective
  - Single episode → dietary change or psychogenic

Constipation

- Diagnosis is difficult because daily bowel opening is NOT required for normal bowel habit.
  - Constipation = ≥2 of the following:
  - <3 stools/week.
  - Hard stools in >¼ of defecations.

- Straining in >¼.
- Incomplete evacuation in >¼.
- Manual maneuvers (digital evacuation/pelvic floor support) in >¼.

Abdominal Pain:

Mainly due to stretching of smooth muscle or organ capsules. Severe acute pain consider acute abdomen ; may also be referred (pneumonia) or metabolic (DKA).

Assess:

intensity, character, duration, frequency, aggravating/relieving factors & associated symptoms.

- Common Causes:
- Epigastric: functional dyspepsia, Peptic ulcer ; heartburn ,GERD.
- Right hypochondrium: biliary disease; hepatitis/cardiac failure.
- Biliary pain: severe, intermittent, lasts hours, recurrent.
- Left iliac fossa: diverticulitis; chronic pain often functional.
- Right iliac fossa: appendicitis/ileocaecal disease.
- Female : consider gynaecological causes.
- Abdominal wall pain: localized tenderness, persistent pain not relieved by tensing muscles.

Anorexia & Weight Loss:

- Anorexia: reduced appetite; common in systemic /psychiatric disease. May occur with cancer but is usually late and nonspecific.
- Weight loss usually due to reduced intake; in malabsorption mainly due to anorexia.
- Weight loss despite normal or increased intake → hyperthyroidism / catabolic states.

GIT Bleeding

- Hematemesis: vomiting blood.
- Melena: black, tarry stool → usually bleeding proximal to right colon; requires >50 mL blood.
- Massive upper GI bleed: fresh blood per rectum + shock.
- Hematochezia: bright-red/maroon rectal bleeding → usually lower GI source.

Investigations of gastrointestinal diseases

Investigations are selected according to the clinical presentation and include laboratory tests, stool studies, endoscopy, and imaging.

1. Laboratory & Stool Investigations:

- Routine blood tests: CBC, renal function, liver function.
- Stool examination: useful for detecting blood, fat (steatorrhoea), parasites and infection.
- Fecal occult blood test (FOBT): detects non-visible GI bleeding.
- Fecal Immunochemical test (FIT) : more accurate less false-positive results.
- Fecal calprotectin elevated in inflammatory bowel disease (IBD).
- Tumor markers:
- CEA → colorectal cancer
- CA 19-9 → pancreaticobiliary cancer
- AFP → hepatocellular carcinoma

Tumor markers are mainly supportive and are not diagnostic alone.

2. Endoscopy:

Upper GI Endoscopy (EGD)

Investigation of choice for most upper GI disorders.

- Allows direct visualization, biopsy and treatment.
- Diagnoses: reflux esophagitis, gastritis, ulcers and malignancy.
- Therapeutic : treatment of upper GI bleeding and obstruction.
- Relative contraindications :recent myocardial infarction and severe COPD.

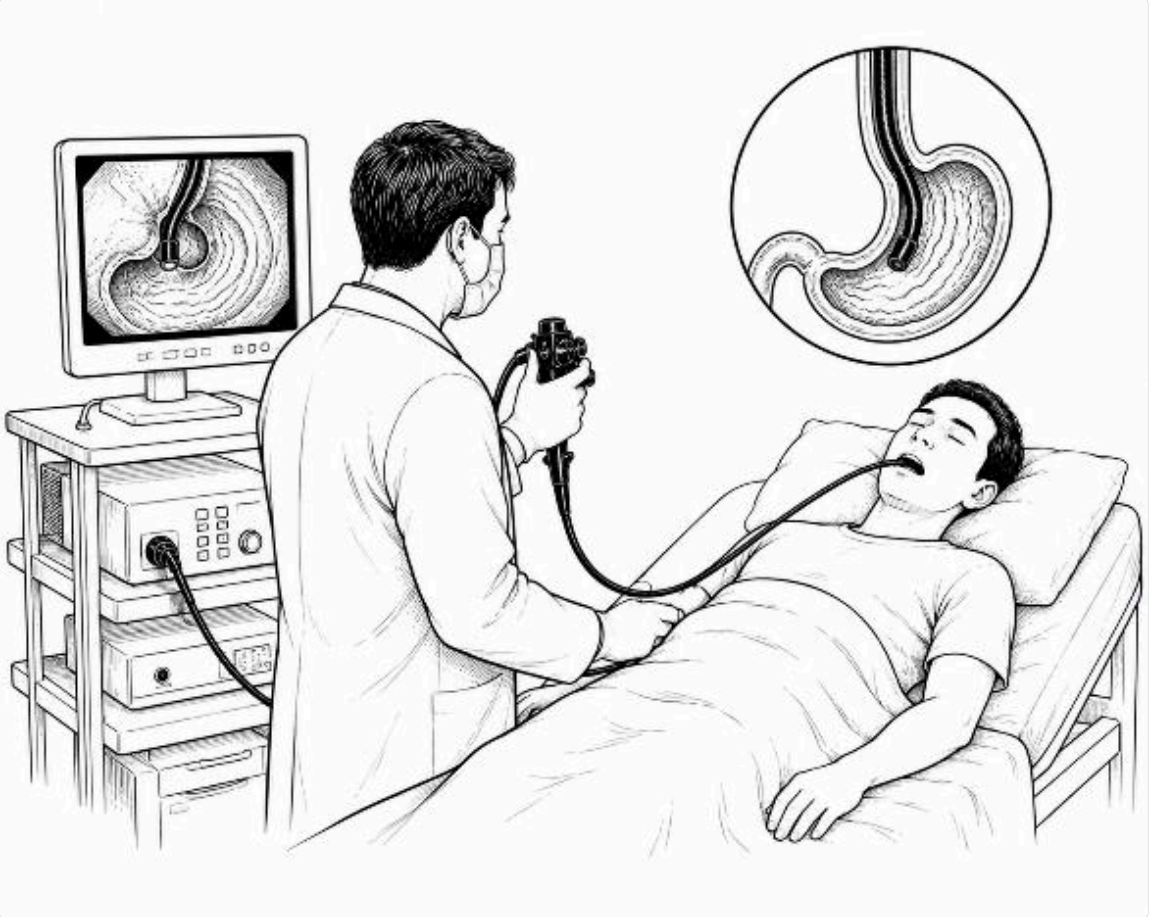


Figure 1.2 — Upper gastrointestinal endoscopy

Colonoscopy

- Visualizes the entire colon and terminal ileum.
  - Allows biopsy, polypectomy, stricture dilatation and stenting.
  - Important findings include cancer, polyps and diverticular disease.
  - Sigmoidoscopy (Rigid / Flexible) : can be use in rectal bleeding.

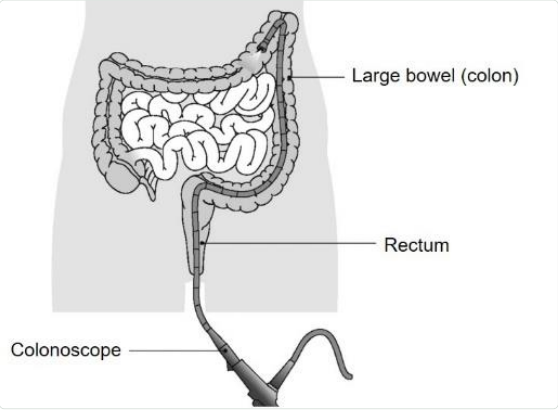


Figure 1.3 — Colonoscopy

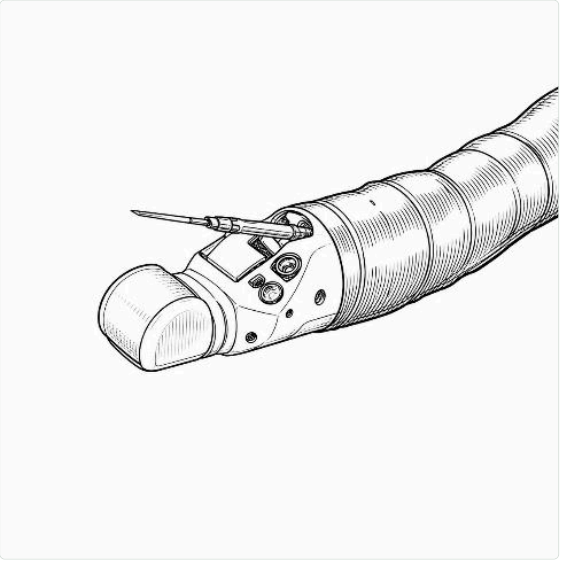


Figure 1.4 — Colonoscope

Small-Bowel Endoscopy

- Balloon enteroscopy: allows examination and treatment of the small bowel.
  - Capsule endoscopy: useful for obscure GI bleeding and small- bowel lesions/IBD after negative gastroscopy and colonoscopy.

Endoscopic Ultrasound (EUS)

Combines endoscopy with ultrasound.

- Useful for GI wall lesions, staging of esophageal/gastric cancer, and detection/biopsy of pancreatic tumors and cysts.

3. Imaging

Plain X-ray

Mainly useful in acute abdomen:

- Bowel obstruction
- Perforation (free subdiaphragmatic air)
- Toxic megacolon
- May show calcification from gallstones or chronic pancreatitis.

Figure 1.5 — Plain abdominal X-ray

Ultrasound

- No radiation; first-line for many abdominal conditions.
  - Useful for ascites, masses, biliary disease, cholecystitis, appendicitis and inflammatory conditions.
  - Can demonstrate bowel dilatation and wall thickening.
  - Allows image-guided biopsy and drainage.

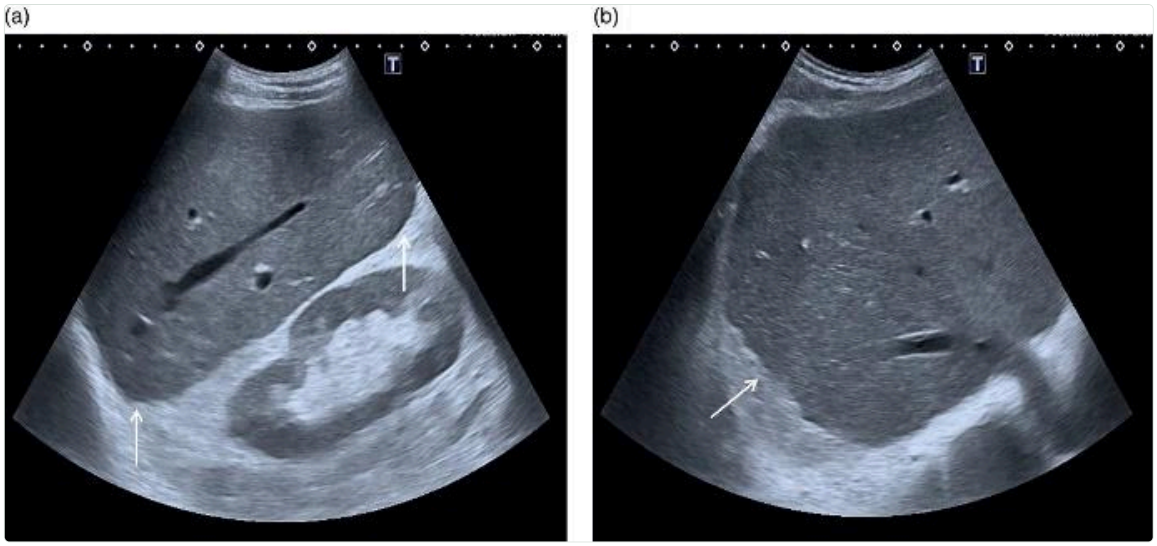


Figure 1.6 — Abdominal ultrasound

CT

A major first-line investigation in acute abdomen.

- Detects perforation, obstruction and its cause.
- Demonstrates appendicitis, diverticulitis, abscesses and Crohn's disease/complications.
- Important for cancer staging and image-guided biopsy.
- Assess renal function before IV contrast.

MRI

Advantages: no ionizing radiation and excellent soft-tissue contrast.

- Particularly useful for rectal cancer, perianal fistulae/abscesses, small-bowel disease, and hepatobiliary/pancreatic disorders.

MR Enterography: small-bowel disease, particularly useful in IBD.

PET

Uses FDG metabolic activity to assess:

- Cancer staging
- Metastatic disease

- Useful in oesophageal, gastric and colorectal cancers.

4. Contrast Studies

- Barium swallow: mainly for dysphagia and esophageal disorders.
- Double-contrast barium meal: evaluates stomach and duodenum; largely replaced by gastroscopy.
- Barium enema: evaluates the colon but has largely been replaced by colonoscopy and CT colonography.
- If perforation is suspected → use water-soluble contrast (e.g. Gastrografin), not barium.

A separate video to be made for extras inshaAllah

المري

CHAPTER 02

THE ESOPHAGUS

23 Pages
internal Medicine & Surgery
"Esophageal Disorders"

- Introduction
  - * Anatomy & Physiology
  - * Symptoms & Signs
  - * Investigations
- → GERD
- → Benign Esophageal Stricture
- → Esophageal Infections
- → Hiatus Hernia
- → Achalasia of the Esophagus
- → Esophageal cancer

General surgical management x2

02

x3 uni lectures

- Digestive 02 | Esophageal Introduction & GERD — Pages 11-16
- Digestive 03 | Hiatal Hernia & Achalasia — Pages 17-19, 21-23
- Digestive 04 | Surgery of the Esophagus — Pages 20-21, 24-32

سبحان الله

STRUCTURE AND PHYSIOLOGY:

- The esophagus is **20 cm** long that connects the pharynx to the stomach.
- The esophagus is lined by **stratified squamous epithelium**.
- The esophagus is separated from the pharynx by the upper esophageal sphincter (UES).

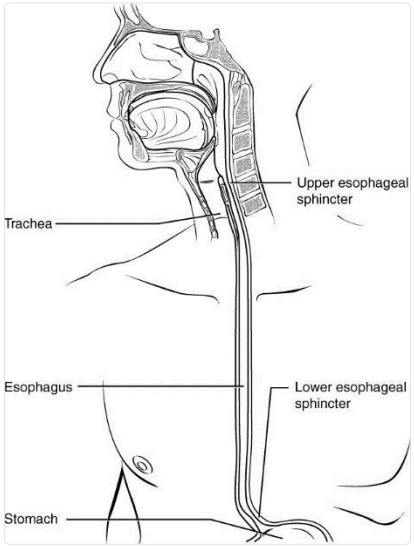


Figure 2.1 — Structure of the esophagus

The lower esophageal sphincter (LES) consists of a 2-4 cm in the distal end of the esophagus that responsible for the prevention of gastric reflux.

SYMPTOMS OF ESOPHAGEAL DISORDERS

1.Dysphagia:

- Difficulty of swallowing.
- The characteristics of the progression of dysphagia can be helpful Progression over a few weeks suggests a **malignant stricture.** The slow onset of dysphagia for solids and liquids at same time suggest **motility disorder** (e.g. Achalasia).

CAUSES **Causes of dysphagia**

- **Disease of mouth and tongue e.g.** tonsillitis .
- **Neuromuscular disorders e.g.** Bulbar palsy , myasthenia gravis .
- **Esophageal motility disorders e.g.** Achalasia ,Scleroderma ,diffuse esophageal spasm, Diabetes mellitus , Chagas' disease.
- **Extrinsic pressure e.g.** Mediastinal glands , goiter ,enlarged left atrium
- **Intrinsic lesion e.g.** Foreign body, Stricture: benign or malignant - carcinoma, esophageal web ,Pharyngeal pouch.

Causes of Dysphagia

Mechanical

A- Painful condition of the Mouth & Pharynx

B- Esophageal
<obstructive>

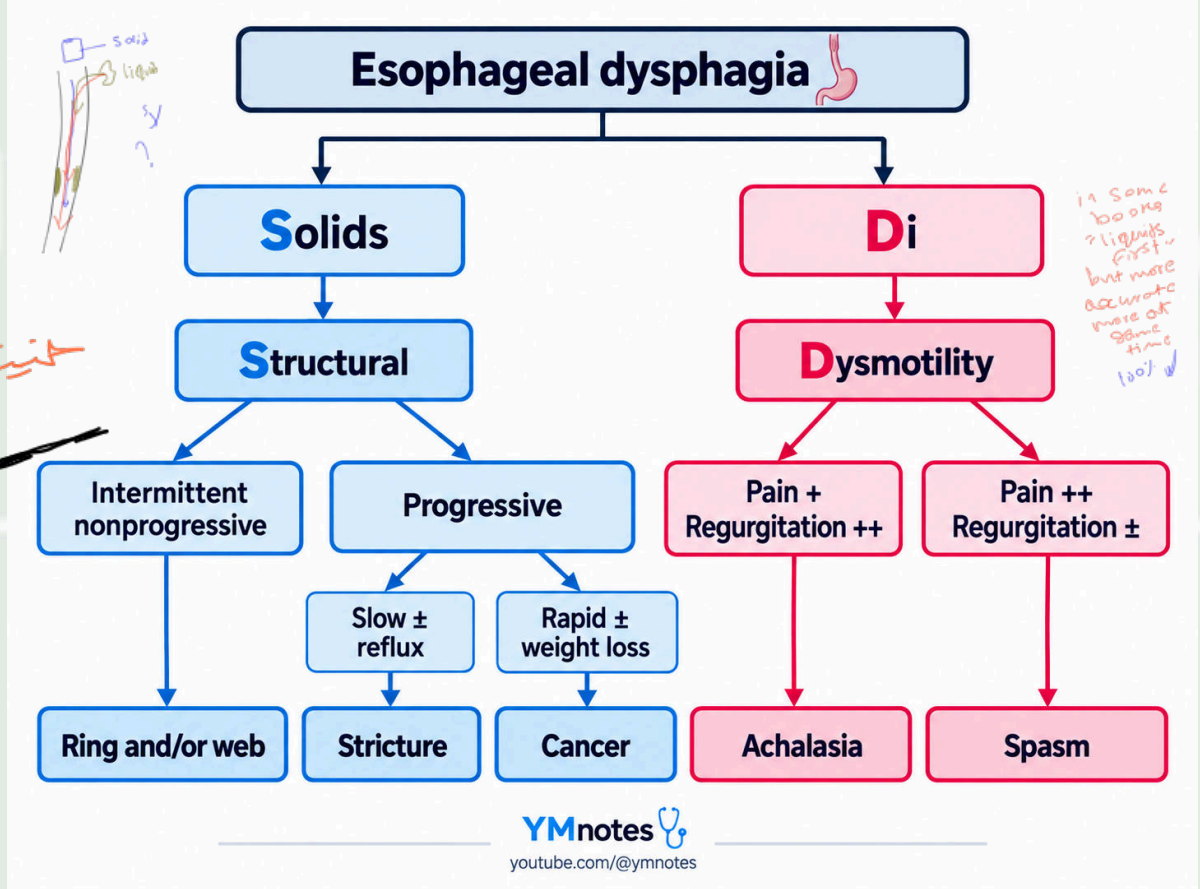
intrinsic

extrinsic

Motor

A- Neurological

B- Functional



2.Odynophagia:

is pain during swallowing and is suggestive of esophagitis.

3.Substernal discomfort, heartburn:

This is a common symptom of reflux of gastric contents into the esophagus.

4.Regurgitation:

is the effortless reflux of esophageal contents into the mouth and pharynx.

SIGNS OF ESOPHAGEAL DISORDERS

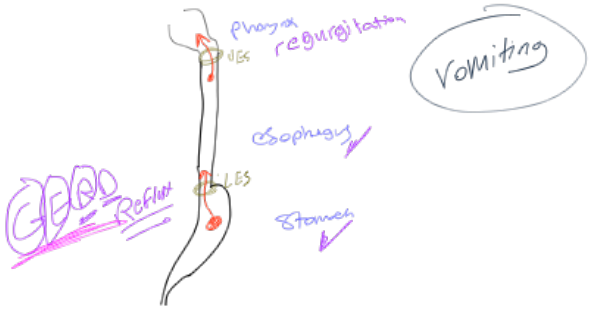
The main sign of esophageal disease is **weight loss** due to reduced food intake.

INVESTIGATIONS FOR ESOPHAGEAL DISORDERS

1. Esophagoscopy.
2. Barium swallow and meal.
3. Manometry

It is used to assess esophageal motor activity.

Figure 2.2 — Esophageal radiographic investigation



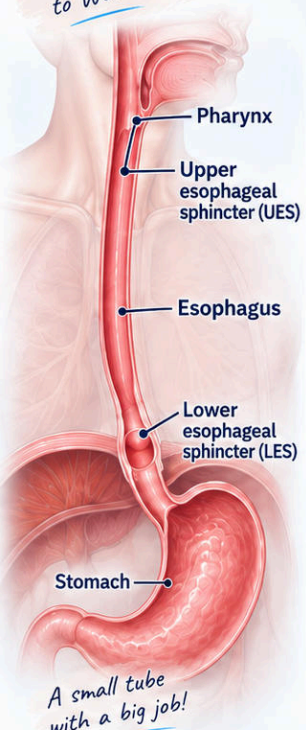
Esophagus Introduction

Summary Before GERD

From Swallowing to Wellness

Same Concepts Stronger Care

YMnotes
youtube.com/@ymnotes
Medicine Made Simple



A small tube with a big job!

1 Normal structure & function

- Muscular tube connecting pharynx to stomach
- Stratified squamous lining
- **UES** and **LES** regulate passage
- **Peristalsis** propels food
- **LES** prevents reflux

2 Key symptoms

- **Dysphagia** = difficulty swallowing
- **Odynophagia** = painful swallowing
- **Heartburn** = retrosternal burning
- **Regurgitation** = effortless return of contents

3 Dysphagia pattern clues

- Solids first → then liquids = **Mechanical obstruction**
- Solids + liquids from onset = **Motility disorder**
- Liquids worse than solids = **Motility disorder**

5 Core investigations

- **Endoscopy** = direct view ± biopsy — Visualize mucosa, take biopsy if needed
- **Barium swallow** = anatomy / passage — Shows structural abnormalities & transit
- **Manometry** = pressure & motility — Assesses esophageal motility and sphincters
- **24-h pH monitoring** = acid reflux — Detects and quantifies pathologic reflux

YMnotes Medical

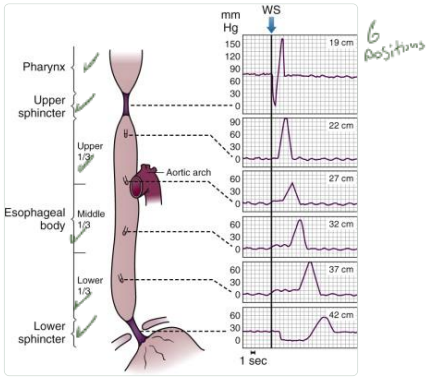


Figure 2.3 — Esophageal manometry

4. pH monitoring

- 24-hour ambulatory monitoring uses a pH-sensitive probe positioned in the lower esophagus

introduction

- **GASTROESOPHOGEAL REFLUX DISEAESE-**

مرض الارتجاع المعدي المريئي

(GERD)

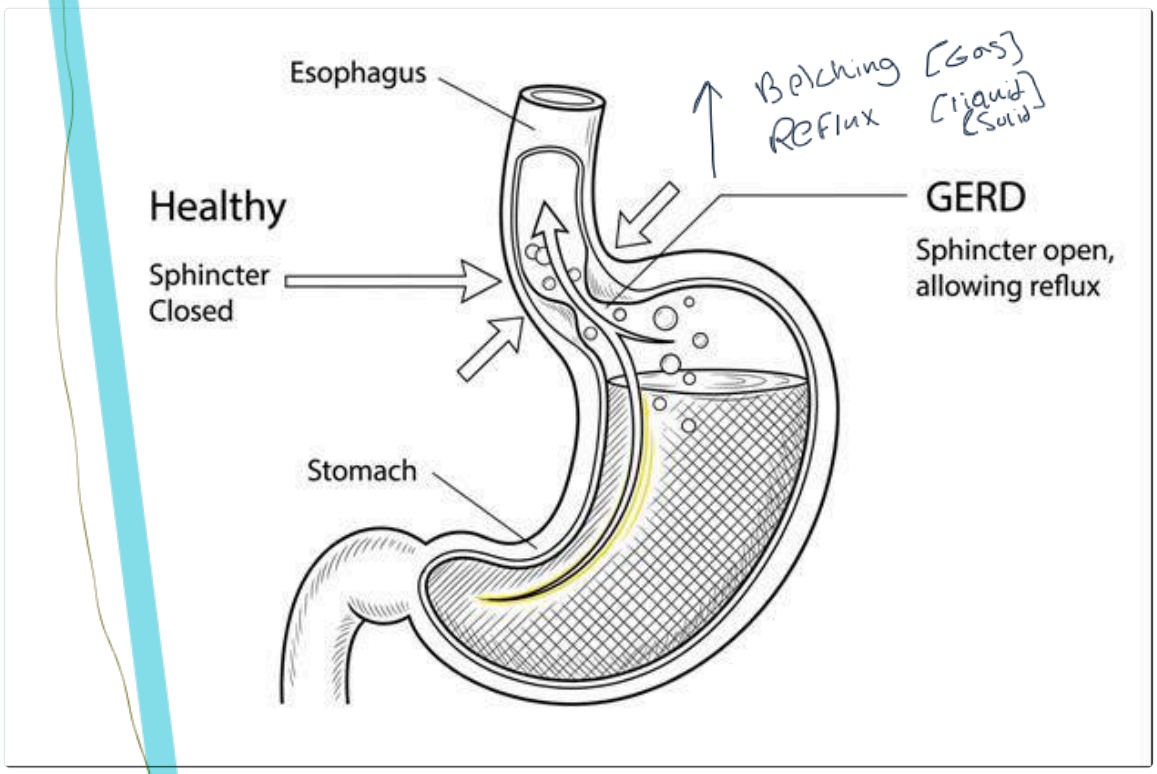


Figure 2.4 — Gastroesophageal reflux disease

PATHOPHYSIOLOGY

- The reflux occur when the anti-reflux mechanisms fail, allowing acidic gastric contents to make prolonged contact with the lower esophageal mucosa.

eg: LES, diaphram, angle of His, ...etc

- Decrease tone of lower esophageal sphincter (LES).

RISK FACTORS:

1) **Increase intra-abdominal pressure** : Pregnancy or obesity

2)Diet:

Fat, chocolate, coffee, alcohol and large meals.

3)Drugs:

antimuscarinic, calcium-channel blockers, nitrates

4) Cigarette smoking

5) Hiatus hernia (described below)

CLINICAL FEATURES

- **Heartburn is the major feature.**

  The burning is aggravated by lying down and relieved by oral antacids.

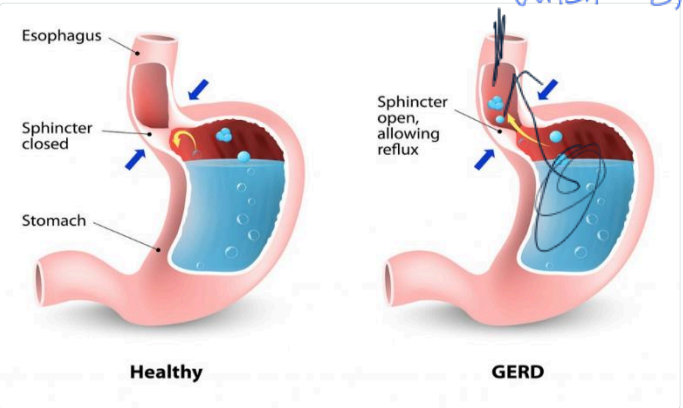


Figure 2.5 — Gastroesophageal reflux

- **Regurgitation** of food and acid into the mouth.
- **Aspiration pneumonia**

DIAGNOSIS AND INVESTIGATIONS

- The clinical diagnosis can usually be made without investigation.
- If there is **alarm signs** **e.g.** weight loss, no response to PPI the following investigations:

1) 24-hour pH monitoring.

2) Upper endoscopy.

3) Manometry.

COMPARE **Esophageal reflux and cardiac ischemia**

- **Reflux pain:** burning, worse on lying down , don't radiates to the arms ,worse with **hot drinks or alcohol and is relieved by antacids**
- **Cardiac ischemic pain:** Gripping or crushing radiates to neck or left arm ,worse **with exercise accompanied by dyspnea**

TREATMENT

- **Simple measures :**
  - ~~Simple antacids.~~ (Repeated)
  - Loss of weight and raising the head of the bed at night.
  - Reduction in caffeine consumption
  - Stop smoking and alcohol intake.
- **Drugs:**
  - **Proton pump inhibitors** e.g. omeprazole, reduce gastric acid secretion and are the drugs of choice for mild cases.
  - **H2 Blockers** e.g. ranitidine.
  - **Antacids.**
  - **Dopamine antagonist** as metoclopramide and domperidone are increase peristalsis and speed emptying .
- **Endoluminal gastroplication:**
  - This recent technique could be performed by endoscopy:
- Endoscopic full-thickness plication **(GERDX)**:
  - A endoscopic device has been designed to inhibit gastroesophageal reflux by placing a transmural plication near the gastroesophageal junction under direct endoscopic visualization.
  - Anti-reflux mucosal ablation

(ARMA):

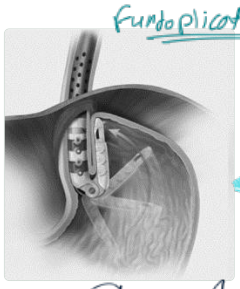


Figure 2.6 — Anti-reflux mucosal ablation

- Mucosal ablation around the gastroesophageal junction to cause a healing process to re-form the cardiac opening
- **Surgery:** management of sliding hiatus hernia (described below)

COMPLICATIONS

- **Peptic stricture:**
  - Since the introduction of PPIs peptic strictures have become less common.
- **Barrett's esophagus :**
  - The normal esophageal squamous epithelium is replaced by metaplastic columnar mucosa.

- Premalignant: esophageal adenocarcinoma.
- Barrett's is commonest in middle-aged obese men.
- Diagnosis: endoscopy, Chromo-endoscopy.
- Treatment : Radiofrequency ablation (RFA).

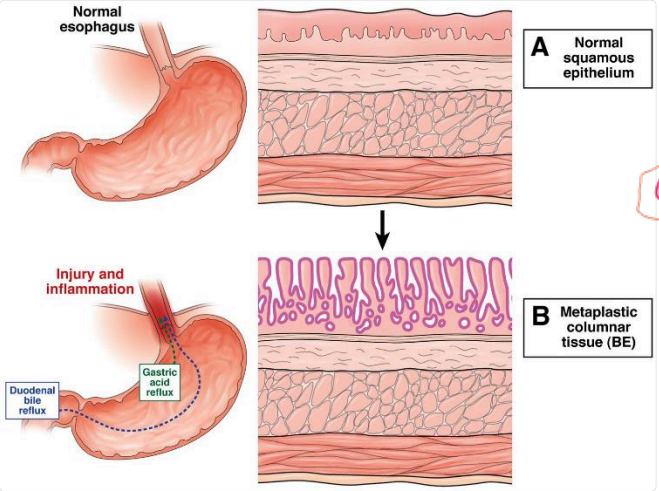


Figure 2.7 — Esophageal epithelial changes

BENIGN ESOPHAGEAL STRICTURE

- Peptic stricture secondary to reflux is the most common cause of benign strictures
- They also occur after the ingestion of corrosives, after radiotherapy, after sclerosis of varices.
- Dysphagia is usually managed by endoscopic dilatation.
- Surgery is sometimes required.

ESOPHAGEAL INFECTIONS

- Infection is a cause of painful swallowing and is seen in immunosuppressed (e.g. on chemotherapy) and AIDS.
- Infection can occur with:
  - Candida - Herpes simplex - Cytomegalovirus - tuberculosis.
- In candidiasis, the characterize by white plaques.
- Most patients on large doses of immunosuppressive agents are treated prophylactically for candida with nystatin or amphotericin.

HIATUS HERNIA- فتق الحجاب الحاجز

- A hiatus hernia describes the **protrusion of an organ** from the abdominal cavity into the chest through the esophageal hiatus.
- This is **typically the stomach** that herniates.
- Extremely common disease.

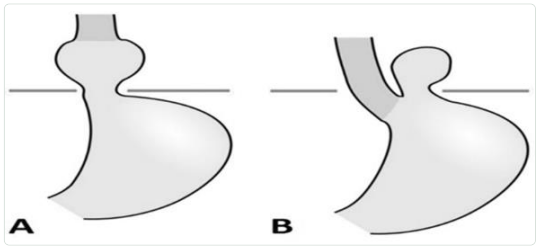
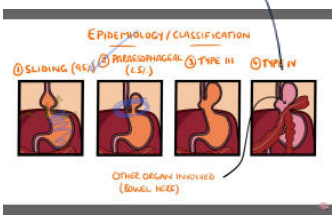


Figure 2.8 — Hiatus hernia

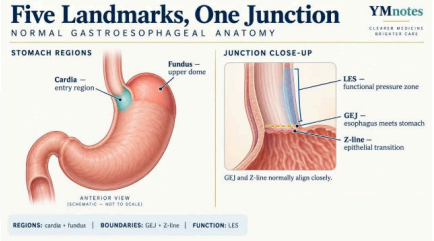


Classification:

3 subtypes:

Sliding hiatus hernia (85 %)	Rolling or Para- esophageal hernia (10%):	A mixed type hiatus hernia
The gastro-oesophageal junction (G O J), the abdominal part of the oesophagus, and the cardia of the stomach 'slide' upwards through the diaphragmatic hiatus into the thorax. (fig. A)	An upward movement of the gastric fundus through esophageal hiatus occurs to lie alongside a normally placed GOJ. This is a true hernia with a peritoneal sac. (fig. B)	which has both a rolling and sliding Components

- RISK FACTORS:
  - **Age:**
    1. Is an important risk factor, due to:
       - A combination of age- related loss of diaphragmatic tone, increasing intra-abdominal pressures (e.g. repetitive coughing), and widening of diaphragmatic hiatus.
  - **Increased intra-abdominal pressure:**
    2. Pregnancy, obesity, and ascites due to upwards displacement of the abdominal viscera.
- CLINICAL FEATURES OF SLIDING H.H.:
  - The majority of hiatus hernias are **asymptomatic**.



- Gastro-esophageal reflux symptoms:
  1) Marked burning epigastric pain

epigastric pain:

- Made worse by lying flat and on leaning forwards.
  - Patient experiences relief on raising head of bed.

2) **Vomiting** – regurgitation.

3) **Bleeding and anemia** secondary to esophageal ulceration leading either melena, occult blood in stool or hematemesis in severe cases.

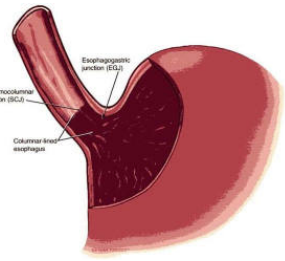
4) **Dysphagia** (esophageal stricture on long-standing cases)

5) Sometimes **Weight loss** in case of long duration vomiting and dyspepsia.

- Epigastric tenderness in case of associated gastritis
- INVESTIGATIONS:

1.Upper endoscopy :

- Is the **standard investigation** for esophageal disorders showing upward displacement of the Gastro-Esophageal Junction also (Z-line).
  - During inspiration the GOJ is opened and closed in expiration
  - It shows evidence of reflux as hyperemia, fissuring, ulceration or stricture of the lower esophagus
  - Biopsy may be taken from suspicious areas (areas of Barett's)



2. Plain x-ray chest:

fundic air may be evident in the chest – detects chest complications as aspiration pneumonia.

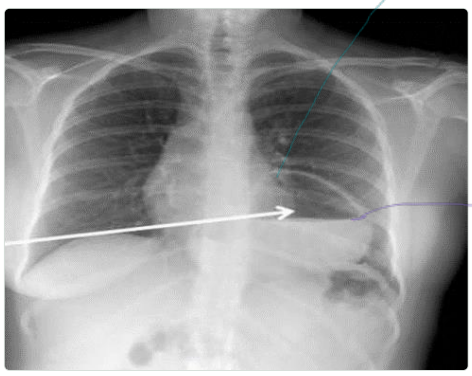


Figure 2.9 — CXR: stomach in the chest

3. Barium swallow:

- Evidence of complication as stricture esophagus.

4. Barium meal in Trendlenberg's position:

- Reflux of the barium to the esophagus in GERD
  - Stomach above the diaphragm in big Hiatus Hernia.

5. CT abdomen and chest

6. pH test:

shows increased acid level in the esophagus.(diagnostic for reflux)

Barium meal: sliding H.H

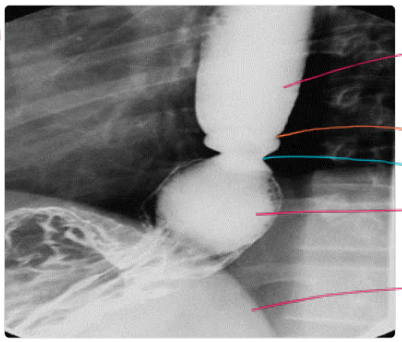


Figure 2.10 — Barium meal: sliding hiatus hernia

- TREATMENT:
  - **Life-style changes as:**
    - Losing weight.
    - Decreasing the size of meals.
    - Avoiding certain acidic foods, such as tomato sauce and citrus
    - Limiting fried and fatty foods, foods containing caffeine (including chocolate), • Eating meals at least three to four hours before lying down, and avoiding bedtime snacks • Keeping head higher than the rest of body when lying on the back.
    - Stop smoking
    - Avoid a tight belt or tight clothing.
  - **Medications:**

- **Antacids** that neutralize stomach acid.
- **H2 blockers**.
- **Proton pump inhibitors**.

PPI stronger acid blockers than H_2 blockers and allow for damaged esophageal tissue to heal.

Surgical management of GERD and hiatal hernia

Aim to restore the intra-abdominal length of the esophagus, repair the diaphragmatic defect, and recreate a functional lower esophageal sphincter (LES) barrier.

Indications for Surgery

Operative repair is recommended for all symptomatic patients

1. Failure of medical treatment for type I
2. due to the risk of volvulus, strangulation, and ischemia. In type II

Principles of Surgical Management

Laparoscopic repair is the gold standard approach:

1. Hernia Sac Reduction & Dissection

- ~~Complete excision of the mediastinal sac~~
- Reduce the herniated stomach into the abdomen and
- completely mobilize and excise the hernia sac from the posterior mediastinum.

2. Hiatal Closure (Cruroplasty)

- Reapproximating diaphragmatic crura

• Close the enlarged diaphragmatic hiatus posteriorly using non-absorbable sutures. Synthetic or biological mesh reinforcement may be used in large defects to reduce recurrence.

3. Anti-Reflux Procedure (Fundoplication)

Wrap the gastric fundus around the distal esophagus to recreate the high-pressure zone:

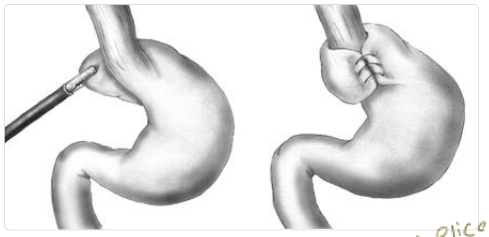


Figure 2.11 — Fundoplication

- Nissen Fundoplication: 360° total wrap (gold standard)
- Toupet Fundoplication: 270° posterior partial wrap
- (preferred if underlying esophageal dysmotility exists to minimize post-operative dysphagia).

Postoperative Complications :

1) **Acute Dysphagia:** tight wrap, or tight crural closure

Treatment: Soft diet for 2–4 weeks. If persistent beyond 6 weeks, perform endoscopic balloon dilation.

2) **Gas-Bloat Syndrome**: Inability to belch/vomit due to hypercompetent 360° wrap treatment: Dietary modification, simethicone, and reassurance; resolves spontaneously in most patients.

3) **Early Hernia Recurrence / Wrap Slippage**

Treatment: revision surgery

4) **Recurrent GERD :** Breakdown or migration of the wrap into the chest

Treatment: revision surgery.

ACHALASIA OF ESOPHAGUS-

- It is a primary esophageal motility disorder characterized by failure of LES relaxation and loss of peristalsis in the distal esophagus.

PATHOPHYSIOLOGY:

- **Vagal degeneration**
- LES contraction and relaxation are regulated through vagus nerve
- Degeneration of ganglion cells in myenteric plexus concerned with the inhibitory neurons → increased tone in LES → severe spasm.

CAUSES OF ACHALASIA:

I- Primary Achalasia:

Etiology unknown, Hereditary, degenerative, autoimmune

II- Secondary Achalasia:

(e.g. secondary to Chaga's disease)

CLINICAL PRESENTATIONS:

- Dysphagia for liquids more than solids most significant symptom. (effect of weight). • Difficulty Belching.
- Regurgitation (especially when lying flat at night)
- Heartburn and Chest Pain in case of esophagitis
- Weight loss is minimal and late
- Hiccups

COMPLICATIONS ASSOCIATED WITH ACHALASIA:

Pneumonic Aspiration

Esophageal cancer

INVESTIGATIONS:

I- Radiographic Evaluation Is the esophagus dilated?

II- Manometric Evaluation Does the LES fail to relax?

III- Endoscopic Evaluation Is there a mechanical obstruction or cancer?

I- Radiographic Evaluation

1- Chest x-ray:

- Markedly dilated Esophagus with widening of mediastinum (From continous Food Retention)
- Absent gastric air bubble: swallowed air does not reach the fundus unlike normally

2- Barium swallows: diagnostic tool

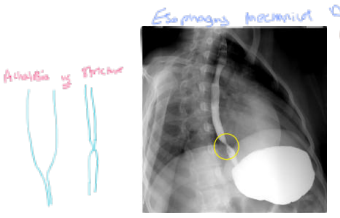
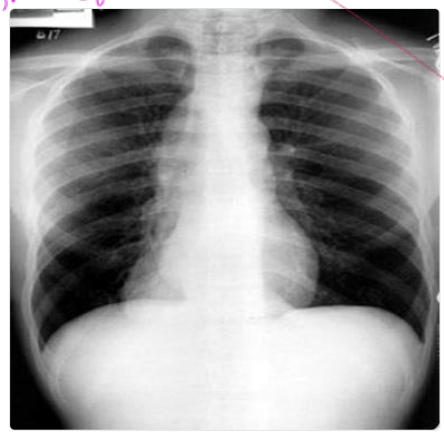


Figure 2.12 — Smooth narrowing of the distal esophagus at the level of the lower esophageal sphincter with the characteristic "bird beak" appearance. or rat-tail appearance

- The proximal esophagus is markedly dilated



Absent gastric air bubble

Figure 2.13 — Radiographic findings in achalasia

II- Esophageal manometry: confirm the diagnosis

= pressure when a person is not swallowing normally

- Elevated resting LES pressure (spasm).
  - Incomplete LES relaxation.
  - Weak peristalsis of esophageal wall (aperistalsis).

	Achalasia
Resting: High	High
During Swallowing low	High

III- Endoscopic Evaluation:

- To exclude cancer of the gastro-esophageal junction.
  - Detects spasm of the LES.

MANAGEMENT:

Pharmacological agents:

Patients who are unable to tolerate more invasive treatment modalities.

1- Calcium channel blockers and nitrates decrease LES pressure.

2- Endoscopic Intra-sphincteric Injection of Botulinum Toxin:

- blocks release of acetylcholine at the level of the LES → relaxation of the sphincter

Endoscopic technics

A- Pneumatic dilatation:

- Is the most effective non-surgical treatment option for patients with achalasia.
- The balloon is inflated at the level of the gastro-esophageal junction to dilate the muscle fibers while leaving the mucosa intact.

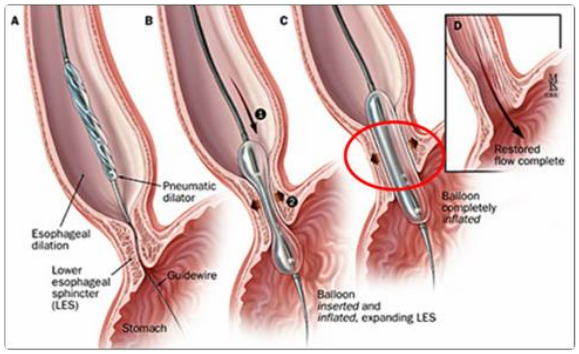
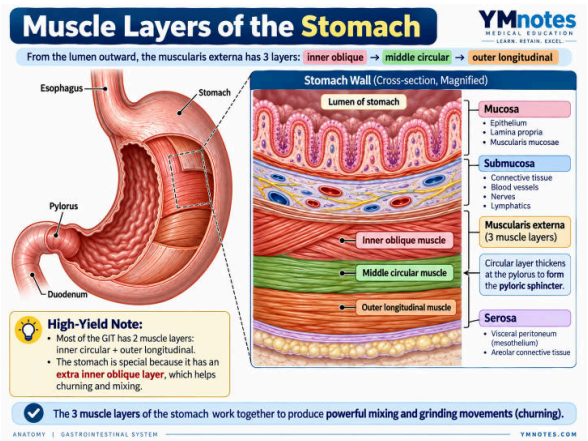


Figure 2.14 — Pneumatic dilatation



B- The Per-Oral Endoscopic Myotomy, or POEM, :

- is a minimally invasive endoscopic procedure for the treatment of achalasia wherein the inner circular muscle layer of the lower esophageal sphincter is divided through a submucosal tunnel.

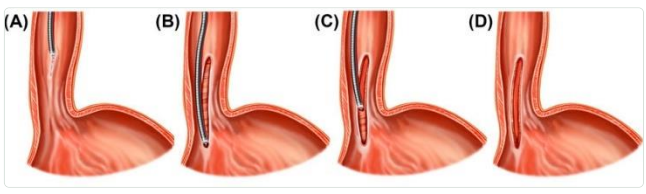


Figure 2.15 — Per-oral endoscopic myotomy

Surgical: Laparoscopic Heller's Cardio-Myotomy

- Weakens LES by cutting muscle fibers through performing an anterior myotomy across the LES.

Combined with Partial Fundoplication: Always paired with a partial fundoplication to prevent postoperative reflux.

c) Refractory / End-Stage (Megaesophagus): Esophagectomy with gastric conduit reconstruction.

Esophageal dysphagia

Primarily caused by

1. Mechanical obstructions
2. Neuromuscular motility disorders.

I- Mechanical Causes

- These conditions physically narrow the esophageal lumen, leading predominantly to dysphagia for solids before progressing to liquids:

1) **Esophageal Carcinoma**: Progressive dysphagia starting with solids and rapidly advancing to liquids, accompanied by weight loss (Squamous Cell Carcinoma or Adenocarcinoma).

2) **Peptic Stricture**: Benign narrowing of the distal oesophagus resulting from chronic gastroesophageal reflux disease (GERD) and scarring.

3) **Esophageal Webs**: constrict the lumen and are often associated with Plummer-Vinson syndrome.

4) **Extrinsic Compression**: Pressure on the esophagus from outside structures, such as a mediastinal tumor, aortic aneurysm, or retrosternal goiter.

5) **Foreign Body Impaction**: Sudden onset of complete dysphagia

6) **Esophageal Diverticula**: Outpouchings like Zenker's (pharyngoesophageal), traction, or epiphrenic diverticula that trap food and cause regurgitation.

7) **Corrosive Stricture**: Severe scarring following ingestion of caustic agents or prior mediastinal radiotherapy.

II- Neuromuscular / Motility Causes

These functional disorders impair peristalsis or lower esophageal sphincter (LES) relaxation, often causing equal dysphagia for both solids and liquids from the onset:

1) **Achalasia**: Failure of the LES to relax alongside absent peristalsis due to degeneration of the myenteric plexus.

2) Distal Oesophageal Spasm (DES): Uncoordinated, high-amplitude contractions causing intermittent dysphagia and chest pain.

3) Hypercontractile Oesophagus: Severe, high-pressure muscular contractions during peristalsis.

4) Systemic Sclerosis (Scleroderma): Atrophy of smooth muscle leading to an incompetent LES and severe aperistalsis.

A definitive diagnosis for esophageal dysphagia relies on a clinical history to endoscopy, radiology, and functional/motility testing.

KEY HIGHLIGHTS diagnostic strategy to identify the most common mechanical and neuromuscular causes:

Step 1: Detailed Clinical History

Step 2: Upper Endoscopy (EGD) — The Initial Diagnostic Procedure

Step 3: Barium swallow

Step 4: Oesophageal Manometry: Gold Standard for Motility

Step 5: Advanced Imaging (CT / Endoscopic Ultrasound EUS)

Used when extrinsic compression or advanced local pathology is suspected

Surgical management of esophageal disorders

Surgical management of esophageal dysphagia depends entirely on whether the primary etiology is a mechanical obstruction, a motility disorder, or an external pathology.

1. Mechanical Obstructive Lesions: Esophageal Carcinoma

2. Benign Strictures (Peptic / Corrosive / Radiation)

•**First-line / Endoscopic** : Serial endoscopic balloon dilation or wire-guided bougie dilation.

• **Refractory Strictures:** Endoscopic steroid injection into the stricture, temporary stent placement, or formal esophageal resection and reconstruction (colon interposition or gastric pullup) for complete caustic destruction.

3. Esophageal Webs & Rings

• Primary Treatment: Endoscopic rupture using balloon dilation or targeted electrosurgical incision (using specialized knife).

4. Distal Esophageal Spasm & Hypercontractile Esophagus

• **Endoscopic (POEM):** Extended Peroral Endoscopic Myotomy targeting the long spastic segment in the mid-todistal esophagus when medical therapy fails.

5. Zenker's Diverticulum (Pharyngoesophageal)

- Endoscopic Approach (Transoral): Diverticulectomy using stapling, laser, or electrocautery (flexible or rigid endoscopic cricopharyngeal myotomy).
- open Surgical Approach: Cricopharyngeal myotomy combined with diverticulectomy (excision of the pouch)

6. Extrinsic Compression & Emergency Impactions

• Extrinsic Tumors / Goiters: Surgical resection of the compressive mediastinal mass or thyroidectomy for retrosternal goiter.

• Foreign Body: Emergent endoscopic removal using snare, basket, or grasping forceps; surgical exploration is reserved solely for cases complicated by perforation.

Cancer Esophagus

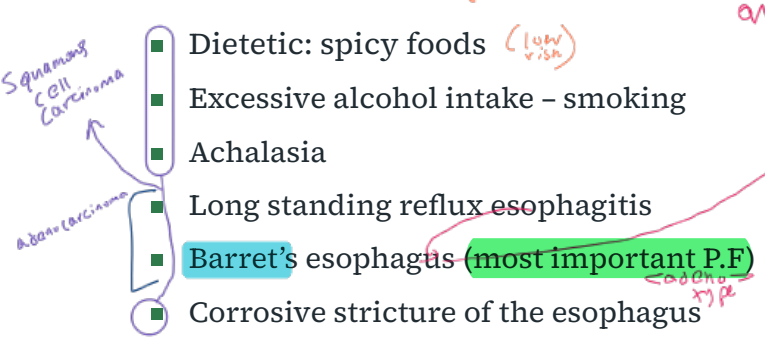
Pathology:

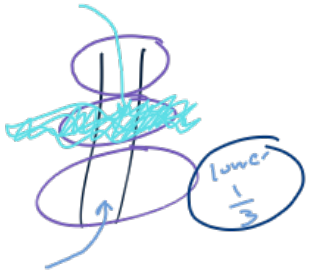
Age: old age > 50 years

Sex: males > females

Predisposing factors: (PF)

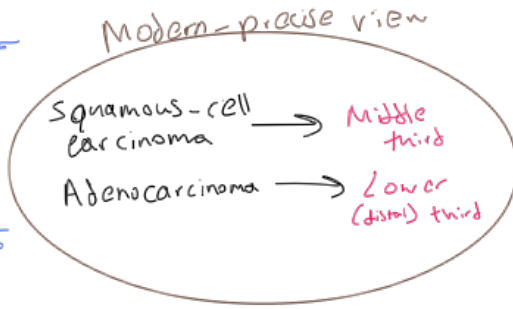
- Dietetic: spicy foods
- Excessive alcohol intake – smoking
- Achalasia
- Long standing reflux esophagitis
- Barret's esophagus (most important P.F)
- Corrosive stricture of the esophagus





Macroscopic picture:

Site: 1- Middle 1/3 → 50 % (COMMONEST)
2- Lower 1/3 → 33 %
3- Upper 1/3 → 10: 15 %



Gross appearance:

1. Malignant ulcer: raised evereted edge, hard base and necrotic floor
2. Stricture: hard, long and irregular
3. Fungating mass: cauliflower, bleeding and necrotic surface

Microscopic picture:

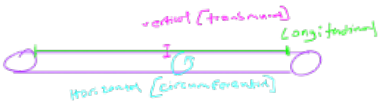
1. Squamous cell carcinoma:

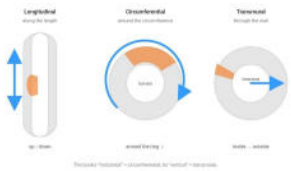
- The most common

  Grade: may be well , moderate or undifferentiated carcinoma o

2. Adenocarcinoma:

- On top of Barett's esophagus
- Local spread from gastric carcinoma at the cardia to lower esophagus.





Spread:

A. local :

- the tumor spread in vertical and horizontal (circumferentially)
- Vertical spread from the mucosa to submucosa to muscle layer to adventitia then penetrate to the surrounding structures as trachea, bronchus, pleura, diaphragm, stomach,....

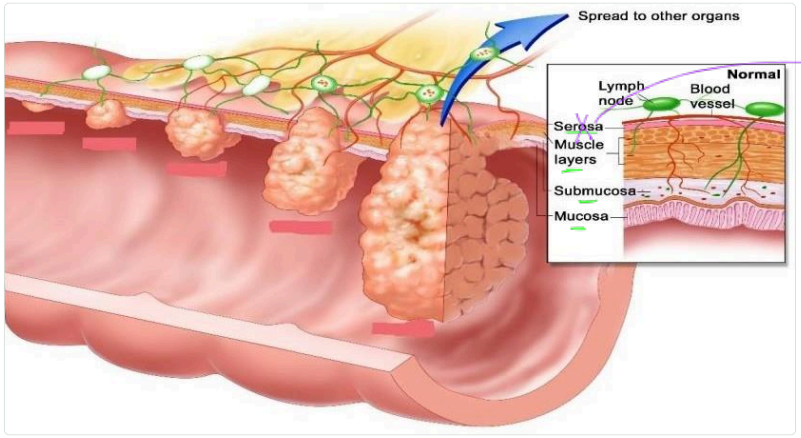


Figure 2.16 — Local spread

B. Lymphatics:

to regional L.N. according to the level:

- Cervical L.N. from upper 1/3
- Hilar and Mediastinal L.N. from the middle 1/3
- Left gastric and celiac L.N. from the lower 1/3

C. Blood :

late: to lung – liver – bone – brain.

Lower third may give early liver metastasis (portal)

Clinical picture:

General symptoms and signs:

- Marked progressive weight loss up to cachexia of rapid onset
  - Enlarged cervical L.N. in upper third cancer
  - Anemia
  - Symptoms of spread as recurrent aspiration in tracheal infiltration, pleural effusion

Local Symptoms:

1) Dysphagia:

- Old aged male – history of P.F. as GERD
- Presented when about 2/3 of the circumference is involved by the tumor
- Progressive and continuous
- First to solids then progresses to fluids as well then event to saliva
- Dribbling of saliva (severe form of dysphagia).

2) Regurgitation → leads to recurrent aspirations and chest infections

3) Hematemesis:

bleeding malignant tissues.

Investigations:

I.Upper endoscopy:

the most valuable and diagnostic

May show malignant ulcer, hard irregular stricture or cauliflower mass

Biopsy is taken for histo-pathology: Confirms the diagnosis.

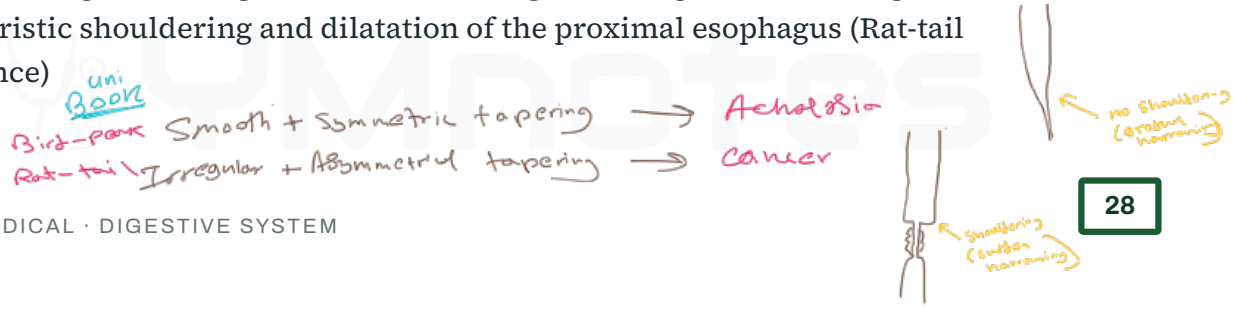
A stent may be introduced to relieve dysphagia (palliative).

- **Endo-luminal ultrasound:** (EUS)
  - To detect infiltration of the tumor within and outside the wall of esophagus to surrounding structure.
  - Detects resectability and operability

III- Radiological:

1. Barium swallow:

- It may show irregular filling defect.
- It may show signs of malignant stricture: elongated, irregular narrowing, characteristic shouldering and dilatation of the proximal esophagus (Rat-tail appearance)



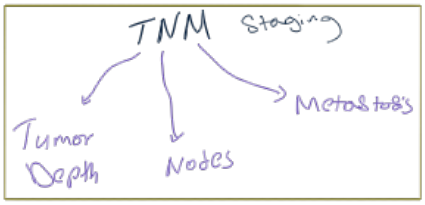


Figure 2.17 — Barium swallow shows irregular stricture and filling defect

2. C.T. scan abdomen and chest:

for staging of the tumor

- Detect local spread and infiltration of surrounding structures
- Detect metastases in the liver, lungs
- Detect enlarged celiac L.N.
- Detects mediastinal L.N.
- Detects pleural effusion
- Evaluates the stage, operability and resectability

IV- LAB.:

- CBC – Liver, kidney functions to evaluate the general condition and fitness for surgery.

Treatment:

Usually the prognosis is bad as the diagnosis is usually late

- **Treatment options:**

1- Palliative 2- Radical Surgery

Palliative treatments:

to relieve dysphagia (temporary effects)

Indications: advanced cases – bad general conditions

a)Intubation:

Insertion of metal stent inside the strictured segment to create a path for feeding

Introduced by endoscopy

Examples: Celestin and Souttar tubes

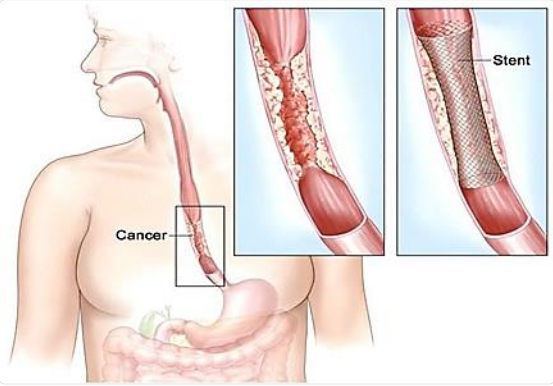


Figure 2.18 — Esophageal stent placement

b)YAG LASER:

Photocoagulation induces coagulation inside the strictured segment to create a path

c) Palliative surgery:

Gastrostomy or jejunostomy used for feeding:

Figure 2.19 — Feeding gastrostomy

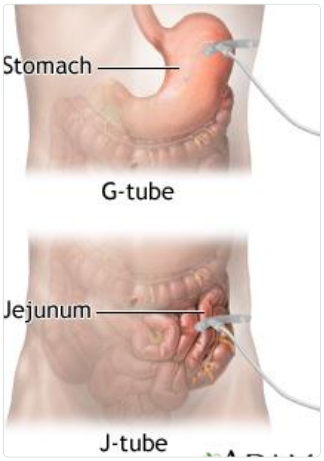


Figure 2.20 — Gastrostomy and jejunostomy

d) Radiotherapy

Updated multimodal management

-A. Early Stage Disease (T1):

- Endoscopic Resection: Endoscopic Mucosal Resection (EMR) or Endoscopic Submucosal Dissection (ESD) is curative for intramucosal (T1) disease without lymphovascular invasion, followed by radiofrequency ablation (RFA) of surrounding Barrett's mucosa.

B. Locally Advanced but Resectable Disease:

Management differs significantly based on pathology:

1.Esophageal Adenocarcinoma (EAC):

- Preferred Standard: Perioperative chem & immunotherapy + surgery.

2.Esophageal Squamous Cell Carcinoma (ESCC):

- Preferred Standard: Neoadjuvant Chemoradiation followed by surgery.

Surgical Approaches (Esophagectomy):

Techniques:

Radical surgery:

Curative surgery – used in early cancer and good general condition. **Aim:** resection of the tumor with safety margin and restore continuity .

- ***Ivor Lewis Esophagectomy***: *Transthoracic approach (right thoracotomy + laparotomy) with gastric conduit reconstruction.*
- ***McKeown Esophagectomy:*** *Three-incision approach (neck, chest, abdomen) for upper or mid-thoracic lesions*
- ***Transhiatal Esophagectomy:*** *Abdominal and cervical incisions without formal thoracotomy.*

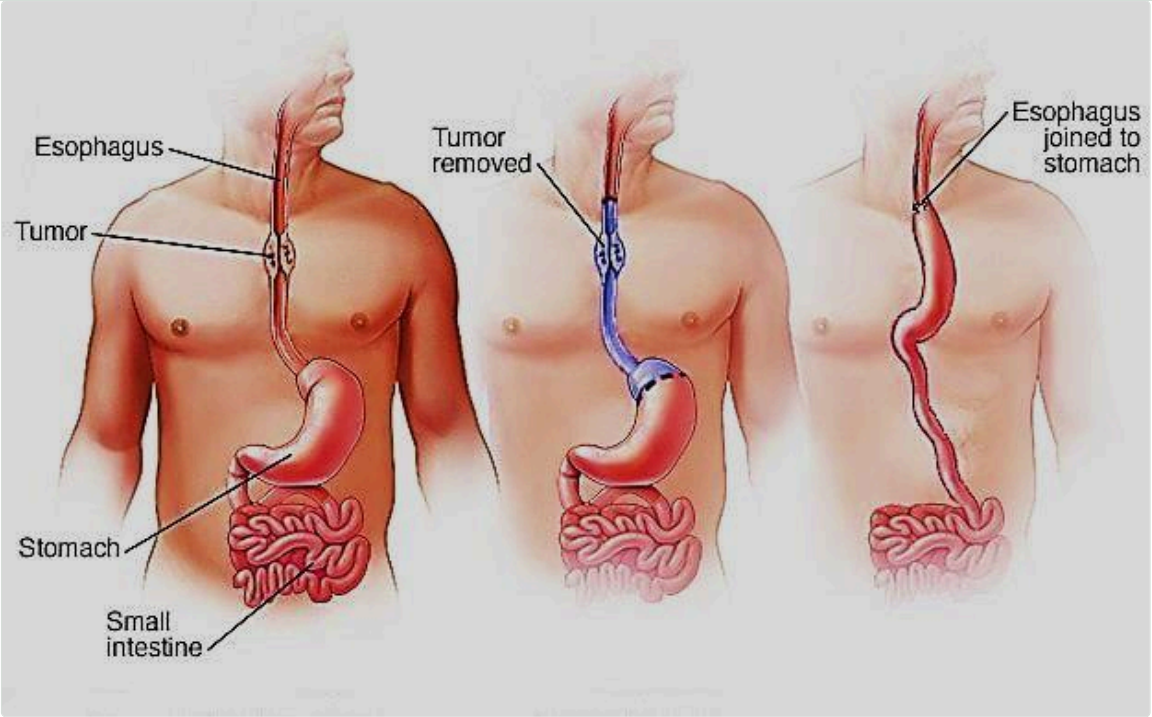


Figure 2.21 — Esophagectomy and gastric reconstruction

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المعدة و الاثنا عشر

CHAPTER 03

THE STOMACH AND DUODENUM

Chapter 37

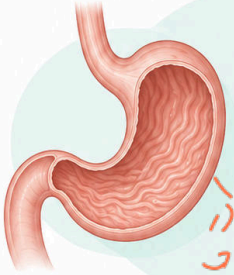
[33 - 69]

YMnotes Medical | Digestive System

Chapter 03

Integrated

The Stomach and Duodenum

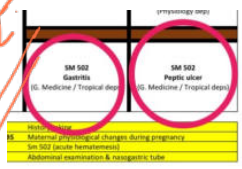


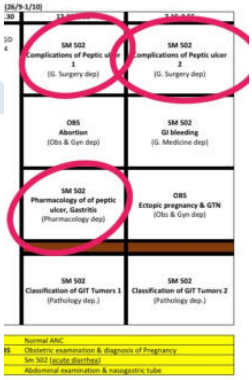
Chapter Outline

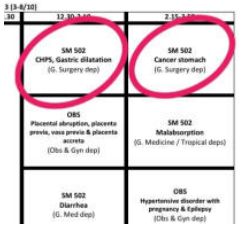
1. **Anatomy and Physiology & Histology**
   - Gastric Anatomy
2. **Gastritis**
   - Acute Gastritis
   - Chronic Gastritis
   - Chronic Autoimmune Gastritis
   - Helicobacter pylori Gastritis — Parasitology
3. **Peptic Ulcer Disease** — Pharmacology
   - Zollinger–Ellison Syndrome
   - Antisecretory drugs: PPIs, P-CABs, H2 blockers
   - Mucosal protectives
   - Anti-H. pylori therapy
   - Antacids
4. **Gastrointestinal Bleeding**
   - Acute Upper GI Bleeding
   - Acute Lower GI Bleeding
   - Chronic GI Bleeding
5. **Complications of Peptic Ulcer**
   - Bleeding peptic ulcer
   - Perforated peptic ulcer
   - Pyloric obstruction / stenosis
6. **Congenital Hypertrophic Pyloric Stenosis (CHPS)**
7. **Cancer Stomach**
   - Gastric cancer
8. **Acute Gastric Dilatation**

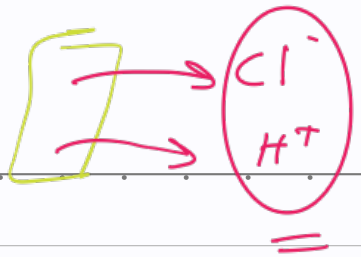
YMnotes Medical | Digestive System

Book → chapters → Sections → Subsections









Some Notes

- Cl^- & H^+ secreted by the parietal cells into the stomach lumen where then they form HCl acid (otherwise, destroyed the cells from inside out)

- HCl Acid production stimulated by 3 receptors on the parietal cells M3 (Ach) H2 (Histamine) G (Gastrin) & Inhibited by Somatostatin from D cells

Learning Objectives

By the end of this chapter, the medical student will be able to:

1. Analyze the clinical manifestations, diagnostic approach, and management of acid- related gastric and duodenal diseases.
2. Evaluate the clinical presentation, diagnosis, and treatment of peptic ulcer disease (PUD).
3. Formulate modern diagnostic pathways and eradication regimens for Helicobacter pylori.
4. Identify clinical manifestations, diagnostic approaches, and surgical indicators for PUD complications.
5. Manage acute and chronic gastrointestinal bleeding.
6. Recognize the clinical presentation, diagnosis, and staging of gastric carcinoma.

Anatomy and Physiology

Gastric Anatomy

The stomach consists of five distinct anatomical regions:

- **Cardia**: Adjacent to the gastroesophageal junction.
- **Fundus**: Dome-shaped region extending above and to the left of the cardia, toward the left hemidiaphragm.
- **Body** (Corpus): Extends from the fundus to the lower end of the lesser curve, known as the **incisura angularis**.
- **Antrum**: Extends from the incisura angularis to the pyloric canal.
- **Pylorus**: The muscular region between the antrum and the duodenum.

The stomach is bordered by two main functional sphincters: the gastroesophageal sphincter and the pyloric sphincter.

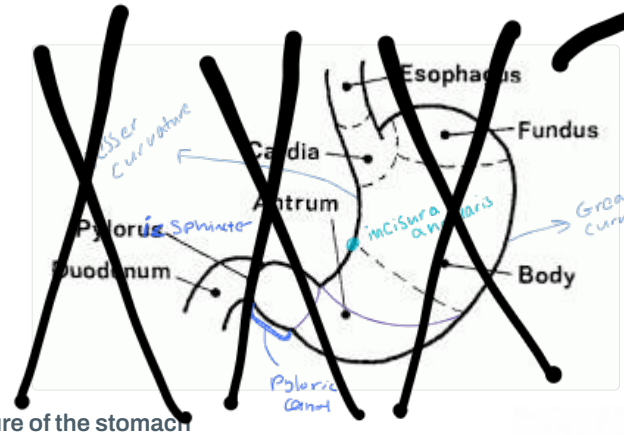
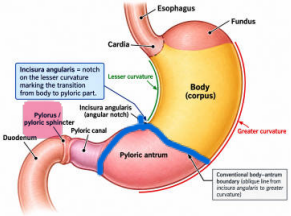


Figure 3.1 — Structure of the stomach



Histological Structure and Secretions

- Upper Two-Thirds (Fundus and Body): The mucosa contains parietal cells that secrete hydrochloric acid and intrinsic factor, and chief cells that secrete pepsinogen to initiate proteolysis.
- Lower One-Third (Antrum): The antral mucosa contains mucus-secreting cells, G cells (which secrete gastrin to stimulate acid production), and D cells (which secrete somatostatin, a potent suppressant of acid secretion).
- Duodenal Mucosa: Features villi and contains Brunner's glands that secrete alkaline mucus. This secretion, alongside pancreatic and biliary inputs, neutralizes acidic gastric chyme entering the duodenum.

Histology

CLINICAL PEARL: Gastric Secretory Regulation

- Acid secretion is under tight neural and hormonal control.
- Acetylcholine (vagal stimulation) and Gastrin stimulate acid production directly, and indirectly by releasing histamine from enterochromaffin-like (ECL) cells.
- Somatostatin acts as a primary negative feedback brake, inhibiting both histamine and gastrin release.

Physiology

The stomach also serves as a protective mucosal barrier composed of cell plasma membranes and a heavy mucus-bicarbonate layer. This barrier shields the epithelium against auto-digestion by acid, pepsin, and ingested irritants (aspirin, NSAIDs, alcohol, and bile salts). Local prostaglandins (PGE2) are critical to maintaining this barrier by stimulating mucus and bicarbonate synthesis.

Gastritis

Definition

Gastritis: Gastritis is inflammation of the gastric mucosa, usually diagnosed histologically, and may result from infectious, immune, chemical, or drug-related causes.

- Gastropathy: A gastric mucosal disorder featuring epithelial injury and subsequent regeneration without associated histological inflammation.

Note: Gastritis and gastropathy are not mutually exclusive and frequently coexist in clinical practice.

Acute Gastritis

Acute gastritis refers to a sudden onset of mucosal inflammation. A disruption in the mucosal barrier triggers an immune response that recruits neutrophils (polymorphonuclear leukocytes) to the site of injury. Severe mucosal damage can progress to acute erosive gastritis, featuring shallow mucosal lesions (erosions), ulcerations, and focal intraluminal bleeding.

Etiology and Pathogenesis

- NSAIDs and Aspirin: Inhibit cyclooxygenase (COX-1), blocking the synthesis of prostaglandin E2. This depletes protective gastric mucus and increases baseline acid output.
- Critical Illness (Stress Ulcers): Extreme physiological stress—such as major surgery, severe trauma, massive burns (Curling ulcers), or central nervous system injury (Cushing ulcers)—induces mucosal ischemia.
- Exogenous Irritants: Elevated intake of alcohol, heavy caffeine consumption, and exposure to cigarette smoke directly irritate and inflame the gastric mucosa.
- Infectious: The early, acute phase of an Helicobacter pylori infection.

Signs and Symptoms

Most individuals are asymptomatic or experience mild upper abdominal discomfort, loss of appetite, belching, nausea, and vomiting. Severe erosive cases can present acutely with upper gastrointestinal bleeding (hematemesis or melena).

Diagnosis and Management

Diagnosis begins with a thorough medical history to pinpoint specific triggers (such as chronic NSAID use or alcohol intake). Because acute gastritis is typically short-lived and self-resolving, removing the offending trigger is often sufficient. If bleeding occurs or the diagnosis remains uncertain, an Upper Endoscopy (EGD) is indicated to visualize the mucosa and obtain biopsy samples.

Acid suppression via Proton Pump Inhibitors (PPIs) or Potassium-Competitive Acid Blockers (PCABs) effectively aids symptom relief and accelerates mucosal healing.

Gastritis and NSAIDs

- Aspirin and other NSAIDs deplete mucosal prostaglandins by inhibiting the cyclo-oxygenase (COX) pathway, which leads to mucosal damage.
- Cyclo-oxygenase occurs in two main forms: COX-1, the constitutive enzyme, and COX-2, inducible by cytokine stimulation in areas of inflammation.
- COX-2 specific inhibitors have less effect on the COX-1 enzyme in the gastric mucosa. It still produce gastric mucosal damage but less than with other conventional NSAIDs. -Because of the large number of patients on NSAIDs including low-dose aspirin for vascular prophylaxis, this is a significant problem, particularly in the elderly.

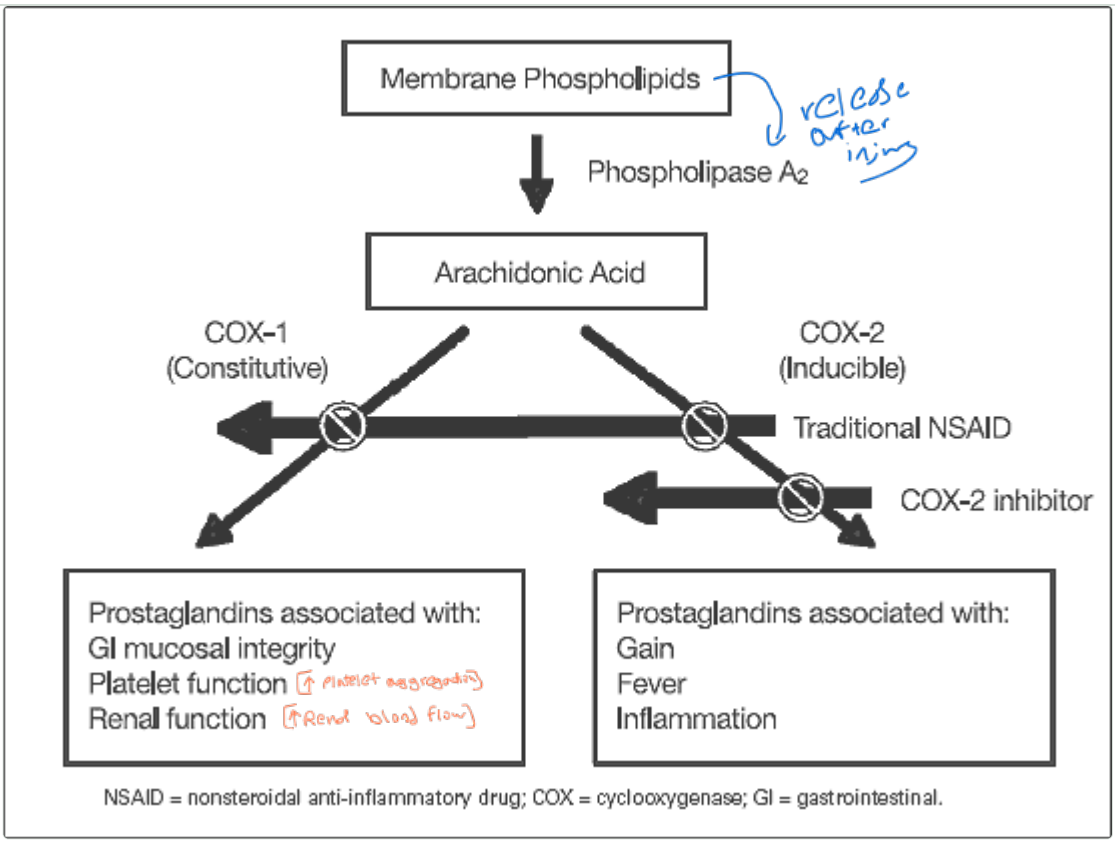


Figure 3.2 — Mechanism of action of traditional NSAID and anti COX-2 inhibitors

Treatment

- Stop the ingestion of NSAIDs.
- A PPI should be given. -H. pylori eradication therapy if H. pylori positive.
- In many people with severe arthritis, stopping NSAIDs may not be possible, therefore use: -A NSAID with low GI side-effects at lowest dose possible or if there is no cardiovascular risk, a COX-2 NSAID can be used

Prophylactic cytoprotective therapy, e.g. PPI or misoprostol (a synthetic analogue of prostaglandin for all high-risk patients, i.e. over 65 years; those with a peptic ulcer history, particularly with complications, and patients on therapy with corticosteroids or anticoagulants. PPIs reduce the risk of endoscopic duodenal and gastric ulcers and are better tolerated than misoprostol, which causes diarrhoea.

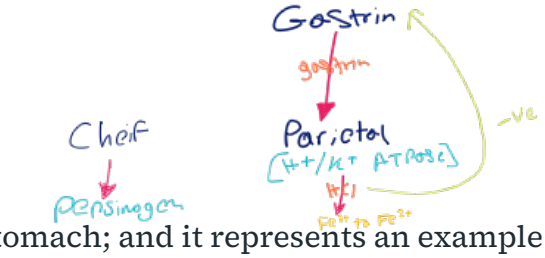
Management of NSAID-Induced Ulcers

1. Discontinue the offending NSAID immediately whenever clinically feasible.
2. Initiate daily healing doses of a PPI.
3. If chronic NSAID therapy cannot be stopped due to severe comorbidities (e.g., severe arthritis), transition the patient to the lowest effective dose, consider a selective COX-2 inhibitor, and co-prescribe daily cytoprotective prophylaxis via a PPI or Misoprostol (a synthetic prostaglandin E_2 analogue).

Chronic Gastritis

Chronic gastritis involves long-standing, persistent inflammation of the gastric mucosa. It is characterized histologically by a lymphocytic and plasma cell infiltrate, and is broadly divided into two major distinct pathological types:

- Chronic autoimmune Gastritis
- Helicobacter Pylori Gastritis



Chronic autoimmune gastritis,

It affects the body and fundus of the stomach; and it represents an example of a type IV hypersensitivity. In chronic autoimmune gastritis, there are autoantibodies against H+/K+ ATPase on parietal cells and intrinsic factor. Autoantibodies against H+/K+ ATPase decrease gastric acid secretion, thereby causing hypochlorhydria, which is associated with impaired iron absorption and hyperplasia of gastrin cells. As a result, there's an increased secretion of gastrin, hypergastrinemia, which is a hormone that has a trophic effect on enterochromaffin-like cells, or ECL cells; therefore, these individuals are at risk for developing neuroendocrine tumors.

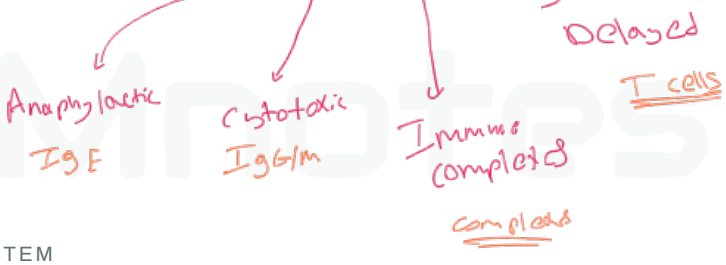
Autoantibodies against intrinsic factor lead to decreased absorption of vitamin B12, resulting in pernicious anemia.

Finally, in chronic autoimmune gastritis chief cells are also lost and this results in decreased levels of serum pepsinogen. Diagnosis of chronic autoimmune gastritis includes endoscopic biopsy, but also detection of anti-parietal cell antibodies, anti- IF antibodies; and levels of serum gastrin and pepsinogen

Helicobacter pylori Gastritis

H. pylori is a slow-growing spiral Gram-negative flagellate urease-producing bacterium which plays a major role in gastritis and peptic ulcer disease.

- It colonizes the mucous layer in the gastric antrum, but is also found in the duodenum in areas of gastric metaplasia.
- Antral gastritis is the usual effect of H. pylori infection. It is usually asymptomatic, although patients without ulcers do sometimes experience relief of dyspeptic symptoms after Helicobacter eradication. Chronic antral gastritis causes hypergastrinaemia due to gastrin release from antral G cells. The subsequent increase in acid output is usually asymptomatic, but can lead to duodenal ulceration



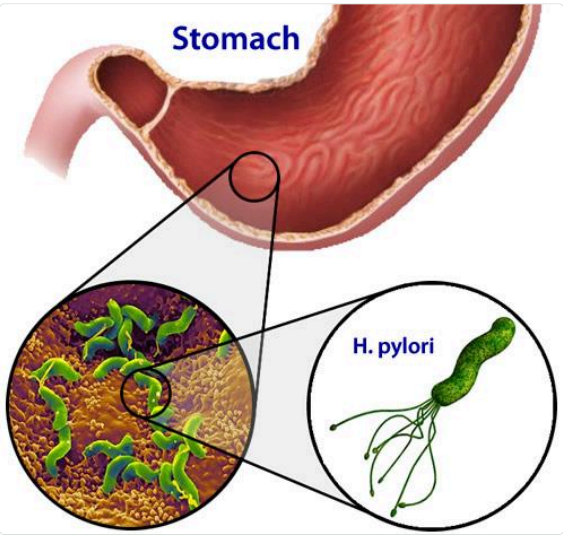


Figure 3.3 — H Pylori gastritis

Epidemiology

The prevalence of H. pylori is high in developing countries (80-90% of the population), and much lower (20-50%) in developed countries.

- Infection rates are highest in lower income groups. Infection is usually acquired in childhood; although the exact route is uncertain, it may be fecal-oral or oral-oral. The incidence increases with age.

Pathogenesis

- The pathogenetic mechanisms are not fully understood, with the majority of the colonized population remaining asymptomatic throughout their life.

  H. pylori is highly adapted to the stomach environment, exclusively colonizing gastric epithelium and inhabiting the mucous layer, or just beneath.
- The production of urease enables the conversion of urea to ammonium and chloride, which are directly cytotoxic.
- Ulcers are commonest when the infecting strain expresses CagA (cytotoxic-associated protein) and VacA (vacuolating toxin) genes secondary to a more pronounced inflammatory and immune response
- The precise mechanism of duodenal ulceration is unclear, as only 15% of patients infected with H. pylori (50-60% of the adult population worldwide) develop duodenal ulcers.
- Factors that have been implicated include: increased gastrin secretion, smoking, bacterial virulence and genetic susceptibility.
- Gastric ulcer (GU) : Gastric ulcers are associated with a gastritis affecting the body as well as the antrum of the stomach (pangastritis) causing parietal cell loss and reduced acid production. The ulcers are thought to occur because of reduction of gastric mucosal resistance due to cytokine production by the H pylori infection or perhaps to alterations in gastric mucus.

- **Other H. pylori associated diseases:**
  - Gastric adenocarcinoma. The incidence of distal (but not proximal) gastric cancer parallels that of H. pylori infection in countries with a high incidence of gastric cancer. Serological studies show that people infected with H. pylori have a higher incidence of distal gastric carcinoma (Gastric B cell lymphoma) . Over 70% of patients with gastric B cell lymphomas (mucosal-associated lymphoid tissue - MALT) have H. pylori.

    H. pylori gastritis has been shown to contain the clonal B cell that eventually gives rise to the MALT lymphoma.

Diagnosis of Helicobacter pylori infection

Diagnosis of H. pylori is necessary if the clinician plans to treat a positive result. This is usually in the context of active peptic ulcer disease, previous peptic ulcer disease or MALT lymphoma, or to 'test and treat' patients with dyspepsia under the age of 55 with no alarm symptoms (i.e. weight loss, anemia, dysphagia, vomiting, or family history of gastrointestinal cancer).

Non-invasive methods

- Stool antigen test. This is beginning to be the method of choice to determine the H. pylori status. A specific immunoassay using monoclonal antibodies for the qualitative detection of H. pylori antigen is now widely available. The overall sensitivity and specificity is over 95% . It is useful in the diagnosis of H. pylori infection and for monitoring efficacy of eradication therapy. Patients should be off PPIs for 2 weeks but can continue with H2 blockers.
- 13 C Urea breath test. This is a quick and reliable test for H. pylori and can be used as a screening test. The measurement of 13 CO 2 in the breath after ingestion of 13 C urea . The test requires a mass spectrometer. The test is sensitive (90%) and specific (96%), but the sensitivity can be improved by insuring the patient has not taken antibiotics in the 4 weeks prior and proton pump inhibitors in the 2 weeks before the test.
- Serological tests detect IgG antibodies are reasonably sensitive (90%) and specific

(83%). They have been used in diagnosis and in epidemiological studies. IgG titres may take up to 1 year to fall by 50% after eradication therapy and therefore are not useful for confirming eradication or the presence of a current infection.

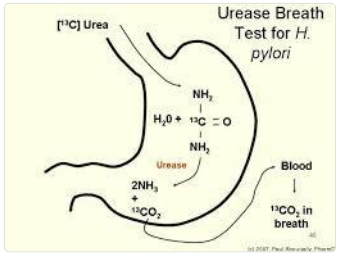


Figure 3.4 — Urea breath test for detection of H pylori

Invasive (endoscopy)

■ Biopsy urease test. Gastric biopsies, usually antral are added to a substrate containing urea and phenol red. If H. pylori are present, the urease enzyme that they produce splits the urea to release ammonia which raises the pH of the solution and causes a rapid colour change (yellow to red). This enables a patient's H. pylori status to be determined before they leave the endoscopy unit. The test may be falsely negative if patients are taking PPIs or antibiotics.

■ Histology. H. pylori can be detected histologically on routine (Giemsa) stained sections of gastric mucosa obtained at endoscopy.

■ Culture. Biopsies obtained can be cultured on a special medium, and in vitro sensitivities to antibiotics can be tested. This technique is typically used for patients with refractory H. pylori infection to identify the appropriate antibiotic regimen and routine culture is rare.

Eradication therapy

Proton pump inhibitor or PCAB twice / day + Amoxicillin 1g twice / day + clarithromycin 500 mg twice /day . Treatment was given for 14 days.

However due to global clarithromycin resistance, classic standard triple therapy is no longer recommended as an empirical first-line choice unless local resistance is confirmed to be low.

1. Bismuth Quadruple Therapy (Preferred First-Line Choice)

- **Acid Suppressant** : Standard-dose PPI twice daily or Vonoprazan (PCAB) 20 mg twice daily.
- **Bismuth Subcitrate** : 120–300 mg four times daily.
- **Tetracycline** : 500 mg four times daily.
- **Metronidazole** : 250–500 mg four times daily.
- *Duration* : 10–14 days

Antibiotics

Protect stomach lining + Antibiotic effect

2. PCAB Dual Therapy

- **Vonoprazan** : 20 mg twice daily.
- **Amoxicillin** : 1g three times daily (total 3g/day).
- *Duration* : 14 days.

Antibiotics

3. Rescue Therapy

- Implements quinolones (e.g., Levofloxacin 500 mg once daily + PPI + Amoxicillin) when standard first-line options fail. Avoid reusing antibiotics previously taken by the patient.

Abs

In developing countries reinfection is common, compliance with treatment may be poor and metronidazole resistance is high (>50%) as it is frequently used for parasitic infections), so failure of eradication is common

SUBTYPES OF CHRONIC GASTRITIS

Parameter	Chronic Autoimmune Gastritis	Helicobacter pylori Gastritis)
Anatomical Site	Fundus and Body (spares the gastric antrum).	Antrum initially; can progress to the body over time(pangastritis).
Mechanism	Type IV Hypersensitivity ; autoantibodies target H^+/K^+ ATPase on parietal cells and Intrinsic Factor(IF).	Chronic active inflammation induced by bacterial virulence factors (CagA, VacA).
Serum Gastrin	Markedly elevated (Severe hypergastrinemia due to lack of acid negative feedback on G cells).	Variable; can cause hypergastrinemia locally if limited to the antrum.
Gastric pH	Elevated (Achlorhydria / Hypochlorhydria) due to parietal cell loss.	Normal to low (antral disease); elevated if it progresses to pangastritis.
Hematologic Signs	Pernicious Anemia (macrocytic anemia due to Vit B_{12} malabsorption) and impaired iron absorption.	Iron Deficiency Anemia (secondary to occult bleeding from associated peptic ulcers).
Neoplastic Risk	Trophic effect of gastrin causes ECL cell hyperplasia → Neuroendocrine (Carcinoid) Tumors ;Adenocarcinoma.	Distal Gastric Adenocarcinoma and MALT Lymphoma (B-cell lymphoma).
Diagnosis	Serology (anti-parietal/anti-IF antibodies), low serum pepsinogen, elevated gastrin, fundic biopsy.	Non-invasive testing (Stool Antigen, Breath Test) or endoscopic biopsy.

ACUTE AND CHRONIC GASTRITIS

Diagnostic Feature	Acute Gastritis	Chronic Gastritis
Onset & Duration	Sudden onset; typically self- limiting within days to weeks.	Insidious onset; long- standing, persistent mucosal inflammation over years.
Primary Cell Type	Neutrophils (Polymorphonuclear leukocytes).	Lymphocytes and Plasma cells .
Dominant Etiology	NSAIDs/Aspirin, alcohol, critical illness (ischemic stress), or early acute H.pylori phase.	2 Types : Autoimmune and . : Chronic active H. pylori colonization.
Pathophysiology	Direct mucosal irritation, toxic injury, or acute ischemia disrupting the surface barrier.	Progressive immune-mediated destruction or persistent bacterial-induced cytokine inflammation.
Endoscopic Findings	Hyperemic mucosa, punctate petechiae, mucosal erosions, and shallow lesions.	Atrophic, thinned mucosa with visible submucosal vessels, or nodular architecture.
Malignancy Risk	No increased risk of malignancy.	Increased risk of gastric adenocarcinoma and MALT lymphoma.

There are two types of chronic gastritis: chronic autoimmune gastritis and Helicobacter pylori-associated gastritis.

PEPTIC ULCER DISEASE

- A peptic ulcer consists of a break in the superficial epithelial cells penetrating down to the muscularis mucosa of either the stomach or the duodenum; there is a fibrous base and an increase in inflammatory cells.

*Etiology of PUD

* Helicobacter pylori (HP)
  (Proinflammatory cytokines & cagA & vacA)
* Non-Steroidal antiinflammatory Drugs (NSAIDs)
  (inhibit cox enzyme)

*ZES

Others:
Smoking
Hereditary

Mechanism

inflammation

↳ Arachnoid Acid

cox → Prostaglandins
lox → Leukotrienes

enzymes

↑ Blood Flow

(∴ ↑Mucus ↑Bicarbonate ...etc. production)

Stomach

- Erosions, by contrast, are superficial breaks in the mucosa alone. Most DUs are found in the duodenal cap; the surrounding mucosa appears inflamed, hemorrhagic or friable (duodenitis). GUs are most commonly seen on the lesser curve near the incisura, but can be found in any part of the stomach.

Epidemiology of peptic ulcer disease :

- Duodenal ulcers affect approximately 10% of the adult population and are two to three times more common than gastric ulcers.
- Ulcer rates are declining rapidly for younger men and increasing for older individuals, particularly women.
- Both DUs and GUs are common in the elderly.
- There is considerable geographical variation, with peptic ulcer disease being more prevalent in developing countries related to the high H. pylori infection.
- In the developed world the percentage of NSAID-induced peptic ulcers is increasing, as the prevalence of H. pylori declines.

Clinical features of peptic ulcer disease

- The characteristic feature of peptic ulcer is recurrent, burning epigastric pain. It has been shown that if a patient points with a single finger to the epigastriurn as the site of the pain this is strongly suggestive of peptic ulcer disease.
- The relation-ship of the pain to food is variable and on the whole not helpful in diagnosis. The pain of a DU classically occurs at night (as well as during the day) and is worse when the patient is hungry, but this is not reliable. The pain of both gastric and duodenal ulcers may be relieved by antacids. Nausea may accompany the pain; vomiting is infrequent but can relieve the pain. Anorexia and weight loss may occur, particularly with GUs. Persistent and severe pain suggests complications such as penetration into other organs.
- Back pain suggests a penetrating posterior ulcer. Severe ulceration can occasionally be symptomless, as many who present with acute ulcer bleeding or perforation have no preceding ulcer symptoms. -Untreated, the symptoms of a DU relapse and remit spontaneously. The natural history is for the disease to remit over many years due to the onset of atrophic gastritis and a decrease in acid secretion.
- Examination is usually unhelpful ; epigastric tenderness is common in non-ulcer dyspepsia.

Investigation of suspected peptic ulcer disease

■ Patients under 55 years of age with typical symptoms of peptic ulcer disease who test positive for H. pylori can start eradication therapy without further investigation.

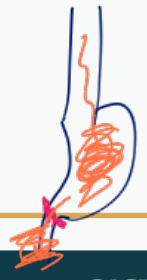
■ Endoscopic diagnosis and exclusion of cancer is required in older patients .All gastric ulcers must be biopsied to exclude an underlying malignancy and should be followed up endoscopically until healing has taken place.

■ Endoscopy is required in all patients with alarm symptoms' .

YMnotes

Peptic ulcer disease

Clinical picture · Duodenal versus gastric ulcer



FEATURE	DUODENAL ULCER	GASTRIC ULCER
Pain site	Upper abdomen; may be right of midline	Upper abdomen; often central (epigastric)
After meals	Often 2–4 hours later	May start sooner (about 30–60 minutes)
Effect of food	Often relieves pain temporarily	May worsen pain; the pattern varies
Pain at night	Common; may wake the patient	Less characteristic, but possible
Vomiting	Usually uncommon	May be more common
Eating pattern	May eat to ease pain	May avoid food because of pain

COMMON PAIN Burning, stabbing, or dull epigastric discomfort.

Patterns are suggestive only; symptoms overlap and cannot confirm ulcer type.

Hunger Pain

(as food used to buffer stomach acid)

غوزن العيان بيزيد

Eating Pain

وزن العيان بيقل

Complications

Any Ulcer complication جراحة

- Hge
- Perforation → peritoneum
- Penetration → pancreas
- Obstruction ↑ inflammation up to obstruction
- Carcinoma

■ Patients with a risk of bleeding or those with complications, i.e. haemorrhage or perforation, should always have a stool test for H. pylori or a 13 C urea breath test 6 weeks after the end of treatment to be sure eradication is successful.

Figure 3.5 — Endoscopic view of a duodenal ulcer

Zollinger-Ellison Syndrome (ZES)

Zollinger-Ellison syndrome is a rare, severe cause of aggressive peptic ulcer disease driven by a gastrinoma. A gastrinoma is a neuroendocrine tumor typically located within the duodenal wall or the pancreas that autonomously secretes abnormal, excessive amounts of gastrin.

- Pathophysiology: Unregulated gastrin hypersecretion continuously over-stimulates parietal cells, causing massive parietal cell hyperplasia and extreme hydrochloric acid output. This overwhelming acid load completely destroys the native mucosal defense mechanisms.
- Clinical Presentation: Multiple, recurrent ulcers extending into atypical structural locations (such as the distal duodenum or proximal jejunum) that remain completely refractory to standard medical management. This is frequently accompanied by chronic watery diarrhea due to structural mucosal injury and acid- induced inactivation of pancreatic digestive enzymes.
- Diagnosis: Confirmed by documenting markedly elevated fasting serum gastrin levels paired with gastric fluid pH verification (≤ 2).
- Genetic Profile: May occur sporadically or as an autosomal dominant manifestation of Multiple Endocrine Neoplasia Type 1 (MEN1).

Pharmacology of treatment of Gastritis and Peptic ulcer

A- ANTI - SECRETORY DRUGS

1. Proton Pump Inhibitors (PPIs)

- Omeprazole - Esomeprazole – Lansoprazole - Dexalansoprazole - Pantoprazole - Rabeprazole (oral – IV - 20-40 mg/d)
- **Mechanism of Action**
- Act on the final step of HCl secretion → Most effective acid suppressants.

PEPTIC ULCER DISEASE

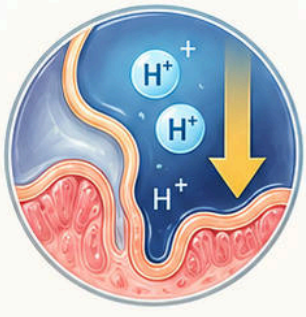
Pharmacological treatment

Pages 44–49

4 main sections • distinct drug classes within each

1 | REDUCE ACID PRODUCTION

Anti-secretory drugs

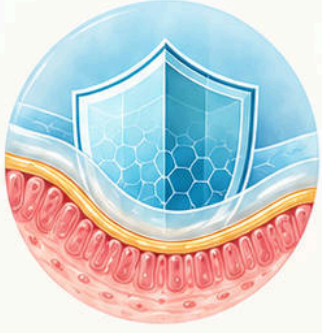


- Proton pump inhibitors (PPIs)
- Potassium-competitive acid blockers (P-CABs)
- H_2-receptor antagonists

M1 - Pyr...

2 | PROTECT THE LINING

Mucosal protectives



- Barrier-forming agents
- Prostaglandin analogues
- Bismuth compounds

3 | ERADICATE *H. PYLORI*

Anti-*H. pylori* therapy

When infection is present

- Combination eradication regimens
- Acid suppression + antibiotics
- ± bismuth, depending on regimen

4 | NEUTRALISE EXISTING ACID

Antacids

- Local antacids
- Systemic antacids
- Rapid, short-term symptom relief

A classification overview — these groups are not interchangeable.

- They are prodrugs → after absorption → parietal cells → secretory canaliculi → converted by acid into active form → inhibits H+/K+ ATPase (proton pump) → Inhibit basal & meal-stimulated acid secretion (98%, 1-2 h after 1st dose).
- They have anti- H. pylori effect (lansoprazole most potent)

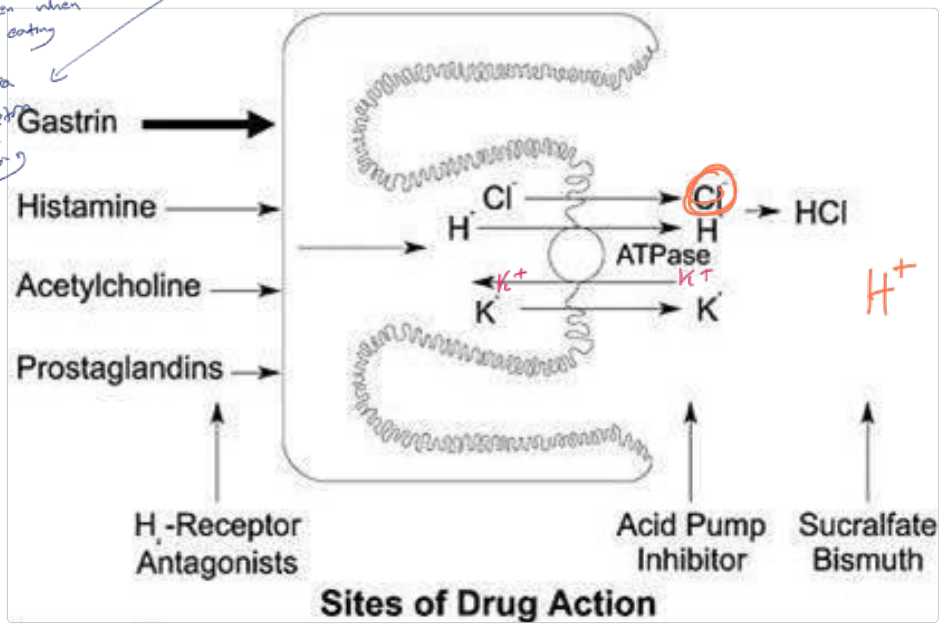


Figure 3.6 — Sites of drug action in gastric acid secretion

Pharmacokinetics

1. Weak bases → destroyed by gastric acidity & ionized in acid stomach (not readily absorbed) → given in enteric-coated form
2. Should be taken 30 – 60 minutes before meals (on empty stomach): only inhibit active pumps - Food →↓ bioavailability by 50%
3. Repeated administration →↓ gastric acidity →↑bioavailability.
4. Long duration (onset 1 hour & duration 24 hours (t1/2 = 1 h but PPIs covalently bind to PP and synthesis of new pump requires 24 hours).
5. Metabolized in the liver.

Adverse effects

1. GIT: abdominal pain, nausea, vomiting, diarrhea or constipation.
2. CNS: headache, dizziness, somnolence.
3. Hypochlorhydria → ↑ Risk of infection (e.g. hospital acquired pneumonia & H. pylori) and Gastric tumor (in animals): bacterial colonization → ↑carcinogenic nitroso compounds.
4. Vitamin B12 deficiency (on long term use)
5. Enzyme inhibition (clinically insignificant but inhibit activation of clopidogrel).

1. Potassium Competitive Acid Blockers (P-CAB)

- **Vonoprazan, revaprazan, tegoprazan (newer version of PPIs): differs from PPIs in:**

Omeprazole	Vonoprazan
Irreversible PP inhibition (↓ active pumps)	Competitive reversible PP inhibition (↓ both active and inactive PP)
Takes 3-5 days for maximal acid suppression	Take 1 day for maximal acid suppression
T1/2 = 1-2 hrs (but longer duration of action being irreversible)	T1/2 = 6-9 hrs
Prodrug	Not prodrug
Not acid stable (Taken 30-60 minutes before meals + need acid for activation)	Acid stable (Taken before or after meals + Not need acid for activation)

2. H2-Receptor Antagonists

- Cimetidine (prototype) – ranitidine (discontinued due to risk of cancer) - famotidine – nizatidine (oral – IV)
- **Mechanism & Pharmacological Actions**
- Reversible competitive antagonists of H2 receptors on parietal cells → Complete inhibition of histamine mediated but partial inhibition of ACh or gastrin mediated HCl secretion.
- Potent inhibition of nocturnal or fasting but less inhibition of daytime (meal-stimulated) acid secretion.
- **Individual H2 Antagonists**

1. Cimetidine: limited use due to its enzyme inhibiting, antiandrogenic & CNS side effects. 2. Ranitidine, Faamotidine, Nizatidine developed to minimize adverse effects of cimetidine (More potent than cimetidine):

Ranitidine (discontinued)	Nizatidine 150 mg	Famotidine 20 mg
Less enzyme inhibiting & antiandrogenic effects.	No enzyme-inhibiting or antiandrogenic effects.	No enzyme-inhibiting or antiandrogenic effects.
Anti H. pylori effect (most efficient).	The most potent.	Cytoprotective effect.
50% bioavailability (↑hepatic first pass effect)	100% bioavailability (no 1st pass effect)	50% bioavailability. 50% decomposed by gastric acidity.
Eliminated by hepatic metabolism → renal excretion of the drugs or their metabolites → dose reduction if ↓ Creatinine clearance or severe liver disease		

- **Adverse Effects of H2 Blockers (Mostly with cimetidine)**

1. Headache, diarrhea or constipation.
2. Tolerance, rebound hyperacidity & recurrence (upregulation of H2 receptors).
3. CNS: sedation, confusion, hallucination (mostly with cimetidine especially in elderly, renal or hepatic dysfunction).

4. Antiandrogenic, ↑ prolactin, gynecomastia, galactorrhea, impotence (mostly with cimetidine).

5. Enzyme inhibition → drug interactions (mostly with cimetidine).

6. Hypotension & bradycardia if given rapidly IV in intensive care (Block cardiac H2 receptors)

Advantages of PPIs and P-CABs over H2 antagonists:

1. Higher efficacy → better symptomatic relief & higher response rates.

2. More prolonged effect (= 24hs → given in single daily dose; versus twice daily dosing with H2 antagonists).

3. More effective in anti H. pylori regimen.

- **Indications of Antisecretory Drugs**

1- Peptic ulcer: PPIs preferred especially in severe cases. H2 antagonists in mild or moderate cases (healing is delayed for 8 weeks & recurrence is common).

2- Anti- H. Pylori regimens: PPIs preferred.

3- GERD: PPIs preferred. H2 antagonists are used to cover the nocturnal HCI secretion. 4- Zollinger Ellison syndrome: PPI are preferred. High dose of parenteral preparations are given early to cover vomiting period.

5- Stress ulcer: to ↓ risk of bleeding.

6- Acute gastritis in children & gastric erosions (Parenteral)

7- Bleeding oesophageal varices: to prevent recurrence (Parenteral)

8- To avoid aspiration pneumonia during surgery e.g. obstetric anesthesia before caesarian section (Parenteral)

MUCOSAL PROTECTIVES (CYTOPROTECTIVES)

	1. Sucralfate	2. Misoprostol	3. Colloidal Bismuth
Mech.	It is an aluminium salt of sulfated sucrose. In acid medium → -ve sulfate groups bind +ve proteins in the ulcer base →protective barrier (also inactivates bile and pepsin ↑ secretion of bicarbonate, mucus & PGs.	Synthetic PGE1 analog cytoprotective through: ↓HCl secretion by ↓ histamine-stimulated c- AMP production). ↑ mucosal blood flow ↑ secretion of mucus & bicarbonate.	Anti- H pylori activity (main effect). Mucosal protective: Chelates proteins in ulcer base → protective coat against acid & pepsin (inactivates pepsin) ↑ secretion of bicarbonate , mucus & PGs.
Uses:	Not used - Stress ulcer prophylaxis (mainly). Antacids or H2 blockers and some drugs e.g. tetracycline are given 1-2 hour apart.	Selective for NSAID-induced gastric ulcer if PPIs fail. Prophylactically with NSAIDs or corticosteroids.	Anti- H. pylori therapy
ADRs	Nausea, vomiting, dry mouth. Flatulence, constipation. Al+3 toxicity: chelates phosphates → osteomalacia & encephalopathy in renal diseases	Diarrhea & colic (most common). Vaginal bleeding & uterine contractions (CI in pregnancy → abortion).	Teeth & tongue discoloration - black stools (confused with bleeding). Encephalopathy and neurotoxicity especially in renal disease → not give >2 months & not restarted within 1 year.

ANTI H PYLORI THERAPY

- Eradication of H **pylori** infection results in rapid healing & ↓ recurrence.
- It is achieved by combination therapy with **High efficacy anti- H. pylori Triple therapy.**
- Triple therapy: 10-14 days: Lansoprazole + Clarithromycin + Amoxicillin (or metronidazole in patients allergic to amoxicillin).
- Quadruple therapy: 10–14 days: PPI + metronidazole 250 mg + bismuth subsalicylate 300 mg + tetracycline 500 mg four times daily
- Sequential therapy: PPI + amoxicillin 1 g twice a day for 5 days followed by PPI + clarithromycin 500 mg and tinidazole/ metronidazole 500 mg twice a day for 5 days
- Or: PPI + amoxicillin 1 g twice a day + levofloxacin 250 or 500 mg twice a day for 10 days
- Followed by 4-6 weeks: PPI therapy to allow ulcer heal.

N.B.: PPIs may be substituted by P-CABs in Peptic acid diseases and eradication of H. Pylori

ANTACIDS

- Antacids are weak bases that reduce gastric acidity by neutralizing HCL
- **Pharmacological Actions**
  1. Neutralize gastric acid (duration 1-2 hrs).
  2. Decrease peptic activity (pepsin is inactive at pH > 4.5).

3. Reduce H. pylori colonization.
4. ↑ mucosal PGs production (promote defense mechanism).
5. Adsorbent (to pepsin, bile &toxins), astringent (Precipitate surface proteins) and demulcent (soothing) effects (with Al+3 & Mg+2 salts only).

- **Therapeutic Uses (taken 1 and 3 hours after meals & at bedtime)**

1. Rapidly relieve pain & help healing of peptic ulcer.
2. Relieve dyspeptic symptoms.
3. GERD (rapid pain relief).

Drug Interactions:

Antacids can affect rates of dissolution, absorption, bioavailability and renal elimination of many drugs through:

1. Changing gastric or urinary pH.
2. Delaying gastric emptying.
3. Chelation: Al+3 salts can form insoluble complexes with fluoroquinolones, tetracyclines, digoxin and iron salts →↓ absorption (do not give antacids within 2 hrs from administration of these drugs).

- **Types of Antacids**

Local Antacids	Systemic Antacids
Al+3 Hydroxide Disadvantage: constipation. Mg+2 Salts Disadvantage: diarrhea. Ca+2 carbonate Disadvantages Constipation. Rebound hyperacidity (Ca+2 →↑ acid secretion) →↓pain relief & healing. Milk-alkali syndrome (systemic alkalosis - hypercalcemia). ↑ CO2 → flatulence & eructation.	Na HCO3 Advantages Rapid onset. Potent (↑ pH > 7). Disadvantages Systemic alkalosis on long term use. Rebound hyperacidity. Na+ content is dangerous in HF, hypertension or renal disease. ↑ CO2 → flatulence, eructation. Alkaline urine → renal stones. N.B.: Sodium bicarbonate is added for effervescent formulations or in immediate release PPIs preparations (e.g. omeprazole).

- **Choice of antacids**
- Al+3 hydroxide & Mg+2 salts do not result in CO2 release or systemic alkalosis thus they are the most commonly used antacids, alone or combined (e.g. Magaldarate - to neutralize the effects of each other on bowel habit).
- Antacids should be cautiously used in elderly & renal impairment since:
- Al+3 antacids → chelate phosphates → osteomalacia, hypophosphatemia, encephalopathy & ↑ risk of Alzheimer dementia.
- Mg+2 antacids → CNS depression.

Precautions to Avoid NSAIDs-induced Ulcer

1. Use low doses and avoid combinations of NSAIDs (↑ toxicity without added benefit).
2. Use topical preparations whenever possible.
3. Co-administer PPIs or misoprostol.
4. Avoid combination with corticosteroids except in selected cases.
5. Use cautiously in the elderly.

Gastrointestinal Bleeding

Acute Upper GI Bleeding

- Hematemesis: vomiting blood.
- Melena: black, tarry stool due to altered blood.
- Fresh blood per rectum may occur with massive upper GI bleeding, usually with shock.

Etiology:

- Peptic ulcer disease : one of the commonest causes.
  - NSAIDs/aspirin → gastric and duodenal ulcers/erosions and bleeding.
  - Anticoagulants increase bleeding severity rather than directly causing bleeding.
  - Other important causes: oesophageal varices, Mallory–Weiss tear, gastric cancer and Dieulafoy lesion.

Clinical approach

a) Initial Management: ABCDE

A – Airway: ensure airway patency. B – Breathing: assess and treat life-threatening respiratory problems. C – Circulation: assume hypovolaemia until proven otherwise; 2 large- bore IV lines and restore circulating volume. D – Disability: assess consciousness and neurological status. E – Exposure: complete examination while maintaining patient dignity and preventing heat loss.

b) Unstable Patient

- Monitor pulse and BP frequently.
- Blood tests: Hemoglobin , urea/electrolytes, liver function, coagulation profile, group & cross-match.
- IV fluids and blood transfusion when indicated.
- Oxygen therapy and keep NPO until endoscopy.
- Stop NSAIDs/aspirin where appropriate; interruption of antiplatelet therapy requires specialist assessment because of thrombosis risk.

- Urgent endoscopy after resuscitation.
- Persistent bleeding → repeat endoscopy → angiographic embolization → surgery if uncontrolled.
- Management is ideally multidisciplinary.

NB : Hemoglobin levels are generally a poor indicator of the need to transfuse because anemia does not develop immediately .

Endoscopic Treatment

Endoscopy provides diagnosis and hemostasis.

High-risk bleeding ulcers may require combination therapy, such as:

- Adrenaline injection + thermal therapy
- Endoscopic clipping

Variceal bleeding → endoscopic band ligation.

For actively bleeding ulcers or a visible vessel, IV PPI therapy reduces rebleeding.

Initial Endoscopy

- Bleeding ulcer / recent bleeding stigmata: Endoscopic hemostasis with ≥2 methods:
  - Diluted epinephrine injection + thermal therapy or clips.
  - H. pylori: Take antral biopsies.
    - Positive urease test → reliable.
    - Negative urease test → confirm with gastric histology.

Drug Therapy

- IV Omeprazole 80 mg bolus → 8 mg/h infusion for 72 h
  - Indicated for active bleeding or visible vessel.
  - Reduces rebleeding and need for surgery.

Persistent / Recurrent Bleeding

- Repeat endoscopy to identify and treat bleeding source.
  - If unsuccessful → angiography + embolization by interventional radiology.
  - Surgery if bleeding remains persistent/uncontrolled → primarily to control hemorrhage.

Causes of Upper GI Bleeding: (x4)

- oesophageal varices.
- Mallory–Weiss Tear
- Mucosal tear at the gastro-oesophageal junction.

- Usually follows retching or coughing.
- Most bleeding stops spontaneously.
- Early endoscopy confirms the diagnosis and allows treatment.

- **Dieulafoy Lesion**
  - Large submucosal vessel that bleeds without an associated ulcer or erosion.
  - May cause intermittent or massive haemorrhage.
  - Requires prompt endoscopic diagnosis and haemostasis.
- **Peptic Ulcer Bleeding**
  - Treat bleeding endoscopically.
  - Eradicate H. pylori when present.
  - Continue PPI therapy to promote ulcer healing.
  - Persistent/recurrent bleeding → angiographic embolization or surgery.

Acute Lower GI Bleeding

- Major bleeding: diverticular disease, ischemic colitis.
- Small-volume bleeding: hemorrhoids, anal fissures.

Assessment

- Start with ABCDE.
  - Unstable patients require resuscitation as for upper GI bleeding.
  - Focused history and examination, including digital rectal examination.
  - Investigations include CBC, coagulation profile, blood grouping/cross- match and appropriate imaging/endoscopy.

Painless hematochezia	Painful hematochezia
• Diverticular bleeding	• Infectious /inflammatory colitis
• Colorectal cancer	• Inflammatory bowel disease.
• Angiodysplasia	• Haemorrhoids /anal fissure
• Dieulafoy lesion	• Radiation proctitis/telangiectasia

Figure 3.7 — Painless and painful hematochezia

Diagnostic Modalities:

- **Proctoscopy**: anorectal causes.
- Flexible sigmoidoscopy/colonoscopy: IBD, cancer, diverticular disease, ischemic colitis and angiodysplasia.
- **Angiography:** selected cases of ongoing bleeding, particularly vascular lesions.

NB: Most acute lower GI bleeding stops spontaneously; surgery is rarely required.

Isolated episodes of rectal bleeding in the young (<45years) usually only require rectal examination and flexible sigmoidoscopy because the probability of a significant proximal lesion is very low unless there is a strong family history of colorectal cancer at a young age, Individual lesions are treated as appropriate.

Chronic GI Bleeding

Usually presents as iron-deficiency anaemia.

- In men and postmenopausal women, unexplained iron-deficiency anaemia should prompt investigation for GI blood loss, particularly malignancy and coeliac disease.
- Common causes include lesions capable of causing acute bleeding; hookworm is an important worldwide cause.
- If the bleeding site is unclear, investigate both upper and lower GI tract.

Investigation

1. Upper GI endoscopy + duodenal biopsy → exclude coeliac disease.
2. Colonoscopy → identify and biopsy/remove lesions.
3. If negative → small-bowel evaluation, with capsule endoscopy as the preferred diagnostic test.
4. Positive capsule findings may be treated using balloon-assisted enteroscopy.
5. CT colonography may be used when colonoscopy is unsuitable.

Treatment

- Treat the underlying cause.
  - Iron replacement for iron-deficiency anemia.
  - IV iron or maintenance transfusion may be required in selected patients.

Complications of peptic ulcer

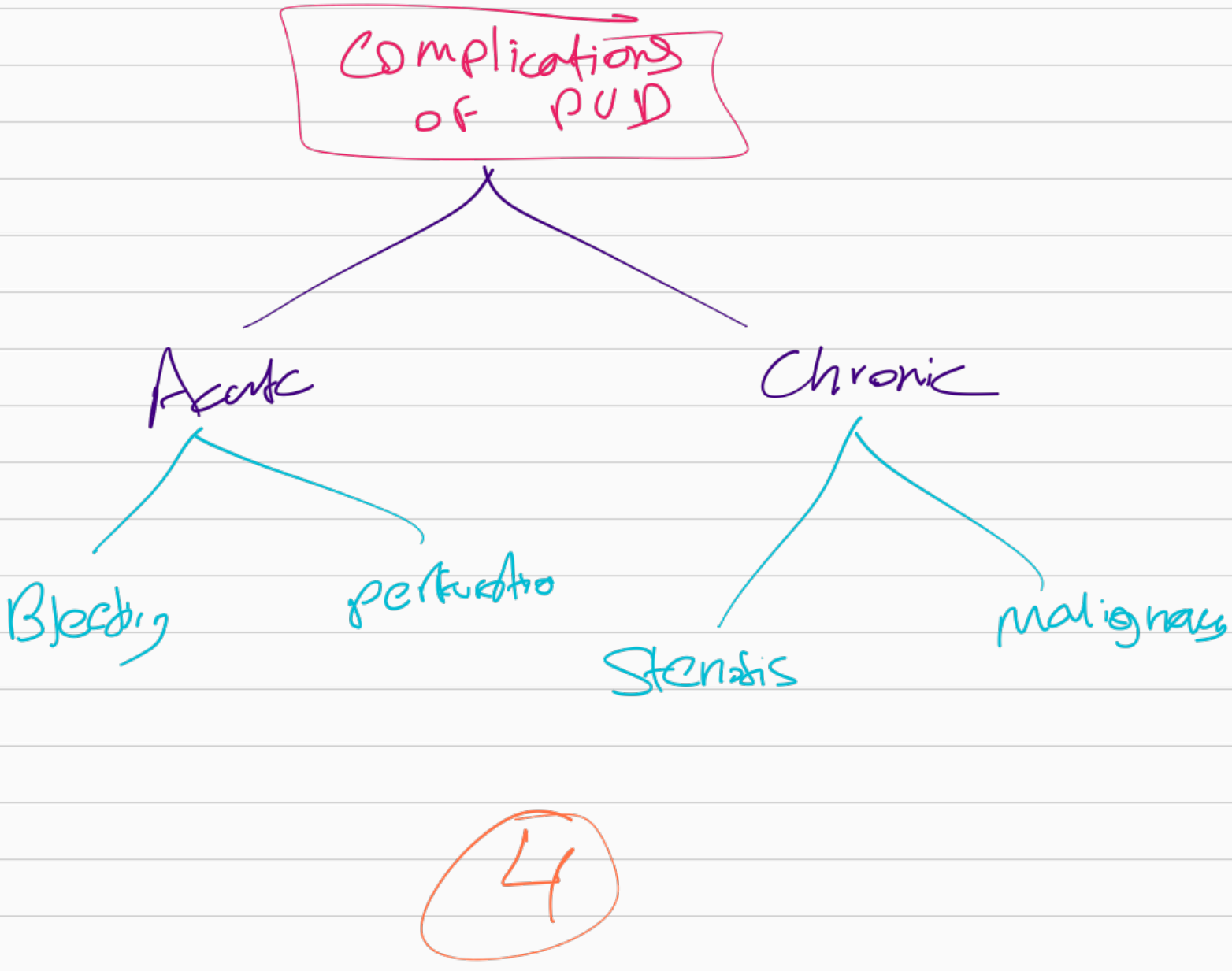
A. Acute complications: (surgical emergency)

1. Bleeding peptic ulcer
2. Perforated peptic ulcer

B. Chronic complications: (surgical indications)

1- Stenosis:

- Pyloric stenosis
- Duodenal obstruction


Complications of PUD
Acute
Chronic
Bleeding
perforation
Stenosis
malignancy

4

- Hour glass stomach

2- Malignant change:

malignant gastric ulcer

I-Perforated peptic ulcer

Pathology:

- Perforation of peptic ulcer occurs due to deepening of the ulcer through the wall
- Perforation occurs due to exposure to ulcerogenic agent
- Chronic DU > Chronic GU
- Anterior wall ulcer > posterior wall
- According to the rate of perforation and the size of perforation the degree of peritonitis develops

Stages:

a.Chemical irritation:

irritation of the peritoneal membrane by the escaped acid contents first at the upper abdomen then escape of the contents to the supra-hepatic, hepato-renal and right para-colic gutter to the right iliac fossa

b.Stage of illusion:

in which the peritoneal membrane secretes a large amount of alkaline fluids to neutralizes the acidic fluid → temporary relieve

c.Stage of septic peritonitis:

infection occurs by the gut bacteria (gram –ve bacilli) → worsen the condition and may lead to septic shock

Clinical presentation:

1. history of peptic ulcer dyspepsia
2. **Chemical irritation :**
   - Sudden severe pain in the epigastrium then becomes generalized
   - General examination: tachycardia, normal temperature, rapid respiration
   - Local exam: severe tender epigastrium then become generalized + rebound tenderness and guarding rigidity extends to right iliac fossa
3. **stage of illusion:**
   - Partial relief of symptoms due to alkaline peritoneal effusion
   - **Generally** : Tachycardia, low grade fever and looks better but still ill.

Local Examination :

Inspection: Abdominal distension. (ileus + effusion)

Palpation: Tenderness, rebound tenderness and guarding decreased but still present Percussion: hyperresonance + shifting dullness

Auscultation: silent abdomen (due to ileus)

4. Stage of septic peritonitis:

- Pain is more severe agonizing and generalized
- **Generally** : Tachycardia, high **fever**, tachypnea and toxic look
- **Local Examination** :
- Inspection: Abdominal distension, Palpation: tenderness, rebound and board like rigidity
- Percussion: hyper-resonance + shifting dullness
- Auscultation: silent abdomen (due to ileus)

How to diagnose perforated peptic ulcer:

1. Clinical suspicion

2. Investigations

How to investigate?

A. Radiological:

1. Plain x-ray abdomen in erect position (very important):

Air under diaphragm (= perforated viscus)

2. Abdominal ultra-sound or C.T. abdomen (basic investigation)

- Detect the presence of peritoneal fluid collection
- Exclude other causes of acute abdominal pain as acute cholecystitis, pancreatitis or leaking abdominal aortic aneurysm

3.Gastro-graffin meal:

may be used to detect escape of the dye into the peritoneal cavity in doubtful cases.(not commonly used)

B. Lab.:

- CBC: shooting Leucocytosis.
- Serum amylase and lipase to exclude pancreatitis.
- Cardiac enzymes: to exclude MI (in doubtful cases)

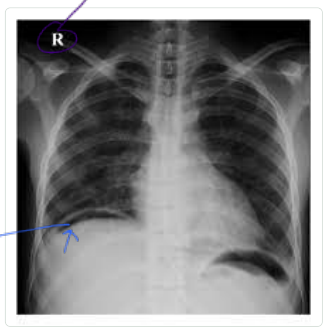


Figure 3.8 — Air under diaphragm

D.D. from all causes of acute abdomen

1- Acute cholecystitis
2- Acute pancreatitis
3- Acute peptic ulceration and gastritis
4- Acute liver abcess
5- Acute pyelonephritis
6- MVO
7- Primary peritonitis
8- Strangulation
9- Others as MI

You need both clinical diagnosis + investigations to reach proper diagnosis.

Management:

General treatment (preparation for surgery):

- Insertion of nasogastric tube
- Intravenous fluids (ringer and saline)
- Broad spectrum antibiotics for infection especially against gram negative bacilli (third generation cephalosporins)
- Intravenous proton pump inhibitor
- ~~Sedative for pain.~~

Surgical exploration of the abdominal cavity:

- Mid line incision
  - Thorough examination of the abdominal cavity to detect other pathologies
  - Peritoneal toilet and irrigation with normal saline and remove all septic collections in different peritoneal spaces.
  - Determine the site of perforation (gastric or duodenal)
  - **Closure of the perforation with omental patch fixed with 3 or 4 overlying**

stitches at the site of perforation ("Graham's operation")

- In case gastric ulcer in suspicious case for malignancy gastrectomy can be done.
- Leave drains inside the peritoneal cavity [↓ postoperative leak like any procedure but]

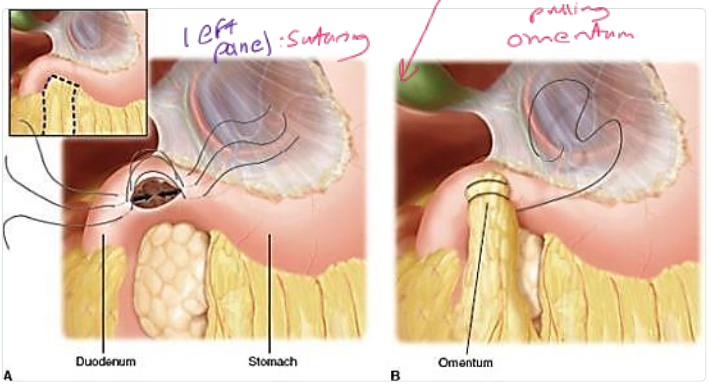


Figure 3.9 — Closure + Omental patch closure

II- Surgery for bleeding peptic ulcer:

GIT bleeding topic

Indications:

1- Failure the above measures to control bleeding with persistent shock state

2- Recurrence of bleedings with failed conservative measures.

3- Old asthero-sclerotic patient (prone for severe and recurrent bleeding).

4- Endoscopic evident of large vessel at the floor of the ulcer (prone for sudden severe bleeding).

① Endoscopy
- Adrenaline injection
- thermal coagulation
- Clipping
or Angiographic embolization

② angiography

③ Surgery [suturing + vagotomy]

NOW: IF failed endoscopic control, consider transcatheter arterial embolization [TAE]

Type of surgery:

- exploration of the **bleeding duodenal ulcer** and **under-running** of the ulcer base with sutures + **vagotomy (commonly used procedure)**
- In bleeding suspicious **gastric ulcer**, partial gastrectomy might be required.

III- Pyloric obstruction (stenosis)

تضيق البواب

Problem solving (history of ulcer dyspepsia + picture of Gastric Outlet Obstruction)

Pathogenesis:

i.e at gastric outlet

long standing pre-pyloric or duodenal ulcer healed by excessive fibrosis leads to narrowing, stenosis and gastric outlet obstruction.

Clinical presentations:

Symptoms:

- History of long duration with ulcer dyspepsia
  - Pain: dull aching, in the central part of the upper abdomen. Pain was more in empty stomach and relieved with intake of food. Pain is periodic in nature.
  - Vomiting: is projectile in nature, contains food materials taken more than 12 hours ago, more in the evening, sour taste (acidic) and is colorless (does not contain bile).
  - Abdominal distension (dilated stomach)
  - Patient complains of sensation of a rolling mass in the abdomen moving from left to right more after taking meals.

General examination:

- In severe cases signs of dehydration (dry tongue, sunken eyes,...)

Local examination:

- Inspection: upper abdominal distension – visible peristalsis running from left to right • Palpation: tender epigastrium – succussion splash (retained fluids)
  - Percussion: hyper-resonance
  - Decreased intestinal sounds

Complications:

- Dehydration
- Persistent and recurrent vomiting lead to hypo-natremia, hypokalemia, hypocalcemia, hypocholeremia and metabolic alkalosis + paradoxical aciduria
- Repeated aspiration with chest infection

Differential diagnosis: (clinical + investigational)

1. Gastric outlet obstruction due to chronic duodenal ulcer
2. Malignant tumor at the pyloric area
3. Extrinsic compression near the pylorus due to enlarged lymph node mass (due to lymphoma for example)
4. Lymphoma of distal stomach
5. Gastric outlet obstruction due to a bezoar ingestion (foreign body).
6. Cancer head of the pancreas causing gastric outlet obstruction.
7. Adult hypertrophic pyloric stenosis.

Investigations:

1) Upper endoscopy:

- Markedly dilated stomach
  - Excess old food residue in the stomach
  - Signs of mucosal irritation, hyperemia, due to stasis
  - Evidence of scarring at the first part of duodenum or pyloric canal
  - Biopsy from suspicious lesions
  - Exclude other luminal causes of gastric outlet obstruction.

2)Barium meal:

- hugely dilated stomach (may reach up to the pelvis)
  - delayed passage of dye to the duodenum
  - Soap-dish appearance (fluid level in the stomach)

3)Abdominal ultra-sound:

detect other causes of Gastric Outlet Obstruction

4) Other radiological investigation as C.T. when suspecting other causes of G.O.O.

5) Lab.:

- Renal functions
- Blood for serum electrolyte Na, K , Cl, Ca, pH

Treatment of G.O.O. due to Chronic Peptic Ulcer:

I. General pre-operative treatment:

- Insert naso-gastric tube: for suction of the gastric contents and relieve distension • Correction of dehydration and electrolytes disturbances by intravenous fluid and crystalloids administration.
  - Treatment of associated chest infections.

II. Surgical management:

1) Truncal vagotomy and gastro-jejunostomy.

2) Highly selective vagotomy with gastro-jejunostomy.

3) Post-operative proton pump inhibitors.

4) Partial gastrectomy + vagotomy + drainage in hugely dilated stomach

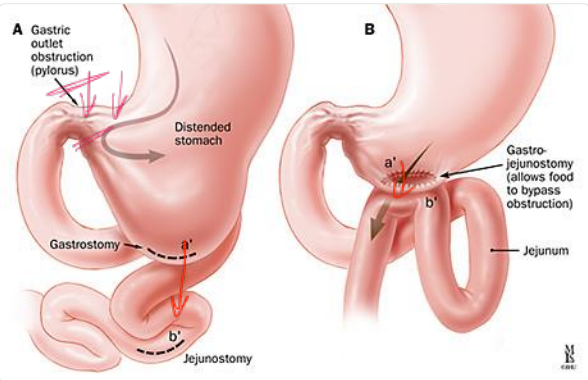


Figure 3.10 — Surgical management of gastric outlet obstruction

Congenital hypertrophic pyloric stenosis (CHPS)

It is a case of congenital hypertrophy of the pyloric sphincter detected in newborn or during infancy.

Incidence: 2 to 5 per 1,000 births /year. The male: female ratio is 4:1.

Etiology: familial inheritance plays a role a) Deficient vagal innervation of the muscular layer leads to failure of relaxation of the pyloric muscle.

b) Increased synthesis of growth factors, and subsequent pyloric muscle hypertrophy, hyperplasia leads to obstruction of the pylorus.

c) The hyper-gastrinemia: increases acid production and repeated pyloric contractions leads to muscle hypertrophy and subsequent obstruction.

Patho-physiology:

- Pyloric obstruction leads to increased gastric contractility to overcome the obstruction → gastric dilatation → projectile vomiting →loss of HCL → Metabolic alkalosis and dehydration and loss of sodium leads to hyponatremia.
- A secondary hyper-aldosteronism develops due to the decreased blood volume. The high aldosterone level causes the kidneys to retain Na+ (to correct the intravascular volume depletion), and excrete increased amounts of K+ into the urine resulting in a low

  blood level of potassium →hypokalemia and increased H+ loss in urine → paradoxical aciduria.
- Net results are hypochloremia, hypokalemia, hyponatremia, metabolic alkalosis and paradoxical aciduria.

Signs and symptoms

- Babies with this condition usually present any time in the first weeks to months of life with progressively vomiting.

- The vomiting is non-bile stained and projectile.
- Constant hunger, belching, colic and the baby is unable to eat properly.
- Dehydration

Abdominal examination:

- Inspection: Epigastric fullness (stomach distension) with visible peristalsis in the upper abdomen from the left to right.
- Palpation: firm olive shaped mass is felt in the epigastrium slightly to the right of midline (hypertrophy of the pylorus)

Complications of CHPS:

1. Dehydration and shock state in severe cases.
2. Electrolyte disturbances (hypokalemia, hyponatremia, hypocalcemia and metabolic alkalosis).
3. Recurrent aspiration and chest infections (pneumonia).

How to investigate?

1- Abdominal ultra-sound is usually diagnostic :

- Thickened pylorus (solid small mass)
- Muscle wall thickness > 3 millimeters (mm)
- pyloric channel length of > 15 mm

2- Plain X-ray abdomen: marked dilation of the stomach.

3- Upper endoscopy and gastro-graffin meal help in the diagnosis.

4- Lab. To detect electrolytes and acid base disturbances (Na, K, Ca, pH,..)

Treatment:

I. Correction of the general condition prior to surgery:

- Introduction of naso-gastric tube to decompress the stomach
- I.V. fluids administration to correct dehydration and electrolyte disturbances (lactated ringer solutions and normal saline)
- Administration of antibiotics to guard against infections.

II. Surgical management: Ramstedt's pyloro-myotomy:

- Laparotomy or laparoscopy
- Open the serosa at the pylorus.
- Dissect the longitudinal muscles without division.
- Expose the hypertrophied circular muscles.
- Partial division of the hypertrophied muscles to relieve the spasm.(care to avoid mucosal injury).

III- Endoscopic pyloro-myotomy: per-oral pyloro-myotomy:

Cancer stomach

- Age: old age
- Sex: males > females

Predisposing factors:

- Dietetic: spicy foods, excess salt intake, foods contain N-nitrosamine preservatives
- Smoking
- Any cause of chronic gastritis: as H.Pylori
- Pernicious anemia
- Genetic predisposition, blood group A
- Benign gastric ulcer – gastric polyp
- Following partial gastrectomy (bile reflux gastritis)

Site:

- Antrum, pyloric area → 60%
  - Body: 15 %
  - Fundus: 25%

Spread:

worsen the prognosis

a.Direct:

longitudinal, circumferential, and vertical within the layers of stomach.
When the tumor gets out of the serosa → it infiltrates the nearby structures as spleen, liver, colon and pancreas

b.Lymphatics:

to the peri-gastric L.N.

- Left gastric, right gastric, gastro-epiploic, pyloric L.N.
- Celiac L.N. → Thoracic duct → left supraclavicular L.N. = Wirchow's L.N.

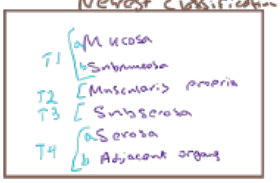
c. Blood spread mainly to the liver (portal circulation) **d. Transcelomic spread** → peritoneal nodules – infiltration of the ovary → **Krukenberg's tumor** or DOUGLAS pouch nodules.

What are TNM definition for carcinoma of stomach?

Hint: افكر ال 4 مستويات وبعدين زود على الرقم واحد

كلشان نعرف ال compare invasion

(T1a) (T1b) (invading mucosa reach up to serosa)



Tumor (T) depth

- T0 – no evidence of primary tumor
- TIS – carcinoma in situ
- T1 – tumor involving mucosa and submucosa
- T2 – tumor invading muscularis propria and extending up to the subserosa
- T3 – tumor invading the serosa but not to adjacent organs
- T4 – tumor invading the adjacent organs.

N (Lymph nodes):

AJCC 8th Edition (Newest)

- N0 – no nodes
- N1 – Metastasis in 1–6 lymph nodes (1-2)
- N2 – Metastasis in 7–15 lymph nodes (3-6)
- N3 – Metastasis in >15 lymph nodes [a: 7-15, b: >16]

M (Metastasis):

- M0 – no metastasis
- M1 – Distant metastasis present.

What are the common tumor markers in gastric cancers?

For prognosis & follow up... not purely diagnostic value

CA → carbohydrate antigen

CEA → Carcinoembryonic antigen

- **CEA, CA 19-9, CA 50, CA 125, CA 72-4** are the important tumor markers to evaluate recurrence and prognosis.
- Most important is **CA 72-4** which disappears with surgery and reappears with regional or distant recurrence.

Clinical presentations of cancer stomach:

A. Dyspeptic symptoms: in male above 50 y is important alarm, progressive:

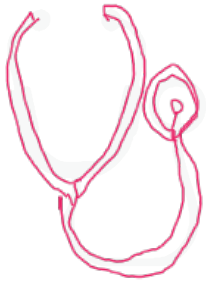
- Anorexia: is marked and more towards meat (descriptive association)
  - Nausea and vomiting
  - Epigastric pain
  - Hematemesis and melena [tumour Friable]

B. Asthenia, Anorexia, Anemia: 3As

وهن عام

Cancer = ↓intake + Systemic catabolic state + Sometimes chronic blood loss

- Unexplained weight loss in old age
  - Marked weight loss
  - Unexplained anemia in elderly



C.Mass group:

epigastric mass ; firm to hard fixed in late cases

↳ invade & adhere

D. Obstructive group:

- **Mass at gastric outlet:**

produces G.O.O. of rapid onset and progressive course with rapid marked weight loss .

→ Gastric outlet obstruction [G.O.O.]

- **Mass at the cardia :**

produces progressive dysphagia

E. Metastatic group:

↗ = i.e metastasis from another primary tumor ↘ developed there e.g. at stomach & transfered to liver

- Nodular hepatomegaly (secondaries), jaundice
  - Malignant ascites [i.e ascitis caused by the cancer] → hyperbilirubinemia due e.g. biliary obstruction
  - Enlarged left supraclavicular L.N. (Troisier sign). (via thoracic duct as taken in lymphatic spread)

Investigations:

Upper endoscopy:

most valuable, visualize the pathology – take a biopsy

Endo-sonar: AKA "Endoscopic ultrasound" (EUS)

determine infiltrations within the gastric wall

Barium meal:

Detect the lesion in the form of filling defect, pyloric obstruction, malignant stricture, linnitis plastica

narrowing

↓ mass occupies space where barium should be

linn- = تشكك
itis = التهاب
plastica = Fibrous الياف
= Rigid inflamed wall (thick)

Poor distensibility on barium video (live - X-ray video) ↙ continuous x-rays

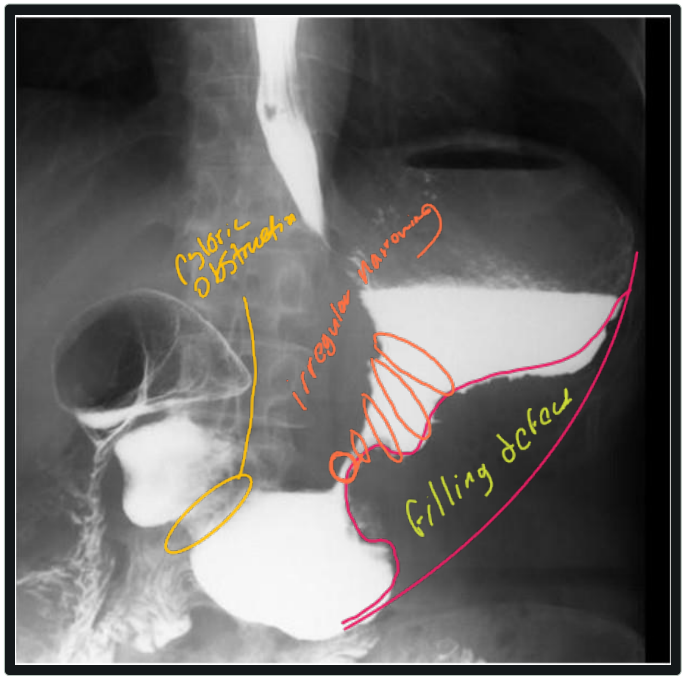


Figure 3.11 — Large gastric cancer

- **Abdominal ultrasound – C.T.** to assess spread outside gastric wall and distant spread.
- **Lab.** : CBC – anemia

Liver and kidney functions

Tumor makers: for prognosis and follow up

Treatment:

I. Early cases:

curative treatment:

1. Minimal surgery in the form of endoscopic **submucosal resection (EMR)** is being tried in some centers; in:

- Tumor smaller than 2 cm
- Tumor confined to the mucosa.
- No lymph-vascular invasion
- No surface ulceration.

2. Tumor larger than 2 cm, submucosal invasion, and surface ulceration or with lympho- vascular invasion should undergo gastrectomy with appropriate lymph node dissection. • Gastric resection involving the tumor with safety margin at least 5 cm proximal and 2 cm of the duodenum distally

- Removal of the omentum
- Removal of the draining L.N.

a)Lower radical gastrectomy:

for tumors of lower 1/3 of stomach + gatro- jejunostomy = Billorth II

b)Total radical gastrectomy:

for proximal gastric tumors + roux en Y esophago- jejunostomy.

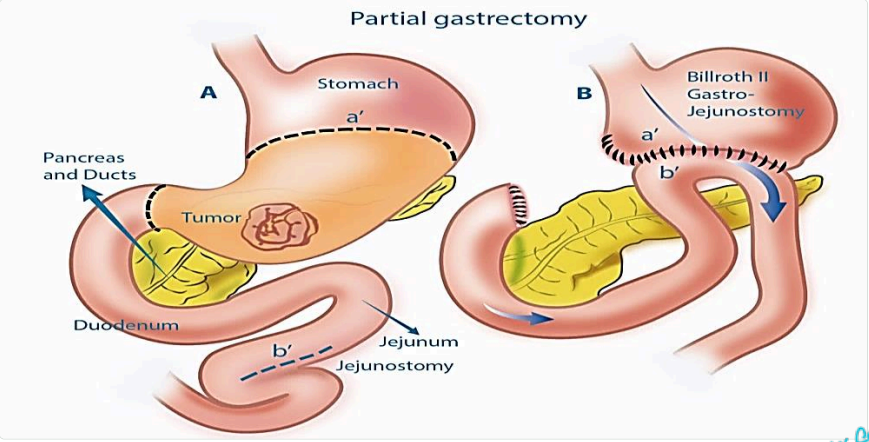


Figure 3.12 — Distal (lower radical gastrerctomy)

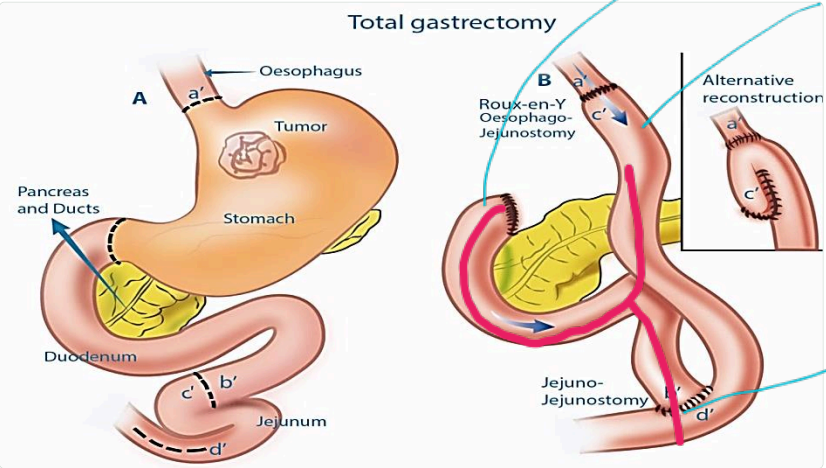


Figure 3.13 — Roux-en-Y anastomosis of the jejunum to the esophagus

II.Late cases of cancer stomach:

with advanced local spread or distant spread: • Radiotherapy and chemotherapy

- Palliative Gastro-jejunostomy in pyloric obstruction
- Feeding jejunostomy in advanced cases

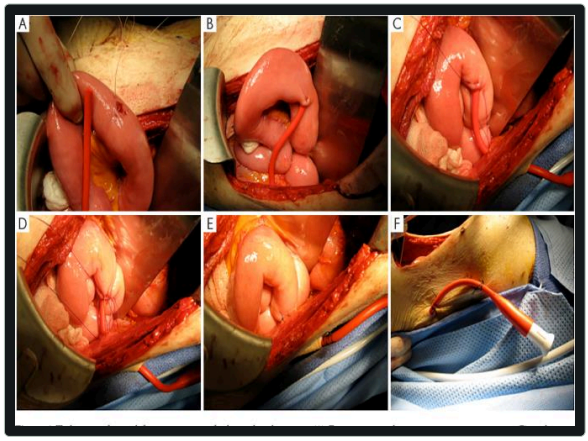


Figure 3.14 — Operative management of gastric cancer

Figure 3.15 — Feeding jejunostomy

Prognosis:

- Depends on tumor grading, stage, state of L.N. and distant spread
- The tumor usually discovered late so it has low 5 years survival rates.

توسع المعدة الحاد

Acute gastric dilatation (AGD)

Definition

Acute Gastric Dilatation (AGD) is a rare, life-threatening emergency characterized by rapid, massive enlargement of the stomach due to the accumulation of fluid, gas, or ingested material.

NB wall perfusion fails on massively distend stomach

Causes & Risk Factors

↑ Distension → ↑ Pressure → ↓ Blood supply → ischemia → necrosis → perforation

I- Mechanical Causes:

- Gastric Outlet Obstruction (GOO): Caused by severe peptic ulcer disease, pyloric stenosis, or periampullary/gastric malignancies.

- Superior Mesenteric Artery (SMA) Syndrome
- Gastric Volvulus

II- Functional Causes

- **Eating Disorders:** Anorexia nervosa, bulimia nervosa
- **Metabolic & electrolyte Disruption:** Acut severe hyperglycemia, & Diabetic Ketoacidosis. Electrolyte disturbances (severe hypokalemia, hypomagnesemia).
- **Iatrogenic & Trauma:** Cardiopulmonary resuscitation, post-operative ileus, abdominal trauma, or spinal cord injuries.
- High-dose anticholinergics, opioids, and anesthetics that suppress GI motility.

Complications: Gastric Perforation (Ischemic Necrosis) & Hemodynamic Collapse

Diagnosis:

Early clinical suspicion is vital, as mortality reaches 80–100% once ischemia or perforation occurs.

Physical Exam & History	Severe epigastric pain, marked abdominal distension, tympanitic percussion, succussion splash, and retching.
Abdominal Plain Film	A massively dilated gas- or fluid-filled stomach occupying the abdomen down to the pelvis, displacing bowel loops.
Abdominal CT with IV Contrast (Gold Standard)	Defines gastric dimensions, confirms NG tube position, evaluates wall enhancement

DIAGNOSIS OF ACUTE GASTRIC DILATATION

Treatment Protocol

Management depends on whether gastric perfusion is intact or compromised.

1. **Immediate Resuscitation & Decompression**
   - **Nasogastric Decompression**
   - **Fluid & Electrolyte Restoration**
   - **Gastric Protection:** IV proton pump inhibitors (PPIs) to reduce gastric secretion volume and mucosal acid injury.
2. **Monitoring:** conservative management with serial abdominal exams and fasting:

   If the stomach successfully decompress via NG tube, pain resolves, and imaging shows no signs of ischemia or perforation
3. **Surgical Intervention:**

   Emergency exploratory laparotomy is indicated if:
   - NG tube decompression fails, CT demonstrates pneumoperitoneum
   - Clinical signs of peritonitis, severe septic shock

Surgical Procedures:

- **Gastrotomy & Decompression:** Direct operative evacuation of unsuctionable food material or bezoars.

- **Resection:** Debridement of all non-viable gastric tissue. Depending on the extent of necrosis, this ranges from partial gastrectomy to total gastrectomy
- **Abdominal Washout & Drainage:** In cases of perforation, irrigate the peritoneal cavity and place abdominal drains.

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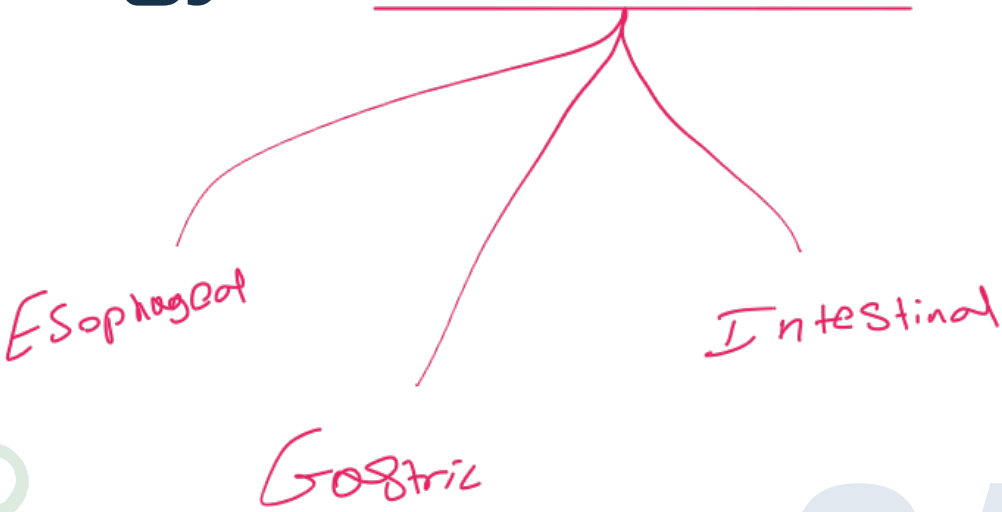
CHAPTER 04

Pathology of GIT tumours



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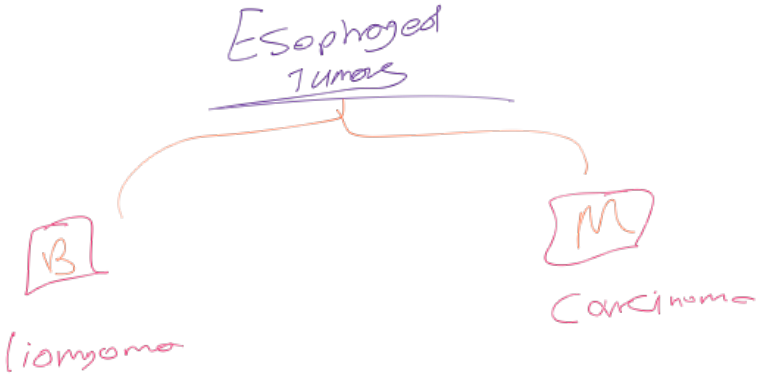


TUMORS OF ESOPHAGUS

- Benign tumors of the esophagus are less common than malignant tumors. The most common are leiomyomas.
- Malignant tumors may be either

a. Carcinoma most common

b. Lymphoma & other sarcomas rare

They present by esophageal obstruction, dysphagia, bleeding, or rarely, paraneoplastic syndrome.

Prognosis of esophageal carcinoma is usually poor (5 year cure rate about 10%)

Carcinoma of esophagus

- Squamous cell carcinoma - More prevalent worldwide 50% in middle 1/3

Risk factors: long-standing esophagitis, achalasia, smoking, alcohol, diet (low vitamins and zinc), genetics

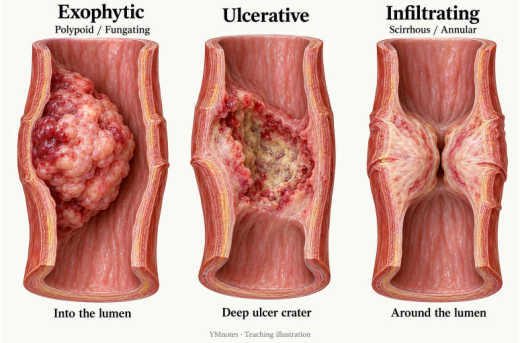
- Adenocarcinoma - More common in USA - More in lower 1/3., Occurs on top of Barrett esophagus, preceded by dysplastic changes

Gross Pathology

1. Exophytic (Polypoid or Fungating). Growth filling the lumen (Most common type).
2. Ulcerative type, which invade deeply into the surrounding structure. With raised reverted edge, rough necrotic floor and indurated base.
3. Infiltrating (Schirrhous) tumor encircling the tube, also called (Annular) & cause constriction (Is the least frequent type).

Histopathology

1. Squamous Cell Carcinoma: Occur in >90 % of cases.
2. Adenocarcinoma (5%): Originating in the cardiac end of the esophagus from dysplastic mucosa in the setting of Columnar Metaplasia
3. Undifferentiated Carcinoma: lacking squamous features & accompanied by lymphocytic infiltration.



Gastric tumors

Benign Neoplasms

Epithelial polyps are rare in the stomach. Three types occur:

1- Hyperplastic polyps (70%),

2- Fundic gland polyps (20%),

3- Adenomatous polyps (10%). Shows dysplasia. The risk of carcinoma is moderate in adenomatous polyp, slight in hyperplastic polyp, and nil in fundic gland polyps.

Gastric polyps appear as small pedunculated lesions often multiple, that can be removed endoscopically. Large polyps are very rare

Mesenchymal Neoplasms : Benign mesenchymal neoplasms are uncommon.

Malignant Neoplasms

Gastric Adenocarcinoma

Sex: male > female Geography: Japan, Chile, Iceland Age > 50

Predisposing factors

Blood group A = 20% increased risk , Adenomatous polyp ,Pernicious anemia (atrophic gastritis type A) , Chronic gastritis associated with Helicobacter pylori infection

Grossly it may appear as: -

-Early gastric cancer. defined as gastric carcinoma restricted to the mucosa and submucosa - Appears as a small, flat mucosal thickening that may have a minimal polypoid and ulcerative component.

- Late gastric cancer... defined as a gastric carcinoma that has invaded the muscle wall. It may present in various ways:

(1) Fungating mass that protrudes into the lumen.

(2) Malignant ulcer with raised, everted edges

(3) Diffusely infiltrating lesion that causes thickening and contraction of the stomach wall with relatively little mucosal involvement (linitis plastica, or leather bottle stomach).

It may take one of two types

1- Intestinal-type

- Risk factors: diet (nitrites, smoked food, increased salt), chronic gastritis (H. pylori), altered anatomy after resection
- On top of intestinal metaplasia

- Histopathology : Glandular morphology

Diffuse-type:

- Undefined risk factors (no known relation to H. pylori)
- Signet cell morphology

Spread of gastric carcinoma –

- Direct spread.... Pancreas, liver, esophagus. –
- Lymphatic spread : Draining lymph nodes Thoracic duct left supra-clavicular L.N. –
- Blood spread: . Liver
- Transcoelomic, ovary (Krukenberg) , peritoneal implants

Tumors of the Colon and Rectum

Neoplastic adenomatous polyp of the colon –

Adenomas of the colon are present in 20–30% of all individuals over the age of 50 years.

1- Tubular adenomas (pedunculated)

2- Tubulo -villous adenomas

3- Villous adenomas – (sessile)

Colonic adenomas are associated with genetic mutations identical to those seen in colon cancer. The number of mutations in adenomas is less than in carcinoma but increases as the adenoma enlarges and becomes more dysplastic.

Tubular adenoma

- Tubular adenomas account for over 90% of colonic adenomas.
- They are commonly multiple, with 10–20 lesions present in some patients,
- They are pedunculated with a well-defined stalk –

Histologically, a tubular adenoma is composed of benign neoplastic glands bunched together above the muscularis. The epithelial cells are dysplastic (hyperchromatic and stratified and show loss of normal mucin content)

Villous Adenoma

Villous adenomas are uncommon, comprising less than 10% of colonic adenomas. - They commonly occur in older individuals as a single large, sessile lesion have a broad base of attachment to the mucosal surface without a defined stalk.

Histologically, villous adenoma is composed of neoplastic proliferation of colonic epithelial cells organized into long finger-like papillary or villous processes. - The covering epithelium is dysplastic.

Tubulo villous Adenoma is a Mix of both

Inherited Polyposis Syndromes

(Adenomatous Polyposis Coli APC - Familial Adenomatous Polyposis (FAP), - Gardner Syndrome. - Turcot Syndrome. - Hereditary Non-polyposis Colorectal Cancer Syndrome (HNPCC)

Familial Adenomatous Polyposis (FAP)

Autosomal dominant - average onset is 25 years old -Adenomatous polyps, mostly colorectal. Patients, on average, have 1,000 polyps -Minimum of 100 polyps are necessary for diagnosis -The genetic defect is on chromosome 5q21, APC (Adenomatous Polyposis Coli) tumor suppressor gene - Precancerous: virtually 100% will develop cancer within 10 to 15 years

Malignant colonic tumors

I- Epithelial Adenocarcinoma and mucoid adenocarcinoma

II- Mesenchymal - Gastro-intestinal Stroma Tumor (GIST) - Lymphoma Colonic

Adenocarcinoma

Risk factors: low fiber, high fat, decreased vit A, C, E, idiopathic inflammatory bowel disease, familial adenomatous polyposis - Several hits to different genes: APC, k-ras, p53; or DNA mismatch repair genes

-Survival depends on stage (depth of invasion and node metastasis) -Incidence - Primarily occurs in patients between 60 and 79 years of age - Ratio of male to female patients 2:1 for rectal cancers, 1:1 before 50)

Gross appearance:- -

Sites:- Rectosigmoid (50%), Coecum any other site

Shape: Fungating -Ulcerating or infiltrating (annular) (some)

↳ Obstruction is uncommon due to large caliber of colon

Histopathology

- Glands have lost polarity, disorganized and adhere to each other
- Epithelial cells are hyperchromatic, tightly packed, and have a higher nucleus-to cytoplasm content (appear more malignant).
- The malignant acini are separated by desmoplastic stroma.
- Tumor invades deeply into colonic mucosa
- Tumor cells may continue to produce mucin; in 10-15% of patients, mucin lakes can be seen.

Course: tumor invades bowel wall and lymphatics/blood vessels with metastasis to lymph nodes, liver, lungs, and bones.

25% of patients have metastatic disease at presentation.

Diagnosis is based on :

- Endoscopy and biopsy.
- Increased serum CEA which is more valuable in follow up after surgical removal of a tumor.

(as non-specific)

Intestinal carcinoid

tumor

Carcinoid tumors arise from neuroendocrine cells present in the mucosa throughout the gastrointestinal tract.

Grossly,

- Carcinoid tumors are firm and yellow. They arise in the submucosa, elevating and ulcerating the mucosa and locally infiltrating the wall. –

Multiple lesions are present in 25% of cases.

Microscopically - Sheets of monotonous, polyhydral cells with deeply stained nuclei. - Stained with silver or by IHC for chromogranin or neurone specific enolase.

↳ contents of the cell < Affect tricuspid valve of heart as well >

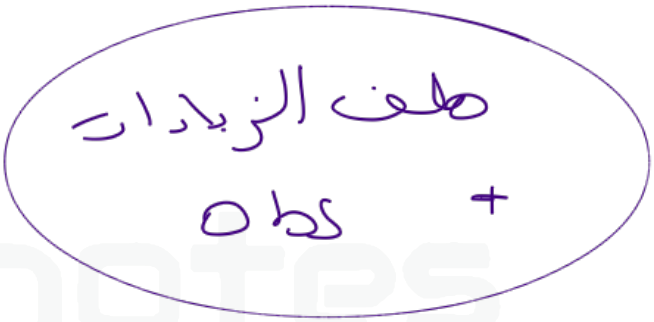
Gastrointestinal stromal tumor (GIST)

Gastrointestinal stromal tumors are the most common mesenchymal neoplasms of the gastrointestinal tract. GISTs arise in the smooth muscle pacemaker interstitial cell of Cajal.

- Grossly: large, bulky, intramural masses with tannish brown cut section with hemorrhage, necrosis and cystic softening.
- Microscope: Mesenchymal tumor usually centered in the muscularis propria
- There are two types:

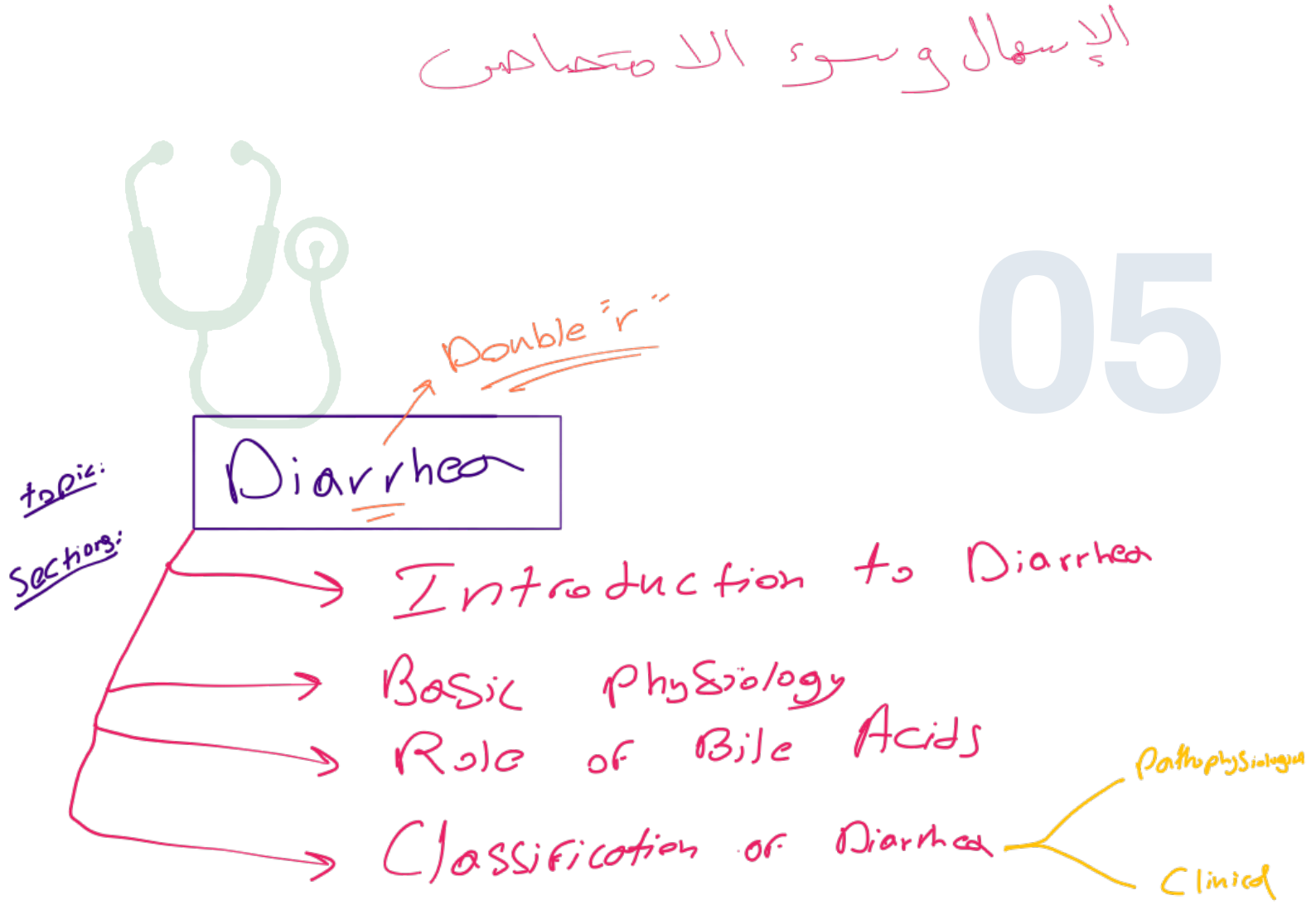
1-Spindle cell GISTs: intersecting fascicles of plump spindled cells with eosinophilic cytoplasm within variably hyalinized or edematous stroma

2- Epithelioid GISTs: rounded epithelioid cells



CHAPTER 05

DIARRHEA AND MALABSORPTION



Objectives

The student will

1) Know the definition of acute and chronic diarrhea

2) Understand the basic mechanisms and causes underlying acute and chronic diarrhea 3) Know the clinical presentation of various forms of acute and chronic diarrhea

4) Know the diagnostic studies for acute and chronic diarrhea

5) Understand the basic treatment options for acute and chronic diarrhea

6) Understand the basic concepts and mechanisms for malabsorption.

7) Know the various causes of malabsorption

8) Know the clinical features of various types of malabsorption

9) Know the diagnostic tools for malabsorption

10) Know the basic treatment options for different types of malabsorption.

Diarrhea

- Diarrhea is defined as an increase in the fluidity, frequency, and volume of daily stool output.
- Normal stool frequency ranges from three times a week to three times a day.
- Perception as normal bowel pattern is related to consistency and frequency.
- Consistency is defined as the ratio of fecal water to the water-holding capacity of fecal insoluble solids.
- Acute diarrheas are those lasting less than 2 to 3 weeks or, rarely, 6 to 8 weeks.
- Approximately 80% of acute diarrheas are due to infections with viruses, bacteria, and parasites.
- Diarrhea lasting >4 weeks warrants evaluation to exclude serious underlying pathology. • In contrast to acute diarrhea, most of the causes of chronic diarrhea are noninfectious.

Basic physiology

- During fasting, the intestine contains very little fluid.
- On a normal diet of three meals a day, about 9 L of fluid is delivered to the small intestine.
- 90% is absorbed in the small intestine.
- One to two liters is presented to the colon, and approximately 90% is absorbed.
- The colon has the capacity to absorb all of the fluid presented to it.
- This results in 100 to 200 mL of fluid in the stool.
- Feces on the average contain 100 mL of water, 40 mEq/L of sodium (Na+), 90 mEq/L of potassium (K+), 16 mEq/L of chloride (Cl-), and 30 mEq/L of bicarbonate (HCO3-). • Water transport across the intestinal epithelium is passive.

- Na+ is the major ion actively absorbed, and Cl- is the major ion secreted.
- Water is absorbed along with Na+.
- Active Cl- secretion or inhibited Na+ absorption also creates an osmotic gradient favorable for the movement of fluids from blood to lumen and is the basis of the secretory diarrheas.

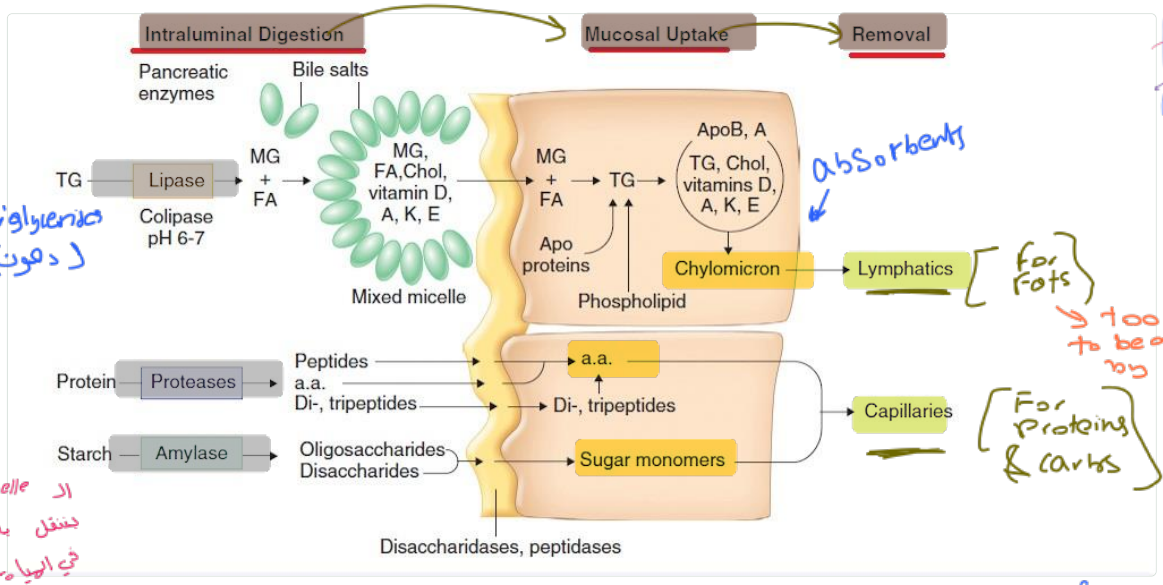


Figure 5.1 — Digestion and absorption

Role of bile acids

- Bile acids are not present in the diet but are synthesized in the liver by a series of enzymatic steps that also include cholesterol catabolism.
- Bile acids are either primary or secondary: primary bile acids are synthesized in the liver from cholesterol, and secondary bile acids are synthesized from primary bile acids in the intestine by colonic bacterial enzymes.
- The primary functions of bile acids are (1) to promote bile flow, (2) to solubilize cholesterol and phospholipid in the gallbladder by mixed micelle formation, and (3) to enhance dietary lipid digestion and absorption by forming mixed micelles in the proximal small intestine.

Defects Affecting Bile Acids

Process	Pathophysiologic Defect	Disease Example
Synthesis	Decreased hepatic function	Cirrhosis
Biliary secretion	Altered canalicular function	Primary biliary cirrhosis
Maintenance of conjugated bile acids	Bacterial overgrowth	Jejunal diverticulosis
Reabsorption	Abnormal ileal function	Crohn's disease

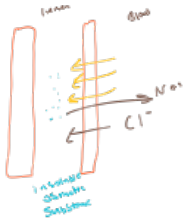
Figure 5.2 — Defects affecting bile acids

Classification of diarrhea

pathophysiologic classification			
Osmotic diarrhea	Secretory diarrhea	Exudative diarrhea	Altered motility

Osmotic diarrhea:

There is increased amounts of poorly absorbable, osmotically active solutes in the gut lumen.



Secretory diarrhea:

There is, increased Cl- and water secretion with or without inhibition of normal active Na+ and water absorption.

Exudative diarrhea:

There is mucus, blood, and protein from sites of active inflammation into the bowel lumen

Altered motility:

There is increased or decreased contact between luminal contents and mucosal surface.

clinical classification		
acute	chronic	refractory

Causes of osmotic diarrhea

Carbohydrate malabsorption:

1) Disaccharidase deficiency in the small-intestinal mucosa:
   - **Primary** (e.g., idiopathic lactase, sucrase-maltase deficiency)
   - **Secondary** (e.g., after infectious gastroenteritis and with mucosal inflammation)
2) Ingestion of mannitol, sorbitol
3) Lactulose therapy

General malabsorption

Sprue (idiopathic gluten enteropathy, tropical)

Ingestion of sodium sulfate (Glauber's salt), sodium phosphate, magnesium sulfate

Secretory Diarrhea

- Diarrhea of 1 L or more per day results from secretion of fluid across the intestinal mucosa.
- In most cases, a pathophysiologic event causes small intestinal secretion by simultaneously stimulating active secretion and partial inhibition of intestinal absorption.
- Often, the intestinal mucosa is intact and has normal histologic findings
- Clinically, there are features that characterize secretory diarrhea:
- Stool volume is usually large (>1 L per day).
- Stools are watery in consistency.
- Stools do not contain pus or blood.
- Diarrhea typically continues while the patient fasts for 24 to 48 hours.

Causes of secretory diarrhea

Enterotoxins

Vibrio cholerae

Staphylococcus aureus

Hormonal secretagogues

- Vasoactive intestinal polypeptide (pancreatic cholera syndrome)
- Serotonin (carcinoid)

Gastric hypersecretion

Zollinger-Ellison syndrome

Laxatives

Fatty acids

Pancreatic insufficiency

Exudative Diarrhea

- If the intestinal mucosa is inflamed and ulcerated, mucus, blood, and pus leak into the lumen and are discharged as stool.
- This may also create an increased osmotic load.
- If a large surface area of the bowel lumen is involved, absorption of ions, solutes, and water will also be impaired, and patients may have large-volume diarrhea.
- Inflammation may generate prostaglandins, which stimulate secretion and may increase bowel motility, thus compounding the diarrhea.
- The severity of the diarrhea and the systemic signs and symptoms depends on the extent of bowel involvement.

Causes of exudative diarrhea

Idiopathic (e.g., Crohn's disease, ulcerative colitis)

Infectious (e.g., from invasive organisms or cytotoxins: Shigella, Salmonella, Campylobacter, Yersinia, tuberculosis, amebae, and Clostridium difficile)

Ischemic

Vasculitic

Due to radiation injury

- **Motility Disturbances**
  - Both reduced and increased motility of the intestine may result in diarrhea.
  - Increased Motility Of The Small Intestine: Results in decreased contact time of chyme with absorptive surfaces.
  - Large amounts of fluid delivered to the colon may overwhelm its absorptive capacity and result in diarrhea.
  - The reduced contact time in the small intestine may interfere with absorption of fatty acids and bile salts, allowing them to reach the colon where they induce a secretory diarrhea.
  - Decreased Motility Of The Small Intestine May allow colonization of the small intestine with colonic-type bacteria. (leading to decreased absorption)

Clinical classification of diarrhea

Acute diarrhea

- Approximately 80% of acute diarrheas are due to infections with viruses, bacteria, helminthes, and protozoa.
- The remainder are secondary to the ingestion of medications, poorly absorbed sugars

(fructose polymers or sorbitol.

- Most infectious diarrheas are acquired through fecal-oral transmission from water, food, or person-to-person contact.
- Patients with infectious diarrhea often complain of nausea, vomiting, and abdominal pain and have watery, malabsorptive, or bloody diarrhea and fever (dysentery). (= Bloody diarrhea)
- Food poisoning: is produced by a preformed bacterial toxin that contaminates the food. • Bacterial replication in the host is not necessary for the development of disease.
- Diarrhea resulting from multiplication of organisms in the intestine may be divided into inflammatory-invasive versus noninflammatory-noninvasive categories.

Causes of Acute Diarrhea According to the mechanism

Small intestine is generally minimally inflamed (or non-inflammatory) due to superficial epithelial affection

Colon is generally highly inflamed due to deep mucosal invasion (with the role of immunity)

Causes	Mechanism	Organisms
Viral Infections Small intestine	Mucosal invasion absent. Noninflammatory Watery diarrhea. Fecal leukocyte absent.	Rotavirus (children, adults) Norwalk virus
Bacterial infections Small intestine	Mucosal invasion absent Noninflammatory Watery diarrhea Fecal leukocytes absent	Vibrio cholerae Toxigenic Escherichia coli
Colon	Mucosal invasion present Inflammatory Bloody diarrhea Fecal leukocytes present	Salmonella Shigella Invasive E. coli
Parasitic infections Small intestine	Mucosal invasion absent Noninflammatory Watery diarrhea Fecal leukocyte absent	Giardia lamblia Cryptosporidium
Colon	Mucosal invasion present Inflammatory Bloody diarrhea Fecal leukocytes present	Entamoeba histolytica
Food poisoning Small intestine	Toxin induced Mucosal invasion absent Noninflammatory Watery diarrhea Fecal leukocytes absent	Staphylococcus aureus Clostridium botulinum

Remarks about treatment of acute diarrhea

- Acute diarrhea with fluid and electrolyte depletion is a major cause of mortality, especially in children in developing countries of the world.
- Fluid repletion by intravenous or oral routes can prevent death.
- Oral rehydration therapy can be accomplished with a simply prepared oral rehydration solution.
- The use of Opiate derivatives include paregoric (tincture of opium), diphenoxylate with atropine (Lomotil), and loperamide (Imodium) may help in certain selected cases.
- They should not be used in patients with fever, systemic toxicity, or bloody diarrhea.
- Antimicrobials should be only used when there is evidence of mucosal invasion.

Chronic Diarrhea

- Diarrhea lasting >4 weeks warrants evaluation to exclude serious underlying pathology. • In contrast to acute diarrhea, most of the causes of chronic diarrhea are noninfectious. **CAUSES**

Infections.

- Giardia lamblia.
- Entamoeba histolytica.
- Tubercle bacillus.
- Clostridium difficile.

Inflammatory bowel disease:

Ulcerative colitis and Crohn's disease may result in diarrhea of varying severity, depending on the extent and degree of bowel involvement. Diarrhea in Crohn's disease of the small bowel may be compounded by concomitant bile salt and fat malabsorption.

Malabsorption syndromes

Endocrine disorders:

Zollinger-Ellison syndrome. Hyperthyroidism. Carcinoid. Non-β- cell pancreatic tumor.

Neoplasms

Drugs and laxatives

Irritable bowel syndrome

Refractory diarrhea

Refractory chronic diarrhea unresponsive to specific antimicrobial therapy or standard unspecific medication may present a challenging and serious clinical problem.

Causes

- Protracted diarrhea in infants and children (more prone to infection + weaker immune system)
- Tropical sprue (Unknown cause)
- Persistent diarrhea in travelers (Stress-infection)
- Protracted infectious diarrhea in adults: Organisms that are difficult to diagnose and are known to cause protracted or prolonged diarrheas include enteropathogenic

(enteroadherent) E. coli , Giardia , amoeba , Cryptosporidium , Aeromonas , Yersinia enterocolitica , Clostridium difficile infection is often difficult to clear, and five or more relapses have been observed .

- Post-infectious diarrhea induced irritable bowel syndrome: After documented infectious diarrhea, 25% of patients experience pain, bloating, urgency, a sense of incomplete evacuation, and loose stools for 6 months or longer.

This syndrome of infectious diarrhea–induced irritable bowel syndrome, also called

Brainerd's diarrhea if it is particularly prolonged and accompanied by severe diarrhea.

MALABSORPTION

- Malabsorption is caused by many different diseases.
- Dietary fat is the nutrient most difficult to absorb.
- Fat is predominantly insoluble in the aqueous milieu of the intestine and critically depends on all phases of digestion, absorption, and delivery for its assimilation.
- Steatorrhea (excess fat in the stool) is the hallmark of malabsorption.

Malabsorption Syndromes

- Malabsorption is characterized by suboptimal absorption of fats, fat-soluble and other vitamins, proteins, carbohydrates, electrolytes and minerals, and water.
- At the most basic level, it is the result of disturbance of at least one of these normal digestive functions:

Intraluminal digestion.

Terminal digestion.

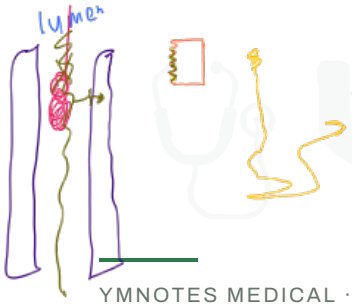
Transepithelial transport.

Intraluminal digestion
The process begins in the mouth with saliva, receives a major boost from gastric peptic digestion, and continues in the small intestine, assisted by pancreatic enzyme secretion and the emulsive action of bile.

Terminal digestion
which involves the hydrolysis of carbohydrates and peptides by disaccharidases and peptidases, respectively, in the brush border of the small intestinal mucosa

Transepithelial transport
in which nutrients, fluid, and electrolytes are transported across the epithelium of the small intestine for delivery to the intestinal vasculature or to the intestinal lymphatic system

Overall digestive and absorptive process in small intestine



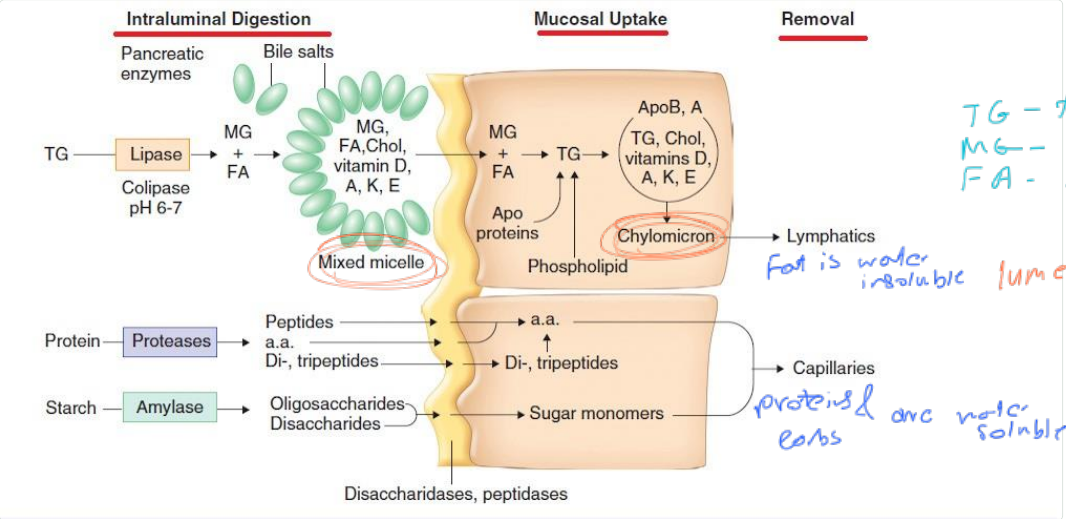


Figure 5.3 — Overall digestive and absorptive process

The Major Malabsorption Syndromes

- Disorders of absorption constitute a broad spectrum of conditions with multiple etiologies and varied clinical manifestations.
- Most, but not all, malabsorption syndromes are associated with steatorrhea, an increase in stool fat excretion of >6% of dietary fat intake.
- Some malabsorption disorders are not associated with steatorrhea.
- Disorders of absorption (malabsorption) must be included in the differential diagnosis of chronic diarrhea.
- Most patients will indicate that they have diarrhea, not that they have fat malabsorption.

MECHANISM OF MALABSORPTION	CONDITIONS
Impaired mixing	Partial/total gastrectomy Gastric bypass surgery
Impaired lipolysis	Chronic pancreatitis Pancreatic cancer
Impaired micelle formation	Severe chronic liver disease Cholestatic liver disease Bacterial overgrowth Crohn disease Ileal resection
Impaired mucosal absorption	Giardiasis Celiac disease Tropical sprue Radiation enteritis Whipple disease Lymphoma
Impaired nutrient delivery	Lymphoma Tuberculosis Constrictive pericarditis Severe congestive heart failure

CLINICAL CONSEQUENCES OF MALABSORPTION

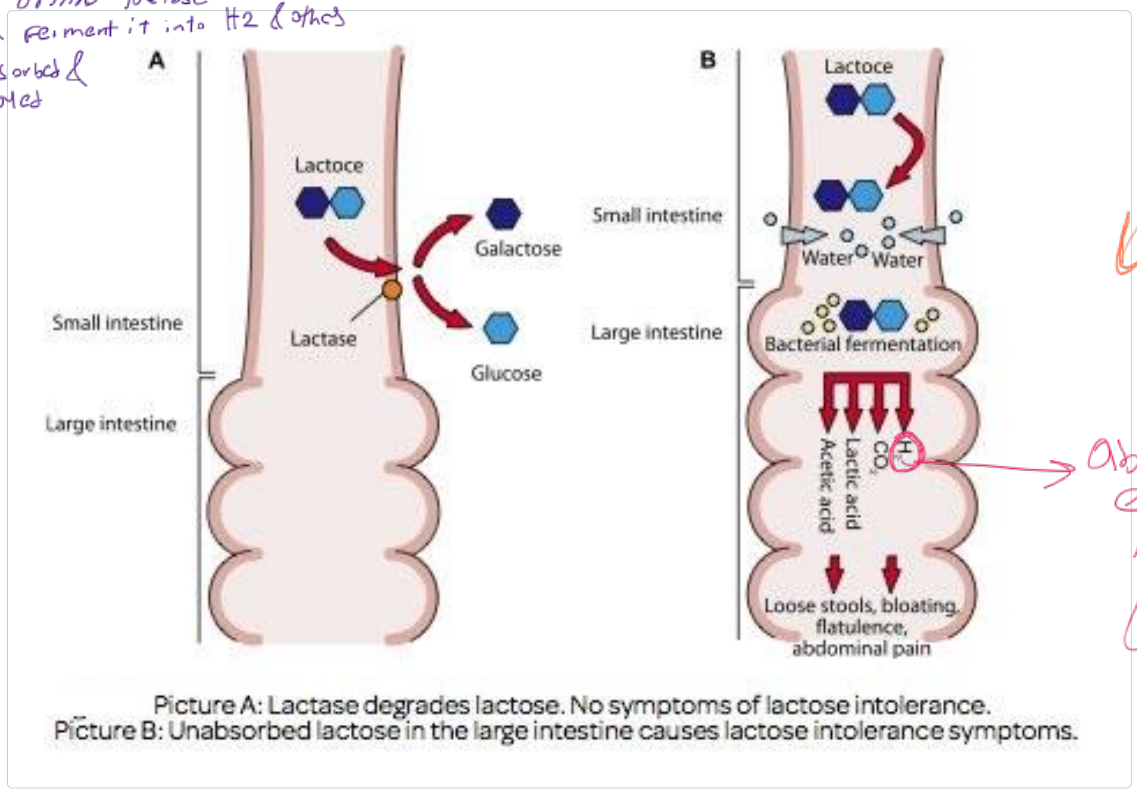
Nutrient Malabsorbed	Clinical Manifestation
Protein	Wasting, edema
Carbohydrate and fat	Diarrhea, abdominal cramps and bloating, weight loss/growth retardation
Fluid and electrolytes	Diarrhea, dehydration
Iron	Anemia, cheilosis, angular stomatitis
Calcium/vitamin D	Bone pain, fractures, tetany
Magnesium	Paresthesia, tetany
Vitamin B12/folate	Anemia, glositis, cheilosis, paresthesias, ataxia (vitamin B12 only)
Vitamin A	Night blindness, xerophthalmia, hyperkeratosis, diarrhea
Vitamin K	Ecchymoses
Riboflavin	Angular stomatitis, cheilosis
Selenium	Cardiomyopathy
Essential fatty acids	Dermatitis

SPECIFIC DISEASE EXAMPLES

Lactase Deficiency (lactose intolerance)

- Acquired lactase deficiency is the most common cause of selective carbohydrate malabsorption.

- The prevalence of lactase deficiency is highest (85 to 100%) in persons of Asian, African, and Native-American descent.
- Adults with lactase deficiency typically complain of gas, bloating, and diarrhea after the ingestion of milk or dairy products but do not lose weight.
- Unabsorbed lactose is osmotically active, drawing water followed by ions into the intestinal lumen.
- On reaching the colon, bacteria metabolize lactose to short-chain fatty acids, carbon dioxide, and hydrogen gas.
- Short-chain fatty acids are transported with sodium into colonic epithelial cells, facilitating the reabsorption of fluid in the colon.
- If the colonic capacity for the reabsorption of short-chain fatty acids is exceeded, an osmotic diarrhea results.
- The diagnosis of acquired lactase deficiency can be made by empirical treatment with a lactose-free diet, which results in resolution of symptoms; by the hydrogen breath test after oral administration of lactose.
- Many intestinal diseases cause secondary reversible lactase deficiency, including viral gastroenteritis, celiac disease, giardiasis, and bacterial overgrowth.



Picture A: Lactase degrades lactose. No symptoms of lactose intolerance.
Picture B: Unabsorbed lactose in the large intestine causes lactose intolerance symptoms.

Figure 5.4 — Lactase deficiency and lactose intolerance

Intestinal Bacterial Overgrowth

- In health, only small numbers of lactobacilli, enterococci, gram-positive aerobes, or facultative anaerobes can be cultured from the upper small bowel lumen.
- Motility And Acid are the most important factors in keeping the number of bacteria in the upper small bowel low.
- Any condition that produces local stasis or recirculation of colonic luminal contents allows development of a predominantly "colonic" flora (coliforms and anaerobes, such as Bacteroides and Clostridium) in the small intestine.
- This will cause impaired micelle formation by releasing cholylamidases, which deconjugate bile salts.
- As a result, the concentration of bile salts decreases in the intestinal lumen causing malabsorption of fats and fat-soluble vitamins.
- Vitamin B12 deficiency and carbohydrate malabsorption also can occur with generalized bacterial overgrowth.
- Anaerobic bacteria ingest vitamin B12 and release proteases that degrade brush-border disaccharidases.
- Individuals with bacterial overgrowth usually have low serum vitamin B12 levels but normal or high folate levels.
- This helps distinguish bacterial overgrowth from tropical sprue.

Some Causes of Bacterial Overgrowth

STRUCTURAL

- **Surgical**

Ileocecal valve resection

- **Anatomic**

Duodenal and jejunal diverticula

Gastrojejunocolic fistulas

MOTOR

Scleroderma

Diabetes mellitus

HYPOCHLORHYDRIA

Atrophic gastritis

Proton-pump inhibitors

MISCELLANEOUS

Cirrhosis

Chronic renal failure

Celiac Disease

- Celiac disease, also called celiac sprue, nontropical sprue, and gluten-sensitive enteropathy, is an inflammatory condition of the small intestine precipitated by the ingestion of wheat, rye, and barley in individuals with certain genetic predispositions. • It is the prototype of a noninfectious cause of malabsorption resulting from a reduction in small intestinal absorptive surface area.
- The basic disorder in celiac disease is sensitivity to gluten, the component of wheat and related grains (oat, barley, and rye) that contains the water-insoluble protein gliadin
- The small intestinal mucosa, when exposed to gluten, accumulates large numbers of B cells and plasma cells sensitized to gliadin; accumulation of lymphocytes in gastric and colonic mucosa also may occur.
- Total flattening of mucosal villi (and hence loss of surface area) is the outcome, affecting the proximal more than the distal small intestine.
- Patients with celiac disease have increased levels of serum antibodies to a variety of antigens, including IgA antiendomysial , anti gliadin and anti tissue transglutaminase antibodies (TTG Ab).
- The age of presentation with symptomatic diarrhea and malnutrition varies from infancy to mid-adulthood.
- About half of adults with celiac disease in the present with anemia or osteoporosis without diarrhea or other gastrointestinal symptoms.
- These individuals most likely have proximal disease that impairs iron, folate, and calcium absorption but an adequate surface area in the remaining intestine for absorption of other nutrients.
- Other Extraintestinal Manifestations Of Celiac Disease include rash (dermatitis herpetiformis), neurologic disorders (peripheral neuropathy, ataxia, epilepsy), psychiatric disorders (depression, paranoia), reproductive disorders (infertility, spontaneous abortion), short stature, dental enamel hypoplasia, chronic hepatitis, or cardiomyopathy.
- Individuals with significant mucosal involvement present with watery diarrhea, weight loss or growth retardation, and the clinical manifestations of vitamin and mineral deficiencies.

- Diarrhea is caused by many mechanisms, including a 1) decreased surface area for water and electrolyte absorption, 2) the osmotic effect of unabsorbed luminal nutrients, and 3) increased surface area for chloride secretion (crypt hyperplasia), and the stimulation of intestinal fluid secretion by 4) inflammatory mediators and unabsorbed fatty acids.
- Recently The perception that gluten causes gastrointestinal and extra gastrointestinal symptoms in patients who do not have coeliac disease (CD), and are not allergic to wheat, is an increasing clinical problem.
- This has been called Non-coeliac Gluten Sensitivity (NCGS)
- It is purely a clinical diagnosis as, in contrast to CD, it has no biomarker or characteristic intestinal biopsy findings.

Celiac Disease vs. Wheat Allergy vs. NCWS

	Celiac Disease	Wheat Allergy (IgE-mediated)	Wheat Allergy (Non-IgE)	Non-Celiac Wheat Hypersensitivity
Route of exposure	Oral	Inhalation or oral	Oral	Oral
Onset after exposure	Hours / months	Immediate	Days / weeks	Hours / days
Extraintestinal features	Liver, joint/bone, skin, blood, CNS	Respiratory tract, skin, anaphylaxis	None	Skin, joint/muscle, CNS
Antibodies	TGA & DGP (IgA/IgG)	IgE	None	Possibly antigliadin
Complications	Poor growth, micronutrient deficiency, infertility, osteoporosis, lymphoma	None	Esophageal strictures (EoE)	–
Comorbidities	Type 1 diabetes, immune liver & thyroid disease	Other allergies	Atopic dermatitis	Thyroiditis

Figure 5.5 — Celiac Disease vs. Wheat Allergy vs. NCWS

Classical, Non-Classical & Subclinical Disease

Coeliac disease is a *"clinical chameleon"* that can present at any age with widely varying manifestations — a major reason for delayed or missed diagnosis.

Classical

More common in children < 5 years

- Chronic diarrhea
- Poor appetite
- Weight loss
- Abdominal distension
- Muscle wasting
- Mood changes

Non-Classical

The most common presentation overall

- Recurrent abdominal pain, bloating
- Diarrhea or constipation
- Persistent iron deficiency
- Chronic fatigue
- Elevated liver enzymes
- Arthralgia, alopecia, urticaria

Subclinical (Silent)

Found only through screening

- No overt symptoms or signs
- Identified via population screening
- Or casefinding in at-risk relatives
- Villous atrophy still present on biopsy

Figure 5.6 — Celiac disease: classical, non-classical and subclinical presentations

Manifestations Across the Lifespan

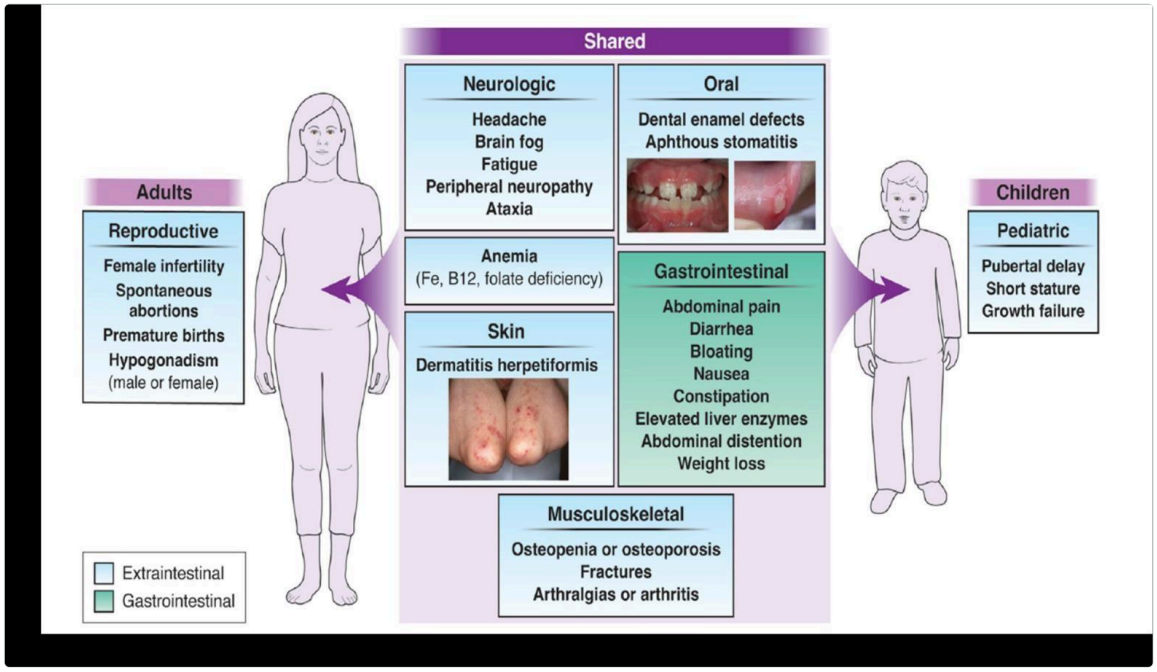


Figure 5.7 — Manifestations across the lifespan

Serologic Testing

tTG-IgA is the single preferred test in adults, measured concurrently with total IgA to catch IgA deficiency (then use DGP -IgG).

Test	Sensitivity	Specificity	Notes
tTG-IgA	90–98%	95–97%	Preferred single first-line test
EMA-IgA	85–98%	97–100%	Highest accuracy; costlier, less available
DGP-IgA	94%	99%	Useful adjunct / IgA-deficient patients
DGP-IgG	92%	100%	Used when IgA deficient
AGA (IgA/IgG)	Lower	Lower	No longer recommended — 15–20% false positive

Figure 5.8 — Serologic testing

Tropical Sprue And Whipple Disease

- Are two disorders that exemplify malabsorption syndromes arising from intestinal infection.
- Tropical sprue resembles celiac disease in symptomatology but occurs almost exclusively in persons living in or visiting the tropics.
- Small intestinal changes vary from near normal to a severe diffuse enteritis with villus flattening.
- In contrast to celiac disease, injury is seen at all levels of the small intestine.
- It is caused by overgrowth of predominantly coliform bacteria.
- Individuals with tropical sprue classically present with diarrhea and megaloblastic anemia secondary to vitamin B12 and folate deficiency, but some have anemia only.
- Intestinal biopsy characteristically shows subtotal and patchy villous atrophy in the proximal and distal small intestine.
- Treatment is a prolonged course of tetracycline (250 mg PO four times daily) or doxycycline (100 mg PO two times daily), folic acid (5 mg/day PO), and, with coexistent deficiency, vitamin B12 injections (1000 µg weekly)

Whipple disease is a rare, systemic infection that may involve any organ of the body but principally affects the Intestine, Central Nervous System, And Joints.

- The hallmark of Whipple disease is a small intestinal mucosa laden with distended periodic acid-Schiff (PAS)-positive macrophages in the lamina propria.
- The organism responsible for causing Whipple disease is a gram-positive actinomycete, Tropheryma whippelii.
- Approximately one third of patients have cardiac involvement, most commonly culture- negative endocarditis.
- Occasionally, individuals present with ocular or neurologic disease without gastrointestinal symptoms.
- Men are affected more commonly than women, particularly white men.
- Treatment is with a prolonged course of broad-spectrum antibiotics (e.g., ceftriaxone, 2 g/day intravenously (IV) or meropenem 1 g IV three times daily; then trimethoprim 160 mg and sulfamethoxazole 800 mg PO two times daily for 1 year.

DIAGNOSTIC TESTS FOR MALABSORPTION

GENERAL TESTS OF ABSORPTION

Quantitative stool fat test Gold standard test of fat malabsorption. Requires ingestion of a high-fat diet (70- 100 g) for 2 days before and during the collection. Stool is collected for 2-3 days. Normally, <7 g/24 hr is excreted on a high-fat diet

Qualitative stool fat test Sudan stain of a stool sample for fat.

Hydrogen breath test Most useful in the diagnosis of lactase deficiency. An oral dose of lactose (25 g) is administered after measurement of basal breath H2 levels. The sole source of H2 in the mammal is bacterial fermentation; unabsorbed lactose makes its way to colonic bacteria, resulting in excess breath H2

SPECIFIC TESTS FOR MALABSORPTION

1) Tests for Pancreatic Function

Secretin stimulation test The gold standard test of pancreatic function

Fecal elastase-1 test Stool test for pancreatic function

2) Tests for Bacterial Overgrowth

Quantitative culture of small intestinal aspirate

Hydrogen breath test The 50-g glucose breath test has a sensitivity of 90% for growth of 105 colonic-type bacteria in the small intestine

3) Tests for Mucosal Disease

Small bowel biopsy

4) Test of Ileal Function

75SeHCAT test This is a test of bile acid absorption.

Seven days after ingestion of radiolabeled synthetic selenium–homocholic acid conjugated with taurine (75SeHCAT), whole body retention is measured by a gamma-counting device. The result is expressed as a fraction of baseline ingestion

Practical approach to malabsorption

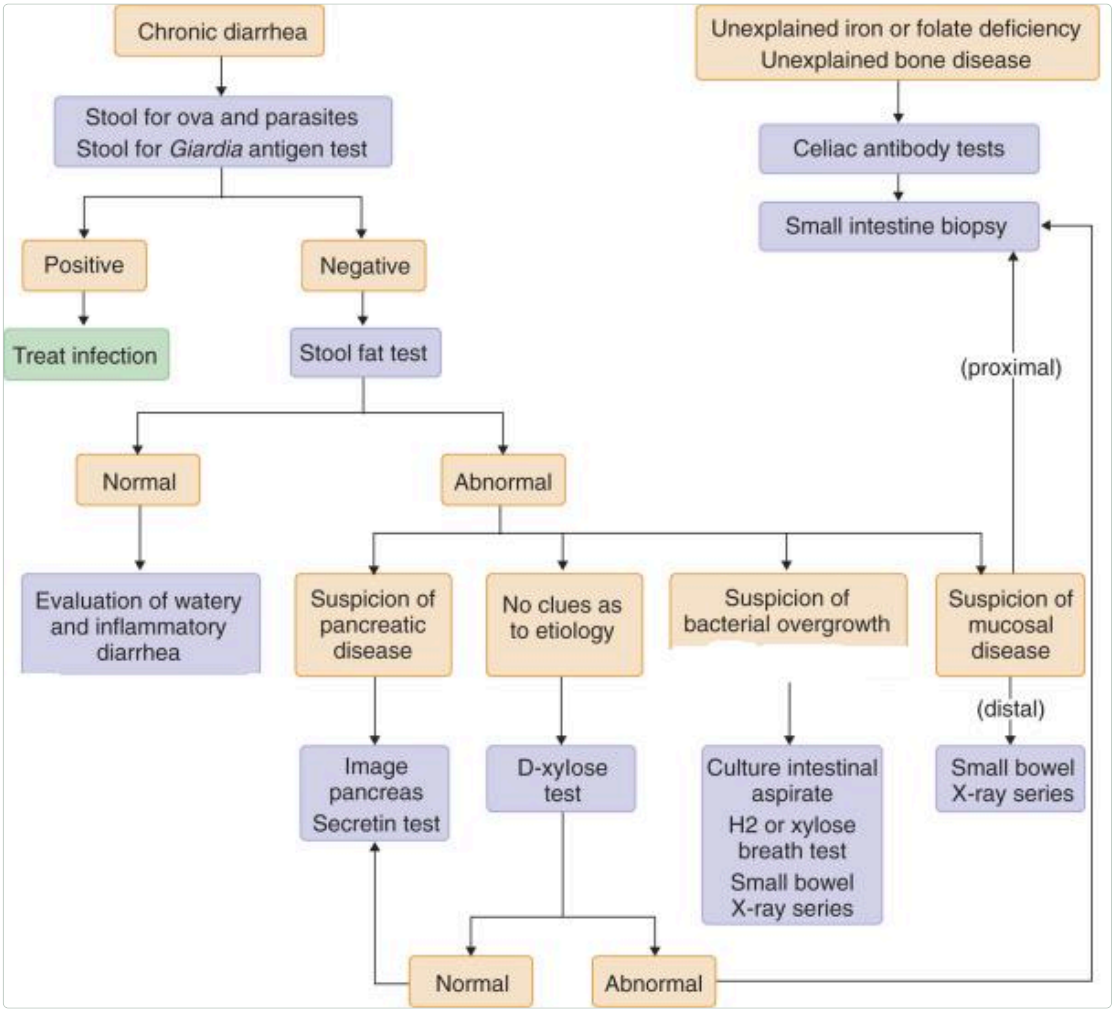


Figure 5.9 — Practical approach to malabsorption

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CHAPTER 06

Constipation

06

Definition

- Constipation is a common gastrointestinal symptom or disorder characterized by infrequent bowel movements, difficult stool passage, excessive straining, hard stools, or a sensation of incomplete evacuation. It results from impaired colonic transit, disorders of anorectal evacuation, or both.
- Prevalence and clinical importance: Constipation affects approximately 16% of adults worldwide and is one of the most common gastrointestinal disorders encountered in clinical practice. Although often regarded as a minor complaint, it may significantly impair quality of life and occasionally indicates serious underlying disease

1. Classification

Primary (functional) constipation	Secondary constipation
Normal-transit constipation; Slow-transit constipation; Defecatory disorders (pelvic floor dyssynergia)	Drugs; Endocrine and metabolic disorders; Neurological disorders; Structural diseases

3. Pathophysiology

Constipation results from disturbances in colonic transit, anorectal evacuation, or both. Lifestyle factors such as inadequate dietary fiber, insufficient fluid intake, and physical inactivity may also contribute to the development of constipation.

Normal colonic motility

Normal colonic transit depends on coordinated colonic contractions that propel fecal material toward the rectum while allowing adequate absorption of water and electrolytes. Defecation requires coordinated relaxation of the pelvic floor muscles and anal sphincters.

Slow-transit constipation

Slow-transit constipation results from reduced colonic motility, leading to prolonged stool transit. Excessive water absorption from the retained stool produces hard, dry stools that are difficult to pass.

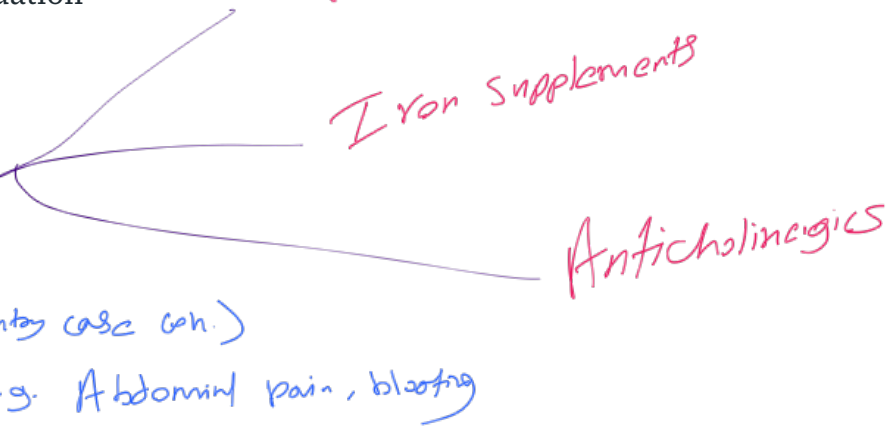
Defecatory disorders

Defecatory disorders (pelvic floor dyssynergia) occur when there is impaired coordination between abdominal muscle contraction and relaxation of the pelvic floor muscles and anal sphincters during defecation. As a result, stool evacuation becomes difficult despite normal colonic transit.

4. Clinical Assessment

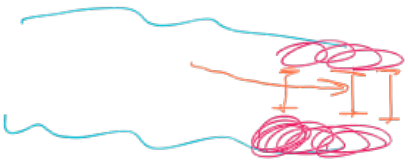
History

- Stool frequency
  - Stool consistency
  - Straining
  - Sense of incomplete evacuation
  - Manual maneuvers
  - Duration
  - Medication history
  - Dietary habits
  - Physical activity
  - Associated symptoms



Alarm features or Red flag features

Based on these history and physical exam findings, it's important to look for red flag features that could indicate underlying serious conditions like malignancy.



Hematochezia

Unintentional weight loss,

Family history of colorectal cancer,

Acute onset of constipation in an older adult,

Change in stool caliber,

Anemia,

The presence of a rectal mass

If any of these red flag features are present, we should consider colorectal cancer and proceed with an endoscopy, such as colonoscopy, with a biopsy to look for obstructive lesions.

Physical examination

- General examination : May be completely normal or there may be alteration of the general condition
  - Abdominal examination : May reveal bloating , tenderness or a mass

- Digital rectal examination: Assessment of perianal sensation and sphincter tone during digital rectal examination may help identify neurological or defecatory disorders . The examination may ay reveal a rectal mass or palpable stool ball , some patients may have anal fissures or hemorrhoids

5. Diagnostic Evaluation

General measures

For patients with no red flag features, exclusion of secondary causes of constipation is mandatory . These include medication side effects, commonly seen in patients taking opioid medications or iron supplements; endocrine or metabolic disorders like hypothyroidism, hypokalemia, or hypercalcemia; and neurogenic conditions, such as spinal cord injury or multiple sclerosis.

In primary constipation a trial of initial conservative treatment , which includes increasing dietary fiber or starting a fiber supplement, advising sedentary patients to increase their physical activity, and starting an over-the-counter laxative, such as an osmotic agent or a stimulant laxative. . If there is an adequate response to empiric therapy no further workup is needed.

On the other hand, if they have an inadequate response to empiric therapy, additional testing will be mandatory as anorectal manometry and a balloon expulsion test or defecography . These tests measure pressures inside the rectum and anus and the ability of the pelvic muscles to expel stool from the rectum. These test are abnormal in defecatory disorder. In this condition constipation arises from difficulty in evacuating stool from the rectum.

However, if the anorectal manometry and balloon expulsion test are normal, a colonic transit study will be recommended. During colonic transit studies, patients swallow radiopaque markers, which are then tracked through the gastrointestinal tract using X-rays. If the markers are not passed within the expected time frame, then the test is abnormal, so you can diagnose slow transit constipation.

On other side, if the colonic transit study results are normal, diagnose normal transit constipation, which is also referred to as a disorder of the gut-brain axis. There are different types of normal transit constipation. One type is irritable bowel syndrome, which might be marked by alternating episodes of diarrhea and constipation, or dominated by either symptom.

6. Management

Management should begin directly with

Life style measures:

- Education [daily defecation is not important but never suppress stool urge]
  - Dietary fiber (retain water) & increase stool bulk
  - Adequate hydration
  - Physical activity
  - Scheduled toileting [After breakfast by 15 min is best] Gastrocolic Reflex

Pharmacological treatment

" laxatives = classes of laxatives

Class	Examples	Main indication
Bulk-forming (Absorbs water)	Psyllium	Mild constipation
Osmotic (Retain water also)	Polyethylene glycol, lactulose	First-line
Stimulant (↑ colonic motility)	Bisacodyl, senna	Rescue therapy
Stool softener (Surfactant around lipids)	Docusate	Selected patients
Secretagogues (↑Cl⁻ secretion)	Lubiprostone, linaclotide	Refractory cases
5-HT4 agonist	Prucalopride (improve motor function)	Chronic refractory constipation

(Osmotic: pull water) (Secretagogues: push water)

5-HT4 agonist → Serotonin Receptors → increase propulsive colonic contractions

Biofeedback

Biofeedback is the treatment of choice for pelvic floor dyssynergia.

- pressure measurements during defecation & teach him
- → Normally, the patient increases rectal pressure while relaxing the anal sphincter
- → In dyssynergia, the sphincter may fail to relax or contract paradoxically

laxatives → Soften stool ✓
Biofeedback → Ms coordination ✓ ∴ الحل

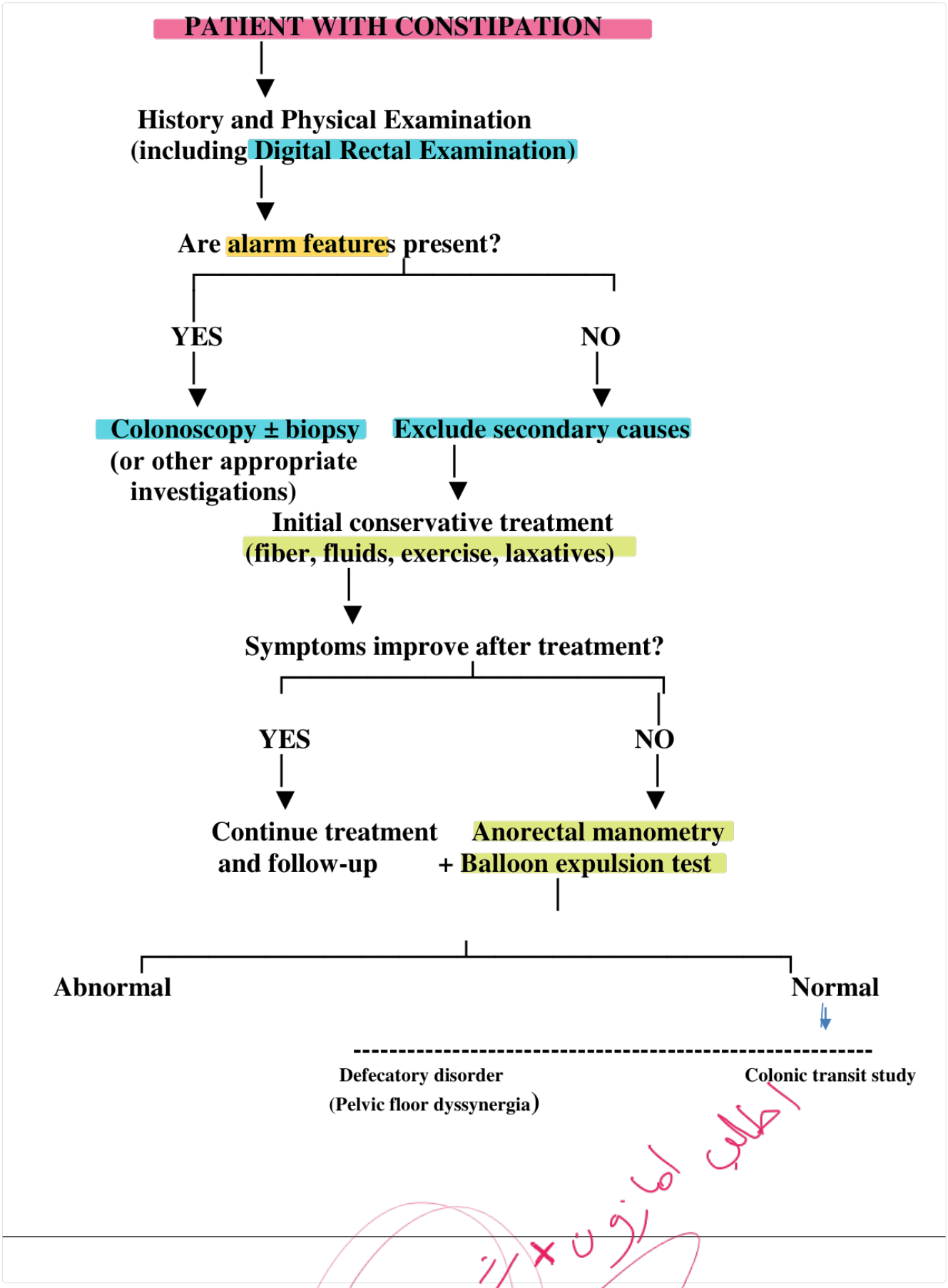
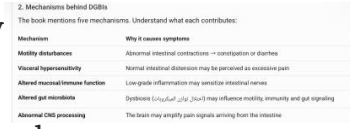


Figure 6.1 — Evaluation of constipation

Disorders of Gut–Brain Interaction (DGBI)

Definition : Disorders of Gut–Brain Interaction (DGBI) are a group of chronic gastrointestinal disorders characterized by persistent or recurrent gastrointestinal symptoms resulting from abnormal interactions between the gut and the brain in the absence of structural or biochemical abnormalities that fully explain the symptoms. They are common, frequently overlap, and may significantly impair quality of life.


2. Mechanisms behind DGBIs

The book mentions five mechanisms. Understand what each contributes:

Mechanism	Why it causes symptoms
Motility disturbances	Abnormal intestinal contractions → constipation or diarrhea
Visceral hypersensitivity	Normal intestinal distension may be perceived as excessive pain
Altered mucosal/immune function	Low-grade inflammation may sensitize intestinal nerves
Altered gut microbiota	Dysbiosis (اختلال توازن البكتيريا) may influence motility, immunity and gut signaling
Abnormal CNS processing	The brain may amplify pain signals arriving from the intestine

Symptoms can be related to any combination of motility disturbances, visceral hypersensitivity, altered mucosal and immune function, altered gut microbiota, or disordered central nervous system processing (formerly known as "functional gastrointestinal disorders) . DGBIs are recognized through clinical symptom rather than structural biomarkers.

Bidirectional Signaling occur

- Pathophysiology relies on the gut-brain axis, where psychological factors (like stress, anxiety, or trauma) directly alter gut motility and processing, also peripheral gut factors (such as post-infectious low-grade inflammation or dysbiosis) alter signaling pathways back to the brain.
- Diagnostic Approach: Standard medical tests—including endoscopies, blood work, and imaging—typically yield normal results, making clinical criteria (such as the Rome IV or Rome V guidelines) essential for diagnosis.

3. Common DGBIs

There are many recognized adult and pediatric DGBIs, categorized by the region of the gastrointestinal tract they impact:

normal reflux = excessive relaxation

GI Tract Region	Common DGBI Examples	Core Clinical Presentation
Esophageal	Functional Heartburn, Reflux Hypersensitivity	Chest burning or pain despite normal acid profiles and endoscopy.
Gastroduodenal	Functional Dyspepsia (FD)	Early satiety, postprandial fullness, or epigastric burning.
Bowel	Irritable Bowel Syndrome (IBS) قولون عصبي	Abdominal pain related to defecation, accompanied by altered bowel habits (IBS-C, IBS-D, IBS- M).
Anorectal	Functional Constipation, Fecal Incontinence	Difficult, infrequent, or incomplete bowel movements unrelated to structural blockages.

4. Clinical Importance

- DGBIs represent a massive public health burden, affecting over 40% of the population worldwide
- Diminished Quality of Life: The chronic, unpredictable nature of symptoms significantly impairs daily functioning, causing a reduction in quality of life

- Psychiatric Comorbidities: There is a high overlap with anxiety, depression, and somatization. This requires a multidisciplinary approach
- Healthcare Utilization Costs: Because these disorders lack physical biomarkers, patients frequently undergo unnecessary, invasive diagnostic testing and surgeries (e.g., appendectomies or cholecystectomies), driving up healthcare expenses

Irritable Bowel Syndrome (IBS)

Irritable Bowel Syndrome (IBS) is a classic example of a Disorder of Gut–Brain Interaction (DGBI) characterized by recurrent abdominal pain associated with altered bowel habits.

1. Definition (Rome V Updates)

To make a clinical diagnosis of IBS, a patient must present with recurrent abdominal pain or discomfort that satisfies the following conditions:

- Frequency Threshold: No continuous abdominal pain but occurring, on average, at least 3 days per month over the last 3 months.
- Duration Frame: IBS requires an overall symptom onset of at least 6 months prior, with active symptoms over the last 3 months,
- Bothersomeness: Symptoms must be sufficiently bothersome to interfere with daily activities or prompt the individual to actively seek healthcare.

2. Required Symptom Associations

The abdominal pain or discomfort must be associated with two or more of the following characteristics:

1. Relation to Defecation: The pain/discomfort is modified (either relieved or exacerbated) by bowel movements.

2. Change in Stool Frequency: Onset of symptoms is linked to a change in how often the patient passes stool (e.g., more or less frequent).

3. Change in Stool Form: Onset of symptoms is linked to a change in the appearance or consistency of stool,

3. Mandatory Clinical Exclusions - Ruling Out "Red Flags"

A clinician can apply the Rome V clinical criteria only if standard medical history and a basic physical examination rule out these warning signs:

- Symptom onset after 50 years of age.
- Unintentional weight loss or unexplained fever.

- Rectal bleeding (hematochezia) or melena not attributed to hemorrhoids.
- Nocturnal symptoms that awaken the patient from sleep.
- Unexplained iron-deficiency anemia.
- Family history of colorectal cancer, celiac disease, or inflammatory bowel disease (IBD).

4. Epidemiology

- Prevalence Rebound: at approximately 8.5% to 10% and higher in women than men (approx. 2:1 ratio) and is most frequently diagnosed in individuals under the age of 50.

5. IBS Subtypes

- IBS with Predominant Constipation (IBS-C)
- IBS with Predominant Diarrhea (IBS-D):
- IBS with Mixed Bowel Habits (IBS-M):]
- IBS Unclassified (IBS-U): Meets diagnostic criteria for IBS, but stool consistency cannot be accurately categorized into any of the three main subtypes.

6-Pathophysiology of IBS

1. Altered Gut–Brain Interaction
2. Visceral Hypersensitivity
3. Altered Gastrointestinal Motility

Disruptions in the transit time and contraction patterns of the muscular layers of the GI

4. Altered Intestinal Microbiota (Dysbiosis)

Patients with IBS consistently exhibit a loss of microbial diversity and stability.

5.Low-Grade Immune Activation & Increased Intestinal Permeability

6. Psychosocial Factors

Psychological status acts as both an initiator and a modifier of IBS severity.

. 7- Clinical Assessment

History

- Abdominal pain
  - Relationship to defecation

- Stool frequency
- Stool consistency (Bristol Stool Form Scale)
- Bloating
- Symptom duration
- Dietary factors
- Psychological factors
- Previous gastrointestinal infection

Physical examination

- Usually normal
  - Mild abdominal tenderness and tympanitic resonance of the abdomen
  - Digital rectal examination when indicated

8-Rome V Clinical Diagnostic Criteria

IBS is a positive clinical diagnosis based on the Rome V clinical criteria rather than a diagnosis of exclusion.

We can rely only on limited laboratory investigations

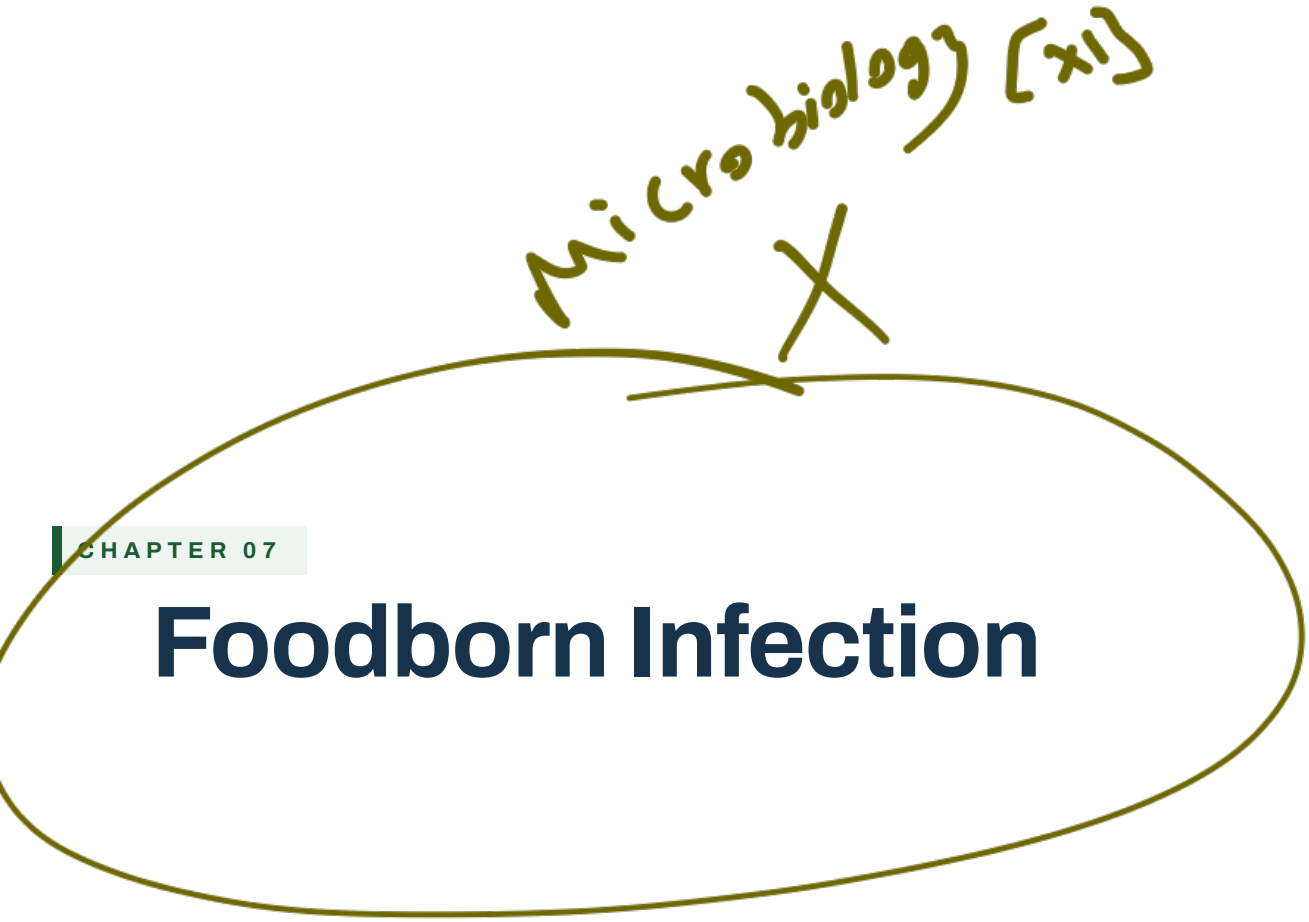
- Fecal calprotectin (to exclude IBD)
- Celiac serology (mainly in IBS-D)
- Colonoscopy only when alarm features or other indications are present

Management of IBS

General measures

- Patient education
  - Reassurance
  - Regular physical activity
  - The patient should be advice for diet modification, which includes consuming soluble fiber, and a trial of a diet low in a specific set of carbohydrates, and low in high FODMAP including fructans and oligosaccharides found in onions, garlic, wheat, rye, and legumes.

- The use of peppermint oil, has been shown to relax smooth muscle and target spasming of the GI tract; as well as tricyclic antidepressants, like amitriptyline or nortriptyline, which are beneficial in relieving IBS- associated abdominal pain. Finally, gut-directed psychotherapy can be beneficial since IBS has a strong association with mental health.
- No routine recommendation of Probiotics
- For patient with the diarrhea-predominant type, the goal is to increase stool firmness or decrease bowel movement frequency. Treatment can be achieved by the non-absorbed antibiotic rifaximin, which works by addressing bacterial growth associated with diarrhea.
- On the other hand, there's the constipation-predominant type. Treatment for these individuals is geared towards stool softening and increasing bowel movement frequency. There are two options available for treatment. The first one includes chloride channel activators, like lubiprostone; while the second as prucalopride which is a selective, high-affinity serotonin type 4 (5-HT_4) receptor agonist. It acts as a prokinetic agent



CHAPTER 07

Foodborn Infection

07

Learning objectives

Explain the terminology related to gastrointestinal infection.

Discuss the etiology, pathogenesis ,epidemiology and prevention of microbial infection of gastrointestinal infections

Discuss virulence mechanisms and correlate virulence mechanisms with specific pathogens

Describe the laboratory diagnosis for bacteria and viruses

Discuss different methods for preventing foodborne illness and food poisoning

Foodborne illness: are infections caused by a variety of microorganisms mainly

Salmonella, Clostridium perfringens, Campylobacter, and Staphylococcus aureus

Norovirus

Salmonella is a member of Enterobacteriacae family,it is the causative agent of:

- Enteric Fever (typhoid) or Salmonellosis ,there is bacterial invasion of bloodstream by Salmonella .
- Acute gastroenteritis, resulting from a food-borne infection mainly contaminated **eggs** and intoxication.

  Morphology: Gram negative bacilli, **motile,** non spore forming.

  Culture: non-lactose fermenter on MacConkey agar .Special culture media include SS agar , deoxycholate citrate agar (DCA) and broth selenite and tetrathionate are used for **its isolation .**

  Biochemical reactions: partial sugar fermentation ,**urease negative** ,oxidase negative, and H2S positive

  Antigenic structure: O (somatic) antigens which subdivide the Salmonellae into groups, and H antigen (flagella) antigens which determine the type within the groups.Virulent antigen (Vi Ag) is anti-phagocytic and is present only in S. typhi.

  Pathogenesis is due to endotoxins and exotoxins .The exotoxins such as enterotoxins is responsible for diarrhea and body fluid loss .

  Resistance:Killed by heating at 60°C for 30 min, antibiotics and other chemotherapy agents .Resist the action of bile salts.

Diseases caused by Salmonella:

1. Enteric fever is caused by S. typhi and the milder paratyphoid fever is caused by

Paratyphi A, B and C. Enteric fever is defined as generalized infection of reticuloendothelial system and lymphoid tissue accompanied by sustained fever and **bacteremia. Enteric fevers are endemic in Egypt.**

Pathogenesis:.

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After multiplication in reticuloendothelial system it is released into bloodstream and results signs of septicemia mainly fever. The organism is transported to the gall bladder to be excreted with bile. Salmonella multiply in intestinal lymphoid tissue and are excreted in **stools at** the second week.

Enteric fever is a strictly human disease, from case or carrier stool and urine. The organisms enter the human digestive tract with contaminated food or drink e.g raw shell fish ,milk products and raw vegetables. The organism adhere to the mucosa of small intestine by pili then penetrate to submucosal layer causing no lesion. In lymphoid tissue the organism invade macrophages where bacterial multiplication occurs . They are transported to mesenteric lymph nodes and lymphatics, released into the bloodstream and are carried to many organs such as bone marrow, kidneys intestine, spleen and liver where replication of salmonella inside the macrophages occurs.

Salmonella organisms **survives intracellularly** and **replicates in macrophages** due to decrease fusion of lysosome with phagosomes as well as due to the presence of surface proteins **defensins** which allow the bacteria to withstand oxygen-dependent and independent intra-cellular killing. When released from macrophages, the Vi capsular antigen (only in S. typhi) withstand complement - mediating killing

After multiplication in reticuloendothelial system it is released into bloodstream and results signs of septicemia mainly fever. The organism is transported to the gall bladder to be excreted with bile. Salmonella multiply in intestinal lymphoid tissue and are excreted in stools **at the second week**.

- The incubation period of the disease is 10-14 days .The fever is **gradually** elevated during the first week then become steady during the second week. There is enlargement of spleen and liver .The white cell count is normal **or low**. Salmonella excretion by patients may continue long after clinical cure.

  Outcome: 5% of patients cured from typhoid remain carriers for months or even years. Urinary carriers are common in Egypt because of bilharziasis.

- Laboratory diagnosis: Specimens are collected according **to the stages** of the disease. A- Isolation of Salmonella during the first week by doing blood culture

  B- Second week , Salmonella is isolated from stool and after the second week from urine by direct culture on selective media.

  C- Serological diagnosis by Widal agglutination test:This test demonstrates agglutination of serum antibodies of the patients against O-somatic antigens of Salmonella and H-flagellar antigens of S.typhi and Paratyphi A,B,C. Antibodies rise sharply during the second and third weeks of enteric fevers. Demonstration of a rising titers in antibodies in the paired samples indicates acute recent infections. The results should include the titer of both O and H antibodies and Vi antibodies. The presence of **IgM** is indicative of acute recent infections.

  Interpretation of Widal test:

  1- High or rising titer of antibody to O antigens (≥1:160) suggests that active infection is present.

  2- High titer of antibody to one type of H antigen (≥1:160) suggest recent infection with this type of Salmonella.

  3- High titer of antibody to the **Vi antigen** occur in some carriers.

  4- High titer of antibodies to O antigens and more than one H antigen indicate recent immunization. With old immunization or past infection O antibodies disappear and only antibodies to H antigens remain.

Widal test has many limitations as a diagnostic serological test for enteric fever. **Several** factors need to be considered for interpretation:

1- **Endemicity of enteric fever** in an area of investigation. Healthy persons have antibody titer due to subclinical infection. Titers up to 1:160 are considered normal in Egypt.

False positive results may occur in an endemic area, immunization with any typhoid vaccine or a previous infection or infection with other related members of **Enterobacteriacae**

2- False negative in case of administration of antibiotics and if widal test is done in early stage of disease where low antibody titers are found .

- Antibiotics may be not effective on Salmonella carriage because the site of carriage may **not allow penetration by the antibiotic e,g due to the presence of fibrous tissue**

Immunity:

Persistent immunity is acquired after enteric disease. **CMI** is the main immunity. Humoral immunity destroy the organism in the blood. Antibody titer can continue for a long time after recover.

Prevention and Control by

1- Immunization: several typhoid vaccines are available for use :

a. Parenteral heat-phenol-inactivated vaccine contains S.typhi and paratyphi A,B,,C hence the name TAB vaccine . The vaccine is given as 2 subcutaneous (SC) doses. This is the current vaccine used in Egypt.

b. Other vaccines include :acetone-inactivated parenteral vaccine,oral live-attenuated vaccine for S. typhi. Living vaccine should not be given to children younger than 6 years of age. Vi capsular polysaccharide (ViCPS) is a newly licensed capsular polysaccharide vaccine for parenteral (intramuscular) use.

No typhoid vaccine is 100% effective and is not a substitute for good hygiene

2.Salmonella food poisoning:

a- Salmonella typhimurium foodborne toxic infections. Symptoms (diarrhea, vomiting, fever) appear after 12-24 hours following ingestion of **meat** contaminated **with rat excreta which contain large number of organisms. It last for 2-5 days.**

b- Salmonella enteritidis infection or Egg-associated salmonellosis The egg can be inside perfectly normal. Incubation period from 12 to 72 hours after consuming a contaminated food. The illness usually lasts 4 to 7 days, and most persons recover without antibiotic treatment.

Campylobacter :12-15 species are associated with human disease: Campylobacter jejuni (poultry; humans), Campylobacter coli (porcine; humans) and Campylobacter fetus (cattle; sheep).

Morphology: They are gram negative curved bacilli (S-shaped or “gull-winged” appearance),non-spore forming, have a characteristic rapid darting motility by means of long sheathed polar flagellum at one (polar) or both (bipolar) ends. Motility slows **quickly in wet mount microscopy preparation due to aerotoxicity (oxygen sensitivity).**

a- Culture: Micro-aerophilic thermophilic and capnophilic (CO2 :10%) on Selective media with added antibiotics as Skirrows medium at **42°C** which is the natural body **temperature of avian species.**

Biochemical reactions: **Oxidase positive**, catalase positive, do not utilize carbohydrates. Pathogenesis: Virulence factors include:

a- Cellular components: Endotoxin(LPS), flagellum,adhesins and invasins.

b- Extracellular components: Enterotoxins (LT) and cytotoxins.

Transmission: feco-oral route.Campylobacter is usually acquired from food (specially poultry),milk,or contact with infected animals (e.g. dogs). Children under 5 years and young adults are the most frequently affected age groups.

Incubation period of 1-7 days with acute onset, abdominal cramps, nausea but not vomiting. The bacteria damage the mucosal surfaces of jejunum; ileum and colon. Stools may contain mucus or gross blood. Generally self-limiting, symptoms may last for a week or longer. Bacteremia is common and Campylobacter is isolated from blood.

Important late onset complications following gastroenteritis include: Guillain-Barre

Syndrome; Reiter's syndrome; reactive arthritis and spontaneous abortion .

Laboratory diagnosis: Specimen collection and processing stool should be examined within few hours. Care to avoid oxygen exposure while taking sample or better a special transport medium is used.

A- Direct examination: by Gram stain and observation of darting motility in wet mount preparation

B-Culture: on Skirrows medium.

Antibiotic resistance: Fluoroquinolone resistance has developed rapidly related to unrestricted use and the use of enrofloxacin in poultry. The organism is inherently resistant to trimethoprim and most cephalosporins.

Brucella is the cause of brucellosis, a zoonotic disease.It is a virulent, highly infectious bacteria. They are facultative intracellular pathogen surviving in macrophages. There are different species of Brucella which include:Brucella abortus (cattle)Brucella melintensis (goats, sheep)Brucella suis (swine,pigs) Brucella canis (dogs).Species are differentiated by production **of urease and H2S,** dye sensitivity, cell wall antigens and phage sensitivity.

Morphology: Gram-negative small bacilli, non-spore forming, non-motile and non- capsulated. Culture: Aerobic, require enriched media containing serum e.g. serum dextrose agar. Colonies appear after 48 to 72 hours. Biochemical reactions: Positive for catalase, oxidase ,urease and reduce nitrates.

Brucellosis (Malta fever or undulant fever): This a true zoonotic disease as human to- human transmission has not been identified. The disease is characterized by acute undulating fever i.e bouts of fever 3-4 weeks then 3-4 weeks without fever.

Transmission: Direct contact through skin abrasions ,mucous membrane with infected animals or animal products (farmers, butchers and veterinarians), Consumption of unpasteurized milk products and meat, inhalation of aerosols from infected animal or laboratory-associated.

Laboratory diagnosis:

Specimen: blood is taken during acute disease. Other sample according to site of infection such as joint fluid, CSF, pus .Gram stain is of no significance.

A- Culture:Blood culture is the method of choice early in the disease .Cultures need to be incubated for up to 4-6 weeks as they are slow -growing. It is important to differentiate Brucella from Salmonella by urease test as brucella produce urease (+ve) while Salmonella not (-ve).

B- Serology: Is the main method of laboratory diagnosis.Two serum samples is collected 14-21 days apart. A wide range of serum dilutions must be done in order to avoid false negative result due to prozone effect that is commonly

present in cases of brucellosis as large amount of antibodies are produced. Dilutions are done from 1:20 to 1:520. ELISA is used to differentiate between specific IgM and IgG antibodies.

D- Brucellin test: intradermal test based on delayed type hyper-sensitivity.

Interpretation of the result is difficult in persons frequently exposed to infection.

Prevention:Wearing protective impermeable clothing, rubber boots, gloves and face masks when exposed to animals. Pasteurization of milk. Treatment of domestic animals and Live attenuated vaccines for animals(cattle)

Treatment: using combinations of antibiotics for 4 to 6 weeks.

Food born poisoning:

Staphylococcus aureus :are catalase positive Gram-positive cocci

.Staphylococcal enterotoxins do not change the taste, color or odor of food. They are heat stable for 30 minutes of boiling and are not inactivated by brief cooking.The source of food contamination is a food handlers carrying the organism in nose, hands, or having superficial lesions as boil (abscess) . The common types of food are carbohydrate rich food e.g.cake, koshari and protein rich food e.g milk and its product .The incubation period is short ,1-6 hours, due consumption of preformed enterotoxins. The food poisoning is characterized by vomiting and watery non bloody diarrhea.

Laboratory diagnosis: Specimen include food, vomitus microscopic examination of

Direct examination: Gram positive cocci in grape like clusters.Culture is done on blood agar to isolate beta (complete) hemolytic colonies and by using selective mannitol-salt agar. Typing of S .aureus is done for epidemiologic purposes, in food poisoning outbreaks by phage typing and PCR.

Salmonella food poisoning: under Salmonella

Spore forming gram positive bacilli : The gram positive spore forming bacilli include Bacillus and Clostridium species. Bacillus species are aerobes while clostridia are anaerobes.

1. B.cereus cause two forms of food poisoning a. The “short-incubation” or emetic form of the disease is mediated by preformed heat stable enterotoxin usually in improperly stored, cooked rice.It has an incubation period of 1-6 hours. It resembles S. aureus food poisoning b. The “long-incubation” or diarrheal form: caused by heat labile enterotoxin present in meat or vegetables with an incubation period of 8-16 hours. This type of exotoxin activate adenylate cyclase, increase cAMP cause profuse fluid secretion. In either type, the illness usually lasts for less than 24 hours after onset.

2. Clostridium botulinum:is tha causative agent of Botulism which is a rare but fatal form of food poisoning, is caused by a potent botulinum toxin.

Transmission and pathogenesis : Toxin is produced within the contaminated canned food and ingested preformed then becomes activated by proteases in the gastric fluid and by gastric acidity then absorbed in the intestine and is transported systemically via the blood stream. it binds to receptors on peripheral nerves, inhibits the release of acetylcholine and lead to flaccid paralysis. Botulinum toxin is rapidly acting and signs and symptoms appear within 12 hours after ingestion . Antitoxin is only effective before the toxin binds the neuromuscular junction within 12 h after ingestion. Prognosis is poor for patients diagnosed after this time period.

Laboratory diagnosis: Diagnosis depend on clinical manifestations .The presence of **exotoxins in canned food, is detected by by ELISA .**

Trivalent anti-toxin serum(types A,B and E) is given IV along with respiratory support are used as treatment .Antibiotics are not used to treat botulism, Prevention: Swollen cans must be discarded.

Other GIT pathogens include

Leptospira are spirochetes that live in stagnant water, lakes .

Morphology: Spiral-shaped, motile ,one or both ends are hooked. Because they are so thin, live Leptospira are best observed by dark-field microscopy.Culture: Aerobic, at 30 °C, need enriched media (Stuart 's media) to enhance growth that appear after 1-2 weeks.

Leptospirosis (Weil's disease,7-day fever): is a bacterial zoonotic disease caused by

Leptospira. The infection is commonly transmitted to humans by **allowing fresh water** that has been contaminated by animal urine to come in contact with breaks in the skin, eyes or with the mucous membranes. Occupational risk factors include veterinarians, slaughter house workers, farmers, mine, and sewer workers.The disease is not known to be spread from person to person.

Pathogenesis : Incubation period: 4-14 days.It is a biphasic disease, the patient is asymptomatic in between phases: Variable febrile onset: last for 4-7 days .There is hepatitis and jaundice and the second phase is manifested by meningitis (in 40%) with intense headache and stiff neck. Hepatitis, nephritis may recur.

Laboratory diagnosis: Specimen: Organisms are present in blood for the first 7 to 10 days. After this period the microorganism can be found in fresh urine, kidney biopsy.Direct examination: Dark-field microscope and PCR. and Culture: Colonies are identified by agglutination test, DNA probes.Serologic diagnosis: By ELISA for Ig M and total antibody.There are no human vaccines. Animal vaccines are effective for a few months.

Yellow Fever (YF): It is a haemorrhagic fever caused by yellow fever virus which is a single-stranded enveloped RNA arboviruses (arthropod borne virus) .The disease is present in equatorial countries in Africa and South America (Amazon basin). Man usually forms an accidental, non-normal host for the virus .**The liver and stomach are** highly impacted

Pathogenesis and disease: Incubation period:3-5 days.

- Human activity can bring man into contact with mosquitos eg. clearing of forests, agriculture. The virus multiply in the reticuloendothelial cells, then viremia occur to be localized in liver,spleen,kidneys and bone marrow. This is a severe systemic disease in which hemorrhages, degeneration of the liver, kidney and heart occur.

  Jaundice is a prominent feature due to liver necrosis. The case-fatality rate can be 50%.

Life cycles:

1- Jungle (sylvatic) yellow fever : Monkey is the main host and the virus is transmitted between monkeys by Aedes Africans mosquito in Africa .Forest worker/visitor are infected accidentally and carries infection to urban areas.

2- Urban yellow fever: The disease is introduced into towns; so the cycle is: man - mosquito-man.The mosquito is Aedes egypti.

Laboratory diagnosis:

1- Intra-cerebral inoculation of mice with blood sample.

2- Direct detection of RNA virus by PCR.

3- Detection of rising antibody titre by neutralizing test, complement fixation,haemagglutination test.

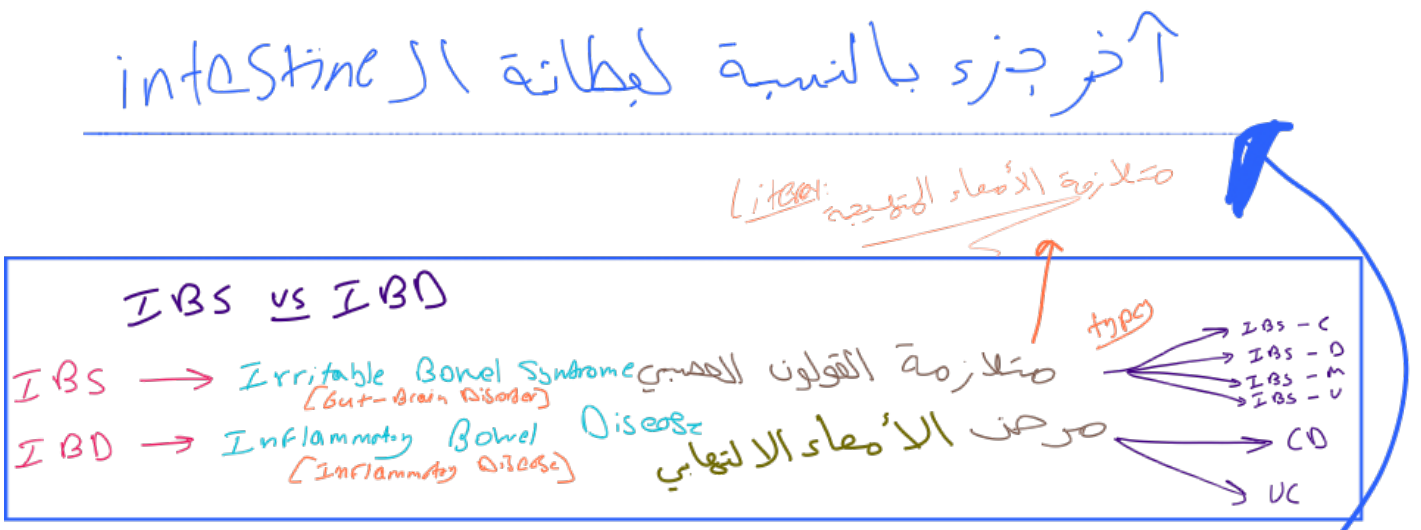
Prevention and control:

1- Use of insecticide to kill mosquitoes and their eggs.

2- Vaccine: “17 D strain” live attenuated virus, given as a single dose subcutaneously. Safe long lasting protection is obtained for 10 years. Vaccination is required for **travelers to endemic equatorial zones.**

Dengue, and Ebola all affect the liver and the gastrointestinal (GI) system, but with varying severity.

- Dengue virus : RNA enveloped viruses .The natural hosts are monkeys and man, mosquitoes are the vector. Dengue virus has an urban and a jungle cycle. Symptoms include fever, headache, rash, bone pain (“breakbone fever”),myalgias and arthralgia. Dengue Hemorrhagic Fever is a deadly complication of dengue.Case fatality rate is 50%.
- Ebola and Marburg viruses are RNA enveloped viruses cause fatal disease in man, there is haemorrhage, fever, diarrhea, vomiting and muscle pain. No known reservoir/ **vector hosts. They are transmitted by direct contact with infected body fluids.**



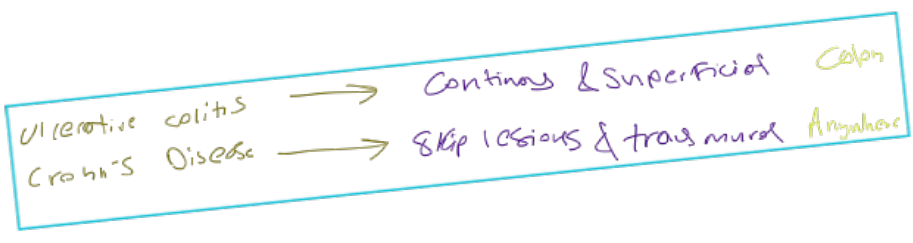
CHAPTER 08

INFLAMMATORY BOWEL DISEASE (IBD)

YMnotes

Ulcerative Colitis vs Crohn's Disease

Scan to open the one-page comparison



YMnotes

ULCERATIVE COLITIS vs CROHN'S DISEASE

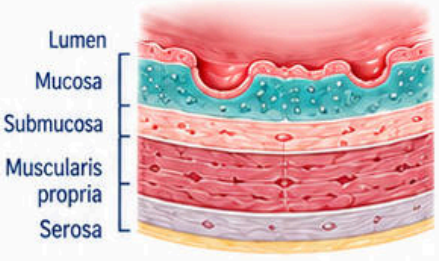
ONE-PAGE IBD COMPARISON

Both: chronic, relapsing, immune-mediated inflammation; often begins in young adults.

ULCERATIVE COLITIS

COLON • CONTINUOUS • SUPERFICIAL

Colon + rectum; starts at rectum and extends proximally

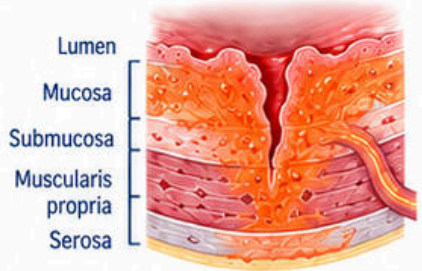


Mainly mucosa → bleeding
(superficial inflammation and ulceration)

CROHN'S DISEASE

ANYWHERE • SKIP LESIONS • TRANSMURAL

Mouth to anus; terminal ileum / ileocolon common



Full wall → fistulas / strictures
(deep, transmural inflammation)

FEATURE	ULCERATIVE COLITIS	CROHN'S DISEASE
Symptoms	Bloody diarrhea, urgency, tenesmus	Diarrhea, pain, weight loss; may also bleed
Endoscopy	Continuous friable mucosa; superficial ulcers; pseudopolyps	Skip lesions; deep linear ulcers; cobblestoning
Histology	Crypt distortion; crypt abscesses	Noncaseating granulomas may occur
	Crypt abscesses can occur in both; granulomas are often absent in Crohn's.	
Complications	Severe bleeding; toxic megacolon; perforation	Fistulas, abscesses, perianal disease; strictures → obstruction
	Toxic megacolon and perforation can occur in either. Longstanding colitis increases colorectal cancer risk inn BOTH.	
Nutrition	Iron deficiency from blood loss	Ileal disease / resection → B12 and bile salt malabsorption; extensive disease → steatorrhea, fat-soluble vitamin deficiency
Outside the gut	Eyes, skin, joints; stronger PSC association	Eyes, skin, joints; ileal disease → gallstones / oxalate stones
Serology clues	pANCA more frequent	ASCA more frequent
	Neither antibody confirms or excludes IBD.	
Classic X-ray signs	Lead-pipe colon = loss of haustra	String sign = marked ileal narrowing

INVESTIGATIONS IN BOTH

CBC ± anemia; CRP / ESR may rise; fecal calprotectin reflects intestinal inflammation.
Exclude stool infection, including *C. difficile*. Ileocolonoscopy + biopsies establish pattern.
MR / CT enterography maps small-bowel Crohn's and complications.

Lead-pipe / string signs are classic exam signs; barium studies are not routine first-line tests.

TREATMENT — MATCH SEVERITY + LOCATION

Both: nutrition and deficiency replacement; steroids for induction ONLY; biologics / small molecules for moderate–severe disease. Selected immunomodulators help maintain remission.

ULCERATIVE COLITIS

- **Mild–moderate:** oral / rectal 5-ASA (mesalamine); sulfasalazine is an alternative.
- **Selected maintenance:** azathioprine. Advanced options include infliximab, vedolizumab and other targeted drugs.

CROHN'S DISEASE

- **Mild ileocecal:** budesonide for induction. Mesalamine is NOT routine therapy.
- **Selected maintenance:** azathioprine / methotrexate; biologics or small molecules for higher-risk disease.
- **Perianal:** drain abscesses; combined medical–surgical care, often infliximab; antiboitics are adjuncts.

Avoid antimotility drugs in severe active colitis.

SURGERY	Remove colon + rectum → eliminates colitis	Treat complications / refractory disease → recurrence possible
	Surgery may be urgent for bleeding, perforation, toxic megacolon or obstruction.	

SMOKING	Lower UC incidence; increases Crohn's risk and severity. Never a treatment.
OTHER POINTS	Typical patterns, not absolute rules. PSC = primary sclerosing cholangitis.

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Objectives

The student will

1) Understand the definition , basic concepts of inflammatory bowel disease
2) Understand the differences between types of inflammatory bowel disease
3) Know the clinical features of inflammatory bowel disease
4) Know the diagnostic tools for inflammatory bowel disease
5) Know the basic treatment options for inflammatory bowel disease

Definition

- Chronic inflammatory diseases of the gastrointestinal tract.
- They are diagnosed by a set of clinical, endoscopic, and histologic characteristics, but no single finding is absolutely diagnostic for one disease or the other.
- Moreover, some patients have a clinical picture that falls between the two diseases and are said to have indeterminate colitis.

Epidemiology

- The highest rates occur in white populations in northern Europe and North America, where the incidence for each disease is about 5 per 100,000 and the prevalence is approximately 50 per 100,000.
- The peak age at onset is between 15 and 25 years of age, with a second, lesser peak between 55 and 65 years.

Clinical features

Ulcerative Colitis [UC]

- The dominant symptom in ulcerative colitis is diarrhea, which is usually associated with blood in the stool.
- Other symptoms include fever and pain, which may be in either lower quadrant or in the rectum.
- The initial attack of ulcerative colitis may be fulminant with bloody diarrhea, but more commonly the disease begins indolently, with non-bloody diarrhea progressing to bloody diarrhea.

Crohn's Disease [CD]

- Crohn's disease is marked by one of three major patterns: (1) disease in the ileum and cecum (40% of patients), (2) disease confined to the small intestine (30%), and (3) disease confined to the colon (25%).
- The predominant symptoms are diarrhea, abdominal pain, and weight loss.
- In patients with colonic disease, especially with rectal involvement, diarrhea is of small volume and associated with urgency and tenesmus.

- Prolonged inflammation and scarring in the rectum can leave it so rigid and nondistensible that the patient is incontinent.
- In disease confined to the small intestine, stools are of larger volume and not associated with urgency or tenesmus.
- Patients with severe involvement of the terminal ileum and those who have undergone surgical resection of the terminal ileum may have bile salt diarrhea or steatorrhea.

Feature	Crohn's Colitis	Ulcerative Colitis
Mucosal lesions	Aphthous ulcers are common in early disease; late disease is notable for stellate, rake, bear-claw, linear, or serpiginous ulcers and cobblestoning	Micro ulcers are more common, but larger ulcers are seen Pseudopolyps are more common
Distribution	Often discontinuous and asymmetric, with skipped segments of normal intervening mucosa, especially in early disease	Continuous, symmetric, and diffuse, with granularity or ulceration found in the entirety of involved segments; however, peri-appendiceal inflammation (cecal patch) is common, even when the cecum is not involved
Rectum	Complete, or more often relative, rectal sparing	Typically involved with variable proximal distribution
Ileum	Often involved (≈75% of cases)	Not involved, except as backwash ileitis in panulcerative colitis
Depth of inflammation	Mucosal, submucosal, and transmural	Mucosal; transmural only in fulminant disease
Serosal findings	Marked erythema and creeping fat (the latter is virtually pathognomonic)	Absent except in severe colitis or toxic megacolon
Perianal complications	Often prominent, including large anal skin tags, deep fissures, perianal fistulas, that are often complex	Not prominent (fissure or fistula if present, should be uncomplicated)
Strictures	Often present	Rarely present; when present, suggests adenocarcinoma
Fistulas	Perianal, enterocutaneous, rectovaginal, enterovesicular, and other fistulas may be present	Absent, except for the rare occurrence of rectovaginal or perianal fistula
Histopathology	Granulomas are present in 15%-60% of patients (higher frequency in surgical specimens than in mucosal pinch biopsies) Crypt abscesses may be present Focally enhanced inflammation, often on a normal background, is the hallmark	Granulomas should not be present (microgranulomas may be associated with ruptured crypt abscess) Crypt abscesses and ulcers are the defining lesion Ulceration on a background of inflamed mucosa
Serology	pANCA in 20%-25%, ASCA in 41%-76%	pANCA in 60%-65%, ASCA in 5%

Figure 8.1 — Crohn's disease and ulcerative colitis: comparison

Extra-intestinal Manifestations

COMPLICATIONS	CROHN DISEASE	ULCERATIVE COLITIS
Ocular disorders (uveitis, episcleritis)	+	+
Arthropathy	+	+
Oral ulcers	+	–
Skin disorders (pyoderma gangrenosum, erythema nodosum)	+	+
Nephrolithiasis	+	+
Primary sclerosing cholangitis	+	+
Bone disorders (osteoporosis, osteomalacia)	+	–
Thromboembolic disease	+	+
B_{12} deficiency	+	–

Figure 8.2 — Extra-intestinal manifestations

Diagnosis

Laboratory Studies

A complete blood count (CBC), urinalysis, stool analysis and culture and serum chemistry tests are appropriate during the initial evaluation.

Serologic examination of blood for specific antibodies , perinuclear antineutrophil cytoplasmic antibodies (pANCA)[frequent in UC], antibodies to Saccharomyces cerevisiae (ASCA) [frequent in CD].

A newly recognized and clinically useful inflammatory marker for UC disease activity is fecal calprotectin, which is a protein secreted by neutrophils in the feces and is therefore a marker of intestinal inflammation.

- **Barium enema and barium follow through**
- **Endoscopy**

Differential diagnosis

Diagnosis	Clinical Features	Radiologic and Colonoscopic Features	Histologic Features
UC	Bloody diarrhea	Extends proximally from rectum; fine mucosal ulceration	Distortion of crypts; acute and chronic diffuse inflammatory cell infiltrate; goblet cell depletion; crypt abscesses; lymphoid aggregates
Crohn's colitis	Perianal lesions are common; may be associated with ileitis; frank bleeding is less common than in UC	Segmental disease; rectal sparing; strictures, fissures, ulcers, fistulas; small bowel involvement	Focal inflammation; submucosal involvement; granulomas; goblet cell preservation; transmural inflammation; fissuring
Ischemic colitis	Occurs in older adults; sudden onset, often painful; usually resolves spontaneously in several days	Segmental splenic flexure and sigmoid involvement are most common, with thumbprinting early and ulceration after 24-72 hr; rectal involvement is rare	Mucosal necrosis with ghost cells; congestion with red blood cells; hemosiderin-laden macrophages and fibrosis (when disease is chronic)
Microscopic colitis	Watery diarrhea; normal-appearing mucosa at colonoscopy	Usually normal	Chronic inflammatory infiltrate; increased intraepithelial lymphocytes (lymphocytic colitis) and/or subepithelial collagen band (collagenous colitis)
Infectious colitis	Sudden onset; identifiable source in some cases (e.g., *Salmonella* spp.); pain may predominate (e.g., *Campylobacter* spp.); pathogens are present in stool	Nonspecific findings	Crypt architecture is usually normal; edema, superficial neutrophilic infiltrate, crypt abscesses
Amebic colitis	History of travel to endemic area; amebae may be detected in a fresh stool specimen but ELISA for amebic lectin antigen is the preferable diagnostic test	Discrete ulcers; ameboma or strictures	Similar to UC; amebae present in lamina propria or in flask-shaped ulcers, identified by periodic acid–Schiff stain
Gonococcal proctitis	Rectal pain; pus	Granular changes in rectum	Intense polymorphonuclear neutrophil infiltration; purulent exudate; Gram-negative diplococci
Pseudomembranous colitis	Often a history of antibiotic use; characteristic pseudomembranes may be seen on sigmoidoscopy; *Clostridium difficile* toxin is detectable in stools	Edematous; shaggy outline of colon; pseudomembranes may be identified radiologically or seen at colonoscopy	May resemble acute ischemic colitis; summit lesions of fibrinopurulent exudate

Figure 8.3 — Differential diagnosis of inflammatory bowel disease

Treatment

Diet and nutrition

- Patients with mild symptomatic IBD usually are able to take food orally.
- The diet should be nutritious.
- Patients with Crohn's disease who have terminal ileal involvement and steatorrhea may require supplemental fat-soluble vitamins, medium-chain triglycerides, and parenteral vitamin B12.
- Replacement iron may be indicated in patients who are iron-deficient.

Corticosteroids

- Historically, corticosteroids have been used in patients with severe UC or Crohn's disease to induce a remission.
- Intravenous (IV) steroids (i.e., hydrocortisone 100 mg IV q6h or methylprednisolone 10-30 mg IV q6-8h) are usually used in such patients.

- When patients can take oral medications, prednisone at doses of 40 to 60 mg per day is usually given.

Antibiotics

- Bacteria is known to play an important role in the pathogenesis of Crohn's disease and it may play a role in UC.
- In Crohn's diseases, the indications for the use of antibiotics include perianal disease, localized peritonitis due to microperforation, bacterial overgrowth secondary to a chronic stricture, and as an adjunct to drainage therapy for abscesses and fistulas.
- Metronidazole ,Ciprofloxacin and Rifaximin can be used.

Immunomodulators and biologic therapy

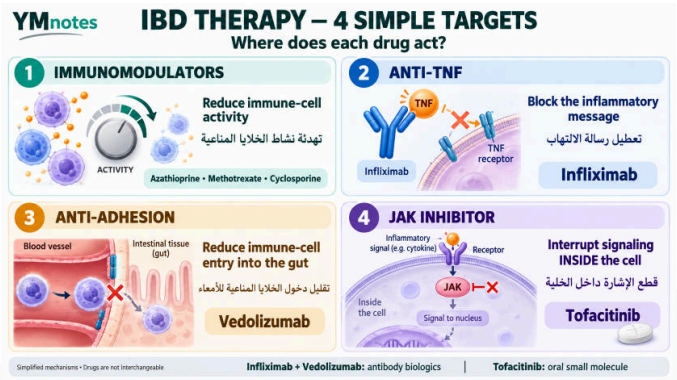
- Immunomodulator drugs act by blocking lymphocyte proliferation, activation, or effector mechanisms
- Azathioprine Methotrexate Cyclosporin can be used
- Anti–Tumor Necrosis Factor Antibody (biologic therapy)
- TNF is a key inflammatory cytokine and mediator of intestinal inflammation. The expression of TNF is increased in IBD.
- Infliximab is a chimeric mouse-human monoclonal antibody against TNF that is effective in CD.

Anti-Adhesion Molecules

- Adhesion molecules are important in cellular trafficking in IBD and other diseases, in which immune and inflammatory cells from the periphery are recruited into sites of inflammation.
- Gut-selective anti-adhesion molecule, vedolizumab shows promising results.

Kinase Inhibitors

- The Janus kinase (JAK) family of kinases, which include JAK1 and JAK3 which are critical for lymphocyte proliferation, function, and activation.
- Tofacitinib, a novel oral inhibitor of JAK1 and JAK3, was recently shown to be safe and effective for the treatment of moderate to severe active UC.



Indications for surgical intervention

Ulcerative Colitis	Crohn's Disease
Intractable disease	**Small Intestine**
Fulminant disease	Stricture and obstruction unresponsive to medical therapy
Toxic megacolon	Massive hemorrhage
Colonic perforation	Refractory fistula
Massive colonic hemorrhage	Abscess
Extracolonic disease	**Colon and Rectum**
Colonic obstruction	Intractable disease
Colon cancer prophylaxis	Fulminant disease
Colon dysplasia or cancer	Perianal disease unresponsive to medical therapy
	Refractory fistula
	Colonic obstruction
	Cancer prophylaxis
	Colon dysplasia or cancer

Figure 8.4 — Indications for surgical intervention

Complications

Toxic megacolon

- Toxic megacolon is a condition in which the colon becomes atonic and dilated because of transmural inflammation.
- It is characteristically associated with severe ulcerative colitis, but it may complicate any severe inflammatory condition of the bowel, including Crohn's disease, bacterial colitis, amebiasis, pseudomembranous colitis, and ischemic colitis
- Worsening of the patient's clinical condition and the development of fever, tachycardia, and leukocytosis.
- Medical therapy is designed to reduce the likelihood of perforation.
- If the patient does not begin to show signs of clinical improvement during the first 24 to 48 hours of medical therapy, the risk for perforation increases markedly, and surgical intervention is indicated.

Malignancy

- Patients with extensive ulcerative colitis have a markedly increased risk for colon cancer in comparison to the general population beginning 8 to 10 years after diagnosis and increasing with time.
- There should be surveillance colonoscopy with random biopsies in patients with long- standing ulcerative colitis beginning 8 to 10 years after the onset of disease and repeated every 1 to 2 years.
- If the specimens show dysplasia the patient is sent for colectomy.

CHAPTER 09

Parasitic GI diseases

X Microbiology (x2)

09

BOOD FLUKES

SCHISTOSOMA

Schistosomes are the causative agents of the disease schistosomiasis. Three species of schistosomes that parasitize humans are of major importance: S.mansoni, S. haematobium and S.japonicum.

The morphologic characteristics of the three species are given in the following table:

	S.mansoni	S.japonicum
Disease	Intestinal bilharziasis	Intestinal bilharziasis
Distribution	Nile Delta, Africa, South America, Middle East.	Far East.
Intermediate host	Biomphalaria alexandrina	Onchomelania species.
Reservoir host	Monkeys, rodents.	Dogs, cats, rodents, pigs and cattle
Habitat	Inferior mesenteric plexus of veins	perior mesenteric plexus of veins

Lifecycle:

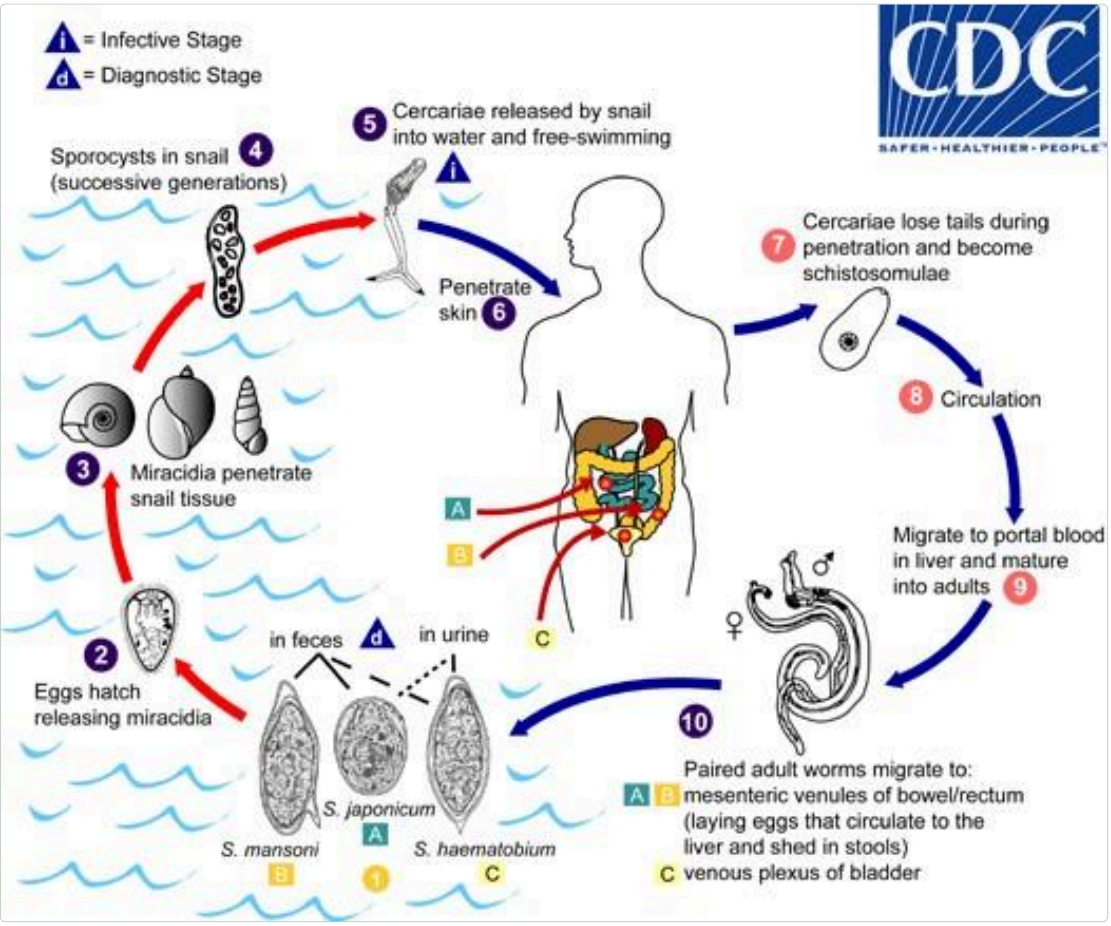


Figure 9.1 — Schistosoma life cycle

Eggs escape to the perivascular tissue due to:

i. pressure within the venule.

ii. effect of the spine.

iii. the oozing lytic secretion of the miracidium within the egg. From the perivascular tissue eggs pass to the lumen of the intestine in case of S.mansoni and S.japonicum and to the lumen of the urinary bladder in case of S. haematobium.

When the embryonated eggs (diagnostic stage) discharged in faeces enter fresh water, miracidia hatch from the eggs, rapidly swim by means of cilia (survive for 24 hours). Then they penetrate the body of snail (intermediate host).

The cercariae infect the definitive host by penetrating the skin, aided by:

i. proteolytic enzymes secreted from penetration glands.

ii. the surface tension of drying droplets of water.

iii. the strong lashing movements of the tail pressing the body of the cercaria into the skin. Drinking water may lead to infection when cercariae penetrate the mucosa above gastric acidity that kills them.

They lose their tails out and transform into the schistosomula.

Schistosomula migrate to the lungs via the venous circulation. They break out from the - lungs Through the pulmonary capillaries and are carried through the left heart into the systemic circulation. They finally reach the intra-hepatic branches of the portal veins in the liver where they settle down and mature.

Within several weeks, pairing of the worms takes place and they migrate to the mesenteric - veins (S.mansoni and S. japonicum) or to the vesical plexus (S. haematobium), where the females lay eggs.

The prepatent period between penetration by cercariae and the appearance of eggs in the - excreta is around 35 days for S.munsoni and Sjaponicum and about 70 days for S.haematobium.

PATHOGENESIS AND CLINICAL PICTURE:

Intestinal schistosomiasis is 7-8 weeks (for S. mansoni infection) and 5-6 weeks (for S. japonicum infection).

Pathological changes are caused mainly by schistosome eggs deposited in various tissues. The antigens excreted from embryonated eggs result in formation of granulomas that lead eventually to fibrotic changes in the tissues involved in chronic infections. Thus, the lesions produced by the infections are essentially the same among the 3 major species of schistosomes. There are 4 progressive stages:

1. Stage of invasion (1 - 4 days):

- Skin lesion due to cercarial penetration.
- Local dermatitis, irritation, papular rash.

2. Stage of migration (3 — 4 weeks):

- Lung: verminous pneumonitis, minute haemorrhages, cough and haemoptysis.
- Liver: enlarged and tender.

Metabolic products of migrating parasites: result in toxic and allergic manifestations de - urticarial, eosinophilia, leukocytosis, fever, and headache and muscle pain.

3. Stage of egg deposition and extrusion (acute stage: 1-2 month):

Eggs deposited in the venous plexus escape from the veins into the perivascular tissue and finally find their way to the outside with urine and stool (open lesion stage). With extrusion of eggs through the wall of the intestine or bladder, there may be again generalized malaise, fever, rigors, urticaria, abdominal pain and liver tenderness.

- **In schistosomiasis mansoni and schistosomiasis japonicum (intestinal** bilharziasis):

Active egg deposition results in tissue damage and haemorrhage.

symptoms:

dysentery with blood and mucus in stooliaca

- abdominal pain.
- Sjaponicum may present with Katayama fever

4. Stage of tissue reaction (chronic stage):

Eggs trapped in the tissue secrete soluble egg antigen that stimulate inflammatory reactions (cosinophilic granulomas) and fibrosis leading to formation of bilharzial nodules, papiliomata, sandy patches and reduced egg output (closed - lesion stage).

- **In schistosomiasis mansoni and schistosomiasis japonicum:**

The intestinal wall becomes fibrosed, thickened and may be complicated with strictures, sinuses, fistulac and prolapse.

Eggs that fail to be fixed to venules of intestinal walls fall in the lumen and are swept by the blood to the liver. This results in periportal fibrosis, portal hypertension, hepatosplenomegaly, followed by shrunken fibroscd liver.

This results in portal hypertension with oesophageal varices, haemorrhoids, and asetes. This is manifested by haematemesis, melena and distended abdomen. This stage ends in hepatocellular failure (impaired liver function) and fatal haemorrhage The enlarged spleen and the hypersplenism lead to destruction of the red blood cells that add i the existing anaemia. Later on it leads to pancytopenia.

II. Intestinal schistosomiasis:

1.Clinical diagnosis:

Living in or visiting endemic areas with history of exposure to canals water and complaining of dysentery or hepatosplenomegaly is suggestive.

2.Parasitological diagnosis:

Stool examination for recovery of the eggs (diagnostic stage) by:

a. Direct smear.

b. Concentration techniques may be required when infection is light.

3.Immunodiagnosis:

Decection of antigens and antibodies as in urinary schistosomiasis.

4. Other tests:

a) Complete blood count (CBC): Shows eosinophilia up to 40 % and anaemia.

b) Rectal swab **c)** colonoscopy and sigmoidoscopy for visualizing lesions e.g. papillomata, cologic peiyrosis, and for biopsies.

d) Ultrasonography and other imaging techniques: show periportal fibrosis and heptesplenomegaly.

TREATMENT:

1. Praziquante! (Biltricide), Oxmoquine (Vansil) and Metriphonate

PREVENTION AND CONTROL:

1. Mass treatment and follow up of infected persons.
2. Health education
3- Snail control

TAENIA SAGINATA

(BEEF TAPE WORM, BALD TAPE WORM)

GEOGRAPHICAL DISTRIBUTION:

Cosmopolitan especially in cattle raising countries.

1. Adult lives in the small intestine of man who is the only definitive host attaching itself to the mucosa.
2. Mode of infection: by ingestion of undercooked beef containing viable cysticerci usually in the form of hamburgers or grilled meat (Kebab).

In the small intestine, the scolex is evaginated and gets attached to the mucosa and matures in about 3 months.

Habitat: small intestine.

DH: man.

IH: cattle.

Infective stage: cysticercous bovis in cattle meat.

Diagnostic stage : mature egg in stool.

PATHOGENESIS AND CLINICAL PICTURE:

1. Intestinal disturbances, hunger pains and diarrhea.
2. Neurologic manifestations.
3. Intestinal obstruction.
4. Loss of weight as the parasite consumes much of the food of the patient leading to matnutrition and avitaminosis.
5. Appendicitis or cholangitis caused by stray segments of the worm.

6. Migrating segments creeping out of the anus cause irritation, itching and worry of the patient

DIAGNOSIS:

1. Clinical diagnosis: Suggestive symptoms as diarrhoea and weight loss, with passage of gravid segments through the anus.

2. Parasitological diagnosis:

- Stool examination to find eggs and proglottids (diagnostic stagesRecovery of eggs from perianal region using a swab.
- Gravid segments in faeces, if not found give a saline purge.

Compress the segments between two slides and count the main lateral branches.

3. Immunodiagnosis: Detection of coproantigen in patients' stool by ELISA.

TREATMENT:

1. Praziquantel is the drug of choice for both Taenia saginata and Taenia solium; it causes worm paralysis, in a dose of 10mg/Kg as a single oral dose.

2. Yomesan in a single oral dose of 2 gm on empty stomach.

CONTROL:

1. Treatment of the infected persons.
2. Avoid grazing cattle on fields fertilized by human manure.
3. Proper meat inspection in slaughter houses.
4. Refrigeration of meat at -10°C for 5 days.
5. Thorough cooking of beef.

TAENIA SOLIUM

(Pork tape worm)

GEOGRAPHICAL DISTRIBUTION:

Cosmopolitan where pork is eaten. Thus it is rare in Islamic and Jewish communities.

- **Similar to T.saginata but differs in:**

1. Pig is the intermediate host.

2. Larval stage is the cysticercus cellulosae which differs from cysticercus bovis in having invaginated scolex with four suckers and hooks (infective stage).

3. Man is infected by ingestion of undercooked pork

PATHOGENESIS AND CLINICAL PICTURE:

1- Intestinal disturbances, neurologic manifestations, intestinal obstruction or loss of weight.

2- When man ingests eggs of T.solium, the larval stage (cysticercus cellulosae) develops in the extraintestinal tissues leading to a condition known as cysticercosis. This condition is likely to occur with T.solium not with T.saginata, a fact which makes T.solium runs a more serious course.

DIAGNOSIS:

1- Clinical diagnosis.

2- Laboratory diagnosis:

a) Examination of stool to detect the gravid segments .The eggs are rarely present in stool and are similar to those of T.saginata.

TREATMENT:

1. Similar to that of T.saginata.

2. Yomesan in a single oral dose of 2gm on empty stomach. It is recommended to give a saline purge 1-2 hours later to avoid cysticercosis from eggs of disintigerated segments. Also Quinacrine hydrochloride (atebrine) can be added as it leads to expulsion of the parasite in an intact form.

3. Giving antiemetic before treatment to avoid vomiting.

Life cycle: (Taenia saginata and solium)

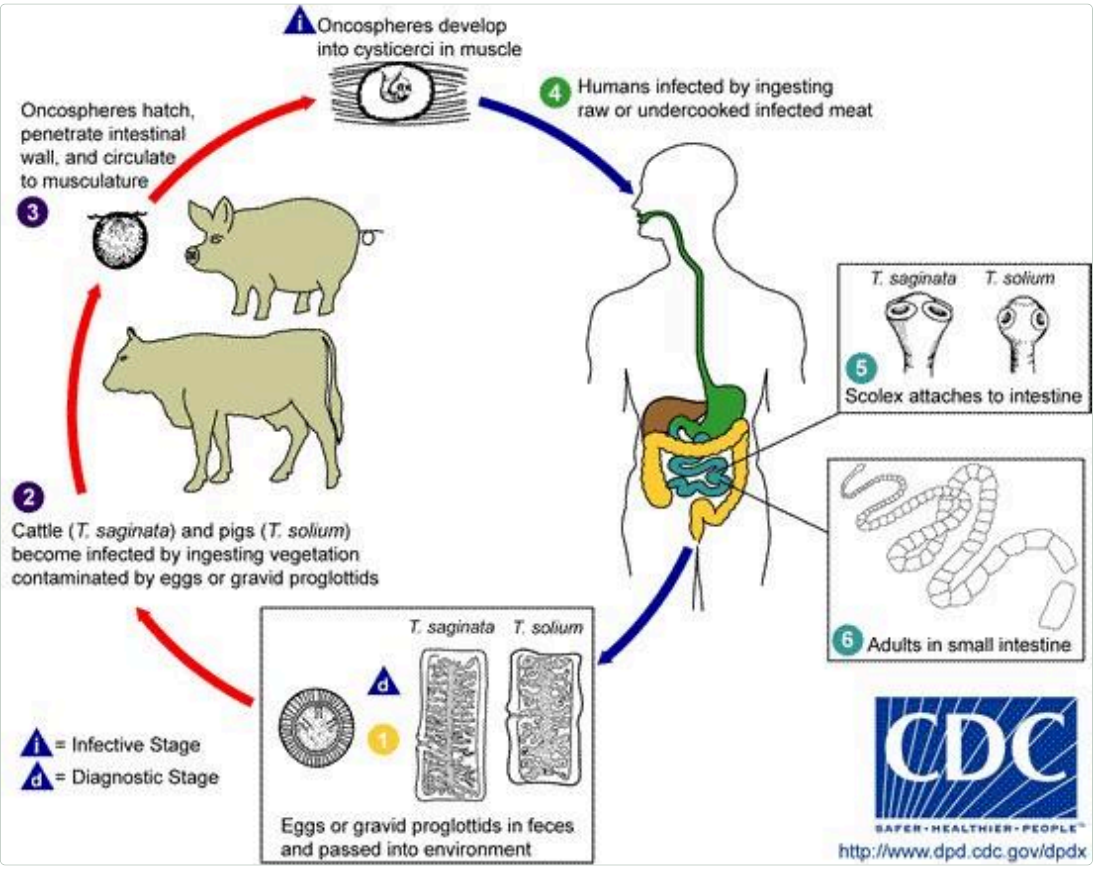


Figure 9.2 — Taenia saginata and solium life cycle

intestinal Nematodes

Ascaris lumibricoides

(Giant intestinal round worm)

GEOGRAPHICAL DISTRIBUTION:

Cosmopolitan. More prevalent in warm countries with bad sanitation. Commn in Egypt (especially children).

- **Host: Man.**
- **Habitat: Small intestine (free in the lumen).**

Life cycle:

(Geohelminthic infection

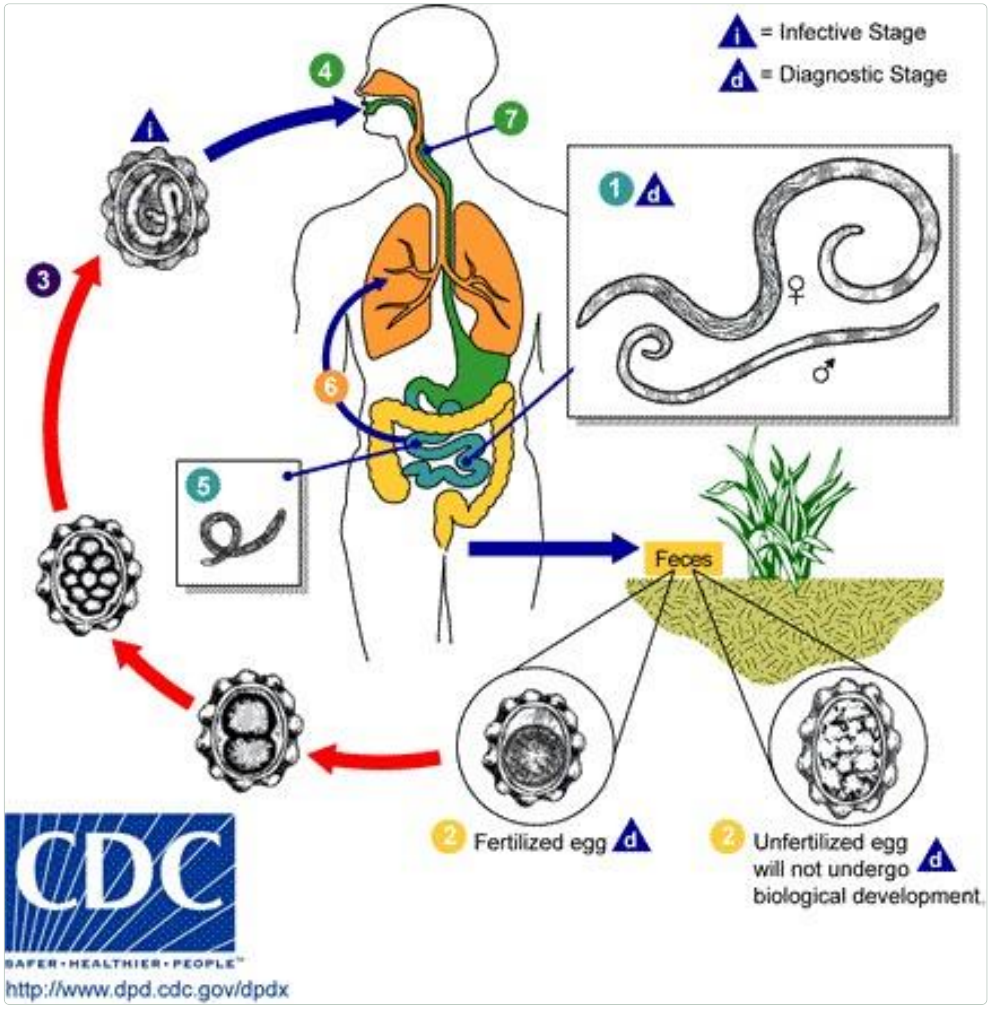


Figure 9.3 — Ascaris life cycle

Figure 9.4 — Ascaris: male and female worms

Figure 9.5 — Ascaris egg

Ascaris egg infection of man takes place by swallowing water or raw vegetables polluted with embryonated eggs containing the infective larva or through contaminated hands by polluted soil.

PATHOGENESIS AND CLINICAL PICTURE:

Tissue damage may be due to:

1. Large size of Ascaris (largest intestinal nematode).
2. Adults do not attach to intestinal wall, thus they may go to ectopic places.
3. Toxic products stimulate immune response.

CLINICAL PICTURE:

The usual infection (10 - 20 worms) often passes unnoticed. In heavier infection the following occurs:

1. During larval migration:

There is intense local inflammatory reaction around the larvae leading to pneumonitis manifested by fever, cough, dyspnea and eosinophilia.

- **In the intestine:**

Abdominal pain, nausea, vomiting, colic, distention or dyspepsia due to production of - anti. enzymes that interfere with protein digestion leading to malnutrition and underdevelopment of children.

- Changes of the bowel movements (diarrhoea or constipation).

2. Complications:

Traumatic effects:

due to irritation of adult worms by fever, drugs or anaesthesia. -

Intestinal obstruction, volvulus and intussusception occur by a mass of worms. Worms are not fixed to intestine so may be sloughed to:

- Bile ducts causing obstructive jaundice.
- Liver (carrying intestinal bacteria) resulting in liver abscesses.
- Appendix leading appendicitis.
- Ampulla of vater leading acute haemorrhagic pancreatitis.
- Peritoneum leading to peritonitis.
- Stomach leading to vomiting or escape through external nares.
- Very rarely to the trachea leading to suffocation.
- **Toxic effects:**

The by-products of living and dead worms may produce toxic symptoms in sensitized persons such as oedema of the face, urticaria, asthma, insomnia, nervous irritability and convulsions.

DIAGNOSIS:

1- Clinical diagnosis: non specific.

2- Parasitological diagnosis:

Detection of eggs in the stool.

Detection of larvae in the sputum and Charcot Leyden crystals early in the disease.

3- other tests:

- High cosinophilia.
- X-ray on abdomen after barium meal reveals worms as filling defects

TREATMENT:

Mebendazole or flubendazole: 100 mg twice per day for 3 days.

PREVENTION AND CONTROL:

1- Purification and chlorination of water.

2- Proper washing of raw vegetables and washing hands before meals.

3- Proper disposal of sewage (human faeces).

4- Mass treatment, personal hygiene and education of people.

Enterobius vermicularis

(Pin worm, Seat worm)

Geographical distribution:

Cosmopolitan, more in temperate and cold climates.

Final host:

Man especially children.

Habitat:

Large intestine mainly caecum, colon, rectum and appendix.

Mode of infection:

1- Ingestion of infective eggs through:

- Autoinfection (contaminated hand to mouth).
- Contaminated food or drinks.

2- Air-borne infection through inhalation of eggs in dust particles.

Pathogenesis and symptoms:

1) Irritation, itching and scratching of the peri-anal skin (pruritis ani), lead to insomnia, nervous manifestations and secondary bacterial infection.

2) Worms in young girls may deposit eggs on vulva producing pruritis vulvae.

3) Worms may migrate to the vagina (vaginitis), uterus, urinary bladder (urethritis) leading to irritation and inflammation.

4) Appendicitis is due to migrating adults.

5) Due to the easy spread of infection from one member of the family to another, all the family should be examined and all infected ones should be treated at the same time to get rid of the infection among the whole family.

6) Sometimes, the vagina and bladder are invaded in female patients causing irritation.

7) Migration to the peritoneal cavity (through uterine tubes) will cause pelvic peritonitis.

Diagnosis:

1- Clinically: Infection is suspected in children with Pruritus ani at night.

2-Laboratory:

- Adult worms may be seen in stool or anal area.

- Eggs are rarely found in stool (about 5 % only).

Swabbing of anal or perianal area by:

a) N.I.H. swab (National Institute of Health) **b)** Scotch adhesive tape swab:

Treatment:

1. Albendazle, Mebendazole

2. Local application of white oxide of mercury (white precipitate ointment) around the anus at night (relieves itching, kills female worms coming out and prevents dispersal of eggs).

Prevention and control:

- Personal hygiene.
- Infected persons should sleep alone, their underwear and bedding carefully handled & washed.
- Food and drink should be protected from dust & hand.

Life cycle:

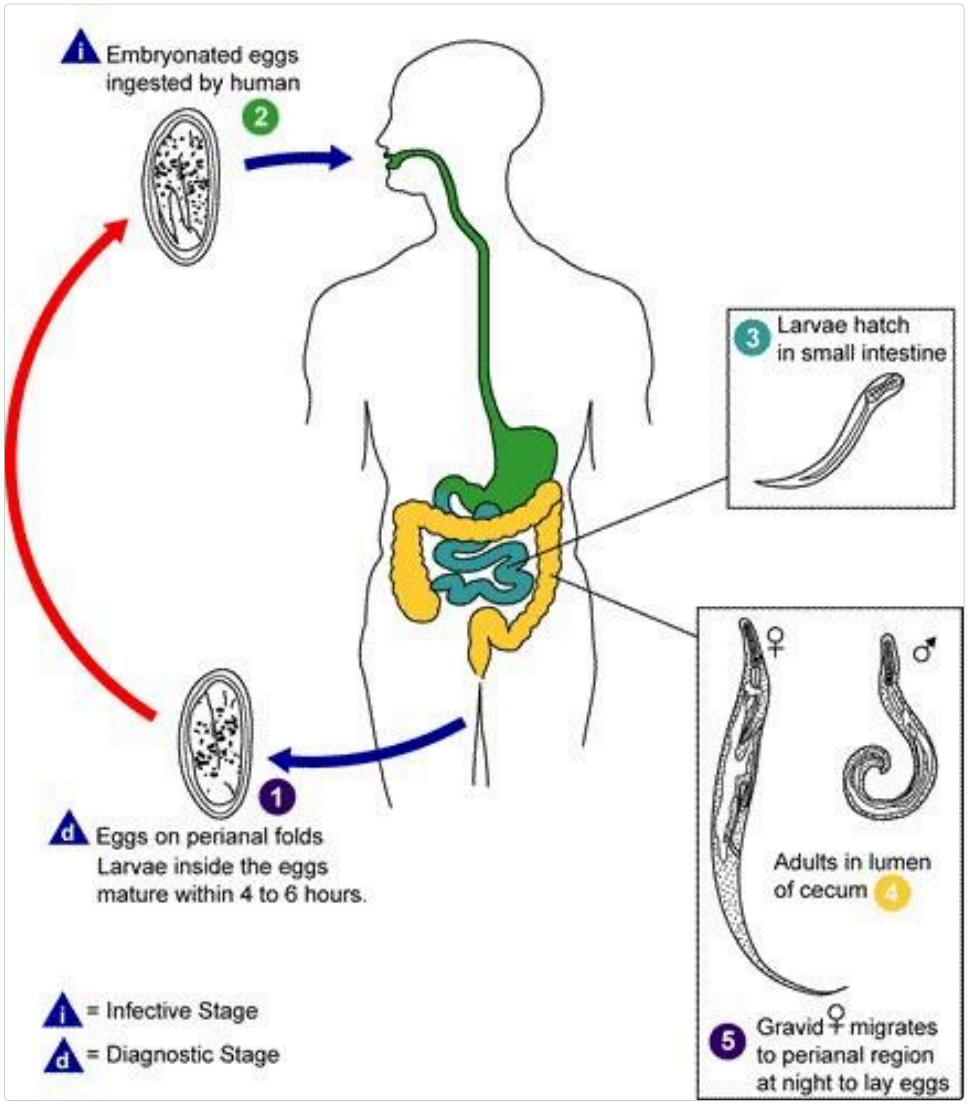


Figure 9.6 — Enterobius vermicularis life cycle

EXTRA-INTESTINAL AMOEBIASIS

A. HEPATIC AMOEBIASIS

PATHOGENESIS

Hepatic amoebiasis may occur due to either:

a. **Direct spread** of amoebic trophozoites from a primary intestinal lesion to surrounding tissues.

Or b y

b. **Haematogenous** spread through invasion of blood , Portal circulation draining the intestine carries amoebic emboli mainly to the right lobe of the liver.

Hepatic amoebiasis:

liver affection may be in the form of diffuse amoebic hepatitis or liver abscess.

a.Diffuse Amoebic Hepatitis:

In considerable number of patients with amoebic colitis there is an associated enlargement of the liver. This condition starts with constant invasion of the liver by amoebae from an active intestinal process via the portal circulation.

b.Amoebic liver Abscess:

In foci where amoebic colonization continues, there is lytic necrosis of hepatic parenchyma. The lesions coalesce forming typically a solitary lesion. It begins as a small area of necrosed hepatic tissue , which increases in size with liquefaction in the centre and thickening of the wall. Its contents are lysed liver cells, bile,fat,RBCs and collagen strands; all forming a viscid chocolate red or creamy mass(anchovy-sauce pus).Amoebic trophozoites are only found in the edge of the lesion where they multiply. The commonest site is the posterior part of the right lobe (silent area) causing upward bulging of the diaphragm. The hepatic abscess may become chronic, absorbed, fibrosed, calcified or may extend to other organs.

Clinical picture:

a. Symptoms:

There is usually an acute onset of pain in the upper abdomen with high fever. Th pain is intense and constant, radiating to the scapular region and right shoulder; it increases with deep breathing or coughing.

Fever is varying from 380 and 400 C. There is anorexia , nausea , vomiting and fatigue.

b. Signs:

1. Enlarged tender liver
2. Pallor
3. Earthy look
4. Fever
5. Tinge of jaundice

Differential diagnosis

Other causes of hepatomegaly and cystic lesions in the liver (hydatid disease)

Diagnosis:

a.Clinical diagnosis:

clinical picture

b. Laboratory diagnosis:

1. Indirect by serological tests: to detect specific antibodies. The indirect haemagglutination test (IHAT) and ELISA technique and both sensitive and specific but cannot be used to assess cure because antibodies persist for a longtime.

2. Direct diagnosis by:

i. Aspiration of the abscess: specimen is taken from the border of the abscess for detection of trophozoite.

ii. Stool examination: may be positive in only one third of cases.

iii. Detection of copro-antigen (in stool) by enzyme immunoassay.

c. Radiology:

i. X-ray shows high density in hepatic shadow and abnormal elevation of the right dome of diaphragm

ii. Ultrasonography, computed tomography (CT) and magnetic resonancemay visualize the amoebic liver abscess.

d. **Complete blood count (CBC)** shows polymorphonuclearleucocytosis in Amoebic

Liver Abscess (differentiated from viral hepatitis).

Prevention and control:

1. personal hygiene
2. Washing of fresh-eaten vegetables and fruits.
3. Safe water supply, water from polluted source should be boiled.
4. Insect control or protection of food from insects
5. Treatment of patients and examination of contacts

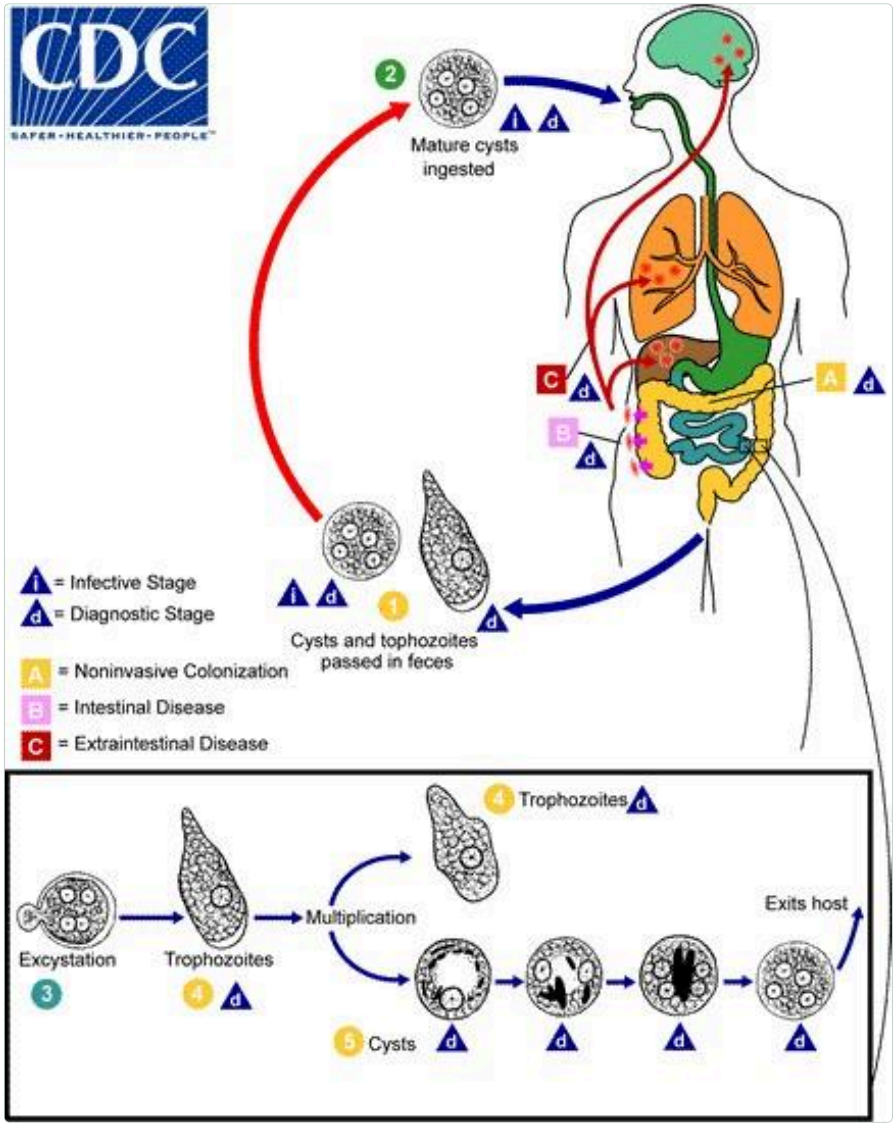


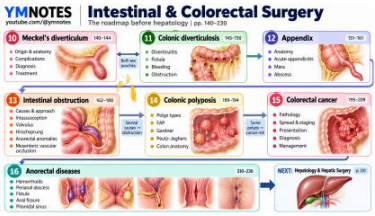
Figure 9.7 — Entamoeba histolytica life cycle

First 13 chapters → are the content of Mid-Module

Digestive 19 | Diverticular Disease

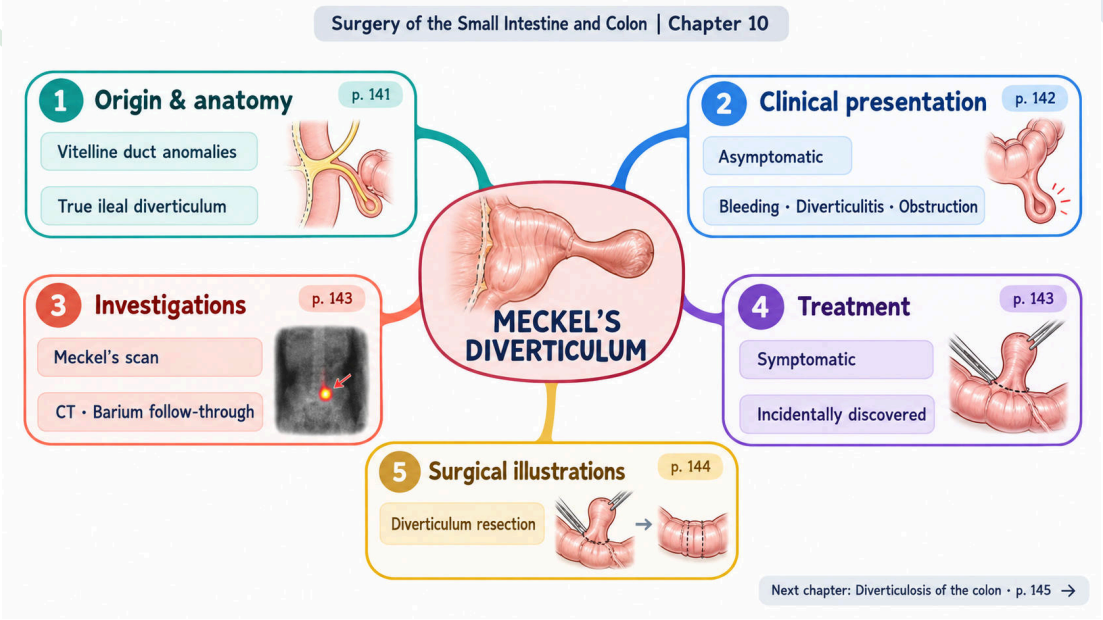
CHAPTER 10

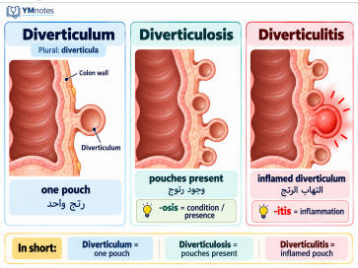
~~Surgery of the Small Intestine and Colon~~



Ch:10 Meckel's Diverticulum

10





Diverticular diseases

1- Meckel's diverticulum

2- Diverticulosis of the colon

↳ Diverticulitis

Meckel's diverticulum

- It is the most common congenital abnormality of the small intestine.
- It is caused by an incomplete obliteration of the vitello-intestinal duct.

Patho - physiology:

1- In embryonic life, the fetal mid-gut receives its nutrition from the ~~placenta~~ via the vitello- intestinal duct.

2- The duct then undergoes progressive obliteration and usually disappears by 7 weeks' gestation.

3- When the duct fails to fully obliterate, different types of vitelline duct anomalies appear.

(a) Umbilical fistula: a patent whole vitelline duct

(b) A fibrous band that connects the ileum to the inner surface of the umbilicus.

(c) Enterocele or cyst in the middle part

(d) Umbilical polyp patent distal part.

(e) **Meckel diverticulum**, most common (97 %):

- Patent proximal part of the duct.
- A blind-ending true diverticulum that contains all layers of the ileum.
- Meckel diverticulum almost always occurs on the anti-mesenteric border of the ileum.
- **Rule of 2 :**
- It is usually 2 feet (40-60 cm) proximal to the ileocecal valve,
- 2 cm wide
- 2 inches long
- found in 2 % of the population,
- often presents before age 2 years
- May contain 2 ectopic mucosa (gastric – pancreatic).
- Symptomatic in 2 %
- 2 times common in males than females

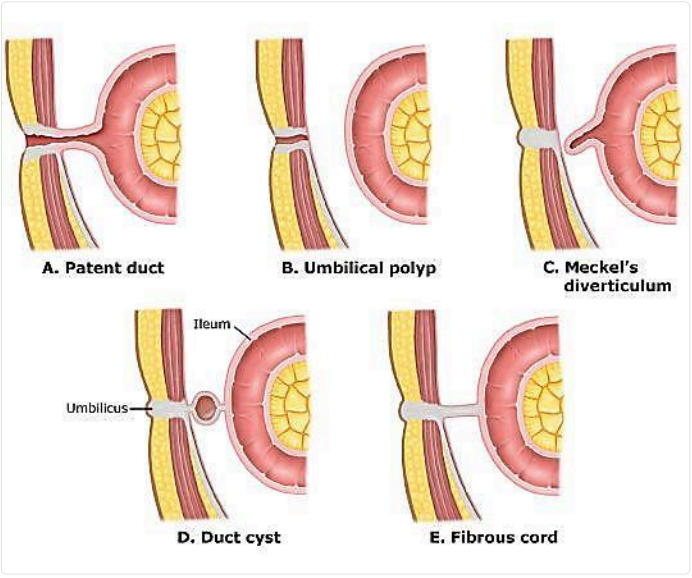


Figure 10.1 — Anomalies of vitello-intestinal duct

Clinical presentation and complications:

I- Aymptomatic in 98 % (Accidentally discovered)

II - lower duodeno-jejunal bleeding

Lower GI bleeding: Most common cause for bleeding per rectum in infants Menorrhagia in adults

- 25-50% (of symptomatic cases)
- Due to ileal ulceration by heterotopic **gastric mucosa:** ectopic (HCl secretion)
- presented with painless bleeding per rectum mostly in children below 5 y III -
  Local inflammation = **Mickel's diverticulitis** : 10 - 20%
- Clinically resembling appendicitis. (Rt iliac fossa)
- Usually in adult patients
- Usually due to ectopic acid producing **gastric mucosa.**

IV - **Intestinal obstruction** :

- Due to internal segmental volvulus or intussusception or by congenital band.
- Picture of acute intestinal obstruction (Colics – distension –vomiting – absolute constipation)

Painless if ulcers without significant inflammation

All prevent intestinal content from moving forward

التواء الأمعاء

انغلاق الأمعاء

bands or compress on intestinal wall

(no gas or fluid)

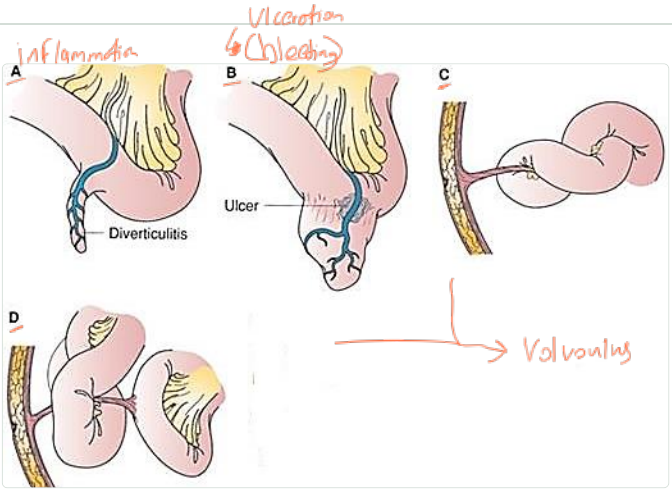


Figure 10.2 — Complications of Meckel's diverticulum

Investigations:

1- Difficult diagnosis preoperatively

2- "**Meckel's scan**" by -Tc-99:

- Most accurate test, especially in children.
- Taken up by gastric mucosa.

(● Accuracy 90% in pediatric patients.

3- Abdominal CT scan is often helpful in cases of obstruction

4- Barium meal and follow through.

Treatment

- ● **If symptomatic**, surgical intervention to resect the diverticulum or segment of ileum containing the diverticulum.
- ● If bleeding, the source of bleed is often in the segment of ileum adjacent to the diverticulum.
- ● **If asymptomatic** and found incidentally at surgery:
- In children, resection is recommended.
- In asymptomatic adults, resection is controversial since only about 2% of these patients will become symptomatic and there is about a 2% incidence of complications (stenosis, adhesions) after prophylactic resection.

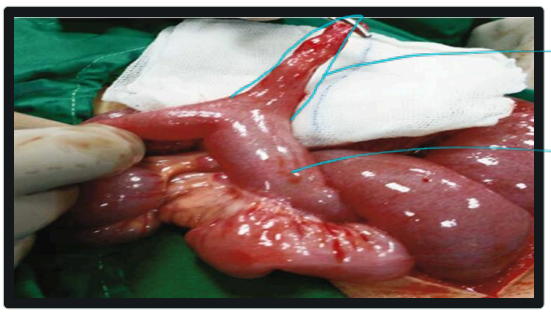


Figure 10.3 — Meckel's diverticulum

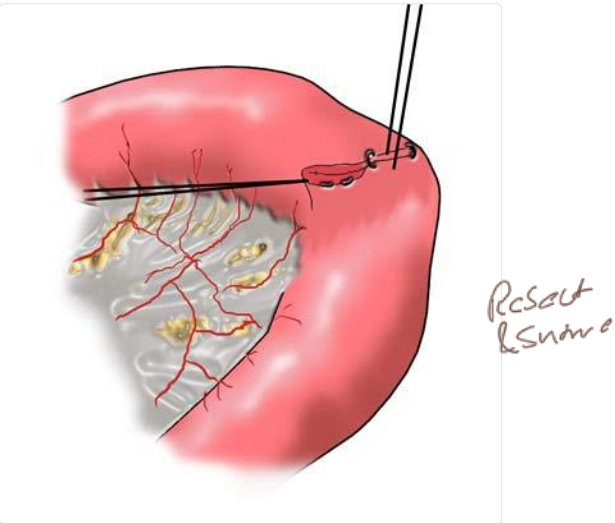


Figure 10.4 — Diverticulum resection

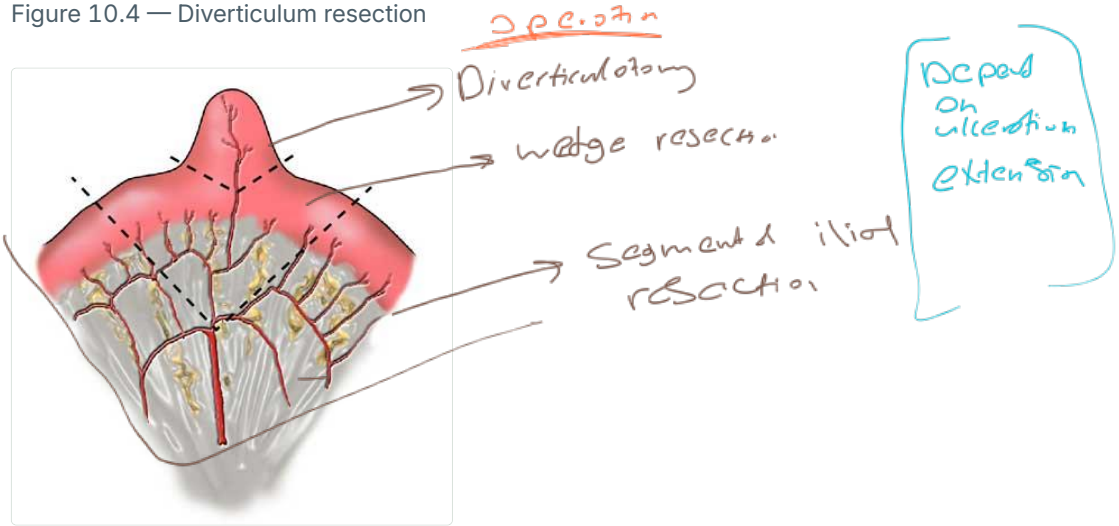


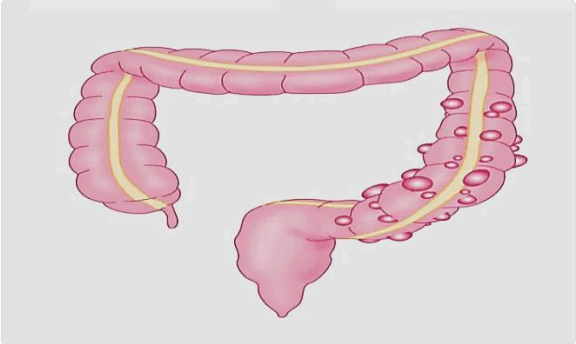
Figure 10.5 — Resection of the diverticulum

CHAPTER 11

Diverticulosis of the colon

Diverticulosis = Colonic Diverticula
↳ usually pseudodiverticula

11



N.B. Small intestine has complete muscle coat
Large intestine has incomplete muscle coat (x3 bands of taeniae coli)
Rect

Figure 11.1 — Diverticulosis of the colon

Definition:

A diverticulum is a sac like protrusion in the colonic wall that develops as Result of herniation of the mucosa and submucosa through points of weakness in the muscular wall of the colon.

افكر الطبقات من جوا لبرا
+circular muscle coat
(between the tenia coli muscle fibers)
[not true diverticulum as it doesn't contain all layers]

Predisposing factors: (increased intra-luminal pressure)

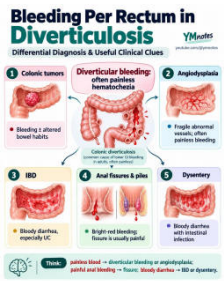
- Low-fiber diet and high carbohydrates
- Obesity
- Decreased physical activity
- Cigarette smoking
- Disordered motility of the colon
- Chronic constipation
- Increasing age

from increased segmental contractions in response to those factors
Goal is to move stool

How it is formed ? Increased intra-luminal pressure as in chronic constipation → protrusion of the mucosa through the points of entry of the arteries in the muscle wall (weak points). where vasa recta penetrate the muscular wall

Which parts of the colon affected?

- Any part could be affected except the rectum [Mostly Sigmoid colon]
- The rectum has complete muscle coat (no taenia coli) prevent protrusion of the mucosa



Clinical presentation and complications

I- Aymptomatic (non-complicated)
II- Acute diverticulitis (due e.g. fecal impaction)
III- Fistula (colo-vesical or colo-vaginal)
IV- Colonic bleeding
V- Obstruction

Studied in details

Common disease association:
IBS (C - predominant type)

A- Acute diverticulitis

- **Stages:**

1- Acute diverticulitis → just inflammation of the diverticulum

2- Perforation of the diverticulum → peri-colic collection of pus and fecal matter

3- peri-colic abcess formation

4- Generalized peritonitis

General Symptoms and signs:

- Fever (high with perforation and abcess)
  - Toxemia in case of peritonitis.

Local symptoms:

1- **Pain**

- Mainly in left side of the abdomen (more in left iliac fossa).
- Dull aching and fixed

2- Anorexia and vomiting

3- History of chronic constipation of long duration

Local signs:

1- Abdominal distension

2- Tenderness in left side of the abdomen and left iliac fossa (diverticulitis) 3- Tenderness, rebound tenderness and guarding rigidity in case of perforation and abcess formation

4- Picture of generalized peritonitis may occur in neglected cases.

(same as in 3 but generalized).

B- Picture of fistula

- Colo-vesical fistula presented with pneumaturia, and recurrent urinary infections.
- Colo-vaginal fistula presented with recurrent vaginitis.

C- Bleeding per-rectum

D.D : common causes of bleeding per/rectum

1- Tumours of the colon

2- Angiodysplasia

3- Inflammatory bowel diseases

4- Fissures and piles

5- Dysentery

6- Diverticulosis

Investigations of diverticulosis:

Lab. 1- CBC → Leucocytosis (↑neutrophils)

2- Liver, renal functions (Routine For general condition)

Radiology:

1- Plain X-ray abdomen erect → air under diaphragm in perforation (Pneumoperitoneum)

2- Abdominal ultra-sound to detect fluid collections in perforation e.g. abscess, inflammatory exudate

3- C.T. Scan → detect the diverticulae, collection and peri-colic abcess 4- Barium enema is done in non-complicated diverticulosis → Saw-tooth appearance

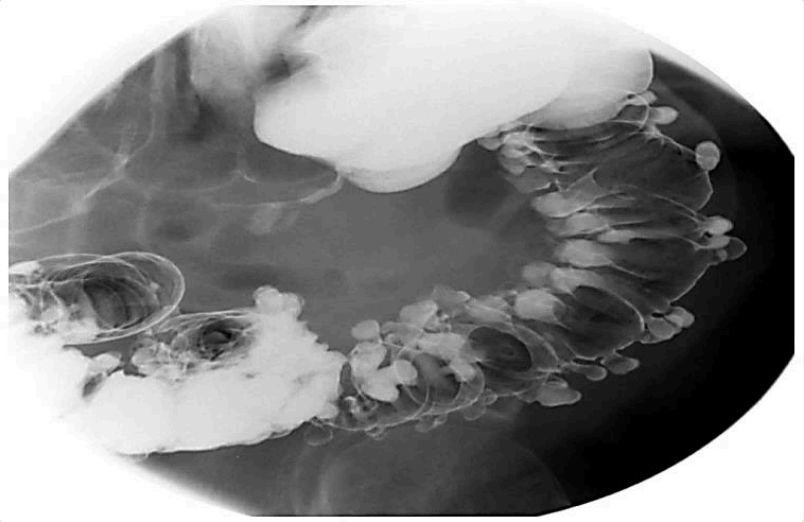


Figure 11.2 — Barium enema: multiple diverticulae

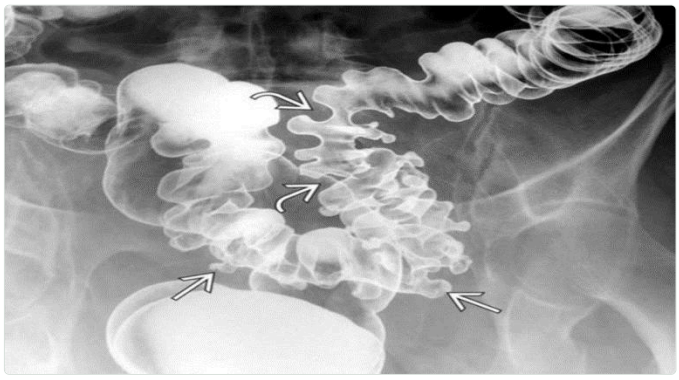


Figure 11.3 — Barium enema: multiple diverticulae, saw tooth

Treatment:

I- Conservative treatment: in NON-Complicated cases

1- Treatment of chronic constipation

2- High fiber diet

3- Antibiotics in acute diverticulitis without peritonitis or abcess will succeed in more than 85 % of cases

I-Per-cutaneous drainage:

ultrasound or CT guided in localized small abcess with drainage catheter left inside.

III- Surgical Treatment

- **Indications:**

1- Diffuse peritonitis (signs & symptoms).

2- Failure of conservative treatment

3- Failure of percutaneous drainage of peri-colic abcess.

- **Types of surgery:**

1- Hartmann" procedure:

1 st stage:

resection without anastomosis + end colostomy + closure of rectal stump •

2 nd stage:

colorectal anastomosis after 6 - 8 weeks

2-1 st stage:

resection + colo-rectal anastomosis + temporary ileostomy

2 nd stage:

closure of ileostomy.

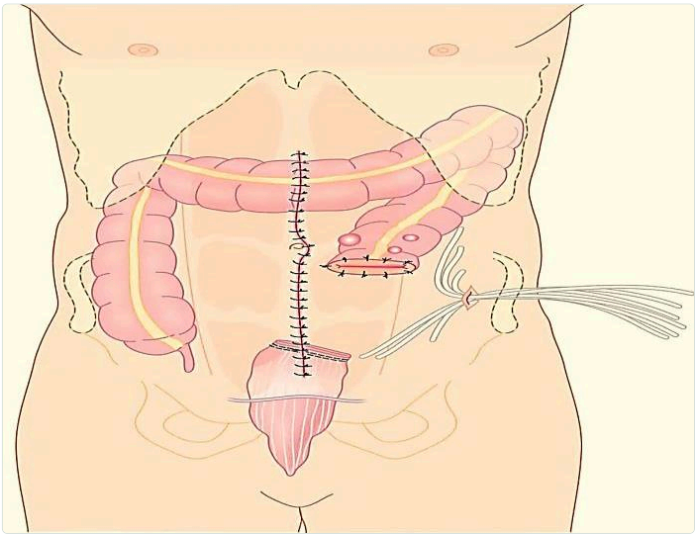


Figure 11.4 — Hartmann procedure

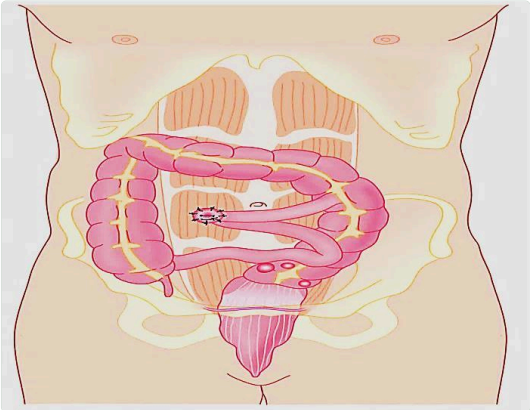
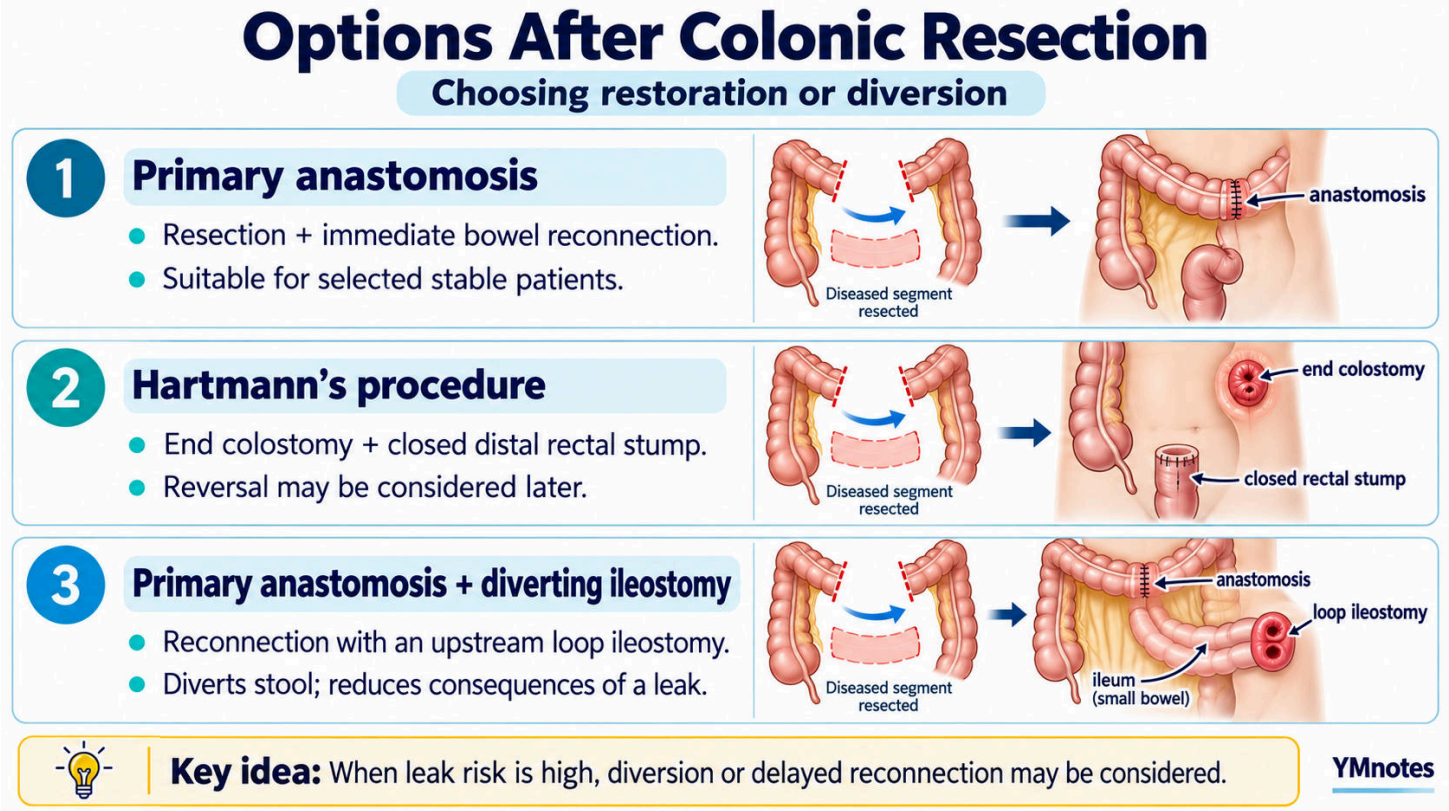
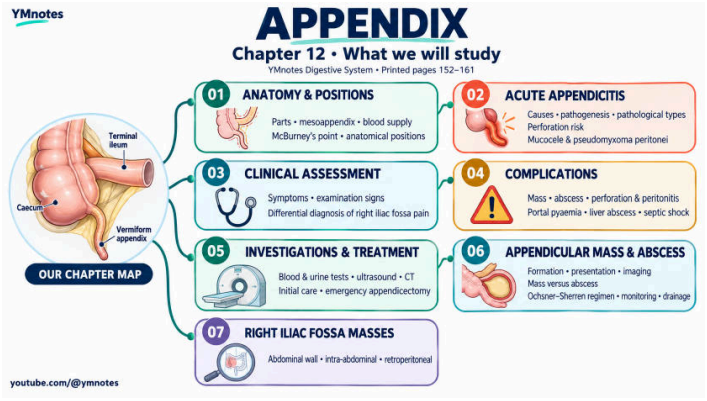


Figure 11.5 — Hartmann procedure (resection + temporary colostomy) Resection + colo-rectal anastomosis + diverting ileostomy





"attached projection from the cecum"

جزء ملتصق إلى الـ cecum أو كيس الفائدة

الزائدة الدودية

لا يخلق الله شيئاً عبثاً تعالى الله أحسن الخالقين

Latest Available knowledge:

مخزن للطلايا الهاضمة و البكتريا النافعة

After diarrhea, reservoir is needed.

CHAPTER 12

Appendix

12

سُورَةُ المُؤمِنُونَ

بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

أَفَحَسِبْتُمْ أَنَّمَا خَلَقْنَاكُمْ عَبَثًا وَأَنَّكُمْ إِلَيْنَا لَا تُرْجَعُونَ

﴿١١٥﴾ فَتَعَالَى اللَّهُ الْمَلِكُ الْحَقُّ ۖ لَا إِلَٰهَ إِلَّا هُوَ رَبُّ الْعَرْشِ

الْكَرِيمِ ﴿١١٦﴾

- The human appendix ranges from **2 to 20 cm.** (Length)
- The diameter of the appendix is usually between **7 and 8 mm**. (Diameter)
- The appendix is connected to the mesentery in the lower region of the ileum, by a meso- appendix containing appendicular artery and vein
  ↳ peritoneal fold

Parts of appendix:

The appendix has a base, body and tip. The mesentery attached to the appendix is known as mesoappendix.

- **Where the base of appendix is located?**
- The base of the appendix is attached to the posteromedial wall of the cecum 2 cm below the ileocecal junction.
- If the taenia coli in the cecum are traced downwards all the three taenia coli converges to the base of the appendix and continue as the longitudinal muscle coat of the appendix. (Done in surgery if appendix hidden)
- On the skin, the base of the appendix lies at Mac-Burney's point which is located at the right spino-umbilical line (Line joining between the anterior superior iliac spine and the umbilicus) at the junction of medial two-thirds and lateral one-third.
- **What is meso-appendix ?**
- This is the mesentery of the appendix attached to the mesenteric border of the appendix. • The mesoappendix contains the appendicular vessels.
- The appendicular vessels stop just before the tip of the appendix.
- The tip of the appendix is the least vascular area, and in obstructive type of appendicitis the commonest site of gangrene is the tip of the appendix.
- The appendix is supplied by appendicular artery which is a branch of inferior division of ileo-colic artery. The appendicular artery is an end artery.
  SMA → ileocolic artery → inferior division → appendicular artery (End artery)
  (other arteries cannot adequately compensate what this artery supplies if blocked [no collaterals])
- Sometimes an accessory appendicular artery may arise from the posterior cecal artery and may supply the appendix in addition to appendicular artery.
- The vein drains into the superior mesenteric vein (portal).
- **Why appendix is known as abdominal tonsil?**

An artery is considered an end artery when its blockage cannot be adequately compensated by other 4

The submucous coat of appendix contains lots of lymphoid follicles. The presence of lymphoid follicles is one important etiological factor for development of appendicitis.

- **What are the different locations of appendix?**

The base of the appendix is usually located at the Mac-Burney's point. The body and the tip of the appendix may lie at different locations and hence named according to the location.

No anastomosis

Hence, appendicitis doesn't always present with same clinical presentation

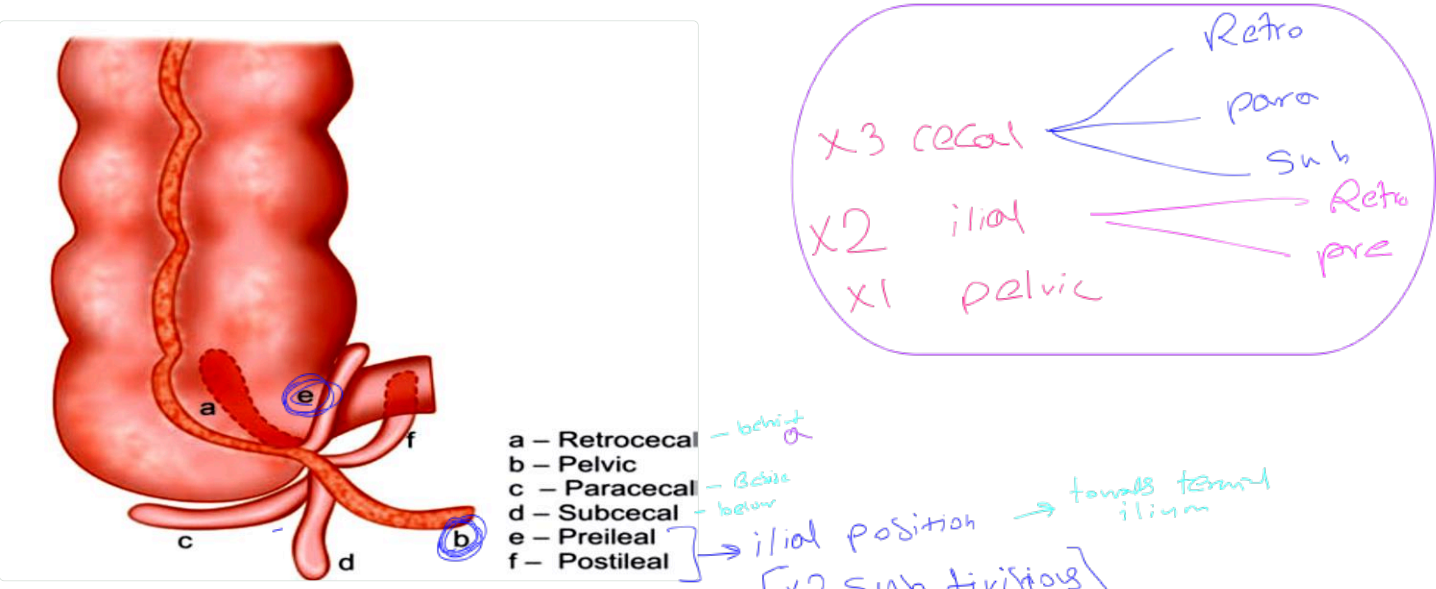


Figure 12.1 — Positions of the appendix

Retrocecal (60–70%):

commonest location, lies behind the cecum.

Pelvic (20–30%):

Second commonest location lies towards the pelvis.

Paracecal (1–2%):

Lies along the side of the caecum.

Subcecal (1 %):

Lies below the cecum.

Ileal (1–2%):

Lies towards the terminal ileum. May be:

- **Pre-ileal:** Runs infront of the terminal ileum
- **Post-ileal:** Runs behind the terminal ileum.

Ectopic appendix:

Due to mal-rotation of the gut, the appendix along with cecum may be located in **left iliac fossa or in the right sub-hepatic region**.

Acute appendicits

Acute inflammation of the appendix

Pathophysiology:

1- Appendicitis is caused by obstruction of the appendiceal lumen.

2- The most common causes of luminal obstruction include:

- lymphoid hyperplasia secondary infections (more common during childhood and in young adults), • Faecoliths, foreign bodies, worms, adhesions causing kink of the appendix and neoplasms as carcinoid tumor or cancer cecum.

3- Bacterial infection:

The commonest organism is Escherichia coli.

Other organis: Include enterococci, bacteroides, anaerobic streptococci, clostridia.

4- Dietary factors:

Low residue, high protein diet is associated with increased incidence of acute appendicitis. This has been related to constipation because of such diet.

Pathological types:

A-Acute catarrhal appendicitis:

Obstruction causes an increase in pressure within the lumen due to continuous secretion of fluids and mucus from the mucosa.

B-Mucocele of appendix:

A condition where appendix is swollen with mucoid secretion due to occlusion of its lumen proximally.

C-Acute suppurative appendicitis:

If appendicular obstruction persists, Intra-luminal pressure rises. As a consequence, appendicular wall ischemia begins, resulting in bacterial invasion of the appendicular wall.

D-Gangrene and perforation:

due to thrombosis of the appendicular artery and Veins.

E- **Appendicular abscess and peritonitis** in neglected cases

N.B.

The mucocele may rupture and the epithelial cells may be implanted on the peritoneal surface with formation of large mucoid mass. This condition is known as **pseudo-myxoma peritonii**.

Risk factors for perforation of acute appendicitis

- Extremes of age.

- Fecolith obstruction.
- Diabetes mellitus.
- Administration of purgative.
- Immunosupressed patients—chronic renal failure, steroid

Clinical picture:

General symptoms:

1- Fever → low grade unless complicated by suppuration and peritonitis. 2- Toxic look

Local symptoms:

1- Pain

- Often begins in the center of the abdomen, (peri-umbilical)
- This pain is typically a dull, and poorly localized,
- As the inflammation progresses, the pain begins to localize more clearly to the right iliac fossa, as the peritoneum becomes inflamed .
- Typically, point pain is not present until the parietal peritoneum is Inflamed. **2- Anorexia, nausea and vomiting.**

General signs:

- Fever – tachycardia
- Toxic look

Local signs:

1- Localized tenderness, rebound tenderness and guarding rigidity:

- In the right iliac fossa mainly at **Mc-Burney's point** (1/3 of the way along a line drawn from the anterior superior iliac spine to the umbilicus).

2- Rovsing's signs:

deep palpation in left iliac fossa elicit pain in right iliac fossa

3- Psoas sign:

pain in right iliac fossa on extension of the right hip due to inflammation of peritoneum over psoas muscle

4- Obturator sign:

internal rotation of the flexed hip causing pain in right iliac fossa

5- Silent abdomen due to ileus

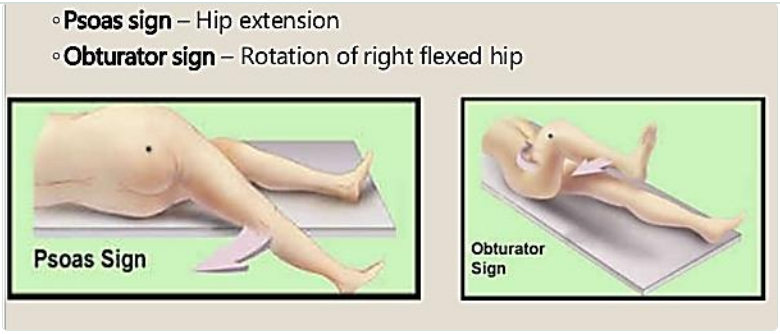
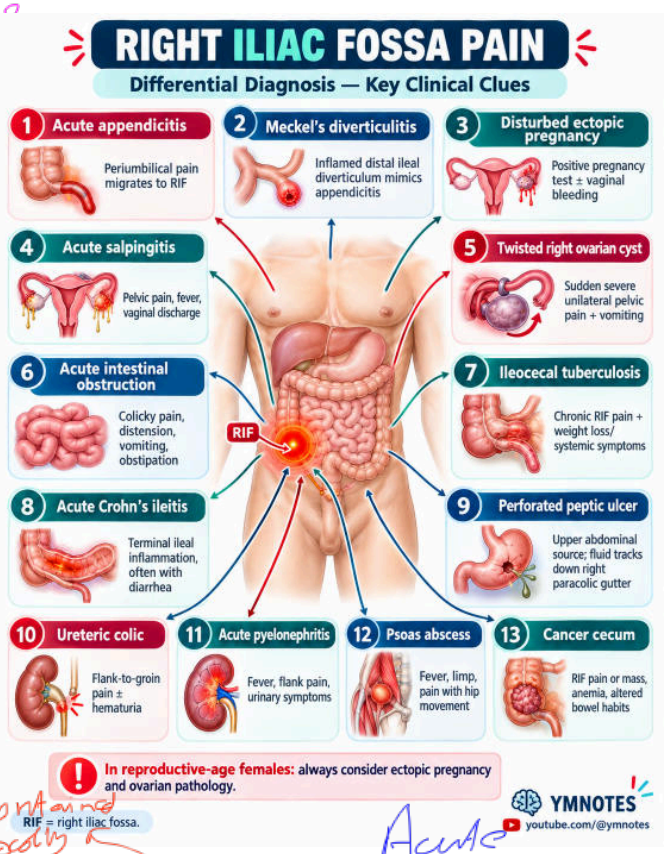


Figure 12.2 — Psoas and obturator signs

D.D. of pain in right iliac fossa

1. Acute appendicitis
2. Meckel's diverticulitis
3. Right-sided disturbed ectopic pregnancy
4. Acute salpingitis
5. Twisted right ovarian cyst
6. Acute intestinal obstruction
7. Ileo-cecal tuberculosis
8. Acute Crohn's ileitis
9. Perforated peptic ulcer
10. Ureteric colic
11. Acute pyelonephritis
12. Psoas abscess.
13. Cancer cecum



Complications of acute appendicitis:

1- Appendicular mass
2- Appendicular abcess
3- Perforated appendix and generalized peritonitis 4- Portal pyemia and liver abcess.
5- Septic shock

Management of non-complicated acute appendicitis:

Investigations helpful for diagnosis of acute appendicitis:

1. CBC: polymorphonuclear leukocytosis.
2. Urine analysis:

for suspected urinary causes as UTI → **PYURIA**

3. **C-reactive protein** may be elevated.

4. **Ultrasonography**—sensitivity of 86% for diagnosis of acute appendicitis and exclude other causes as gynecological causes.

5. **CT scan of abdomen**—sensitivity of 94% for diagnosis of acute appendicitis and exclude other causes.

6. In women **of child-bearing age— beta-HCG to exclude ectopic pregnancy.**

Treatment:

A.Initial treatment:

- Nothing by mouth.
- Intravenous fluid: Ringer's lactate and normal saline
- Broad spectrum antibiotics—2nd generation cephalosporin and metronidazole.

B. Emergency Appendicectomy:

- The operation should preferably done within 24 hours of admission.
  - The risk of perforation increases with time.
  - After **36 hours** the risk of perforation is up to **36%** and the risk increases by **5 %** for every 12 hours passed.
  - The procedure either done laparoscopically or by open technique.

Appendicular mass

It is an inflammatory mass composed of:

1- Edematous omentum,

2- ileal loops and cecum

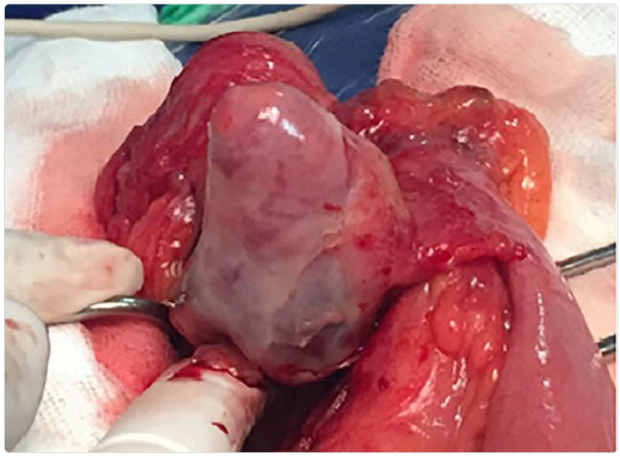
3- Inflammed appendix.

Clinically

- If the patient presents with classic features of acute appendicitis and then presents with a lump in right iliac fossa, this is likely to be an appendicular mass.
- Detected as tender mass in the right iliac fossa with guarding rigidity.
- Toxic look, high fever,...

How is the appendicular mass formed?

- Appendicular lump is a conglomerate consisting of inflamed appendix, adjacent terminal ileum, cecum and the greater omentum.



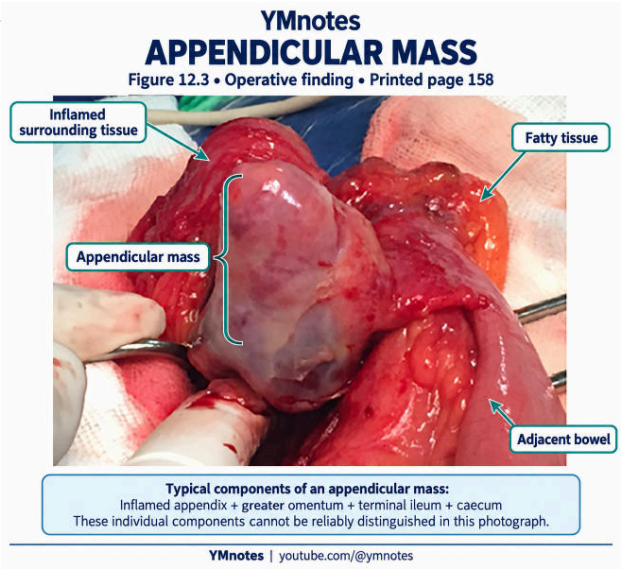


Figure 12.3 — Appendicular mass(operative finding)

How to investigate?

1- pelvi-abdominal ultrasound to exclude other causes (adnexal mass):

it reveals inflammatory mass in the right iliac fossa+ pelvic fluid collections

2- CT SCAN abdomen and pelvis with contrast:

more valuable in diagnosis and differentiate between appendicular mass and other pathologies

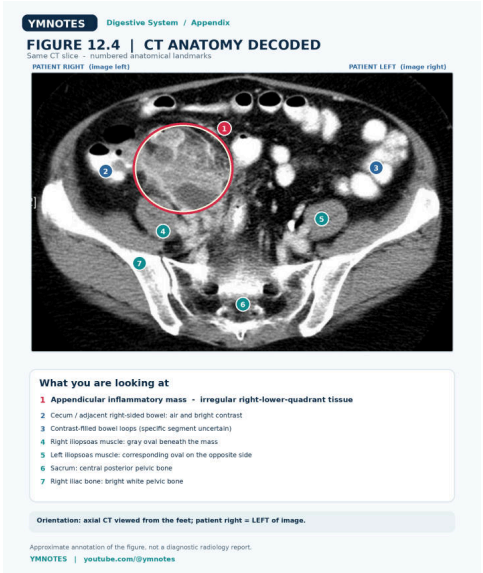


Figure 12.4 — Appendicular mass

How to differentiate an appendicular mass and an abscess?

- In a classical case the appendicular mass forms around the third day of acute attack of pain and develops into an abscess around fifth to tenth day.
- Pyrexia, aggravation of the local signs and a rising leukocyte count are indicators of abscess formation.
- Ultrasound and CT scan may demonstrate pus within an appendicular mass.

Figure 12.5 — Appendicular collection: CT view

Figure 12.6 — CT appendicular abcess

How do you plan management of appendicular mass?

- If appendicular mass is palpable on admission: conservative treatment

(Ochsner-Sherren regimen) is the preferred treatment

- Immediate appendicectomy is technically difficult.
- There is increased chance of fecal fistula postoperatively due to bowel injury
- **Operation should be done if conservative treatment fails with persistent picture.**
- **Components of conservative treatment Ochsner's Sherren regime:**

For pain:

Nonsteroidal anti-inflammatory drugs or narcotic analgesics as Pethidine.

For vomitting:

Stop oral fluids, and insert nasogastric tube for suction.

Fluid administration:

Intravenous fluids to maintain fluid electrolyte balance.

(Ringer's and normal saline)

- **If patient is not vomiting**—oral fluids may be administered.

Control infection:

3rd generation cephalosporin along with metronidazole administered parenterally.

Monitoring is the most important component of conservative treatment: • Symptomatic improvement—pain, vomiting.

- Hourly monitoring of pulse, blood pressure and respiration.
- Monitoring of temperature 4 hourly.
- Tenderness in right iliac fossa
- Progress of the mass (is to be marked)
- If the patient responds well to conservative treatment: Lump may regress, pain decreases, pulse and temperature normalizes → Continue antibiotics for 7–10 days - Allow oral fluid and then food when patient has the desire to eat

(4th– 5th day).

- Complete disappearance of the mass takes 6–8 weeks.
- **Interval appendicectomy** should be considered after 6–8 weeks.
- If with the above regimen; the mass increasing in size, pain and tenderness aggravating, rising pulse rate and fever with chills and rigor—this indicates **appendicular abscess** formation and early intervention for drainage is indicated.

Management of a case of appendicular abscess

- The patient may show features of septicemia.
- Formation of pus is confirmed by USG or CT scan.

I- Resuscitation followed by drainage of abscess is the treatment of choice.

1. Intravenous fluids.
2. Nasogastric suction
3. Catheterization (for monitoring urine output)
4. Intravenous antibiotics—3RD generation cephalosporins+ Metronidazole.

II- Steps of drainage of appendicular abscess:

a) General anesthesia with endotracheal intubation.

b) A muscle cutting incision. Skin and external oblique incision like Mc-Burney's incision. But the internal oblique and transversus abdominis muscles are cut in the same line of incision for better drainage.

c) The abscess wall is incised and abscess drained without opening the peritoneum.

d) A tube drain is inserted through separate wound and the main wound is closed in layers. e) if the appendix is accessible in the operative field it should be removed.

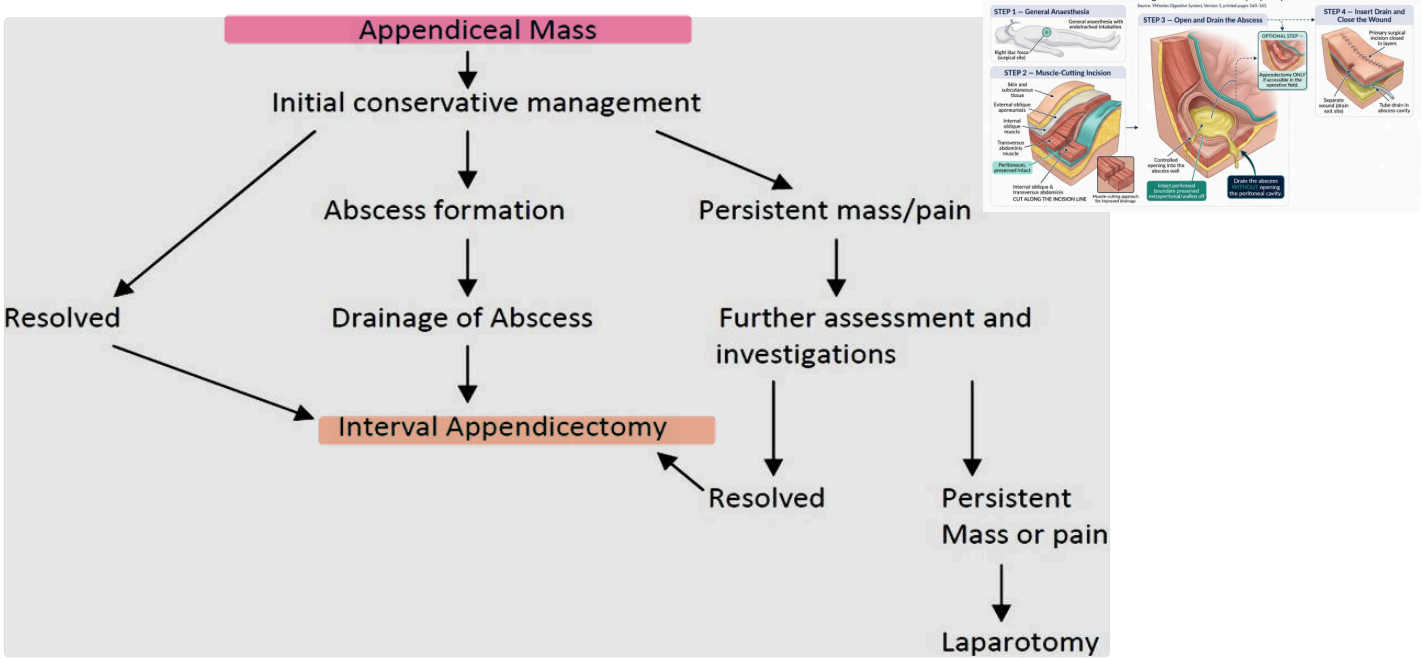


Figure 12.7 — Management of appendicular mass

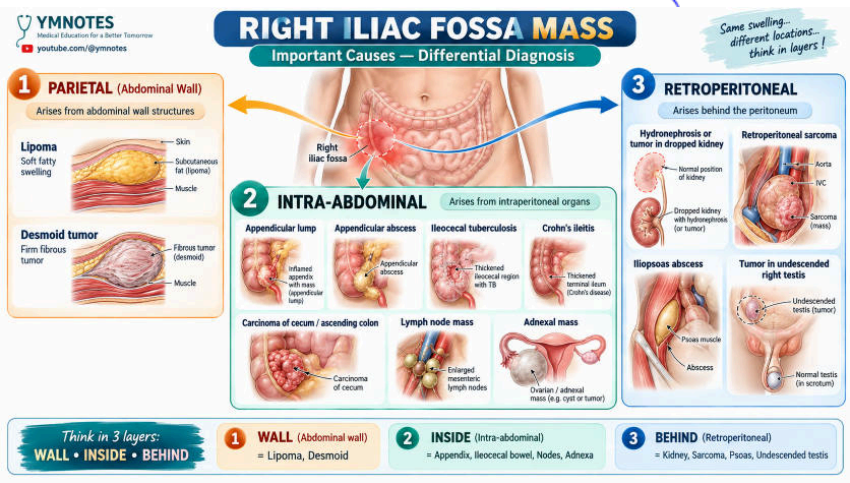
Important causes of mass in right iliac fossa:

1. Parietal swelling:

- Lipoma
- Desmoid tumor

2. Intra-abdominal swelling:

- Appendicular lump.
- Appendicular abscess.
- Ileocecal tuberculosis.
- Crohn's ileitis.
- Carcinoma cecum or ascending colon
- Lymph node mass
- Adnexal mass



3. Retroperitoneal swelling:

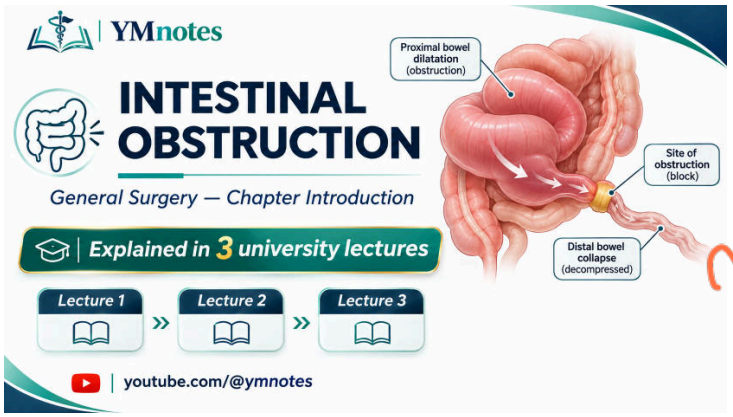
- Hydronephrosis or tumor in an unascended or dropped kidney.
- Retroperitoneal sarcoma.
- Iliopsoas abscess.
- Tumor in an undescended right testis.

انسداد الأمعاء

CHAPTER 13

Intestinal obstruction

(Small / large intestine)



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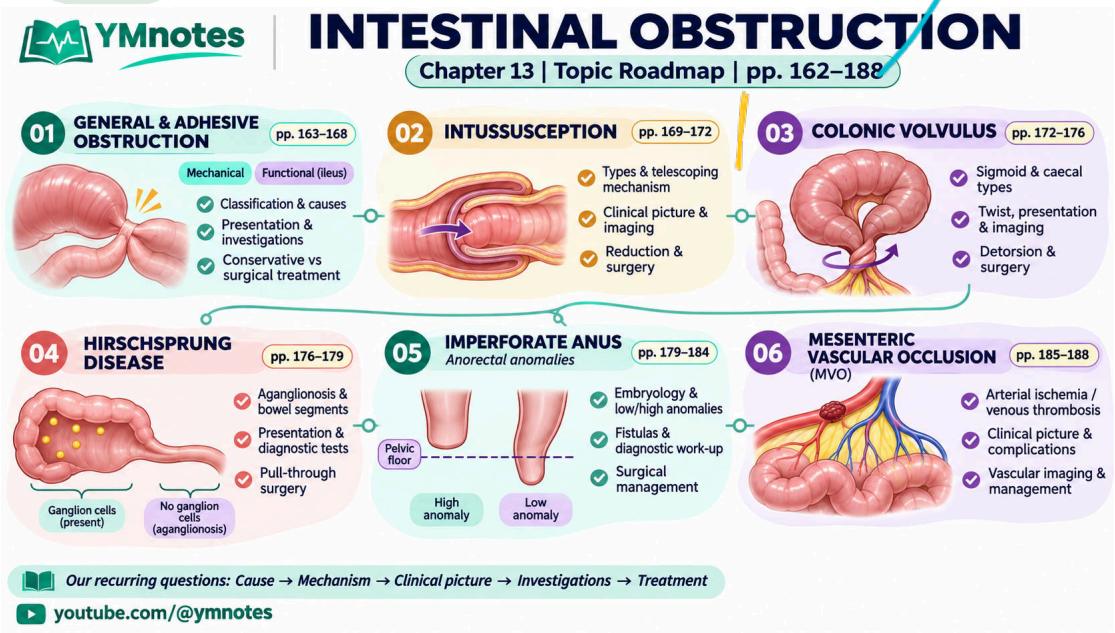


Definition:

Any condition interferes with normal forwards passage of intestinal contents.

Classification:

Cause of obstruction: mechanical or functional (Ileus).

Duration of obstruction: acute or chronic.

Degree of obstruction : partial or complete

Type of obstruction (complicated or not): (simple or strangulation) .i.e. Simple vs complicated

Mechanical obstruction

It means obstruction due to occlusion of the lumen of the bowel

According to the duration :

1- Acute (small intestine)

2- Chronic (colon)

Complicated or simple :

1- Simple →intact blood supply and wall of the intestine.

2- Closed loop as in volvulus = **complicated**

3- Strangulated (interfere with the blood supply) = **complicated**

4- Perforation = **complicated**

- **Important Facts:**
- Adhesion is the most common cause of intestinal obstruction
- 80 % of bowel obstruction is due to small bowel obstruction ; 20% colonic obstruction • The most common causes of small bowel obstruction are **adhesions and hernias** and less common neoplasms
- Commonest causes of colon obstruction are **Colo-Rectal cancer** in 60-70% while 30% are **diverticular disease**, **volvulus and others.**
- Mortality rate ranges between 3 % for simple bowel obstruction to 30% when there is strangulation or perforation.

Etiology:

Mechanical bowel obstruction:

A. Small bowel obstruction:

1. Adhesion 60% (postoperative – congenital)
2. Hernia 20% (femoral, inguinal H.)
3. Neoplasm 5%

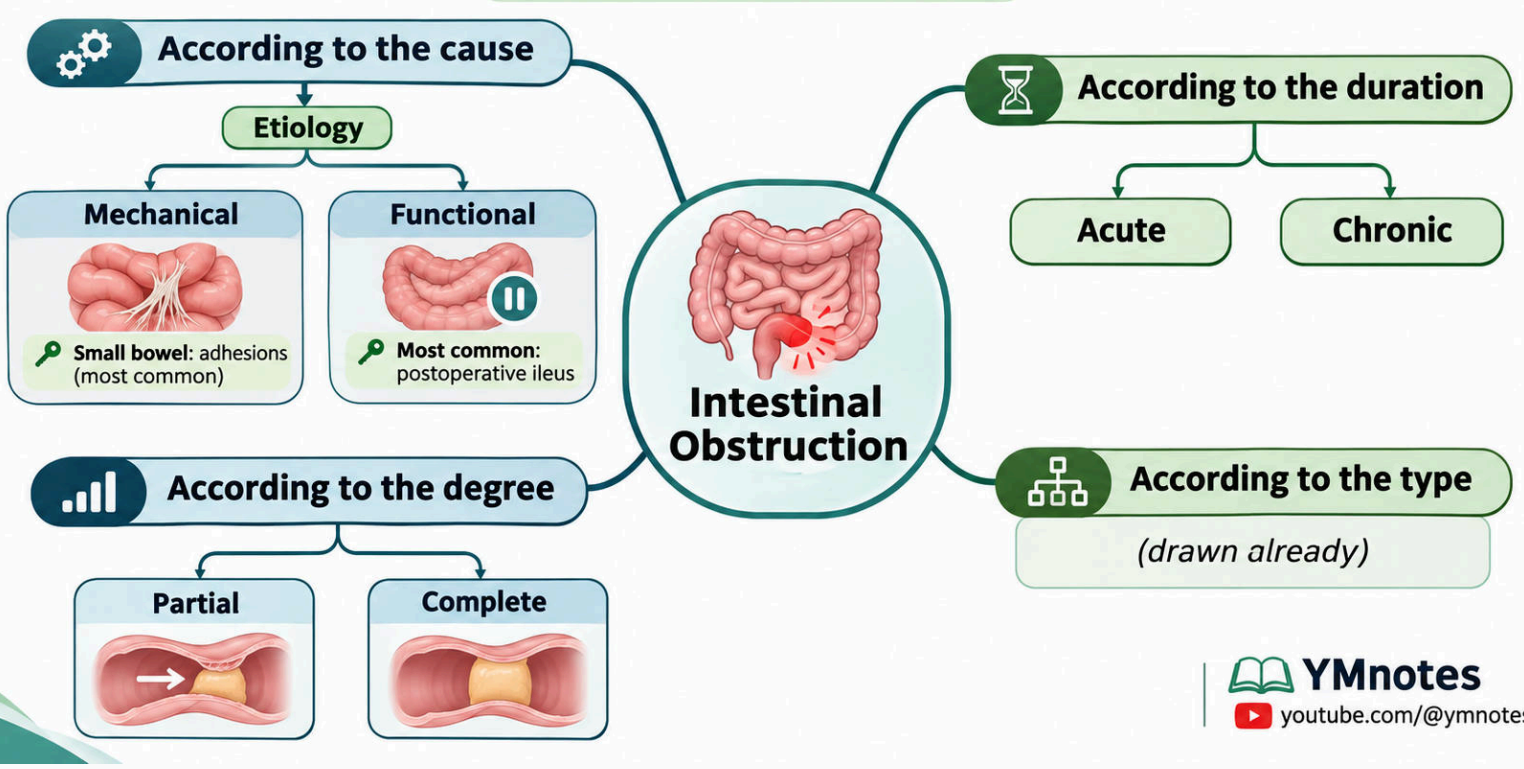
Intestinal obstruction Classification

→ Type

- Simple
  - intact wall & blood supply
- Complicated
  - Closed-loop
    - obstructed at 2 points
  - Strangulated
    - ischemic
  - perforated
    - ruptured

INTESTINAL OBSTRUCTION

Classification & Etiology



YMnotes
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4. Volvulus 5%.

5. Others: intussception, foreign bodies, gall stones

B. Large bowel obstruction :

1) Cancer colon 60 %.

2) Diverticular disease 15 %.

3) Volvulus (sigmoid or cecum) 15 %.

4) Others: hernia –fecal impaction – inflammatory strictures.

Functional bowel obstruction:

Adynamic ileus = Paralytic ileus

1. Postoperative mostly after abdominal surgery is the most common cause
2. Metabolic causes: DKA - hyponateremia - hypokalemia
3. Intra-abdominal inflammation—sepsis → peritonitis
4. Retroperitoneal hematoma
5. Mesenteric vascular occlusion
6. Fracture spine
7. Uremia.
8. Drugs: morphine - anticonvulsant.

Clinical picture of mechanical obstruction

Symptoms:

1. **Colicky abdominal pain** (due to contraction of the proximal intestinal segment against the obstruction)
   - Pain is central abdominal
   - Pain occurs in attacks few minutes in between
   - Pain gradually increasing in severity
2. **Vomiting**
   - Common in proximal intestinal obstruction and late in colonic obstruction
   - The colour of vomitus is whitish in gastric obstruction, greenish in upper intestinal obstruction and brownish or feculent in colonic obstruction.
3. **Distension** due to proximal loop dilatation
4. **Absolute constipation** in complete obstruction.
5. History of previous abdominal operation **(Adhesions) in adhesive I.O.** 6. Red currant jelly stool in **intussception**
7. Symptoms of associated cancers (in colonic obstruction) in **obstructed** carcinoma.

8. History of tender swelling in anatomical site of hernia (Femoral- inguinal) in **obstructed hernia.**

General Signs:

1. Signs of dehydration in prolonged cases (dry tongue, sunken eyes)
2. Fever, tachycardia, toxic look in cases complicated with strangulation and peritonitis.

Local signs:

1- Inspection:

- Scars of previous operation
- Hernia swelling
- Abdominal distension

2- Palpation:

- Examine for hernias
- Tenderness, rebound tenderness and rigidity in cases of complicated peritonitis.
- Palpable mass in obstructed carcinoma

3- Percussion:

hyper-resonance

4- Auscultation:

- Exaggerated intestinal sounds (mechanical obstruction)
- Silent in associated peritonitis

5- P/R:

empty rectum

Investigations:

Laboratory:

1- Serum electrolytes (Na & K) decreased

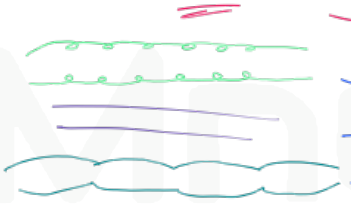
2- Renal function: impaired in neglected cases (pre-renal failure).

3- CBC → Leucocytosis in strangulations or perforations and peritonitis

Radiology:

1- Plain X-Ray abdomen erect and supine

- Supine → dilated loops: detect the level of obstruction
- Concertina effect (jejunal)
- Featureless tubes (ileum)
- Haustrations (colon)



- Erect →
- Multiple air and fluid levels

• Air under diaphragm in perforated viscus (closed loop obstruction or strangulation)

2 - Computed tomography

- Used to detect obstruction, strangulation and fluid collection following perforation and peritonitisAccuracy 94% in diagnosis of obstruction

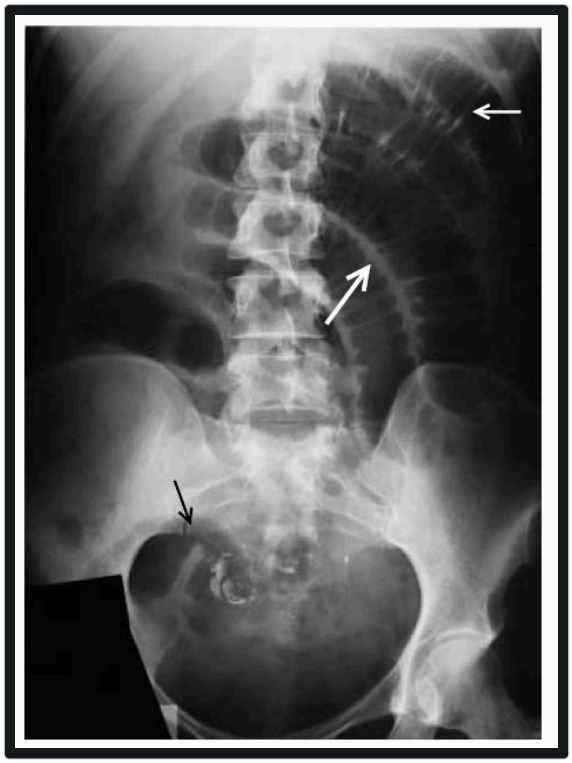
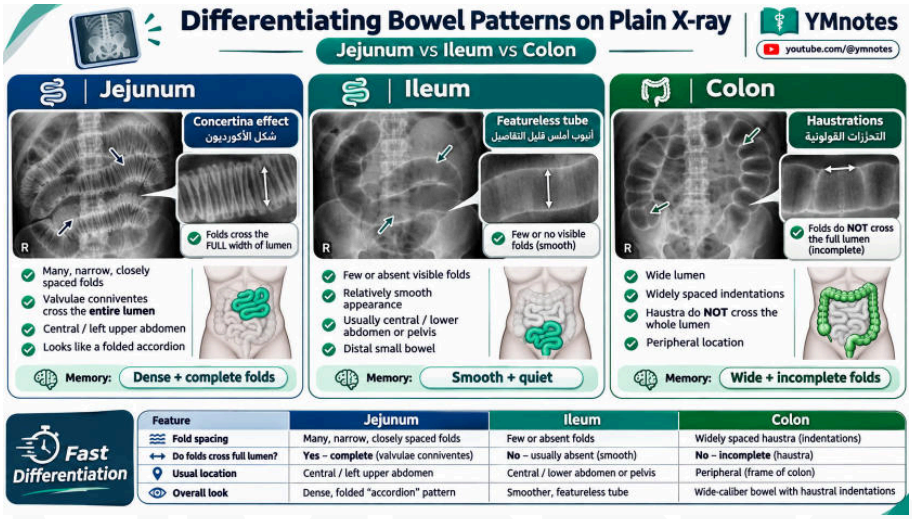


Figure 13.1 — Plain abdominal radiograph in obstruction



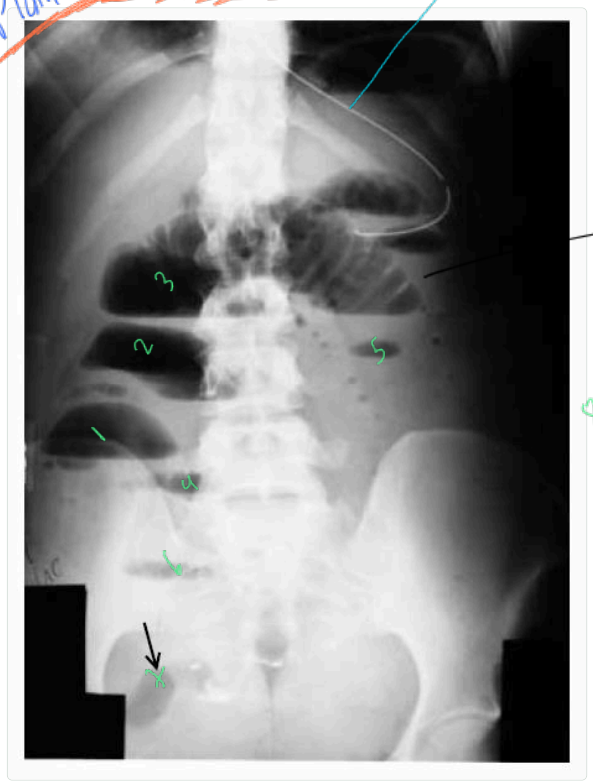


Figure 13.2 — Plain x-ray Supine and erect films

Treatment of mechanical intestinal obstruction (adhesive)

I- Conservative treatment

Indications:

1- All cases with adhesive intestinal obstruction

2- Preoperative preparation

Contraindications:

1- Evidence of peritonitis or strangulation

2- Obstructing carcinoma

3- Obstructed hernia

Lines of treatment:

I.Conservative treatment:

This treatment continues for 48 hour

1- **Intravenous fluids** administration to correct dehydration (Ringer solution) and protect the kidneys

2 - Naso-gastric tube:

- Suction of the fluids from the proximal segment

- Decreases the distension
- Decreases the colicky pain
- Gives an idea about the level of obstruction through the colour of aspirate

3- **Antibiotics** mainly against gram - ve and anaerobes (3rd Generation Cephalosporin and Metronidazole I.V.)

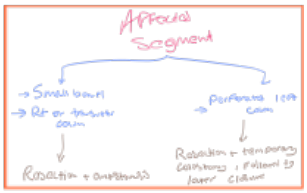
4- **Urinary catheter** to monitor urine output

- **Indications for surgery in obstruction:**

1. Failure of conservative treatment 48 hours
2. Development of peritonitis
3. Proved or suspected strangulation
4. Obstructed hernia
5. Obstructed cancer
6. Gut perforation
7. Closed loop obstruction

- **Types of surgical procedures:**

1- **In adhesive intestinal obstruction →Exploration and adhesiolysis**



- Explration of the intestine from the duodeno-jejunal flexure till ileo-cecal junction
- The adhesive band is identified and divided.
- The band is identified at the junction of the dilated proximal segment and narrow collapsed distal segment.
- In cases of strangulation, or gut perforation → proceed for resection and anastomosis in case of small bowel, right and transverse colon
- In case of perforation of left colon → resection and temporary colostomy (1 st stage) and Closure of colostomy (2nd stage)

2-Obstruction with specific causes:

- **Obstructed hernia →exploration of the hernia content**
  - If viable return inside the abdomen and repair of the hernia
  - If non-viable resection and anastomosis and repair of the hernia
    - **Obstructed carcinoma** of colon →look cancer colon
    - Intussception
    - Volvulus of the sigmoid colon
    - Hirschsprung disease

Intussuception

- A condition in which part of the intestine slides into an adjacent part of the intestine • Intussusception is the most common cause of intestinal obstruction in children younger than 3 years.

Types:

1- ileo-cecal →the commonest
2- ileo-ileal
3- colo-colic

Etiology and Pathophysiology:

- The etiology is unknown in children, sometimes hypertrophied lymphoid tissue in the intestinal submucosa initiate the process
- In adults it might be due to tumor in the intestine

Parts of the Intussusception:

1. The entering = invaginating loop = Intussusceptum
2. The outer = recipient loop = Intussusceptient

- The intussusceptum proceed along the lumen of the intussuscipient from proximal to distal.
- The mesentery of the intussusceptum is invaginated with the intestine, leading to bowel obstruction.
- Compression of the vessels in the mesentery leading to bowel ischemia
- The intestinal mucosa sloughs and ulcerate due to ischemia, leading to bleeding and "currant jelly stool" (a mixture of sloughed mucosa, blood, and mucus).
- Neglected cases leads to gangrene and perforation at the head of Intussusceptum

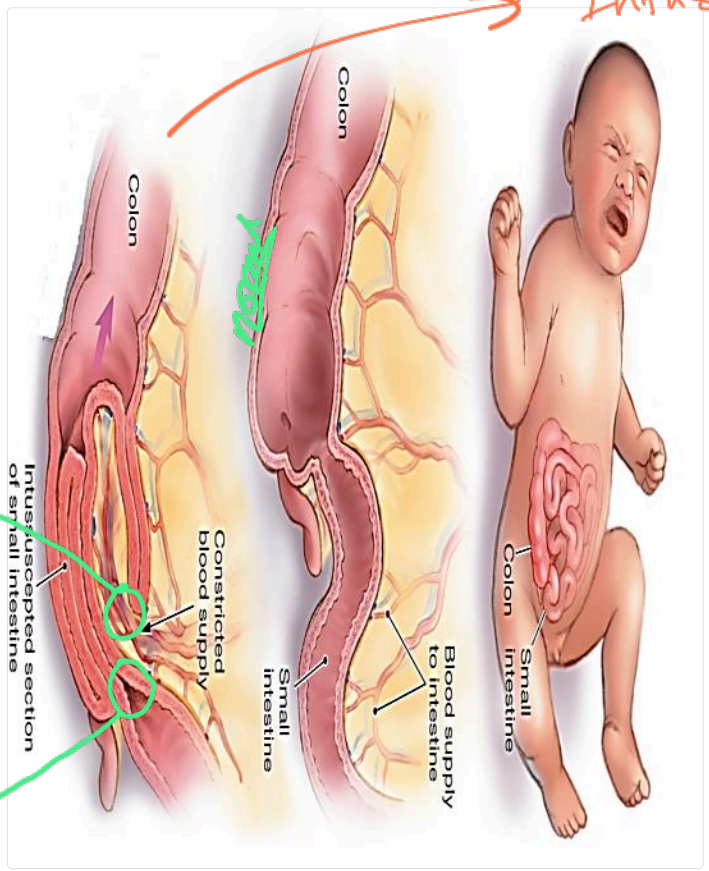


Figure 13.3 — Intussusception

Clinical picture:

Symptoms

1- The patient with intussusception is usually an infant,

2- Has an upper respiratory infection

3- Vomiting: Initially, vomiting is non-bilious, but when the intestinal obstruction occurs, vomiting becomes bilious

4- Abdominal pain: colicky, severe, and intermittent

5- Passage of blood and mucus: Parents notice the stools like currant jelly. (This is a mixture of mucus, sloughed mucosa, and blood.).

6- Palpable abdominal mass

7- Absolute constipation

8- Abdominal distension

Signs:

1- Signs of dehydration in neglected cases + fever in case of peritonitis

2- Abdominal distension

3- Right hypochondrium sausage-shaped mass and empty right iliac fossa

- (Dance sign)

4- Tenderness, rebound tenderness and rigidity: If intestinal gangrene and peritonitis occur

Investigations:

Lab. : CBC → Leucocytosis in strangulation or perforation

Rad.:

1- Plain x-ray abdomen erect and supine

Dilated proximal intestinal loops and air/fluid level

2 - Ultrasound:

a) Reveal intestinal obstruction caused by intussusception.

b) **Target sign** : Typically show a "bu ll's - eye," representing the intestine coiled within intestine.

3 - Barium enema Claw sign

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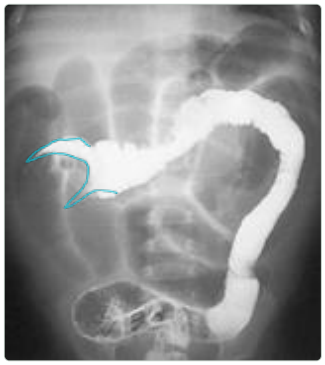


Figure 13.4 — Claw sign

Treatment:

A - Initial treatment: as in adhesive I.O.

1- Intravenous fluids to correct dehydration

2- Nasogastric tube for suction

3- Start antibiotic therapy

B - Hydrostatic reduction using barium enemas succeed in many cases

→

Indications: All early and uncomplicated cases (no signs of peritonitis and or strangulation)

- Success known by passage of the contrast to the ileum

C - Surgery

1- When hydroststic reduction fails

2- Presence of complications as peritonitis or gut perforation.

- **Procedures:**

1- In non-complicated cases: manual reduction of the intussuception by compression not pulling to avoid tearing of the mesentry (due to presence of edema & fragile tissue)

2- In case of perforation or gangrene → resection and anastomosis:

In ileocecal intussuception : right hemicolectomy and ileo-transverse anastomosis (m/c)

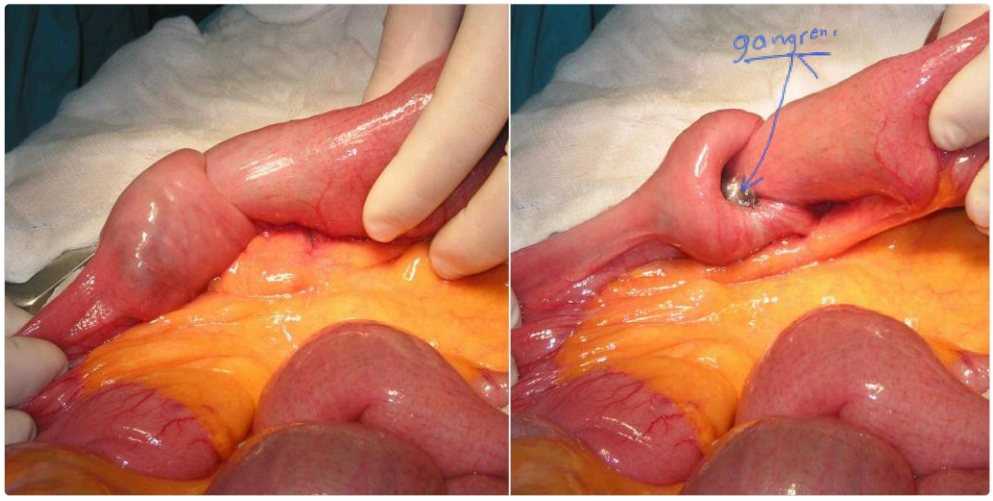


Figure 13.5 — Manual reduction of the intussception

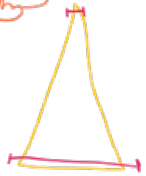
Colonic Volvulus

- The term volvulus = twist → twisting may compress both the bowel lumen & its mesenteric B.vs
- A colonic volvulus occurs when a part of the colon twists on its mesentery, resulting in acute, sub-acute, or chronic colonic obstruction. (more gradual development)

Types of colonic volvulus:

1- Sigmoid volvulus (most common = longer mesentery) → (Abnormally long, mobile Sigmoid mesentery)

2- Caecal volvulus. (less common) (when cecum is abnormally mobile)



Etiology of Sigmoid volvulus:

1. The presence of a long mesentery with a narrow base of fixation to the retroperitoneum 2. Elongated, redundant colon predisposes to the formation of volvulus.

Redundant = excessively long, with extra looping

Long mesentery = Mobility

narrow base = easy twisting (narrower axis)

3. Chronic constipation → loaded sigmoid → elongation → twist along its mesentery
4. Old age → colonic hypotonia → chronic constipation → increases intra-luminal pressure.
5. Postoperative adhesions
6. Intestinal malrotations
7. Pelvic tumors
8. Mega-colon

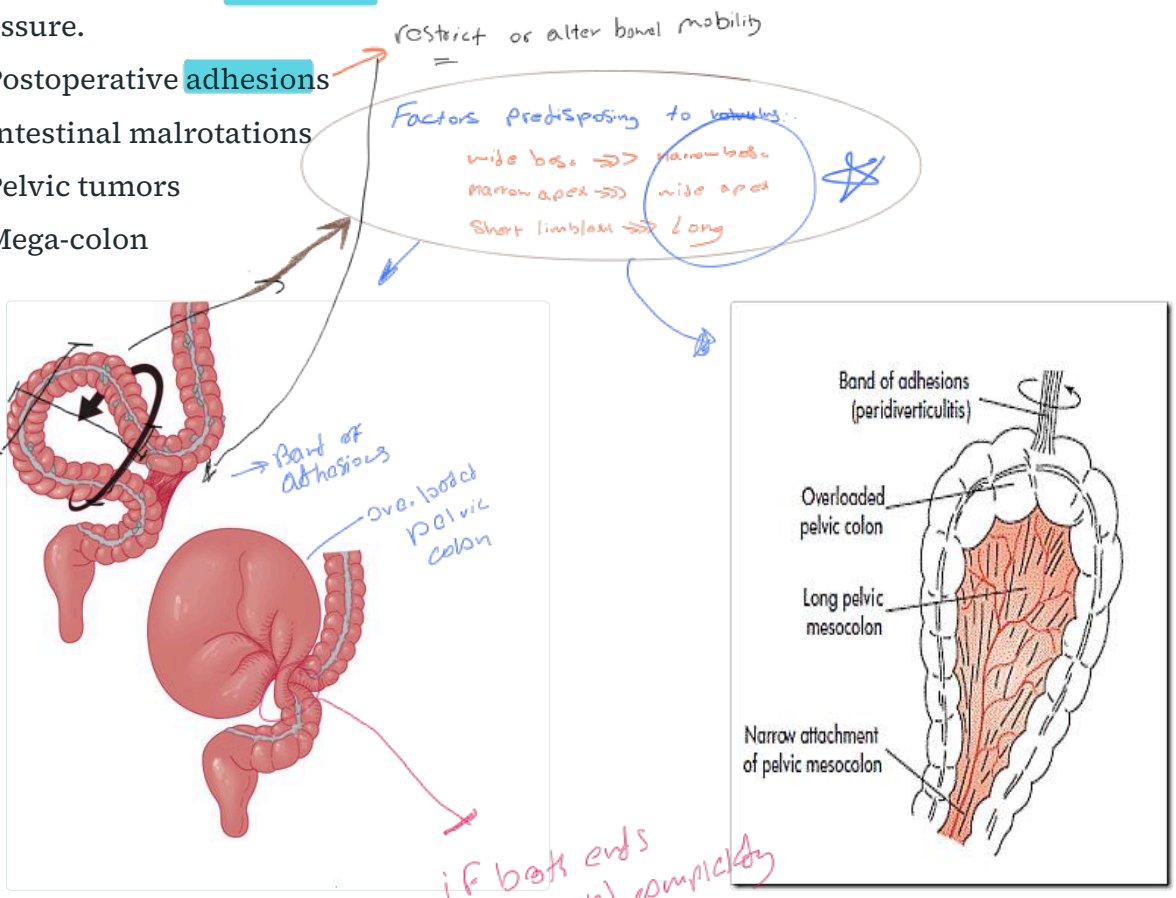


Figure 13.6 — Etiology of volvulus

Pathophysiology:

- The presence of long mesentry, narrow attachment and elongated loaded colon → anti-clockwise rotation of the sigmoid along its mesentry → occlusion of the lumen → obstruction (Mixed Strangulation)
- Occlusion of the blood vessels → ischemia, gangrene and perforation

Clinical picture:

Symptoms :

(+) Four cardinal signs of obstruction

1- History of chronic constipation + old age
2- Colicky abdominal pains – constant severe abdominal pain indicates closed loop obstruction and impending perforation (due to ischemia) (before was intermittent) (left sided)
3- Progressive abdominal distension (Rapid due to rapid fermentation & gas production in closed loop)
4- Progressive to absolute constipation
5- Late vomiting [The more distal the later vomiting]

Signs:

1- Fever, tachycardia and toxic look in cases with peritonitis

2- Massive abdominal distension

3- Tenderness, rigidity and rebound tenderness in associated peroration and peritonitis

4- Percussion → tympanitic resonance

5- P/R → empty rectum

Investigations :

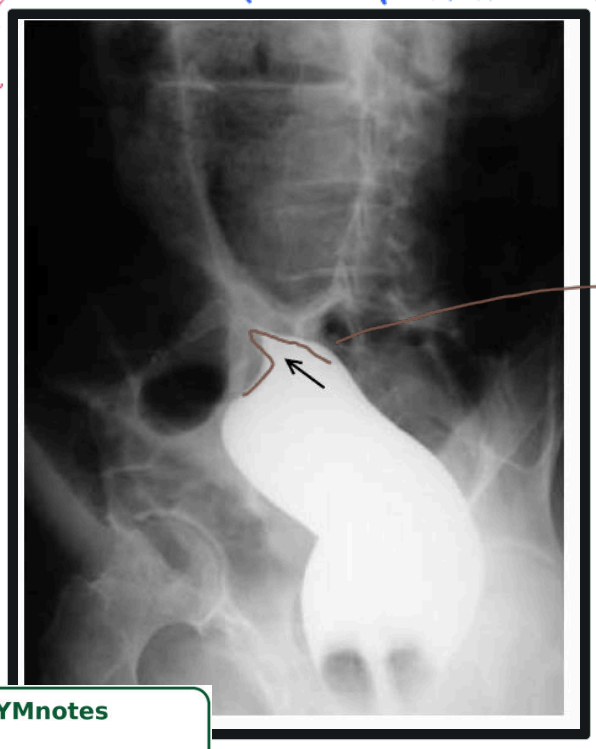
A- Lab. → CBC for leukocytosis in peritonitis

B- Rad.

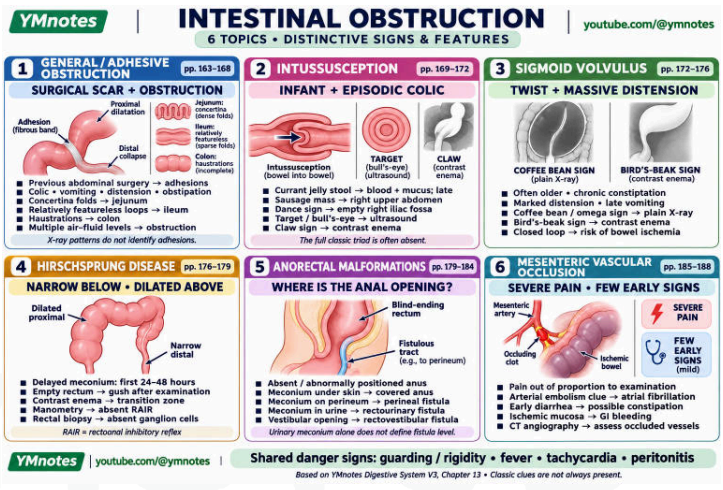
1- Plain X-Ray abdomen → huge distended sigmoid colon filling most of the abdomen

(omega sign = coffee bean sign)

2- Barium enema → "Birds beak"



ntrast study in sigmoid volvulus



YMnotes

Intestinal Obstruction: Distinctive Signs

Scan to open the image

Figure 13.8 — Sigmoid volvulus (omega sign) Coffee bean

3- Abdominal ultra-sound → Abdominal collection in perforation

+ CT ?

Treatment Sigmoid Volvulus

1 - **Sigmoidoscopic de - tortion** : detorsion [untwist]

clock-wise and decompression and placing rectal tube succeeds in most cases

- First choice initial treatment in uncomplicated cases
- Contraindicated in suspected perforation

2- Surgery:

- **Elective:** Resection of redundant colon and re-anastomosis after colonic preparation
- **Urgent:** in case of gangrene or colon perforation: **Hartmann' procedure**

1 st stage:

Resection and colostomy

2 nd stage:

Closure of colostomy

YMnotes

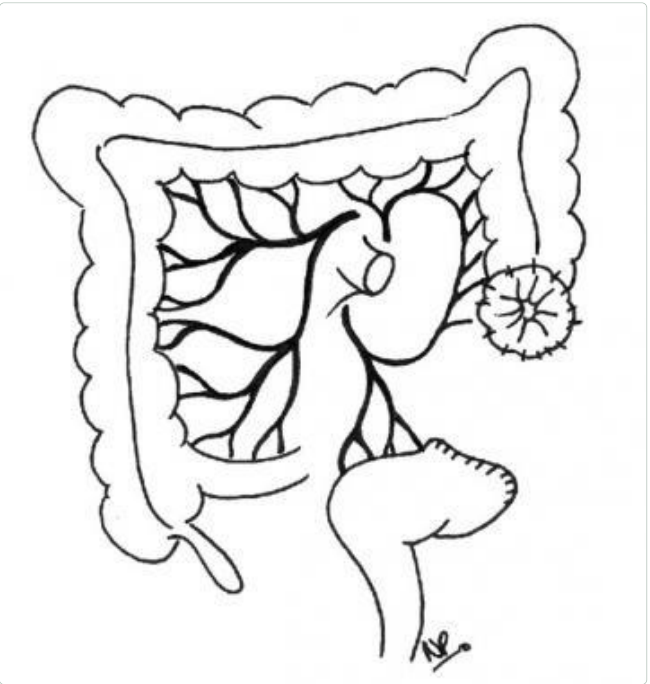


Figure 13.9 — Hartmann's procedure

Hirschsprung Disease = Congenital megacolon

Hirschsprung disease is a congenital disorder characterized by the absence of ganglia in the distal colon, resulting in a functional obstruction

Etiology :

genetic predisposition

Pathophysiology:

- The wall of the colon contains 2 enteric plexus
- Meisener plexus in the submucosa and Auerbach plexus in

The muscle layer

- Both causing colonic wall relaxation
- The wall of the colon supplied by parasympathetic and sympathetic
- Parasympathetic is predominant and causes wall contraction
- In Hirschsprung Disease: the ganglion cells in the meisener and Auerbach's plexus are absent (aganglionic segment)
- So there is predominat action of the parasympathetic nerves leading to spasm in the segment of the colon affected → functional obstruction
- **Parts of the colon**

1. Proximal dilated colon = normal

2. Transition zone = hypo-ganglionic

3. Distal zone = spastic aganglionic

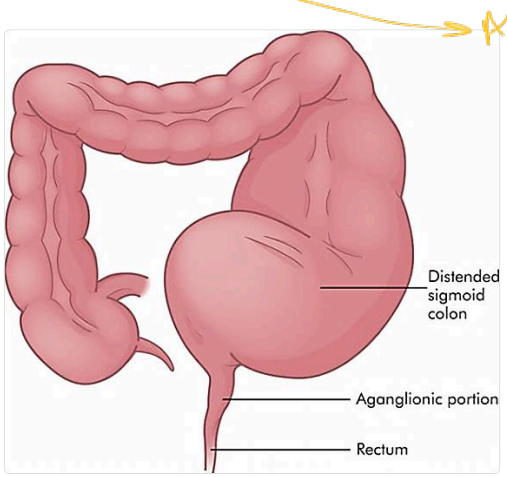


Figure 13.10 — Hirschsprung disease: bowel segments

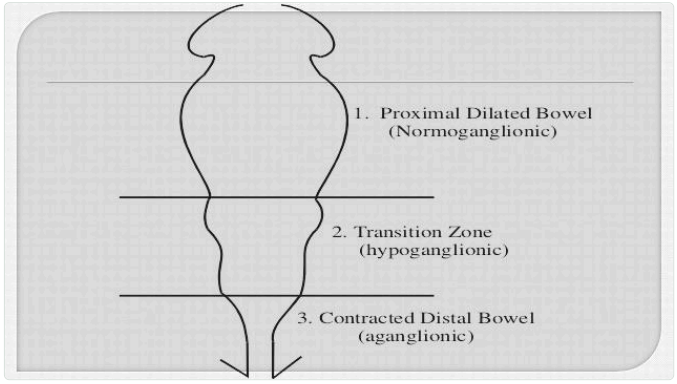


Figure 13.11 — Hirschsprung disease: transition zone

Sites:

H.D usually starts in the rectum → sigmoid → and proceeds proximally to involve variable parts of the colon

- Male is affected more than females (4: 1)

Clinical presentation:

Most cases of Hirschsprung disease are diagnosed in the **newborn** that **fails to pass meconium** within 24-48 hours of birth.

Abdominal distension

Vomiting

Poor nutrition and **failure to gain weight**

Older infants and children typically present with **chronic constipation** poorly responds to laxatives and distension

Hirschsprung enterocolitis : it is a severe form of entero-colitis:

1. Bloody diarrhea- foul smelling

2. Abdominal pain
3. Fever
4. Vomiting
5. Fatal if untreate

On examination:

1. Marked abdominal distension
2. Tympanitic resonance
3. Increased intestinal sounds
4. P/R: reveals an empty rectum and may result in gush of stool upon withdrawal of the finger.

Investigations

A- Lab. 1- CBC → anemia and infection

2- Serum electrolytes (Na and K)

B - Rad. : Plain X - Ray abdomen erect and supine **dilated left colon** &

→

- No rectal gas

1-Barium enema:

Shows **narrow distal segment**, **funnel-shaped dilatation** at level of transition zone with **marked dilatation of** the proximal colon.

see Fig. 13.11

Figure 13.12 — Barium enemaCongenital mega-colon

3- Rectal biopsy:

Rectal biopsy is the **best diagnostic** test and demonstrates absence of ganglion cells (done through endoscopy).,

4- Electro-manometry:

The classic finding is the absence of the recto-anal inhibitory reflex when the rectum is distended.. (Transient involuntary relaxation of the internal anal sphincter in response to distention of the rectum help to discriminate between the gas, fluid and stool at anal canal)

Management:

Manegement of HD differs according to the presentation:

1- If the patient presents with acute intestinal obstruction in the early life the management will be:

Resuscitation, IVF

Naso-Gastric Tube

Antibiotics

The initial surgery requires performing **a temporary colostomy**

2- Definitive procedures = pull through operation

- Resection of aganglionic segment and anastomosing the two normal ganglionic ends. They give excellent result in 90%.
- **Swenson.** The aganglionic segment is resected down to the sigmoid colon and rectum, and anastomosis is performed between the normal colon and the low rectum.

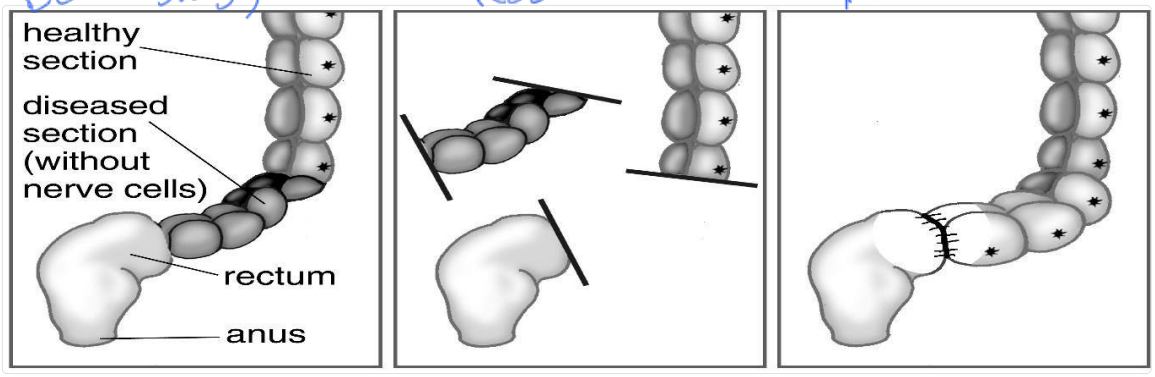


Figure 13.13 — Pull-through operation

...

Imperforate anus

Embryology of the rectum and anal canal:

1- The lowest part of the hindgut (cloaca) is divided by uro-rectal septum into:

A- Anorectal canal → posteiorly

B- Uro-genital sinus → anteriorly

2- This Ano-rectal canal will form the rectum and upper ½ of anal canal

3- The anus and lower ½ of anal canal develop by an external invagination from skin, known as the proctodeum, which deepens toward the upper part of anal canal to join it but remains separated from it by the anal membrane. This membrane should disintegrate at 8 weeks' gestation.

The causes of Imperforate anus are unknown

Classifications: in relationship to the level of the pubo-rectalis portion of the levator ani muscles

I Low anomaly :

Ano - rectal canal descends below the pubo - rectalis muscle a- Covered anus b- Ectopic anus (perineum- low vaginal fistula) c- Anal stenosis

II - High anomaly :

The ano-rectal canal ends above pubo-rectalis muscle with or without fistulae formation a- Ano-Rectal agenesis + recto vaginal, or recto-vesical fistula

b- Ano-Rectal agenesis (atresia) → the rectum is blind ended above pubo-rectalis with no fistula formation.

Clinical presentation:

The term imperforate anus is clinical term: it means that the anus is not evident in its normal position at birth

I low anomaly:

a- The anus is missing or stenotic b- Failure to pass meconium after birth (picture of intestinal obstruction). c- Meconium evident at the perineum → a recto-perineal fistula = (Ectopic anus)

d- Meconium bulge at skin at site of anus → covered anus e- If meconium is in the urine, the diagnosis of a recto-urethral fistula f- Anal dimple and median raphe usually present.

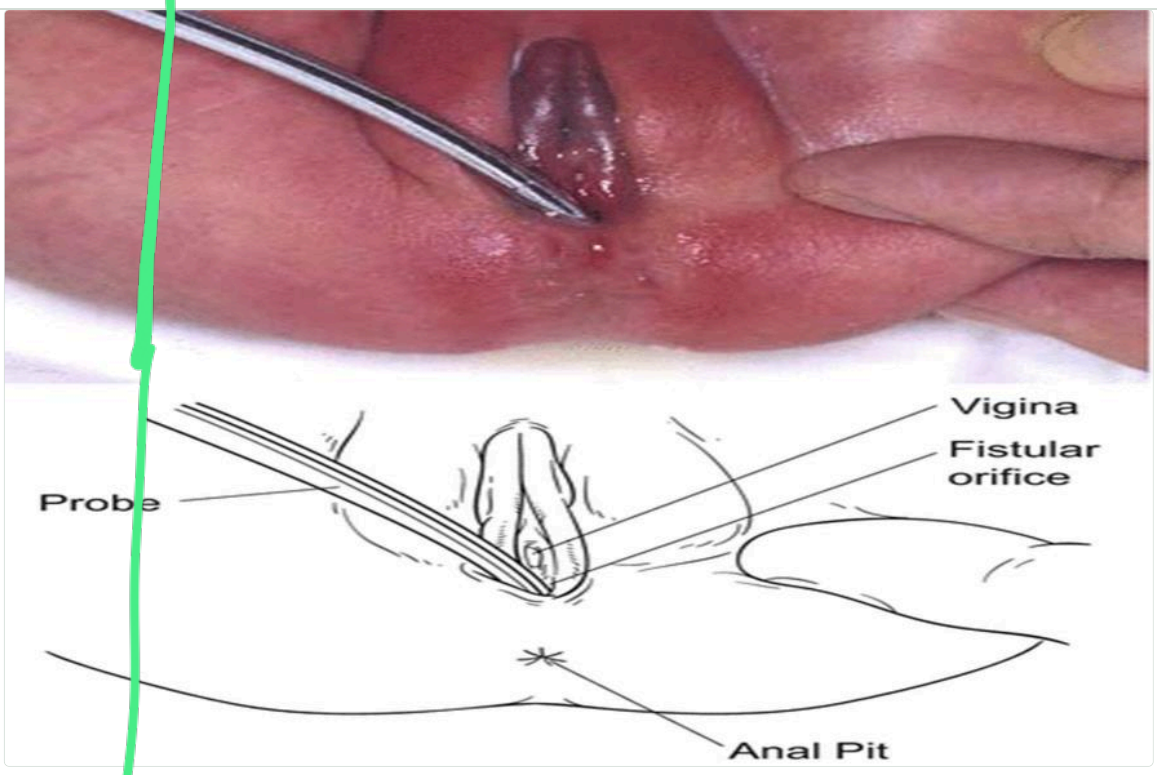


Figure 13.14 — Rectovestibular fistula

Rectovestibular fistula

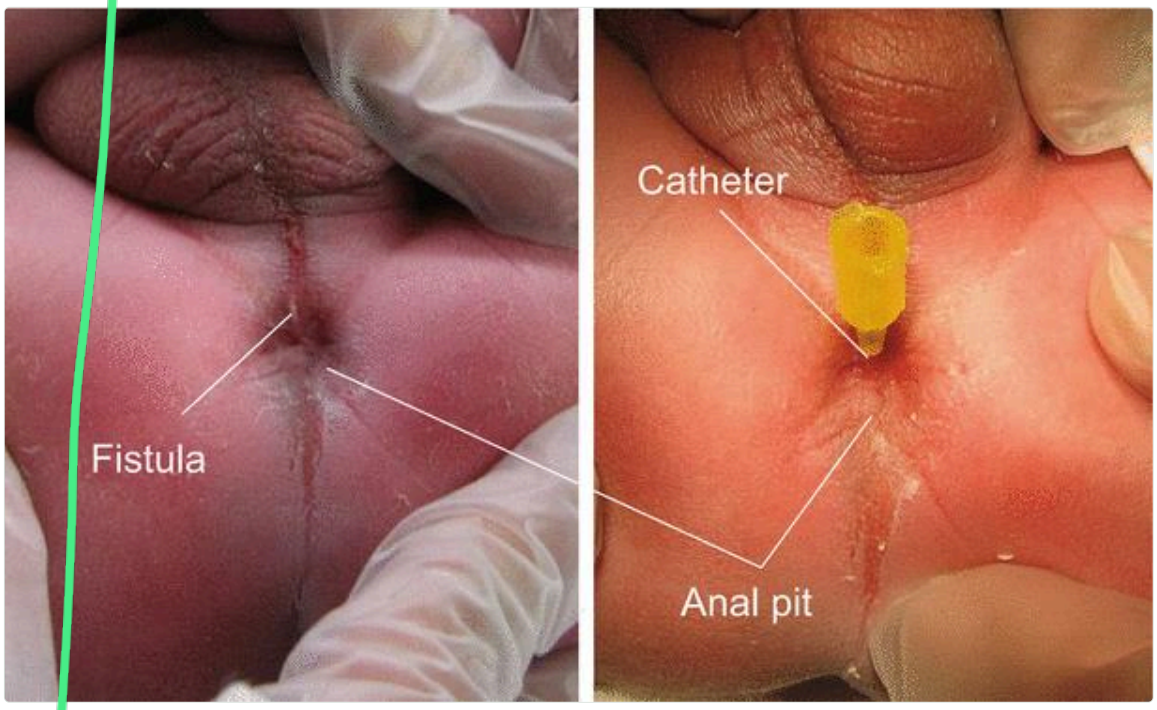


Figure 13.15 — Perineal fistula

Perineal fistula

Figure 13.16 — Covered anus

Covered anus

I- High anomaly:

a- The anus is missing b- Failure to pass meconium after birth (picture of intestinal obstruction) c- No Meconium evident at the perineum d- Meconium in urine → high recto- vesical fistula e- Recto-vaginal fistula → meconium comes out of the vagina. f- No anal dimple & No median raphe

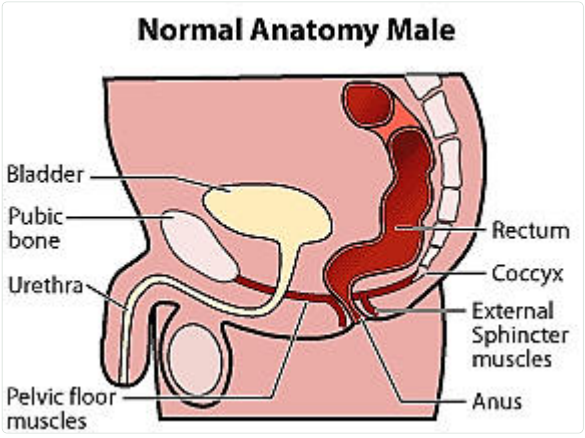


Figure 13.17 — Normal male anatomy

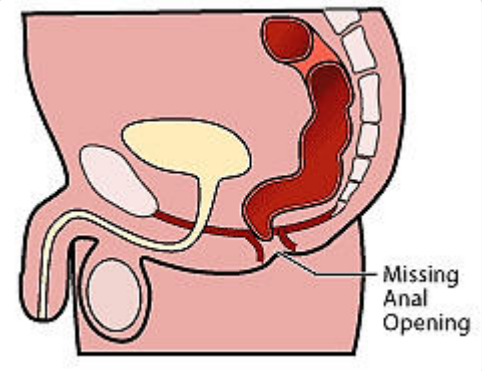


Figure 13.18 — Low anorectal anomaly

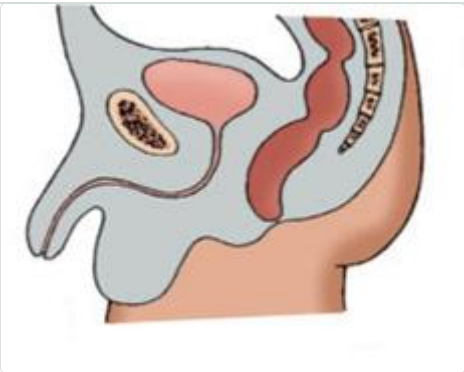


Figure 13.19 — Low anorectal anomaly

Low anomaly

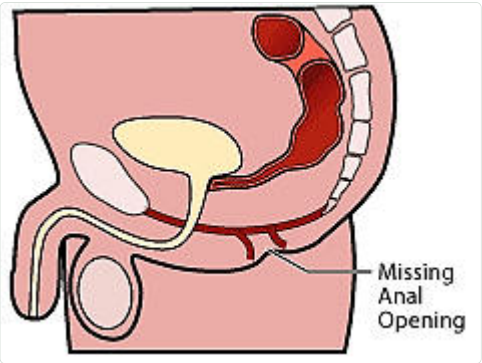


Figure 13.20 — Agenesis with fistula

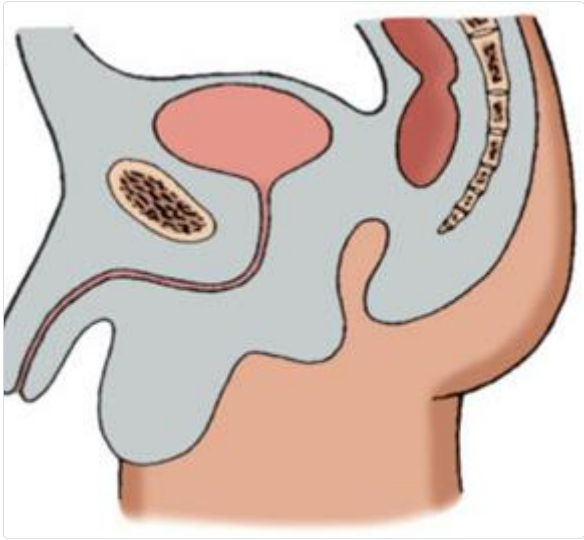


Figure 13.21 — Rectal atresia with no fistula

High anomaly

agenesis with fistula Rectal atresia with no fistula

Investigations:

1. Invertogram:

- Holding the baby upside down,
  - a lateral plain x-ray is performed to observe the distance between the level of gas in the distal rectum and a metal piece placed over the expected anus.
  - And the relation of recta gas to PC line (pubo-coccygeal line).
  - If the rectal Gas is below the P.C line or < 2.5 cm from the skin →it is low anomaly
  - If the rectal gas ends above the P.C. line or > 2.5 cm from the skin → it is high anomaly

2. C.T scan invertography: better evaluating tool

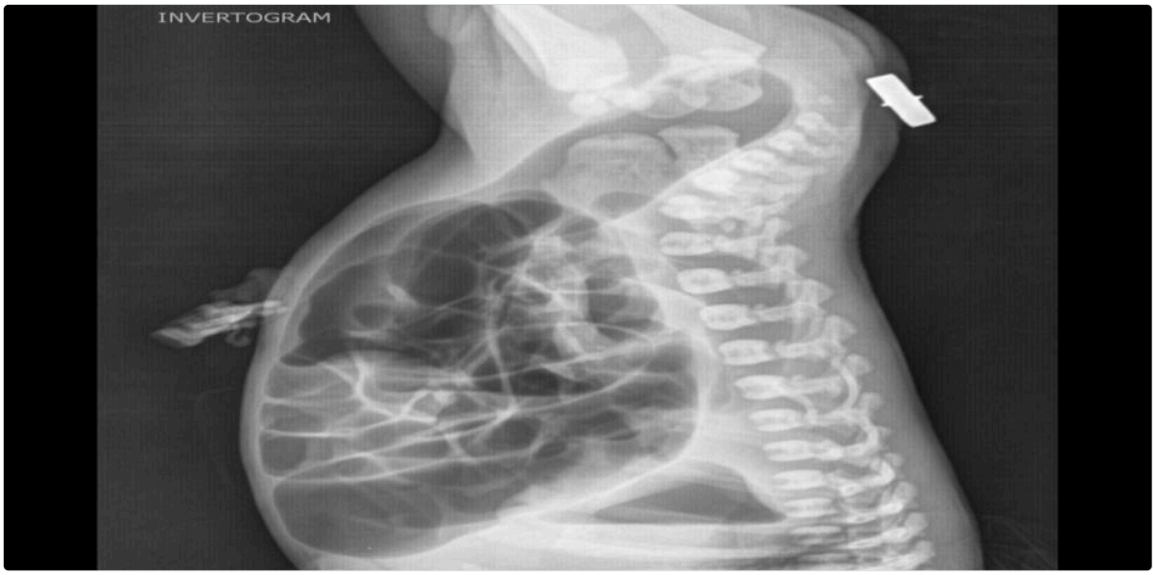


Figure 13.22 — Invertogram

Low anomaly

Figure 13.23 — Distal rectal gas level

High anomaly

Treatment:

- Colostomy for cases presented with intestinal obstruction
- Ano-plasty

Mesenteric vascular occlusion

It is a syndrome caused by inadequate blood flow through the mesenteric vessels, resulting in ischemia and gangrene of the bowel wall.

Types:

A- Mesentric Arterial ischemia

B- Mesentric Venous thrombosis

- **Acute mesenteric arterial embolism = 50 %**
- Mural thrombus after myocardial infarction, • Atrial thrombus associated with mitral stenosis and atrial fibrillation
- Septic emboli from infective endocarditis
- Emboli from fragments of proximal aortic thrombus **(30 %)**

B- Mesentric Venous thrombosis = 10 %

- Hypercoagulability
- Venous congestion from cirrhosis (portal hypertension)
- Infection

Pathophysiology:

- S.M. Vessels are more frequently involved than the I.M. vessels (larger diameter and better collaterals with inferior vessels).
- **Mucosal sloughing** may cause **bleeding** into the GIT
- If ischemia persists **disruption of mucosal barrier** occurs and bacteria, toxins, and vasoactive substances are released into the systemic circulation
- Damage may range from ischemia to necrosis, gangrene and perforation
- **Bowel necrosis** occurs in **8-12 hours** from onset of symptoms
- High mortality up to **90 %**

Sloughing: انسلاخ الطبقة المخاطية

SMA	IMA
Larger, more likely to receive emboli	Smaller, less likely to receive emboli
Supplies most of the small intestine and right colon	Supplies left colon and upper rectum
Common site of acute embolic occlusion	Territory has important collateral connections

Exam takeaway: SMA is more frequently involved because of its anatomy favoring embolic entry, while the IMA territory has substantial collateral blood supply.

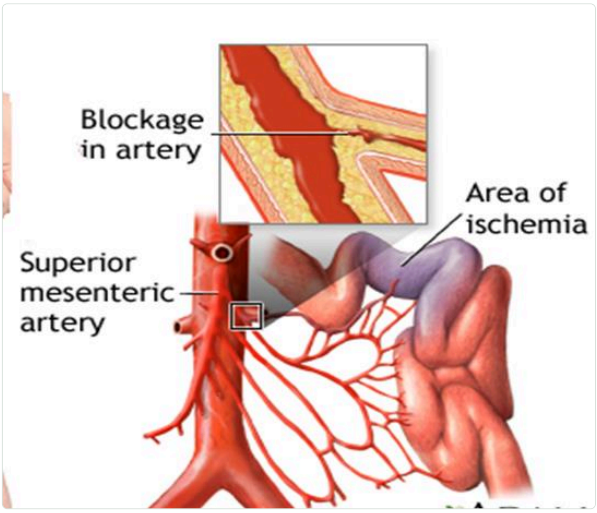


Figure 13.24 — Mesenteric arterial occlusion

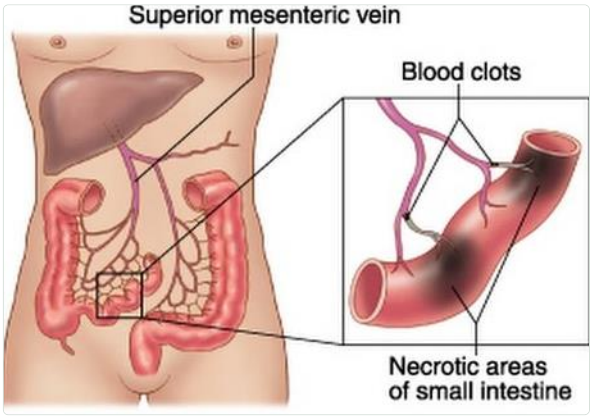


Figure 13.25 — Mesenteric venous occlusion

SM Venous occlusion SMA occlusion

Clinical presentation:

- History of cardiac disease (AF – Mitral stenosis)
- Usually old age
- The most important finding is **pain** that is disproportionate to physical examination findings.
- Typically, **pain is severe**, and diffuse not responding to Opioid
- Nausea and vomiting
- **Diarrhea** at first followed by **constipation**
- Abdominal distention
- Gastrointestinal **bleeding**
- As the bowel becomes gangrenous, rectal bleeding and sign of sepsis (tachycardia, hypotension, fever)

Abdominal examination:

In early cases no specific findings present although the severe abdominal pain

In cases complicated with **gangrene and peritonitis** → tenderness, rebound, rigidity distension and silent abdomen

Complications of MVO:

Bowel gangrene and peritonitis

- septic shock

Investigations:

I- Lab :

CBC – Serum electrolyte (Na & K) - Renal functions

II-Radiology:

A- Plain X-Ray abdomen erect and supine

1- Dilated intestinal loops + air and fluid levels

2- **Air under diaphragm** in perforation

B- C.T. or MRI angiography:

detect occluded mesenteric vessels and edematus bowel loops

Figure 13.26 — Thickened edematous bowel loops

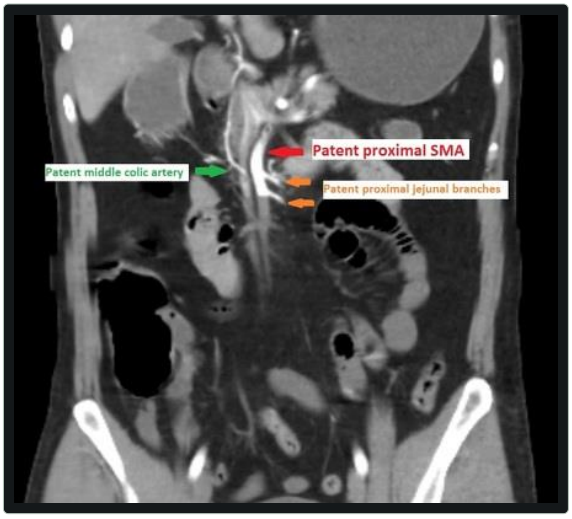


Figure 13.27 — Occlusion of the distal SMA

C- Duplex scan on mesenteric vessels

D- ECG: detect the cause as AF.

Treatment

I- Initial treatment:

1- Intra-venous fluids

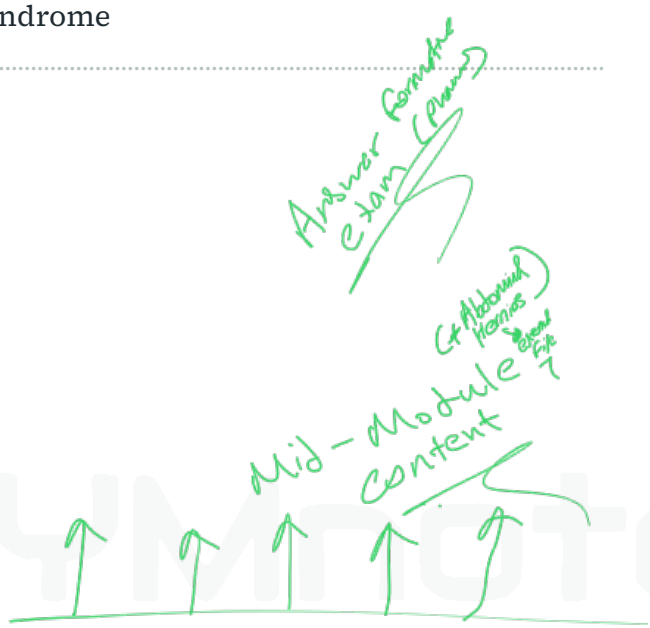
2- Naso-gastric tube and urinary catheter

3- Antibiotics

II- Surgery: time factor is very important

1- Exploration and **mesenteric embolectomy** (in acute embolism)

2- Assess bowel viability: in ischemic, gangrenous cases → **Resection anastomosis** according to extentof ischemic bowel. Extensive resection may lead to short bowel syndrome



CHAPTER 14

Polyposis of Colon

14

What is a polyp?

Polyp is a tissue growth protruding from the mucous membrane into the lumen of the gastrointestinal tract.

Types of colonic polyps

A. Benign non-neoplastic polyp:

- Juvenile polyps
- Peutz Jegher's syndrome
- Inflammatory polyp
- Hyperplastic polyp.

B. Benign neoplastic polyp:

- Adenomatous polyp
- Familial adenmatous polypsis = Familial Polyposis Coli
- Gardner's syndrome

Familial Polyposis Coli (F P C)

- This is an inherited disorder transmitted by autosomal dominant genes
- There are more than 100 polyps in the colon or any number of polyps in patients with family history of familial adenomatous polyposis
- It appears in second decade of life but may appear earlier
- If left untreated the chance of malignant transformation is 100%
- The mean age of cancer diagnosis is around 40 years
- These are adenomatous polyps.

Types of adenomatous polyps:

Tubular adenoma:

commonest type. Either pedunculated or sessile.

Villous adenoma:

These are usually sessile and has a velvety surface and there are finger like processes on the surface

Tubulo-villous adenoma:

characteristics in-between the tubular and villous adenoma.

Clinical presentation:

1. Asymptomatic ; discovered on routine screening of a family with a known patient with familial polyposis coli
2. Bleeding per rectum.
3. Lower abdominal pain.
4. Mucus discharge.
5. Diarrhea.

What is Gardner's syndrome?

Familial polyposis coli with some extra-colonic manifestation.

a) Osteomas, epidermoid cyst and congenital hypertrophy of retinal pigment epithelium.

b) Polyps of stomach and small intestine.

c) Retroperitoneal fibrosis and desmoid tumor

Investigations of F P C:

1.Colonoscopy:

- is the most valuable investigation with biopsy to confirm the diagnosis and detect malignant changes in any suspicious polyp
  - The disease involves the colon, rectum and anal canal with innumerable polyps within the lumen.
  - The intervening mucosa appears normal. The serosal surface of the colon is also normal. • Screen family members of patients with familial polypsis coli at the age of 10 years, clinically and with colonoscopy.
  - Colonoscopy should be done yearly till the age of 20 years as most of the patients develop these polyps by the age of 20 years.
  - If there are no polyps at the age of 20 years, 2 yearly colonoscopy is to be done till the age of 40 years.

Figure 14.1 — Colonoscopy in familial polyposis

2. Barium enema:

shows multiple small filling defects

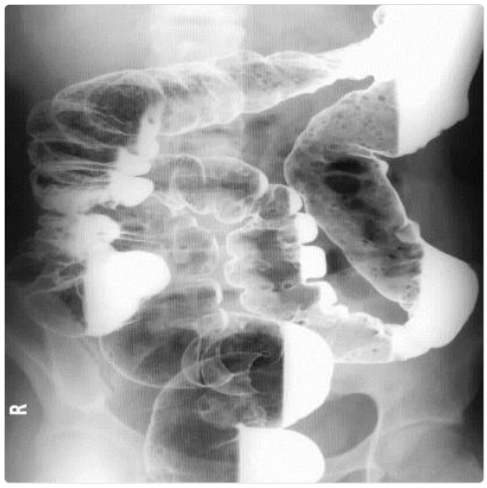


Figure 14.2 — Familial polyposis coli in contrast study

Management of familial adenomatous polyposis coli:

Surgery is mandatory for these patients as there is 100% chance of malignant transformation.

Options for surgical treatment:

1. **Total proctocolectomy with ileostomy.** Patient has a permanent ileostomy.

2. **Total colectomy with excision of rectal polyps and ileo-rectal anastomosis**. Needs regular follow up as the rectum is retained.

3. Proctocolectomy with ileo-anal anastomosis with an ileal pouch reservoir.

N.B. **Peutz Jegher's syndrome**

- Transmitted by autosomal dominant gene
- Gastrointestinal polyposis with pigmentation of oral mucous membrane or pigmentation in perianal region
- Polyps may be present in the whole gastrointestinal tract
- These are hamartomatous polyps
- Onset of symptoms is usually around the age of 10–30 years
- Patient may present with GI bleeding, intestinal obstruction due to intussusception

Colon and rectum

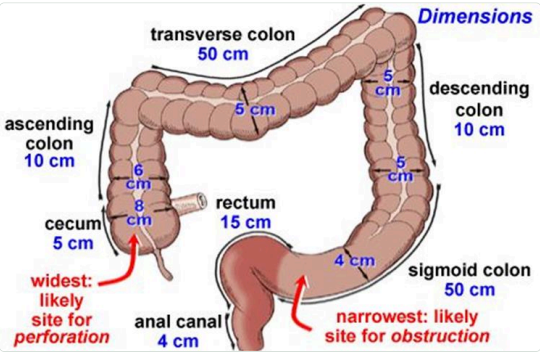


Figure 14.3 — Dimensions of the colon

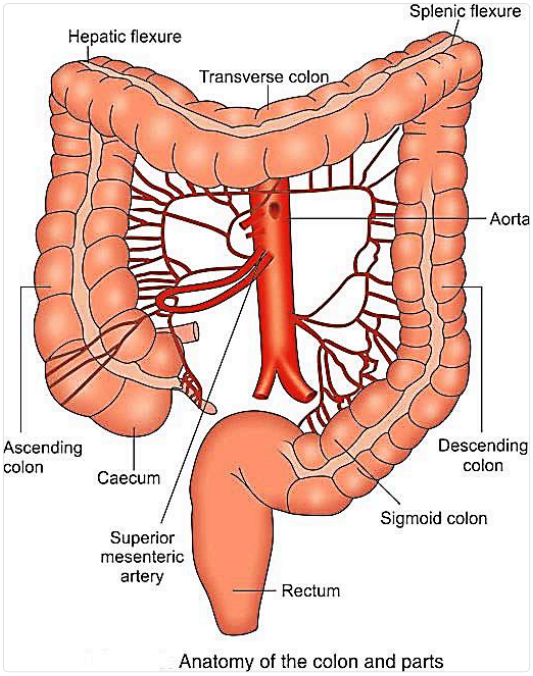


Figure 14.4 — Anatomy of the colon

Diameters of different parts

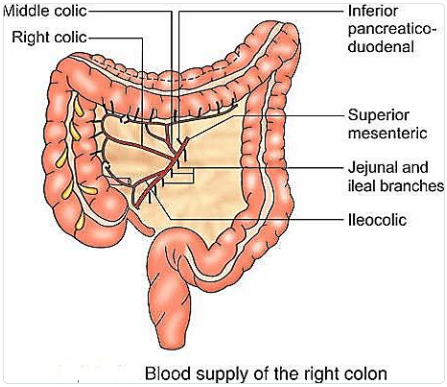


Figure 14.5 — Blood supply of the right colon

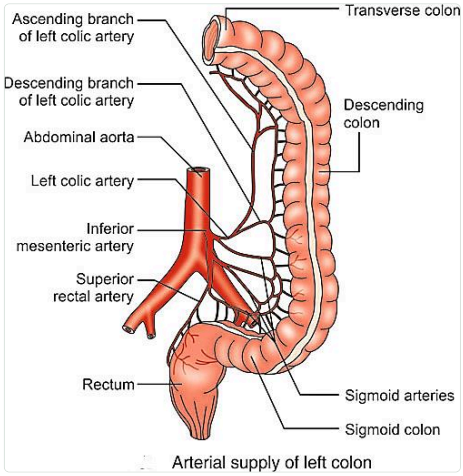


Figure 14.6 — Arterial supply of the left colon

CHAPTER 15

Colorectal Cancer

15

Pathology of colorectal carcinoma

Age:

old age , however no age is immune

Sex :

more common in males than females

Aetiology:

1- Predisposing factors:

- Personal history of colorectal polyps or previous cancer
- Colo-rectal cancer family history
- Red meat, processed meat, fatty meals
- Obesity
- Smoking
- Alcohol

2- Precancerous lesions:

- FPC : (familial polyposis of colon)

  Autosomal Dominant – 100 % turn malignant.
  - Inflammatory bowel disease (UC > Chron's)
  - Adenomatous polyp.

3- Microscopic picture:

Adenocarcinoma : More than 90% of colorectal carcinomas are adenocarcinomas originating from epithelial cells of the colorectal mucosa

- Mucinous adenocarcinoma (>50% mucinous material) ia a variant of adenocarcinoma • According to the glandular contents :

a) Grade 1: Well differentiated carcinoma

b) Grade 2: Moderate differentiated (most common) c) Grade 3: Undifferentiated (poor prognosis)

Macroscopic picture: Naked eye:

- Cauliflower mass → with ulcerating bleeding surface
- Malignant ulcer → raised everted edge, hard base and necrotic floor
- Malignant stricture: hard and irregular

- Multicentricity is common

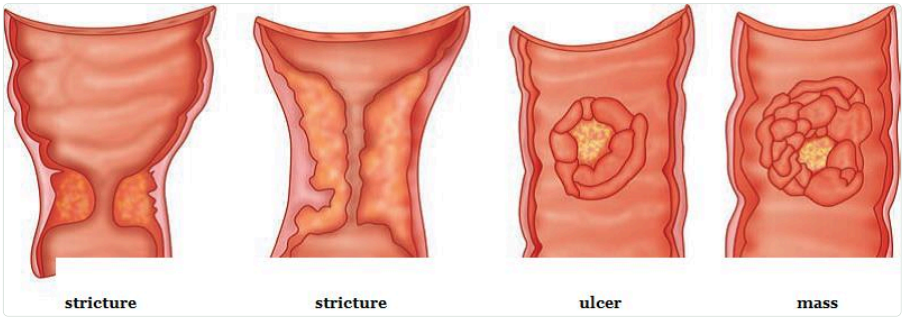


Figure 15.1 — Macroscopic appearances of colorectal cancer

Incidence of colorectal cancer in different parts:

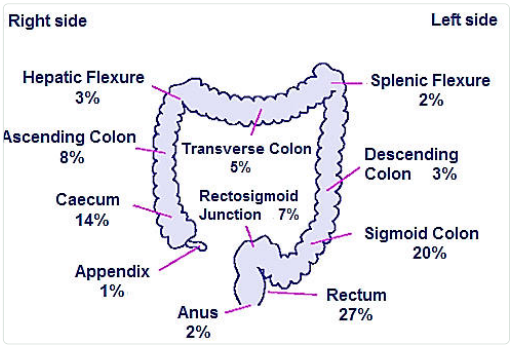


Figure 15.2 — Distribution of colorectal cancer

The most common site incidence is rectosigmoid part of the colon

Spread:

A- Direct: - mucosa → submucosa → muscle layer → serosa → nearby structures or Organs. It may lead to internal fistula as colo-vesical or recto-vaginal fisualae.

- **Right colon** → right ureter, abdominal wall, gonadal vessels, duodenum, small intestine, peritoneum, liver.
- **Transverse colon** → duodenum, stomach, liver, intestine, peritoneum.
- **Left colon** → spleen, abdominal wall, intestine, left ureter, gonadal vessels, peritoneum.

Rectum:

to urinary bladder, vagina → causing fistula →to sacrum, prostate, seminal vesicles, and peritoneum**.**

B- Lymphatic spread:

1- Epicolic lymph nodes

2- Para-colic lymph nodes

3- Mesocolic lymph nodes

4- Mesenteric lymph nodes: along the mesenteric vessels

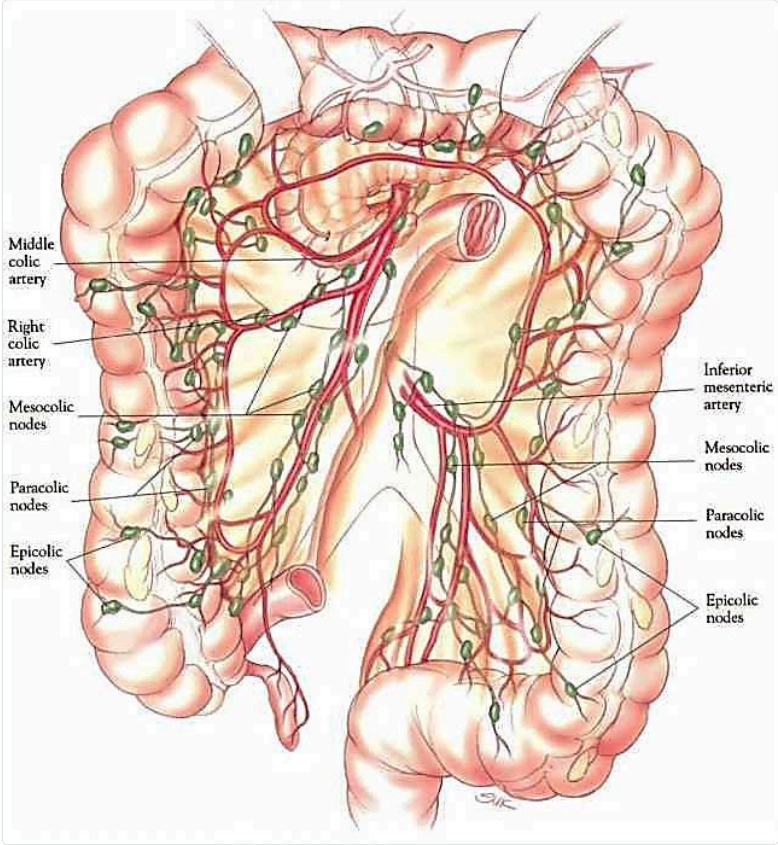


Figure 15.3 — Lymphatics of the colon

C- Blood spread:

- Liver → most common (Portal circulation).
  - Lung, bone, brain.

Staging:

Duke's staging for carcinoma of colon

Stage A:

Tumor in the bowel but not beyond muscularis propria.

Stage B:

Trans-mural invasion of bowel wall.

Stage C:

Any degree of bowel wall invasion with lymph node metastasis.

- C1—Epicolic and paracolic nodes.

- C2—Intermediate and proximal lymph nodes (mesenteric lymph nodes).

In modified Duke's :

Stage D : Presence of distant metastasis; is added

TNM classification of cancer colon:

T= Tumour

- **T1:** The tumor has grown through the muscularis mucosa and extends into the submucosa.
- **T2:** The tumor has grown through the submucosa and extends into the muscularis propria .
- **T3:** The tumor has grown through the muscularis propria and into the outer muscle layers of the colon or rectum but not through them. It has not reached any nearby organs
- **T4a:** The cancer has grown through the serosa the outermost lining of the intestines.
- **T4b:** The cancer has grown through the wall of the colon or rectum and is attached or invades into nearby tissues or organs.

N= nodes

N0: No cancer in nearby lymph nodes.

N1: Cancer cells are found in or near 1 to 3 nearby lymph nodes

N2: Cancer cells are found in 4 or more nearby lymph nodes

M= metastasis

To The **liver**, lungs, bones , brain.

M0: No distant spread is detected.

M1a:

The cancer has spread to ONE distant organ

M1b:

The cancer has spread to MORE than ONE distant organ or it has spread to distant parts of the peritoneum.

Differences between cancer right and left colon:

Right-side colorectal cancer characteristics:

Colorectal cancer that starts on the right side (Cecum, Ascending Colon, Right Half of Transverse Colon, Hepatic Flexure)

is seen more often in women and people over 65

- is diagnosed more frequently in African Americans
- often begins with flat, sessile polyps that go unnoticed until they are fairly large • is usually diagnosed at a later stage

- often presents with symptoms such as anemia and bleeding
- has a higher occurrence of mucinous tumors, BRAF-mt or KRAS-mt mutations, MSI-H • responds well to immunotherapy

- **Left-side colorectal cancer characteristics:**

Colorectal cancer that begins on the left side (Splenic Flexure, Descending and Sigmoid Colon, Left Half of Transverse Colon, Rectosigmoid and Rectum) occurs more often than right-sided colorectal cancers

- is seen more often in men and in people under 65
- has tumors that grow along the wall of the colon
- often presents with constipation, narrow stools, and other bowel habit changes • causes bowel obstructions more often than right-sided colorectal cancer
- is usually diagnosed in its early stages
- is associated with FAP (familial adenomatous polyposis)
- has a higher occurrence of HER2 amplifications
- overall better prognosis than right-sided tumors
- responds better to standard chemotherapy and targeted therapies

Clinical picture of colorectal cancers:

- **General symptoms for all:**

1- old aged male

2- history of FAP of colon (mainly left colon)

3- history of inflammatory bowel disease

4- loss of weight, anaemia, cachexia

I- Cancer of the cecum

The right colon is larger in diameter, contain fluid stool, low bacterial load and low pressure. The cancer comes more in the form of mass or ulcer

- Weight loss
- Unexplained anaemia
- Pain in the right iliac fossa (D.D.)
- Bleeding per-rectum (occult or obovious)
- Irregular bowel habits
- Late → intestinal obstruction: Colicky abdominal pain – Distension - progressive constipation - Repeated vomiting
- Firm to hard fixed right iliac fossa mass of variable size.

II. Cancer left colon & sigmoid:

- The left colon is narrower than right, contain more solid stool and the pressure is higher • The cancer lesion is commonly stricture (growth along the wall).

1. Pain in left side of the abdomen:

- Left hypochondrium (splenic flexure)
- Left iliac fossa (sigmoid colon)
- Ana pain (rectum)

2. Altered bowel habits (intermittent diarrhea and progressive constipation (common) 3. Bleeding per–rectum more frequent

4. Attacks of colonic obstruction

- Recurrent colicky pains
- Abdominal distension
- Rarely vomiting
- Constipation

5. Intestinal obstruction full picture

6. Mass is felt → hard & fixed

III- Cancer rectum

1- **Tenesmus** (sense of incomplete evacuation of rectum+ mucus and blood with Little stoo)l

2- Bleeding per-rectum

3- Anal pain

4- Mass can be felt on P/R examination in low rectal cancers →hard fixed mass with bleeding surface.

5- Altered bowel habits

6- Colonic obstruction

Investigations:

I- Lab.

1- CBC for the presence of anaemia and increased ESR

2- Stool analysis for occult blood in stool

3- **CEA (Carcino-Embryonic Antigen)** for follow up: A rising CEA level is an indication for colonoscopic examination or CT scan abdomen (For local recurrence or metastasis)

4- Other routine lab. Investigations

II-Endoscopy:

for viualization of the lesions size and site and **multi-centricity** and take biopies for histo-pathology (most valuable investigation).

a)Sigmoidoscope:

to examine the rectum and sigmoid colon b) **Colonoscopy** to examine the entire colon till the cecum

III-Radiological:

1- Barium enema → detects the lesion as:

a) mass as filling defect b) stricture as apple- core ,shouldering, irregularities and proximal dilatation c) ulcer as filling defect

2- C.T.

scan with contrast:

for staging (detect extra-luminal spread and metastases in liver).

3- Endo- rectal ultra-sound for rectal carcinoma spread locally

Figure 15.4 — Colonoscopy: source comparison view

Figure 15.5 — Colonoscopy: source mass image

Normal colonoscope mass

Figure 15.6 — Barium enema: source stricture image

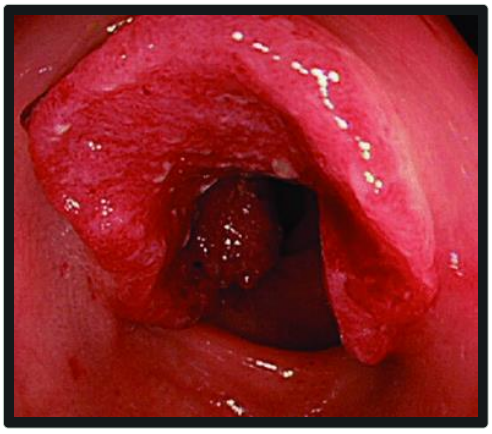


Figure 15.7 — Endoscopy: source ulcer image

Malignant ulcer barium enema; stricture

Management

- **Management of cancer colon depends on some factors:**

1- Site of the cancer (right, left, transverse, sigmoid, rectum, anal canal).

2- Multi-centricity of the tumour.

3- Stage of the tumour and the presence of Metastases

4- Operable or non-operable tumour.

5- Presence of complications as intestinal 0bstruction or Perforation.

6- General condition of the patient.

I- Cancer cecum or ascending colon

- **Operable + non- obstructed Or obstructed carcinoma**
- **Steps:**

1- Colonic **preparation 3 days before surgery**. Done in elective cases

- Oral fluids intake

- Laxatives
- Neomycin & metronidazole oral
- Repeated enemas
- Colonic preparation

- **one day before surgery:**

total bowel irrigation with normal saline through naso-gastric tube

- On table irrigation during surgery

Right hemi-colectomy and ileo-transverse anastomosis:

a. Ligation of the ileo-colic and right colic vessels that supply the Right colon

b. Resection of the whole right colon.

c. Removal of the regional lymph nodes.

d. Ileo-transverse anastomosis (anastomose the terminal ileum to the right transverse colon.

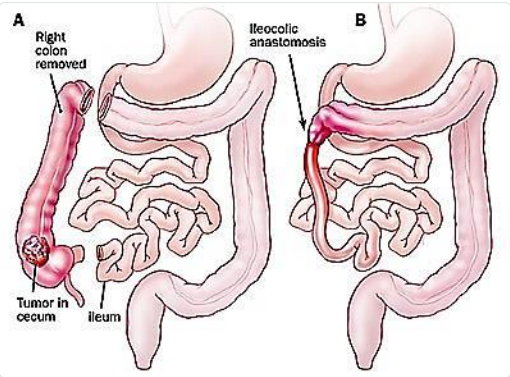


Figure 15.8 — Right hemi-colectomy

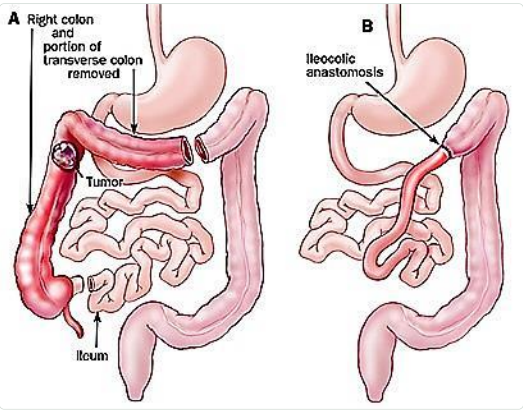


Figure 15.9 — Cancer ascending colon- extended right hemi-colectomy

II - Cancer transverse colon:

A - Operable non - obstructed case Or obstructed cases:

Surgery :

1- Transverse colectomy + colo-colic anastomosis

2-transverse colectomy and ileo-colic anastomosis (In tumour in the right transverse colon)

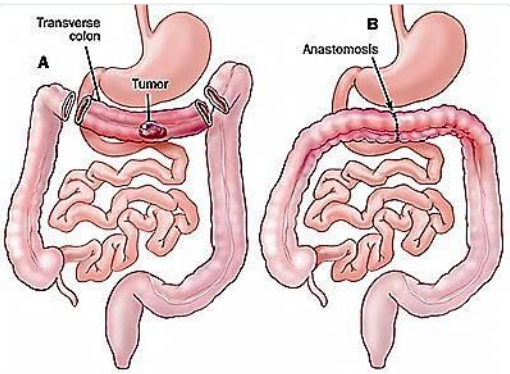


Figure 15.10 — Cancer transverse colon

III - Cancer left colon

A - Non - obstructed Carcinoma

Steps:

good colonic preparation due to the presence of high bacterial load in the left side.

- **Colonic preparation:**
- Oral fluids + low residue diet for 3 days + purgatives
- Repeated daily enemas
- Oral Flagyl and neomycin (intestinal antiseptic)
- Total bowel irrigations by normal saline 24 hours before surgery through naso-gastric tube.(rapid preparation)
- Not done in obstructed patients

Values of colon preparation:

- The idea of gut preparation is to get an empty colon with reduced bacterial load.
- This reduces contamination during surgery and thereby incidence of wound infection and leakage following surgery are minimized

- **Surgery: using surgical stablers (EEA)**

1- **Descending colectomy + colo-colic anastomosis**

2- In cancer sigmoid colon → sigmoid colectomy + Colo-rectal anastomosis.

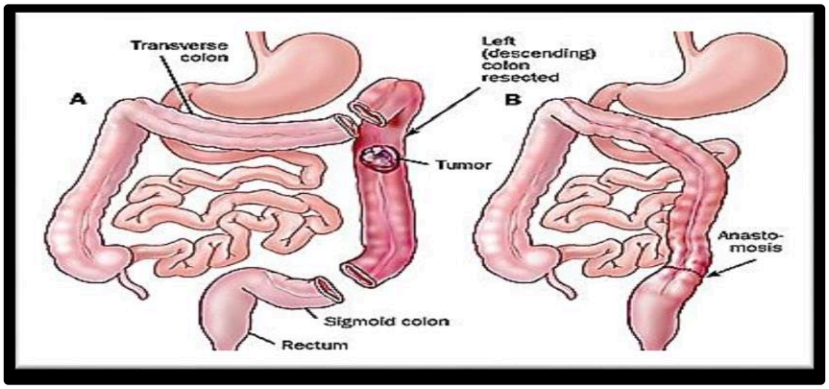


Figure 15.11 — Descending colon cancer

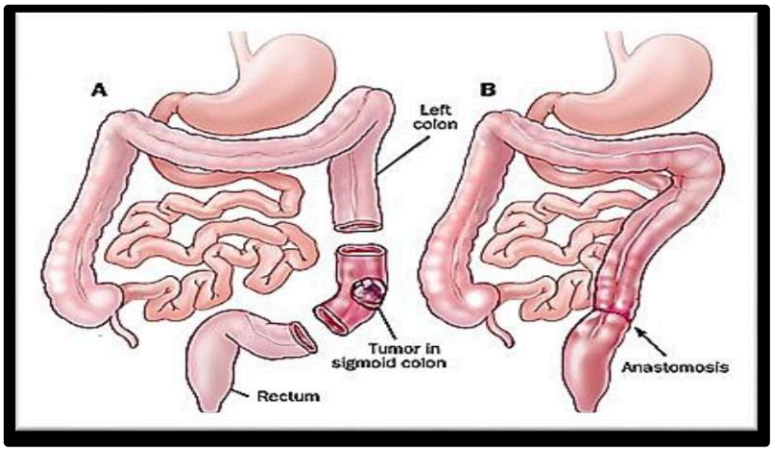


Figure 15.12 — Sigmoid colon cancer

B - Obstructed carcinoma of left colon:

- **Steps:**

1- No colonic preparation

2- Naso- gastric tube suction

3- Intravenous fluids and correction of electrolyte disturbances. 4- Antibiotic are started pre-operatively to control infections

(3rd generation cephalosporins and Metronidazole I.V.).

5 - Surgery:

- **First stage: (Hartmann's procedure)**

a) Resection:

1- Descending colectomy in cancer descending colon

2 -Sigmoid colectomy in cancer sigmoid colon.

b) Proximal colostomy or ileostomy → to divert stool away from site of surgery **(temporal colostomy)**.

Second stage: 6 to 8 weeks later: Using surgical staplers

1- Restoration of bowel continuity by anastomosis of the two ends.

2- Closure of the colostomy.

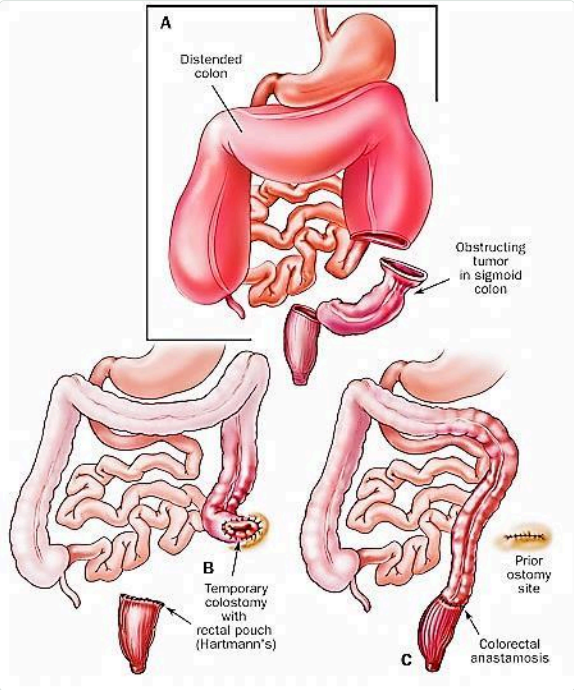


Figure 15.13 — Obstructing cancer sigmoid

IV - Cancer rectum

A - Non - obstructed case:

1- As above in cancer left colon

2- Surgery:

- **In upper and middle rectal cancers → Low anterior resection. Using**
- **surgical staplers:**

Resection of lower sigmoid + rectum till 2 cm below the lesion + colo-rectal or colo-anal anastomosis.c3eeef

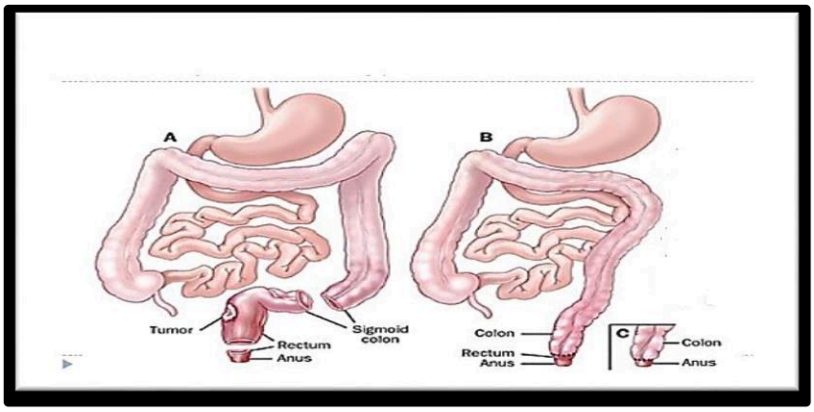


Figure 15.14 — Low anterior resection

- **Low rectal cancer:**
- **Abdomino - Perineal Resection (APR)**

a) Mobilization of the lower sigmoid and rectum (abdominal) b) Mobilization the anal canal (trans-perineal) c) Removal of the whole specimen through the abdomen.

d) Terminal end-colostomy (permenant colostomy).

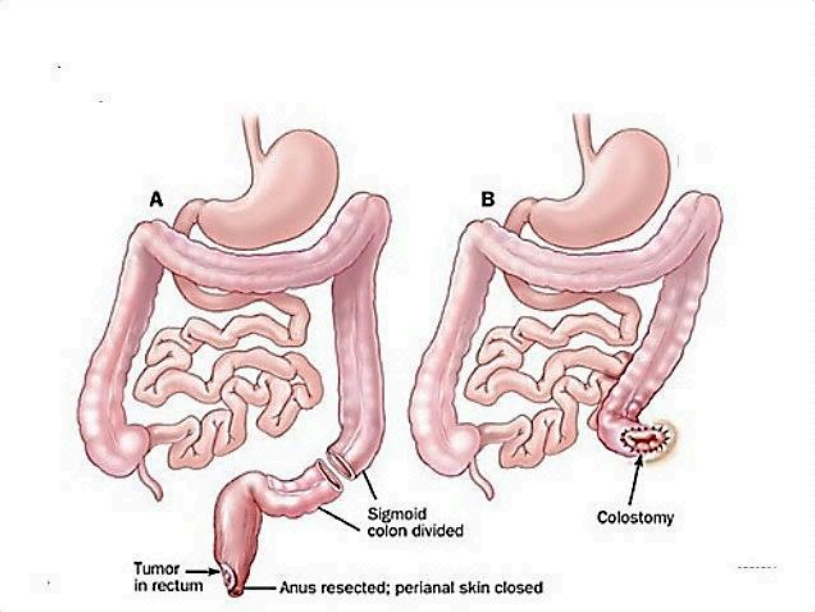


Figure 15.15 — Abdomino-Perineal Resection (APR)

B - Obstructed carcinoma: upper rectum : using staplers:

First stage:

Hartmann's procedure as before.

Second stage:

Reversal of Hartmann's (re-anastomosis and closure of colostomy)

N.B. indications of Total colectomy in cancer colon:

1- Done in multi- centric cancers

2- Done in Cancer complicating familial adenomatous polyposis

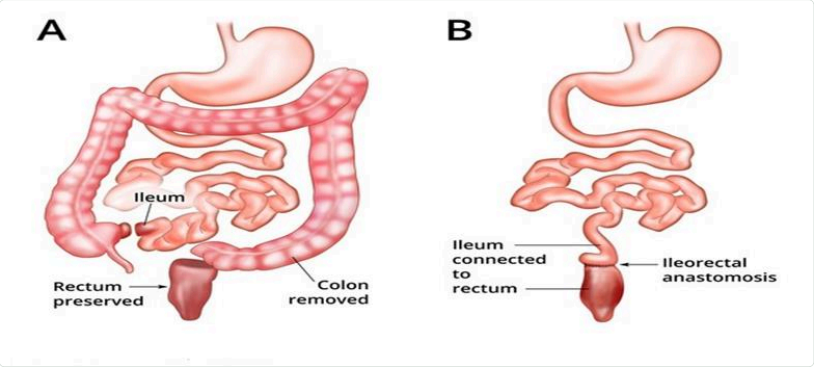


Figure 15.16 — Total colectomy and ileo-rectal anastomosis

CHAPTER 16

Anorectal diseases

16

Anatomical points:

Venous drainage of rectum and anal canal:

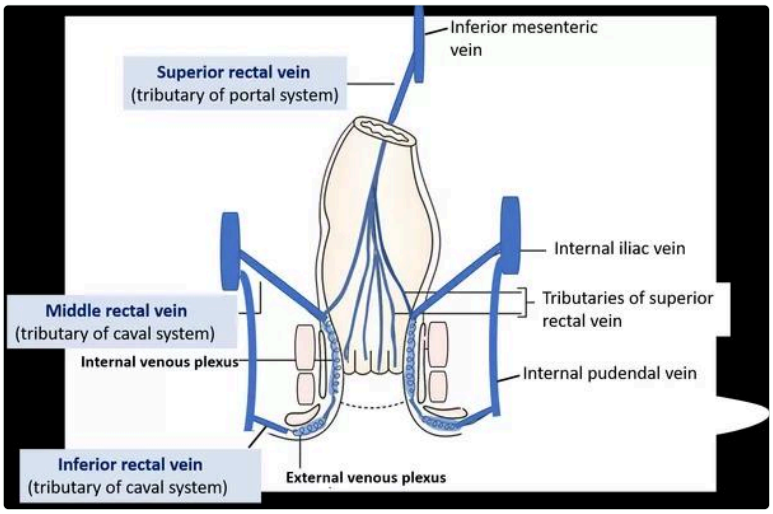


Figure 16.1 — Venous drainage of the rectum and anal canal

Hemorrhoids = piles

Haemorrhoides → Haima = blood Roos = flowing

Piles → Pila = ball

Definition:

- These are the dilated veins within the anal canal in the sub-epithelial region formed by radicals of Superior, Middle and Inferior rectal veins.
- Piles can be described as masses ("cushions") of tissue within the anal canal that contain blood vessels and the surrounding connective tissue (hemorrhoidal cushions).

Classifications:

I- According to Location-

1. Internal Piles –It is covered with mucous membrane.

It arises from Internal Hemorrhoidal plexus & above dentate line.

It is insensitive (autonomic).

2. External piles – It is situated outside the anal orifice & is covered by skin. It arises from External Hemorrhoidal plexus & below dentate line – it is sensitive (spinal nerves)

3. Internal + External haemorrhoids.

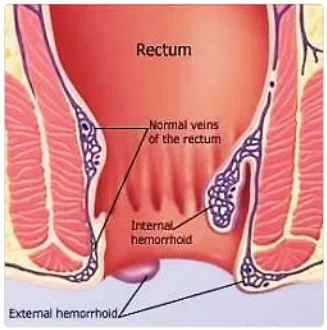


Figure 16.2 — Position of internal and external piles

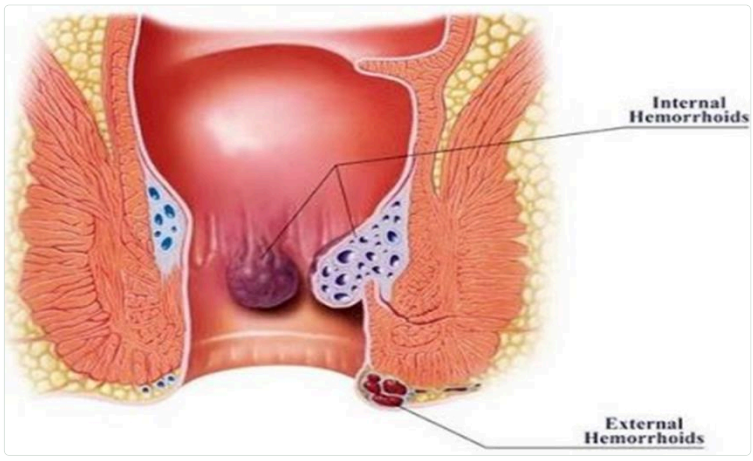


Figure 16.3 — Internal and external hemorrhoids

II- According to etiology-

1. Primary – Due to diets & habits

2. Secondary – Due to some other underlying disease

Primary piles:

predisposing factors:

- The haemoroidal veins are situated in anal sub-mucosa in longitudinal direction & does not have support of any other surrounding tissue
- Long term standing or sitting (gravity)
- Straining at defecation or micturition
- Chronic constipation
- Diarrhea → mucosal irritation and prolapsing down
- Low fiber diet prone to constipation

Secondary piles:

- Pregnancy: pressure of the gravid uterus on pelvic veins → venous stasis and congestion • labor → high anal pressure → prolapsed piles and anal fissure
- Portal hypertension: dilated porto-systemic collaterals in lower rectum

Degrees of internal piles:

1- 1st-degree

Piles Project into anal lumen internally

2- 2nd-degree

Protrusion outside anal canal at defecation with spontaneous Reduction

3- 3rd-degree

Protrusion outside anal canal at defecation or straining needs digital repositioning

4-4th-degree:

Permanently prolapsed outside the anus = irreducible piles – prone for strangulation.

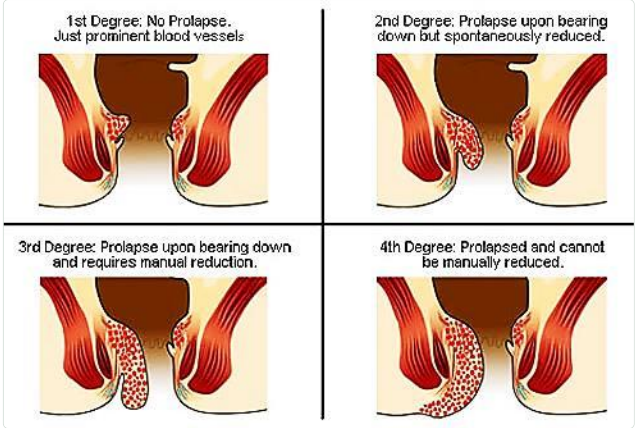


Figure 16.4 — Degrees of piles

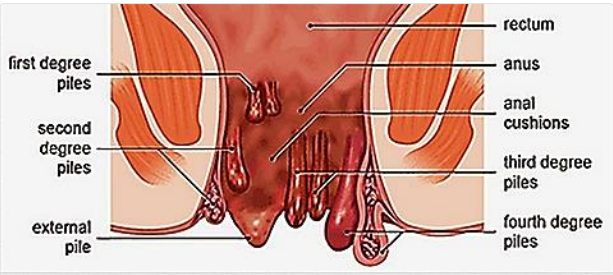


Figure 16.5 — Degrees of piles

Position of piles intra-anally (mother piles):

- Right anterior (11- o'clock)
- Right posterior (7- o'clock)
- Left lateral (3- o'clock)

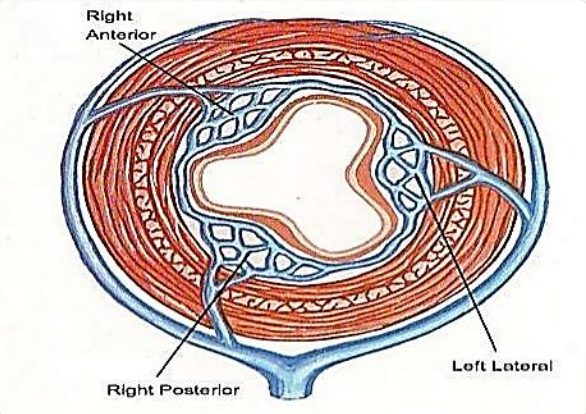


Figure 16.6 — Positions of internal piles

Clinical picture of internal piles:

- Bleeding (bright red) caused by hard stool and straining.
- Swelling / Prolapse of soft tissue (according to the degree)
- Straining / Pain / Discomfort
- Constipation
- Itching, Irritation and discharge
- Incomplete evacuation
- Anemia in recurrent prolonged cases

Anal Examination:

1- Position of the patient: Sim's = left lateral - Lithotomy

2 - Separate the buttocks apart gently - Focus the light source

Observe the peri-anal region anal verge

3- 3rd-degree = Inner red/purple mucosa & outer skin covered pile

4th-degree = Evident irreducible prolapsed pile

External piles → appear like peri-anal hematoma at anal verge

4- P/R examination: detect piles, detect associated pathology as rectal mass 5- Proctoscope: to detect site and degrees of pile and associated pathology

Figure 16.7 — Bleeding third-degree piles

Figure 16.8 — External piles: source clinical image

External piles: (peri - anal hematoma – Thrombosed external pile):

- It is situated outside the anal orifice & is covered by skin.
- It arise from External Hemorrhoidal plexus & below dentate line
- it is sensitive
- Thrombosed external piles sometimes referred to as 5 days' disease
- It usually follow severe constipation and straining at defecation with rupture of small venules causing hematoma.

- **Clinically:**
- Patient presented with painful peri-anal swelling
- P/R: very tender, tense and bluish swelling at the anal verge about 1 or 2 cm

Figure 16.9 — Thrombosed external pile

Complications of internal piles:

1- Anemia: due to chronic blood loss

2- Strangulation in 4th degree piles

Treatment of piles

- **Grade I, II internal piles :**

A - Medical treatment

1- Treatment of constipation and the Predisposing Factors

2- Venotonics to stop bleeding

3- Hemostatic agents

4- Decongestant local ointment

B - Out - patient procedures:

1- Sclero- therapy (induction of fibrosis of the pedicle) using phenol in almond oil

2- **Rubber band ligation** of the pile pedicle → ideal for grade II

The pile sloughs after 7 - 10 days

- **Grade III, IV:**
- **Surgical management:**

1- open hemorrhoidectomy:

Electro-cautery:

Dissection of the pile till its pedicle intra-anally with electro-cautery device with double ligation and division.

2- Stapling of hemorrhoids using stapler

3- LASER hemorroidectomy

- **Treatment of external thrombosed piles:**

1.Conservative:

local decongestant – venotonics – analgesics – laxatives

2.Surgical:

incision and drainage of the hematoma under local or regional anesthesia

Peri - anal abcess

Types:

1- subcutaneous peri-anal abcess

2- submucous abcess

3- Inter-sphincteric abcess

4- supra-levator abcess

5- ischio-rectal abcess → present in the ischio-rectal fossa

Causes and risk factors:

1- Blocked anal glands is the most important starting triggering factor for abcess formation

2- Chron's disease

3- Diabetics and other immune-compromized situations

4- Infected anal fissure

5- Diarrhea – constipation

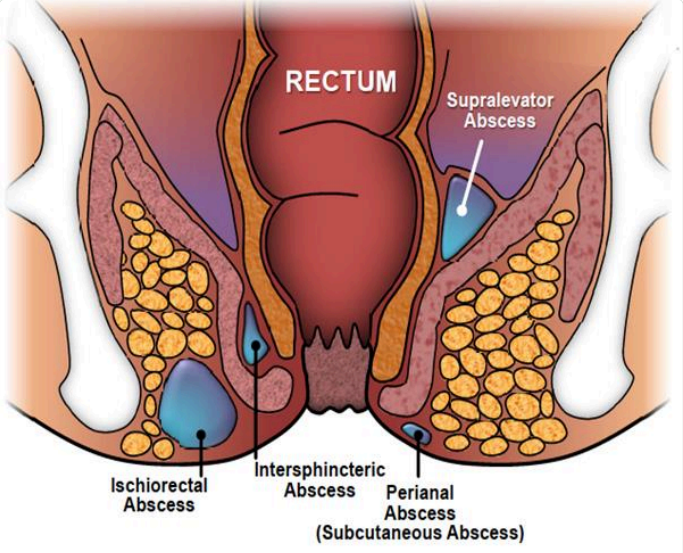


Figure 16.10 — Types of peri-anal abcesses

Clinical picture :

- fever, severe anal pain
- external tender peri-anal swelling is evident in 1 & 2
- tender swelling intra-anal in 3 & 4
- ischiorectal swelling tender red edematous in 5
- pus may come out from the anus in opened abcess
- sometimes the patient may get reflex difficult urination
- sometimes thepatient needs examination under anaesthsia to diagnose higher abcesses

Figure 16.11 — Perianal abscess: source clinical view

Figure 16.12 — Perianal abcess

Treatment:

- Incision and Drainage under G.A. with proper irrigation with saline and anti-septic solutions.

Complications → peri-anal fistula – recurrent abcess

Peri - anal fistula

Anatomy :

1- The external sphincter muscle

- Is a striated muscle under voluntary control
- Formed of three components: subcutaneous, superficial, and deep muscle. • Its deep part is continuous with the puborectalis and forms the anorectal ring, which is palpable upon digital examination.

2- The internal sphincter muscle is a smooth muscle under autonomic control and is an extension of the circular muscle of the rectum.

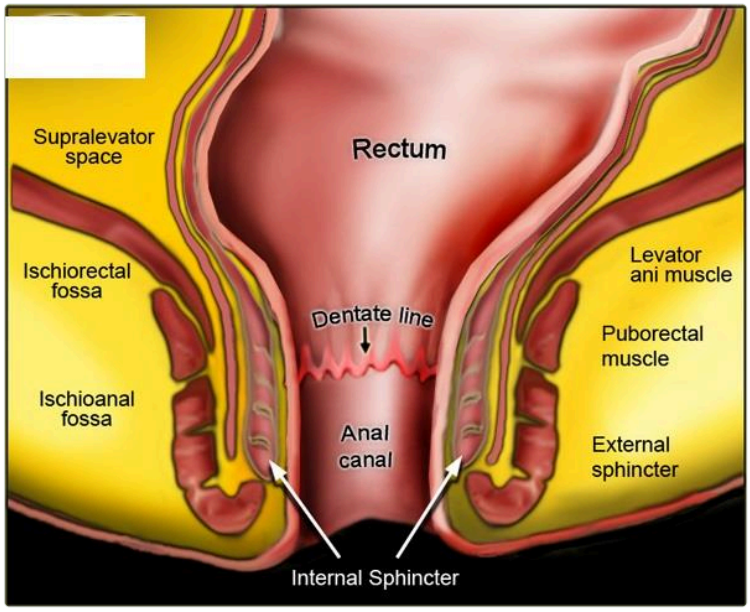


Figure 16.13 — Anal sphincter anatomy

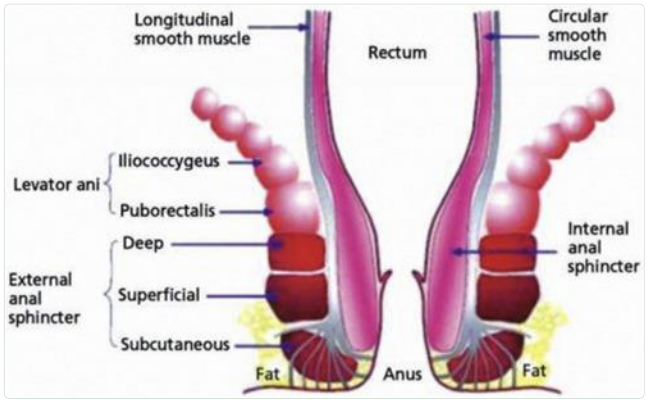


Figure 16.14 — Anatomy of anal sphincters

Etiology of anal fistula :

1 - previous anorectal abscess.

- There are 8 to 10 anal crypt glands at the level of the dentate line in the anal canal, arranged circumferentially.
- These glands penetrate the internal sphincter and end in the Inter-sphincteric plane.
- An infection begins in the anal canal glands and extends into the anal sphincters to cause an anorectal abscess.
- After surgical or spontaneous drainage in the perianal skin, A granulation tissulined tract is occasionally left behind, causing fistula symptoms.

2 - Crohn disease:

multiple and recurrent fistulae

3- Chronic Anal fissures

4- Actinomycoses, tuberculosis infections.

C lassifications of anal fistula

Parks Classification

The most commonly used classification for anal fistula. There are four types of fistulae that result from crypto-glandular infections:

1. Inter-sphincteric

2. Trans-sphincteric

3. Supra-sphincteric

4. Extra-sphincteric

1- An intersphincteric fistula:

a) It is the result of a perianal abscess b) **Common course** - It begins at the dentate line, then pass via the internal sphincter to the inter-sphincteric space between the internal and external anal sphincters, and finally terminates in the perianal skin c) Incidence – 70 % of all anal fistulas

2- A trans-sphincteric fistula:

a) **Common course** - It passes from the internal opening at the dentate line through the internal and external anal sphincters into the ischiorectal fossa and then terminates in the perianal skin b) Incidence – 25 % of all anal fistulas

3- A supra-sphincteric fistula:

a) Common course - It passes from the internal opening at the dentate line to the intersphincteric space, pass superiorly to above the puborectalis, and then curves downward lateral to the external anal sphincter into the ischiorectal fossa and finally to the peri-anal skin b) Incidence – 5 % percent of all anal fistulas

4- An extra-sphincteric fistula:

a) **Common course** - It runs from the perianal skin via the ischiorectal fossa, tracking upward and through the levator ani muscles to the rectal wall, completely outside the sphincters, with or without connection to the dentate line b) Incidence - 1% of all anal fistulas

Old classifications:

1- Low peri-anal fistula → internal opening below the anorectal ring

2- High peri-anal fistula → internal opening above the anorectal ring

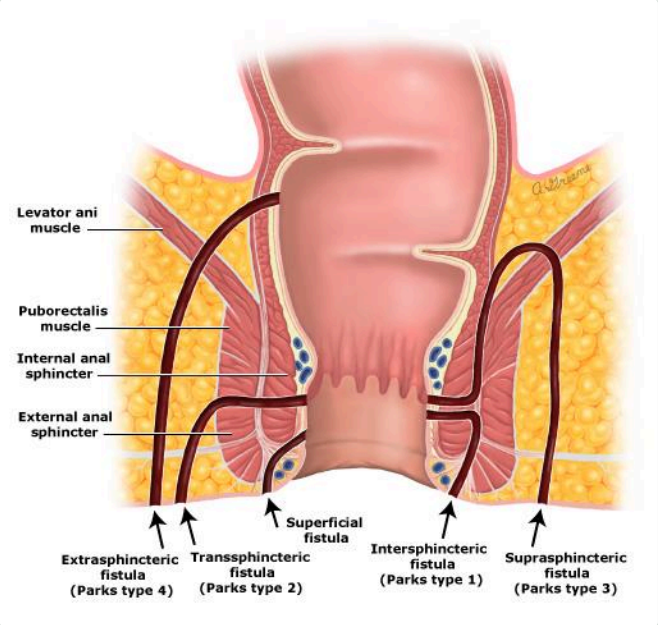


Figure 16.15 — Different types and anatomy of anal fistulae

Clinical presentation:

- History of previous pain, swelling, and spontaneous or surgical drainage of an anorectal abscess.
- History of Inflammatory bowel disease (Chron's)

Signs and symptoms of fistula:

- Perianal discharge

- Pain
- Swelling
- Bleeding
- Skin excoriation
- External opening on peri-anal skin

Examination:

1. Looking for an external opening that appears as an open sinus or elevation of granulation tissue.

2. Spontaneous discharge of pus or blood via the external opening may be apparent or expressible on digital rectal examination.

Figure 16.16 — Peri-anal fistula: external opening

3. Goodsall's law:

- An external opening opens in posteriorly has curved track that opens in the Midline
- An anterior opening has straight course and the internal opening to one side of the midline

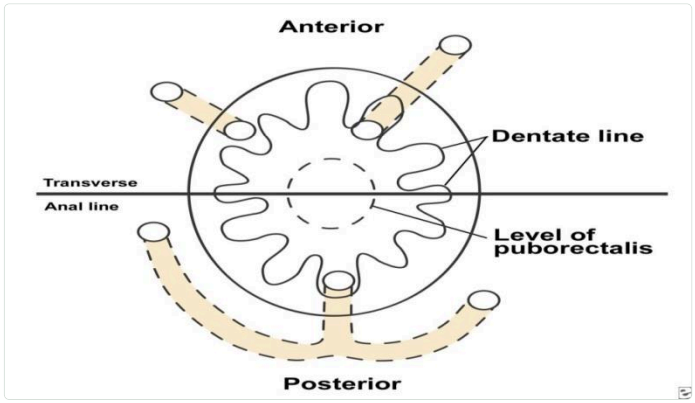


Figure 16.17 — Goodsall's law

4. Digital rectal examination (DRE):

a) may reveal a fibrous tract or cord beneath the skin

b) determine the relationship between the anorectal ring and the position of the tract

c) The sphincter tone should be assessed

5. **Anoscopy** is usually required to identify the internal opening.

6. **Proctoscopy** is also indicated in the presence of rectal disease (eg, Crohn disease)

Investigations:

1- Fistulography:

A dye is injected through the external opening to outline the course of the fistula tract (of low values nowadays).

2 - Endoanal or endorectal ultrasonography

Help to define the sphincter anatomy and thereby help to differentiate inter-sphincteric from trans-sphincteric lesions.

3 - Magnetic resonance imaging Fistulogram : MRI

- The most accurate in detecting the fistulous tracts.
- Detects the relation of the track to the sphincters
- Detects the presence of peri-anal or peri-rectal abcesses

Management of peri - anal fistula:

Intraoperative considerations include the following:

Examine the patient under anesthesia to confirm the extent of the fistula

Identifying the internal opening to prevent recurrence is imperative by probing and

- hydrogen peroxide injection through the external opening

Fistulotomy:

1- The laying-open technique (fistulotomy) is useful for 85-95% of primary fistulas (ie, submucosal, inter-sphincteric, and low trans-sphincteric)

2- A probe is passed into the tract through the external and internal openings.

3- The overlying skin, subcutaneous tissue, and internal sphincter muscle are divided with electrocautery, and the entire fibrous tract is opened and curetted.

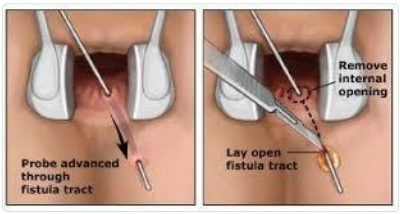


Figure 16.18 — Fistulotomy

Seton Placement:

This technique is useful in patients with the following conditions:

1. **Complex fistulas** (i.e, high trans-sphincteric, supras-phincteric, extra-sphincteric) or multiple fistulas

2. **Recurrent fistulas** after previous fistulotomy

3. Poor sphincter tone:

- The purposes of setons are to drain, to promote fibrosis, and to cut through the fistula. • Setons can be made from large silk suture.
- The seton is tightened on subsequent visits until it is pulled through over 6-8 weeks.

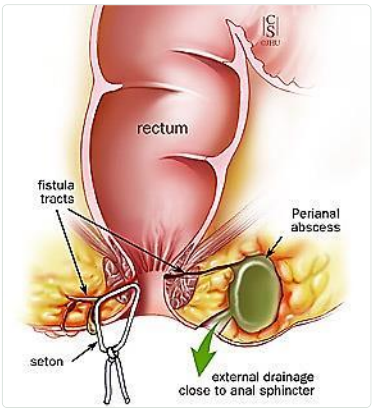


Figure 16.19 — Seton placement: anatomy

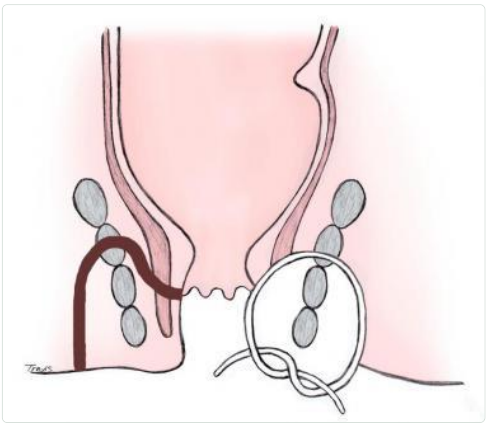


Figure 16.20 — Seton placement

Other lines :

1. Proximal colostomy temporary in high and complex anal fistula
2. Local fibrin glue injection inside the fistulous tract.
3. Anal fistula plug placed inside the tract to induce fibrosis

Anal fissure

Anal fissures are tears in the mucosa of anal canal.

Types:

They may be **acute or chronic**

Causes:

1- trauma to the anal mucosa →fissure in midline posterior

- Hard stool
- Straining in chronic constipation
- Diarrhea
- Instruments (thermometer, enema tip, endoscope, or ultrasound probe
- During labour

2- Other causes:

fissure is placed in abnormal sites

- Chron's disease
- Syphilis
- Anal cancer

Site of anal fissure:

In the lower anal canal in the **midline posterior** at the **crypta magna** of the mucosa. **why ?**

a) because of the configuration of the muscle that surrounds the anus are oval shaped with better support at the sides and weakest posteriorly b) posterior mucosa most commonly exposed to trauma c) posterior part of the anal canal has less blood flowing to it than the other parts of the anal canal preventing fissures from healing

Patho-physiology of anal fissure:

- Trauma to the anal mucosa → tear → infection → fissure → lead to spasm of the internal anal sphincter → failure to relax which may lead to pulling of the fissure edges apart → **gapping and failure of healing**.

Clinical picture of anal fissure:

A- symptoms:

1- Anal pain:

a) it is more severe in acute than chronic fissure b) increase with defecation and relived after defecation c) may lead to constipation due to spasm of internal sphincters

d) may lead to dysuria

2- Constipation due to spasm or stenosis

3- Anal pruritis and discharge

4- Bleeding per-rectum

B- Anal examination:

- **An acute anal fissure** looks like a linear tear with hyperemic edematous edges + tight anal spasm.
  - **A chronic anal fissure** frequently is associated with a **triad of findings** that includes :

1- A tag of skin at the edge of the anus (**sentinel pile**)

2- Thickened edges of the fissure with muscle fibers of the internal sphincter visible at the base of the fissure, (button hole)

3- Enlarged **anal papilla** at the upper end of the fissure inside the anal canal.

c- Anoscope or proctoscopy for associated lesions as piles or masses

Figure 16.21 — Anal fissure: source clinical view

Figure 16.22 — Anal fissure

General treatment for anal fissures

- In acute fissures, medical (non-operative) therapy is successful in the majority of patients.
- Of acute fissures, 80% to 90% will heal with conservative measures as compared with chronic (recurrent) fissures, which show only a 40% rate of healing.

1- Excess oral fluids

2- Treatment of constipation

3- High fiber diet

4- Bulk forming stools

5- Avoid spicy foods

6- Avoid sharp foods

7- Local anaethetic ointment

8- Local decongestant (ointment and suppository)

9- Nitroglycerine preparation locally

10- Botulinum toxin locally to relief the spasm of the sphincter and allows the healing of the fissure

- **Surgery:**
  - In failure of medical treatment
  - **It includes partial lateral internal sphincterotomy** to relief the internal sphincter spasm allowing healing of the fissure

Figure 16.23 — Lateral sphincterotomy: operative view

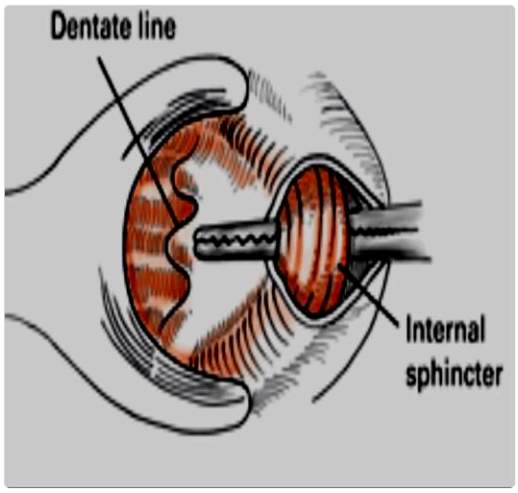


Figure 16.24 — Partial internal lateral sphincterotomy

Pilonidal sinus

Pilonidal means a 'nest of hairs

A pilonidal sinus (PNS) is a track in the natal cleft between the buttocks over the sacrococcygeal area

Types:

1- congenital 2- Acquired

Risk factors include:

- Male gender
- **Extensive hairy individuals** (presence of hair within the natal cleft).
- **Sitting occupations**

- Presence of a deep natal cleft, • **Family history** is seen in 38% of patients with pilonidal disease.
- Obesity is a risk factor for recurrent disease (deep natal cleft)

Patho - physiology

(1) The invader , hair

(2) The force , causing hair penetration; occupation

(3) The nature of the skin, extensive hairy

a- Upon prolonged sitting, hair follicles can break and open a pit in the skin b- Debris may collect in this pit, followed by development of a sinus with a short tract c- Hair penetrates into the subcutaneous tissues through dilated hair follicles → initiates inflammatory process in the subcutaneous tissue d- Pilonidal abcess will form e- Improper treatment leads to PNS f- A PNS usually contains hair, dirt, and debris

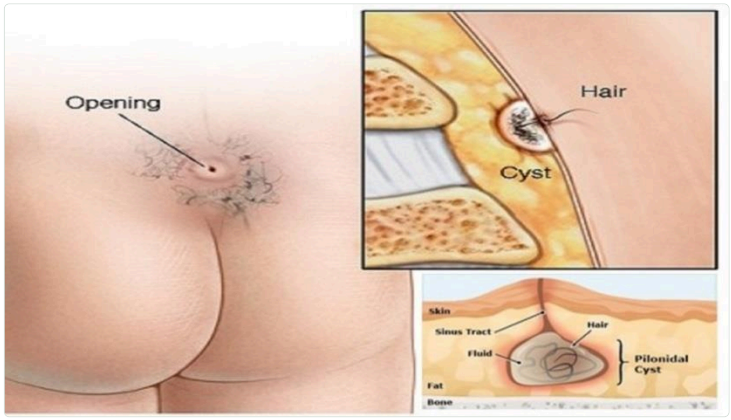


Figure 16.25 — Pilonidal sinus

Clinical picture:

A - Acute pilonidal abcess

- Typical findings of an abscess, including severe pain redness, warmth, local tenderness, and fluctuation.

b- Chronic cases

Patients may present with intermittent swelling and drainage, including purulent, mucoid, or bloody fluid from this area.

- Chronic pilonidal disease often manifests as recurrent or persistent discharge from a sinus at sacrococcygeal region.
- Loose hair may be seen projecting from the site of sinus

Treatment:

A- Pilonidal abcess:

incision and drainage under G.A

B- Pilonidal sinus

1- Non-surgical treatment

- 80% phenol is injected into the sinus, left there for 1 minute, and then expressed out of the cavity. The sinus is then curetted.
- Fibrin glue placed inside the sinus track allows healing in some cases

2- Surgical treatment:

Surgical options for management of a chronic pilonidal sinus include the following: a) **Excision and lying open of the sinus tract** → mainly used in markedly infected sinuses followed by regular dressings with healing by 2nd intension.

The average time for wound healing to occur is about 6 weeks

b) Excision with primary closure

Used in low infected sinuses

Wide and deep excision to the sacrum

Then closure of the wound leaving drain inside

Healing occurs by primary intension

c)Using flaps techniques:

used commonly in recurrent cases d) After management the hair at the lower back and sacro-coccygeal area should be permanently shaved to avoid recurrence

N.B. D.D. Of painful anal conditions:

1. Acute anal abcess
2. Peri-anal hematoma
3. Anal fissure
4. Thrombosed or strangulated prolapsed piles
5. Anal carcinoma
6. Proctalgia
7. parasitic infestations

CHAPTER 17

Hepatology and Hepatic Surgery

17

Anatomy of the Liver

- **Surgically:** the liver divided into 8 segments (4 for each lobe)

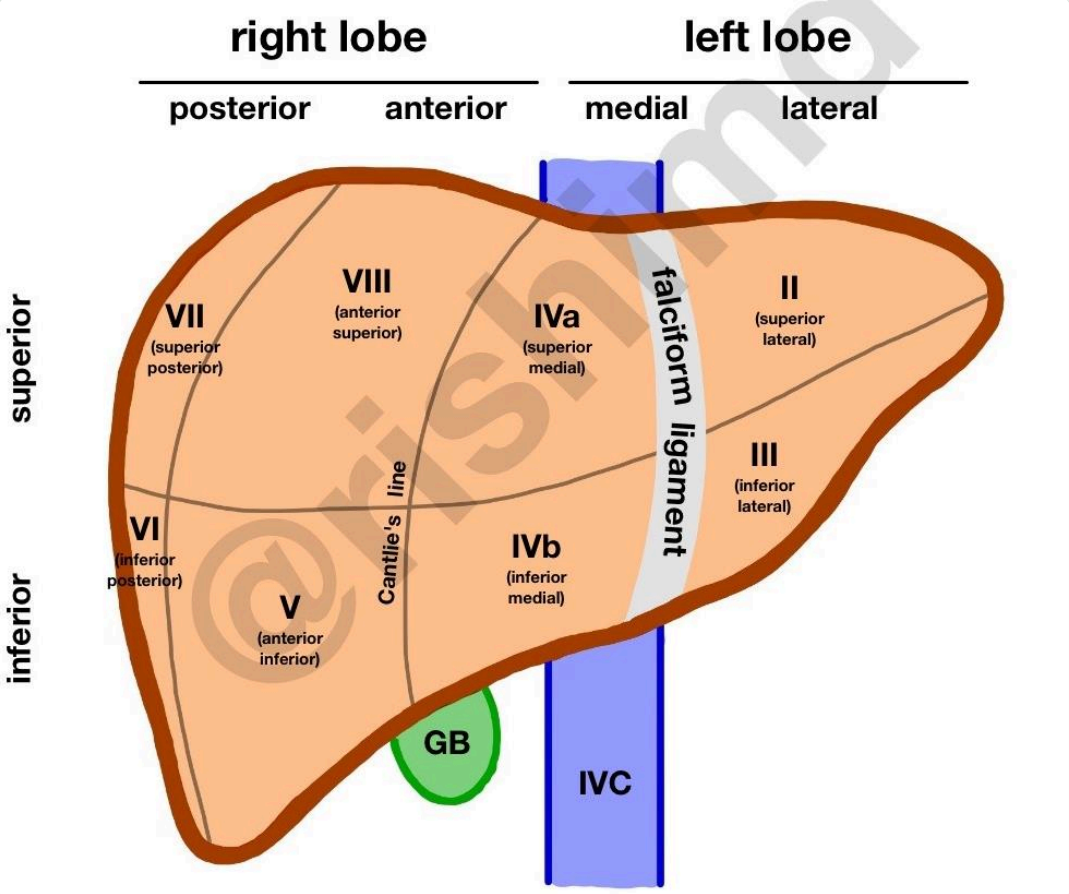


Figure 17.1 — Segments of the liver

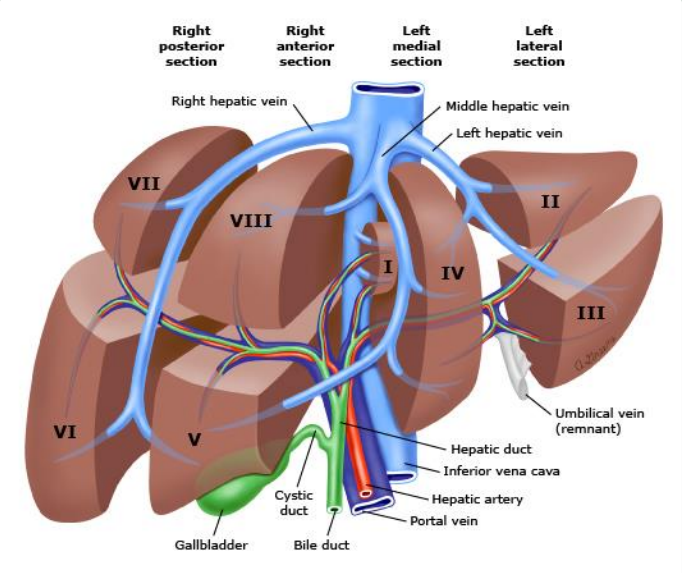


Figure 17.2 — Segments of the liver

Blood supply of the liver:

The liver has rich blood supply

1)Portal vein blood:

through the **portal vein** (portal circulation) =

80%

2)Systemic blood:

through the **common hepatic artery** = 20%

Branch of the celiac trunk (pure oxygenated blood)

3) The blood enters the liver passes to the portal vein and hepatic artery branches in the portal tracts → to the hepatic sinusoids between the hepatocytes then to the central veins then collected to emerges through the hepatic veins.

4) The blood leaves the liver through the 3 hepatic veins to the IVC

(systemic)

5) The arrangement of the structures at the free border of lesser omentum:

- Common bile duct (ant. to right)
- Common hepatic artery (ant. to left)
- Portal vein (most posterior)

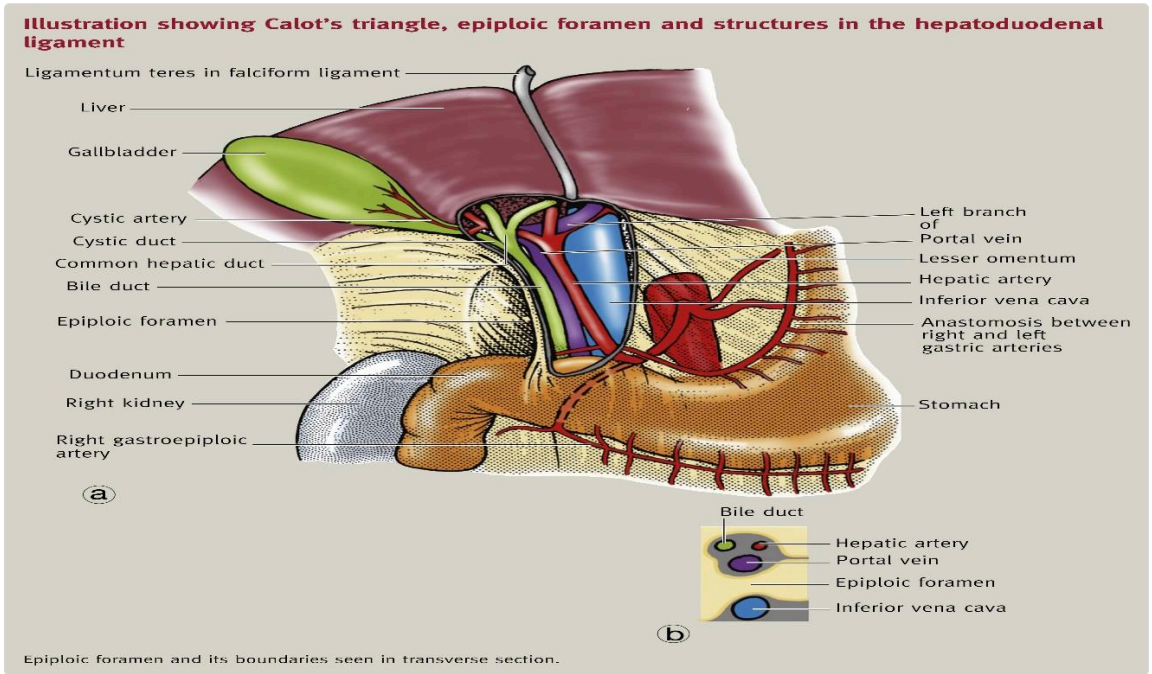


Figure 17.3 — Hepatoduodenal structures

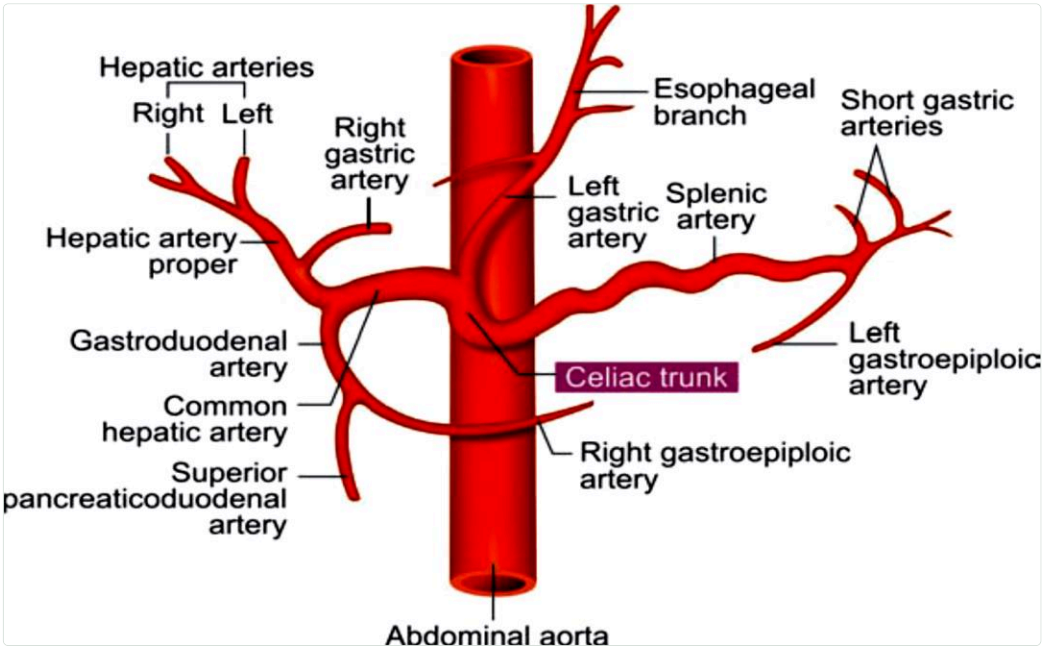


Figure 17.4 — Branches of the celiac trunk

Branches of the celiac trunk

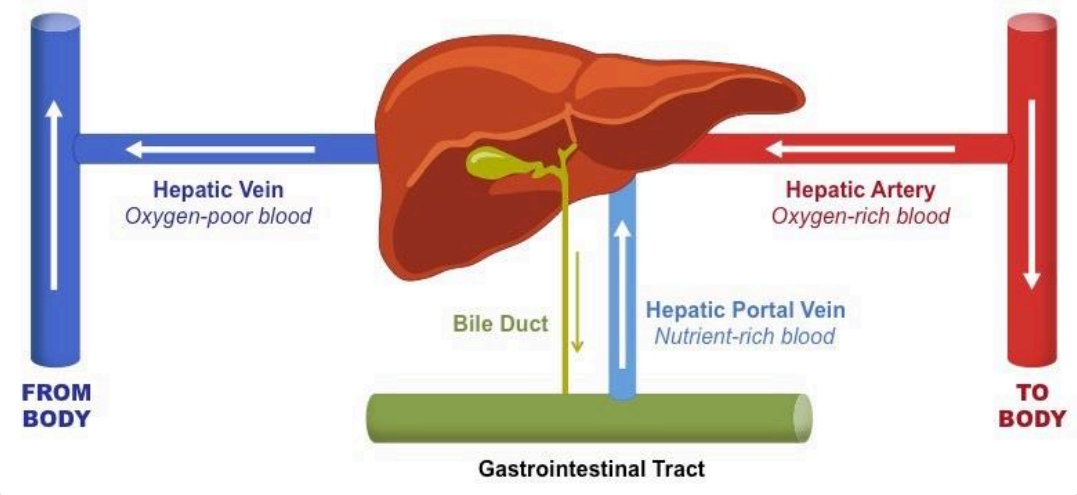


Figure 17.5 — Hepatic blood flow

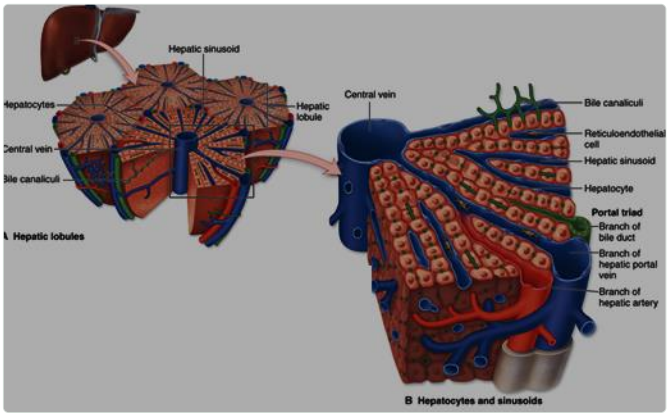


Figure 17.6 — Hepatic microcirculation

The hepatic lobule : the hepatic lobule is the basic unit of the liver. In the centre, there is a central tributary of the hepatic vein and at the periphery, the portal tracts containing branches of the portal vein, hepatic artery and the bile duct. Cords of liver cells and sinusoids extend between these two systems. The portal tracts are surrounded by a limiting plate of liver cells.

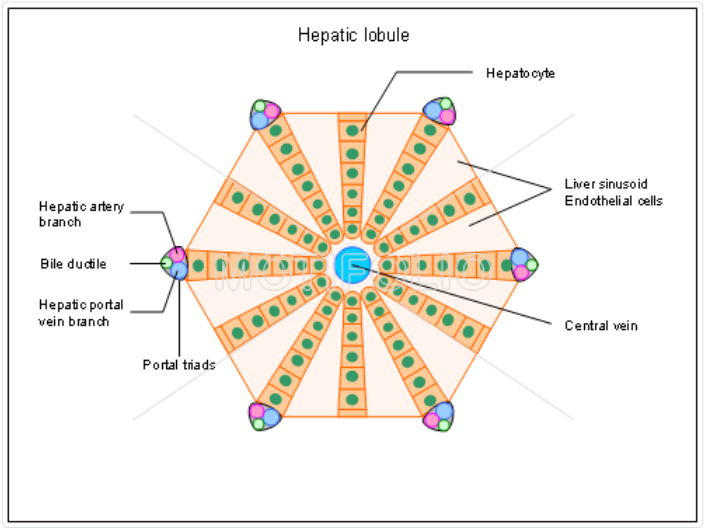


Figure 17.7 — Diagram of a hepatic lobule

The acinus is the functional hepatic unit. This consists of parenchyma supplied by the smallest portal tracts containing portal vein radicals, hepatic arterioles and bile ductules . The hepatocytes near this triad (zone 1) are well supplied with oxygenated blood and are more resistant to damage than the cells nearer the terminal hepatic (central) veins (zone 3).

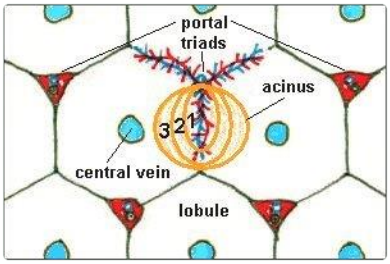


Figure 17.8 — Diagram of an acinus. Zones 1, 2 and 3 represent areas supplied by blood, with zone 1 being best oxygenated.

The sinusoids lack a basement membrane and are loosely surrounded by fenestrated endothelial cells and Kupffer cells (phagocytic cells). Sinusoids are separated by plates of liver cells (hepatocytes). The subendothelial space between the sinusoids and hepatocytes is the space of Disse, which contains a matrix of basement membrane and stellate cells. Activated stellate cells produce extracellular matrix components including collagen and participate in the fibrosis of the liver.

CHAPTER 18

SYMPTOMATOLOGY AND INVESTIGATIONS OF LIVER DISEASES

18

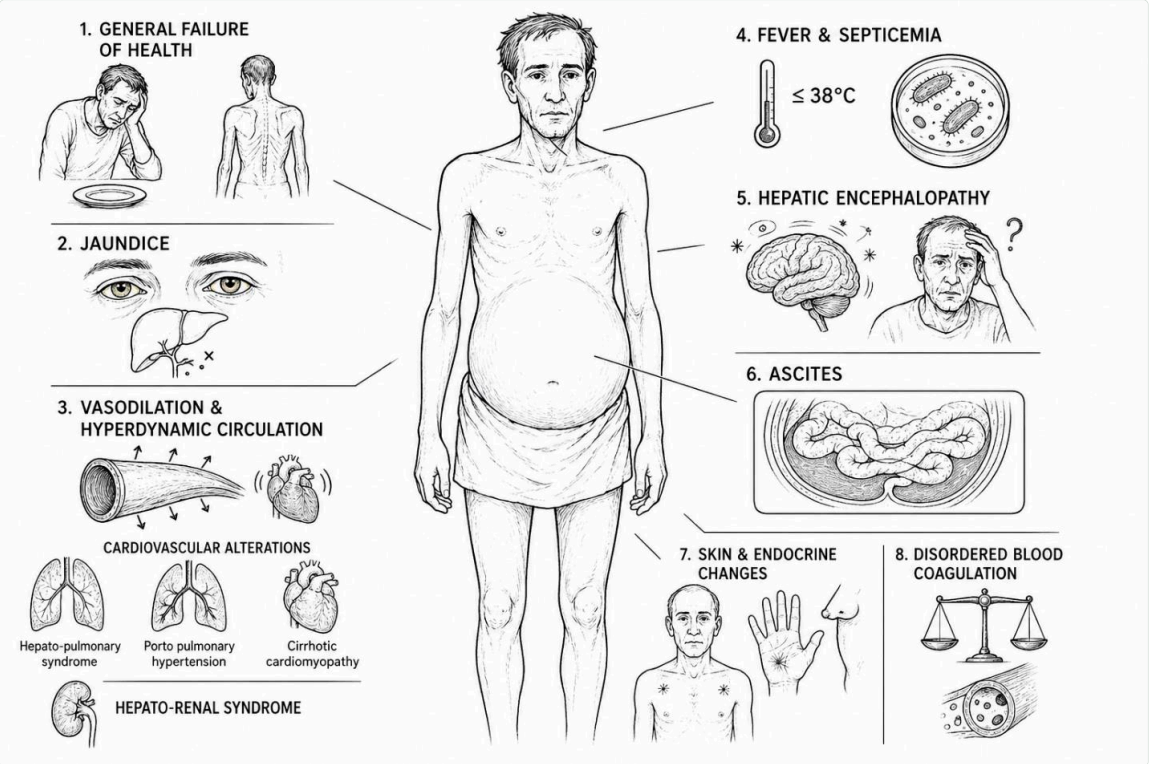


Figure 18.1 — Physical signs of liver disease

General failure of health

- The most conspicuous feature is easy fatigability.
- Muscle wasting, anorexia and malnutrition.

Jaundice

- due to failure of the liver cells to metabolize bilirubin.

Vasodilation and hyperdynamic circulation

- Cardiovascular Alterations:
- Hepato-pulmonary syndrome.
- Porto pulmonary hypertension.
- Cirrhotic cardiomyopathy.
- Hepato-renal syndrome.

Fever and septicemia

- Patients with decompensated cirrhosis show a continuous low-grade fever which rarely exceeds 38°C.
- Septicemia is frequent in terminal hepato-cellular failure.

Hepatic Encephalopathy

- Neuropsychiatric abnormalities occurs in patients with acute or chronic liver disease.

Ascites

is free fluid within the peritoneal cavity.

Skin and endocrine changes.

Disordered blood coagulation.

INVESTIGATIONS OF LIVER DISEASES

Blood tests

Liver function tests :

- Serum albumin:
  - Is the most important serum protein synthesized by the liver.
  - It has a long half-life (21 days).
  - Not a good indicator of acute liver injury, decreased only in chronic liver diseases.
  - Total protein in chronic liver diseases is not of diagnostic value (may be↑ or ↓ or normal).
  - Prothrombin time (PT): Prolongation of PT in liver disease may be due to impaired synthesis of clotting factors or impaired absorption of vitamin K.

Liver biochemistry :

- Bilirubin : increased serum bilirubin in liver disease.
  - Aminotransferases :
  - These enzymes are present in the hepatocytes and thus their elevations indicate liver cell damage.
  - Alanine aminotransferase (ALT) more specific to the liver so that a rise only occurs with liver disease.
  - Aspartate aminotransferase (AST) is also present in heart, muscle, kidney and brain. High levels are seen in hepatic necrosis, myocardial infarction, muscle injury.

Marked elevation of ALT and AST 10-100 times in acute viral hepatitis.

Mild to moderate elevation of ALT and AST (2-5 folds) in liver cirrhosis.

Mild to moderate elevation with ALT/AST ratio >1 in chronic active hepatitis.

- Alkaline phosphatase (ALP) and gamma glutamyl transferase (GGT):
- Raised in obstructive jaundice.
- Slight to moderate elevation occur in hepatitis and cirrhosis.
- GGT elevated in alcoholic even with normal liver.
- GGT is more specific for liver disease than ALP.

Additional blood investigations :

- Viral markers : including hepatitis A, B, C, D, and E.
  - Biochemical tests :
  - α-Fetoprotein.
  - Increased in hepatocellular carcinoma , slightly raised in patients with hepatitis and chronic liver disease.
  - α1Anti-trypsin deficiency can produce cirrhosis.
  - Raised urinary copper, low serum copper and ceruloplasmin occur in Wilson's disease

Immunological tests :

- Serum immunoglobulins :
  - In primary biliary cirrhosis, raised IgM.
  - Autoimmune hepatitis , raised IgG.
  - In liver cirrhosis IgA is mainly increased.
  - In chronic hepatitis there is a main increase in IgG.
  - Serum autoantibodies:
  - Anti-mitochondrial antibodies (AMA's) found in 95% of patients with primary biliary cirrhosis (PBC), and are also found in autoimmune hepatitis.
  - Anti-nuclear, anti-smooth muscle, liver/kidney microsomal antibodies increased in autoimmune hepatitis.
  - Anti-nuclear cytoplasmic antibodies (ANCA)
  - found in primary sclerosing cholangitis.

Genetic analysis : e.g. haemochromatosis "HFE gene".

Imaging techniques

Ultrasound examination (US)

- The first imaging modality used to investigate liver disease.
- non-invasive, safe cheap technique.
- The normal liver appears as a relatively homogeneous structure.

Figure 18.2 — Liver ultrasound

- **Color Doppler** will demonstrate vascularity within a lesion.

Figure 18.3 — Liver ultrasonography

Computed tomography (CT)

- **Triphasic CT** is the investigation of choice for the diagnosis of hepatocellular carcinoma.

Figure 18.4 — Computed tomography of the liver

Figure 18.5 — Abdominal CT

Magnetic resonance imaging (MRI)

- MRI is the most sensitive investigation of focal liver disease
- **Magnetic resonance cholangiopancreatography (MRCP)** is usually the next test if a biliary abnormality is present on US examination.

Plain X-ray:

rarely used but may show gallstones or pancreatic calcification

Fibroscan:

(Elastography)

- is a special ultrasound technology that measures liver stiffness (hardness) which reflect liver fibrosis , also it measures fatty changes in the liver (sensitivity and specificity 80-95% compared to liver biopsy).

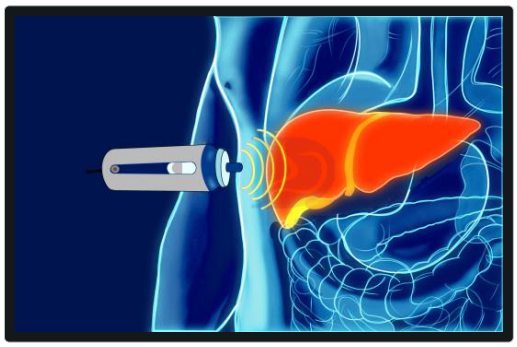


Figure 18.6 — Fibroscan

Figure 18.7 — FibroScan

Endoscopy

Upper GI endoscopy

- Is used for diagnosis and treatment of varices, detection of portal hypertensive gastropathy, associated lesions such as peptic ulcers.

  Endoscopic Retrograde Cholangiopancreatography (ERCP)

  This technique outlines the biliary and pancreatic ducts. It involves the passage of an endoscope into the second part of the duodenum and cannulation of the ampulla.

  Contrast is injected into both systems and the patient is screened radiologically.

Figure 18.8 — Endoscopic retrograde cholangiopancreatography

Therapeutic ERCP involves the following:

- Removal of common bile duct stones.
- Drainage of the biliary system by a stent through an obstruction.

Endoscopic ultrasound (EUS)

A less invasive method for bile duct imaging. EUS-guided fine-needle aspiration of tumors e.g. pancreatic tumors.

Percutaneous transhepatic cholangiography (PTC) : A fine needle is passed into the liver, contrast is injected until a biliary radicle is identified PTC is performed if

ERCP fails

Liver biopsy :

Histological examination of the liver is valuable in the differential diagnosis of diffuse or localized parenchymal disease.

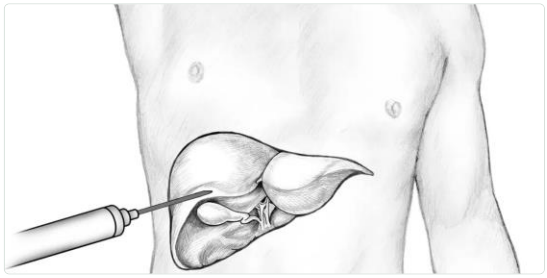


Figure 18.9 — Liver biopsy

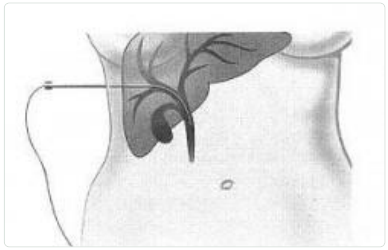


Figure 18.10 — Percutaneous transhepatic cholangiography

Biopsy PTC

Laboratory diagnostic findings of liver diseases

The most important liver function tests include the following:

1-Albumin and other proteins:

Albumin Is the most important serum protein synthesized by the liver. It has a long half-life (21 days) and is therefore not a good indicator of acute liver injury. Furthermore, there is a high reserve of hepatic albumin synthesis and thus albumin is decreased only in chronic liver diseases with severe cells destruction.

As regards immunoglobulins, there is a polyclonal increase of gamma globulin (due to ↓clearance or ↑synthesis) in chronic liver disease with reversed albumin/globulin ratio. In liver cirrhosis IgA is mainly ↑and to a less extent other immunoglobulins. In chronic hepatitis there is a main ↑ in IgG. Total protein in chronic liver diseases is not of diagnostic value (may be↑ or ↓ or normal).

2- Liver enzymes:

I- Aminotransferases:

They include alanine aminotransferase (ALT or SGPT) and aspartate aminotransferase (AST or SGOT). These enzymes are present in the hepatocytes and thus their elevations indicate liver cell damage. ALT is found primarily in the liver whereas AST is found in many other tissues including heart, muscles, and

RBCs. Thus, ALT measurement is more specific for liver diseases than AST. In hepatocytes ALT is found exclusively in cytoplasm while AST exists in both cytoplasm and mitochondria. In acute viral hepatitis, there is marked elevation of ALT and AST

(10-100 times). The rise of ALT is usually more than AST (ALT/AST ratio>1) because more damage of cytoplasm and slower clearance of ALT. Serum AST has a shorter half- life (17 hour) than ALT (47 hour). Hence, the serum ALT level takes a longer time to normalize than the AST after the injury has subsided. In liver cirrhosis, there is mild to moderate elevation of ALT and AST (2-5 folds). The rise of AST is more than ALT

(ALT/AST ratio is <1) because the damage affecting the mitochondria of liver cells with release of more AST. In chronic active hepatitis, there is also mild to moderate elevation of ALT and AST with ALT/AST ratio >1.

II-Alkaline phosphatase (ALP) and gamma glutamyl transferase (GGT): These enzymes are found in biliary tract epithelium. In cholestasis (obstructive jaundice), there is marked elevation of these enzymes (more than 3 folds). This marked rise is due to increase enzyme synthesis by cells lining biliary canaliculi. Slight to moderate elevation (1-2-fold) occur in patients with other liver diseases as hepatitis and cirrhosis. GGT is more specific for liver disease than ALP. The latter is also elevated in non-hepatic disorders as bone diseases. GGT also provides a more sensitive index of cholestasis than ALP. Elevated levels of GGT are also seen in alcoholic even with normal liver.

3-Bilirubin:

Bilirubin is the end product of heme degradation. Heme oxygenase oxidizes heme to biliverdin, and biliverdin reductase reduces biliverdin to bilirubin.

Hepatocellular processing of bilirubin involves: Carrier mediated uptake by hepatic cells, Conjugation to glucuronic acid and Excretion into bile. The gut bacterial deconjugate most bilirubin to colorless urobilinogen. The urobilinogen is largely excreted in feces as stercobilinogen and stercobilin.10-20% of urobilinogen are reabsorbed in the ileum and colon and escape enterohepatic circulation and pass to the kidney to be excreted in urine.

Unconjugated (indirect) bilirubin is insoluble in water and is tightly complexed to serum albumin and not excreted in urine. Conjugated bilirubin(direct) is water soluble and only loosely bound to albumin and excreted in urine. Jaundice occurs when the equilibrium between production and clearance of bilirubin is disturbed

A-Unconjugated hyperbilirubinemia (unconjugated bilirubin is usually > 80%): it is due to:

I-Overproduction as in hemolytic anemia (hemolytic jaundice)

II-Decrease uptake as in Gilbert syndrome. This syndrome is relatively common and has a benign course. It results from defect of hepatic uptake and/or decrease of conjugation. Bilirubin level is increased but rarely exceeds 5 mg/dl. Liver enzymes are normal, with no evidence of hemolysis (normal reticulocyte count, no hemolytic anemia...).

III- Decrease conjugation as in Physiologic jaundice of newborn and Crigler-Najjar syndrome type. The severe form of this syndrome is characterized by complete absence of UDGT (uridine diphosphate glucuronosyl transferase) which is the enzyme responsible for bilirubin conjugation. It might be fatal within 18 months of birth. Total Bilirubin is more than 20 mg/dl.

Laboratory findings of hemolytic jaundice:

- Increase bilirubin (mainly indirect)
- Increase urobilinogen in urine; Urine is normal in color (absent bilirubin)
- Stool is dark (increase stercobilin),

- Normal liver enzymes or only slight ↑of AST (present in RBCs)
- Evidence of hemolysis: as ↑reticulocytes

B-Conjugated hyperbilirubinemia (Conjugated bilirubin is usually > 50%):

It is due to:

A. **Impaired secretion** (Defect of bilirubin transfer into bile as in Dubin Johnson syndrome)

B.Cholestasis:

due to intrahepatic or extrahepatic obstruction.

Intrahepatic due to hepatic malignancy, primary biliary cirrhosis or other causes

Extrahepatic due to gallstones, tumors or other causes

Laboratory findings of obstructive jaundice(cholestasis):

- Increase bilirubin (mainly direct).
- Marked rise of GGT and ALP
- Normal ALT and AST except in late stages with secondary liver cell damage from bile
- Urine is dark (positive for bilirubin)
- Urobilinogen is decreased or absent
- Stool is clay color
- prothrombin time (PT) is prolonged
- Increase cholesterol.

C- Conjugated and unconjugated hyperbilirubinemia (hepatic jaundice): it is due to hepatic diseases as acute viral hepatitis. In hepatic jaundice both types of bilirubin are increased (direct and indirect). Liver enzymes are increased. Urine is dark due to the presence of direct bilirubin. Urobilinogen increases in early stage and then decrease

4- Prothrombin time (PT):

PT is dependent on normal hepatic synthesis of coagulation factors. Many of these factors are vitamin K dependent (factor II, VII,IX,X). Prolongation of PT in liver disease may be due to impaired synthesis of clotting factors or impaired absorption of vitamin K. In the latter PT is normalize after parenteral injection of vitamin K.

Acute viral hepatitis:

1 pre-icteric stage:

ALT is increased early and is more sensitive and specific than AST.

Bilirubin is still normal.

2- Icteric stage:

Marked elevation (10-100 times) of ALT and AST and ALT/AST ratio is usually >1. There is Increase both direct and indirect bilirubin (few cases are Anicteric, and bilirubin is normal). ALP and GGT showed mild elevations. Serum albumin is

normal. PT is prolonged in severe cases. Urine: is dark brown (Positive for bilirubin and bile salts).

Immunological (serological) Diagnosis of different types of viral hepatitis:

Hepatitis A:

is a single stranded RNA picornavirus. Incubation period is 2-6 week.

Usually benign, self-limited disease. Rarely causes fulminant hepatitis; and never causes chronic hepatitis. Mode of transmission is Fecal/oral. There are two types of Hepatitis A virus (HAV) antibodies:

First:

is IgM, which appears 2 to 4 weeks after infection and is present for 2 to 6 months.

Second:

is IgG, which appears a few weeks after IgM and is present for the rest of the person's life.

The presence of Hepatitis A virus IgM diagnoses recent infection. Testing HAV total antibodies (IgG plus IgM), measures both IgG and IgM antibodies, but does not differentiate between them. A positive HAV-Ab total test (without isolated testing for

IgM) indicates prior HAV exposure but does not differentiate current from previous

HAV infection. A negative HAV-Ab total test means there has been no prior exposure to the virus and the hepatitis A vaccination should be administered

Hepatitis B:

It is double stranded -DNA virus. Incubation period is 2-6 months.

Chronic hepatitis occurs in 5-10% of cases. It has increased risk of hepatocellular carcinoma. Mode of transmission is parenteral (as contaminated blood transfusion or contaminated syringes), sexual or perinatal from infected mother to their newborn.

Hepatitis B serological markers include:

Hepatitis B surface antigen (HBsAg): Is the earliest marker, appears in the late days of incubation period and denotes infectivity.

Hepatitis B surface antibody (HBsAb or anti-HBs): The presence of anti-HBs is generally interpreted as recovery (the acute infection is completely resolved) and immunity from hepatitis B virus infection. Anti-HBs also develops in a person who has been successfully vaccinated against hepatitis B. HBsAb does not develop in those with chronic hepatitis. Those who are immune due to natural infection will have HBsAb and HBcAb. Immunity due to vaccination will have only HBsAb.

Hepatitis B core antibody (HBcAb or anti-HBc):

Appears at the onset of symptoms in acute hepatitis B and persists for life. This is the only positive marker in window period (gap between disappearance of HBsAg and the appearance of HBsAb). Two types are measured in blood:

First the anti-HBc total antibodies test which detects both IgM and IgG antibodies, thus a positive result indicates acute, chronic, or past infection.

Second is IgM antibody to hepatitis B core antigen (HBcAb IgM or IgM anti-HBc)**:** Its positivity indicates recent infection with hepatitis B virus. It is detected within 2 weeks after HBsAg and persists for 3-6 months. Hepatitis B core antigen is not detected in blood. The most important markers for diagnosis of recent infection are HBsAg and

HBcAb IgM.

Hepatitis B envelop antigen (

HBeAg):

it is not found in every case. If positive this denotes rapid viral multiplication and high infectivity

Hepatitis B e antibody (HBeAb): is not found in every case. It appears after disappearance of HBeAg

HBV DNA by PCR:

hepatitis B virus DNA (HBV DNA) is now the most important and reliable marker for monitoring viral replication. The main advantages of PCR are its extreme sensitivity and quantitative PCR assays help to monitor viral replication and response to treatment. It can diagnose some cases of early acute HBV infection (before the appearance of HBsAg)

Hepatitis C:

is a single stranded -RNA virus. More than 50% of cases develop chronic hepatitis. Incubation period varies from 1-6 month. Mode of transmission is parenteral. There is increased risk of hepatocellular carcinoma. Diagnosis of hepatitis C is by detection of Hepatitis C Antibody (HCV-Ab) which is

usually positive within 2-5 months after infection. For early diagnosis, Hepatitis C RNA by PCR is done. It may be positive 1-2 weeks after infection. Quantitative PCR has a higher value for monitoring of therapy.

Hepatitis D (Delta virus):

It is RNA defective or incomplete virus. It cannot replicate without the presence of HBV. It is diagnosed by detection of HDV IgM antibody or by PCR. Transmission is parenteral or sexual. Incubation period varies from 4-8 weeks. Chronicity and hepatocellular carcinoma could occur.

Hepatitis E:

it is a single stranded RNA virus. Mode of transmission is fecal/oral. It never causes chronicity. Incubation period varies from 2-8 weeks. It has a high mortality rate among pregnant women reaching 20%. Diagnosis of recent infection is by detection of HEV-IgM antibody or by PCR.

Hepatitis G:

It is a single stranded RNA virus. Mode of transmission is parenteral.

Most cases recovered completely, and chronicity is very rare or absent

Laboratory findings of liver cirrhosis:

- Albumin is low and gamma globulin is high with reversed A/G ratio. Total protein is variable
- Mild to moderate elevation of AST and ALT (AST>ALT).
- ALP is normal or mildly elevated.
- GGT is mildly elevated (especially alcoholic).
- Bilirubin is usually normal or elevated in late stages.
- PT is prolonged ant not corrected by Vit K
- Decreased platelets count (due to hypersplenism and or decreased thrombopoietin production from the liver). The latter is a glycoprotein hormone produced by the liver and kidney, which stimulates platelets production from megakaryocytes in the bone marrow.
- Serum protein electrophoresis (SPE): polyclonal increase of gamma globulin with beta gamma bridge (due to increase of IgA mainly). However, nowadays SPE is not commonly used for diagnosis of liver cirrhosis.
- Mild Increases in α-fetoprotein (AFP) concentrations are common in cirrhotic and chronic hepatitis patients, even in the absence of hepatocellular carcinoma (HCC). AFP is the most common tumor marker used for screening of HCC and is usually present at higher concentrations in HCC.

Chronic hepatitis:

Definition: chronic inflammation of liver cells for at least 6 months

It is commonly due to viral infection (HBV or HCV or HBV +HDV) or autoimmune or drug induced. Clinical and laboratory features of chronic hepatitis are also observed occasionally in patients with such hereditary/metabolic disorders as Wilson disease

(copper overload).

Laboratory findings of chronic hepatitis

- Albumin is low and gamma globulin is high (IgG mainly).
- Mild to moderate elevation of ALT and AST (ALT>AST)
- Positive HBsAg, IgG anti-HBc and HBV DNA in chronic hepatis caused by HBV. In some cases, HBeAg may be also positive.
- Positive HCV-Ab and HCV RNA in chronic hepatitis caused by HCV
- Auto antibodies may be found particularly in autoimmune type (ANA, ASMA, LKM). In chronic hepatitis due to viral infection, autoantibodies are uncommon or found at low titer
- Antinuclear antibody (ANA) and Anti smooth muscle antibody (ASMA) are commonly found in type 1 autoimmune hepatitis (often seen in adult particularly young women. It is the most common type)
- Anti-liver kidney microsome antibody type 1 (LKM 1) and anti-liver cytosol (anti-liver cytosol 1) are commonly found in type 2 autoimmune hepatitis (often affects pediatrics from 2-18 years. It is less common than type 1)

Diagnosis of Wilson disease:

- Slit-lamp examination of the eyes for Kayser-Fleischer rings
- Blood tests to measure levels of ceruloplasmin (a copper-carrying protein). Serum ceruloplasmin is usually low in Wilson disease.
- Measurement of copper excreted in the urine: In Wilson disease, 24-hour urinary copper excretion (normally, ≤ 30 mcg/day) is increased and usually > 100 mcg/day. If serum ceruloplasmin is low and urinary copper excretion is high, diagnosis is clear
- Liver biopsy: If the diagnosis is still unclear (indeterminate values for ceruloplasmin and urinary copper), the diagnosis is made by doing a liver biopsy to measure hepatic copper concentration. A diagnosis of Wilson disease is established if the hepatic copper concentration is ≥250 mcg/g dry weight.
- Serum copper levels are sometimes measured; however, they may be variable. In general, Total serum copper (which includes copper incorporated in caeruloplasmin) in Wilson disease is usually decreased in proportion to the decreased caeruloplasmin in the circulation but serum free non-ceruloplasmin bound copper concentration is elevated.

CHAPTER 19

ACUTE HEPATITIS

19

Acute damage of the liver cells can be caused by several agents:

- Viral infections : Virus A, B , D , C , E , Epstein- Barr virus , Cytomegalovirus.
- Non viral infection : Toxoplasmosis – Leptospirosis – Q fever.
- Drugs : e. g .Paracetamol.
- Alcohol.
- Poisons : e.g. Amanita phalloides (mushrooms).
- Other : Pregnancy , circulatory insufficiency , Wilson's disease .

Acute Viral Hepatitis

Acute viral hepatitis is the most common cause of liver disease in the world. With its sequalae, Viral hepatitis is responsible for 1-2 million deaths annually.

Etiology:

- Enterically transmitted viruses: HAV, HEV. - Blood-borne viruses: HBV, HDV and HCV.

Hepatitis A virus (HAV):

Hepatitis A is the most common viral hepatitis occurring worldwide, often in epidemics. The disease is commonly seen in the autumn and affects children and young adults.

Spread of infection is mainly by the feco-oral route and arises from the ingestion of contaminated food.

HAV is a picornavirus, It replicates in the liver, and is excreted in bile and feces for about 2 weeks before the onset of clinical illness and for up to 7 days after. The disease is maximally infectious just before the onset of jaundice.

Investigations :

Liver biochemistry : in the prodromal stage: the serum bilirubin is usually normal, however, there is bilirubinuria and increased urinary urobilinogen. A raised serum AST or ALT, which can sometimes be very high, precedes the jaundice. In the icteric stage, the serum bilirubin reflects the level of jaundice.

Haematological tests :There is leucopenia with a relative lymphocytosis. The prothrombin time (PT) is prolonged in severe cases. The erythrocyte sedimentation rate (ESR) is raised.

. Viral markers:

see Figure 19.1. An anti-HAV IgM means an acute infection , HAV IgG antibodies are common in the general population over the age of 50 years and has no role in diagnosis of acute infection , also in areas of high prevalence most children have antibodies following asymptomatic infection.

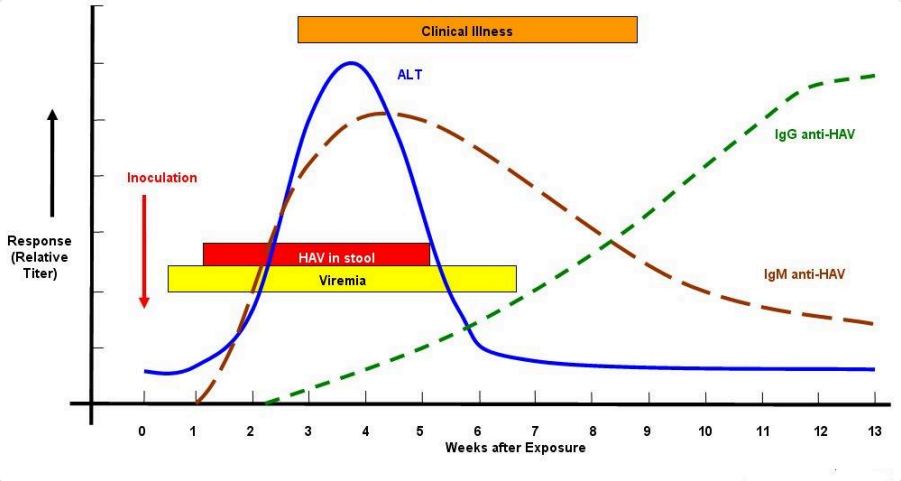


Figure 19.1 — Time course of the events and serological changes of acute Hepatitis A infection

Other tests : Further tests are not necessary in the presence of an IgM antibody, but liver biochemistry must be followed to establish a return to normal levels.

Differential diagnosis:

is from all other causes of jaundice, but in particular from other types of viral and drug-induced hepatitis.

Course and prognosis : The prognosis is excellent, with most patients making a complete recovery. The mortality in young adults is 0.1 % but it increases with age.

Death is due to fulminant hepatic failure . During convalescence, 5-15% of patients may have relapse of the hepatitis but this settles spontaneously. Occasionally, a more severe jaundice with cholestasis will run a prolonged course and is called `cholestatic viral hepatitis'. Treatment is by reassurance. HAV hepatitis never progresses to chronic liver disease.

Treatment :

There is no specific treatment, and rest and dietary measures are unhelpful. Corticosteroids have no benefit. Admission

Prevention : Hygiene (e.g., hand washing) and sanitation (e.g., clean water sources) Boiling, chlorination of water

Vaccination -

Active immunization:

HAV vaccine (inactivated vaccine) , it is effective& safe, the protection starts after 15 days and lasts for more than 20 years. It is given in two doses followed by a booster at 6 or12 months. It is recommended for travelers to endemic areas, patients with CLD and close contacts of a patient.

Post-exposure: Theoretically immunoglobulins can be given to contacts of patients with acute HAV infections

Hepatitis E virus (HEV) :

HEV is an RNA virus causing hepatitis clinically very similar to hepatitis A. It is entericaly transmitted, usually by contaminated water. Epidemics have been seen in many developing countries and sporadically in developed countries, in patients who have had contacts with farm animals or travel abroad. It has mortality from fulminant hepatic failure of 1-2%, which rises to 20% in pregnant women especially in the second and third trimester. There is no carrier state and it does not progress to chronic liver disease except in some immunosuppressed patients in solid -organ transplant recipients. An ELISA for IgG and IgM anti-HEV is available for diagnosis. HEV RNA can

be detected in the serum or stools by PCR. Prevention and control depend on good sanitation and hygiene; a vaccine has been developed.

Hepatitis B Virus (HBV)

- It is a DNA, hepadna virus (see Figure 19.2) , there are 8 genotypes (A to H).
- It has an outer envelope: HBsAg.
- The HBV core contains: -HB core antigen (HBcAg) -HBe antigen (HBeAg) -Partially double-stranded DNA -DNA polymerase/reverse transcriptase enzyme.

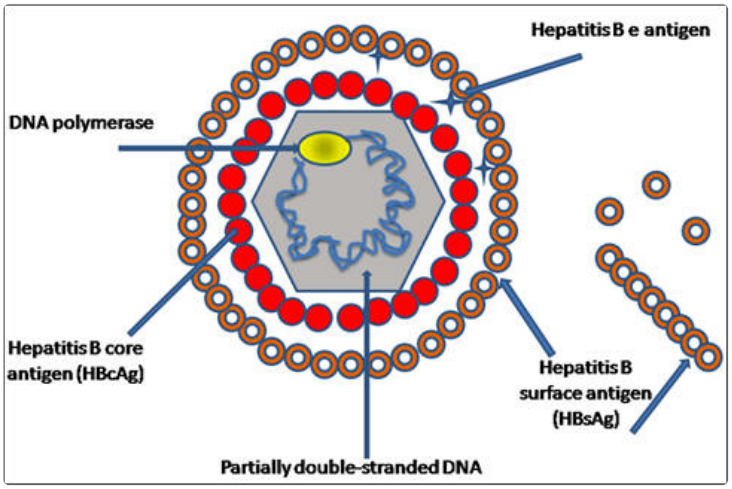


Figure 19.2 — Hepatitis B Virus

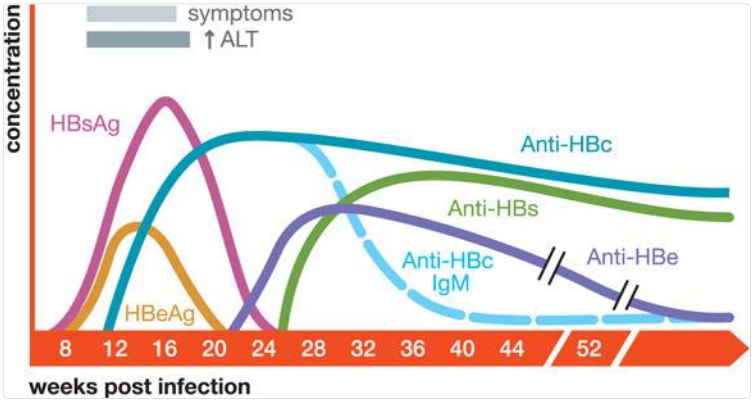


Figure 19.3 — Time course of the events and serological changes of Hepatitis B infection

- The transmission is by the parenteral, sexual and vertical route (at time of birth & close contact afterwards). Interfamily spread can also occurs by household articles .
- Incubation period: 2-6 months. The clinical attack may be mild or severe.

Serological diagnosis in the acute stage depends on IgM anti- HBV , HBVDNA and HBsAg (Figure 19.3) .After the attack, persistent infection may occur and can result in chronicity and cirrhosis and hepatocellular carcinoma.

Most patients require only supportive care, however antiviral therapy is recommended for severe acute hepatitis B, especially if there is

- INR >1.5
- prolonged jaundice
- acute liver failure
- severe coagulopathy

Preferred drugs are

- Tenofovir and Entecavir

Prophylaxis: hepatitis B immunoglobulin and vaccination.

Hepatitis D virus (HDV) : It is an incomplete RNA virus, it is unable to replicate on its own but is activated in the presence of HBV .

Mode of transmission: blood- born, sexual, vertical.

Incubation period: 4-7 weeks. HDV infection is declining due to HBV vaccination. It is endemic in Mediterranean basin.

Infection occurs as:

1- Co-infection with HBV at the same time.

2- Super infection of a chronic HBV carrier. Synthesis of HDV depresses the appearance of HBV viral markers. Prophylaxis: vaccination against hepatitis B.

Hepatitis C virus (HCV):

- HCV is an important cause of chronic liver disease worldwide, it was highly prevalent in Egypt .
- HCV is a single-stranded RNA of the flaviviridae family ,the RNA genome is encoding a polyprotein product consisting of structural (capsid and envelope) and non structural viral protein .(Figure 19.4)

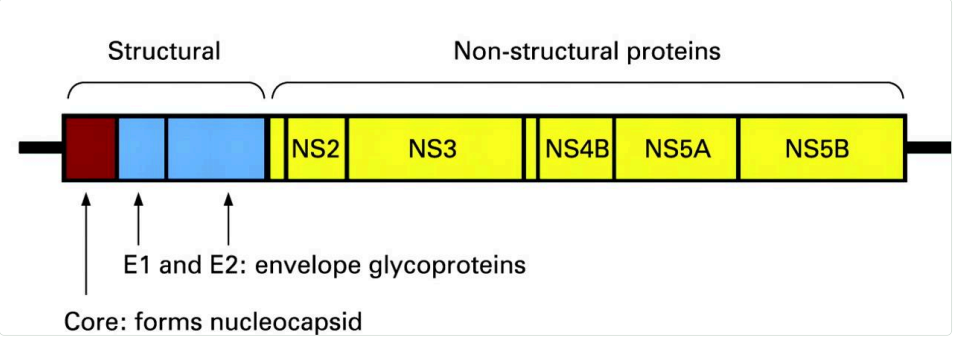


Figure 19.4 — HCV genome

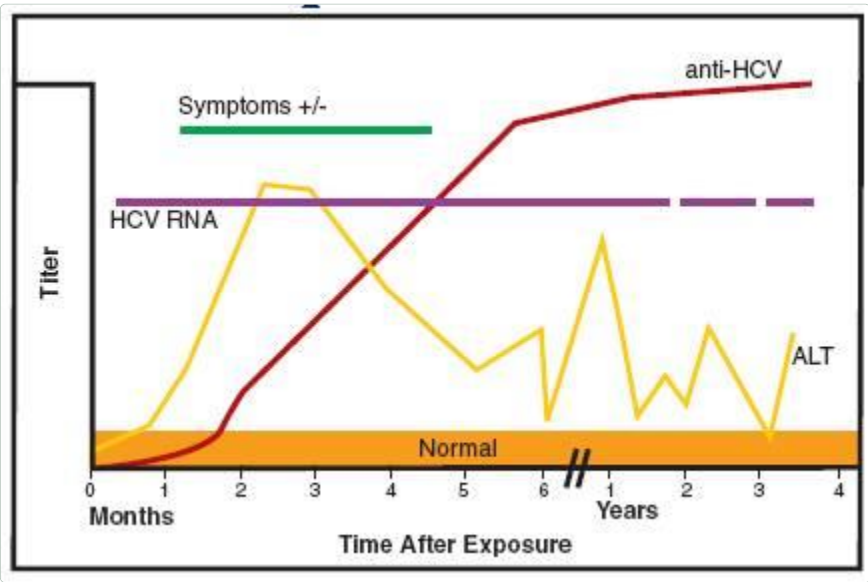


Figure 19.5 — Serology and Virology of Acute and Chronic HCV

- HCV has at least 8 major genotypes. Genotype 4 is the most prevalent in Egypt .
  Incubation period: 1-6 months.

Transmission:

mainly parenteral. There is a low frequency of sexual and vertical transmission

Age:

usually above 16 years.

Clinical attack:

Usually mild and sub-clinical . The elevation of transaminases and the appearance of HCVRNA precede the appearance of HCV antibodies (Figure 19.5) .

Persistent infection is common (about 80%). Infection is linked with chronic hepatitis, cirrhosis and HCC.

- Prophylaxis: Prevention of nosocomial infection and supervision of blood donation

General clinical Picture of viral hepatitis

1- Anicteric Hepatitis : Mild attack with no jaundice , it may presents as gastroenteritis and is usually underdiagnosed

2- Icteric hepatitis:

Preicteric stage: It may lasts for few days. Patient feels generally unwell with marked malaise. Anorexia & nausea Pain may occurs in right hypochondrium associated with fever. This is followed by darkening of urine & light stool. In hepatitis B serum sickness-like syndrome occurs in less than 10% of cases with manifestations of fever, maculopapular rash and polyarthritis of the small joints. Icteric stage: 2-4 weeks, with the appearance of jaundice, symptoms decrease: fever decreases, appetite improves and vomiting stops. Persistent vomiting and/ or drowsiness or confusion indicates urgent hospital a as they may reflect liver failure. Liver is enlarged in 70% of patients, soft & tender, splenomegaly and lymphadenopathy may develop in some patients .

Convalescence stage: Recovery is rapid in children. Fatigue persists for some weeks. Clinical & biochemical recovery is usual within 6 months

Sequela of an acute attack of viral hepatitis:

1-Complete recovery is the rule in most HAV& HEV infections, also in more than 90% of HBV inections in afdults . Only 20% of HCV patients recover spontaneously. 2- Relapse: - Especially in HAV infections. The attack is usually milder attack, and is marked by elevation of transaminases and serum bilirubin. Recovery usually follows.

3-Prolonged cholestasis: commonly in HAV infection, the jaundice persists for several months. Pruritus is prominent. Abdominal U/S is usually performed to exclude surgical obstructive jaundice. Complete recovery is the rule .

4-Acute liver failure: Jaundice is deep, drowsiness, confusion , flapping tremors develop and coagulopathy.

5-Persistent infection: - HBV carrier rate depends on age (see later).

- HDV super-infection causes chronic hepatitis in about 70%.

- HCV infections persist in about 80%.

TABLE: SOME OF THE CHARACTERISTICS OF HEPATITIS VIRUSES

	HAV	HBV	HCV	HDV	HEV
Genome	RNA	DNA	RNA	RNA	RNA
Incubation period	2-6 w	2-6M	1-6 M	4-7 w	2-6 w
Transmission	Feco - oral	Blood- sexual – vertical	Blood	Blood – sexual	Feco- oral
Serologic diagnosis	IgM anti HAV	IgM anti HBV HBVDNA HBsAg	Anti HCV HCVRNA	Ig M anti HDV	IgM anti HEV
Prevention	Vaccine	Vaccine	-	Vaccine against HBV	-
Chronicity	-	+	+	+	It does not progress to chronic liver disease except in some immunosuppressed patients in solid - organ transplant recipients
Treatment	Non specific	Non specific + antiviral	Non specific + antiviral	Pegylated IFN - Bulevirtide	Non specific
				-	

Pathology:

- All forms of acute viral hepatitis have a basic pathology.
- Acute inflammation of the entire liver
- Centrizonal necrosis & cholestasis are most marked in zone 3.
- Cellular infiltration (mainly lymphocytes) of portal tracts (zone 1) and bile duct proliferation.
- The reticulin framework is usually preserved & liver cells regenerate from zone 1.
- Other organs: There may be regional lymph nodes enlargement. Splenomegaly and hypoplasia in bone marrow

Investigations:

1- Liver function tests: Marked elevation of ALT, AST (500-5000U/L). Serum bilirubin is increased (both direct and indirect). Alkaline phosphatase and prothrombine time are normal or mildly increased. Albumin is normal. ɣ globulins are increased.

2-Blood changes: - Leucopenia with relative lymphocytosis.- ESR is increased.

3-Urine and stool : Bilirubin appears early (dark urine). - Urobilinogen is variable.

the Stools appears pale due to decreased sterchobilinogen.

5- Viral markers:

- **HAV:**
  HAV IgM: indicates recent infection and is positive for 3-6 months thereafter.
  HAV IgG: indicates old infection and immunity.
- **HBV:**
  HBsAg: appears 6 weeks after infection & disappear in 3 months. Persistence for more than six months indicates the development of a carrier state.
  Anti-HBs: appear after 3months. It means recovery & immunity.
  HBc IgM: is present in acute phase & in severe chronic hepatitis.
  HBc IgG: Past exposure.
  HBeAg: viral replication and infectivity.
  Anti-HBe: low infectivity.
  HBV-DNA: most sensitive marker of viral replication, measured by PCR.
- **HDV:**
  HDV IgM and HDV RNA in acute infections. HDV IgG in chronic phase.
- **HCV:**
  HCV antibodies become positive after 6-8 weeks of infection. HCVRNA can be detected 1-2 weeks after infection by PCR.
  –

HEV:

HEV IgM: indicates recent infection. HEV IgG: old infection.

Treatment (acute attack):

Normal activity as tolerated, patient should be adequately hydrated and advised for nutritional support and avoidance of hepatotoxic medications

: Traditionally the patient was given a low fat, high CHO diet as it is said to be the most appetizing.

Prevention :

Pre-exposure: HAV vaccine (inactivated vaccine) , (previously discussed)

HEV : Efficiency of immunoglobulins is uncertain. An effective vaccine has been developed.

HBV : Pre-exposure HBV recombinant vaccines (or Plasma derived) are both effective & safe. - They contain HBsAg as the immunogen. For the recombinant vaccine: the dose is lml IM in the arm at 0-1-6 months. Indications: - Surgical, dental, laboratory staff & medical students.

- Patients: dialysis, oncology, CLD, mental patients.
- Neonates of HBsAg mothers.
- Accidental exposure.
- Drug users & homosexuals.

  Post-exposure: both HBV vaccine and HBimmunoglobulin (HBIG) are given in different arms at the same time to: neonates of HBsAg positive -mothers, accidental exposure

  HBsAg +ve blood, to sexual contacts of acute sufferers and to prevent reinfection of a donor liver inserted into an HBV DNA positive patient.

HCV

Work on HCV vaccine is in progress. Screening of blood donors and blood banks for

HCV antibodies.

HERPES VIRUS HEPATITIS

- A large family of enveloped DNA viruses including cytomegalovirus (CMV), Epstein-Barr virus (EBV), herpes simplex virus (HSV).
- Very common infections ,90 to 95% of adults are (+) for EBV, and CMV serology, and up to 80% for HSV

- Unlike hepatitis viruses A, B, C, D and E, herpes viruses are not hepatotropic, and do not replicate in liver cells but can cause hepatitis by proliferation of infected cells and release of cytokines. They do not cause chronic hepatitis or cirrhosis. EBV and CMV hepatitis are very common and usually mild, subclinical and self-limited. Herpesvirus hepatitis can present as a primary infection or reactivation of latent infection

  EBV and CMV transmission : Person to person transmission, uses of contaminated objects .Blood transfusion, and organ transplantations have also been reported as rare modes of transmission.

  EBV symptoms : Non-specific: malaise, anorexia, right sided abdominal pain ,characteristic: Infectious mononucleosis (IM) syndrome - sore throat with exudative pharyngitis, enlarged lymph nodes. EBV is especially associated with left sided abdominal pain with splenomegaly.

  Diagnosis by Serology : eg viral capsid antibody (VCA IgM) - Heterophil Ab (Specific but not very sensitive) EBV DNA by PCR .Lymphocytosis, atypical lymphocytes (not specific)

  CMV Symptoms : similar to EBV (infectious mononucleosis), sore throat, enlarged lymph nodes. Exudative pharyngitis and splenomegaly are less common than with EBV. Diagnosis by serology , IgM Ab to envelope (+) ,Heterophile (-) ,CMV DNA by PCR ,Lymphocytosis, atypical lymphocytes – not specific

  EBV and CMV biochemistry : usually only mildly elevated ALT and usually ALT >AST. Elevated alkaline phosphatase in 60%, hyperbilirubinemia in 45%, but jaundice in usually <5% of cases .The laboratory abnormalities typically normalize in a few weeks, but can rise again during reactivation. Severe cholestatic hepatitis, granulomatous hepatitis, and rarely, acute fulminant hepatic failure - possible especially in immunocompromised.

  Most cases of EBV, and CMV hepatitis are mild, resolve spontaneously - only supportive treatment is needed. In cases of severe hepatitis, valganciclovir or ganciclovircan be used

  HSV Hepatitis :

  Rare, usually a result of disseminated herpes sepsis. Recipients of organ transplants on immunosuppressive therapy, chemotherapy and persons with immunodeficiency diseases like HIV – high risk.

  Pregnancy, 2nd or 3rd trimester – high risk

- Common cause of fulminant hepatic failure even in immunocompetent patients.

  Herpetic cutaneous, muco-cutaneous, mucosal lesions.(cold sores) patient presents with severe hepatitis - AST, ALT – very high, >10 X ULN .

  Fulminant hepatic failure frequent especially, but not only, in immunocompromised

  Because of potential rapid deterioration and high mortality of HSV hepatitis ,empiric treatment with acyclovir or Foscarnet is recommended as soon as possibe after diagnosis to decrease mortality and the need for liver transplant

CHRONIC HEPATITIS

Chronic hepatitis is a group of liver diseases characterized by persistent hepatic inflammation with hepatocellular injury lasting for more than six months. If left untreated, chronic inflammation may progress to hepatic fibrosis, cirrhosis, portal hypertension, liver failure, and hepatocellular carcinoma, depending on the underlying cause.

Although the causes are diverse, the histopathological features of chronic hepatitis share many common characteristics.

Etiology

- Chronic viral hepatitis
- Hepatitis B virus (HBV), with or without hepatitis D virus (HDV) coinfection
- Hepatitis C virus (HCV)
- Autoimmune hepatitis
- Drug-induced chronic hepatitis
- Methyldopa
- Isoniazid
- Methotrexate
- Nitrofurantoin
- Alcohol-associated steatohepatitis (ASH)
- Metabolic liver diseases
- Metabolic dysfunction-associated steatohepatitis (MASH)

(formerly non-alcoholic steatohepatitis, NASH)

- Wilson disease
- Hereditary hemochromatosis
- Alpha-1 antitrypsin deficiency

With the availability of highly effective direct-acting antiviral therapy, chronic hepatitis C has become curable in most patients.

Pathology

The characteristic histological findings include varying degrees of portal inflammation, interface hepatitis, lobular inflammation, and fibrosis.

In mild disease:

- Expansion of the portal tracts by inflammatory infiltrates composed mainly of lymphocytes, with variable numbers of plasma cells (particularly prominent in autoimmune hepatitis).

- Interface hepatitis (formerly called piecemeal necrosis) characterized by inflammatory destruction of hepatocytes at the junction between the portal tract and hepatic lobule.
- Mild lobular hepatitis, with scattered foci of hepatocyte necrosis (spotty necrosis) and inflammatory cell infiltration.

In more advanced disease:

- More extensive interface hepatitis.
- Confluent hepatocyte necrosis may occur.
- Progressive deposition of collagen leading to fibrous septa connecting portal tracts and/or central veins (bridging fibrosis).
- Regenerating hepatocytes may form regenerative nodules.
- Continued fibrosis eventually results in cirrhosis.

Histological Staging

Liver biopsy may be graded and staged using the METAVIR scoring system, particularly in chronic viral hepatitis.

- Activity (A) reflects the degree of necroinflammatory activity:
- A0: No activity
- A1: Mild activity
- A2: Moderate activity
- A3: Severe activity
- Fibrosis (F) reflects the stage of fibrosis:
- F0: No fibrosis
- F1: Portal fibrosis without septa
- F2: Portal fibrosis with few septa
- F3: Numerous septa without cirrhosis
- F4: Cirrhosis

Important note: Although liver biopsy remains the reference standard for histological assessment, it is now performed much less frequently than in the past. In current clinical practice, liver fibrosis is commonly assessed using non-invasive methods, particularly transient elastography (FibroScan) and validated serum fibrosis scores such as FIB-4 and the APRI score.

CHAPTER 20

CHRONIC HEPATITIS B

20

Chronic hepatitis B (CHB) is a major global health problem affecting approximately **250–300 million people worldwide** . It remains highly prevalent in many developing countries. In Egypt, the prevalence has declined considerably following the introduction of universal infant vaccination in 1992 and is currently estimated to be **below 2%** in the general population. **HBV genotype D** is the predominant genotype in Egypt.

Natural History

The likelihood of developing chronic HBV infection depends largely on the **age at infection** and the **host immune response** .

Following acute HBV infection:

- **Neonates:** approximately **90%** develop chronic infection.
- **Children <5 years: 20–30%** develop chronic infection.
- **Adults:** only **<5%** develop chronic infection because most adults successfully clear the virus.

Most liver injury results from the **host immune response against HBV-infected hepatocytes** , rather than from a direct cytopathic effect of the virus.

Clinical Features

Many patients remain asymptomatic for years.

Symptoms, when present, may include:

- Fatigue
- Malaise
- Right upper quadrant discomfort
- Hepatomegaly

Advanced disease may present with complications of cirrhosis such as:

- Ascites
- Variceal bleeding
- Hepatic encephalopathy
- Hepatocellular carcinoma (HCC)

Phases of Chronic HBV Infection

Current international guidelines classify chronic HBV infection into **five phases** , replacing the older terms “immune tolerant”, “inactive carrier”, etc.

- **“Chronic Infection”** = Chronic Infection = Characterized by low or stable viral replication and normal ALT, though periodic subclinical inflammation can still occur.”
- **“Chronic Hepatitis”** = Elevated ALT (Active liver inflammation).

Phase Breakdown Summary Table

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Phase	Old Nomenclature	New Nomenclature	HBeAg	HBV DNA	ALT Level
Phase 1	Immune Tolerant	**HBeAg-Positive Chronic Infection**	Positive	Very High (>10^7 IU/mL)	**Normal**
Phase 2	Immune Clearance	**HBeAg-Positive Chronic Hepatitis**	Positive	High (Variable)	**Elevated**
Phase 3	Inactive Carrier	**HBeAg-Negative Chronic Infection**	Negative	Low (<2,000 IU/mL)	**Normal**
Phase 4	Reactivation	**HBeAg-Negative Chronic Hepatitis**	Negative	Moderate (>2,000 IU/mL)	**Elevated**

1. HBeAg-positive chronic HBV infection

(formerly **immune-tolerant phase**)

- HBsAg positive
- HBeAg positive
- Very high HBV DNA levels (usually >10^7 IU/mL)
- Normal ALT
- Minimal hepatic inflammation or fibrosis

This phase is most common in individuals infected at birth or during early childhood.

Most patients are monitored without treatment, although treatment may be considered in selected older patients with significant fibrosis or other risk factors.

2. HBeAg-positive chronic hepatitis

- HBeAg positive
- Elevated ALT
- High HBV DNA
- Active hepatic inflammation
- Progressive fibrosis may occur

These patients are usually candidates for antiviral therapy.

3. HBeAg-negative chronic infection

(formerly **inactive carrier state**)

- HBsAg positive
- Anti-HBe positive
- HBV DNA usually **<2,000 IU/mL**
- Persistently normal ALT
- No significant liver inflammation

Most patients remain stable, although lifelong follow-up is necessary because reactivation may occur, especially during immunosuppressive therapy.

Spontaneous HBsAg clearance occurs in approximately **1% per year** .

4. HBeAg-negative chronic hepatitis

- HBeAg negative
- Anti-HBe positive
- HBV DNA usually >2,000 IU/mL
- Elevated ALT
- Active hepatic inflammation

This phase usually results from precore or basal core promoter mutations preventing HBeAg production.

5. HBsAg-negative phase

(Occult HBV infection)

is defined by undetectable HBsAg, positive anti-HBc, and detectable HBV DNA in the liver or blood.

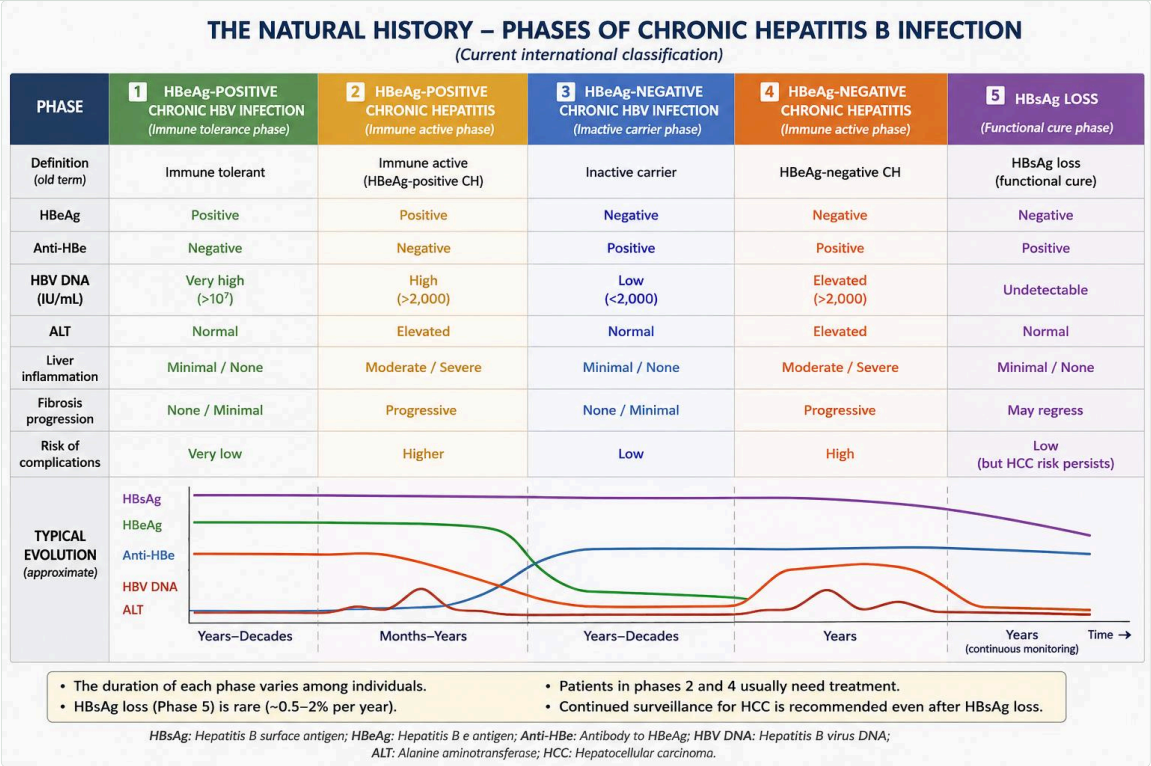


Figure 20.1 — Natural history and phases of chronic hepatitis B infection

Reactivation may occur during chemotherapy or immunosuppressive therapy.

Investigations

Diagnosis is established by detection of **HBsAg** for more than **6 months** .

Routine investigations include:

- Liver function tests (ALT, AST, bilirubin, albumin)
- Complete blood count
- Prothrombin time/INR
- HBeAg and anti-HBe
- Quantitative HBV DNA by PCR
- Abdominal ultrasound
- Serum alpha-fetoprotein (AFP) for HCC surveillance when indicated
- *Abdominal ultrasound every 6 months, with or without serum alphafetoprotein (AFP), for hepatocellular carcinoma (HCC) surveillance in high-risk patients."*

Assessment of Fibrosis

Transient elastography (FibroScan) has largely replaced liver biopsy for fibrosis assessment.

Liver biopsy is now reserved for selected patients in whom the diagnosis or fibrosis stage remains uncertain.

Typical histological findings include:

- Chronic hepatitis

- Interface hepatitis
- Fibrosis
- **Ground-glass hepatocytes** , "Histology may reveal ground-glass hepatocytes, which reflect the accumulation of HBsAg within the cytoplasm."

Indications for Treatment

Treatment decisions are based on several factors:

- HBV DNA level
- ALT level
- Severity of liver fibrosis
- Presence of cirrhosis
- Age
- Family history of hepatocellular carcinoma
- Extrahepatic manifestations
- Planned immunosuppressive therapy

Treatment is recommended in:

- Patients with chronic hepatitis and:
  - HBV DNA >2,000 IU/mL
  - Elevated ALT
  - Significant fibrosis
- **All patients with compensated or decompensated cirrhosis** and detectable HBV DNA, regardless of ALT level.
- Selected patients with normal ALT but significant fibrosis or a strong family history of HCC.

Goals of Therapy

The aims of treatment are to:

- Suppress HBV replication
- Prevent progression to cirrhosis
- Prevent hepatic decompensation
- Reduce the risk of hepatocellular carcinoma
- Improve survival

The ideal endpoint is **loss of HBsAg** , with or without development of anti-HBs antibodies (functional cure). However, this is achieved in only a small proportion of patients.

In HBeAg-positive patients, **HBeAg seroconversion** is an important treatment milestone.

Antiviral Therapy

First-line Oral Antiviral Agents

Tenofovir disoproxil fumarate (TDF)

- 300 mg orally once daily
- Highly potent
- Extremely low resistance

Long-term therapy requires monitoring of:

- Renal function
- Serum phosphate
- Bone mineral density in at-risk patients

Tenofovir alafenamide (TAF)

- 25 mg orally once daily

TAF has similar antiviral efficacy to TDF but causes less renal and bone toxicity, making it preferable in patients with chronic kidney disease or osteoporosis.

Entecavir

- 0.5 mg once daily
- 1 mg daily in patients with previous lamivudine resistance

Highly effective with a very low resistance rate in nucleos(t)ide-naïve patients.

Drugs No Longer Recommended as First-Line Therapy

> The following agents are no longer recommended because of high resistance rates:

- Lamivudine ,Adefovir Telbivudine

They have largely been replaced by tenofovir or entecavir.

Duration of Therapy

Treatment with nucleos(t)ide analogues is usually **long-term** , and many patients require lifelong therapy, especially those with cirrhosis.

Treatment discontinuation may be considered in carefully selected non-cirrhotic patients under close specialist follow-up.

Pegylated interferon-α2a:

- **180 µg subcutaneously once weekly**
- Duration: **48 weeks**

Advantages:

- Finite treatment duration
- Higher likelihood of HBsAg loss than oral therapy

Best candidates include:

- Younger patients
- Elevated ALT
- Lower HBV DNA levels
- Genotypes A and B respond better than genotype D

Common adverse effects include:

- Influenza-like symptoms
- Depression
- Bone marrow suppression
- Thyroid dysfunction
- HBV genotype D is the predominant genotype in Egypt. Because Genotype D responds poorly to interferon, so oral nucleos(t)ide analogues (TDF/TAF/Entecavir) are the preferred first-line choice in the Egyptian clinical context.

Hepatitis D (HDV)

Patients with HBV should be screened for HDV when clinically indicated.

Treatment options include:

- **Pegylated interferon-α**
- **Bulevirtide** , the first approved entry inhibitor for chronic hepatitis D, which has significantly improved treatment outcomes in patients with compensated liver disease.

CHRONIC HEPATITIS C

Clinical features

Acute HCV infection is usually asymptomatic. Approximately 20–30% of infected individuals clear the virus spontaneously, whereas 55–85% develop chronic infection.

- Most acute infections are asymptomatic.
- Patients are identified years later and usually accidentally (elevated liver enzymes or with positive HCV antibody).
- HCV Infection has the propensity to become a chronic hepatitis in 80% of cases.
- Patients may be asymptomatic or present with fatigue or upper abdominal pain; they may also present with manifestations of liver cirrhosis and its complications. There is a long period between infection and cirrhosis (usually 20 years). Regression is usually slow. Cirrhosis develops in approximately 15–30% of untreated patients over 20–30 years, depending on age, alcohol consumption, obesity, diabetes and other cofactors.

- Hepatocellular carcinoma occurs in about 15% of cirrhotic patients.
- Extrahepatic manifestations may be due to deposition of immune complexes:
  - Mixed cryoglobulinemia (most common)
  - Membranoproliferative glomerulonephritis
  - Porphyria cutanea tarda
  - B-cell non-Hodgkin lymphoma
  - Vasculitis
  - Insulin resistance and type 2 diabetes mellitus

Diagnosis

HCV antibody alone is NOT diagnostic of active infection.

Current approach:

1. Anti-HCV antibody
2. HCV RNA PCR

Quantitative HCV RNA before treatment

4. Fibrosis assessment

Liver biopsy is now rarely required because liver fibrosis can usually be assessed non-invasively using transient elastography (FibroScan) and serum fibrosis scores (FIB-4 or APRI).

-Viral markers: HCV Ab by ELISA to indicate infection or exposure.

- HCV RNA by PCR to indicate viral replication.
- Liver biopsy is variable: and may show minor pathological changes or show cirrhotic changes . Fatty changes and lymphoid follicles in the portal tracts are frequently present.
- ALT may be normal/ elevated or fluctuating.

Treatment

Direct-acting antiviral agents (DAAs) have revolutionized the treatment of chronic hepatitis C. They are orally administered, well tolerated, and achieve sustained virological response (SVR) rates exceeding **95%** in most patients.

Current first-line pangenotypic regimen :

Sofosbuvir/Velpatasvir for 12 weeks

This regimens is effective against all HCV genotypes.

Earlier combinations such as Sofosbuvir/Daclatasvir and Sofosbuvir/Ledipasvir played an important role in the initial DAA era but have largely been replaced by newer pangenotypic regimens in many countries.

Ribavirin

Ribavirin is now used only in selected difficult-to-treat patients such as those with decompensated cirrhosis or previous DAA failure.

Pegylated interferon

Pegylated interferon is no longer recommended for routine treatment of chronic hepatitis C because highly effective interferon-free DAA regimens are available..

Cure

Sustained virological response (SVR12) is defined as undetectable HCV RNA **12 weeks after completion of therapy** and is considered a virological cure.

Vaccination

Patients with chronic HCV should be vaccinated against hepatitis A and hepatitis B if they are not already immune.

Egypt and HCV

Egypt previously had the highest prevalence of hepatitis C worldwide, predominantly genotype 4. Much of this epidemic resulted from the use of inadequately sterilized needles during the national parenteral antischistosomal therapy campaigns conducted between the 1950s and 1980s.

Beginning in 2014, Egypt implemented one of the world's largest hepatitis C elimination programs using locally produced direct-acting antivirals.

Starting in 2018–2019, Egypt launched the nationwide **"100 Million Healthy Lives"** initiative, which included large-scale screening of adults for HCV infection. Individuals diagnosed with HCV were offered highly effective directacting antiviral (DAA) therapy free of charge through the national program. To date, **approximately 60 million people have been screened for HCV** , making Egypt's program one of the largest population-based HCV screening and treatment initiatives worldwide

As a result, Egypt became the **first country to achieve the WHO Gold Tier status on the path toward hepatitis C elimination** , with a dramatic reduction in the national prevalence of active HCV infection.

CHAPTER 21

Autoimmune hepatitis

21

It is a an immune-mediated chronic inflammatory liver disease of unknown exact etiology occurring in genetically susceptible individuals. It is characterized the presence of circulating autoantibodies, increased gammaglobulins and interface hepatitis.

Near 50% increase in incidence over the last few decades

- Occurs at all ages and within all ethnic groups around the world

Types of autoimmune hepatitis (based on serological markers):

Type 1: Characterized by detection of serum anti-nuclear antibody (ANA), anti- smooth muscle antibody (ASMA). It accounts for approximately 80–90% of cases

Type 2: Characterised by detection of serum anti- liver- kidney microsomal antibody(LKM-1), anti-liver cytosole antibody-1(LC I)

Clinical Picture and diagnosis:

Autoimmune hepatitis (AIH) usually occurs in females (70%) mostly between 15- 25 years or at menopause.

- Broad range from asymptomatic to acute/severe disease
- Diagnosis of AIH relies particularly on the presence of autoantibodies, hypergammaglobulinaemia and typical or compatible histology
- Elevated IgG levels, especially in the absence of cirrhosis, are a distinctive feature of AIH
- AIH should be considered in any patient with acute or chronic liver disease particularly if hypergammaglobulinaemia is present and if the patient has features of other autoimmune diseases

The diagnosis is based upon

- Autoantibodies
- Elevated IgG
- Typical biopsy
- Exclusion of viral hepatitis

Typical symptoms are fatigue, joint pains, hepatomegaly and jaundice.

Associated autoimmune manifestations: polyarthritis, hemolytic anemia, purpura, DM, autoimmune thyroiditis.

-γ globulins are increased mostly IgG (>1.5 folds ULN).

- AST & ALT are elevated. Alkaline phosphatase is not markedly increased
- Serum auto antibodies: ANA, SMA, LKM-1, LC-1.
- Liver biopsy: interface hepatitis with a lymphoplasmacytic infiltrate rich in plasma cells is the characteristic histological finding., no biliary lesions..

- Other causes of liver disease must be excluded : viral hepatitis, drug induced liver diseases (DILI) ,Wilson's disease, consumption of alcohol, ,NASH , alpha- 1 antitrypsin deficiency, hepatotoxic drugs, PBC, PSC, genetic hemochromatosis must be excluded .

Treatment :

Aims:

-To obtain complete remission of the disease and to prevent its further progression , this requires complete normalization of aminotransferases and IgG levels . Generally the treatment requires permanent maintenance therapy

Absolute treatment indications :

- **Clinical:**
  - Incapacitating symptoms
  - Progression of disease
- **Laboratory**
  - AST ≥ 10 x ULN
  - AST ≥ 5 x ULN + GG ≥ 2 x ULN
- **Histology**
  - Bridging necrosis
  - Multilobular necrosis

* Corticosteroids improve symptoms, signs, LFTs & histology. Dose: prednisone 0.5 – 1 mg/ Kg Bwt or 20-40 mg / day for 2 weeks then reduce to a maintenance dose for 4weeks in combination with azathioprine (1-2 mg /Kg Bwt or 50-150mg/day) . Budesonide (9 mg/day) may be used instead of prednisone in selected non-cirrhotic patients because it causes fewer systemic adverse effects

Azathioprine , cyclosporine and other immunosuppressive drugs have been tried alone or with steroids as steroid sparing agent

Disease remission :

Disappearance of symptoms GGT and bilirubin: normal

65% achieve remission within 18 months, and 80% within 3 yrs of initiation of Rx

10 year survival: 90%

Liver transplantation: is another option , however the condition may recur.

Treament endpoint : Instead of AST <2×ULN

Current guidelines recommend:

Complete biochemical remission is defined by normal ALT, AST and IgG levels.

Second-line therapy

Patients intolerant of azathioprine or with inadequate response may receive:

- Mycophenolate mofetil
- Tacrolimus
- Cyclosporine
- Other immunosuppressive agents in specialized centers

Prognosis

Untreated autoimmune hepatitis has a poor prognosis, whereas appropriate immunosuppressive therapy results in excellent long-term survival, with a 10-year survival exceeding 80–90%.

Metabolic dysfunction-associated steatotic liver disease (MASLD) and metabolic dysfunction-associated steatohepatitis (MASH)

Metabolic dysfunction-associated steatotic liver disease (MASLD) is defined as hepatic steatosis in individuals with at least one cardiometabolic risk factor and without another predominant cause of hepatic steatosis such as significant alcohol intake or chronic liver diseases.

Metabolic dysfunction-associated steatohepatitis (MASH) is characterized by hepatic steatosis associated with hepatocyte injury (ballooning) and lobular inflammation, with or without fibrosis

MASLD and MASH are the current internationally accepted terms that replaced NAFLD (Non alcoholic fatty liver disease) and NASH (Nonalcoholic steatohepatitis) in 2023.

- Steatosis: accumulation of fat in the liver (mainly triglycerides). Simple steatosis usually has a favorable prognosis, although some patients may progress to steatohepatitis and fibrosis. Hepatic steatosis results mainly from insulin resistance, which increases the delivery and synthesis of fatty acids in the liver.

Inflammation and necrosis of liver cells may be caused by the toxicity of excess free fatty acids, oxidant stress, or mitochondrial dysfunction. This may lead finally to fibrosis, cirrhosis and even hepatocellular carcinoma.

MASLD is the most common chronic liver disease worldwide as it is associated with the global epidemic of obesity. **MASLD is strongly associated with metabolic syndrome, which includes:**

- Central obesity

- Type 2 diabetes mellitus
- Insulin resistance
- Hypertension
- Dyslipidemia

Clinical picture

- Typically asymptomatic. Most patients are diagnosed incidentally during ultrasound or routine liver function tests.
- Occurs in both sexes
- Fatigue/ right upper quadrant pain
- Hepatomegaly: ALT, AST: normal or increased. ALT is often greater than AST in early disease.

U/S: Increase hepatic echogenicity (bright liver).

- Assess fasting blood glucose ,HbA1c and a lipid profile..
- -Exclude viral, drug, autoimmune & other causes.

Diagnosis

Fibroscan device Transient elastography measures liver stiffness to estimate fibrosis. It also estimates hepatic steatosis using the Controlled Attenuation Parameter (CAP).

- can distinguish between advanced cirrhosis and no cirrhosis

Liver biopsy - Liver biopsy remains the reference standard for diagnosing MASH and grading fibrosis but is reserved for selected patients because it is invasive.

Non-invasive fibrosis assessment using FIB-4 followed by transient elastography is recommended before considering liver biopsy.

Management:

Lifestyle modification and diet manipulation are the cornerstone of treatment and includes:

- Weight loss of 7–10%
- Mediterranean diet
- Regular aerobic and resistance exercise

Gradual and sustained weight loss of 7–10% of body weight improves steatosis, inflammation, and fibrosis.

Increase vegetables& fruits/ decrease fat. Mediterranean diet

- Rapid weight loss should be avoided because it may worsen hepatic inflammation
- Pioglitazone may be considered in selected patients, particularly those with type 2 diabetes.
- GLP-1 receptor agonists (e.g. semaglutide) improve weight loss and liver steatosis and are increasingly used in obese or diabetic patients.
- New therapies for MASH continue to emerge. Metformin should be used to treat diabetes but is not recommended specifically for treating MASH
- Management of co-morbid factors: DM, hypertension, hyperlipidemia
- Bariatric surgery for morbidly obese patients (BMI>35-40). According to current obesity guidelines after multidisciplinary assessment.

Surveillance : Patients with cirrhosis require

- HCC surveillance
- screening for esophageal varices

Prognosis

Most patients with simple steatosis have a favorable prognosis. Patients with MASH and advanced fibrosis are at increased risk of cirrhosis, liver failure, and hepatocellular carcinoma. Cardiovascular disease remains the leading cause of death in patients with MASLD.

CHAPTER 22

LIVER CIRRHOSIS

22

It is a diffuse process characterized by necrosis of liver cells, followed by fibrosis, regeneration and nodule formation. The liver architecture is diffusely abnormal and this interferes with liver blood flow and function. This finally leads to impaired liver function.

Liver cirrhosis is the end stage of chronic liver disease, characterized by diffuse hepatic fibrosis associated with the formation of regenerative nodules, resulting in distortion of the normal hepatic architecture and impairment of liver function and intrahepatic blood flow.

Etiology :It is caused by chronic non self-limited injury of the liver :

-Viral hepatitis (chronic hepatitis C, Chronic hepatitis B±D).

- Metabolic dysfunction-associated steatohepatitis (MASH) (formerly NASH) - Alcohol-associated liver disease (ALD): Prolonged (>10 years) and regular alcohol intake (>30 ml/day).
- Metabolic : Wilson's disease, hemochromatosis, α1 antitrypsin deficiency.

Primary biliary cholangitis (PBC).

- Chronic right-sided heart failure and hepatic venous outflow obstruction (congestive hepatopathy) -Autoimmune hepatitis. -Toxins & drugs: Methotrexate- Isoniazide- methyl dopa

Morphological classification:

Morphological classification has largely been replaced by etiological and clinical classifications because mixed patterns are common in advanced disease.

1- Micronodular cirrhosis: small nodules, nearly same size (<3mm) usually caused by ongoing alcohol damage .

2- Macronodular cirrhosis: big nodules of variable size (>3mm), usually follow chronic hepatitis

3- Mixed type.

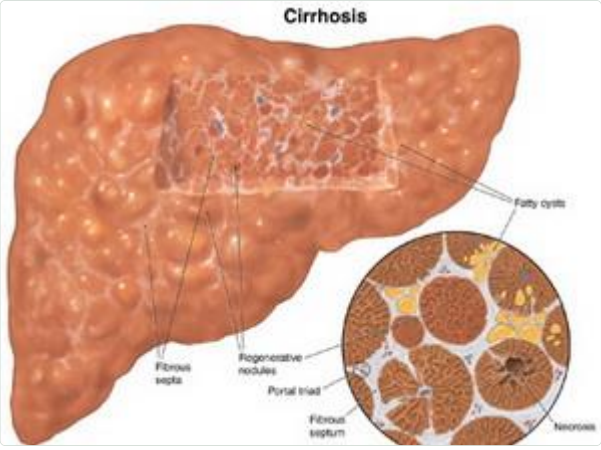


Figure 22.1 — Gross and microscopic picture of liver cirrhosis

Gross and microscopic picture of liver cirrhosis

Clinical picture

1- Compensated cirrhosis:

- The disease is discovered accidentally on routine examination and by screening. Examination may reveal an enlarged firm liver, splenomegaly, palmer erythema, spiders. Liver function tests are nearly normal.

1- Decompensated cirrhosis: There are several clinical manifestations of cirrhosis that include liver cell failure: ascites, jaundice, encephalopathy... The transition from compensated to decompensated cirrhosis is marked by the development of ascites, variceal bleeding, hepatic encephalopathy or jaundice.

Symptoms: There are usually three modes of presentation:

1-Asymptomatic: The patient may remain non symptomatizing for years to be discovered by chance during routine check-up, pre-employment or pre-operative assessment

2-Symptoms of conditions commonly associated with chronic liver disease such as gall stones, parotid enlargement, Dupuytren's contractures and muscle cramps

3- Symptoms of complications e.g. ascites, bleeding, encephalopathy, recurrent lower limb edema, spontaneous bacterial peritonitis or renal impairment. Fever due to infections: UTI, SBP, chest and skin infections.

Signs:

1- Protein-calorie malnutrition : found in 20-40%. of patients as shown by muscle wasting and decreased fat stores

2-Skin: Spider naevi and Palmer erythema may be detected .

3- Nail changes:

a. Disappearance of nail lacunae

b. White opaque nails (leuconychia)

c. Clubbing of fingers is more commonly found in biliary cirrhosis and in cirrhosis associated with large pulmonary arterio-venous shunt

4-Manifestations of hyperdynamic circulation due to the presence of excess vasodilator materials (VDM) leading to warm hands, high volume pulse, tachycardia

2- Manifestations of hormonal disturbance: manifestations of hypogonadism and hyperestrogenism including gynecomastia, testicular atrophy and loss of body

3- Sarcopenia (progressive loss of skeletal muscle mass and function) as one of the most common and clinically vital manifestations of advanced chronic liver disease. It affects between 30% to 70% of patients with cirrhosis.

Abdominal examination:

1- Abdominal wall hernias are common particularly umbilical hernia.

2- Liver is usually palpable with a sharp edge and firm consistency. The surface is usually smooth except in cases of macronodular cirrhosis or carcinoma. The liver size may become shrunken in late cases.

3- Spleen is usually palpable, it varies in size, and its lower end may reach to iliac fossa in cases of hepatosplenic schistosomiasis.

4- Ascites may be detected.

Chest examination

- Pleural effusion in association with cirrhosis and in the absence of primary pulmonary or cardiac disease. It is usually right sided (70%) with ascites but sometimes without ascites.
  - The hepatopulmonary syndrome may be present

Central cyanosis, clubbing, and dyspnea due to:

Pulmonary AV shunts Porto- pulmonary shunts basal lung collapse due to tense ascites.

- Fetor hepaticus: Fetor hepaticus is a sweet, musty odor caused by volatile sulfur-containing compounds .

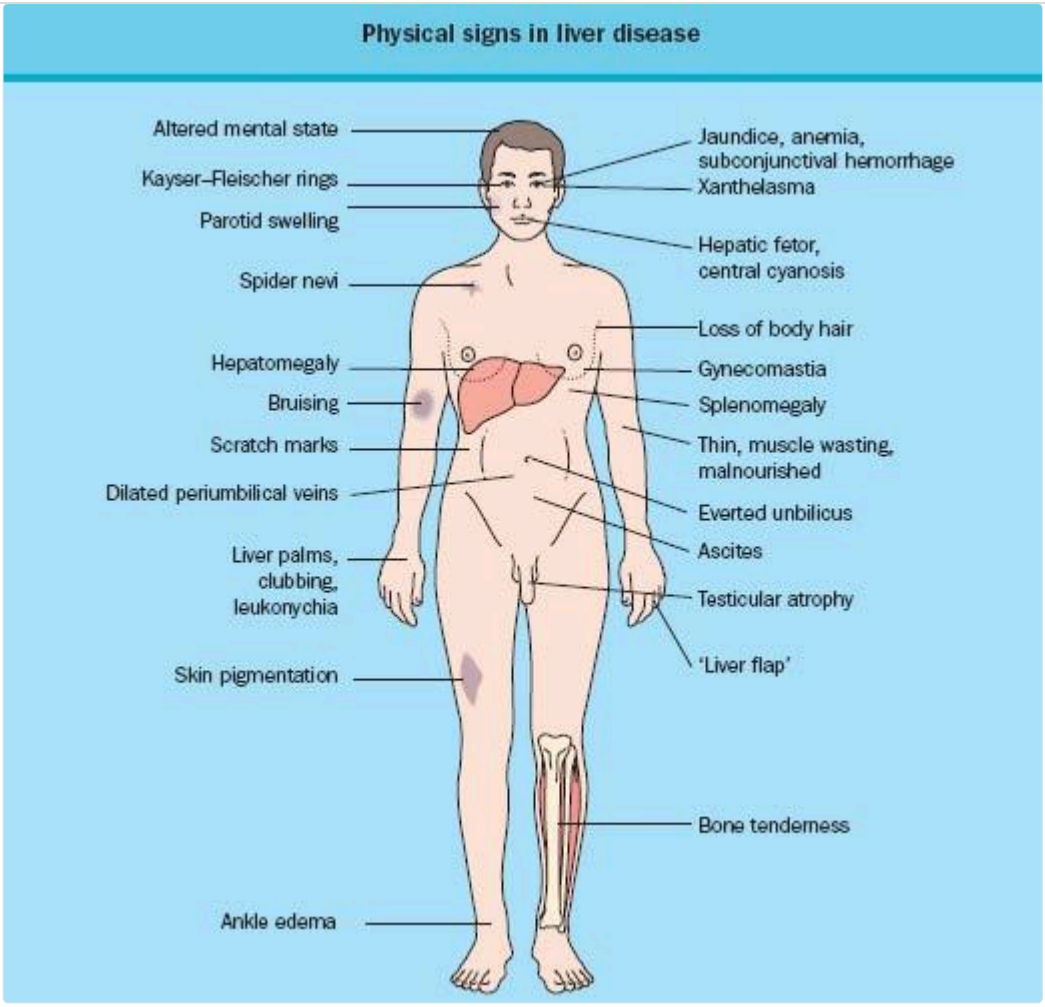


Figure 22.2 — Physical signs in liver cirrhosis

Physical signs in liver cirrhosis

Investigations:

A- Laboratory investigations:

1- Blood picture:

-Microcytic hypormoic anemia is common in patients having gastrointestinal hemorrhage.

- Normochromic macrocytic anemia may be found. -Leucopenia and thrombocytopenia in patients with hypersplenism. -Prolonged prothrombin time that is not corrected by vitamin K injections indicating failure of the liver to synthesize prothrombin complex.

2- Serum albumin is decreased, serum gamma globulins are increased, and bilirubin is increased..

3-Serum aminotransferases (ALT and AST) are mildly elevated.

4-Blood glucose: impaired glucose tolerance may be found in 50% of patients and frank diabetes in 15-30% of patients, particularly in post-hepatitis C cirrhosis .

B- Imaging

-Abdominal ultrasound

The liver may show a coarse echogenic pattern. The edge may be irregular and in advanced cases, the liver may be shrunken and portal hypertension may be shown by the dilated portal vein. Splenomegaly and ascites may be detected. Ultrasonography is best combined with Doppler evaluation of the hepatic vasculature

- Elastography (FibroScan)

C- Upper endoscopy for detection of varices screening endoscopy is guided by Baveno VII criteria, and some patients can safely avoid endoscopy.

D- Investigations for the cause:

Viral markers, serum autoantibodies, serum iron-and ferritin, copper and caeruloplasmin, liver biopsy could be done to reach the possible cause of cirrhosis

Non invasive tests for the diagnosis of hepatic fibrosis, cirrhosis and steatosis

Noninvasive tests has now replaced the role of liver biopsy in the diagnosis of fibrosis. These approaches can overcome the limitations of liver biopsy and have become more widely used in routine clinical practice Imaging-based elastography and serum biomarkers are currently the two main noninvasive tests for liver fibrosis staging. -Liver stiffness measurement by imaging techniques such as transient elastography (Fibroscan). -Simple biomarker blood tests for fibrosis, including the fibrosis-4 index (FIB-4) score and aspartate aminotransferase-platelet ratio index (APRI)

Treatment of cirrhosis:

Actually the condition of management of hepatic fibrosis and portal hypertension has changed with the development of 2 new definitions - Compensated advanced chronic liver disease (cACLD)

- Clinically significant portal Hypertension (CSPH).

. Transition from the compensated to the decompensated stage in chronic liver disease is clinically marked by the development of complications such as ascites, variceal hemorrhage and overt hepatic encephalopathy.

Among patients with cACLD, at least two different stages have been identified based on the presence or absence of clinically significant portal hypertension (CSPH)

The use of Fibroscan (elastography) in clinical practice has enabled the early identification of patients who are at risk of having CSPH and decompensation to be treated by non selective beta blocker like Carvedilol

Although the development of cirrhosis is an irreversible process, many lines of treatment can be provided to the patient ensuring a better quality of life and may promote regeneration of normally functioning hepatocytes.

Nutrition:

Protein restriction is no longer recommended.

Current guidelines recommend Protein 1.2–1.5 g/kg/day even in patients with hepatic encephalopathy.

Temporary reduction may occasionally be necessary during severe overt encephalopathy, but prolonged protein restriction should be avoided because it worsens sarcopenia

- Calories: caloric requirements are supplied as normal individuals. Good intake of carbohydrates and moderate amounts of fat should be supplied.

-Salt restriction in ascitic patients.

Late evening carbohydrate snack.

- Treatment of underlying cause: -Antiviral drugs in early cases with active inflammation. -Abstinence of alcohol intake.
- Chelation of copper and iron in Wilson's disease and hemochromatosis respectively.
- Corticosteroids in autoimmune hepatitis.
- Treatment of complications: ascites, encephalopathy, bleeding, spontaneous bacterial peritonitis and renal impairment.
- Liver transplantation is the treatment of choice if available.

Course and Prognosis: This is extremely variable and depends on several factors such as the etiology, and the presence or absence of complications. The Child-Turcotte Pugh (CTP) Score is used to assess the prognosis. It is proposed to assess survival, risk of general anesthesia, and operative mortality. Patients are classified into A, B &C according to 5 parameters: The level of albumin, bilirubin, prothrombin time & the presence of ascites& encephalopathy

TABLE (2). CHILD-TURCOTTE PUGH (CTP) SCORE

Parameter	1 point	2 points	3 points
Encephalopathy	None	Grade 1-2	Grade 3-4
Ascites	None	Medically controlled	Uncontrolled
Albumin	>3.5	2.8-3.5 g/dl	< 2.8 g/dl
Bilirubin	<2	2-3 mg/dl	>3mg/dl
International normalized ratio (INR)	< 1.7	1.7-2.3	> 2.3

A patient is considered to be :

Child A = 5–6

Child B = 7–9

Child C = 10–15

. The Model for End-Stage Liver Disease incorporating serum sodium (MELD- Na) is a prognostic scoring system based on serum bilirubin, serum creatinine, INR, and serum sodium. It is used to assess disease severity and prioritize patients for liver transplantation.

PRIMARY BILIARY CHOLANGITIS (PBC)

Primary biliary cholangitis (PBC) is a chronic autoimmune cholestatic liver disease characterized by progressive immune-mediated destruction of the small intrahepatic bile ducts, leading to chronic cholestasis, fibrosis and, if untreated, cirrhosis.

The cause is not yet identified but it is said to be an immunological disease also there is a familial background The exact cause remains unknown. It is believed to result from an interaction between genetic susceptibility and environmental triggers, leading to autoimmune destruction of the small intrahepatic bile ducts.

It is sometimes associated with autoimmune manifestations in the form of rheumatoid arthritis, and Sjogren's syndrome.

- Sjögren syndrome
- Autoimmune thyroid disease
- Systemic sclerosis
- Rheumatoid arthritis
- Celiac disease

Pathogenesis: - Autoreactive T lymphocytes target cholangiocytes in genetically susceptible individuals after exposure to environmental triggers. It is proposed that an environmental factor acts on a genetically predisposed host. Infective agents have been accused as the triggers of the autoimmune process e.g. E.coli, enterococci and retroviruses. There may be a defect in T suppressor cells or a defect in immunoregulation producing damage to the duct system.

Clinical picture: Females are more affected than males especially in their fourth to sixth decades of life. Pruritus is the usual presenting symptom and is present in 50% of symptomatic patients , fatigue is usually present . However, the condition may be asymptomatic. Hepatomegaly and jaundice then develop. The symptoms may resemble those of obstructive jaundice. In later stages the patient may have xanthelasmas on the eyelids due to cholesterol deposition. Malabsorption of fat soluble vitamins and steatorrhea due to deficiency of bile salts may occur in advanced stages. Osteoporosis is one of the complications.

Many patients are now diagnosed incidentally because of isolated elevation of alkaline phosphatase.

Diagnosis :

Current diagnosis requires two of three:

1. Persistent cholestatic liver enzymes (especially ALP)
2. Positive AMA (or PBC-specific ANA if AMA negative)
3. Compatible liver biopsy

Laboratory tests:

Elevated alkaline phosphatase is often the only and the early abnormality in liver biochemistry. Levels of aminotransferases and GGT are also raised. Specific autoantibodies are present: Anti Mitochondria' antibodies (AMA) especially M2 are specific and are present in over 95% of patients. Antinuclear antibodies, smooth muscle antibodies may be present and are not specific. Serum cholesterol is usually raised.

Imaging:

- Abdominal ultrasound is indicated to exclude extrahepatic biliary obstruction and if there is doubt. MRCP can be done .

Liver biopsy:

Damage and loss of the small bile ducts with Florid duct lesion are characteristic of the disease.

N.B.: Other causes of hepatic granulomas include; sarcoidosis, tuberculosis, schistosomiasis, and drug reactions.

Treatment

Ursodeoxycholic acid (UDCA) 13–15 mg/kg/day

It is considere the first-line treatment. It improves liver biochemistry , slows fibrosis progression and improves survival

Patients are evaluated after one year.

If response inadequate

Second-line therapy

Obeticholic acid or Fibrates (bezafibrate/fenofibrate)

depending on availability.

- Pruritus is difficult to treat; First-line is Cholestyramine or second-line drugs as

Rifampicin , Naltrexone , Sertraline

Cholestyramine and naloxone have been tried with mild improvement.

- Control of signs and symptoms of liver cirrhosis and liver cell failure in later stages.
- Liver transplantation is the definite treatment.

N.B.: Secondary biliary cholangitis can result from prolonged (months to years) large duct obstruction by stones or strictures

PRIMARY SCLEROSING CHOLANGITIS (PSC)

Primary sclerosing cholangitis (PSC) is a chronic progressive cholestatic liver disease characterized by immune-mediated inflammation and fibrosis of the intrahepatic and/or extrahepatic bile ducts, resulting in multifocal biliary strictures, cholestasis, and eventually biliary cirrhosis and liver failure.

Pathogenesis:

Chronic inflammation of the bile ducts leads to concentric periductal fibrosis and the formation of multiple short strictures alternating with mildly dilated segments, producing the characteristic "beaded" appearance of the biliary tree

- The exact cause is unknown. Genetic predisposition has been proposed - -These changes are characteristic and are seen on MRCP and ERCP.
- The changes may range from mild inflammation to established cirrhosis.

Clinical picture:

- Males are commonly affected. Age of onset is usually the forties.
- Manifestations of cholestasis are usually present e.g. pruritus and jaundice . The course of the disease is usually fluctuating.
- Approximately 70–80% of patients have inflammatory bowel disease, most commonly ulcerative colitis. .
- Later on, picture of end stage liver disease eventually takes place.

Many patients are discovered incidentally because of isolated elevation of alkaline phosphatase.

Patients may experience fatigue and recurrent bacterial cholangitis

Investigations:

-Elevated bilirubin and alkaline phosphatase are the earliest abnormalities. -The autoantibody p-ANCA is found in the serum of 60% of patients. p-ANCA is not diagnostic because of poor specificity. Magnetic resonance cholangiopancreatography (MRCP) is the preferred diagnostic imaging modality and demonstrates multiple short strictures alternating with dilated segments ("beading"). ERCP is reserved for therapeutic intervention or evaluation of dominant strictures. -However, the diagnosis is mainly based on

imaging studies; MRCP and or ERCP show attenuation and the characteristic beading of the extra and intra hepatic biliary radicals. -Confirmation may need liver biopsy which shows the changes mentioned above.

Biopsy is not routinely required.

It is useful when: small-duct PSC is suspected also when PSC-AIH overlap syndrome is suspected

Characteristic histology

"Onion-skin" periductal fibrosis

PSC may overlap with autoimmune hepatitis, particularly in younger patients.

Treatment:

- The only proven treatment is liver transplantation. Five-year survival exceeds 80%.
- Ursodeoxycholic acid (UDCA) Supplementary treatment as in cases of obstructive jaundice. Ursodeoxycholic acid may improve liver biochemistry in some patients, but it has not been shown to alter disease progression and is not routinely recommended solely to improve prognosis.
- Control of manifestations of end stage liver disease.

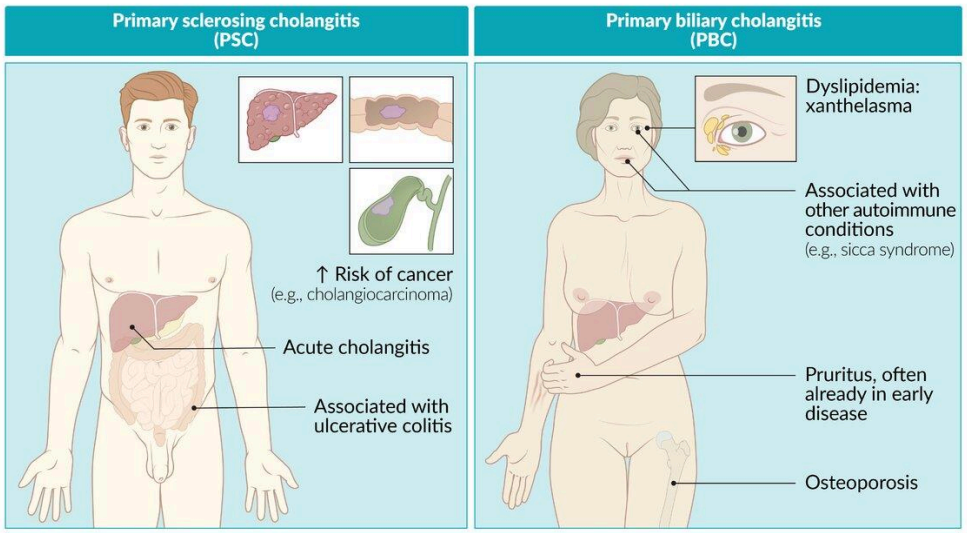
ERCP may benefit in balloon dilatation or temporary stenting to relieve cholestasis , pruritus or recurrent cholangitis.

Surveillance

Patients require surveillance for:

- Cholangiocarcinoma
- Gallbladder carcinoma
- Hepatocellular carcinoma (if cirrhotic)

Patients with PSC and ulcerative colitis require annual colonoscopic surveillance because of the markedly increased risk of colorectal cancer.



	Primary sclerosing cholangitis (PSC)	Primary biliary cholangitis (PBC)
Epidemiology	♀ < ♂ (1:2)	♀ > ♂ (9:1)
Antibodies	p-ANCA	AMA-M2
Pathophysiology	Progressive inflammation and fibrosis of **intrahepatic and extrahepatic** bile ducts	Autoimmune destruction of small **intrahepatic** bile ducts
Most accurate diagnostic test	MRCP: multifocal bile duct strictures and dilatations (beading)	Liver biopsy: lymphocytic infiltration of portal areas and periductal granulomas

Figure 22.3 — Primary sclerosing cholangitis and primary biliary cholangitis

HEREDITARY HEMOCHROMATOSIS

Definition:

This is an inherited autosomal recessive disease characterized by iron deposition in various organs leading to eventual fibrosis and functional organ failure. It is due to mutation of HFE gene on the short arm of chromosome 6.

Pathophysiology:

The HFE gene mutation results in decreased hepatic production of hepcidin, the principal hormone regulating iron homeostasis.

Low hepcidin levels increase the activity of ferroportin, leading to excessive intestinal iron absorption and increased iron release from macrophages. Because the body has no physiological mechanism for iron excretion, excess iron progressively accumulates in tissues.

Initially, iron is deposited predominantly in periportal hepatocytes. With disease progression, iron deposition extends throughout the liver and later involves:

- Kupffer cells
- Biliary epithelium

- Fibrous tissue
- Pancreas
- Heart
- Pituitary gland
- Joints
- Skin

Persistent iron overload causes oxidative stress, hepatocyte injury, fibrosis, cirrhosis, and increases the risk of hepatocellular carcinoma (HCC)

Clinical Features

Many patients are now diagnosed before symptoms develop through family screening or incidental discovery of elevated serum ferritin and transferrin saturation.

Symptomatic disease usually appears after the age of 40–50 years in men and later in women because menstruation and pregnancy delay iron accumulation.

Clinical manifestations include:

- Fatigue and weakness (most common early symptoms)
- Hepatomegaly
- Chronic liver disease and cirrhosis
- Skin hyperpigmentation ("bronze diabetes")
- Diabetes mellitus due to pancreatic iron deposition
- Arthropathy, especially involving the second and third metacarpophalangeal joints
- Hypogonadotropic hypogonadism
- Loss of libido and erectile dysfunction
- Cardiomyopathy
- Cardiac arrhythmias
- Heart failure

Patients with established cirrhosis remain at increased risk of hepatocellular carcinoma, even after successful iron depletion.

Investigations

Laboratory studies

- Increased transferrin saturation (>45%) (earliest biochemical abnormality)
  - Elevated serum ferritin
  - Elevated serum iron
  - Reduced or normal total iron-binding capacity (TIBC)

- Mild elevation of liver enzymes

Genetic testing

Testing for HFE mutations (C282Y and H63D) confirms the diagnosis in most patients.

Imaging

MRI is the preferred non-invasive method for quantifying hepatic iron concentration and assessing extrahepatic iron deposition.

Liver biopsy

Liver biopsy is no longer routinely required for diagnosis. It is reserved for patients in whom:

- The diagnosis remains uncertain
- Non-HFE hemochromatosis is suspected
- The degree of fibrosis requires assessment
- Another liver disease is suspected

Treatment:

Therapeutic phlebotomy (Venesection)

This is the treatment of choice.

- Approximately 500 mL of blood is removed weekly or every two weeks during the induction phase.
- Treatment continues until serum ferritin reaches 50–100 ng/mL.
- Maintenance phlebotomy is then performed every 2–4 months according to iron status.

Iron chelation

Iron chelators (e.g., deferoxamine, deferasirox, or deferiprone) are reserved for patients who cannot undergo phlebotomy, such as those with severe anemia or transfusion-dependent conditions. They are not first-line therapy in hereditary hemochromatosis.

Lifestyle measures

Patients should be advised to:

- Avoid iron supplements
- Avoid vitamin C supplements in large doses
- Limit alcohol consumption
- Avoid raw shellfish because of the risk of Vibrio vulnificus infection

Liver transplantation Liver transplantation is indicated for patients with decompensated cirrhosis or HCC meeting transplant criteria. It corrects liver failure but does not correct the underlying genetic defect.

Family Screening

First-degree relatives should undergo screening with:

- Transferrin saturation
- Serum ferritin
- HFE genetic testing when appropriate

WILSON DISEASE (Hepatolenticular Degeneration)

Definition

Wilson disease is a rare autosomal recessive disorder of copper metabolism caused by mutations in the ATP7B gene on chromosome 13. The disease results in impaired biliary copper excretion and defective incorporation of copper into ceruloplasmin, leading to progressive copper accumulation, mainly in the liver, brain, cornea, kidneys, and other organs.

A history of parental consanguinity may be present.

Pathophysiology

Dietary copper is absorbed mainly from the stomach and proximal small intestine and is transported to the liver bound to albumin.

Within hepatocytes, copper normally has two major fates:

- Incorporation into ceruloplasmin, which is released into the circulation.
- Excretion into bile, which is the major route of copper elimination.

In Wilson disease, mutations of the ATP7B gene impair both:

- Incorporation of copper into ceruloplasmin.
- Biliary excretion of copper.

As a result, copper accumulates initially in the liver. Once hepatic storage capacity is exceeded, free copper is released into the bloodstream and deposited in other organs, particularly:

- Brain (especially the basal ganglia)
- Cornea
- Kidneys
- Bones and joints

Copper-induced oxidative injury leads to progressive hepatic inflammation, fibrosis, cirrhosis, and neurological damage.

Clinical Features

The disease usually presents between 5 and 35 years of age, although presentation outside this age range may occur.

Clinical manifestations vary widely.

Hepatic manifestations

- Asymptomatic elevation of liver enzymes
  - Acute hepatitis
  - Chronic hepatitis
  - Compensated or decompensated cirrhosis
  - Acute liver failure (particularly in children and young adults)

Neurological manifestations

- Tremor
  - Dysarthria
  - Dystonia
  - Parkinsonian features
  - Chorea
  - Incoordination
  - Drooling (unintential loss of saliva)
  - Dysphagia

Psychiatric manifestations

- Personality changes
  - Depression
  - Anxiety
  - Behavioral disturbances
  - Cognitive impairment

Ophthalmic manifestations

Kayser–Fleischer rings are caused by copper deposition in Descemet's membrane of the cornea. They are best detected by slit-lamp examination and are present in nearly all patients with neurological disease.

Other manifestations

- Coombs-negative hemolytic anemia

- Renal tubular dysfunction
- Osteopenia or osteoporosis
- Premature arthritis
- Cardiomyopathy (uncommon)

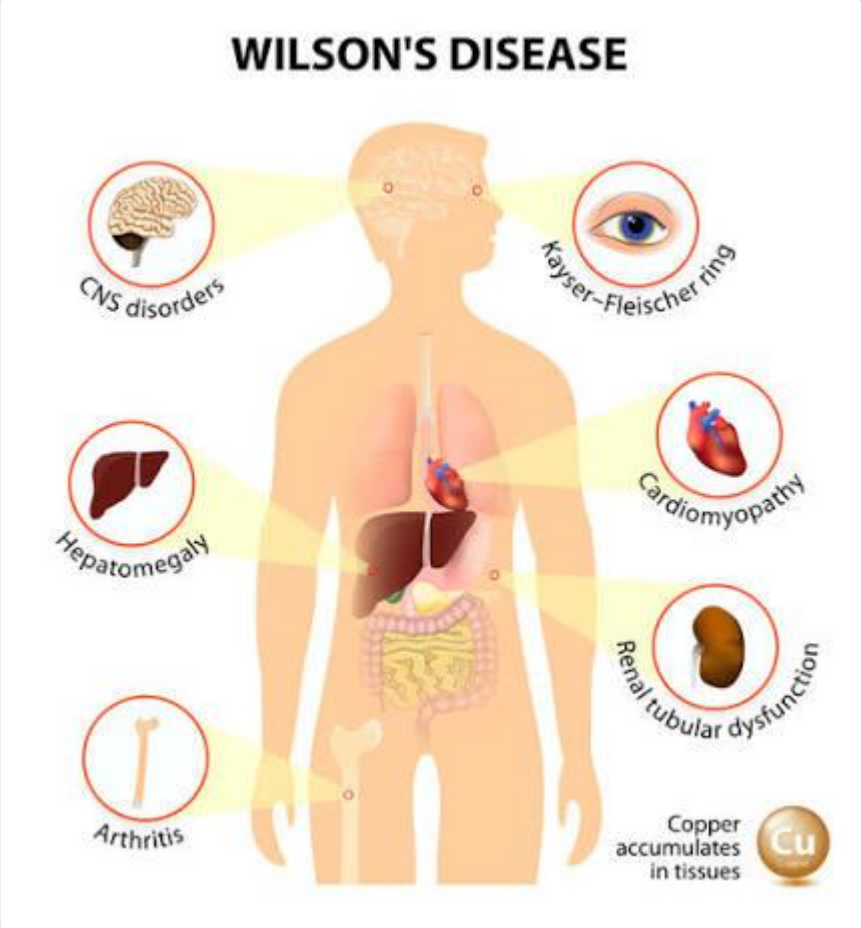


Figure 22.4 — Wilson disease

Investigations

Laboratory studies

- Low serum ceruloplasmin (usually <20 mg/dL)
  - Increased 24-hour urinary copper excretion (>100 µg/day in symptomatic adults)
  - Elevated non-ceruloplasmin ("free") copper
  - Liver function tests may be abnormal

Ophthalmologic examination

- Slit-lamp examination for Kayser–Fleischer rings

Imaging

- Brain MRI may show characteristic abnormalities of the basal ganglia in neurological disease.

Liver biopsy

Liver biopsy is reserved for selected cases when the diagnosis remains uncertain. Quantitative hepatic copper measurement (>250 µg/g dry weight) supports the diagnosis.

Genetic testing

Identification of ATP7B mutations confirms the diagnosis and is increasingly used, especially for family screening.

Diagnosis

The diagnosis is based on a combination of:

- Clinical findings
- Serum ceruloplasmin
- 24-hour urinary copper excretion
- Kayser–Fleischer rings
- Hepatic copper quantification (when needed)
- ATP7B genetic testing

Treatment

Treatment is lifelong.

Copper chelation

- D-Penicillamine (first-line in many patients)
  - Trientine (alternative with fewer adverse effects)

Zinc salts

Zinc reduces intestinal copper absorption and is commonly used for:

- Presymptomatic patients
- Maintenance therapy
- Patients intolerant of chelators

Dietary measures

Patients should initially avoid foods rich in copper, including:

- Liver
- Shellfish
- Nuts
- Chocolate
- Mushrooms

Liver transplantation

Indicated for:

- Acute liver failure
- Decompensated cirrhosis unresponsive to medical therapy

Liver transplantation corrects the underlying metabolic defect because the transplanted liver contains normal ATP7B function.

Family Screening

All first-degree relatives, particularly siblings, should be screened by:

- Clinical assessment
- Serum ceruloplasmin
- 24-hour urinary copper excretion
- Liver function tests
- ATP7B genetic testing when available

Patients with confirmed Wilson disease should receive lifelong treatment even if asymptomatic.

Heterozygous carriers do not develop Wilson disease and do not require treatment.

Alcohol-Associated Cirrhosis

Etiology

Alcohol-associated cirrhosis develops after prolonged excessive alcohol consumption, usually for more than 10 years. The risk increases with both the amount and duration of alcohol intake. Daily drinking is more harmful than intermittent drinking.

Alcohol may also enhance the hepatotoxicity of certain drugs, particularly paracetamol (acetaminophen).

Pathology

- Diffuse micronodular cirrhosis that may become mixed micro- and macronodular in advanced disease.
- Macrovesicular hepatic steatosis.
- Hepatocyte ballooning with Mallory–Denk bodies.

Clinical Features

Alcohol-associated liver disease includes a spectrum of:

- Hepatic steatosis (fatty liver)
- Alcohol-associated hepatitis

Alcohol-associated cirrhosis

Patients with chronic alcohol misuse may also have:

- Gastrointestinal: gastritis, peptic ulcer disease, pancreatitis, malabsorption.
- Cardiovascular: alcoholic cardiomyopathy.
- Neurological: peripheral neuropathy, cerebellar ataxia, myopathy.
- Hematological: macrocytic anemia, leukopenia, thrombocytopenia.
- General: parotid enlargement and Dupuytren's contracture.

Investigations

- Mild to moderate elevation of liver enzymes with AST typically higher than ALT (often AST/ALT ratio >2).
- Elevated γ-glutamyl transferase (GGT).
- Abdominal ultrasonography to assess liver morphology and portal hypertension.
- CT or MRI and liver biopsy are reserved for selected cases.

Treatment

- Complete abstinence from alcohol (the most important treatment).
- Adequate nutritional support, including vitamin supplementation (especially thiamine).
- Management of cirrhosis and its complications.

Alpha-1 Antitrypsin Deficiency

Definition

Alpha-1 antitrypsin (AAT) deficiency is an autosomal recessive inherited metabolic disorder caused by mutations in the SERPINA1 gene (formerly known as the Pi gene). It results in reduced circulating levels of alpha-1 antitrypsin and accumulation of abnormal AAT protein within hepatocytes.

Pathophysiology

Alpha-1 antitrypsin is synthesized mainly by the liver and acts as an inhibitor of several proteolytic enzymes, particularly neutrophil elastase.

Deficiency of circulating AAT allows unopposed neutrophil elastase activity, resulting in progressive destruction of alveolar walls and early-onset emphysema.

The abnormal AAT protein is retained within hepatocytes, leading to hepatocellular injury, chronic hepatitis, fibrosis, cirrhosis, and occasionally hepatocellular carcinoma.

Clinical Features

The clinical manifestations involve mainly the lungs and liver.

Pulmonary manifestations

- Early-onset emphysema (usually in the third or fourth decade, especially in smokers)
  - Chronic obstructive pulmonary disease (COPD)

Hepatic manifestations

- Neonatal cholestatic jaundice (in some infants)
  - Chronic hepatitis
  - Cirrhosis
  - Portal hypertension
  - Hepatocellular carcinoma (in patients with cirrhosis)

Diagnosis

- Low serum alpha-1 antitrypsin level.
- Phenotyping or genotyping of the SERPINA1 gene (the PiZZ genotype is the most severe and clinically important).

Treatment

- Smoking cessation.
- Standard treatment of COPD.
- Intravenous alpha-1 antitrypsin augmentation therapy (plasma-derived) for selected patients with emphysema.
- Management of chronic liver disease and cirrhosis.
- Liver transplantation for end-stage liver disease.

Hepatic Venous Outflow Disorders

Hepatic venous outflow disorders are conditions that impair the drainage of blood from the liver.

The main conditions are:

- **Congestive hepatopathy** – due to increased venous pressure from cardiac disease
- **Budd–Chiari syndrome** – due to obstruction of the hepatic veins
- **Sinusoidal obstruction syndrome (SOS)** – injury and obstruction of hepatic sinusoids
- **Inferior vena cava (IVC) obstruction** – obstruction of the IVC affecting hepatic venous drainage

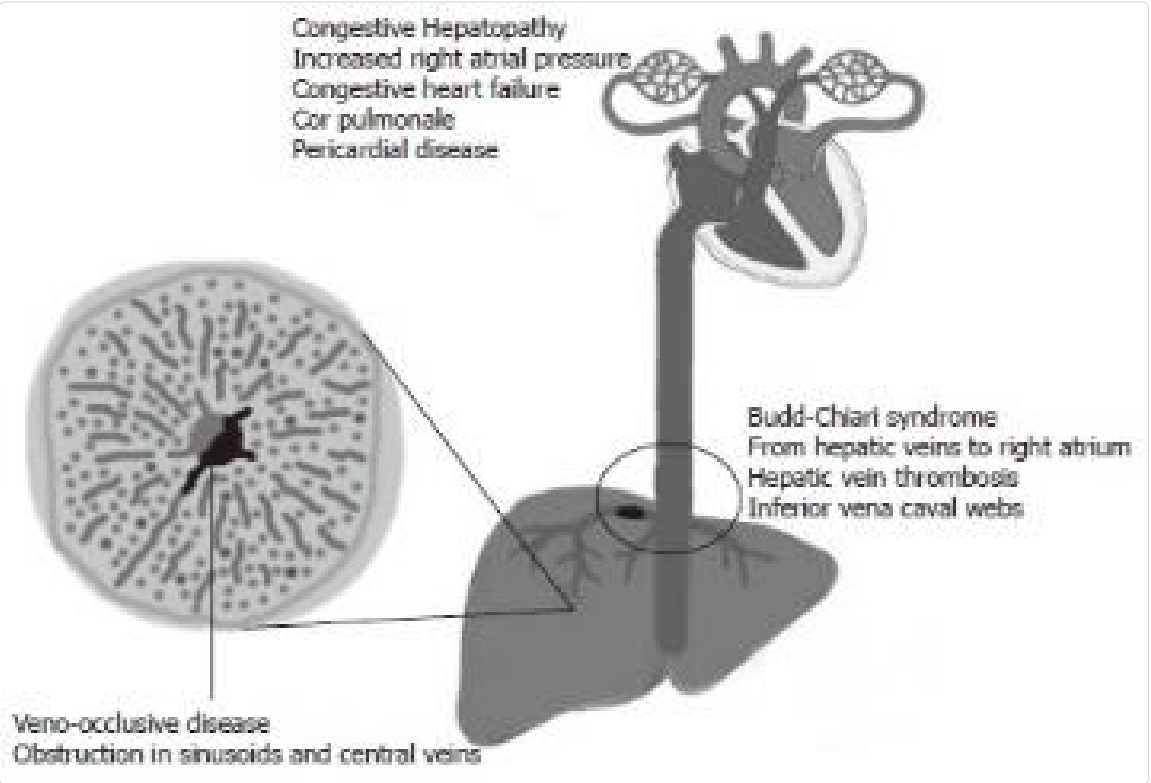


Figure 22.5 — Hepatic venous outflow disorders

1. Congestive Hepatopathy

Definition

Congestive hepatopathy is chronic liver congestion caused by increased central venous pressure, most commonly due to **right-sided heart failure**.

Causes

- Right-sided heart failure
- Severe tricuspid regurgitation
- Constrictive pericarditis

Pathophysiology

Increased right-sided cardiac pressure is transmitted backward to the hepatic veins and sinusoids, causing **hepatic congestion** , particularly around the central veins.

Clinical Features

- Hepatomegaly, often tender
- Right upper quadrant discomfort
- Peripheral edema
- Ascites, usually in advanced disease
- Clinical features of right-sided heart failure

Diagnosis

- Liver function tests may show mild abnormalities
- **Echocardiography** to assess the underlying cardiac disease
- **Doppler ultrasound** can demonstrate hepatic venous congestion

Treatment

The main treatment is **treatment of the underlying cardiac disease** .

- Optimize treatment of heart failure
- Diuretics when appropriate
- Treat significant valvular disease

Key concept:

Congestive hepatopathy = the heart cannot pump effectively → blood backs up into the liver.

2. Budd–Chiari Syndrome

Definition

Budd–Chiari syndrome is **obstruction of hepatic venous outflow** , usually involving the hepatic veins, in the absence of cardiac disease as the primary cause.

Causes

Most cases are related to **thrombosis or a prothrombotic state** .

Important associated conditions include:

- Myeloproliferative disorders
- Other thrombophilic disorders
- Pregnancy and oral contraceptive use

- Malignancy

Clinical Presentation

Budd–Chiari syndrome may be acute, subacute, or chronic.

The classic clinical triad is:

- **Abdominal pain + hepatomegaly + ascites**

Chronic disease may lead to:

- Portal hypertension
- Splenomegaly
- Esophageal varices
- Chronic liver disease/cirrhosis

Diagnosis

Doppler ultrasonography is the first-line investigation.

CT or MRI may be required for further assessment.

Typical Imaging Findings

- Reduced or absent hepatic venous flow
- Hepatic vein obstruction
- Collateral venous circulation
- Caudate lobe enlargement

Treatment

Treatment depends on the severity and underlying cause and may include:

- **Anticoagulation**
- Treatment of the underlying prothrombotic condition
- Angioplasty/stenting in selected patients
- TIPS in patients with persistent portal hypertension
- Liver transplantation in severe cases

- **Key concept:**

Budd–Chiari syndrome = the liver cannot drain → hepatic veins are obstructed.

3. Sinusoidal Obstruction Syndrome

Sinusoidal obstruction syndrome (SOS), formerly called **veno-occlusive disease** , results from injury to the hepatic sinusoids and small hepatic venules.

It occurs mainly after **hematopoietic stem-cell transplantation or certain chemotherapy regimens** .

Clinical Features

- Painful hepatomegaly
- Jaundice
- Ascites
- Rapid weight gain due to fluid retention

Treatment

Management is mainly supportive. **Defibrotide** is used in severe cases.

Remember: SOS is mainly associated with **hematopoietic stem-cell transplantation/chemotherapy** .

4. Inferior Vena Cava Obstruction

IVC obstruction can impair hepatic venous drainage and cause hepatic congestion.

Causes

- Thrombosis
- Membranous webs
- Tumor compression or invasion

Clinical Features

- Hepatomegaly
- Ascites
- Lower-limb edema
- Dilated abdominal wall veins

Diagnosis

- Doppler ultrasound
- CT or MRI when required

Treatment

Treatment depends on the cause and may include **anticoagulation or angioplasty/stenting** .

Feature	Congestive hepatopathy	Budd–Chiarisyndrome
Basic problem	Increased venous pressure from the heart	Obstruction of hepatic venous drainage
Main cause	Right-sided heart disease	Thrombosis/prothrombotic disorders

Feature	Congestive hepatopathy	Budd–Chiarisyndrome
Important clinical clue	Raised JVP + peripheral edema	Abdominal pain + hepatomegaly + ascites
Hepatic veins	Congested/dilated	Obstructed
Main treatment	Treat cardiac disease	Anticoagulation ± restoration of venous drainage

- **Congestive hepatopathy:**

Heart → pressure backs up → liver becomes congested

Budd–Chiari:

Liver → hepatic veins blocked → blood cannot leave

Drug-Induced Liver Injury (DILI)

Definition

Drug-induced liver injury (DILI) is liver injury caused by a **medication, herbal product, or dietary supplement** . It may range from mild asymptomatic elevation of liver enzymes to **acute liver failure** .

DILI should be suspected when liver injury develops after exposure to a potentially hepatotoxic drug and other common causes have been excluded.

Types of DILI

1. Intrinsic DILI

- Usually **dose-dependent and predictable**
- Can occur in most individuals at sufficiently high doses
- **Paracetamol overdose** is the classic example

2. Idiosyncratic DILI

- Usually **unpredictable**
- Occurs only in susceptible individuals
- Most clinically important DILI is idiosyncratic

Patterns of Liver Injury

DILI may have three biochemical patterns:

Hepatocellular

- Predominant elevation of **ALT/AST**

- May present with hepatitis and jaundice

Cholestatic

- Predominant elevation of **ALP and bilirubin**
- May cause jaundice and pruritus

Mixed

- Features of both hepatocellular and cholestatic injury

The R-ratio can be used to classify the pattern of liver injury.

Clinical Manifestations

DILI may present with:

- Asymptomatic elevation of liver enzymes
- Acute hepatitis
- Cholestatic jaundice
- Hypersensitivity features such as fever, rash and eosinophilia
- Acute liver failure

Diagnosis

There is **no single diagnostic test** for DILI.

Diagnosis is based on:

- Detailed history of medications, herbal products and supplements
- Appropriate temporal relationship between drug exposure and liver injury
- Identification of the biochemical pattern
- Exclusion of other causes of liver disease
- Improvement after withdrawal of the suspected drug (**dechallenge**)

Liver biopsy is not routinely required.

Paracetamol Hepatotoxicity

Paracetamol overdose is the classic example of **intrinsic, dose-dependent DILI** .

In overdose, increased formation of the toxic metabolite **NAPQI** leads to depletion of glutathione and hepatocellular necrosis.

Treatment : N-acetylcysteine (NAC) is the specific antidote.

Treatment of DILI

The most important treatment: Stop the suspected offending drug , when clinically appropriate.

Other measures include:

- Monitor liver enzymes, bilirubin and INR
- Provide supportive treatment
- Avoid unnecessary hepatotoxic medications
- Treat complications

Patients who develop **acute liver failure** require specialist management and may require **liver transplantation** .

Important Examples

Drug	Typical association
Paracetamol	Dose-dependent hepatocellular injury
Isoniazid	Hepatocellular injury
Amoxicillin–clavulanate	Cholestatic/mixed injury
Diclofenac	Idiosyncratic DILI
Methotrexate	Chronic liver injury/fibrosis
Amiodarone	Hepatocellular/chronic liver injury

Key Points

DILI may be **intrinsic or idiosyncratic** .

- The three patterns are **hepatocellular, cholestatic and mixed** .
- DILI is mainly a **clinical diagnosis and diagnosis of exclusion** .
- Always ask about **medications, herbal products and supplements** .
- **Withdrawal of the offending drug is the main treatment.**
- **Paracetamol overdose → NAC.**
- Severe DILI can cause **acute liver failure** and may require transplantation.

CHAPTER 23

PORTAL HYPERTENSION

23

- The portal vein is formed by the union of the superior mesenteric vein and splenic vein.

It enters the liver at the porta hepatitis and divides into right and left branches supplying the corresponding hepatic lobes.

- Normal portal pressure: 5–10 mmHg.
- Clinically significant portal hypertension develops when portal pressure exceeds approximately 10–12 mmHg.
- Increased resistance to portal blood flow results in the development of portosystemic collaterals, which divert portal blood into the systemic circulation, bypassing the liver.

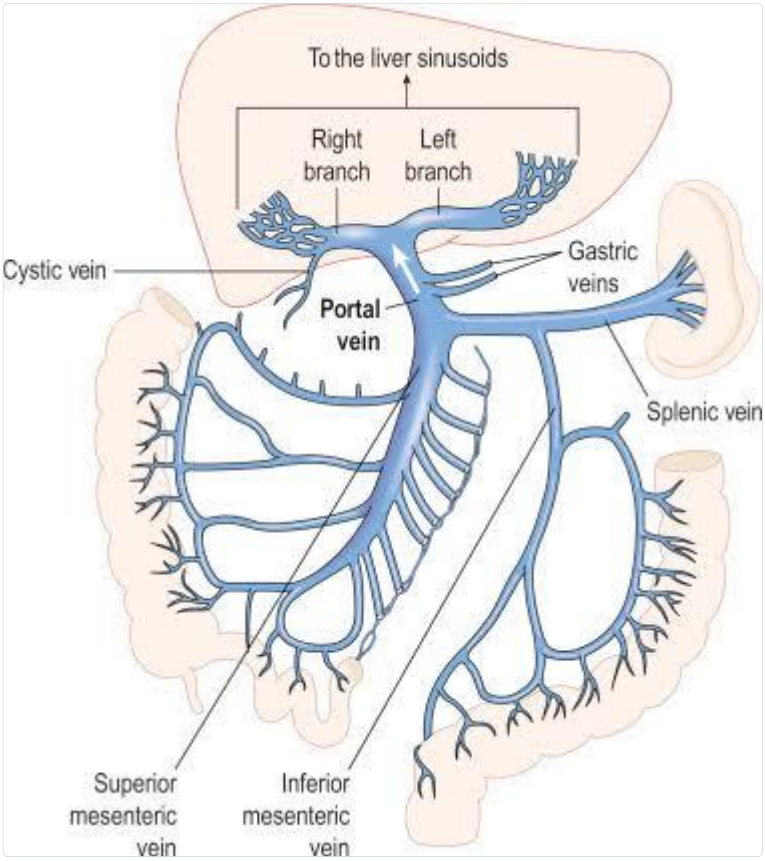


Figure 23.1 — Portal venous circulation

Causes and Classification

- Portal hypertension is classified according to the site of obstruction into prehepatic, intrahepatic, and posthepatic causes.

1. Prehepatic Portal Hypertension

- Caused by obstruction of portal blood flow before entering the liver, most commonly due to portal vein thrombosis.

Causes of portal vein thrombosis include:

- Idiopathic causes.

- Neonatal umbilical sepsis.
- Intra-abdominal infection.
- Increased blood viscosity, e.g. dehydration.
- Malignancy, e.g. hepatocellular carcinoma.
- Congenital portal vein stenosis or narrowing.

2. Intrahepatic Portal Hypertension

- Results from increased resistance within the liver due to distortion or obstruction of the intrahepatic portal circulation.

A. Presinusoidal

- Schistosomiasis with periportal fibrosis.
- Hodgkin's disease.
- Leukemia.
- Sarcoidosis.
- Congenital hepatic fibrosis.
- Toxins, e.g. arsenic.

B. Sinusoidal and Postsinusoidal

- Cirrhosis — the most common cause.
- Alcoholic hepatitis.
- Hepatic veno-occlusive disease.

3. Posthepatic Portal Hypertension

- Caused by obstruction of venous outflow after the liver, including:
  - Budd–Chiari syndrome.
  - Right-sided heart failure, particularly with tricuspid incompetence.
  - Constrictive pericarditis.

Cirrhosis is the most common cause of portal hypertension worldwide, while schistosomiasis is an important cause in endemic areas such as Egypt.

Portosystemic Collaterals

- When portal circulation is obstructed and portal pressure rises, particularly above 10–12 mmHg, extensive collateral circulation develops to divert portal blood into the systemic venous circulation.

1. Gastroesophageal junction

- Esophageal varices arise from anastomoses between the left and short gastric veins (portal system) and the azygos system (systemic circulation).

- Gastric varices may also occur, particularly in the fundus.
- Esophageal varices are superficial and therefore prone to rupture and life-threatening hemorrhage.

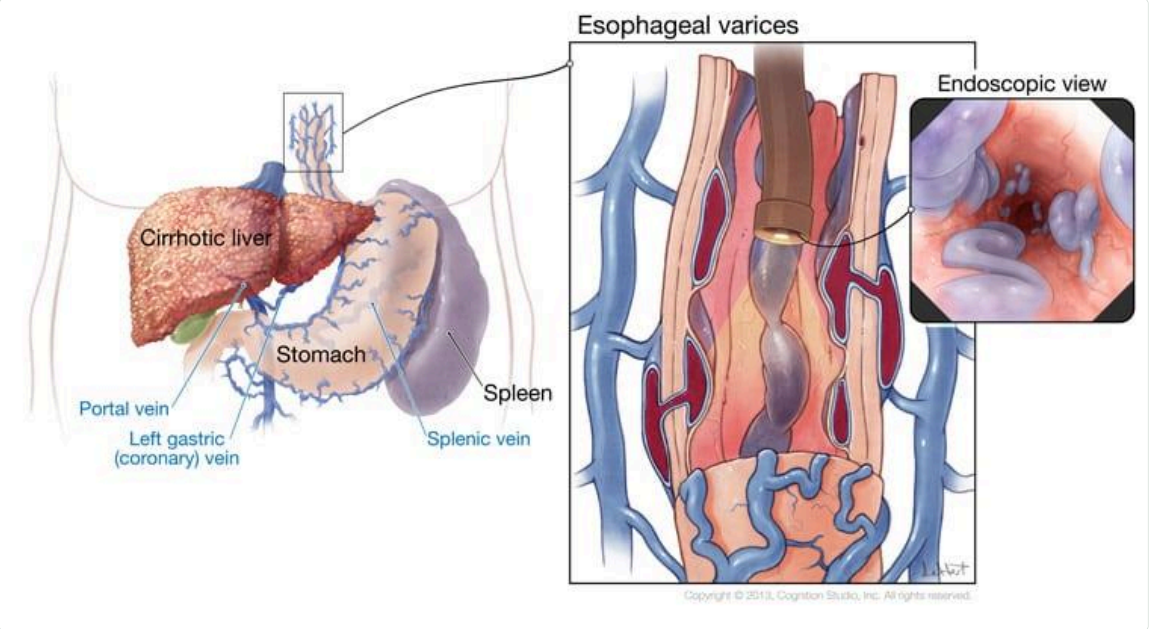


Figure 23.2 — Esophageal varices

2. Rectum

- Rectal portosystemic collaterals occur between the superior hemorrhoidal vein (portal) and the middle and inferior hemorrhoidal veins (systemic).

3. Anterior abdominal wall

- Results from recanalization of the umbilical vein with communication with the superior and inferior epigastric veins.
  - Prominent abdominal wall veins may produce caput medusa.

4. Other sites

Collateral vessels may develop between:

- Liver and diaphragm.
- Spleen and diaphragm.
- Spleen and kidney, producing lienorenal collaterals.
- Other retroperitoneal sites.

Clinical Features

Patients may remain asymptomatic, and splenomegaly may be the only clinical finding, particularly in presinusoidal portal hypertension.

Clinical manifestations:

- Splenomegaly.

- Hypersplenism, particularly thrombocytopenia.
- Hematemesis and melena due to rupture of gastroesophageal varices or bleeding from portal hypertensive/congestive gastropathy.
- Hepatic encephalopathy due to passage of gut-derived toxins into the systemic circulation through portosystemic shunts.
- Ascites.

Investigations

1. Upper Gastrointestinal Endoscopy

- Esophageal varices.
  - Gastric varices.
  - Grade/size of varices.
  - Stigmata or active bleeding.
  - Portal hypertensive gastropathy.

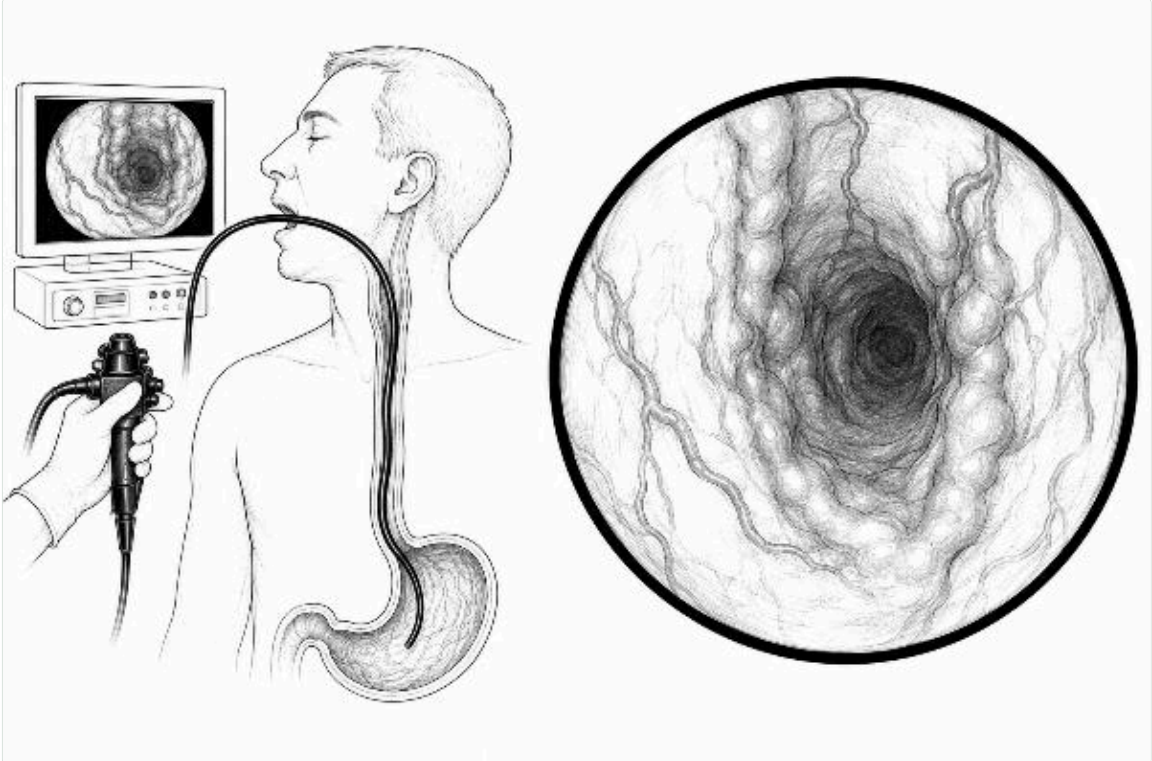


Figure 23.3 — Esophageal varices on endoscopy

2. Abdominal Ultrasonography and Doppler

- A non-invasive investigation used to:
  - Assess the portal and splenic veins.
  - Measure portal and splenic vein diameter.
  - Detect portal vein thrombosis or other vascular abnormalities.

- Identify portosystemic collaterals, including lienorenal collaterals.
- Assess the liver and spleen.
- Detect ascites.
- Help determine the underlying cause of portal hypertension.

3. Laboratory and Histological Assessment

- Liver function tests.
  - Other relevant laboratory investigations according to the suspected cause.
  - Liver biopsy may be required to determine the etiology and assess hepatic pathology in selected cases.

Variceal Hemorrhage

Approximately 90% of patients with cirrhosis may develop gastroesophageal varices over 10 years, although only about one-third develop variceal bleeding.

Esophageal variceal hemorrhage accounts for a significant proportion of deaths in patients with cirrhosis and portal hypertension.

Mortality depends on the patient's underlying hepatic function, general clinical status, and severity of the bleeding episode.

Management is divided into:

1. **Primary prophylaxis —** prevention of the first bleeding episode.
2. Management of **acute variceal bleeding.**
3. **Secondary prophylaxis** — prevention of recurrent bleeding.

Prevention of First Variceal Bleeding

Patients with cirrhosis should undergo assessment for clinically significant portal hypertension and varices using appropriate non-invasive criteria and/or upper GI endoscopy.

For patients with significant esophageal varices who have not previously bleed, prophylactic treatment includes:

1. Non-selective β-Blockers

- Carvedilol is a non-selective β-blocker with additional α1-blocking activity and is commonly used for prevention of decompensation and variceal bleeding , 6.25 mg twice daily.
  - They reduce the risk of variceal bleeding and may also reduce bleeding from portal hypertensive gastropathy.

2. Endoscopic band Ligation

- used for patients who have an indication for endoscopic prophylaxis, particularly when β-blockers are contraindicated or not tolerated, such as in:
  - Bronchial asthma.
  - Significant heart block.
  - Hypotension.
  - The varix is suctioned into the endoscope cap and a rubber band is deployed around it, producing thrombosis, necrosis, and subsequent fibrosis.

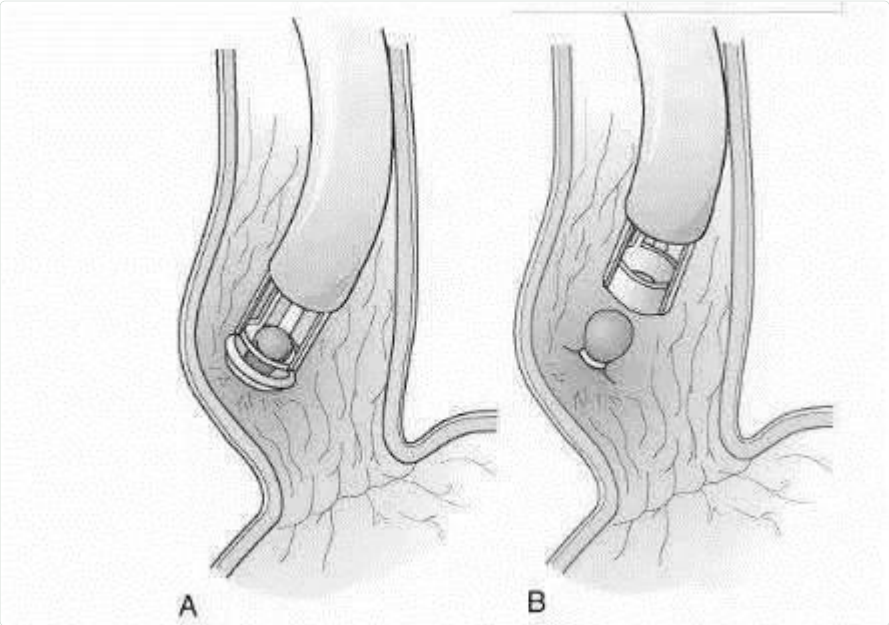


Figure 23.4 — Endoscopic variceal band ligation

Management of Acute Variceal Bleeding

1. Initial Resuscitation:

- Assess airway, breathing, circulation and general condition.
  - Monitor pulse and blood pressure.
  - Establish adequate intravenous access.
  - Obtain blood for:
  - CBC.
  - Blood grouping and cross-matching.
  - PT/INR.
  - Liver function tests.
  - Renal function.
  - Electrolytes.

- Blood glucose.
- Blood transfusion should be restrictive, with a target hemoglobin generally around 7–8 g/dL, individualized according to the patient's clinical condition.

- Avoid over-transfusion because excessive volume replacement may increase portal pressure and precipitate rebleeding.
- A nasogastric tube may be considered selectively for aspiration or gastric lavage, but it is not routinely required in every patient.

2. Urgent Upper GI Endoscopy:

- Endoscopy should be performed after initial stabilization, preferably within 12 hours in suspected acute variceal bleeding.
  - It is performed to:
    - Confirm variceal bleeding.
    - Exclude other causes of upper GI bleeding, such as peptic ulcer disease.
    - Identify gastric varices.
    - Detect portal hypertensive gastropathy.
    - Provide definitive endoscopic hemostasis.

Endoscopic Treatment

- **Endoscopic variceal ligation** the preferred endoscopic treatment.
- **Sclerotherapy:** Injection of a sclerosant, such as ethanolamine oleate, produces thrombosis and obliteration of the varix.
- **Gastric varices:** Large gastric varices may require injection of tissue adhesives (Histoacryl)

Band ligation and Sclerotherapy are the treatment of choice as they arrest bleeding in 80% of cases

3. Vasoactive Therapy

- Vasoactive drugs should be started as soon as variceal bleeding is suspected, ideally before endoscopy, and continued for several days.
- They reduce portal pressure by decreasing splanchnic arterial blood flow and portal venous inflow.
  - **Terlipressin**
    - Usually continued for 2–5 days.

  - Possible adverse effects include abdominal cramps, pallor, and ischemic complications.
  - Particular caution is required in patients with ischemic heart disease or other vascular disease.

- **o Somatostatin and Octreotide**
  - Reduce splanchnic blood flow and portal pressure.

- Have relatively few adverse effects.
- Octreotide is commonly used when terlipressin is contraindicated or unsuitable.

4. Additional Measures

- Prophylactic antibiotics : oral or IV.
- Measures to prevent encephalopathy.
- Vitamin K IV .
- PPI IV or H2 blockers.
- Sucralfate 1 g/ 6 hours to reduce esophageal ulceration after endoscopic therapy.

Balloon tamponade is now rarely used because of serious complications, particularly aspiration and esophageal injury.

Management of Failure or Early Rebleeding

Patients may rebleed within 10 days and in this case endoscopic therapy is repeated.

Transjugular Intrahepatic Portosystemic Shunt (TIPS)

- TIPS creates a channel between the portal vein and a hepatic vein, thereby decompressing the portal circulation.

Technique

- A guidewire is introduced through the jugular vein into the hepatic venous system.
- The portal vein is accessed through the liver.
- The tract between the hepatic and portal veins is dilated.
- A covered expandable metal stent is deployed to establish the portosystemic shunt.
    - TIPS markedly reduces portal pressure and is highly effective in controlling refractory variceal bleeding.
    - increased risk of hepatic encephalopathy.

Prevention of Recurrent Variceal Bleeding

Following an episode of variceal hemorrhage, the risk of recurrent bleeding is approximately 60–80% within 2 years, making secondary prophylaxis essential.

1. Endoscopic Variceal Ligation

- Repeated EVL sessions are performed, commonly at approximately 2-week intervals, until variceal obliteration.
- Follow-up endoscopy is performed to assess eradication and recurrence.

2. Non-selective β-Blockers

3. TIPS

4. Surgical Portosystemic Shunts

Surgical shunts decompress the portal circulation and reduce rebleeding but may produce significant hepatic encephalopathy.

5. Liver Transplantation

Measurement of Portal Pressure

Patency of the portal vein should be assessed before measurement of portal pressure by **Doppler ultrasound**

- **Direct measurement** of portal pressure is **invasive,** expensive and complicated (percutaneous transhepatic measurement - transjugular neasurement)
- **Indirect measurement** of portal pressure is safer and less invasive.

Hepatic vein catheterization: Cannulation of the hepatic vein, balloon occlusion of the hepatic vein and measurement of wedged vein pressure (WHVP). This actually measures the sinusoidal pressure.

Portal venous pressure gradient (PVPG): difference between portal pressure and pressure in the inferior vena cava.

-Hepatic venous pressure gradient (HVPG): is the difference between the wedged (occluded hepatic venous pressure and the free hepatic venous pressure.

- HVPG has a high correlation with PVPG (when portal hypertension is due to sinusoidal post sinusoidal lesion).

HVPG>12 mmHg is necessary for esophageal varices to form and bleed.

CHAPTER 24

Hydatid cyst

24

A common parasitic disease that is endemic in sheep-raising areas.

Mode of infection

1- Humans become infected by ingestion of eggs of the E- Granulosus, either by eating contaminated food or from contact with dogs .

2- The ingested embryos invade the intestinal mucosal wall and proceed to the liver via the portal venous system.

Structure of hydatid cyst:

1. Adventitia: layer of fibrous tissue surrounding the cyst = **Pseudocyst or**

- **pericyst**

2. Laminated membrane: the outer surrounding true layer **= Ectocyst**

3. Inner germinal cell layer: the most active layer of the cyst **= Endocyst**

4. Hydatid fluid containing hydatid sand, scolices, daughter cysts, and brood cysts – it is highly antigenic fluid

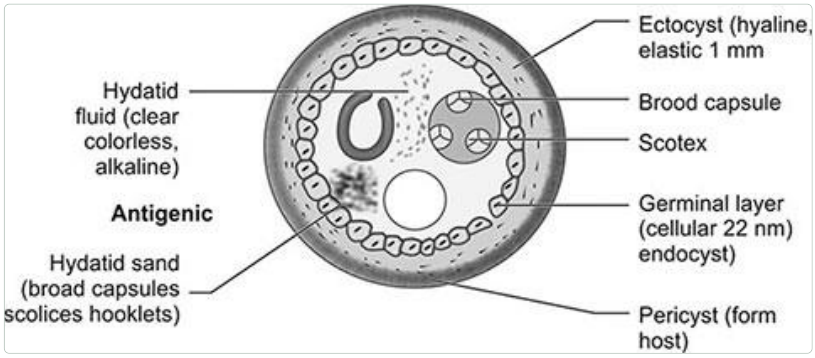


Figure 24.1 — Hydatid cyst

Sequelae of hydatid cyst:

1. The cyst may gradually enlarge and produces symptoms.

2. The parasite may die, fluid is absorbed, and the laminated membrane may calcify. A completely calcified cyst indicate a dead non- reactive cyst

3. May lead **to some complications.**

Clinical picture:

1- Most patients may be asymptomatic for years .

2- Non- specific picture:

a) Right upper quadrant or epigastric pain b) On examination → enlarged liver and a palpable mass

3- One third may present with

complications:

as infection, pressure or rupture.

A- Rupture of the cyst:

1- Intra-peritoneal rupture (spontaneous or post-traumatic).

2- Rarely, intra-thoracic or intra-pericardial rupture.

Clinically:

- Cyst rupture can be associated with allergic reaction as Urticaria, itching and erythema.
- Shock in cases of generalized anaphylactic reaction.
- With biliary rupture the classic triad of jaundice, biliary colic and urticarial rash occur.
- Multiple intra-peritoneal cysts → abdominal pain, peritonitis and ascitis.

Figure 24.2 — Hydatid cyst complications: source imaging and specimen

B- Secondary infection: infected cyst (picture of abcess)

- Tender hepatomegaly, rigors and fever.

C- Pressure complications:

- Pressure on main biliary ducts causing obstructive Jaundice.

Investigations:

1- ELISA has a sensitivity of 90% and is the initial screening tests of choice

Casoni test:

not used now.

2 - Ultrasound is the screening method of choice.

It is currently the primary diagnostic technique and has diagnostic accuracy of 90%

3- C.T. and MRI:

- A well-defined cyst with thick wall

- Dense wall calcifications are present in 50% of cases.
- Daughter cysts are identified inside.

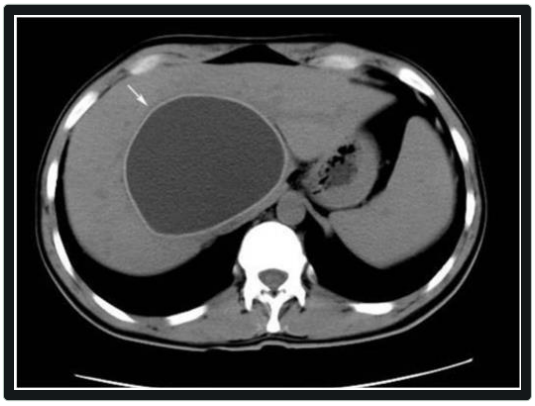


Figure 24.3 — Hydatid cyst: CT view

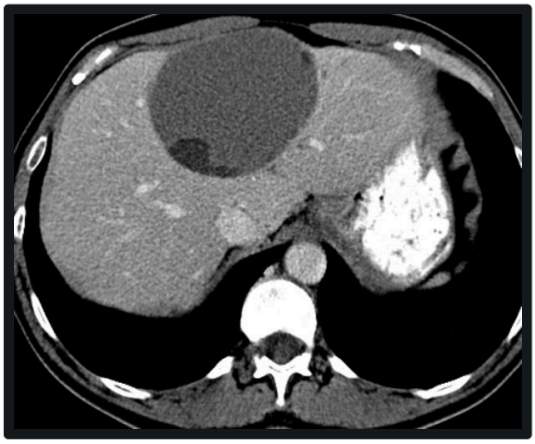


Figure 24.4 — CT: Hydatid cyst

Management of hydatid cyst in the liver:

Lines of treatment:

in symptomatic cases

1- Medical treatment → Albendazole

2- Surgical → is the main line used in symptomatic cases not responding to medications

3- Per-cutaneous aspiration. Needs special precautions.

- **Surgery:**
  - Surgery remains the gold standard treatment for hydatid liver disease.
- **Types of surgical procedures:**

1- Hydatid cystectomy (removal of the cyst only).

2- Peri-cystectomy (Cyst + part surrounding liver tissue)

3- Liver resection mainly in left lobe cyst

Aim of surgery:

- Injection of scolicidal solution inside the cyst as hypertonic saline or formaline
- Aspirate the content before removal.

The surgery Comprising removal of:

- Laminar layer
- Germinal layer
- Cyst contents (daughter cysts and brood capsules).
- Preventing spillage of the cyst contents in the peritoneal cavity (Green towels were placed around the liver)
- Obliterate the residual cavity.

Figure 24.5 — Hydatid cyst surgery

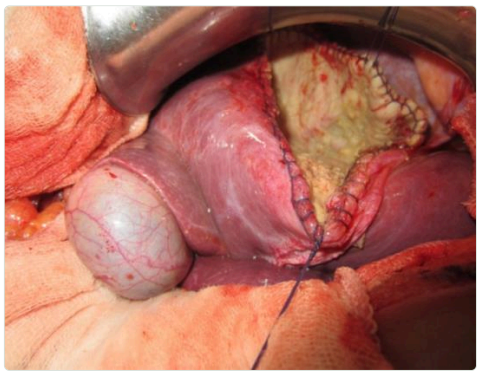


Figure 24.6 — Hydatid cystectomy

- **Percutaneous Treatment of Hydatid CYST:**

PAIR:

(puncture, aspiration, injection, and re-aspiration) is a percutaneous treatment technique for hydatid disease.

- This technique has become more common and its safety and efficacy have been reported.
- Done under radiological guidance as U/S or C.T. scan
- It has been increasingly accepted as a treatment option for hydatid disease in special centers.
- Needs well trained persons and precautions to avoid peritoneal rupture

CHAPTER 25

Pyogenic liver abscess

25

Sources (routes) of infection:

1- Portal vein :

Infection causes portal pyemia reaching the liver producing abscess.

- Acute suppurative appendicitis and appendicular abscess **(most common).** • Acute diverticulitis of the colon.
- Diverticular abscess.
- Infected Meckel 's diverticulum
- Inflammatory bowel disease
- Perforated hollow viscus.

2- Bile duct: (the most common)

- Extra-hepatic biliary obstruction leading to ascending cholangitis and abscess formation is the most common cause.

  e.g. CBD stones – stricture – malignant obstruction

  - Biliary-enteric anastomoses (choledocho-duodenostomy or choledocho-jejunostomy)

3- Hepatic artery:

Infection reaches the liver through systemic bacteremia caused by:

- Bacterial endocarditis
- Renal infection

4- Sub phrenic abcess :

directly reach the liver

5- Umbilical sepsis :

infection reaches liver via para-umbilical veins

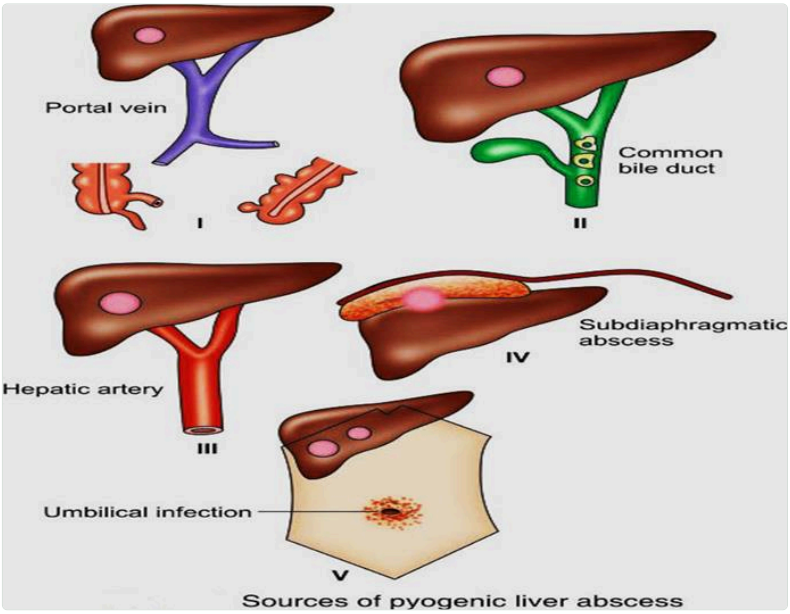


Figure 25.1 — Sources of pyogenic liver abscess

Causative Organisms:

gram –ve bacilli

The microorganisms most commonly isolated from blood and abscess cultures are as follows:

- **E coli – The commonest**
- Klebsiella pneumoniae
- Bacteroides species
- Streptococci and staph.
- Mixed infection is common

Prognosis

1- When left untreated, pyogenic liver abscess is associated with a mortality of 100 % . 2- With early diagnosis, appropriate drainage, and proper antibiotic therapy, the

Prognosis has improved markedly.

3- Poor prognostic factors are as follows:

- Age older than 70 years
- Multiple abscesses
- Poly-microbial infection
- Presence of associated malignancy or immunosuppressive disease

Clinical picture:

patient is markedly toxic and ill

General symptoms and signs

1- Fever and rigors 2- Toxic look and pallor 3- Weight loss

Local symptoms:

1- Right hypochondrial pain

2- Repeated vomiting

3- History of biliary disease as gall stones or jaundice.

4- History of G.I.T. infection as appendicitis.

Local signs:

1- Enlarged tender liver

2- Tenderness and rigidity in right hypochondrium.

Complications:

1- Septic shock → (fever, tachycardia and hypotension) fatal if neglected.

2- Rupture into the pleural cavity → empyema.

3- Peritoneal rupture → generalized peritonitis (generalized tenderness and rigidity).

Investigations:

Laboratory:

- CBC → Leucocytosis – increased ESR
  - Blood cultures for bacteria to identify the bacteria and determine which antibiotic is needed
  - Renal and liver function tests.

Radiological:

1- Abdominal ultrasound to check for an abscess in the liver

2- **CT scan** with intravenous contrast, to detect site and size of the abscess cavity.

3- Diagnostic Aspirations through US or CT scan for culture and sensitivity

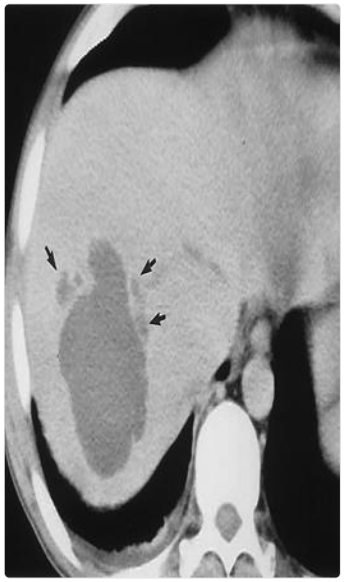


Figure 25.2 — Acute abcess within the right lobe

Management of pyogenic liver abcess

- **Treatment approaches:**

1- Initiation of Antibiotics therapy → a combination of two or more antibiotics is given intravenously (e.g. Third generation cephalosporins + Metronidazole). 2- **Percutaneous drainage** is the treatment of choice:

- Must be guided by US or CT scan.
- Useful in critical ill patient.
- Used in solitary and superficial abscess.
- Usually followed by placement of a drainage catheter.
- The drain is removed when the abscess cavity collapses, as confirmed by CT.

Contra-indications:

- Ascites
- Coagulopathy (Bleeding tendency)
- Associated malignancy
- Multiple abcesses.

3- Surgical drainage:

- Used in failure of medical and per-cutaneous drainage.
- Used in multiple abcesses.
- In complicated cases with **peritonitis**.
- Open drainage of pyogenic liver abscess may be:

a) Extra-peritoneal approach.

b) Trans-peritoneal approach.

Amebic liver abscess

- It is the most frequent extra-intestinal manifestation of Entamoeba histolytica infection.
- This infection is caused by the protozoa E- histolytica, which enters the portal venous system from the colon.

Pathophysiology

E- Histolytica exists in 2 forms:

- The cyst stage is the infective form.
- The trophozoite stage causes invasive disease.

Mode of infection:

- Feco-oral route → cysts are ingested → reach the small intestine → trophozoites are released and settle in the cecum → Amebae then enter the portal circulation andtravel to the liver where they form one or more amebic abscesses.
- The abcess cavity has thick wall and contain necrotic tissues and brown, thick fluid (called “anchovy paste”).

Right or left lobe ?

- The right lobe of the liver is more commonly affected than the left lobe.
- This has been attributed to the facts that:

1- The right lobe portal blood flow is supplied predominantly by the superior mesenteric vein, whereas the left lobe portal blood flow is supplied mainly by the splenic and inferior mesenteric vein.

2- The right lobe receives more blood than left so the chance of infection is more.

Clinical picture:

A) General symptoms:

1- Fever is the most common symptom

2- Sometimes rigors (in acute cases)

3- Weight loss is more in chronic cases

B) Local symptoms:

1- Some patients give a history of dysentery (bloody diarrhea) within the previous few months (key for diagnosis) .

2- Abdominal pain:

- The most common symptom.
- Pain is usually constant and dull aching, frequently located in the right upper quadrant.

3- Nausea and vomiting.

4- Diarrhea sometimes present

C) General signs:

1- Fever present in all patients

2- Toxic look

D) Local signs:

1- Hepatomegaly

2- Right upper abdominal tenderness is a typical finding.

- When the abscess is located in the left lobe, epigastric tenderness is noted.

3- Signs of peritonitis:

As rebound tenderness, guarding, and absence of bowel sounds, are present when the abscess ruptures into the peritoneal cavity.

Complications:

1- Rupture in the pleural cavity → pleural effusion or empyema

2- Rupture in peritoneal cavity → peritonitis.

3- Rupture in the pericardial sac. → with pericardial effusion in left lobe abcess

4- Jaundice when large abscess compressing the biliary tract.

Investigations:

A- Radiological:

1- **Abdominal ultrasound** : detect cystic lesion (abcess) in the liver

2- Abdominal CT scan or MRI:

- Amebic abscesses usually appear as rounded, well-defined lesions with the presence of complex fluid
- Thick wall (3–15 mm) and a peripheral zone of edema around the abscess are common and characteristic with sometimes multiple septa are present inside.
- Usually solitary.

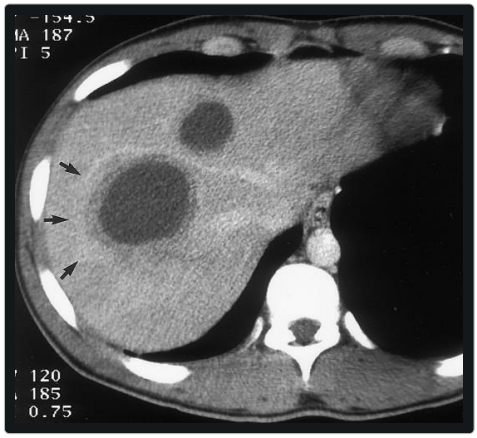


Figure 25.3 — Amebic abcess

B- Liver abscess aspiration with C/S:

- Check for bacterial infection in the amebic liver abscess
- The aspirate is brown, thick fluid (called “anchovy paste”).

C-Lab:

- CBC: Leukocytosis and esinophilia
  - Liver function tests: are elevated in case of marked amebic hepatitis.
  - Serological tests for amebiasis: ELISA
  - Patients with amebic liver abscesses will develop antibodies to E. histolytica.
  - Stool testing for amebiasis: Non-specific.

Treatment:

1- Amebicidal therapy is generally highly effective. Used in combination with broad spectrum antibiotics

2- Per-cutaneous catheter drainage of amebic abscess is rarely necessary and used only in impending rupture of an abcess. Done under guidance of C.T. or U/S.

3- Surgical drainage: extra-peritoneal or trans-peritoneal approaches

Amebicidal therapy:

- The treatment of choice is **Metronidazole**, • The cure rate for which is **> 90%.**
- Dose: 500-750 mg IV every 8 hours for 10 days.
- Resolution of symptoms is fairly rapid and is observed within 3 days in most patients.

CHAPTER 26

Hepato-cellular carcinoma (HCC)

26

Definition

Hepatocellular carcinoma (HCC) is the most common primary malignant tumor of the liver and arises from hepatocytes. It usually develops in a chronically diseased liver, particularly in patients with cirrhosis, but may also occur without cirrhosis, especially in chronic hepatitis B.

Risk Factors

The major risk factors include:

1. **Liver cirrhosis** — the major risk factor regardless of cause.
2. **Chronic viral hepatitis**
   - Chronic hepatitis B — HCC may occur with or without cirrhosis.
   - Chronic hepatitis C — risk is particularly high with advanced fibrosis or cirrhosis.
3. **Alcohol-associated liver disease**
4. **Metabolic dysfunction-associated steatotic liver disease (MASLD)** , particularly with advanced fibrosis or cirrhosis.
5. **Aflatoxin B1** exposure. 6. Other chronic liver diseases associated with advanced fibrosis or cirrhosis.

Clinical Presentation

HCC may be **asymptomatic** , particularly in its early stages, and may be detected during surveillance.

When symptomatic, patients may present with:

- Right upper quadrant or upper abdominal pain
- Weight loss and anorexia
- Fatigue
- Hepatomegaly or a palpable liver mass
- Jaundice
- Ascites
- Deterioration of previously compensated chronic liver disease

New deterioration of liver function or new portal vein thrombosis in a patient with chronic liver disease should raise suspicion of HCC.

HCC Surveillance

The aim of surveillance is to detect HCC at an **early stage when curative treatment may be possible** .

- Patients at sufficiently high risk, particularly those with cirrhosis, should undergo surveillance.

- **Abdominal ultrasound every 6 months** is the standard surveillance approach.
- AFP may be used as an additional biomarker in selected settings.
- Surveillance is not itself diagnostic; a suspicious lesion requires further evaluation.

Diagnosis

1. Ultrasound

Ultrasound is mainly used for surveillance.

A new or enlarging focal liver lesion in an at-risk patient requires further evaluation.

2. Multiphasic CT or MRI

HCC can often be diagnosed **without biopsy** in an appropriate high-risk patient when characteristic imaging findings are present.

The typical pattern is:

- **Arterial phase hyperenhancement (APHE)**
- Followed by **washout** in the portal venous or delayed phase.

3. Alpha-fetoprotein (AFP)

AFP may be elevated in HCC but has limited sensitivity and specificity.

- Normal AFP does **not exclude HCC** .
- Elevated AFP alone does **not establish the diagnosis** .

4. Liver Biopsy

Biopsy is not routinely required when characteristic imaging establishes the diagnosis in an appropriate high-risk patient.

It may be required when imaging is indeterminate or atypical.

Staging

The **Barcelona Clinic Liver Cancer (BCLC)** system is commonly used to stage HCC and guide treatment.

It considers:

- Tumor extent
- Liver function
- General condition/performance status
- Vascular invasion or extrahepatic spread

HCC can be broadly classified into:

- **Early stage** — potentially curable.

- **Intermediate stage** — mainly treated with locoregional therapy in selected patients.
- **Advanced stage** — usually associated with vascular invasion or extrahepatic spread and treated mainly with systemic therapy.
- **Terminal stage** — severe liver dysfunction and/or poor general condition; treatment is mainly supportive.

Treatment

Treatment depends on **tumor stage, liver function, general condition, vascular invasion, and extrahepatic spread** .

1. Curative-Intent Treatment

Liver resection

May provide a potential cure in selected patients with resectable HCC and preserved liver function.

Liver transplantation

Treats both the HCC and the underlying diseased liver and is particularly valuable in selected patients with HCC and significant cirrhosis who meet transplant criteria.

Thermal ablation

Radiofrequency ablation (RFA) and microwave ablation (MWA) are important potentially curative treatments for selected small HCCs.

2. Locoregional Treatment

Transarterial chemoembolization (TACE) is an important treatment for selected patients with intermediate-stage, liver-confined HCC.

3. Systemic Treatment

Systemic therapy is mainly used for advanced or unresectable HCC, particularly in the presence of vascular invasion or extrahepatic spread.

Modern systemic treatment is based mainly on **immunotherapy and targeted therapies** .

Key Points

- HCC is the most common primary malignant tumor of the liver.
- **Cirrhosis is the major risk factor** , but HCC can occur without cirrhosis, particularly with chronic HBV.
- Surveillance aims to detect HCC at an early, potentially curable stage.
- **Ultrasound every 6 months** is the standard surveillance approach in high-risk patients.

- HCC can often be diagnosed non-invasively by characteristic findings on **multiphasic CT or MRI** .
- **AFP alone cannot diagnose or exclude HCC.**
- Treatment depends on **tumor stage and the condition of the underlying liver** .
- Curative treatments include **resection, transplantation, and thermal ablation** .
- **TACE** is an important locoregional treatment for selected intermediate-stage disease.
- Advanced HCC is mainly treated with **systemic therapy** .

CHAPTER 27

ASCITES

27

Ascites is the pathological accumulation of fluid in the peritoneal cavity , it is either due to a general cause as liver cirrhosis or right ventricular failure or due to a local cause as tuberculosis or malignant invasion of the peritoneum .

Cirrhosis is responsible for 80% of causes of ascites .

Several contributing factors are responsible for ascites formation , they are listed below.

A) Increased hydrostatic pressure:

1. Cirrhosis
2. Hepatic vein occlusion(Budd-Chiari syndrome)
3. Inferior vena cava obstruction
4. Constrictive pericarditis
5. Congestive heart failure I

B) Diminished colloid osmotic pressure:

1. End-stage liver disease with poor protein synthesis
2. Nephrotic syndrome with protein loss
3. Malnutrition

4 Protein-losing enteropathy

C) Increased permeability of peritoneal capillaries:

I . Tuberculous peritonitis

2. Bacterial peritonitis
3. Malignant disease of the peritoneum

D)Leakage of fluid into the peritoneal cavity: .

1. Bile ascites
2. Pancreatic ascites(secondary to a leaking pseudocyst)

3.Chylous ascites

4. Urine ascites

Miscellaneous causes:

1. Myxedema
2. Ovarian disease(Meigs' syndrome)
3. Chronic hemodialysis

Mechanism of ascites formation in liver cirrhosis :

- Portal hypertension leads to transudation of fluid in the peritoneal cavity .
- Renal retention of sodium.
- Splanchnic and arterial vasodilatation .

- Increase hepatic lymph production due to congestion of sinusoids .
- Hypoalbuminemia due to decrease hepatic synthesis .

  The mechanism involve inappropriate renal sodium and water retention either secondary to vascular changes (underfill and peripheral arterial vasodilatation theory) or as a primary event (overfill theory)

  In the underfill theory : stimulation of the rennin- angiotensin – aldosterone system (RAAS) occurs as result of reduced effective circulating volume

  Peripheral vasodilatation → diminished effective arterial blood volume → rise in rennin , aldosterone , vasopressin , and sympathetic stimulation → sodium and water retention and renal vasoconstriction —→ plasma volume expansion + portal hypertension → Ascites

  In the overfill theory : The liver send a signal which leads to renal sodium and water retention and in the presence of portal hypertension , fluid will accumulates in the peritoneal cavity

Clinical Picture:

The onset may be sudden due to a precipitating factor or may be gradual

Ascites may be graded as follow:

Grade 1 :Mild ascites: only detectable by ultrasound examination

Grade 2 : Moderate ascites : manifest by moderate symmetrical distention of abdomen Grade 3 : Large or gross ascites provokes marked abdominal distension

Examination :

- Physical signs suggestive of cirrhosis: Jaundice , palmer erythema, abdominal collaterals.....
- Abdominal distention & thin limbs “spider man’.
- Wide subcostal angle, divarication of recti, umbilicus is shifted downwards, everted ± hernia.
- Scrotal oedema and hydrocele .
- Abdominal wall veins may be apparent due to: caput medusa (portal hypertension) or due to compression of the IVC
- Palpation : Liver and spleen could be palpated by dipping. Fluid thrill could be felt
- Percussion : dullness in the flanks. Shifting dullness.

  Secondary effects : Pleural effusion may occur in 5-10% of patients usually right sided this is due to defect in the diaphragm. hydrothorax may be seen in the absence of ascites, due to negative intrathoracic pressure during breathing.

  Edema in lower limb: follow ascites, due to hypoalbuminemia.
- Cardiac apex is shifted up & out.

  Neck veins could be seen congested: due to increase intrapleural pressure & raised diaphragm.

Investigations :

- Ultrasonography: to diagnose ascites, whether fluid is free or localized, to detect liver cirrhosis

  Diagnostic paracentesis and ascitic fluid analysis:

  a) Cells: endothelial cells/lymphocytes/PNLs/malignant cells.

  b) Culture.

  c) Protein: ascitic fluid total protein (AFTP) < 2.5 g/dL.

  d) Albumin: is important to calculate Serum-Ascites Albumin Gradient (SAAG)

  SAAG = Serum albumin — ascitic fluid albumin.

  In cirrhosis, SAAG is > 1.1g/dL.

  Differential diagnosis of ascites based on SAAG

Low gradient SAAG < 1.1 g/dl	High gradient ≥ 1.1 g/dl
TB peritonitis	Cirrhosis
Pancreatic Ascites	Fulminant liver failure
Peritoneal carcinomatosis	Alcoholic hepatitis
Biliary ascites	Cardiac ascites
Nephrotic syndrome	Budd – Chiari syndrome

Differential Diagnosis of ascites and its causes:

1-Tuberculous ascites :

- History of TB toxemia.

  Matted intestinal loops.

  Abdominal lymph nodes may be enlarged.
- Ascitic fluid: Ascitic fluid total protein (AFTP) > 2.5 g/dL, SAAG < 1.1 g/dL , excess lymphocytes. Smear & culture for TB. 4- PCR.
- Laparoscopy and biopsy.

2 -Malignant ascites:

- Symptoms & signs of primary tumor.
- Ascitic fluid: - AFTP > 2.5 g/dL
- SAAG < 1 . 1 g/dL
- Malignant cells. - ”

  Increase Lactic acid dehydrogenase.

  3-**Chylous ascites** : It may be caused by malignant lymphoma. The paracentesis sample is characteristic as it is milky white due to ++ fat & clears with ether.

4-Constrictive pericarditis:

-Increase jugular venous pressure. - Chest X-ray :

calcification. - ECG & CT of the heart.

5-Budd- Chiari Syndrome:

Increased protein of ascitic fluid. Abdominal ultrasound: congested veins, caudate lobe is prominent. Doppler U/S: shows obstruction of hepatic veins.

6-Ovarian tumor:

- Resonance in the flanks, central distension. - Umbilicus is shifted upwards.

7-**Full bladder, pregnancy, flatulence, fatty**.

Management of cirrhotic ascites :

- Body weight and urinary volume are measured daily
- Measurement of serum and urinary electrolytes
- Diagnostic paracentesis
- Bed rest is advised

Salt restriction : This is the main stay of therapy in cirrhotic patients with ascites
-

- Diuretics :

Aldosterone antagonist : Spironolactone (Aldactone) , amiloride : They block

- sodium reabsorption in the distal and collecting tubules , and they conserve potassium

Loop diuretic : Furosemide (Lasix) inhibits sodium , potassium and chloride

- reabsorption from the ascending limb of loop of Henle .
- Two therapeutic approach :

Spironolactone alone : This can be used alone in mild ascites with a dose of 100 –
-

200 mg daily

- Combination therapy Spironolactone 100 mg and furosemide 40 mg
- Sufficient response means loss of 0.5 Kg/ day or a maximum of 1 Kg /day in the - presence of lower limb edema

  If the response is insufficient the dose of spironolactone and furosemide can be - increased gradually on the expense of development of side effects , maximum of dose of furosemid is 160 mg /day and of spironalactone is 400 mg / day .

- Salt-free albumin I.V followed by IV Furosemide increases intravascular volume - and increases diuresis

  Small volume paracentesis: for < 5 liters removed, I.V. and in this case volume - expanders are unnecessary .
- **Complications limiting diuretic therapy of ascites**
- Development of hepatic encephalopathy , in the absence of any other precipitating factor (It is better to leave the patient wet and wise than dry and demented !)

  Renal impairment: increase of serum creatinine by >100% to a value >2 mg/dl in patients with ascites responding to treatment

  Hyponatraemia: a decrease of serum sodium by >10 mmol/L to a serum sodium of <125 mmol/L

  Hypo- or hyperkalaemia: a change in serum potassium to <3 mmol/L or >6 mmol/L despite appropriate measures

  Incapacitating muscle cramps

Treatment options for refractory ascites

- An ascites is said to be resistant , if there is lack of response after dietary salt restriction& intensive diuretic therapy
- **Large volume paracentesis**: removal of > 5 L of fluid by therapeutic paracentesis with administration of IV albumin (8gm/ L removed) may decrease the post paracentesis hypovolemia. It is safe& effective.
- **TIPS**: bridges a branch of the hepatic vein with a branch of the portal vein . It may increase frequency of hepatic encephalopathy.
- **Liver transplantation**.

Bacterial infection in patients with liver cirrhosis

- Patients with cirrhosis are at risk of developing bacterial infections due to various factors including liver dysfunction , portosystemic dysfunction , bacterial translocation and immune dysfunction
- Bacterial infection in cirrhotics should be treated quickly and effectively with potent antibiotic
- The most common sites for bacterial infection in patients with liver cirrhosis that requires potent empirical antibiotic treatment are : the ascitic fluid (spontaneous bacterial peritonitis) , lung (pneumonia) , Urinary tract (pyelonephritis) and skin (cellulitis)

Spontaneous Bacterial Peritonitis (SBP)

- It represents a serious complication of ascites with cirrhosis , and it affects 10% of cirrhotic patients. The prognosis has improved over time, but mortality rates remain relatively high (about 20% for SBP]

Cause:

Root of infection:

a) Hematogenous spread b) Translocation of bacteria through the intestinal wall

- Organisms: monomicrobial (single organism), E-coli, pneumococci, Klebsiella and gram-positive organisms.
- Abnormal host defenses: due to .decrease ascitic opsonins and abnormal neutrophil function.
- SBP is an indication to list the patient for liver transplantation

Clinical Picture & Diagnosis:

- The condition should be suspected in any cirrhotic patient with ascites who clinically deteriorates . The patient becomes drowsy and confused .
- Symptoms and signs of infection : fever, abdominal pain, rebound tenderness, nausea, vomiting& may be diarrhea.
- Diagnostic paracentesis: polymorphs in ascitic fluid > 250 cells/mm3. Ascitic fluid culture.

Treatment:

- Albumin has been recommended in association with antibiotics for the treatment of SBP
- Third-generation cephalosporins are recommended as first-line antibiotic treatment for community-acquired SBP in countries with low rates of antibiotic resistance
- In countries with high rates of antibiotic resistance piperacillin/tazobactam or carbapenem should be considered
- Antibiotic efficacy should be checked with a second diagnostic paracentesis at 48 hours from starting treatment .Suspicion of failure of first-line antibiotic if worsening clinical signs and symptoms and/or increase, or no marked reduction in leucocyte count (at least 25%) in 48 hours
- Example of antibiotics used in SBP : Cefotaxime 2g/8h IV , ceftriaxone 1g/ 12 h ,Ciprofloxacin 500mg/12hours

Prognosis: 25% of patients die. 70% recurrence in the first year.

Prophylaxis:

Primary prophylaxis : Patients with cirrhosis and low ascitic fluid protein concentration (<10 g/L) and/or high serum bilirubin levels are at high risk of developing a first episode of SBP , they can be given primary prophylaxis by norfloxacin 400 mg/day .

Secondary prophylaxis : Norfloxacin 400mg/day for life, after the first attack

Patients doing sclerotherapy should receive norfloxacin 400mg/12 hours days.

Norfloxacin prophylaxis should be stopped in patients with long-lasting improvement of their clinical condition and disappearance of ascites

CHAPTER 28

Renal impairment in liver cirrhosis

28

Hepatorenal syndrome (HRS) : It is a functional liver failure occurring in patients with liver cirrhosis , portal hypertension and ascites , the histology of the kidney is normal , The hallmark of HRS is renal vasoconstriction, although the pathogenesis is not fully understood , however the arterial vasodilatation theory is the most accepted Peripheral and splanchinic vasodilation → Decreased effective blood volume

→ Activation of RAAS, ADH, sympathetic → Increase Renal vasoconstrictors (rennin , ADH, angiotensin II, sympathetic stimulation) this is associated also with decrease renal vasodilators (PGE2, NO)

- **Acute kidney injury (AKI)** : It is a common complication of advanced cirrhosis, and is predictive of a worse outcome.
- Early recognition of AKI and early identification of the phenotype of AKI are essential to initiate appropriate therapy at an early stage and to improve outcome.
- Patients with advanced cirrhosis have reduced muscle mass (especially women) and decreased creatine production. Thus, patients with serum creatinine within the normal range may have markedly impaired renal function.
- The definitions of AKI in cirrhosis have recently been revised to take into account these difficulties. .AKI in cirrhosis is now defined by an increase in serum creatinine ≥ 0.3 mg/dL within 48 hours or a percentage increase of serum creatinine ≥ 50% from baseline which is known or presumed to have occurred within the prior 7 days.

  AKI is then categorized into 3 stages of increasing severity :

Differentiating types of AKI

- All types of AKI can occur in patients with cirrhosis (Pre-renal, HRS, acute tubular necrosis (ATN) and post-renal)
- Key point is to differentiate HRS-AKI from ATN
- Classification of HRS was recently revised

  – HRS-AKI (formerly type I HRS)

  – HRS includes renal impairment that fulfills the criteria of HRS but not of AKI (non-AKI-HRS or NAKI) (formerly type II HRS)

Diagnostic criteria for HRS-AKI

Cirrhosis and ascites

No response after 2 consecutive days of diuretic withdrawal and plasma volume expansion with albumin 1 g per kg of body weight

Absence of shock

No current or recent use of nephrotoxic drugs (NSAIDs, aminoglycosides, iodinated contrast media, etc.)

– No macroscopic signs of structural kidney injury, defined as: Absence of proteinuria (>500 mg/day) and absence of microhaematuria (>50 RBCs per high power field) and normal findings on renal ultrasonography

All patients meeting the current definition of **HRS-AKI** should be treated with

vasoconstrictors and albumin

.A distinction must be made between HRS and acute tubular necrosis (ATN) because vasoconstrictors such as terlipressin are only justified in the presence of HRS. Besides criteria for the clinical definition, urinary neutrophil gelatinase (NGAL) associated lipocalin, a biomarker of tubular damage, is significantly higher in patients with ATN compared to those with HRS.

Terlipressin can be administered by IV boluses (1 mg every 4–6 hours) or by continuous IV infusion (2 mg/day)†

In case of non-response (decrease in SCr <25% from the peak value) after 2 days, the dose of terlipressin should be increased in a stepwise manner to a maximum of 12 mg/day

Albumin solution (20%) should be used at 20–40 g/day

Serial measures assessing central blood volume can help to titrate the dose of albumin to prevent circulatory overload

Noradrenaline can be an alternative to terlipressin‡

Requires a central venous line often in an ICU

Midodrine + octreotide can be an option when terlipressin or noradrenaline are unavailable (but efficacy is much lower)

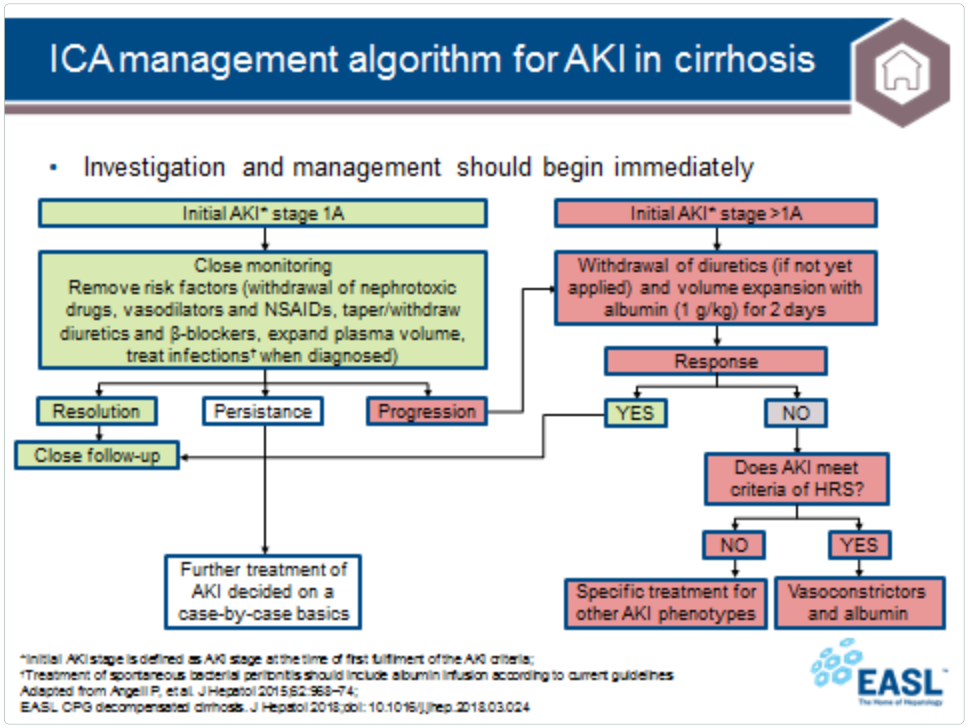


Figure 28.1 — ICA management algorithm for AKI in cirrhosis

CHAPTER 29

JAUNDICE

29

- Jaundice, from the French jaune (yellow), is the yellow-orange discoloration of the skin, conjunctivae, and mucous membranes that results from elevated concentrations of bilirubin in plasma.
- Jaundice becomes evident at plasma bilirubin concentrations of 3 to 4 mg/dL, depending on the patient's normal pigmentation.

Approach to the patient with jaundice

- The bilirubin present in serum represents a balance between input from production of bilirubin and hepatic/biliary removal of the pigment.
- Hyperbilirubinemia may result from (1) overproduction of bilirubin; (2) impaired uptake, conjugation, or excretion of bilirubin; or (3) regurgitation of unconjugated or conjugated bilirubin from damaged hepatocytes or bile ducts.
- In other words hyperbilirubinemia and jaundice may result from A) isolated disorders of bilirubin metabolism, B) liver disease, or C) Obstruction of the biliary tract.
- An increase in unconjugated bilirubin in serum results from either 1) overproduction, 2) impairment of uptake, or conjugation of bilirubin.
- An increase in conjugated bilirubin is due to 1) decreased excretion into the bile ductules or 2) backward leakage of the pigment.
- The initial steps in evaluating the patient with jaundice are to determine (1) whether the hyperbilirubinemia is predominantly conjugated or unconjugated in nature (fractionation of Bilirubin), and (2) whether other biochemical liver tests are abnormal.

Differential Diagnosis of Jaundice and Hyperbilirubinemia

Isolated Disorders Of Bilirubin Metabolism

Unconjugated Hyperbilirubinemia

- Increased bilirubin production

Examples: hemolysis

- Decreased hepatocellular uptake

Examples: drugs (e.g., rifampin)

- Decreased conjugation

Examples: Gilbert's and Crigler-Najjar syndromes

Conjugated Or Mixed Hyperbilirubinemia

Decreased canalicular transport: Dubin-Johnson syndrome

Gilbert's Syndrome

- It is defined as benign, familial, mild, unconjugated hyperbilirubinaemia not due to haemolysis and with normal routine tests of liver function and hepatic histology.
- Jaundice is mild and intermittent.
- Deepening may follow an intercurrent infection or fasting and is associated with malaise, nausea and often discomfort over the liver.
- Patients with Gilbert's syndrome have a deficiency in hepatic bilirubin glucuronidation.
- Hyperbilirubinaemia is life long and not associated with increased morbidity.

Hepatocellular Conditions That May Produce Jaundice

Viral Hepatitis

Hepatitis A, B, C, D, and E

Epstein-Barr virus

Cytomegalovirus

Alcohol

Drug toxicity

Environmental Toxins

Vinyl chloride

Wilson's disease

Autoimmune hepatitis

Cholestatic Conditions That May Produce Jaundice

- Cholestasis is defined as the failure of normal bile to reach the duodenum.
- This may be due to pathology anywhere between the hepatocyte and the ampulla of Vater.
- The term 'obstructive jaundice' is not used, as in many instances no mechanical block can be shown in the biliary tract.

Syndrome Of Cholestasis

- Cholestasis is interference with bile flow or formation.
- This can occur anywhere between the hepatocyte and the ampulla of Vater.
- Functionally, cholestasis is defined as a decrease in canalicular bile flow.
- Morphologically, cholestasis is defined as the accumulation of bile in liver cells and biliary passages.
- Clinically, cholestasis is the retention in the blood of all substances normally excreted in the bile.

- The importance of the distinction between extra and intra-hepatic cholestasis is that symptoms and biochemistry may not separate them.
- There is a need for a diagnostic algorithm to differentiate between the two.

Classification

Extra-hepatic Cholestasis

- Encompasses conditions where there is physical obstruction to the bile ducts.
- Usually this is outside the liver.

Intra-hepatic cholestasis

- Includes those conditions where there is no demonstrable obstruction.

Intrahepatic Conditions Producing Cholestasis

Viral hepatitis

Fibrosing cholestatic hepatitis—hepatitis B and C

Hepatitis A, Epstein-Barr virus

Alcoholic hepatitis

Drug toxicity

Pure cholestasis—anabolic and contraceptive steroids

Cholestatic hepatitis—chlorpromazine

Chronic cholestasis—chlorpromazine

Primary biliary cirrhosis

Primary sclerosing cholangitis

Cholestasis of pregnancy

Benign postoperative cholestasis

Extrahepatic Conditions Producing Cholestasis

Malignant

Cholangiocarcinoma

Pancreatic cancer

Ampullary cancer

Malignant involvement of the porta hepatis lymph nodes

Benign

Choledocholithiasis

Postoperative biliary structures

Chronic pancreatitis

Clinical features of cholestasis

- Prominent features of cholestasis, both acute and chronic, are itching and malabsorption.
- Bone disease (hepatic osteodystrophy) and cholesterol deposition (xanthomas, xanthelasmas) are seen with chronic cholestasis, which is also associated with skin pigmentation.
- In contrast to the patient with hepato-cellular disease where there is malaise and physical deterioration, the cholestatic patient feels well.
- On examination, the liver is usually enlarged.
- Splenomegaly is unusual except in biliary cirrhosis where portal hypertension has developed.
- Steatorrhoea is proportional to the degree of jaundice. It is due to the lack of sufficient intestinal bile salts for the absorption of dietary fat and fat-soluble vitamins.
- Hepatic osteodystrophy: Bone disease is a complication of chronic liver disease and in particular chronic cholestasis.
- Bone pain and fractures occur.
- Possible mechanisms are osteomalacia and osteoporosis.
- Bone disease manifests as loss of height, back pain (usually mid-thoracic or lumbar), collapsed vertebrae and fractures particularly of ribs with minimal trauma.

Extra-hepatic Effects Of Cholestasis

- Itching and jaundice are self-evident, but there are numerous other less obvious effects of cholestasis.
- Cardio-vascular responses are abnormal and peripheral vaso-constriction in response to hypotension is impaired.
- The kidneys have an increased susceptibility to hypotension and hypoxic damage.
- The gastric mucosa is more susceptible to ulceration
- Thus although deeply jaundiced patients with cholestasis may appear well apart from itching, there are metabolic and functional changes that under the stress of surgical and non-surgical procedures may result in: acute renal failure, haemorrhage, wound dehiscence and an increased risk of sepsis

Laboratory findings in cholestasis

- All the constituents of the bile show an increased level in the serum. Conjugation of biliary substances is intact but excretion defective
- The serum conjugated bilirubin level is raised
- The serum alkaline phosphatase level is raised, usually to more than three times the upper limit of normal

- Serum γ glutamyl- transpeptidase levels are raised.
- Serum albumin and globulin concentrations are normal.
- The serum aspartate transaminase is usually less than 100iu/l.
- Urine. Conjugated bilirubin is present.

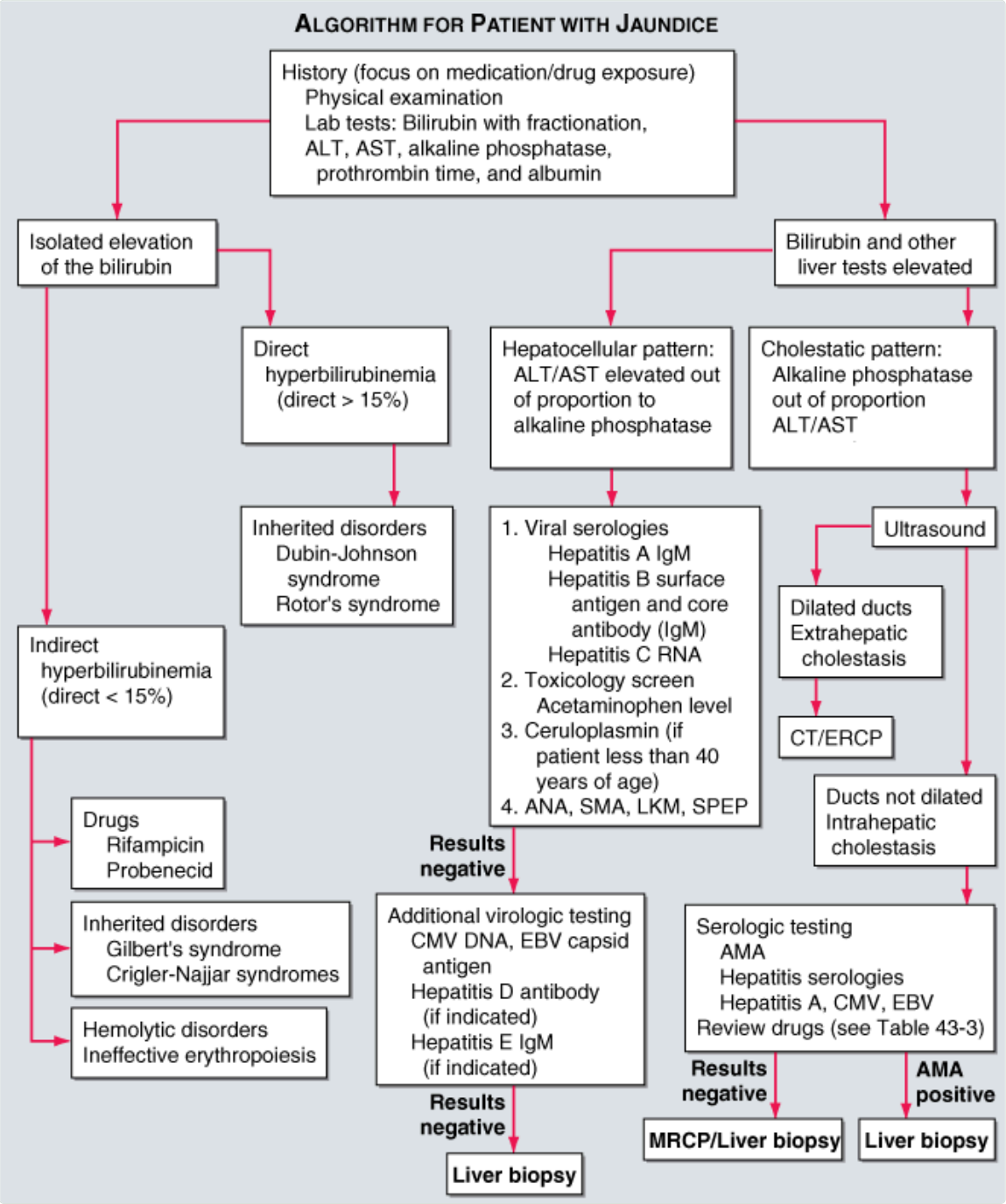


Figure 29.1 — Approach to the patient with jaundice

CHAPTER 30

HEPATIC ENCEPHALOPATHY

30

Objectives

The student will

1) Understand the definition and various forms of hepatic encephalopathy
2) Know the various causes of hepatic encephalopathy
3) Understand the pathogenesis of hepatic encephalopathy
4) Know the clinical presentation of hepatic encephalopathy
5) Know the diagnostic tools for hepatic encephalopathy
6) Understand the basic treatment options for hepatic encephalopathy

- Hepatic encephalopathy is a reversible neuropsychiatric state that complicates liver disease.
- A spectrum of syndromes exists, and it may accompany various forms of liver diseases.
- Types of encephalopathy

Type of encephalopathy	% Survival	Aetiological factors
Acute liver failure	20*	Viral hepatitis Alcoholic hepatitis Drug reactions and overdose
Cirrhosis with precipitant	70–80	Diuresis Haemorrhage Paracentesis Diarrhoea and vomiting Surgery Alcoholic excess Sedatives Infections Constipation
Chronic portal-systemic encephalopathy	100	Portal-systemic shunting Dietary protein intake Intestinal bacteria

Figure 30.1 — Types of encephalopathy

Clinical features

- The picture is complex and affects all parts of the brain.
- There are neurological and psychiatric components.
- For descriptive purposes, features of encephalopathy can be separated into changes in consciousness, personality, intellect and speech.

- Disturbed consciousness:

Hypersomnia appears early and progresses to reversal of the normal sleep pattern. Further deterioration results in reaction only to intense or noxious stimuli.

- Personality changes:

These include childishness, irritability and loss of concern for family.

- Intellectual deterioration

Varies from slight impairment of organic mental function to gross confusion.

- Speech

Is slow and slurred and the voice is monotonous.

- The most characteristic neurological abnormality is the 'flapping' tremor

(asterixis).

This is due to impaired inflow of joint and other afferent information to the brainstem reticular formation resulting in lapses in posture.

Sometimes arms, neck, jaw, protruded tongue, retracted mouth and tightly closed eyelids are involved, and the gait is ataxic.

Precipitants Of Acute Hepatic Encephalopathy In The Cirrhotic Patient

Electrolyte imbalance

Diuretics

Vomiting

Diarrhoea

Bleeding

Oesophageal and gastric varices

Gastro-duodenal erosions

Mallory–Weiss tear

Drugs

Infection

Spontaneous bacterial peritonitis

Urinary

Chest

Constipation

Large protein meal

Pathogenetic mechanisms

- The basic processes are 1) failure of hepatic clearance of gut derived substances, either through hepato-cellular failure or shunting, and 2) altered amino acid metabolism, both of which result in changes in cerebral neurotransmission. 3) Ammonia is thought to play an important role; other neurotransmitter systems are strongly implicated.

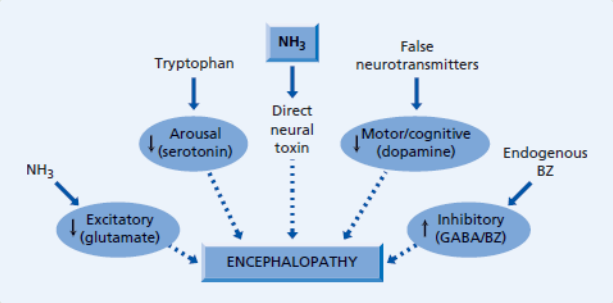
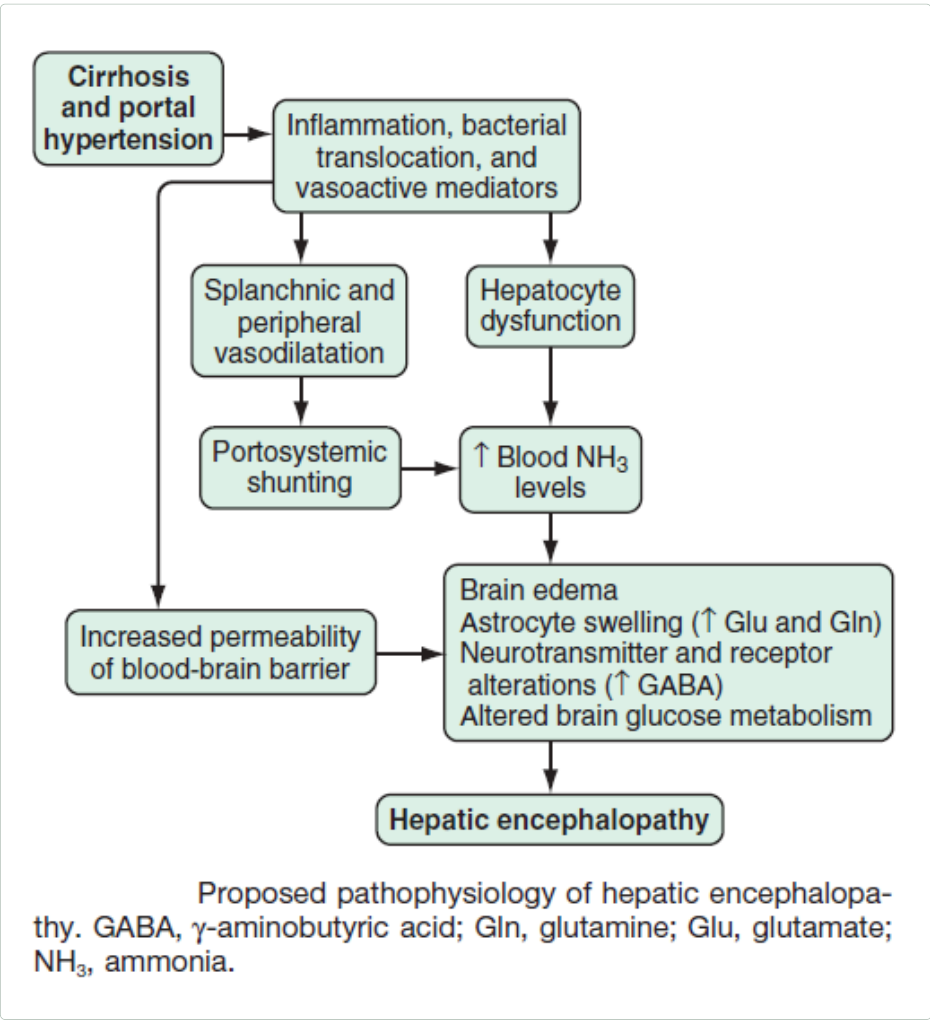


Figure 30.2 — Pathogenetic mechanisms of encephalopathy



Proposed pathophysiology of hepatic encephalopathy. GABA, γ-aminobutyric acid; Gln, glutamine; Glu, glutamate; NH_3, ammonia.

Figure 30.3 — Proposed pathophysiology of hepatic encephalopathy

Treatment of hepatic encephalopathy

Diet

> **University source discrepancy.** Handout p341 gives 20 g/day during an acute attack. The same handout's p279 nutrition update states 1.2–1.5 g/kg/day even with hepatic encephalopathy and permits occasional temporary reduction during severe overt encephalopathy. The 20 g/day statement below is retained as an unresolved source claim, not an editorial endorsement. See Cirrhosis — Nutrition.

- In the acute attack dietary protein is reduced to 20 g/day.
- In patients after an acute episode of coma, a normal protein intake is soon achieved.
- It is important in cirrhotic patients to avoid protein restriction for any longer than is necessary since these patients have a higher than normal protein requirement (1.2 g/kg/day) to remain in positive balance.
- Vegetable protein may be used. The latter is less ammoniagenic and contains small amounts of methionine and aromatic amino acids.

Antibiotics

- Neomycin is a nonabsorbable aminoglycoside given orally in doses of 2 to 6 g per day.
- Metronidazole has become the oral antibiotic of choice for acute management of hepatic encephalopathy when lactulose alone does not achieve adequate results.
  • Rifaximin (Xifaxan) has recently shown to be effective in the treatment of PSE in the dose range of 400 mg 2-3 times daily. It is a nonabsorbable, gut-specific antibiotic that alters the intestinal flora with any systemic side effects.

Lactulose and lactilol

- The human intestinal mucosa does not have an enzyme to split these synthetic disaccharides.
- When given by mouth lactulose reaches the caecum where it is broken down by bacteria predominantly to lactic acid.
- The osmotic volume of the colon is increased and the faecal pH drops.
- The colonic fermentative bacteria prefer lactulose to blood when both are present.
- The growth of lactose-fermenting organisms is favored and organisms such as bacteroides, which are ammonia formers, are suppressed.
- Faecal acidity would reduce the ionization and hence absorption of ammonia

(also amines and other toxic nitrogenous compounds).

- Side-effects include flatulence, diarrhea and intestinal pain.

Purgation

Lactulose or lactose enemas may be used and are superior to water

Sodium benzoate and L-ornithine -L-aspartate

Sodium benzoate promotes urinary excretion of ammonia and is as effective as lactulose and is less expensive.

L-ornithine-L-aspartate treatment promotes hepatic removal of ammonia by stimulating residual hepatic urea cycle activity and promoting glutamine synthesis, particularly in skeletal muscle.

Levodopa and bromocriptine

If portal-systemic encephalopathy is related to a defect in dopaminergic neurotransmission then replenishment of cerebral dopamines should be beneficial

Dopamine does not pass the blood–brain barrier, but its precursor, levodopa, does and can cause temporary arousal in acute hepatic encephalopathy.

Bromocriptine is a specific dopamine receptor agonist with a prolonged action.

Branched-chain amino acids

In cirrhotic patients the serum branched chain amino acids valine, leucine and isoleucine are low, and aromatic amino acid levels are increased.

Infusions of solutions containing a high concentration of branched-chain amino acids have been used to treat acute and chronic hepatic encephalopathy.

Hepatic transplantation

This may be the ultimate answer to the problem of chronic hepatic encephalopathy

CHAPTER 31

Acute Hepatic Failure

31

ACUTE HEPATIC FAILURE (FULMINANT HEPATITIS) AND JAUNDICE

Objectives

The student will

1) Understand the definition of fulminant hepatitis
2) Know the various causes of acute hepatic failure
3) Know the clinical features of acute hepatic failure
4) Know the diagnostic studies in acute hepatic failure
5) Know the basic treatment options for acute hepatic failure
6) Know the basic mechanisms of jaundice
7) Know the different causes of jaundice
8) Know the clinical presentation of jaundice
9) Know the basic diagnostic studies for jaundice

ACUTE LIVER FAILURE (FULMINANT HEPATITIS)

- Acute liver failure describes the clinical syndrome of severe impairment of liver function (Encephalopathy, Coagulopathy And Jaundice) Within 6 Months Of The Onset Of Symptoms.
- Coagulopathy and Encephalopathy are the twin cardinal features that reflect the severity of liver injury, and both are required to make the clinical diagnosis of

ALF.

- The vast majority of patients have no evidence of preexisting liver disease, but 2 exceptions to this requirement have been made: (1) an acute presentation of

Wilson disease in a young adult that presents in the setting of previously asymptomatic cirrhosis and (2) Reactivation of HBV infection, either spontaneously or as a consequence of immunosuppression or chemotherapy.

Classification

Classification of acute liver failure

Liver failure subcategory	Jaundice to encephalopathy	Clinical presentation	Common aetiologies	Prognosis
Hyperacute	0–7 days	Cerebral oedema common	Paracetamol, hepatitis A, ischaemia	Fair
Acute	8–28 days	Cerebral oedema less common	Hepatitis B, drugs	Poor
Subacute	29 days to 12 weeks	Cerebral oedema rare; ascites, peripheral oedema and renal failure more common	Drugs, indeterminate	Very poor

Figure 31.1 — Classification of acute liver failure

Causes:

Causes of acute liver failure
Infections
Hepatitis A, B, C, D, E
Herpes simplex
Epstein–Barr virus
Cytomegalovirus
Transfusion-transmitted virus (TTV)
Dengue fever
Drugs and toxins
Paracetamol
Carbon tetrachloride
Idiosyncratic drug reactions*
Mushroom poisoning
Sea anemone sting
Ischaemic
Cardiogenic shock
Hypotension
Heat stroke
Cocaine, methamphetamines, ephedrine
Vascular
Acute Budd–Chiari syndrome
Sinusoidal obstruction syndrome
Miscellaneous
Wilson's disease
Acute fatty liver of pregnancy
Eclampsia/ HELLP syndrome
Malignancy
Primary graft non-function after liver transplantation

Figure 31.2 — Causes of acute liver failure

Clinical Features And Diagnosis

The diagnosis of ALF is clinical and based on the presence of encephalopathy in a patient with acute liver injury manifested by coagulopathy.

The encephalopathy is usually overt and ranges from confusion to coma.

1) Encephalopathy

Encephalopathy is mandatory for a diagnosis of ALF. It is classically graded on a scale of 1 to 4

2) Intracranial Hypertension and Cerebral Edema

Cerebral edema has been a hallmark complication of ALF, major cause of death.

The clinical features of cerebral edema include systemic hypertension, decerebrate posturing, hyperventilation, abnormal pupillary reflexes, and ultimately impairment of brainstem reflexes and functions.

3) Hemodynamic Changes and Circulatory Failure

The initial profile is of a hyperdynamic circulation, increased cardiac output, and reduced systemic peripheral vascular resistance.

Progressive disease leads to circulatory failure either as a result of a falling cardiac output or an inability to maintain an adequate mean arterial pressure.

This is a common cause of death in ALF.

4) Infection

Patients with ALF are notoriously prone to infection The predominant bacteria were

Staphylococcus aureus and Streptococci.

5) Renal Failure

In most non-acetaminophen cases, renal failure tends to be a consequence of liver failure and develops later in the course of the disease.

6) Hematologic Abnormalities

In addition to decreased synthesis of coagulation factors by the liver, there is evidence of increased peripheral consumption.

Quantitative and qualitative defects in platelet function are well described in ALF.

Diagnostic Studies

Haematology
Complete blood count: white blood cells, haemoglobin, haematocrit, anti-hepatitis B core
Coagulation panel: prothrombin time/INR, factor V
Blood group

Biochemical
Serum chemistries: sodium, potassium, bicarbonate, chloride, urea, creatinine, calcium, magnesium, phosphate, glucose
Hepatic panel: aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase, albumin, total protein, total bilirubin

Arterial blood gas
pH, $Paco_2$, Pao_2, lactate, ammonia

Virology
Hepatitis B surface antigen and IgM anticore
Hepatitis A (IgM) antibody
Hepatitis C antibody (for underlying chronic infection)
HCV RNA
IgM hepatitis E antibody (in endemic areas)
Hepatitis D antibody if hepatitis B positive
HSV, CMV, EBV PCR (if history of immunosuppression)
Human immunodeficiency virus (if considering transplantation)

Figure 31.3 — Diagnostic studies: laboratory tests

Autoimmune markers
Antinuclear antibody (ANA)
Antismooth-muscle antibody (SMA)
Antiliver/ kidney microsome 1 (ALKM1)
Immunoglobulins

Toxicology
Paracetamol level
Blood alcohol
Urine drug screen

Miscellaneous
Urine copper
Pregnancy test

Microbiology
Blood culture, aerobic and anaerobic
Urine culture and microscopy
Sputum culture and microscopy

Other studies
Chest X-ray, electrocardiogram
Liver ultrasound with Dopplers
Electroencephalogram (EEG), non-contrast head CT (in some cases)
Liver biopsy

Figure 31.4 — Diagnostic studies: additional investigations

Management

1) Neurologic Complications

- Management of the portosystemic encephalopathy of cirrhosis has centered on lactulose and non – absorbable antibiotics. However, lactulose has shown no benefit in acute liver failure.
- There is not enough evidence to recommend non - absorbable antibiotics in acute liver failure. In cirrhosis, L-ornithine and L-aspartate (LOLA) treats encephalopathy by increasing muscle metabolism of ammonia but a recent randomized controlled trial showed no benefit in acute liver failure.
- Therefore, the management of the encephalopathy is essentially that of the underlying liver disease.

2) Treatment of cerebral edema

- Mannitol (0.25 – 1 g/kg i.v. bolus): use for severe elevation of ICP or first clinical signs of herniation, monitor urine output and serum osmolarity.
- Hypertonic saline (30% 5 – 10 mL/h): goal serum sodium 145 – 155 mmol/L, avoid rapid correction.
- Hyperventilation: effects short - lived; may use for impending herniation.

3) Infection

- Aseptic techniques.
- Surveillance for and prompt antimicrobial treatment of infection required.

- Antifungal coverage for patients not responding to broad spectrum antibiotics. • Antibiotic prophylaxis possibly helpful but not proven.

4) Coagulopathy

- Vitamin K (10 mg i.v. or subcutaneous).
- FFP: give only for invasive procedures or active bleeding.
- Platelets: give for platelet counts < 10 000/mm 3 or invasive procedures.
- Recombinant activated factor VII (40 μ g/kg bolus): give 30 min before procedure if coagulopathy refractory to FFP.
- Cryoprecipitate: for fibrinogen < 100 mg/dL and bleeding.
- Exchange plasmapheresis: allows transfusion of large amount of FFP.
- Prophylaxis for stress ulceration: give H2 blocker or PPI.

5) Haemodynamics/ renal failure

- Arterial line.
- Pulmonary artery catheterization.
- Volume replacement.
- Check cortisol and cosyntropin stimulation test; hydrocortisone 200 – 300 mg/day for adrenal insufficiency.
- Pressor support (noradrenaline preferred over dopamine or adrenaline) as needed to maintain adequate mean arterial pressure.
- Avoid nephrotoxic agents.
- Continuous modes of haemodialysis if needed, such as venovenous haemodialysis.

6) Metabolic concerns

- Follow closely: glucose, potassium, magnesium, phosphate.
- Consider nutrition: enteral feedings preferred over total parenteral nutrition.

Paracetamol hepatotoxicity

N-acetylcysteine is well established in the management of an acetaminophen overdose and effective in preventing liver injury when administered within 15 hours of acetaminophen Ingestion.

NAC is the antidote for paracetamol poisoning, repleting glutathione that detoxifies the harmful metabolite.

7) **Liver transplantation**

Hepatic transplantation has to be considered for patients reaching grade 3 and 4 coma due to acute liver failure.

CHAPTER 32

Anatomy of the biliary system

32

B iliary surgery

The gall bladder is second intra-abdominal organ most commonly requiring surgical intervention (after the appendix)

extrahepatic biliary system:

The extra-hepatic biliary tree consists of:

- Right and left hepatic ducts.
- Common hepatic duct. (CHD)
- Gallbladder and cystic ducts.
- Common bile duct (CBD)
- Ampulla of Vater.

parts of gallbladder:

- The gallbladder lies in the gallbladder fossa on the undersurface of the liver.
- This is a pear shaped structure with an average length of 7–12 cm and capacity of 30–50 mL.

The gallbladder consists of:

1. Fundus: Part of the gallbladder projecting beyond the inferior border of the liver.

2. Body: Part of the gallbladder extending from the end of the fundus to the neck of gallbladder.

3. Neck: The neck of the gallbladder forms an S-shaped curve and connects the body of the gallbladder to a narrow infundibulum which is continued as cystic duct.

4. From the neck a small diverticulum may project towards the duodenum. This is known as **Hartman's pouch**.

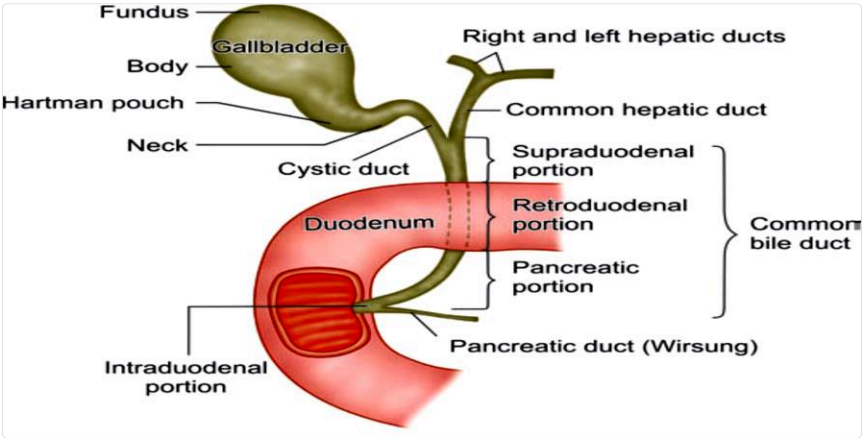


Figure 32.1 — Biliary system anatomy

Extra-hepatic biliary ducts:

- **The right and left hepatic duct**s emerge at the porta hepatis.
- **The left hepatic duct** has a longer course than the right hepatic duct.
- These two ducts join at the confluence to form the common hepatic duct.
- **The common hepatic duct** is 3 cm in length and runs in the free margin of the lesser omentum.
- **The cystic duct** emerges from the neck of the gallbladder is 3 cm long and joins the right margin of the CHD to form the common bile duct.
- **The common bile duct is** 7.5 cm long and is divided into four parts:

1.Supra-duodenal part:

2.5 cm long runs in the free margin of lesser omentum lying to the right of hepatic artery and infront of portal vein.

2.Retro-duodenal part:

Lying behind the first part of the duodenum.

3. **Infra-duodenal part** lies in the groove or in a tunnel in the posterior surface of the head of pancreas.

4.Intra-duodenal part:

Runs obliquely through the wall of the second part of the duodenum, dilates to form the ampulla of vater and joined by the pancreatic duct opens at the posteromedial wall of the duodenum at the major duodenal papilla.

Sphincters around the bile duct:

1. Sphincter choledochus (Sphincter of Boyden): Sphincter around the terminal part of the common bile duct.

2. Sphincter pancreaticus: This is the sphincter muscle around the terminal part of the pancreatic duct.

3. Sphincter of Oddi: Sphincter around the ampulla of Vater.

Blood supply of gallbladder and biliary tree :

- The gallbladder is supplied by the cystic artery which is usually a branch of right Hepatic artery. The cystic artery arises behind the common hepatic duct, crosses behind and enters the gallbladder.
- An accessory cystic artery arising from the gastro-duodenal artery may also supply the gallbladder.
- The bile duct is supplied by two vertical arteries arising from the hepatic artery running along the bile duct at 3 and 9 o'clock position and giving off circumferential arteries anteriorly and posteriorly.
- The cystic veins draining the gallbladder do not accompany the cystic artery. These veins pierce the fossa for the gallbladder and drains into the intrahepatic part of portal vein.

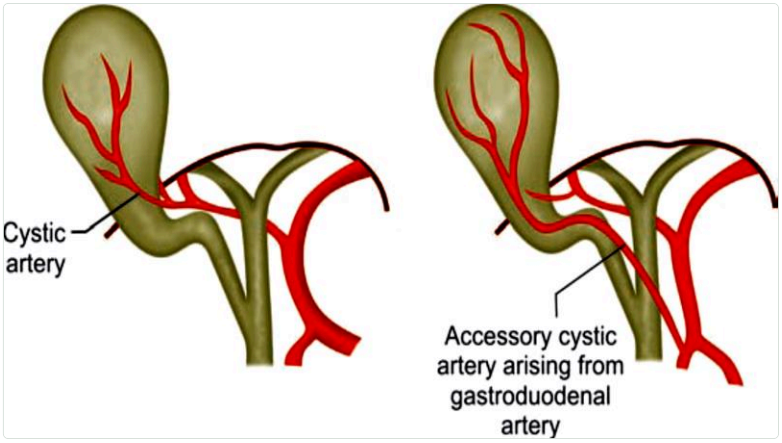


Figure 32.2 — Cystic artery anatomy

N.B. Moynihan's hump:

- This is one of the anomalies in relation to the course of hepatic artery.
- The hepatic artery makes a tortuous course in front of the bile duct near the entry of cystic duct.
- The cystic artery arising from this hump or the hump of right hepatic artery is usually short.
- This is a dangerous anomaly as hepatic artery may be confused with cystic artery and may be clipped during cholecystectomy.

Calot's triangle:

- Superiorly limb: cystic artery
- Right limb: cystic duct
- Left limb: CHD

Important to be identified during cholecystectomy for safety ligation or clipping of the cystic duct and artery and to decrease incidence of bile duct injury

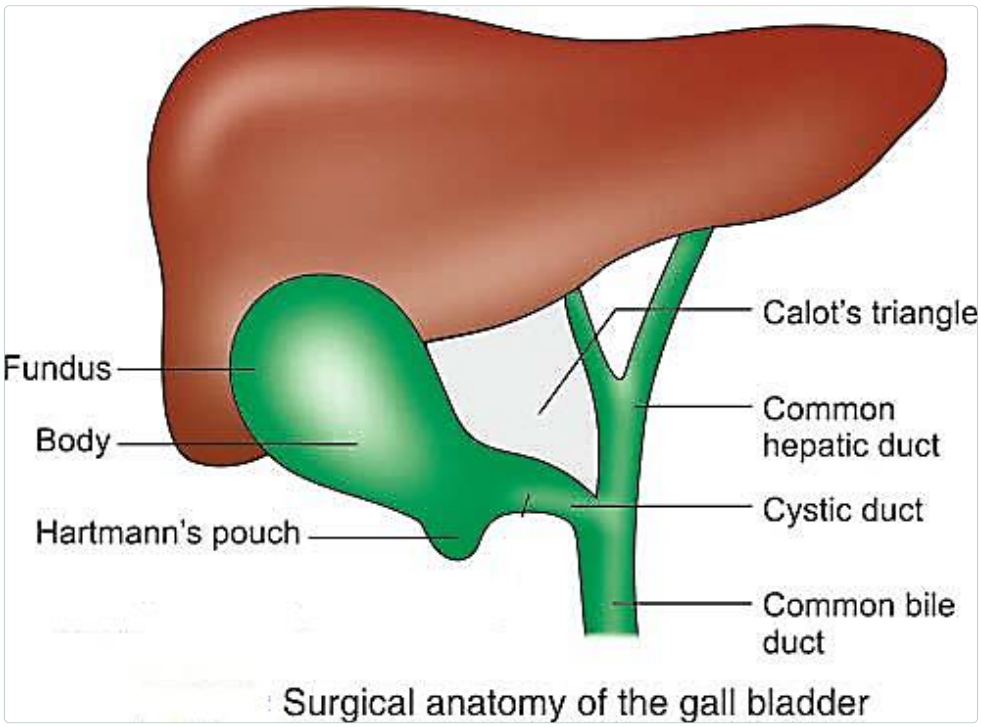


Figure 32.3 — Surgical anatomy of the gallbladder

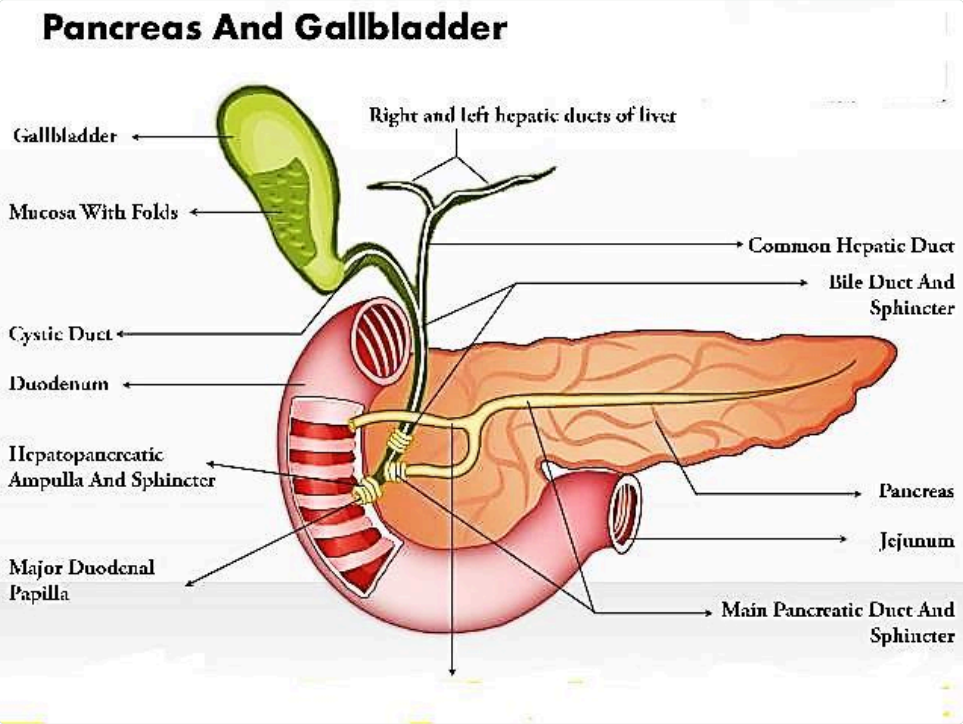


Figure 32.4 — Pancreas and gallbladder

N.B. Arrangement of the structures at free border of lesser omentum =hepato-duodenal ligament:

Formed of 2 layers and contains 3 important structures:

a. CBD most anteriorly to the right b. Common hepatic artery in the middle to the left c. Portal vein: posterior d. Epiploic foramen (Winslow): most posteriorly

Clamping of the hepato-duodenal ligament = Pringel's maneuver in severe bleeding from the liver or vascular injury in the liver hilum, hepatic artery or portal vein.

CHAPTER 33

Diseases of the gall bladder

33

- Congenital diseases
- Gall bladder stones
- Inflammations (acute and chronic cholecystitis)
- Complications (e.g. obstructive jaundice)
- Cancer G.B.

Congenital anomalies of the gall bladder

1. Agenesis: absent gall bladder
2. Double gall bladders
3. Ectopic (intra-hepatic , left side,..)

4. Phrygian cap:

- It the most common congenital anomaly related to the shape of gall bladder
- There is a septum or fold between the fundus and body

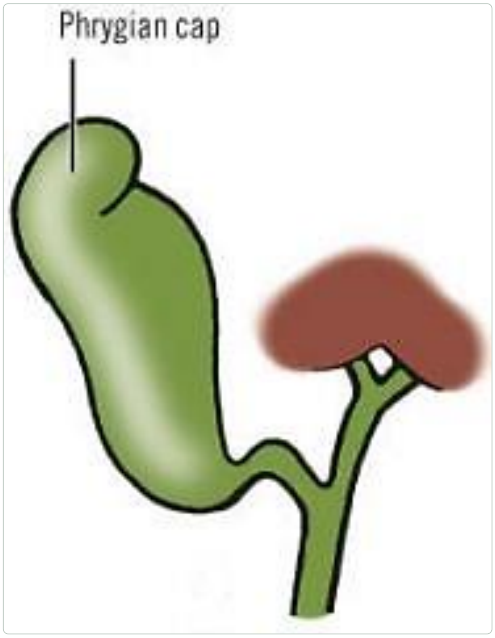


Figure 33.1 — Phrygian cap

5. Anomalies related to the position of the cystic duct and artery

6. Choledochal cyst:

- It is congenital cystic dilatation of the extra-hepatic bile duct mostly the CBD
- Clinically presented usually before 10 years.

Complications:

- Rupture with biliary peritonitis
- Stasis and cholangitis

- Liver abcess
- Stone formation
- Malignant transformation with cholangiocarcinoma
- Portal hypertension and portal vein thrombosis

Types of choledochal cysts:

Type	Description
Type I	Fusiform dilation of the common bile duct (80-90% of the cases).
Type II	Bile duct diverticulum (0-2% of the cases).
Type III	Saccular dilation of the common bile duct inside the duodenum wall, or choledochocele.
Type IV	Multiple intra- and extrahepatic bile dilations (10-15%). Variant A: intra- and extrahepatic affectation. Variant B: involves only the extrahepatic bile duct.
Type V	Fusiform or saccular cystic dilation of the intrahepatic bile ducts, whether or not associated to hepatic fibrosis (Caroli disease).

Figure 33.2 — Types of choledochal cysts

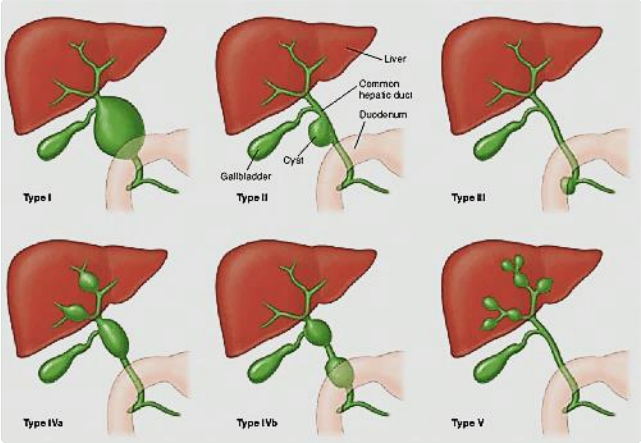


Figure 33.3 — Choledochal cyst classification diagrams

Treatment of choledochal cyst:

- Excision of the cyst with biliary reconstruction according to the type of the cyst e.g. excision with bilio-enteric anastomosis.

CHAPTER 34

Gall stones and chronic cholecystitis

34

Stone means solid substance formed from the constituents of bile within the biliary system.

Pathology

Incidence:

Gall stones are common; they are more common in fatty fertile females around 40 years, with the incidence rising with age.

Types of stones:

Gallstones are composed of two basic components, **cholesterol and bile pigment**.

Cholesterol stones:

1- **Pure cholesterol stones**

2- Mixed stones

I - Pure Cholesterol stones (20%)

They are whitish, usually solitary and big (2.5 cm).

- **The process of cholesterol stone formation:**

1- Cholesterol elimination occurs as a result of its secretion into bile as:

a- Cholesterol itself b- bile salt (25:1)

2- In bile, the cholesterol is kept in solution by secreted **phospholipids**.

- **Causes of Cholesterol stones:**
  - Gall bladder stasis → precipitation of cholesterol
  - Prolonged fasting → decreases water and increases concentration
  - **Any factor that increases plasma cholesterol** will increase bile cholesterol as:
    a) Obesity: (common in females and older ages) b) High fat in the diet c) Metabolic conditions (diabetes, familial hyper-lipidaemias)
    - **Pregnancy** → stasis → Precipitation
    - Stones may also occur as a result of **bile salt depletion** which occurs in mal-absorption of bile salts in the **distal ileum**, and cirrhosis.
- **Cholesterol is prevented from precipitation by:**
  - Bile salts (25 : 1)
  - Non-steroidal anti-inflammatory drugs
  - Caffeine

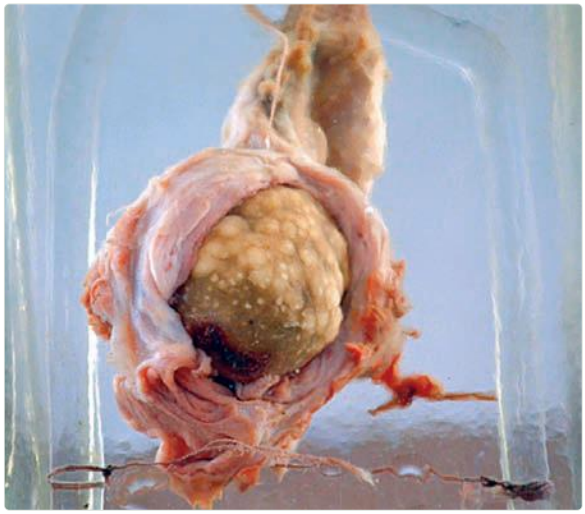


Figure 34.1 — Cholesterol stone

Cholesterol stone

II - Mixed stones

- Formed of cholesterol + bilirubin + debris + bacteria
- Main causes are stasis and super-added infection and hypercholesterolemia.
- Most common type of stones → **75 % or more.**
- **Mixed stone** s are **usually multiple dark or light brown and faceted.**

III - Pigment stones (5 – 10%)

Stones may occur as a result of increase in **bilirubin** load due to **haemolytic anaemias** .

- Stones are multiple small and black or brown.
  a) Brown pigment stones formed of bilirubin + calcium soaps+ bacteria

b) Black pigment:

high bilirubin + low calcium.

Figure 34.2 — Pigment and mixed stones

Clinical picture of gall stones

1- Non complicated gall stones are **Asymptomatic**

2- Fatty dyspepsia (common in fatty, fertile females around 40 years).

3 - Biliary pain (colic):

Biliary pain is caused by: **Obstruction** or **inflammation.** • Pain occurs when the neck of the gall bladder is obstructed by a stone and gall bladder contraction continues under the influence of cholecystokinin.

Character:

- The resultant visceral pain is therefore dull, poorly localised, • usually extends across the right hypochondrium, epigastrium and may radiate to the back, and can be felt at lower end of right scapula.
- This pain is of rapid onset and usually lasts a number of hours
- Relief occurs only when the stone either dislodges or, passes through the cystic duct into the bile duct. The patient remains unwell for 1–2 days.

4- Patient may be presented with complication as jaundice, pancreatitis, acute cholecystitis, ...

Abdominal examination:

- During the attack → tender epigastrium and right hypochondrium

Investigations

1 - Ultrasound:

a) Ultrasound has a major role in the diagnosis of gallstones.

b) The typical appearance is acoustic shadowing (dark area) radiating beyond the stone c) Determine the state of gall bladder (inflamed or not) and wall thickness d) Determine the diameter of the CBD and IHBR e) Determine the state of liver (cirrhotic or fibrotic)

2- Plain X-Ray abdomen → stones are mostly radiolucent (not used nowadays)

3 - Lab.

a) CBC → leukocytosis in associated acute cholecystitis b) Liver enzymes c) Total and direct bilirubin (for biliary obstruction) d) ALP & GGT (for biliary obstruction) e) PT & PTT (for bleeding disorders)

Treatment:

Gallstones: clinical situations and recommended management:

Incidental (asymptomatic):

elective laparoscopic cholecystectomy (in diabetics or multiple small stones).

Incidental (at laparotomy):

remove the gall bladder if technically suitable.

Biliary pain:

Elective laparoscopic cholecystectomy.

Acute cholecystitis:

Urgent cholecystectomy or delayed 6 weeks.

Post-endoscopic sphincterotomy for common bile duct stone: elective laparoscopic cholecystectomy.

Gallstone pancreatitis:

laparoscopic cholecystectomy

- **Types of operations:**

1- Open cholecystectomy

2- Laparoscopic cholecystectomy (best)

- **Why do you prefer laparoscopic cholecystectomy?**

Laparoscopic cholecystectomy has been established as gold standard for the treatment of Gall stone diseases

- Surgery is safe in the hands of a trained surgeon
- Less pain, less hospital stays
- Better Cosmetic
- Early return to work

Complications of Gall Stones

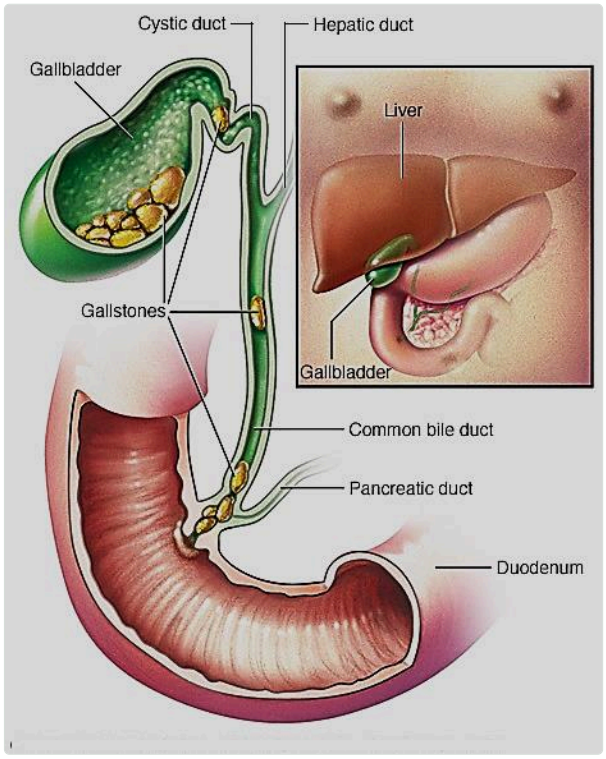


Figure 34.3 — Gallstone complications: biliary anatomy

A. Stones remaining in the gall bladde r

1. Biliary colic

2. Acute cholecystitis and its sequelae—gangrene/perforation/local abscess/biliary peritonitis

3. Chronic cholecystitis

4. Mucocele of gallbladder

5. Empyema of gallbladder

6. Carcinoma of gallbladder.

B. Stones migrated into the bile duct

1- Obstructive jaundice

2. Recurrent cholangitis

3. Acute pancreatitis.

C. Stones migrated into the intestine Gallstone ileus.

→

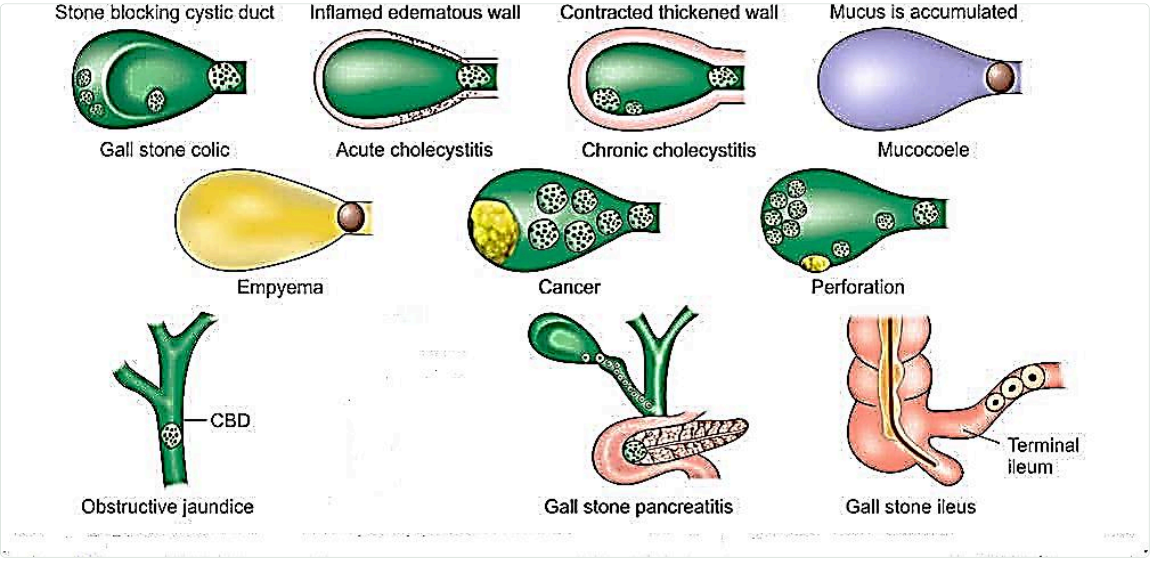


Figure 34.4 — Complications of gall stones

Acute cholecystitis

It means acute inflammations of the gall bladder

Types:

I- **Acute calcular cholecystitis → 90 %**

II- Acute non- calcular cholecystitis → 10 %

Etiology:

Commonest organism causing acute cholecystitis is E. coli

Routes of infection in acute cholecystitis:

1- Hematogenous— infection reaches the gallbladder via the cystic artery. **2- Through bile—**enteric organisms gains entry into the portal circulation and filtered by the liver and excreted in bile and reaches the gallbladder.

Risk factors for calcular cholecystitis

- Female sex
- Obesity
- Drugs (especially hormonal therapy in women)
- Pregnancy
- Increasing age (> 40 years)

Risk factors for non-calcular cholecystitis: (low resistance)

Critical illness

Major surgery or severe trauma or burns

Generalized Sepsis

Long-term total parenteral nutrition (TPN)

Prolonged fasting

Diabetes mellitus

Pathology of acute calcular cholecystitis

a) Stone obstructing the cystic duct the most important initiating factor b) Distension of the gall bladder with mucus retention → **Mucocele** c) More distension of the G.B. → ischemia of the wall → infection → **acute empyema** of the G.B.

d) More ischemia of the wall → **gangrene of the G.B. and perforation** → peritonitis

e) The perforation may occur at the fundus of the gallbladder or at the site of impaction of stone

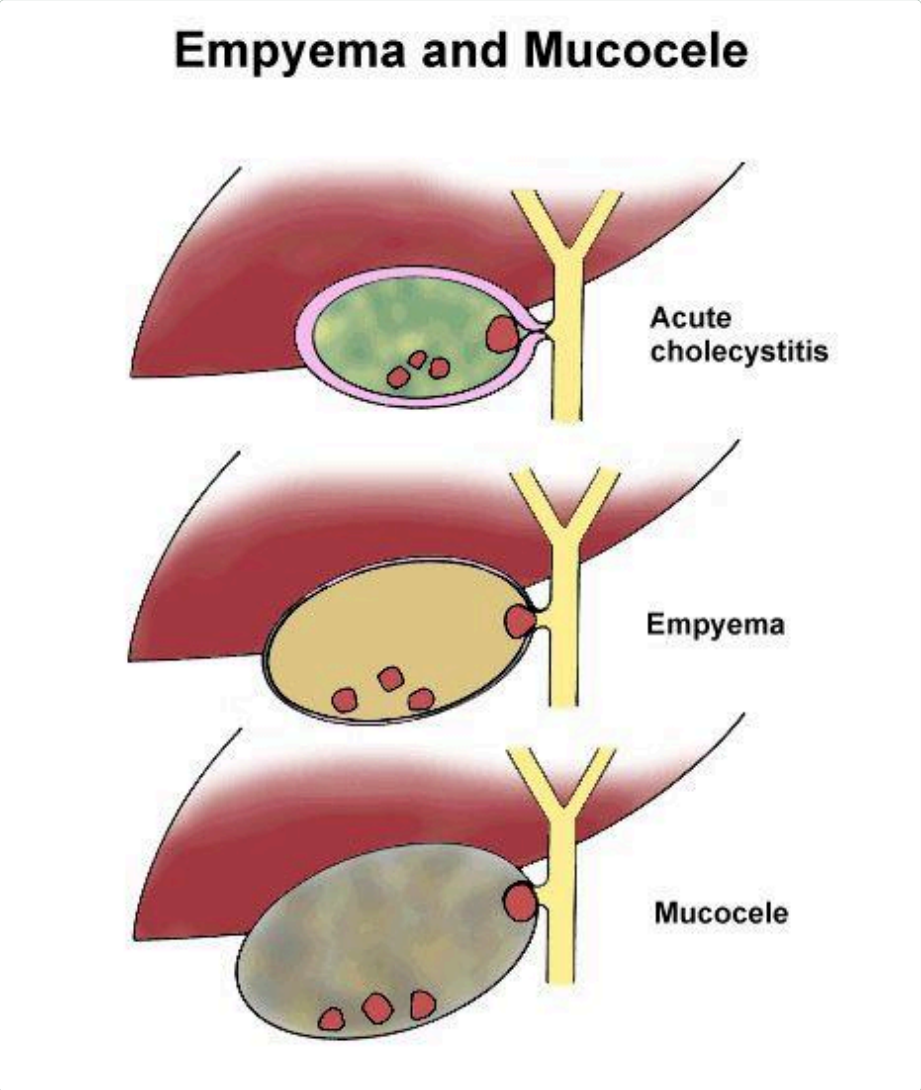


Figure 34.5 — Empyema and mucocele

Symptoms of acute cholecystitis

1- Right Upper abdominal pain is the most common presenting symptom

- The pain may radiate to the right shoulder or scapula
- Pain frequently begins in the epigastric region and then localizes to the right upper quadrant.
- Pain firstly is colicky but almost always becomes constant dull aching

2- Nausea and vomiting.

3- Fever

Signs:

Fever, tachycardia, toxic look

Tenderness in the Right hypochondrium or epigastric region, with guarding • and rebound tenderness

Palpable and tender gallbladder.

+ve Murphy's sign:

patient gets severe pain and holds his breath on dipping • the thumb of the examiner deep to the right costal margin at the region of the gall bladder.

D.D. of acute upper abdominal pain:

1. Acute cholecystitis
2. Acute cholangitis
3. Perforated peptic ulcer
4. Acute pancreatitis
5. Acute liver abcess
6. Acute sub-hepatic appendicitis
7. Intestinal obstruction
8. Acute pyelonephritis and renal colic

The sequelae of acute cholecystitis

Resolution:

Inflammation subsides and patient recovers.

Gangrene:

Infection may lead to gangrenous change in gallbladder manifested by • increasing pain, toxemia and appearence of rebound tenderness.

Perforation of gallbladder. Perforation may be:

- **i. Localized**—localized abscess formation manifested by severe pain, fever with chill and rigors and extreme tenderness in right upper quadrant of abdomen. **ii. Generalized perforation—**leading to generalized biliary peritonitis manifested by generalized pain abdomen, muscle guard or rigidity and extreme tenderness all over abdomen.

iii. Perforation into a neighboring viscus most commonly duodenum

Investigations

CBC → Leukocytosis.

SGPT and SGOT levels may be elevated in cholecystitis or with CBD • obstruction.

High Bilirubin and alkaline phosphatase may indicate CBD obstruction

Amylase/lipase assays are used to assess for pancreatitis; amylase may also be • mildly elevated in cholecystitis. Also differentiate the acute pancreatitis. (as D.D.)

Urinalysis is used to rule out pyelonephritis and renal calculi (as D.D.)

Married females of childbearing age should undergo pregnancy test. (as D.D.) •

Radiological investigations

1- Ultrasonography:

Is the preferred **initial imaging test** for the diagnosis of acute cholecystitis.

- Gallbladder wall thickening and edema (>2 -4 mm)
- Gallbladder distention (diameter > 4 cm, length > 10 cm)
- Peri-cholecystic fluid

2-Hepatobiliary scintigraphy (HIDA SCAN)

- An imaging test used to view the liver, gallbladder, bile ducts, and duodenum.
- If there is no sign of the radioactivity detected in the gallbladder → Diagnostic for acute cholecystitis (edema and cystic duct obstruction)
- Not done routinely

3- CT scan can identify complications of acute cholecystitis (Perforation)

4- Plain x-ray abdomen erect. (To exclude perforated P.U.)

Management

- Treatment of acute cholecystitis depends on the severity of the condition and the presence or absence of complications.
- **The initial treatment** includes:

a) Bowel rest, NPO (nothing per-oral) and nasogastric suction.

b) IV fluids, Analgesia, IV antibiotics:

(3rd generation cephalosporins and Metronidazole)

C) Vomiting can be treated with antiemetics

- **Surgical procedure:**

1- Laparoscopic cholecystectomy:

(standard for surgical treatment of cholecystitis)

2- Open cholecystectomy.

Indications:

- Diagnosis is established within 40 hours of the attack
- Failure of response to medical treatment
- Presence of complications as perforation
- after 6 weeks in postponed cases

  N.B. Surgery could be safely done within 48 hours of diagnosis. After that time the edematous tissues at the gall bladder area makes surgery to be difficult with high incidence of bile duct injury, so some prefer surgery after 6 weeks

Ascending cholangitis

Definition : spreading **infection** of the **bile ducts** (cholangitis)

Pathogenesis :

1 - **Bacteria** ascend through the CBD from its junction with the duodenum.

2- It tends to occur if the bile duct is already obstructed by gallstones, strictures or tumors**.**

3- High pressure →reflux of bacteria to blood → bacteremia → septicemia

Signs and symptoms

- I **Charcot's triad** :
  1- Right upper abdominal pain,
  2- Jaundice,
  3- Fever and rigors
  II- **Reynolds' pentad.**

Charcot's triad + Hypotension + Confusion

Treatment:

1- Combined antibiotics against gram – ve bacilli (3rd generation cephalosporins +

Metronidzole

2- Intravenous fluids

3- Treatment of the cause of bile obstruction (ERCP)

Chronic calcular cholecystitis

Clinical picture:

- May be asymptomatic and discovered accidentally
- Recurrent attacks of epigastric or right hypochondrial pain (persistent pain) • Attacks of severe biliary colic
- Nausea & vomiting
- Flatulent dyspepsia with intolerance to fatty meals
- Tenderness in right hypochondrium (Murphy's sign) with deep inspiration.
  Investigations → as before

D.D.

Chronic duodenal ulcer

Chronic gastric ulcer

Chronic pancreatitis

Recurrent appendicitis

Right-sided renal calculus

Chronic pyelonephritis.

Treatment :

Cholecystectomy either open or laparoscopic is the ideal treatment for symptomatic patients**. Or** in those who are diabetics or having small G.B. stones that are liable for migration and producing complications

CHAPTER 35

Obstructive Jaundice

35

CAUSES OF OBSTRUCTIVE JAUNDICE:

According to the pathology:

1. Congenital:

Biliary atresia - choledochal cyst.

2 .Inflammatory:

Ascending cholangitis- chronic pancreatitis.

3. Obstructive:

CBD stones- biliary stricture- parasitic infestation (Ascaris).

4. Neoplastic: • Carcinoma of head of pancreas

- Cholangiocarcinomas (CBD)
- Klatskin tumour (CHD).

5. Extrinsic compression of CBD or CHD :

- by lymph nodes or tumours.
- MIRIZZI Syndrome

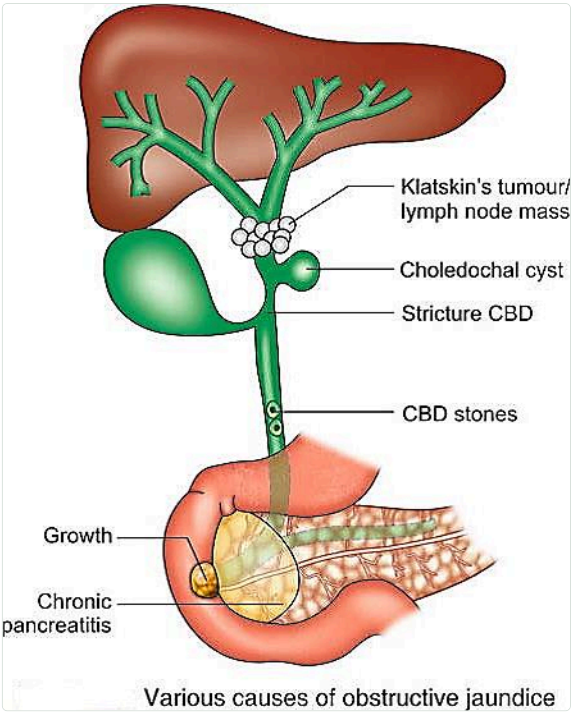


Figure 35.1 — Causes of obstructive jaundice

Another classification: According to site of obstruction:

I-Luminal causes :

CBD stones - parasitic infestation

II-Mural causes:

- Biliary atresia - Choledochal cyst
- Biliary stricture (Inflammtory - iatrogenic – malignant)
- Ascending cholangitis - sclerosing cholangitis
- Cholangio-carcinomas (CBD) - Klatskin tumour

III- Extral-mural: • Compression by lymph nodes

- Cancer head of pancreas
- **MIRIZZI Syndrome** = Hartmann's Pouch stone

  - **Mirizzi syndrome:**
  - The presence of a stone impacted in the cystic duct or the gall bladder neck (Hartmann's pouch), causing inflammation and external compression of the common hepatic duct and thus biliary obstruction and jaundice.

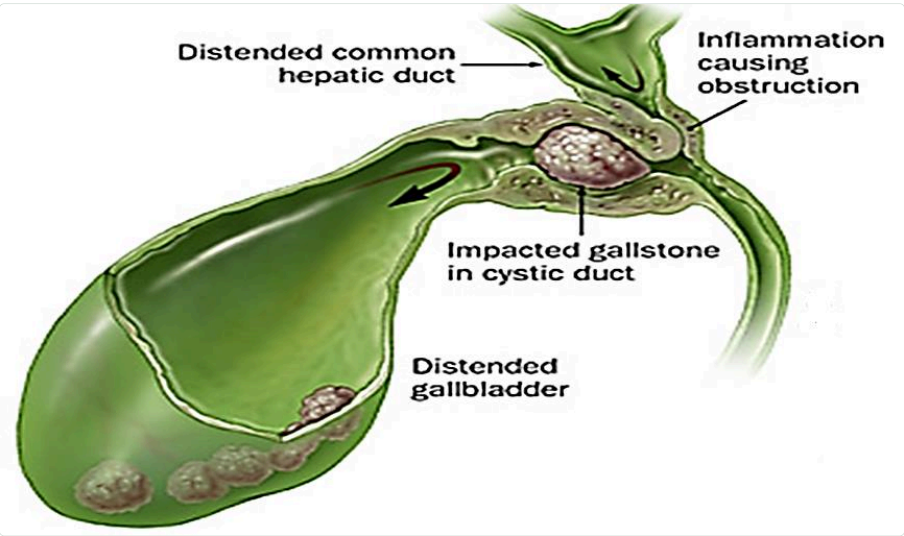


Figure 35.2 — MIRIZZI Syndrome

N.B. The most common causes of biliary obstruction are biliary stones

followed by malignancy (cancer head of pacreas and cholangio- carcinoma)

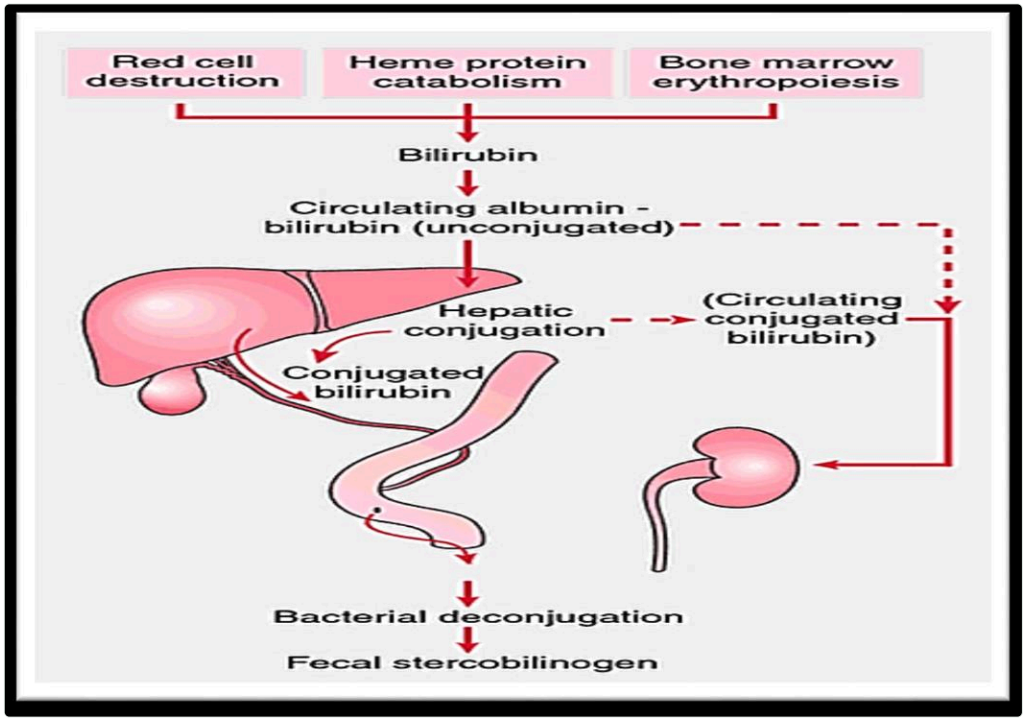


Figure 35.3 — Bilirubin metabolism

Complications of obstructive jaundice

The complications are proportional to the **duration and intensity** of the jaundice.

A. Secondary biliary cirrhosis (liver cirrhosis due to prolonged cholestasis).

B. **Acute cholangitis** is the most common complication, can cause septic shock and death.

C. **Acute renal failure**. may be due to nephrotoxic effects of bile salts, direct bilirubin, endotoxins, or inflammatory mediators.

D. **Failure of bile salts** to reach the intestine results in:

1- **Fat mal-absorption** with steatorrhea.

2- The fat-soluble vitamins **A, D, E, and K** are not absorbed, resulting in **vitamin deficiencies**. Vitamin K deficiency leads to **decreased prothrombin synthesis** → disordered haemostasis with an abnormally prolonged **PT** and bleeding tendency**. 3- Endo-toxemia** → **nephrotoxic** → **Acute renal failure**

.....

Calcular obstructive jaundice

Aetiology:

- GB stone (commonest) → migrate to CBD.

- Primary stones of CBD usually (pigment Brown)
- Meritzzi syndrome

Commonest organisms:

E-coli, klebsilla usually complicate the biliary stasis

Clinical picture:

Symptoms:

- History of fatty dyspepsia
- Yellowish discoloration of skin and sclera (intermittent jaundice)
- Clay colored stool (decreased stercobilin in stool)
- Dark urine (tea- colored) → increased direct bilirubin in urine
- Itching (bile salts)
- Sometimes biliary colic
- **Charcot's triad in case of associated infection (cholangitis):**
  1- Jaundice
  2- Pain
  3-Fever

Raynaud's pentade in associated cholangitis:

a) Charcot's tirade b) Hypotension (septicemia) c) Altered mental status (septicemia)

- The jaundice is not deep usually light yellow (D.D. of olive green in malignant obstruction).
- On examination: G.B not palpable (chronic fibrotic wall) except rarely if its neck is obstructed too (double stones) the back pressure causes bladder distension

Investigations:

Laboratory:

- **Raised :**

Direct and total bilirubin (up to 10 – 12 mg%) - Alkaline phosphatase – Gamma glutamyl transferase.

Mild elevation :

SGOT & SGPT

Prolonged:

- Prothrombin time (due to avitaminosis K)

Decreased urobilinogen in urine.

Radiological:

1- Ultrasound this is most useful initial investigation:

- Most important to show intrahepatic bile ducts dilatation
- Measure the diameter of CBD (normal up to 9 mm) (dilated in OJ)
- Comment on IHBR dilatation (dilated in OJ)
- Comment on the status of the GB and its stones
- Visualize CBD stones or areas of narrowing
- Comment on liver state (cirrhotic or not)

2- CT scan :

Done on suspected cases of malignant O J to detect pancreatic tumors

3- ERCP: the investigation of choice in obstructive jaundice

- ERCP is an endoscopic and radiologic investigation to visualize both the biliary and pancreatic duct systems.
- Endoscopically, the ampulla of Vater is identified and cannulated.
- A contrast agent is injected into these ducts, and x-ray images are taken • It can be also therapeutic for stone extraction by Dormia basket or insertion of a stent in CBD.

Risks of ERCP :

Ascending infections, - perforations, - pancreatitis

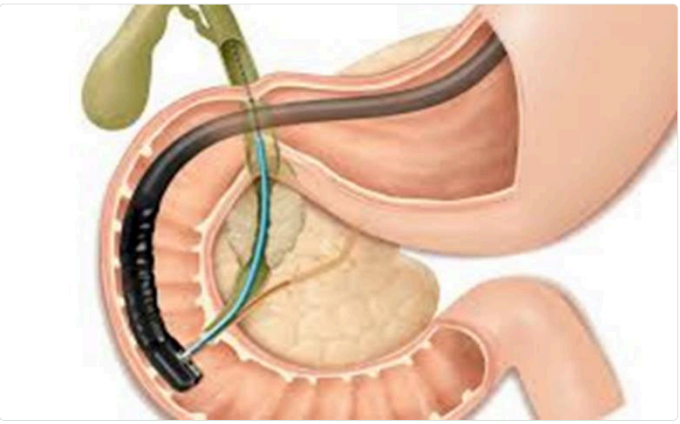


Figure 35.4 — ERCP: procedure diagram

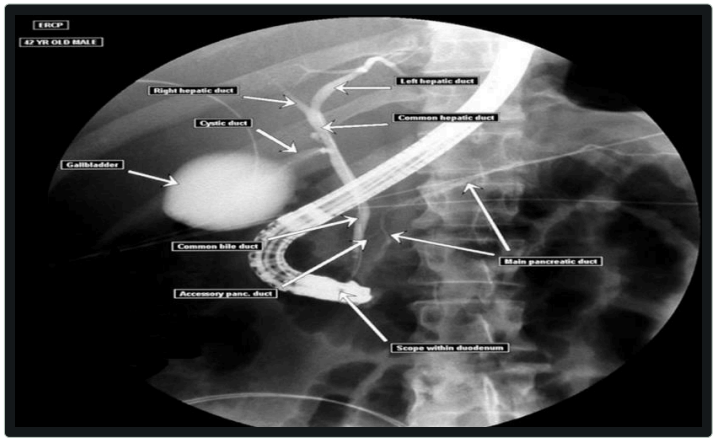


Figure 35.5 — Normal ERCP

4- MRCP

- A sensitive non-invasive method of detecting biliary and pancreatic duct stones, strictures, or dilatations within the biliary system.
- It is also sensitive TO detect cancer.

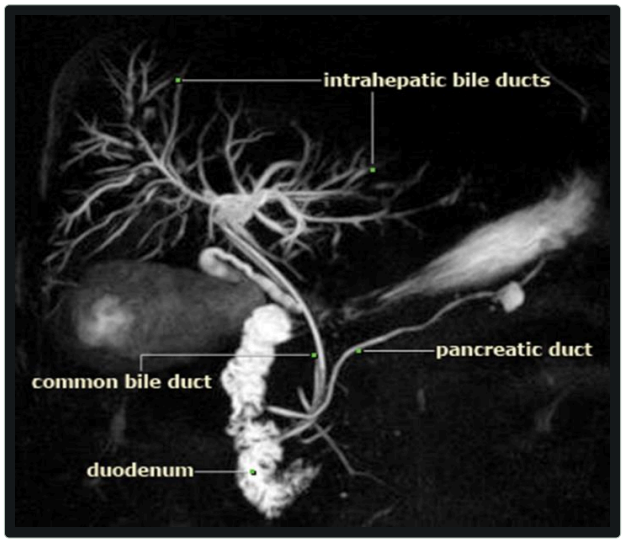


Figure 35.6 — Normal MRCP

5- PTC: percutaneous trans- hepatic cholangiography

- The liver is punctured to enter the peripheral intrahepatic bile duct system. • It is especially useful for lesions proximal to the common hepatic duct.

Management of stone CBD

1) Support the liver by correction of the general condition by I.V. fluids for hydration (glucose 10%, Ringer's lactate)

2) Support the kidney by Mannitol → increases renal blood flow (hypotension and hyperbilirubinemia together causes renal shut down)

3) Prevent infection by antibiotics, using 3 rd generation cephalosporins

4) Prevent bleeding by correction of the avitaminosis (K) by I.V. vitamin K administration.

5) **a) ERCP** extraction of calculi followed by laparoscopic **cholecystectomy**

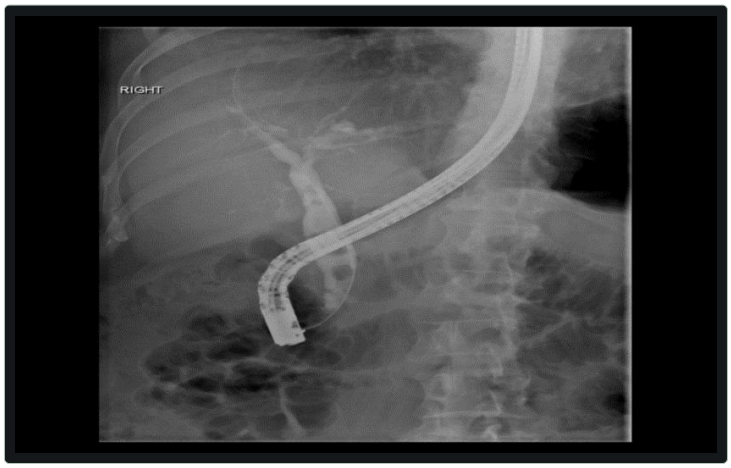


Figure 35.7 — ERCP: Stones in CBD

OR

b) Open cholecystectomy and common bile duct exploration to remove the stone.

Done when ERCP was failed or not available

T- Tube insertion which is made of Latex.

Horizontal limb is suited to lie in the supra-duodenal part of CBD.

Vertical limb comes out straight from the abdomen.

T- Tube cholangiography is done on the 5 th day and tube is extracted at • the **10 th** day after 24-hour of occlusion without problems (fever, leakage, or pain).

Early slippage of the tube (within 3-5 days) should be re-introduced

Excess bile comes out through the tube indicate distal obstruction, and

ERCP is indicated

OR

C) Cholecystectomy and Choledocho- duodenostomy:

- In dilated CBD (more than 1.5 cm)
- Stricture of the lower end of CBD
- Anastomosis is done between the proximal dilated CBD and first part of the duodenum

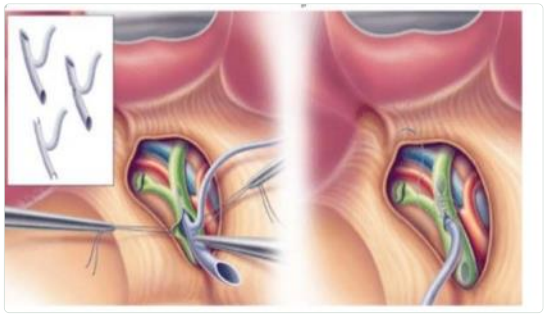


Figure 35.8 — Bile duct exploration

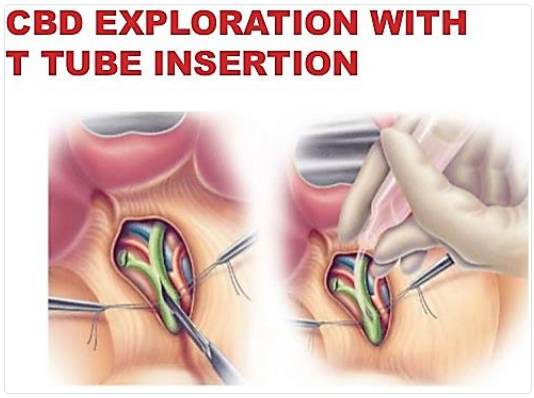


Figure 35.9 — CBD exploration with T-tube insertion

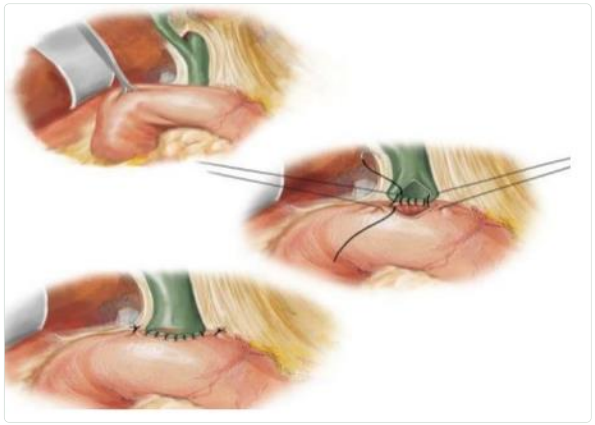


Figure 35.10 — Choledocho-dudenostomy

Differences between calcular and malignant obstructive jaundice:

Calcular obstructive jaundice

- **Usually a fatty fertile female / middle age**
  - Intermittent jaundice (can be progressive if stone is impacted)
  - Jaundice Usually not reaching high levels (deep yellow)
  - Pain is colicky in nature, and typical for biliary colic
  - G.B not palpable except rarely if its neck is obstructed too (double stone)

(Courvoisier law)

- No weight loss.
- Ultrasound is conclusive (stones, thick wall G.B., stone in CBD, Dilated CBD)
- ERCP is diagnostic and therapeutic
- Prognosis is good

Malignant obstructive jaundice:

- **Usually, old age male**
  - Progressive jaundice except very rarely in ampullary tumors where sloughing can give temporary decrease
  - Jaundice usually reaching high levels (deep or olive-green jaundice).
  - Pain is constant and referred to the back in pancreatic tumor, while it is absent in CBD tumors
  - G.B is palpable (healthy G.B. wall) except in Klatskin tumors (tumor of the confluence of CHD)
  - Associated weight loss is severe (Cachexia)
  - Abdominal mass may be felt in the epigastrium, in advanced cases
  - Ultrasound may not conclusive
  - C.T is conclusive and ERCP is diagnostic and can be palliative by inserting a stent in CBD
  - Bad prognosis

D.D. of acute pain in right upper quadrant of abdomen includes:

1. Acute cholecystitis
2. Acute cholangitis
3. Acute pancreatitis
4. Acute peptic ulcer
5. Peptic ulcer perforation
6. Liver abscess
7. Acute retro-cecal sub-hepatic appendicitis
8. Acute intestinal obstruction
9. Right sided renal colic
10. Acute right sided pyelonephritis
11. Acute myocardial infarction
12. Basal pneumonia

CHAPTER 36

Acute pancreatitis

36

It is an acute inflammation with auto-digestion of the pancreas.

Causes:

A- Gall stones: 40 %

B- Alcohol: 35%

C- Idiopathic: 10 – 30 %

D- Post- ERCP: 4%

E- Trauma: external or surgical (during splenectomy)

F- Infections: as mumps, CMV

G- Hyper-calcemia (hyper-parathyroidism)

H- Auto-immune – drug induced

Gall stones:

40% of cases

- Gall stones passing into the bile duct and temporarily lodging at the sphincter of Oddi.
- Transient obstruction of the pancreatic duct by a stone impacted in the ampulla of vater.
- Acinar cell injury occurs secondary to increasing pancreatic duct pressures caused by obstructive biliary stones at the ampulla of Vater, • Reflux of infected bile into the pancreatic duct causes activation of the pancreatic enzyme leading to pancreatitis.
- The risk of a stone causing pancreatitis is inversely proportional to its size

Alcohol:

35 % of cases

- Leads to premature enzyme activation and release.
- Alcohol can cause both acute and chronic pancreatitis.
- Alcohol causes spasm of sphincter of oddi leading to obstruction of flow of pancreatic juice.
- Alcohol is a cellular metabolic poison and causes damage of parenchymal cells.
- Alcohol causes precipitation of proteins and calcium in the pancreatic duct resulting in ductal obstruction.
- Alcohol increases the permeability of the pancreatic duct to the enzymes which leaks out into the surrounding tissues resulting in pancreatic damage.
- Alcohol causes reduction of blood supply of the pancreas resulting in ischemic damage to the pancreas.

Pathogenesis of acute pancreatitis

A. Alcohol use or gall stones can cause Acute pancreatitis by injury of the acinar cells.

B. Once cellular injury occurs, trypsinogen activated to trypsin

C. Premature activation of trypsin →Leads to activation of elastase, chymotrypsinogen, phospholipase A2, complement, and kinins → **Auto-**digestion of the pancreas

D. Elastase will digest wall of capillaries on the surface of pancreas and surrounding tissues → **hemorrhages**

E. Lipases will break down fats into FFA that combine with calcium from the blood → **calcium soaps** (white patches over the pancreas and omentum)

F. Macrophages release cytokines that lead to local and systemic inflammatory responses. (Tumor necrosis factor-alpha (TNF-α), interleukin (IL)-6, and IL-8). G. These mediators of inflammation cause an increased pancreatic vascularPermeability, leading to hemorrhage, edema, and pancreatic necrosis. H. As the mediators are excreted into the circulation, systemic complications **(SIRS)** such as :

- Acute respiratory distress syndrome (ARDS), • pleural effusions, • Renal failure.
- Hypotension and shock state

Pathological types of acute pancreatitis,

- **Acute edematous pancreatitis (80 %)** : parenchymal edema and peri-pancreatic fat necrosis occur first.

Necrotizing pancreatitis (20 %):

When necrosis involves the parenchyma, accompanied by hemorrhage..

- **Pancreatic abscess** : result from infection on top of necrotizing pancreatitis.

Clinical picture of acute pancreatitis:

History of Previous biliary colic OR alcohol consumption (major causes of acute pancreatitis)

1- Abdomenal pain:

- Sever, stay 12-24h after eating a large meal or consuming alcohol, • Radiate to the back
- Pain is worse on laying supine & better after sitting or leaning forward.

2- Bad general condition, hypotension, shock in severe cases, and Toxic look.

3- Nausea and vomiting

4- Jaundice

5- Low grade fever

6- Tachycardia

Abdominal examination:

- Abdominal distension → mainly epigastric
  - Tenderness, guarding rigidity and rebound tenderness more in the epigastrium
  - Hyper resonance (in presence of ileus)
  - Shifting dullness in marked abdominal fluid collection.
  - Silent abdomen by auscultation due to ileus.
  - **Special signs: might be absent**
  - **Grey turners sign.: (**reddish- discoloration along the flanks resulting from retroperitoneal blood dissecting along tissue planes)

Cullen's sign.:

(bluish discoloration around the umbilicus resulting from hemo-peritoneum)

Figure 36.1 — Grey Turner's sign

Figure 36.2 — Cullen's sign

Cullen's sign Grey turner's sign

D.D.

a) Perforated peptic ulcer.

b) Acute cholecystitis & biliary colic.

c) Acute Intestinal obstruction.

d) Renal colic.

Investigations:

Laboratory:

1- Serum amylase:

it remains elevated in the first 48 hours and returns normal (half-life 10 hours)

2- Lipase:

(diagnostic test): It is elevated for 7-14 days.

3- Others:

- ↑ WBC (15000 - 30000). - ↑LDH > 500 U/dl
- ↑ Glucose . - ↓ Albumin.
- ↓ Ca in serum. - ↑ AST. Bilurbin, Alkaline Ph.

- **ABG: show Hypoxia.**

Radiological:

1- Plain x- ray abdomen erect and supine:

- Exclude perforated viscus (air under diaphragm)

Sentinel loop:

air filled loop in the left upper quadrant.(duodenum)

Colon-cut off sign:

gas-filled segment of transverse colon abruptly ending at the area of pancreatic inflammation.

2- U/S:

- Gall stones.
- Biliary obstruction.(dilated CBD)
- Presence free peritoneal collection
- Pseudo-pancreatic cyst.

3- C.T:

- diagnostic even with normal amylase
- Enlarged, oedematous pancreas.
- Abscess & hemorrhage around the pancreas
- Presence of gas bubbles in CT scan indicates pancreatic abscess.
- Pseudo-cyst

N.B. Severe acute pancreatitis

According to the **Atlanta classification**, severe acute pancreatitis is evidenced by the following:

- **Evidence of organ failure**

- Systolic blood pressure below 90 mm Hg
- [P a O 2] 60 mm Hg or lower, • Serum creatinine level 2 mg % or higher, • Local complications (eg, necrosis, abscess, pseudo-cyst)
- Ranson score = 3 or higher.

Ranson's criteria

At admission or diagnosis
- Age over 55 years
- White blood count over 16,000/cu mm
- Blood glucose over 200 mg/dl
- Serum Lactic dehydrogenase (LDH) over 350 U/l
- Serum glutamic oxaloacetic transaminase (AST) over 250 U/l

During initial 48 h
- Hematocrit fall greater than 10% points
- Blood Urea nitrogen rise more than 5 mg/dl
- Arterial Po_2 below 60 mm Hg
- Serum calcium below 8 mg/dl
- Base deficit >4 meq/l
- Estimated fluid sequestration more than 6000 ml

Figure 36.3 — Ranson's criteria

- < 3 criteria: mortality = 1 %
- 3 - 4 : mortality = 15 %
- 5 - 6 : mortality = 40 %
- > 7 : mortality = 100%

Complications of acute pancreatitis:

I- General

- Shock & renal failure:(pancritic necrosis is associated with leakage of fluid in the pancreatic bed & peritoneum
  - Hypocalcemia: sequestration of Ca in fat necrosis.(saponification). • Hypo-albuminemia: Capillary permeability→ascites and edema.. • Hyperglycemia: due to disruption of pancreatic islets (↓ insulin) . • Hypoxia: Respiratory distress → end in respiratory failure.

II- Pancreatic complications

1- Necrosis. - Abscess

2- Pseudo- pancreatic cyst:

- A circumscribed collection of fluid rich in pancreatic enzymes, blood, and necrotic tissue, appear several weeks after pancreatitis.

- Typically located in the lesser sac behind the stomach
- Pancreatic pseudo-cysts are usually complications of pncreatitis.
- It differs from true cysts, which are lined by epithelium; in that they are lined with granulation and fibrous tissues.

Clinically :

- Patient complain of epigastric pain
- History of pancreatitis weeks ago
- Epigastric swelling
- Recurrent vomiting (pressing the duodenum or stomach)
- Fever in case of infection
- C.T. scan is diagnostic

Treatment of acute pancreatitis:

1. Bed rest & Oxygen.
2. NPO & NG tube.
3. IV saline to support the circulation.
4. Pethidine (75-100 mg) to control pain (avoid morphine because it causes spasm of sphincter of oddi)
5. Dopamine (if shock not respond to the fluid replacement)
6. Ca gluconate IV only if there is hypo-calcemia.
7. Fresh frozen plasma for coagulopathy.
8. Serum albumin to correct hypo-albumenemia .
9. Insulin in complicated hyperglycemia.
10. Parenteral nutrition if there are severe pancreatitis & illeus.
11. Antibiotics: prophylaxis to prevent infection
12. ERCP: if sever pancreatitis results from stone in biliary duct.
13. **Treatment of**

pseudocyst:

- Many are resolved spontanously.
- If it is not resolved after 4 - 6 weeks, surgery is done.
- Cysto-gastrostomy or cysto-jejunostomy is used through open surgery.
- Endoscopic: endoscopic cystogastrostomy
- Percutaneous drainage (C.T guided)

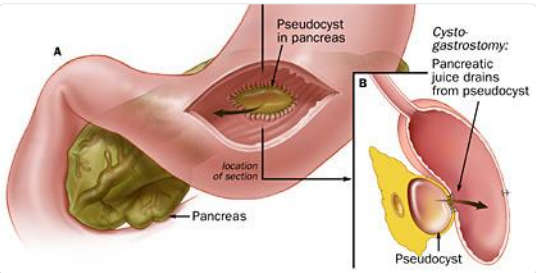


Figure 36.4 — Cysto-gastrostomy.

14- Gallstone pancreatitis

- It is optimal for patients admitted with gallstone pancreatitis to undergo cholecystectomy before discharge.
- Patients discharged with gallstone pancreatitis without a cholecystectomy are at high risk for recurrent bouts of pancreatitis.
- **(ERCP) with sphincterotomy** and stone extraction is indicated in cases with impacted stones.

15- Percutaneous placement of a drainage tube into the fluid collection or abcess under the guidance of ultrasonography or CT scanning.

CHAPTER 37

Cancer pancreas

37

Etiology:

- Pancreatic cancers can arise from the exocrine (95 %) and endocrine portions (5 %) of the pancreas.
- The exocrine portion, include the ductal epithelium and acinar cells.
- Approximately 75 % of all pancreatic carcinomas occur within the head or neck of the pancreas.
- 15-20% occurs in the body of the pancreas.
- 5-10 % occurs in the tail.

Pathology:

- Age: old age
- Sex: more in males

Predisposing and risk factors:

- **Smoking** is the most common environmental risk factor for pancreatic carcinoma.
  - Obesity
  - dietary factors: Consumption of red meat increases the risk
  - Diabetes mellitus
  - Long-standing, chronic pancreatitis increases the risk for the development of pancreatic cancer.
  - Genetic predisposition
  - Multiple endocrine neoplasia.

Microscopic picture:

- 80% are adenocarcinomas of the ductal epithelium
- Endocrine pancreatic tumors include: insulinoma – gastrinoma

Macroscopic:

mass in the head, body or tail of pancreas.

Spread:

1- Local→ nearby organs as stomach, duodenum, sup. mesentric vessels, IVC
2- Lymphatics: peri-pancreatic lymph nodes, celiac and porta-hepatis lymph nodes

3- blood: liver, lungs, bones and brain

Clinical picture: cancer head of pancreas

1- Epigastric pain → radiating to back

2- Obstruction of the bile duct→ **picture of malignant obstructive**

- **jaundice:**
- Old aged male, marked weight loss
- Deep jaundice (olive green)
- Tea-coloured urine, clay stool
- Itching
- Malabsorption symptoms
- Bleeding tendencies
- Superficial thrombophlebitis
- Duodenal or gastric obstruction → gastric outlet obstruction →Vomiting.
- Distended gall bladder (Courvoisier law)
- Hard fixed mass in the epigastrium

Investigations:

I- Lab.

- CBC : anemia
  - Liver function: bilirubin, alkaline phosphatase, Albumin • Renal functions
  - PT- PTT: elevated
  - Glucose : in endocrine tumors
  - CA19-9

II- ERCP →

- diagnostic – biopsy
- put a stent in CBD for drainage (palliative or pre-operative preparation)

III- Ultrasound → primary investigation in obstructive jaundice (state of gall bladder, CBD, Liver, ascites)

IV- Endo-luminal ultra-sound:

to detect local spread

V- PTC (Percutaneous trans-hepatic cholangiography) → Put a stent for drainage.

Figure 37.1 — ERCP: source duct image

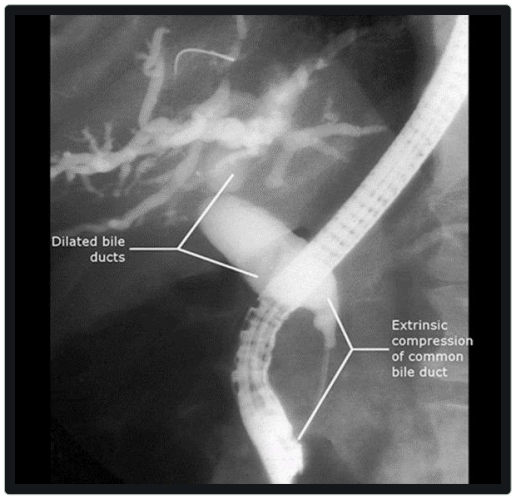


Figure 37.2 — ERCP: malignant stricture ERCP: cancer head

V- CT scan - MRI

- Site and size of mass
  - Detect metastases
  - Detect ascites.

Figure 37.3 — C.T. : cancer head of pancreas

Treatment of cancer head

Operable and resectable cases:

A- Whipple operation

B- Total pancreatectomy

- **Preoperative preparations:**

1- Preoperative stenting of the CBD through the ERCP

To drain the bile →decrease bilirubin →improves the general condition 2- Correct hypo-prothombinemia → by **intravenous vitamin K** administration. 3- Intravenous fluid administration → for proper hydration of the kidneys and improves kidney function

4- Glucose 25% → central line for liver support

5- Broad spectrum antibiotics to guard against infection.

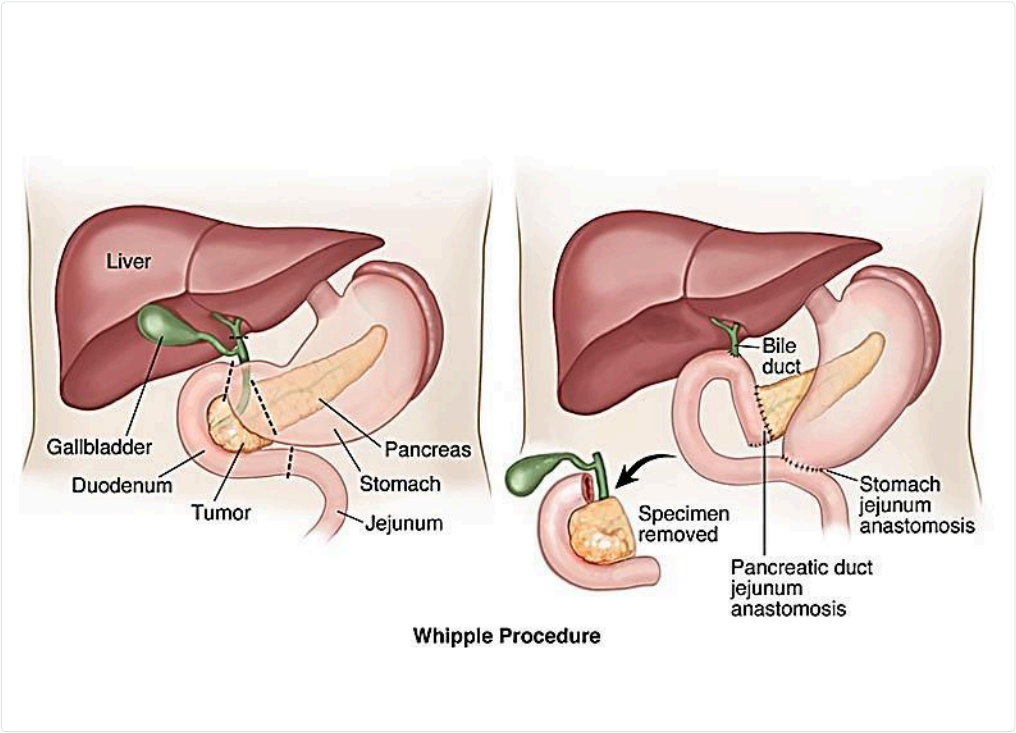


Figure 37.4 — Whipple procedure

Whipple's operation:

1- Resection of head and neck of pancreas including the tumour

2- Resection of duodenum and distal stomach

3- Resection of G.B. and lower CBD

4- Gastro-jejunostomy

5- Pancreatico-jejunostomy

6- Hepatico-jejunostomy

7- Postoperative chemotherapy

Inoperable cases:

Very bad prognosis

1- Palliative bypass:

- Gastro-jejunostomy
- Choledocho-jejunostomy
- Jejuno-jejunostomy

2- Palliative stent in the CBD through ERCP

3- PTC stenting

4- Radio-Therapy and chemotherapy

CHAPTER 38

Acute Abdomen

38

Learning Objectives

The student will

1) Understand the definition of acute abdomen

2) Know the various causes of acute abdomen, both medical and surgical causes

3) Know the clinical presentation of acute abdomen

4) Know the diagnostic studies needed to be done in acute abdomen

Definition

The term acute abdomen describes a syndrome of Sudden Abdominal Pain with accompanying symptoms and signs that Focus Attention On The Abdominal Region and in which the pain has been Present For Less Than 24 H.

The ultimate task for evaluating a case of an acute abdomen is to eventually conclude would the case be managed conservatively (medical acute abdomen) or would it be referred to a surgeon (surgical acute abdomen).

Causes

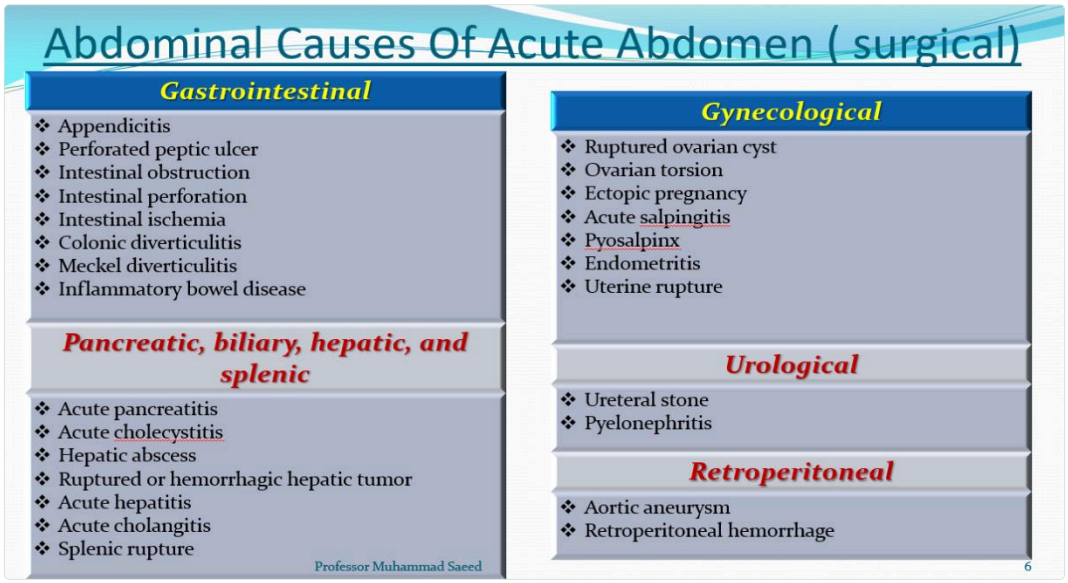


Figure 38.1 — Abdominal causes of acute abdomen: surgical

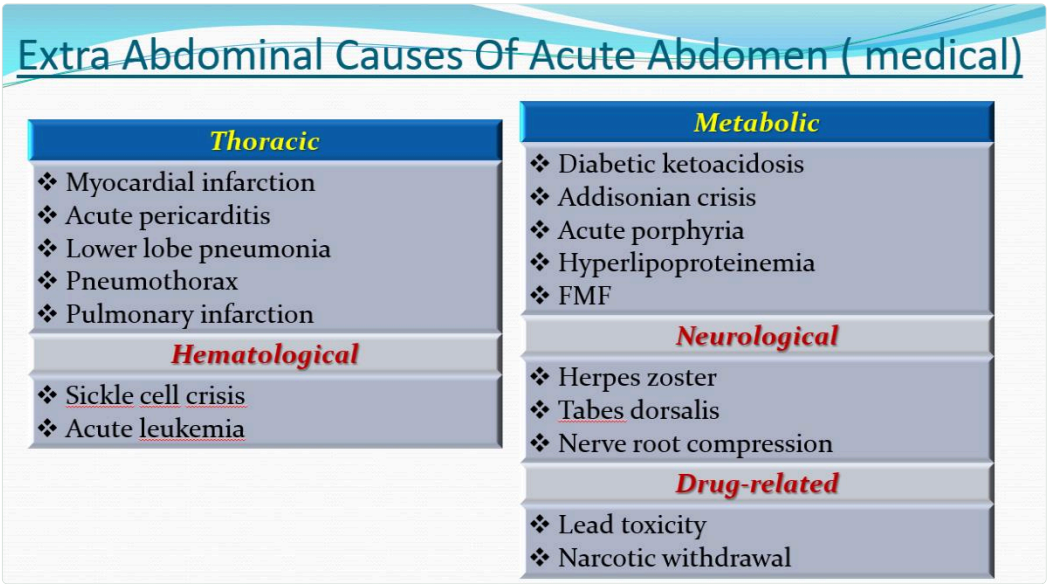


Figure 38.2 — Extra-abdominal causes of acute abdomen: medical

Clinical features:

- Sudden severe pain characterizes such events as a perforated ulcer, a ruptured viscus, a ruptured ectopic pregnancy, a spontaneous pneumothorax, or a dissecting aortic aneurysm.
- More gradual development of pain is typical of acute pancreatitis, acute cholecystitis, intestinal obstruction, bowel perforation, diverticulitis, and intraabdominal abscess.
- Vomiting Usually accompanies an acute abdomen to a variable degree.
- A clinical rule is that pain precedes vomiting in disorders requiring surgical treatment, whereas vomiting precedes pain in medically treated disorders.

Vomiting can be persistent, as with intestinal obstruction.

- Long-standing obstruction may result in feculent vomiting due to the proximal growth of colonic bacterial flora.
- Vomiting of bloody material suggests that the bleeding lesion is above the ligament of Treitz.

Clinical examination:

- Temperature elevation suggests sepsis from intraabdominal inflammation and infection.
  - Tachycardia typically accompanies an acute abdomen.
  - Blood pressure may also be elevated, although hypotension and shock may develop in patients with a perforated viscus or severe sepsis.
  - Local abdominal examination
  - The abdomen may be distended due to ascites or intestinal obstruction.

- In thin patients with complete obstruction, hyperperistalsis may be visible on the abdominal wall.
- Ecchymosis in the flanks (Grey Turner's sign) or around the umbilicus (Cullen's sign) may indicate hemorrhagic pancreatitis or a ruptured ectopic pregnancy. • Peristalsis is increased in partial or complete bowel obstruction; typically there are rushes and high-pitched sounds.
- Hypo- or aperistalsis develops in intestinal ileus from a variety of causes, including peritonitis, electrolyte imbalance, severe inflammation (e.g., toxic megacolon, pancreatitis), and long-standing intestinal obstruction.
- The examiner should listen for vascular bruits and friction rubs.
- Tympany typically is present in the distended abdomen of intestinal obstruction or toxic megacolon.
- Patients with an acute abdomen typically have abdominal tenderness.
- If peritonitis is present, either localized or generalized, there may be guarding and rigidity.
- Rebound tenderness should be elicited by gently depressing the abdominal wall with one or two fingers, then quickly releasing the pressure.
- Cough tenderness also indicates peritonitis.
- One should remember that the signs of peritonitis may be blunted in elderly or severely ill patients.
- Thus, presence of little or no tenderness does not exclude peritonitis in these patients.

Physical signs in case of acute abdomen

Murphy's sign	Cessation of inspiration during right upper quadrant examination	Acute cholecystitis
Cullen's sign	Periumbilical bluish discoloration	Retroperitoneal hemorrhage Pancreatic hemorrhage AAA rupture
Grey Turner's sign	Bluish discoloration of the flanks	Retroperitoneal hemorrhage Pancreatic hemorrhage AAA rupture
Kehr's sign	Severe left shoulder pain	Splenic rupture Ectopic pregnancy rupture
Obturator sign	Pain with flexed right hip rotation	Appendicitis
Psoas sign	Pain when raising a straight leg against resistance	Appendicitis (right side)

Evaluating a case with acute abdomen

- Laboratory tests: Complete blood count, Serum electrolytes, Arterial blood gases, Serum amylase level, Liver profile, and Urinalysis
- Imaging studies
- Ultrasonography
- CT
- IVP

Treatment

- The General Approach:

This includes intravenous fluids, nothing by mouth, and in most instances nasogastric suction to decompress the stomach and prevent additional air from entering the bowel. In some patients, it may be necessary to pass an additional long tube to decompress the bowel.

Careful attention to volume of fluid intake and urine output is important.

Serum electrolytes and arterial blood gases should be monitored

- Specific treatment depends on the cause of the acute abdomen.

CHAPTER 39

Abdominal trauma

39

Liver trauma

Only a major violence usually leads to liver injury as it is well protected

80% of cases of liver injury associated other intra-abdominal injuries.

Grades of liver injury

. Grade-I:

- Hematoma: Subcapsular, nonexpanding hematoma <10% surface area.
- Laceration: Capsular tear, nonbleeding < 1 cm of parenchymal depth.

. Grade-II:

- Hematoma: subcapsular, nonexpanding haematoma, 10–50% surface area.
- Laceration: Capsular tear, active bleeding, 1–3 cm parenchymal depth.

. Grade-III:

- Hematoma: Subcapsular hematoma >50% surface area or expanding, ruptured hematoma with active bleeding.
- Laceration: Parenchymal laceration >3 cm depth.

. Grade- IV:

- Hematoma: Ruptured intraparenchymal hematoma with active bleeding.
- Laceration: Parenchymal disruption involing 25–50% of hepatic lobe.

.

Grade- V:

- Hematoma/laceration: Parenchymal disruption involving >50% of hepatic lobes.
- Injuries of major hepatic veins or juxtahepatic vena cava.

. Grade- VI:

- Vascular injury: Hepatic avulsion.

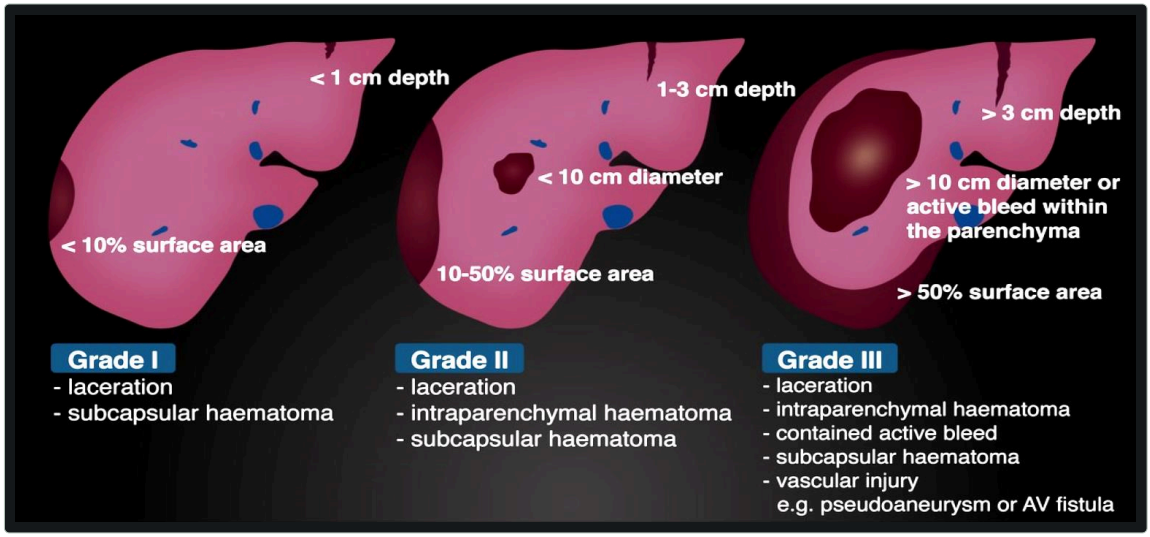


Figure 39.1 — Liver injury grades I–III

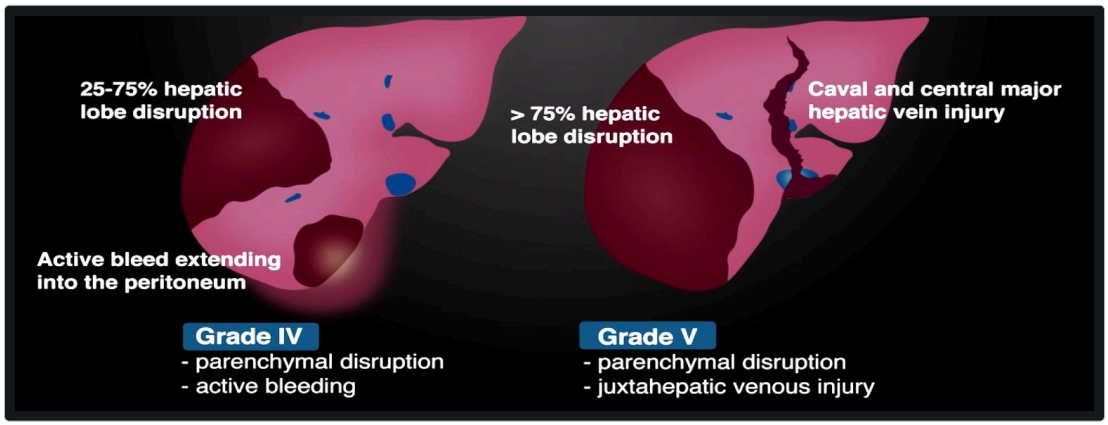


Figure 39.2 — Liver injury grades IV–V

Diagnosis of liver injury

- **Type of trauma:**

1. Severe blunt trauma in the right lower chest with fracture ribs or to the right upper quadrant of abdomen.

2. Penetrating or missile injury in the right upper quadrant.

Clinical picture:

History of abdominal trauma mainly in the right side of the abdomen and • lower rib cage

Asymptomatic cases (sometimes in small sub-capsular hematoma or

- small tear)

Pain in right hypochondrium

Picture of hemo-peritoneum

- Marked abdominal pain

- Abdominal distension (due to ileus and or hemoperitoneum)
- Vomiting
- Generalized abdominal tenderness, rebound tenderness and rigidity.

Picture of hypovolemic shock.

- Drowsiness, loss of consciousness
  - Rapid weak pulse
  - Hypotension
  - Tachypnea

Investigations for diagnosis of liver trauma

- Baseline CBC: for Hemoglobin level (determine severity of bleeding).
- **Diagnostic peritoneal lavage**—positive in patients with intraperitoneal hemorrhage.
- **Ultrasonography** of abdomen—diagnostic in majority of patients.
- **CT scan of abdomen** , accurately diagnose the grades of injury.
  Both detects the presence of hemoperitoneum, liver tears and hematomas and detect associated abdominal injuries as splenic trauma

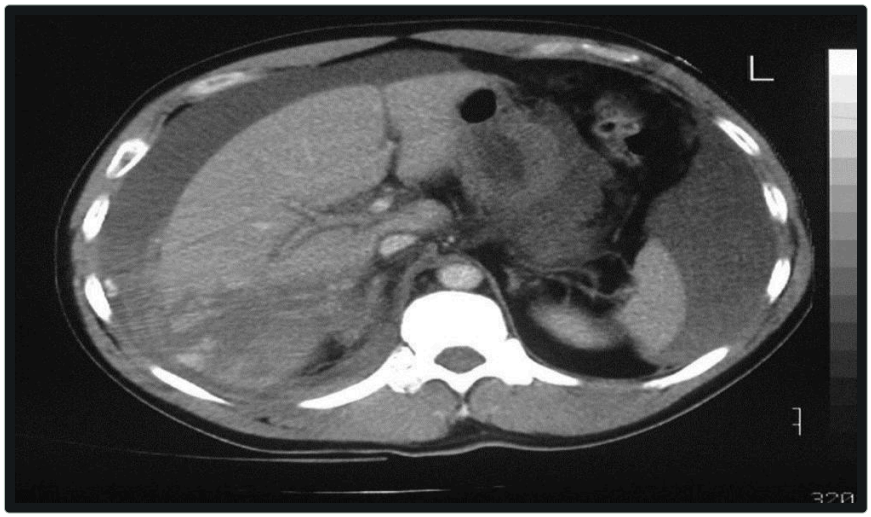


Figure 39.3 — Lacerated liver and splenic hematoma

Indications of conservative treatment of hepatic injuries

1. In majority of hepatic injuries bleeding stops spontaneously.
2. About 80% of hepatic injuries are now treated by non-operative means.
3. Patient hemodynamically stable on admission.
4. Achievement of hemodynamic stability with intravenous fluid.
5. Not requiring excessive blood transfusion for hepatic injury.
6. No other injury which requires laparotomy

Conservative treatment of hepatic injury includes:

- Intravenous fluids infusion
- Blood transfusion if required
- Analgesics
- Antibiotics
- Intensive monitoring every hourly:
  - Pulse, blood pressure, respiration and abdominal signs
    - **Serial contrast CT scan or ultrasonography** to assess the progress of hematoma or to assess continuing bleed. Free contrast in and around the liver indicates continuing bleed.

Indications of surgery for hepatic trauma

- Laparotomy is indicated for persistent or recurrent bleeding.
- Shocked patient with a distended or distending abdomen.
- Temporary response to intravenous fluid followed by hemodynamic instability.
- Failure of conservative treatment due to ongoing or recurrent bleeding. • Associated injury requiring laparotomy as bowel injuries.
- Penetrating trauma

Lines of surgical management:

- Anti-shock measures
  - Local compression
  - Pringle's maneuver
  - Suturing of the tear
  - Peri-hepatic packing
  - Definitive liver surgery as resection

On exploratory laparotomy and evacuation of hemoperitoneum a minor liver injury is found which is not bleeding. What to do?

The strategy in such situation is not to explore the hematoma.

On exploration there is minor liver injury, which is actively bleeding, what to do?

The strategy in such situation is to **compress** the tear and maintain the compression for 15 - 20 minutes. If the bleeding stops the tear sites should not be opened.

What should be the operative strategy in patients with major liver injury and active bleeding, which could not be controlled by compression?

1. When compression fails to stop the bleeding then inflow occlusion is done by applying a non-crushing clamp in the hepato-duodenal ligament to occlude the hepatic artery and portal vein. **(Pringle's maneuver) = Inflow occlusion**

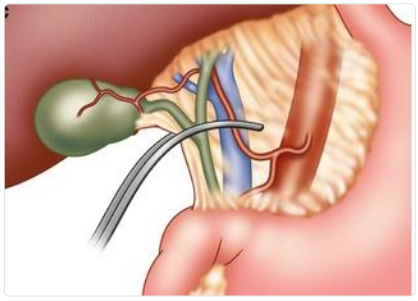


Figure 39.4 — Pringle's maneuver: inflow occlusion

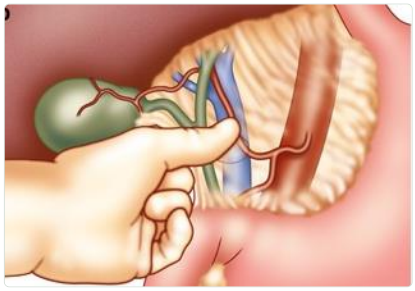


Figure 39.5 — Pringle's maneuver

2. If bleeding is controlled with inflow occlusion, then the possible source of bleeding is either hepatic artery or portal vein branches. Intermittent release of the clamp will allow to identify the site of bleeding, which may then be **suture ligated**. Direct suturing requires a good view into the depths of the injury.

3. If bleeding could not be controlled by suturing, **peri-hepatic packing** would be the approach as a damage control measure. Patient is returned to theater and re- explored after 24–36 hours when he is hemodynamically stable.

4. On re-exploration the pack is removed and bleeding reassessed.

- There may be no bleeding and no evidence of necrosis or bile leaks. No resection is required in such situations.
- The bleeding may stop but there may be non-viable or macerated liver tissue, which require debridement or resection.
- If bleeding recurs on pack removal, decision is to be taken whether to **repack** or to do definitive procedure. If the patient's condition is stable segmental **liver resection** may be done by an experienced surgeon.
- If bleeding is not controlled by inflow occlusion then the likely source of bleeding is either hepatic vein or intrahepatic inferior vena cava injury (high mortality rate).

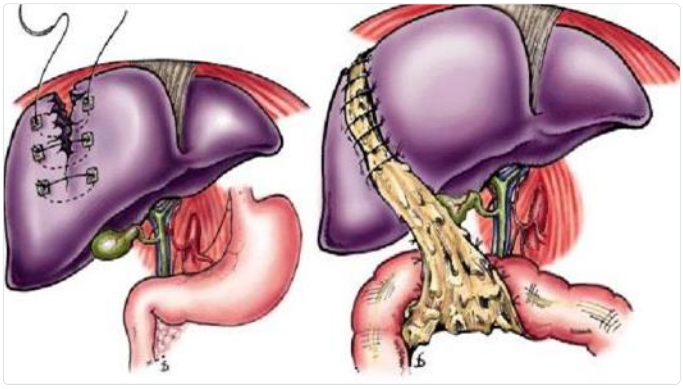


Figure 39.6 — Suturing the tear with omental patch

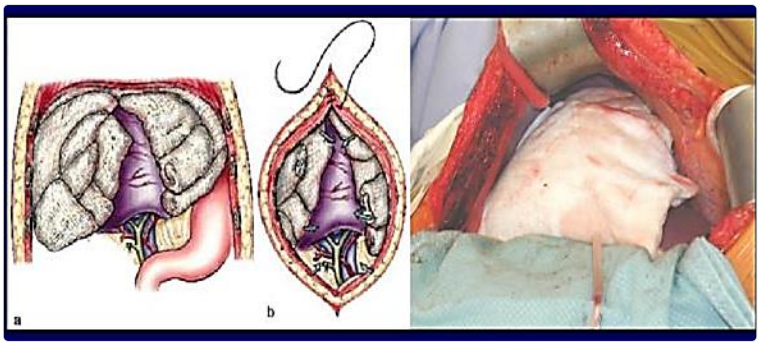


Figure 39.7 — Peri-hepatic packing

Complications following hepatic trauma:

1. Deranged liver function test—bilirubin, transaminase, alkaline phosphatase and GGT level may rise. Serum albumin level may fall.

2. External biliary fistula may occur due to minor or major ductal injury. ERCP and stenting may allow healing. Development of delayed biliary stricture may require appropriate surgery at a later date.

3. Hemobilia is rare complication. Hepatic artery angiography and selective embolization may help.

4. Sub-phrenic abscess and septicemia.

5. Pulmonary complications (ARDS).

6. Multisystem organ failure (irreversible shock).

CHAPTER 40

Splenic trauma

40

Most common indication for laparotomy after blunt trauma

Spleen is the most commonly injured abdominal organ in blunt trauma

Mechanism of trauma:

- **Blunt trauma** as in MCA, falling from height and sports.
- **Sharp penetrating trauma** as in stab wound.
- **Injuries:**
  1- External direct trauma transfer from chest wall.
  2- Direct punctures from rib fracture (9th,10th, 11th ribs).

Clinical Types of splenic injuries:

1- Delayed rupture spleen

a) Patient presented with splenic hematoma and managed conservatively. b) Presented days later signs and symptoms of intra-peritoneal hemorrhage. **2- Classical picture of rupture spleen** → discussed below

3- Fatal rupture spleen:

- Those with severe trauma with main splenic vessels injury and macerated spleen
- Death may occur in delayed management.

Diagnosis of splenic trauma:

I- Clinical II- Radiological III- DPL

I- Clinical: Classic picture of rupture spleen

A. Picture of hypovolemic shock

B. Picture of hemoperitoneum

General Symptoms:

a. History of trauma (**blunt** as MCA or **sharp** as in stab wound) b. Symptoms of hypovolemic shock (cold sensation, thirst, drowsy, loss of consciousness)

c. Symptoms of associated injuries (e.g. fractures).

Local symptoms:

a) Abdominal pains:

1. **Firstly,** left hypochondrial pain

2. Later, generalized abdominal pain (peritoneal irritation due to hemo-peritoneum)

3. Left upper quadrant pain referred to left shoulder.

b) Left sided abdominal swelling (hematoma) c) External wound in case of stab or sharp trauma.

General signs:

a) Signs of hypovolemic shock in

actively bleeding spleen:

1- Rapid weak pulse 2- Hypotension 3- tachypnea

3- Unconsciousness 4- dry mouth b) Signs of associated injuries (fractures, head injury,.....)

Local signs (abdominal examinations):

- **Inspection:**

1- External wound or hematoma

2- Bruising over left upper quadrant.

3-Kehr sign:

left upper quadrant pain, with referred left shoulder Pain when the patient lying down and leg elevated.

4- Abdominal distension (due to ileus)

- **Palpation:**

Signs of peritoneal irritation due to hemo- peritoneum:

1- Abdominal tenderness (firstly left hypochondrial then generalized 2- Rebound tenderness and guarding rigidity

- **Percussion:**

1- Hyper resonance due to ileus

2- Shifting dullness from blood in the peritoneal cavity

- **Auscultation:**

Silent abdomen due to ileus.

Investigations:

I-Radiological:

- **Ultrasound and - CT scan:**

Noninvasive, rapid, low cost

Diagnose liver, spleen, renal traumas.

Presence of intra-peritoneal fluid (blood), Replace DPL

Put grading system for splenic trauma

Figure 40.1 — Splenic trauma: CT view

Figure 40.2 — CT: splenic trauma

- **Splenic injury grading system:**

1- Grades I and II are considered minor injuries**.**

2- Grade III:

moderate injuries

3- Grades IV and V:

severe injuries (fatal)

II. DPL: diagnostic peritoneal lavage

- Blood aspirate from the lower quadrants of the abdomen indicates hemo-peritoneum
  - Sensitive

• Not frequently done nowadays

Grade	Type	Description
I	Tear ————————- Hematoma	Capsular tear, < 1cm ———————————————— – Sub-capsular, < 10% of spleen surface
II	Tear ————————- Hematoma	1–3 cm parenchymal ———————————————— – Sub-capsular, 10–50%, of spleen surface
III	Tear ————————- Hematoma	3 -10 cm parenchymal ———————————————— – Sub-capsular, > 50% of spleen surface
IV	Tear	> 10 cm Involving parenchymal segmental vessels injury
V	Tear	Shattered spleen Hilar injury

III - Laboratory investigations:

1- CBC and hematocrit value →

- Assess blood loss
- Guide for resuscitation and blood volume replacement.

2- ABG, pH and bicarb. assessment

3- Renal functions tests.

4- Routine preoperative investigations.

5- PT & PTT.

Management of splenic injuries

1 - Management of hypovolemic shock

- Insert I.V. line – central venous line and measure C.V.P.
  - Blood samples for cross matching and lab. investigations.
  - insert urinary catheter to assess renal blood flow and degree of shock and response to resuscitation
  - insert naso-gastric tube
  - starts crystalloid fluid resuscitation
  - starts blood transfusion once prepared
  - stop external bleeding wounds
  - ABC system for assessment of this poly-trauma patient (describe)

2 - Indications for Surgery : Urgent laparotomy

In **Hemodynamic instability** and continuous hemorrhage:

- SBP < 90 mmHg

- HR > 120 beats/min
- No response to resusscitation.
- Radiological evidence of major lacerations

- **Steps:**

Midline incision preferred (exploratory)

Rapid evacuation of blood and clots to asses other sources of injury: liver, mesentery, ...

Control of the vascular pedicle of spleen by clamping

Splenic mobilization:

1. Define lieno-renal ligament- divide under direct vision 2. Incision of phrenocolic ligament with the spleen retracted medially.

3. Ligation of short gastric vessels

A- Indications of splenectomy intra-operatively:

1. Unstable patient

2. Extensive injury with continued bleeding

3. Bleeding from hilar injury.

Divide splenic artery + vein after triple ligations individually: Avoid Pancreatic tail injury

Complications after Splenectomy:

1- Pancreatic tail injury → pancreatic fistula

2- Post-splenectomy thrombocytosis

3- Increase in risk for fatal pneumococcal infections and sepsis

More in children than adults, so vaccination is important post-operatively. 4- Other post-operative complications as bleeding, infections, hernias,.....

B- Splenorrhaphy: mainly indicated n children

1- Superficial hemostatic agents as gelatin sponge or topical thrombin for grade I and II 2- Suture repair

3- Mesh Wrapping

N.B. 1/3 of splenic mass needed to maintain immune-competence.

3- Non-operative management

Mainly used in pediatric age group for fear of opportunistic Infection as Pneumococcal

- **Rules:**

a- Hemodynamic stability, b- Adequate monitoring available c- Injury grade: I or II

Penetrating trauma of the abdomen

Causes:

1. Stab wound
2. Gun shot
   - Most commonly associated with vascular and/or visceral injuries
   - Multiple injuries are common
   - The associated injuries depend on the site of penetrating trauma

Lists of tissues injuries in penetrating trauma:

1. Skin- subcutaneous tissues
2. Abdominal wall muscles (tears- hematoma)
3. Parietal Peritoneum (determines the trauma either penetrating or non-penetrating)
4. Liver injuries (epigastrium- right hypochondrium)
5. Duodenum (epigastrium- right hypochondrium)
6. Stomach (epigastrium- left hypochondrium)
7. Colon (anywhere according to the site of trauma)
8. Small intestine (anywhere according to the site of trauma
9. Spleen (left hypochondrium)
10. Kidneys (loins)
11. Vascular injuries: anywhere
    - Retroperitoneal hematoma
    - Major vascular injuries (aorta, iliac, IVC,.....)

Clinical presentations:

Symptoms:

- History of trauma
- External bleeding
- Symptoms of hypovolemic shock (.....)
- Abdominal pain
- Abdominal distension

General Signs:

- Signs of hypovolemic shock (.....)

- Signs of neurogenic shock
- Signs of septic shock (neglected cases)

Local signs:

i- In perforated viscus:

- Abdominal distension
- Tenderness, rebound tenderness & guarding rigidity of the abdomen
- Hyperresonance

- Silent abdomen **ii- In solid organ injuries** (liver – spleen) as before **iii- In vascular injuries: picture of hemoperitoneum**

- Abdominal distension
- Tenderness, rebound tenderness & guarding rigidity of the abdomen
- Hyperresonance + shifting dullness
- Silent abdomen iv- Teat for peritoneal penetration via introduction of sterile catheter through the external wound.

Investigations:

1. CBC

2. Radiological:

a) Plain x-ray abdomen erect: for detection of air under diaphragm in perforated viscus

b) Pelvi-Abdominal ultrasound:

- Fluid and /or blood collections
- Solid organ injuries and grade of injury c) CT scan in stable cases

Treatment

1. Antishock measures in shocked patient (mention in details)

2. Conservative in-patient management with vital stable patients and negative radiological findings

3. Urgent Exploration in:

- Vital unstable irrespective resuscitation
- Evidence of hemoperitoneum or peritoneal irritation clinically
- Radiological evidences of abdominal collections
- Radiological evidence of perforated viscus
- Evisceration of omentum or intestine through the external wound
- Any doubt go for exploration
- Exploration can be done either through laparotomy or via laparoscopy

Procedures:

Deal with the associated injuries accordingly after proper abdominal exploration:

a) Gastric injuries: suturing in 2 layers

b) Small intestine: resection anastomosis – suturing in 2 layers in small injuries

c) Colon according to the site of injury ranging from right hemi- colectomy to proximal diverting colostomy or ileostomy

d) Major vascular injuries: repair accordingly

e) Retroperitoneal hematoma: drains are left inside

f) Finally peritoneal toilet with normal saline and drains are left inside.

رَبِّ زِدْنِي عِلْمًا

[سورة طه : 114]

ymnotes Medical

محاضرات فيديو

مراجعة شاملة

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- Forensic Medicine
- Special Sense Eye
- Ophthalmology
- Rheumatology
- Orthopedic
- Clinical Toxicology
- Cardiovascular System
- Cardiology
- Respiratory System
- Chest
- Gastrointestinal Tract
- General Surgery
- Special Surgery
- Urinary System
- Pediatric 1
- Pediatrics 2
- Surgery
- Neurology
- Endocrinology
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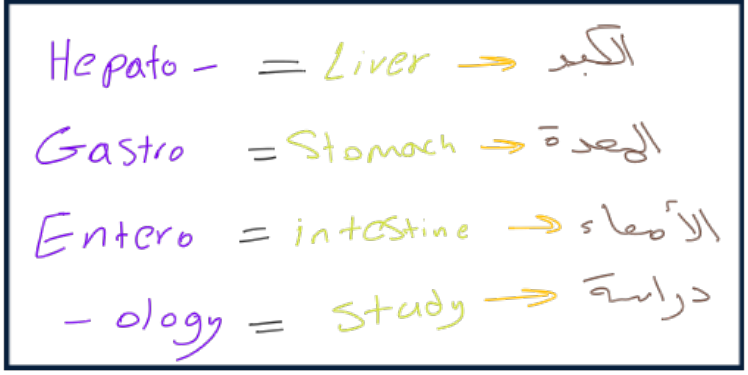

YMnotes

YMNOTES MEDICAL

باطنة جراحة Hepatogastroenterology and Hepatogastrointestinal Surgery

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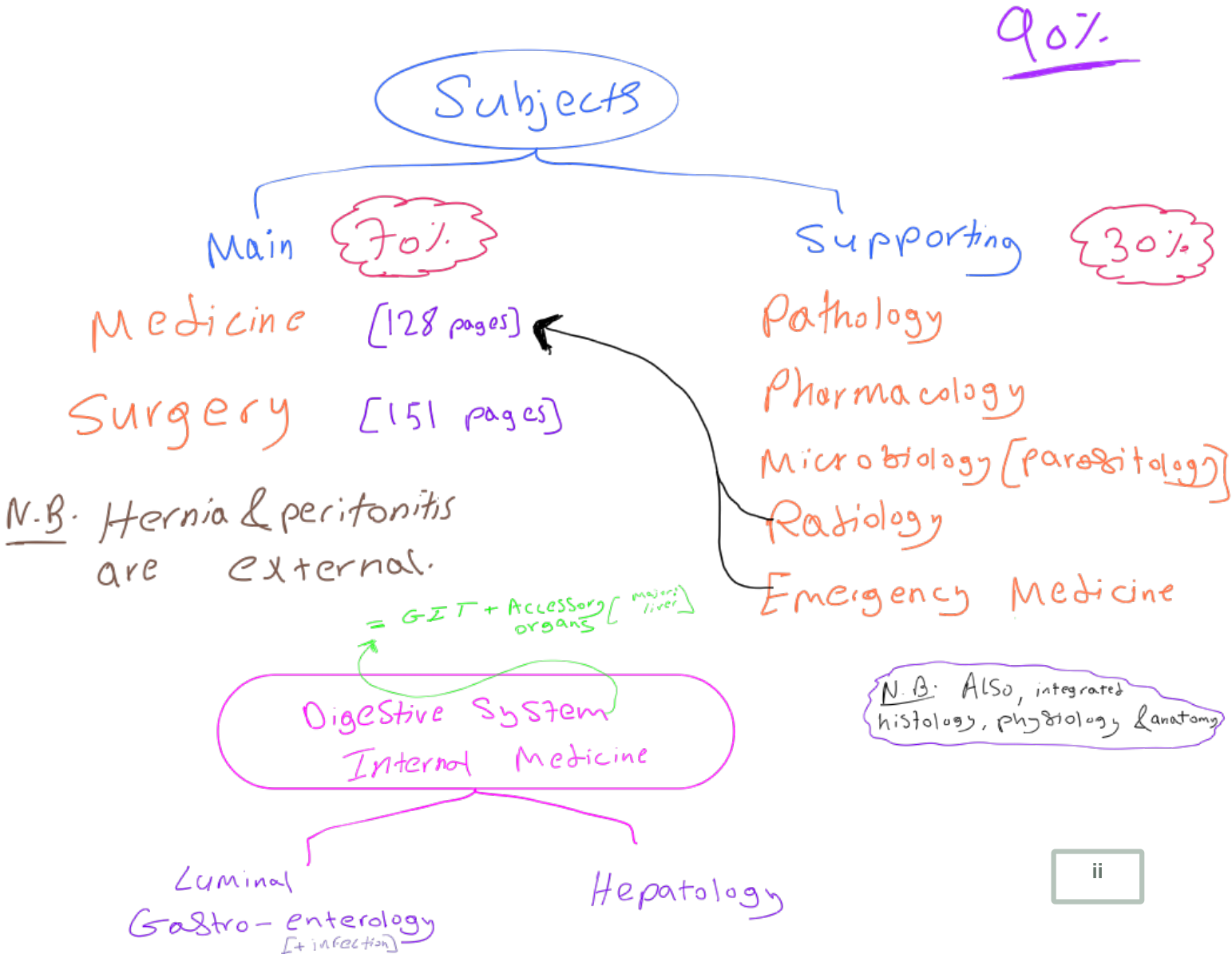

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Shortest Chapter

405 ~~388~~

Average Pages Length 405 ÷ 40 = 10.125 pages

Introduction. Gastrointestinal symptoms and signs and investigations of gastrointestinal diseases

[Clinical Manifestations]

Learning objectives

The student will

1) Know the various symptoms of gastrointestinal diseases

2) Understand the basic mechanism underlying each of the symptoms of gastrointestinal tract

3) Understand the basic diagnostic tools for gastrointestinal diseases

INTRODUCTION

- The gastrointestinal tract has many functions such as digestion, absorption and excretion as well as the synthesis of many hormones.
- The enteric nervous system is present in the gut to control its function and communicate with the central and peripheral nervous systems.
- The gastrointestinal tract ~~by the~~ encounter large foreign antigens that contains share in the expression of the immune response.

Part 1

Clinical Features

GASTROINTESTINAL SYMPTOMS AND SIGNS:

- Many of the gastrointestinal consultations are for non-organic symptoms.
- As 20% of all cancers occur in the gastrointestinal tract, therefore the clinician's main task is to recognize an organic disease among the symptoms, signs and investigations. • In developing countries, malnutrition, viral and bacterial infections and parasitic infestations further add to the picture.
- The physician must sought for 'Alarm' symptoms- Red flag symptoms:

Dysphagia, Weight loss, persistant vomiting, Anorexia, Haematemesis or melaena.

Patients with these features have a higher possibility of significant gastrointestinal pathology and should be investigated properly.

Stomatitis : It is an inflammation in the mouth from any cause (eg: ill-fitting dentures). Angular stomatitis is inflammation of the corners of the mouth .

1) **Dysphagia** :

Difficulty of swallowing (this will be discussed in diseases of the esophagus).

2) **Odynophagia** : Painful swallowing.

3) **Dyspepsia** : It is an inexact term used to describe a number of upper abdominal symptoms such as heartburn, acidity, pain or discomfort, nausea, wind, fullness or belching.

Vomiting :

Vomiting centers are located in the reticular formation in the medulla and are stimulated by the chemoreceptor trigger zone (CTZ), and by the vagus . Direct stimulation of these zones also occurs by drugs, motion sickness and metabolic causes.

Two phases will be experienced by the patient before the actual vomiting :

Nausea : associated with autonomic symptoms such as salivation , pallor , sweating

Retching : strong , involuntary , unproductive effort of vomiting associated with contraction of abdominal muscles

The amount and character of vomiting may suggest the pathology: e.g large in amount and projectile may suggest gastric outlet obstruction or upper intestinal obstruction, feculent vomiting may suggest lower intestinal obstruction

- **Table 1 : Causes of vomiting :**
  - **Any Gastrointestinal disease**
  - **Infections** (viral and bacterial)
  - **CNS diseases**
    - Increase Intracranial pressure
    - Vestibular disease
    - Migraine
  - **Metabolic causes** : Chronic renal failure , diabetic Ketoacidosis
  - **Drugs** : e.g Digoxin
  - **Reflex** : e.g. Myocardial infarction
  - **Psychogenic**
  - **Pregnancy**
  - **Alcohol**

- **Flatulence**

This term describes excessive wind. It is used to indicate belching, abdominal distension, gurgling and the passage of flatus per rectum. Among causes of increase gas production are colonic bacterial breakdown of non-absorbed carbohydrate, high- fiber diet , carbonated drinks

Diarrhea and constipation :

these will be discussed later

- Common complaint in the community and not necessarily due to organic disease.

Diarrhea :

It can be defined as change of normal bowel habits in the form of

- Increase in frequency. (> 3 movements/day)
- Increase in amount.

- Decrease in consistency

Feces exceeding 200g per day when dietary fiber content is low is considered diarrhea'

- The presence of large volume of watery stool is usually due to organic cause.
- Bloody diarrhea is usually due to a colonic cause.
- The presence of fever and acute diarrhea for several days is usually due to an infective cause
- A single episode of diarrhea is usually due to dietary changes or may be psychogenic .

Constipation:

The diagnosis of constipation is difficult as some individuals thought that they must open their bowel on daily basis , the presence of hard stool can be considered also a constipation

Constipation can be defined as the presence of 2 or more of the following:

- Infrequent Stools <3/week.
- Hard Stools >1/4 times of defecation.
- Incomplete evacuation >1/4 times of defecation.
- Straining in >1/4 times of defecation.
- Manual maneuvers as digital evacuation and support of pelvic floor >1/4 times of defecation.

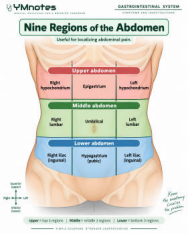
Abdominal pain:

This will be discussed also under the heading (Acute abdomen)

- Pain is stimulated mainly by the stretching of smooth muscle or organ capsules. Severe acute abdominal pain can be due to a large number of gastrointestinal conditions, and normally presents as an emergency. An apparent 'acute abdomen' can occasionally be due to referred pain from the chest, as in pneumonia or to metabolic causes, such as diabetic ketoacidosis or porphyria. .

Check:

- Intensity, character, duration and frequency of the pain
- Aggravating and relieving factors
- Associated symptoms, including non-gastrointestinal symptoms



Upper abdominal pain :

- Epigastric pain is very common and is often related to food intake. Although functional dyspepsia is the commonest diagnosis, the symptoms of peptic ulcer disease can be identical. Heartburn (a burning pain behind the sternum) is a common symptom of gastro-esophageal reflux.
- Right hypochondrial pain may originate from the gall bladder or biliary tract. Biliary pain can also be epigastric. Biliary pain is typically intermittent and severe, lasts a few hours and remits spontaneously to recur weeks or months

later. Hepatic congestion (e.g. in hepatitis or cardiac failure) and sometimes peptic ulcer disease can present with pain in the right hypochondrium. Chronic, persistent or constant pain in the right (or left) hypochondrium in a well-looking patient is a frequent functional symptom.

Lower abdominal pain:

- Pain in the left iliac fossa may be colonic in origin (e.g. acute diverticulitis) but chronic pain is most commonly associated with functional bowel disorders. Lower abdominal pain in women occurs in a number of gynaecological disorders and the differentiation from GI disease is important , but may be difficult. Pain in the right iliac fossa may be due to acute appendicitis or ilececal disease, but may also be functional

Abdominal wall pain: Persistent abdominal pain with localized tenderness, which is not relieved by tensing the abdominal muscles.

- **Anorexia and weight loss**
- Anorexia describes reduced appetite. It is common in systemic disease and may be seen in psychiatric disorders. Anorexia often accompanies cancer but is usually a late symptom and not of diagnostic help. Weight loss is almost always due to reduced food intake and is a frequent accompaniment of gastrointestinal diseases. Weight loss in malabsorption disorders is primarily due to anorexia. Weight loss with a normal or increased dietary intake may occur with hyperthyroidism and other catabolic states. Weight loss should always be assessed objectively as patients' impressions are unreliable.

Gastrointestinal (GIT)bleeding:

- It presents with one of the followings:
- Hematemesis (the vomiting of blood) and melena (the passage of black tarry stools)
- Melena can occur with bleeding from any lesion proximal to the right colon. It needs more than 50 ml blood . the black color due to blood altered by passage through the gut. • Following a bleed from the upper G I tract, unaltered blood can appear per rectum, but the bleeding must be massive and is almost always accompanied by shock. The passage of dark blood and clots without shock is always due to lower GI bleeding.

Hematochezia :

It is bright red or maroon rectal bleeding. It usually occurs with lesion distal to the ligament of Treitz (at junction between duodenum and jejunum) but it can occur with massive upper GIT bleeding with rapid transit time.

INVESTIGATIONS

- Routine haematology and biochemistry. followed by endoscopy and radiology, are the principal investigations.
- Stool examination is useful to confirm a patient's complaint (e.g. passing of blood or steatorrhoea).
- Fecal occult blood test: Can detect blood in stool that is not visibly apparent .. The new Fecal Immunochemical test (FIT) is more accurate and have fewer false positive tests
- Fecal calprotectin: when elevated in stool , it indicates intestinal inflammation , calprotectin increase is seen with inflammatory bowel disease
- Tumour markers of the gastrointestinal tract include CEA for colon cancer, CA 19-9 for pancreatic cancer, also AFP for hepatocellular cancer

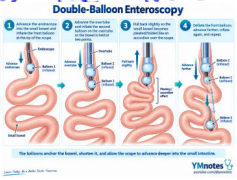
The Endoscopy

Esophagogastroduodenoscopy (EGD)

- It is the investigation of choice for upper GI disorders with the possibility of therapy and mucosal biopsy.
- Findings include reflux esophagitis, gastritis, ulcers and cancer. Therapeutic EGD is used to treat upper GI haemorrhage and both benign and malignant obstruction.
- Relative contraindications include severe chronic obstructive pulmonary disease, a recent myocardial infarction.

Sigmoidoscopy :

- Can be considered as part of the routine hospital examination in cases of diarrhea and in patients with lower abdominal symptoms such as a change in bowel habit or rectal bleeding. The rigid sigmoidoscope allows inspection of a maximum of 20-25 cm of distal colon.
- Flexible sigmoidoscopy can reach up to the splenic flexure
- **Colonoscopy**
  - It allows good visualization of the whole colon and terminal ileum. Biopsies can be obtained and polyps removed. Benign strictures can be dilated and malignant strictures stented Cancer, polyps and diverticular disease are the commonest significant findings.
- **Balloon enteroscopy**, either double or single balloon, can examine the small bowel from the duodenum to the ileum using specialized enteroscopes in expert centers.



- **Capsule endoscopy** is used for the evaluation of obscure GI bleeding (after negative gastroscopy and colonoscopy) and for the detection of small bowel tumors and occult inflammatory bowel disease. It should be avoided if strictures are suspected.

Endoscopic ultrasound (EUS) is performed with a gastroscope incorporating an ultrasound probe at the tip. It is used diagnostically for lesions in the esophageal or gastric wall, including the TNM staging of oesophageal/gastric cancer and for the detection and biopsy of pancreatic tumors and cysts.

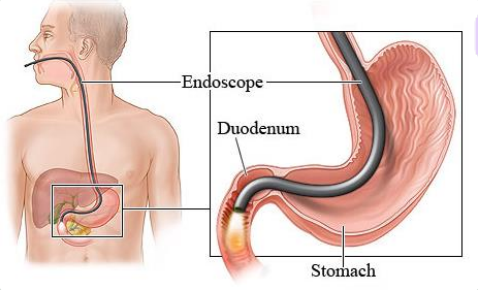


Figure 1.1 — Upper GI endoscopy

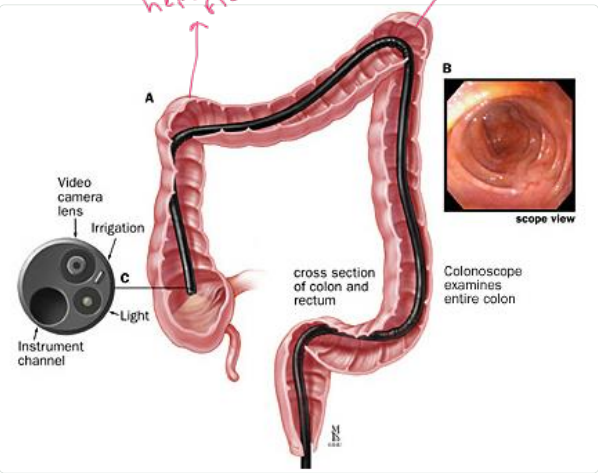


Figure 1.2 — Colonoscopy

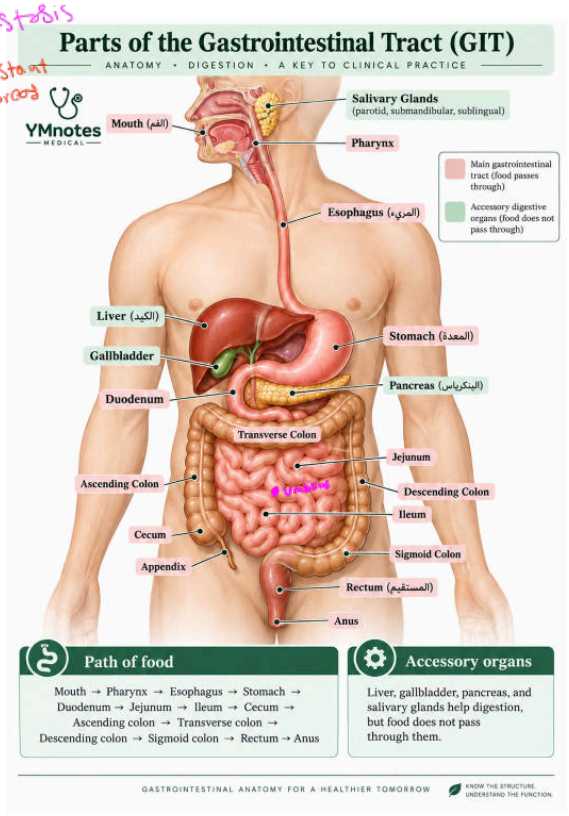


Fig : Upper GI endoscopy and colonoscopy

Imaging

Full clinical information must be provided before the examination, and ideally, the images obtained should be reviewed with the radiologist to aid interpretation. The optimal technique to be used will depend on local expertise.

1. **Plain X-rays** of the chest and abdomen are chiefly used in the investigation of an acute abdomen. Interpretation depends on analysis of gas shadows inside and outside the bowel. Plain films are particularly useful where obstruction or perforation is suspected, to exclude toxic megacolon in colitis and to assess fecal loading in constipation.

Calcification may be seen with gall bladder stones and in chronic pancreatitis, though

CT is more sensitive for both.

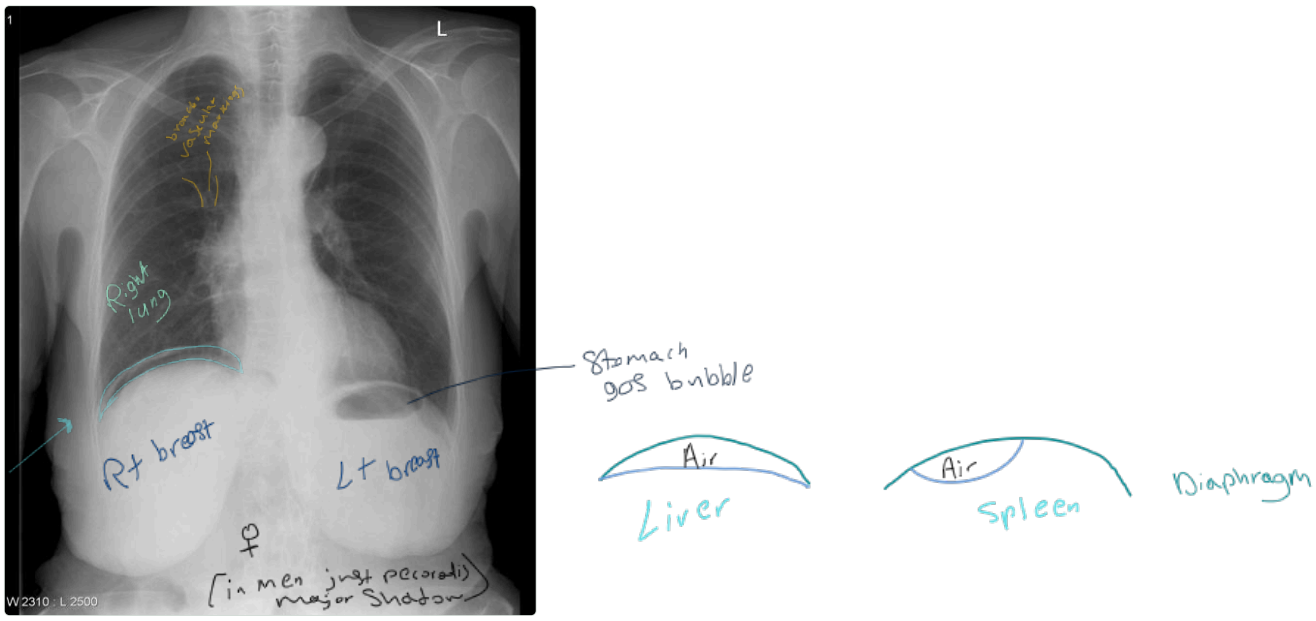


Figure 1.3 — Plain X ray showing air under the diaphragm in a case of perforated duodenal ulcer

2. **Ultrasound** involves no radiation and is the first-line investigation for abdominal distension, e.g. ascites, mass or suspected inflammatory conditions. It can show dilated fluid-filled loops of bowel in obstruction and thickening of the bowel wall. It can be used to guide biopsies or percutaneous drainage. In an acute abdomen, ultrasound can diagnose cholecystitis, appendicitis, enlarged mesenteric glands and other inflammatory conditions.

3. **Computed tomography (CT)** with intraluminal contrast injection (eg gastrografin) and intra-venous contrast injection enhance diagnostic capability (kidney functions must be assessed before the IV injection of the contrast). CT is widely used as a first- line investigation for the acute abdomen. CT is sensitive for small volumes of gas from a perforated viscus as well as leakage of contrast from the gut lumen. Inflammatory conditions such as abscesses, appendicitis, diverticulitis, Crohn's disease and its complications are well demonstrated. In high-grade bowel obstruction, CT is usually diagnostic of both the presence and the cause of the obstruction.

- CT is widely used in cancer staging and as guidance for biopsy of tumor or lymph nodes.
- CT colonography (virtual colonoscopy) after CO2 insufflation into a previously cleansed colon provides an alternative to colonoscopy for diagnosis of colon mass lesions .

.Magnetic Resonance Imaging MRI :

It has the advantage over CT that it uses no radiation and is useful in the evaluation of rectal cancers and abscesses and fistulae in the perianal region. It is also useful in small bowel disease and in hepatobiliary and pancreatic disease.

4. **Magnetic resonance enterography** is a magnetic resonance imaging technique used to evaluate bowel wall features of both upper and lower gastro-intestinal tract, although it is usually used for small bowel evaluation.[It is a less invasive technique with the advantages of no ionizing radiation exposure, multiplanarity and high contrast resolution for soft tissue.

5. **Positron emission tomography (PET)** relies on detection of the metabolism of fluorodeoxyglucose. It is used for staging esophageal, gastric and colorectal cancer and in the detection of metastatic and recurrent disease

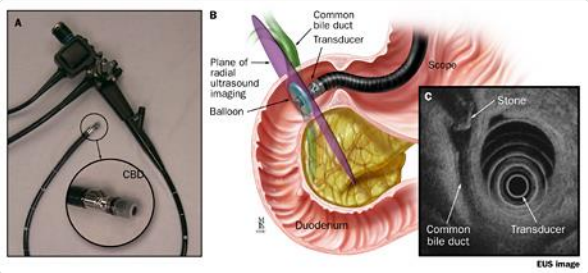


Figure 1.4 — Endoscopic Ultrasound (EUS)

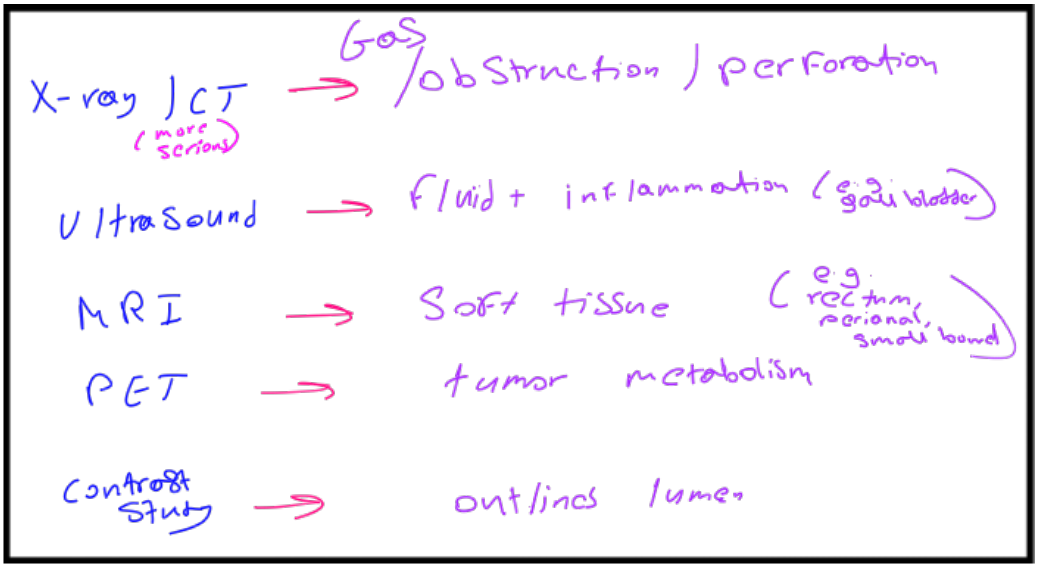
- **6. Contrast studies**
- **Barium** swallow : examines the esophagus and proximal stomach. Its main use is for investigating dysphagia.
- **Double-contrast barium** meal : examines the esophagus, stomach and duodenum, however Gastroscopy is a more sensitive test and enables biopsy of suspicious areas. • **Small bowel follow-through** : specifically examines the fold pattern and calibre of the small bowel . Specific views of the terminal ileum can be obtained and are used to identify early changes in patients with suspected Crohn's disease or TB
- **Small bowel enema** (enteroclysis) . A tube is passed into the duodenum and a large volume of dilute barium is introduced. It is particularly used to demonstrate strictures or adhesions when there is suspicion of intermittent obstruction. Generally, this has been replaced by MR enteroclysis.
- **Barium enema** examines the colon and is used for altered bowel habit. Colonoscopy and CT colonography have largely replaced this examination for rectal bleeding, polyps and inflammatory bowel disease. Absorbable water-soluble (Gastrografin) contrast agents should be used in preference to barium when perforation is suspected anywhere in the gut.

Remember the difference

Investigation	Route	Examines
Barium swallow	Mouth	Esophagus
Barium meal	Mouth	Stomach and duodenum
Small bowel follow-through	Mouth	Small intestine
Barium enema	**Anus**	**Colon**

In a **double-contrast barium enema**, both barium and air are introduced through the rectum to visualize the colonic mucosa more clearly.

Figure 1.5 — Barium enema



Causes of Dysphagia

Mechanical

A- Painful condition of the Mouth & Pharynx

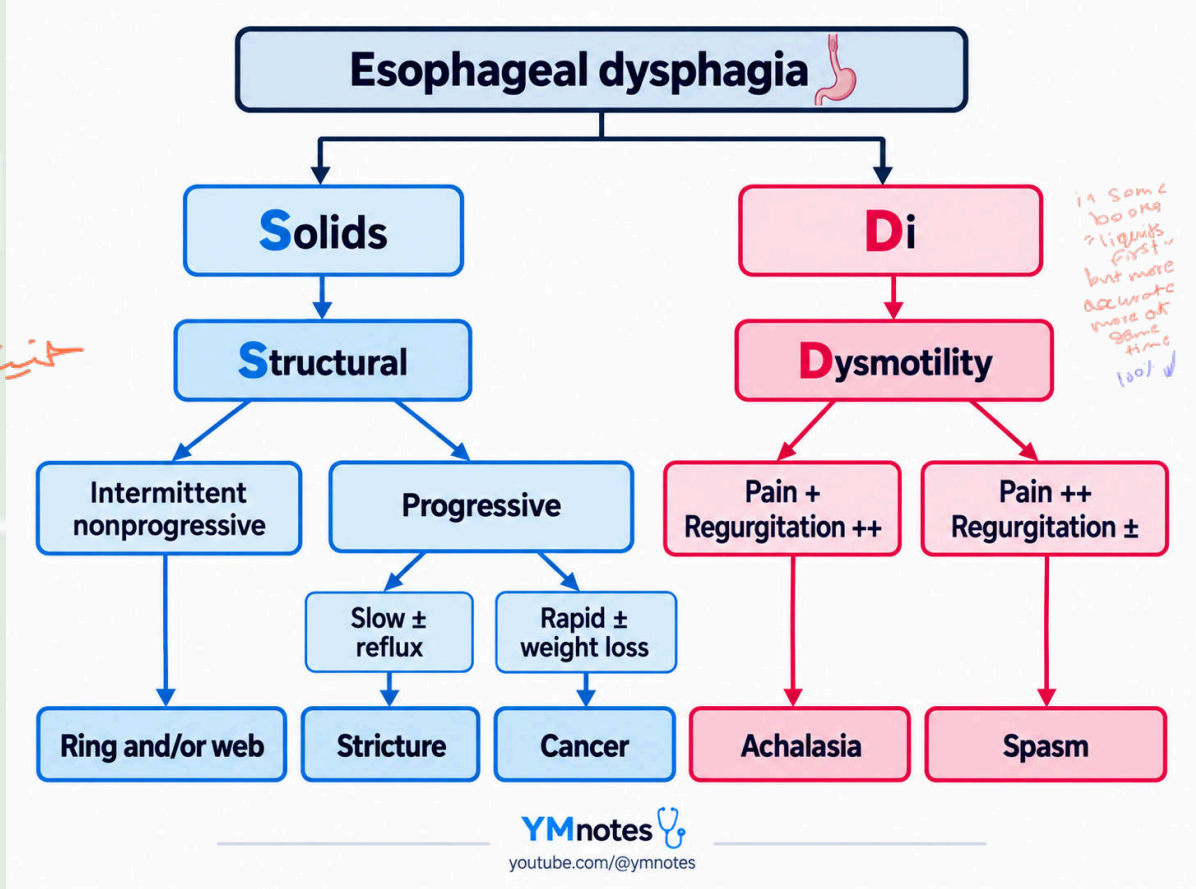
B- Esophageal
<obstructive>

- intrinsic
- extrinsic

Motor

A- Neurological

B- Functional



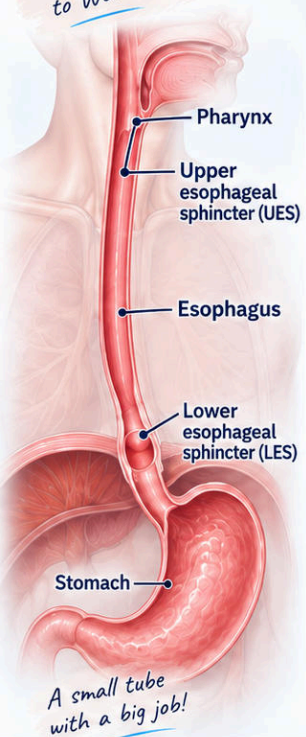
Esophagus Introduction

Summary Before GERD

From Swallowing to Wellness

Same Concepts Stronger Care

YMnotes
youtube.com/@ymnotes
Medicine Made Simple



A small tube with a big job!

1 Normal structure & function

- Muscular tube connecting pharynx to stomach
- Stratified squamous lining
- **UES** and **LES** regulate passage
- **Peristalsis** propels food
- **LES** prevents reflux

2 Key symptoms

- **Dysphagia** = difficulty swallowing
- **Odynophagia** = painful swallowing
- **Heartburn** = retrosternal burning
- **Regurgitation** = effortless return of contents

3 Dysphagia pattern clues

- Solids first → then liquids = **Mechanical obstruction**
- Solids + liquids from onset = **Motility disorder**
- Liquids worse than solids = **Motility disorder**

5 Core investigations

Endoscopy	Barium swallow	Manometry	24-h pH monitoring
= direct view ± biopsy	= anatomy / passage	= pressure & motility	= acid reflux
Visualize mucosa, take biopsy if needed	Shows structural abnormalities & transit	Assesses esophageal motility and sphincters	Detects and quantifies pathologic reflux

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